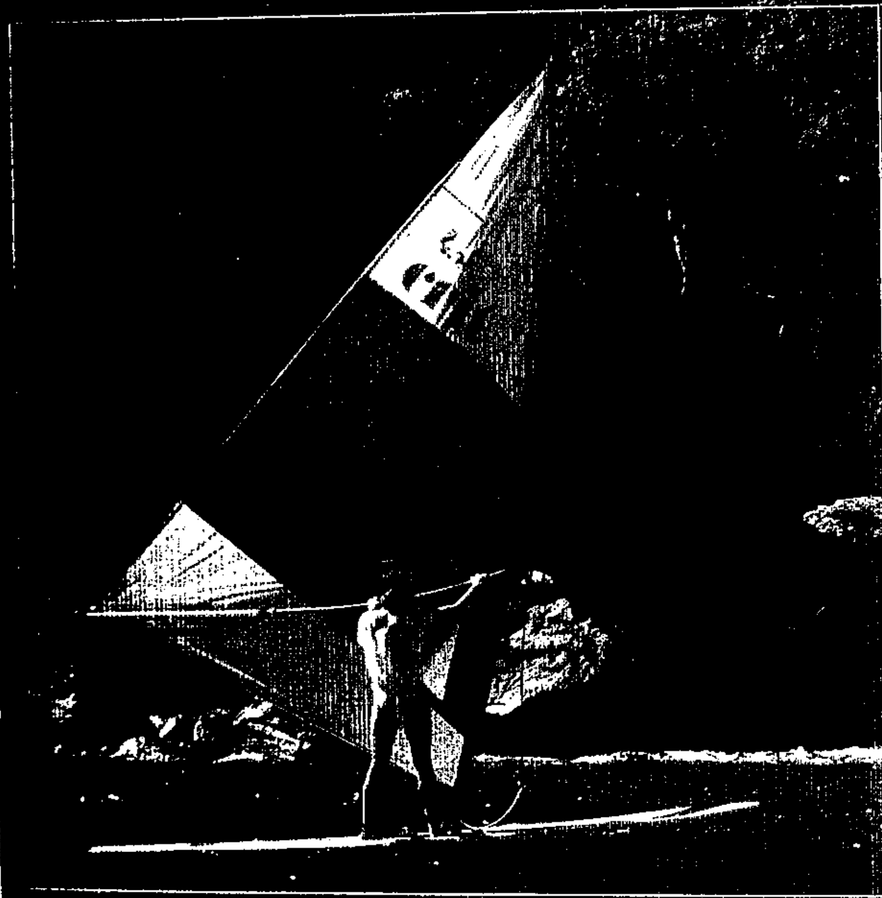
A72935

PROBLEMS IN PHYSICS

E. D. Gardiner
B. L. McKittrick



PROBLEMS IN PHYSICS

E. D. Gardiner
C.M.G., C.B.E., B.Sc., B.Ed. (Melb.), F.A.C.E.

Formerly Head of Science
Melbourne Church of England Grammar School
Victoria, Australia

B. L. McKittrick
B.Sc., B.Ed. (Melb.), Dip.Ed. (W.A.)

Head of Science
Melbourne Church of England Grammar School
Victoria, Australia

McGRAW-HILL BOOK COMPANY Sydney
New York St Louis San Francisco Auckland Bogotá
Hamburg Johannesburg Lisbon London Madrid Mexico City
Montreal New Delhi Panama Paris San Juan São Paulo
Singapore Tokyo Toronto

First published in 1969 by
Angus & Robertson (Publishers) Pty Ltd

SI Edition published in 1971

PROBLEMS IN PHYSICS: SI EDITION
Reprinted in 1975 and 1982 by
McGraw-Hill Book Company Australia Pty Ltd
Third edition 1985

A72935

National Library of Australia
Cataloguing-in-Publication data:

Gardiner, E. D. (Ernest David).
Problems in physics.

3rd ed.
ISBN 0 07 072935 2.

1. Physics—Problems, exercises, etc. I. McKittrick, B. L. (Brian L.). II. Title.

530'.076

Produced in Australia by McGraw-Hill Book Company Australia Pty Limited
4 Barcoo Street, Roseville, NSW 2069
Typeset in Australia by Monoset Typesetters, Brisbane
Printed in Australia by Macarthur Press Sales Pty Limited

Sponsoring Editor: Katrina Brown
Production Editor: Daphne Rawling
Designer: George Sirett
Technical Illustrator: Dianne Booth

CONTENTS

PREFACE

This new edition of the well-known *Problems in Physics* by E. D. Gardiner came about as a result of a request from that distinguished contributor to Australian science education and from the publishers, McGraw-Hill, for an updated edition.

While endeavouring to retain all the best characteristics of the previous edition, it seemed appropriate that this revision ought to include the following:

1. The addition of a number of *new topics* in order to cover the senior secondary syllabuses in all Australian states and New Zealand, at least at the time of writing. This has meant the inclusion of a considerable amount of new material, due to the diversity of these courses and the existence of options. Examples of new topics are household a.c. supply, gravitational potential, electronics, thermodynamics, solar, wind, wave and tidal energy, relativity, fluid dynamics, a.c. circuits, binding energy and stability of nuclei, and nuclear radiation and health.
2. Provision at the beginning of each set of problems of a *classification* of questions by sub-topics to facilitate easy location of problems on a specific sub-topic, thereby accommodating the needs of students and teachers following different syllabuses.
3. Reorganisation in many sections in order to
   (a) improve the grading of problems;
   (b) collect together problems relating to a single sub-topic; and
   (c) subdivide some longer sets into two sets, thereby also suiting the needs of different syllabuses.
4. More "*structured*" problems, with a number of carefully ordered questions designed to encourage appropriate *sequential reasoning*, especially at the beginning of a new sub-topic.
5. An increase in the number of easy, *confidence-building* problems at the beginning of sub-topics.
6. An increase in the number of questions involving the physics of *everyday life*, including some *biophysics*.
7. Inclusion of the *title* and *number* of the set of problems at the top of each page for easy recognition and location of the problem area sought by the student or teacher.
8. The inclusion of a number of "*guestimating*" questions, that is, questions requiring the student *to make sensible estimates* of physical quantities in real-life situations, with answers being expressed as an order of magnitude.
9. With some situations, which at this level *must be idealised* to allow a simple quantitative solution, the inclusion of a final *qualitative* question in order to relate the situation back to real life.
10. Deletion of a few sub-topics no longer in current syllabuses.
11. Deletion of the sets of examination questions which many teachers indicated they rarely used, preferring to concentrate on past papers from their local examining board.
12. A special introductory section of four sets covering *essential background abilities* such as familiarity with mathematical conventions in science, dimensional analysis, graphical analysis of experimental data and the concepts of density and pressure.
13. An expanded and updated description of the SI system of units.
14. Use of the latest recommendations of the appropriate international body, the *Conférence Générale des Poids et Mesures*, for the names of physical quantities, their SI units and the abbreviations for these units, including a number promulgated since the publication of the previous edition of this book.
15. In order to encourage students to think about the expected magnitudes they are *about to determine*, asking them in some questions to predict an approximate answer *before* carrying out the calculation.

ACKNOWLEDGMENTS

A large number of people have assisted with this major revision of Dick Gardiner's widely respected problem book.

First must be thanked those who responded to invitations to make constructive suggestions regarding the previous edition: Tony Brooker, Barbara Lynch, Mike Gunnourie, Terrence Lawrence, Kevin Ryan, Jim Stephens, Alastair Wilson, Sylvia Jennings, Frank Meeking and Kevin Molyneux.

I am grateful to the Incorporated Association of Registered Teachers of Victoria for permission to use questions from the October Tests. As I was a member of the committee which produced these tests for all but three years since 1969, a number of the questions will have been my own. However, many of the questions or ideas used will have originated from co-setters such as Jacki Lang, Andrew

Tait, Tony Brooker, Wes Lofts, Phil Anthony, Bruce Brown and John Morgan. Actually in most cases *ideas* as a source of problems rather than the actual test questions themselves have been used.

Inspiration for many new problems has been gleaned from internal papers set at Melbourne Grammar by Dick Gardiner himself, Gordon Jones, Tony Brooker, Fred Armitage, David Mottram and Steve Higgs.

Thanks must go also to a team of carefully selected tertiary students who worked very thoroughly through the new problems: Rod Vance, Morgan Littlewood, Tim Batten, Peter Youngman, Bruce Davie, Ranil Wickramasinghe, Greg Davis, Ross Wilson and Rowan Doyle. Technical assistance on the biological effects of radiation was provided by Dr Ken Clarke and Michelle Toomey, SRN. Some surviving errors there must be and these must remain my responsibility. Informing the publishers of any detected by teachers or students would be greatly appreciated.

Closer to home I must thank Joanne and Mark, who have given considerable support with diagram preparation and pasting up of the original manuscript, Anita, for checking the manuscript, and Michelle, for assisting with proofreading of the galleys and page proofs.

Finally, I must thank Katie Jackson and Katrina Brown of McGraw-Hill whose patience I have tested severely and who remained unruffled during the extensive period covered by this revision.

B. L. McKITTRICK

THE INTERNATIONAL SYSTEM OF UNITS

The system of units, adopted by the *Conférence Générale des Poids et Mesures* and now widely used by the scientific community throughout the world, is known as *Système Internationale d'Unités* (SI). This system has been employed throughout the book. It uses seven *base units* and two *supplementary units*, with all other *derived units* based on these nine fundamental units.

The base and supplementary units, together with the derived units with special names that are relevant to the problems in this book, are listed below with their internationally recommended abbreviations in Tables I, II and III. In some cases the full names of units have been given for the first problem or two to help students become familiar with them. However, it is recommended that they learn to use the abbreviations as soon as possible.

Table I SI base units

Physical quantity	Name of unit	Abbreviation for unit
length	metre	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
thermodynamic temperature	kelvin	K
luminous intensity	candela	cd
amount of substance	mole	mol

Table II SI supplementary units

Physical quantity	Name of unit	Abbreviation for unit
plane angle	radian	rad
solid angle	steradian	sr

Table III SI derived units with special names

Physical quantity	Special name of unit	Abbreviation for unit	Unit in terms of base, supplementary or derived units
frequency	hertz	Hz	s^{-1}
force	newton	N	$kg\ m\ s^{-2}$
pressure, stress	pascal	Pa	$N\ m^{-2} = kg\ m^{-1}\ s^{-2}$
work, energy	joule	J	$N\ m = kg\ m^2\ s^{-2}$
power	watt	W	$J\ s^{-1} = kg\ m^2\ s^{-3}$
luminous flux	lumen	lm	cd sr
illuminance (illumination)	lux	lx	$lm\ m^{-2} = m^{-2}\ cd\ sr$
electric charge	coulomb	C	As
electric potential, potential difference, electromotive force	volt	V	$J\ C^{-1} = kg\ m^2\ s^{-3}\ A^{-1}$
electric capacitance	farad	F	$C\ V^{-1} = kg^{-1}\ m^{-2}\ s^4\ A^2$
electric resistance	ohm	Ω	$V\ A^{-1} = kg\ m^2\ s^{-3}\ A^{-2}$
electric conductance	siemens	S	$A\ V^{-1} = kg^{-1}\ m^{-2}\ s^3\ A^2$
magnetic flux	weber	Wb	$T\ m^2 = kg\ m^2\ s^{-2}\ A^{-1}$
magnetic flux density	tesla	T	$N\ A^{-1}\ m^{-1} = kg\ s^{-2}\ A^{-1}$
inductance	henry	H	$Wb\ A^{-1} = kg\ m^2\ s^{-2}\ A^{-2}$
source activity	becquerel	Bq	s^{-1}
radiation absorbed dose	gray	Gy	$J\ kg^{-1} = m^2\ s^{-2}$
dose equivalent	sievert	Sv	$J\ kg^{-1} = m^2\ s^{-2}$

The internationally recognised prefixes for the SI units together with their abbreviations are listed in Table IV.

Table IV Prefixes for SI units

Prefix	Abbreviation	Value
exa	E	10^{18}
peta	P	10^{15}
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}
atto	a	10^{-18}

A number of non-SI units are still in use in scientific literature for a variety of reasons. Some of these are gradually being phased out, while some are likely to remain in use. The non-SI units that appear in this book are listed in Table V.

Table V Non-SI units

Physical quantity	Unit	Abbreviation for unit	Units in terms of SI units
time	minute	min	1 min = 60 s
	hour	h	1 h = 3.600×10^3 s
	day	d	1 d = 8.64×10^4 s
	year	y	1 y = 3.16×10^7 s
length	astronomical unit	AU	1 AU = 1.49×10^{11} m
mass	unified atomic mass unit	u	1 u = 1.66×10^{-27} kg
	tonne	t	1 t = 10^3 kg
angle	degree	°	1° = $\pi/180$ rad
energy	electron volt	eV	1 eV = 1.60×10^{-19} J
	kilowatt hour	kW h	1 kW h = 3.6×10^6 J
pressure	millimetre of mercury	mmHg	1 mmHg = 133 Pa
charge	elementary or electronic charge	e	1 e = 1.6×10^{-19} C
source activity	curie	Ci	1 Ci = 3.7×10^{10} Bq
radiation absorbed dose	rad	rad	1 rad = 10^{-2} Gy
dose equivalent	rem	rem	1 rem = 10^{-2} Sv

VALUE FOR *G*, THE GRAVITATIONAL FIELD STRENGTH

Unless instructed to the contrary in the problems throughout this book, the two necessarily equal quantities, the *gravitational field strength* and the *acceleration due to gravity*, are to be taken as 9.80 N kg^{-1} or 9.80 m s^{-2} near the Earth's surface. (In recent years a number of student texts have used the value 10 N kg^{-1} for simplicity. However, with the almost universal access to scientific calculators, this approximation seems no longer necessary.)

PART 1

TOOLS OF TRADE FOR THE PHYSICIST

SET 1 MATHEMATICAL CONVENTIONS IN SCIENCE

Topic	*Problems*
Standard form or powers-of-ten notation	1
Significant figures	2
Significant figures when adding or subtracting	3-4
Significant figures when multiplying or dividing	5-8
Significant figures when using trigonometric functions	9-12
Order of magnitude	13-14
Prefixes to SI units	15-16

1 Write the following quantities in *standard form* (also known as *powers-of-ten notation*):
(*a*) 2230 m, the height of Mt Kosciusko above sea level;
(*b*) 120 000 000 m, the diameter of the planet Saturn;
(*c*) 0.000 84 m, the thickness of a certain piece of wire;
(*d*) 0.000 000 000 25 m, the diameter of a gold atom.

2 State the number of *significant figures* in each of the following:
(*a*) 307 km, the distance from Albury to Melbourne;
(*b*) 5.0 day, the half-life of a radioactive isotope of bismuth;
(*c*) 6.3×10^{17} m, the distance from Earth to the star Regulus;
(*d*) 0.000 902 m, the thickness of a particular sheet of paper;
(*e*) 60 s, the number of seconds in one minute.

3 The lengths of the three sides of a triangular field are measured and found to be 36.27 m, 27.532 m and 30.9 m. Find the perimeter of the field to the degree of accuracy obtainable from these measurements.

4 Paying due attention to the number of significant figures in your answer, deduce how much faster Superman is at 1.4 kilometre/second than a speeding bullet travelling at 0.57 kilometre/second.

5 The length of the sides of a rectangular sheet of aluminium foil are as shown in Figure 1.1. Deduce the area of one face of the piece of foil to a justifiable accuracy.

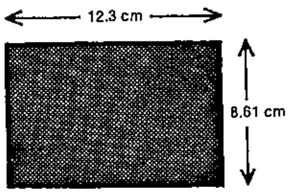


Fig. 1.1

6 The *potential difference* (in volt) across a resistor can be found from the product of its *resistance* (in ohm) and the *current* (in ampere) flowing in it. The resistance of a resistor was measured as 98.6 ohm and an ammeter showed that the current in it was 9.7 ampere. What value can be deduced from these measurements for the potential difference across the resistor? Express your answer in standard form.

7 2.0×10^{-3} kg of hydrogen occupies 2.24×10^{-2} m^3 at a particular temperature and pressure. What is the density of hydrogen, in kg m^{-3}, under these conditions? Express your answer in standard form to the appropriate number of significant figures.

8 The volume of a steel ballbearing is measured and found to be 0.523 cm^3. What is the volume of 12 such ballbearings?

9 (*a*) Using a calculator or mathematical tables, write down (to at least four significant figures) the values of the following trigonometric functions:
sin 52.3°, sin 52.4°, sin 52.5°
(*b*) Which of the following would you then regard as stating sin 52.4° to the appropriate number of significant figures?
0.7923, 0.792, 0.79, 0.8
(*c*) Write down values (to at least four significant figures) for the following:
sin 47°, sin 48°, sin 49°
(*d*) Of the following which would you then regard as stating sin 48° to the appropriate number of significant figures?
0.7431, 0.743, 0.74, 0.7
(*e*) From the two examples given above, what would you say is the appropriate number of significant figures to use for the sine of an angle, if the angle is known
(*i*) to the nearest tenth of a degree?
(*ii*) to the nearest degree?
The results you have deduced provide a *reasonable* working rule when quoting cosines and tangents as well as sines for *most* angles.

10 (*a*) How many significant figures would you quote when writing down the value of the following?
(*i*) sin 42°;
(*ii*) cos 29.7°;
(*iii*) tan 9.4°;
(*iv*) sin 114°.
(*b*) If you were told that sin $A = 0.46$, cos $B = 0.601$ and tan $C = 1.49$, state for each of the angles whether it would be appropriate to quote its size to the nearest degree or the nearest tenth of a degree.

11 Write down the values of the following trigonometric functions to the appropriate number of significant figures:
(*a*) sin 43°;
(*b*) sin 58.3°;
(*c*) cos 69°;
(*d*) tan 14.4°;
(*e*) tan 62.7°.

12 Determine and state to the appropriate number of significant figures the value of x or θ in these equations:
(*a*) $x = \cos 37°$;
(*b*) $\sin \theta = 0.89$;
(*c*) $\sin \theta = 0.891$;
(*d*) $x = \tan 68°$;
(*e*) $0.594 = \tan \theta$.

13 Express the order of magnitude of each of the following quantities:
(*a*) the half-life of uranium 238, which is 4.5×10^9 year;
(*b*) a mass of 2340 kg;
(*c*) the diameter of a wire, 7.42×10^{-4} m;
(*d*) the charge (in coulomb) on an atomic nucleus which contains 3 protons and 3 neutrons—the neutron is uncharged, and a proton carries a charge of 1.6×10^{-19} coulomb;
(*e*) the volume of a hydrogen atom, assuming that the atom occupies a cubical space having a length of 5.0×10^{-11} m on an edge.

14 The diameter of a wire is originally measured as 7.42×10^{-4} m. Heating then increased the diameter by 2×10^{-6} m.
(*a*) What is the number of significant figures in the original diameter?
(*b*) What is the order of magnitude of the original diameter?
(*c*) What was the radius of the wire before heating?
(*d*) What is the new diameter after heating?
(*e*) Find the ratio of the original diameter to the increase in diameter.

15 If you are unfamiliar with the *prefixes to SI units*, consult Table IV on page ix before attempting this question. To answer the question it is not necessary to be familiar with the units N, F, Ω and C. Rewrite and complete the following, giving your answers in standard form:
(*a*) 4 kN = N
(*b*) 5 pF = F
(*c*) 22 MΩ = Ω
(*d*) 12 ms = s
(*e*) 0.7 μC = C
(*f*) 365 nm = m

16 Express in standard form:
(*a*) an area of 5 km^2 in m^2;
(*b*) a volume of 2 cm^3 in m^3;
(*c*) an area of 1.6 μm^2 in m^2;
(*d*) a volume of 25 mm^3 in m^3.

SET 2 USING DIMENSIONAL ANALYSIS

Topic	*Problems*
Dimensions of various physical quantities	1-2
Verifying equations as dimensionally correct	3-6
Determining dimensions of physical constants	7
Deducing the form of a power law	8-11
Deducing units in hypothetical non-SI systems	12-13

While most texts limit the consideration of *dimensional analysis* to the three physical quantities *mass*, *length* and *time*, this set of problems samples more widely the seven basic physical quantities whose SI units are given in Table I on page ix. (The two quantities, *plane angle* and *solid angle*, listed with their SI supplementary units in Table II on page ix, are dimensionless, since each of them is in fact a ratio of two quantities with the same dimensions.) The method of representing the dimensions of the basic physical quantities used in questions and answers in this set is shown in the third column of Table 2.1.

Table 2.1

Basic physical quantity	Symbol for physical quantity	Dimension of physical quantity
length	l	$[l]$
mass	m	$[m]$
time	t	$[t]$
electric current	I	$[I]$
thermodynamic temperature	T	$[T]$
luminous intensity	I_v	$[I_v]$
amount of substance	n	$[n]$

1 Write down the dimensions of the following physical quantities:
(*a*) area;
(*b*) acceleration;
(*c*) force;
(*d*) impulse;
(*e*) momentum;
(*f*) work;
(*g*) refractive index;
(*h*) electric charge;
(*i*) electric field strength.

2 In the physical quantity *velocity*, the dimensions of length and time are 1 and −1 respectively. What is the dimension of each of the following?

(*a*) length in the physical quantity acceleration;
(*b*) length in density;
(*c*) time in pressure;
(*d*) mass in gravitational field strength;
(*e*) electric current in potential difference;
(*f*) time in electrical resistance.

3 Show that the following equations are dimensionally correct.
(*a*) The distance s travelled in time t by a body which starts from rest and travels with a uniform acceleration a is given by
$$s = \tfrac{1}{2}at^2$$
(*b*) The frequency f of vibration of a pendulum of length l is given by the equation
$$f = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$$

4 A sphere falling in a viscous fluid reaches a terminal velocity v_T at which the viscous retarding force and the buoyancy force balance the weight of the sphere. This terminal velocity depends on
- r the radius of the sphere,
- η the viscosity of the fluid,
- g the magnitude of the Earth's gravitational field strength,
- ρ the density of the sphere, and
- ρ' the density of the fluid.

If the dimensions of η are $[m][l]^{-1}[t]^{-1}$, determine by means of dimensional analysis which one of the following relations is correct:

A $v_T = \dfrac{2rg^2}{9\eta}(\rho - \rho')$

B $v_T = \dfrac{2rg}{9\eta^2}(\rho - \rho')^2$

C $v_T = \dfrac{2r^2g}{9\eta}(\rho - \rho')$

5 For a fluid flowing through a tube as in Figure 2.1, the pressure difference at points 1 and 2 is due to two factors: the head $(y_2 - y_1)$ of fluid between the two points and the change in speed v between these points.

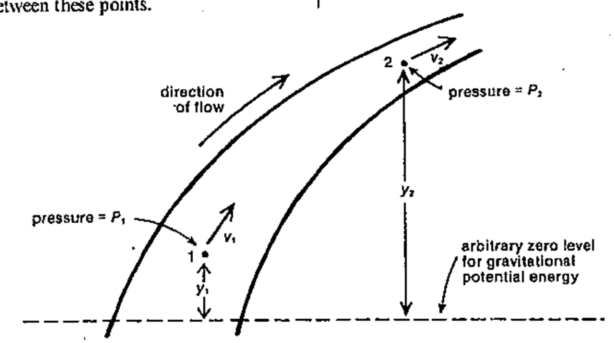


Fig. 2.1

In symbol form
$$P_1 - P_2 = \rho g(y_2 - y_1) + \tfrac{1}{2}\rho(v_2^2 - v_1^2),$$
where ρ is the density of the fluid and g is the gravitational field strength.
Rearranging, we have
$$P_1 + \rho g y_1 + \tfrac{1}{2}\rho v_1^2 = P_2 + \rho g y_2 + \tfrac{1}{2}\rho v_2^2$$
In general,
$$P + \rho g y + \tfrac{1}{2}\rho v^2 = \text{a constant.}$$
This is known as *Bernoulli's equation*. Check that it is dimensionally correct. What are the common dimensions of all three terms?

6 Without looking up texts covering jet streams, determine, *on the basis of dimensional analysis only*, whether any one or more of the four expressions below for the kinetic energy E_k of a jet stream of gas could be correct. (A is the area of cross-section of the jet stream, l its length, v its average speed, and ρ the density of the gas in the stream.)
(*a*) $E_k = 2\pi A\rho v l^2$
(*b*) $E_k = A^2\rho v^2/l$
(*c*) $E_k = 2A\rho v^2 l^2$
(*d*) $E_k = \tfrac{1}{2}A\rho v^2 l$

7 The proportionality constants in some relations are dimensionless, e.g. the "$\frac{1}{2}$" in $E_k = \frac{1}{2}mv^2$ and "$\frac{4}{3}\pi$" in $V = \frac{4}{3}\pi r^3$. However, many constants in equations in science have dimensions.
(*a*) Determine the dimensions of the universal gravitation constant G, which occurs in $F = \dfrac{Gm_1m_2}{R^2}$, where F is the gravitational force between two masses m_1 and m_2, whose centres are distance R apart.
(*b*) What then would be a unit for G in terms of kilogram, metre and second?
(*c*) Determine the dimensions of the gas constant R, occurring in the ideal gas equation
$$PV = nRT,$$
which relates the pressure P, volume V and thermodynamic temperature T of n mole of an ideal gas.

8 The speed v of sound in a gas depends on the density ρ and pressure P of the gas. If this dependence is in the form of a power law, that is,
$$v = k\rho^a P^b,$$
where k, a and b are constants (k a dimensionless one),
(*a*) determine by dimensional analysis the values of a and b, and
(*b*) thereby rewrite the above equation in a simpler form.

9 When a sphere moves through a liquid, it might be reasonable to expect the frictional force F on the sphere to depend on
- η the viscosity of the liquid,
- v the speed of the sphere,
- r the radius of the sphere, and
- ρ the density of the liquid.

$$[\eta] = [m][l]^{-1}[t]^{-1}$$
(*a*) Assuming that
$$F = K\eta^a v^b r^c \rho^d,$$
where K, a, b, c and d are dimensionless constants, use dimensional analysis to simplify the above relation by expressing b, c and d in terms of a.
(*b*) If you are told in addition that the frictional drag is directly proportional to the speed of the sphere, what further deductions can you make?

10 A girl suspects that the period T of a pendulum depends on
- m the mass of the pendulum bob,
- g the magnitude of the Earth's gravitational field strength, and
- l the distance from the point of support to the centre of the bob.

Use dimensional analysis
(*a*) to test whether her predictions are correct, and
(*b*) to find the form of the proportionality between the period and the other variables. (Commence your answer with $T \propto \ldots$)

11 Suppose a physical quantity X has non-zero dimensions of mass, length, time and electric current. If one was endeavouring to use dimensional analysis to determine the algebraic form of the dependence of X on a number of other physical quantities, would it be possible to find the form of the relation if the number of these physical quantities was
(*a*) three?
(*b*) four?
(*c*) five?

12 If the history of science had followed a slightly different development, the unit of time might have been a *click*, the unit of speed a *zoom* and the unit of mass a *lump*.
(*a*) What then would have been the unit of length?
(*b*) What physical quantity would have had the unit zoom click^{-1}?
(*c*) What would have been the unit of density?

13 On the planet Felix the scientists use the FSU system of units rather than our SI system. Their FSU system has the following basic units:
- a *hunk* (abbreviated H) as the unit of mass,
- a *zip* (Z) as the unit of velocity, and
- a *shove* (S) as the unit of force.

(*a*) In the FSU system what physical quantity would have as its unit S/H?
(*b*) For Felicians what would be their unit of time?

SET 3 REDUCTIONS OF OBSERVATIONS

Topic	*Problems*
Mathematical relationships deduced from graphed measurements	1–7
Mathematical relationships deduced from tabulated data	8–12
Relationships deduced from logarithmic graphs	13–15
Miscellaneous questions	16–19

In this set some prior understanding of the relevant physics is required to answer problems where the question is highlighted, e.g. Problems **6** and **7**.

1 In an electric meter or motor a conductor carrying an electric current at right angles to a magnetic field experiences a force. An experiment investigating the relationship between the force and the current yielded the results shown in Table 3.1. When these were graphed, they gave rise to Figure 3.1.

Table 3.1

Current (ampere)	Force (millinewton)
0.30	7.2
0.70	17.1
1.00	23.9
1.20	29.2
1.60	37.8
2.00	48.0

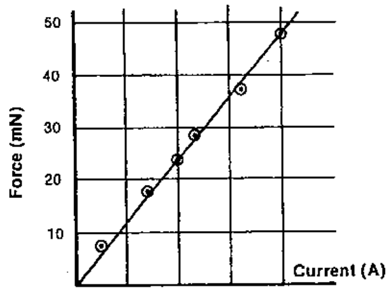


Fig. 3.1

(*a*) What properties of the line of best fit in Figure 3.1 suggest that the force is directly proportional to the current?
(*b*) If I represents the current in ampere and F the force in millinewton, deduce from the graph an equation expressing F in terms of I.

2 The electrical resistance of a coil of copper wire varies with its temperature. In an experiment investigating this effect, the measurements shown in Table 3.2 were recorded.

Table 3.2

Temperature θ (degree Celsius)	Resistance R (ohm)
20	53.7
32	55.9
41	57.5
48	59.4
57	60.9
68	63.1
76	64.9
81	66.0

(*a*) Draw a graph of the resistance R in ohm of the coil against its temperature θ in degree Celsius.
(*b*) Is R directly proportional to θ? What property of the graph supports your answer?
(*c*) From the graph deduce an equation expressing R as a function of θ.

3 A number of possible relationships between two variables d and t are given below. From the graphs in Figure 3.2, select the graph which best shows the relationship in each case. (Only positive values of the variables are to be considered.)
(*a*) d varies directly as t
(*b*) $d = 2t + 6$
(*c*) d is proportional to t^3
(*d*) $\dfrac{d_1}{d_2} = \dfrac{t_2}{t_1}$
d_1, t_1 and d_2, t_2 being two sets of corresponding values of the two variables

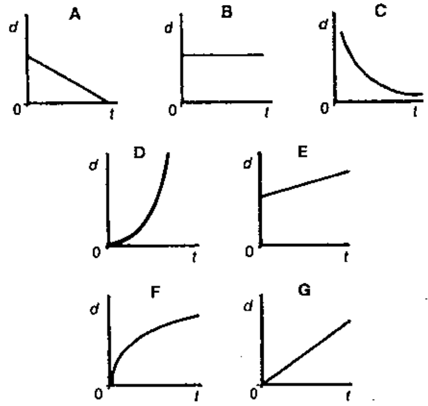


Fig. 3.2

(*e*) d is directly proportional to the square root of t
(*f*) $dt = 8$
(*g*) d is independent of t
(*h*) $d = 5 - 4t$

4 Answer key:
A From the graph obtained, deduce a relation between s and t.
B Plot a second graph of s against t^2.
C Plot a second graph of s against $\sqrt{t}$.
D Plot a second graph of s against $1/t$.
E No next step, since s is independent of t.

Suppose an experiment investigates the relation (if any) between two variables s and t and that, from your observations, you had graphed s against t. Use the answer key above to give your next step if you obtained a result similar to graph (*a*), or (*b*), or (*c*), or (*d*), or (*e*), or (*f*) in Figure 3.3.

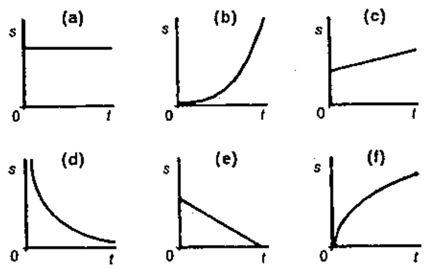


Fig. 3.3

5 F and r are two related variables, corresponding values of which are given in Table 3.3.

Table 3.3

F	r
72.0	2.0
48.0	3.0
24.0	6.0
18.0	8.0

A graph of F against r is given in Figure 3.4. By means of a graphical method, find the mathematical relation between F and r, giving the numerical value of any constants.

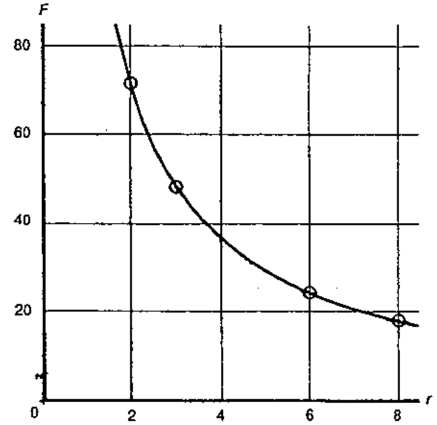


Fig. 3.4

6 The coefficient of friction (μ) for any two given dry surfaces in contact is the ratio of the limiting friction (F) to the normal (i.e. perpendicular) reaction (R) between the surfaces,

$$\text{i.e. } \mu = \frac{F}{R}$$

In an experiment to determine the value of this co-efficient for two wooden surfaces, a block of wood of mass 400 g was placed on a horizontal wooden surface. The force of limiting friction between the surfaces was measured, this measurement being made several times because accidental errors were likely to occur. The experiment was then repeated a number of times, the normal reaction being altered each time by the placing of a load on top of the block. The results obtained are set out in Table 3.4.

Table 3.4

Mass of block (gram)	Load added (gram)	Limiting friction (newton)		
400	zero	1.14	1.16	1.15
400	100	1.40	1.37	1.37
400	200	1.65	1.64	1.66
400	300	1.97	2.00	1.96
400	400	2.23	2.23	2.24
400	500	2.52	2.55	2.52

Use this data to show graphically that the limiting friction is directly proportional to the normal reaction, and deduce the value of the coefficient of friction between the two surfaces.

Be as accurate as possible in plotting the data and in drawing the appropriate curve, and explain how you interpret the graph.

7 From the data given in Table 3.5, you are required to obtain an important physical quantity, Young's modulus E, for a steel wire. This is given by the ratio of the stress in the wire to the strain produced by that stress. Stress is measured as a force per unit area; in this case the stress $= T/A$ where T is the tension imposed on the wire by suitable loading and A is the cross-sectional area of the wire. The strain is measured by the ratio of the increase in length δl produced by this stress to the wire's original length; thus strain $= \delta l/l$.

$$\text{Hence } E = \frac{(T/A)}{(\delta l/l)}$$

In the experiment the load on a vertical steel wire of length 6.0 metre and 0.65 mm in diameter is increased in steps and then decreased in steps by adding or removing masses of 2.0 kilogram from a pan attached to the lower end of the wire, the top end of the wire being fixed to a non-yielding support. The increase or decrease in length at each stage is obtained from a suitably disposed vernier scale.

Table 3.5

Load (kg)	4	6	8	10	12	10	8	6	4
Vernier reading (mm)	4.8	6.5	8.3	10.4	12.3	10.8	8.7	6.8	5.1

From these data deduce the value of E in the SI system.

8 The frequency f of vibration of a stretched string depends on the length L of the string, the tension T in it, and its mass per unit length μ.

Table 3.6 gives corresponding values for two of these variables when the other two are maintained constant. What is the proportional relation between
(*a*) f and L?
(*b*) f and T?
(*c*) f and μ?

Table 3.6

T and μ both constant	f (hertz)	150	300	450
	L (metre)	0.90	0.45	0.30
L and μ both constant	f (hertz)	50	100	150
	T (newton)	60	240	540
L and T both constant	f (hertz)	400	200	100
	μ (kilogram metre^{-1})	0.025	0.10	0.40

9 In investigating the rate P at which heat energy is conducted along a metal rod, a student varies x the length of the rod, and D the temperature difference between its ends. Her seven sets of measurements are shown in Table 3.7.

Table 3.7

Measurement set	x (cm)	D (°C)	P (kJ s^{-1})
1	5	50	0.53
2	5	100	1.05
3	5	150	1.57
4	5	200	2.11
5	10	100	0.51
6	15	100	0.35
7	20	100	0.26

Answer the following *without drawing graphs*.

(*a*) Of the seven sets of measurements, which would you use to investigate the proportionality between P and D?
(*b*) Do the measurements you select suggest a direct or inverse relation between P and D?
(*c*) In order to find the relation between P and D, which of the following would you test for constancy: PD or P/D?
(*d*) Deduce the proportionality existing between P and D.
(*e*) In order to investigate the relationship between P and x, which of the seven sets of readings would you use?
(*f*) Deduce the proportionality between P and x.

10 Water is kept at a constant height in a tank. Water flows from the tank through a number of horizontal tubes of different lengths and cross-sectional areas. These tubes are all attached to the tank at the same level. The volume of water that flows from each of these tubes in 1 minute (the flow rate) is recorded in Table 3.8.

Table 3.8

Length of tube L (cm)	Cross-sectional area of tube S (cm^2)	Flow rate V (cm^3/min)
10	0.010	90
20	0.010	45
30	0.010	30
30	0.020	120
30	0.030	270

(*a*) What graph would be the most useful in predicting the flow rates from tubes of various lengths, all of cross-sectional area 0.010 cm^2, and all at this same level in the tank?
(*b*) What graph would be the most useful in predicting the flow rates from tubes of different cross-sectional areas, all of length 30 cm, and all at this same level in the tank?
(*c*) To what quantity is the flow rate directly proportional?

11 Spheres are allowed to fall vertically through a tank of oil, and are timed between fixed points 1 metre apart. The times of fall for spheres of various radii and densities are given in Table 3.9.

Table 3.9

Density of sphere ρ_s (g cm^{-3})	Radius of sphere r (cm)	Time to fall 1 metre t (s)	$(\rho_s - \rho_0)$ (g cm^{-3})
4	1.00	9	3
4	0.75	16	3
4	0.25	144	3
7	0.25	72	6
10	0.25	48	9

Density of oil, ρ_0, is 1 g cm^{-3}.

(*a*) What graph would be most useful in predicting the time to fall 1 metre for spheres of various radii, all of fixed density 4 g cm^{-3}?
(*b*) What graph would be most useful in predicting the time to fall 1 metre for spheres of various densities, all of fixed radius 0.25 cm?
(*c*) To what quantity is t directly proportional?

12 Four separate flywheels can each be accelerated about its axle by means of a mass m kg, attached to the end of a rope wound around the flywheel, which is allowed to fall.

The number of revolutions through which the wheel turns from rest in 10 s (N_{10}) and in 20 s (N_{20}) are recorded in Table 3.10.

Table 3.10

Wheel	Mass of the wheel M (kg)	Radius of the wheel R (cm)	Number of revs in 10 s N_{10}	Number of revs in 20 s N_{20}
1	2.0	20	20.0	80.0
2	4.0	20	5.0	20.0
3	4.0	10	20.0	80.0
4	8.0	10	5.0	20.0

(*a*) For wheels of the same radius (R), what is the proportionality between N_{10} and M?
(*b*) Let t be the time taken for a flywheel to rotate from rest through a given number of revolutions N. For wheels of the same mass M and radius R, what is the proportionality between N and t?

(*c*) A flywheel has a mass of 1.0 kg and a radius of 20 cm. Starting from rest, through how many revolutions would this flywheel turn in 10 s under the effect of the falling mass of m kg as described above?
(*d*) Starting from rest, through how many revolutions would Wheel 1 turn in 5 s when a mass of m kg is attached to the rope as before?
(*e*) For wheels of the same mass, what is the proportionality between N_{20} and R?

13 In Figure 3.2 graph types B, C, D, F and G all fit the general relation $d = kt^n$, where k and n are constants.

(*a*) In which of the five types would n have the following values?
 (*i*) $n > 1$;
 (*ii*) $n = 1$;
 (*iii*) $1 > n > 0$;
 (*iv*) $n = 0$;
 (*v*) $n < 0$.
(*b*) From the equation $d = kt^n$, deduce an expression for $\log d$ as a function of $\log t$.
(*c*) If $\log d$ was graphed against $\log t$, then in terms of k and n, what would be
 (*i*) the gradient, and
 (*ii*) the intercept (also known as the data zero) on the $\log d$ axis?

14 (*a*) From the information given in Figure 3.5, deduce equations expressing
 (*i*) $\log_{10} P$ as a function of $\log_{10} V$;
 (*ii*) P as a function of V.
(*b*) From Figure 3.6, deduce equations expressing
 (*i*) $\log_{10} T$ as a function of $\log_{10} m$;
 (*ii*) T as a function of m.

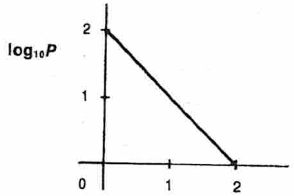


Fig. 3.5

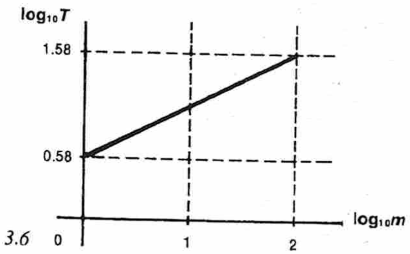


Fig. 3.6

15 An experiment was designed to investigate the dependence of the rate R at which heat energy is given off by a black radiating surface and the thermodynamic temperature T of that surface. R was measured in joule emitted per second from each square metre of the surface, i.e. in watt per square metre. Table 3.11 lists the readings obtained.

Table 3.11

R (W m^{-2})	T (K)
315	273
625	324
1 100	373
2 650	465
6 505	582
17 000	740

Assuming the dependence of R on T is of the form $R = kT^n$, where k and n are constants, draw a graph of $\log_{10} R$ against $\log_{10} T$ to determine the values of k and n (to one significant figure), and deduce an expression for R in terms of T.

16 In an experiment to determine the relation between thermo-e.m.f. and temperature for a thermocouple, a special form of slide wire potentiometer is used to compare the e.m.fs. One junction of the thermocouple is kept at the ice point while the other is immersed in a vessel of water whose temperature is observed with an accurate mercury thermometer. Simultaneous readings of the thermometer and the length of potentiometer wire necessary for balance are listed in Table 3.12.

Table 3.12

Temp (°C)	15.2	27.6	39.7	51.0	62.3	71.9	80.5
Length of wire (cm)	39.7	72.0	102.3	122.1	160.8	184.8	208.0

When balanced against a known e.m.f. of 1.018 millivolt the length of the potentiometer wire is 61.7 cm.

(*a*) Draw a circuit diagram for the arrangement.
(*b*) From a suitably drawn graph, determine a relation connecting the e.m.f. of the thermocouple with the temperature difference between its junctions.
(*c*) What is the melting point of ferrous sulphate if the observed thermo-e.m.f. is 2.74 millivolt?

17 It having been established that the frequency of vibration of a stretched string is inversely proportional to the length of the vibrating segment, the following experiment was performed to investigate the relation between the frequency and the tensile force in the string.

Two wires were stretched across a sounding board. Wire A was adjusted so that the tensile force would remain constant, and the length of the vibrating segment was varied until this wire produced a note in unison with a tuning fork of frequency 192 Hz. The length was found to be 54.1 cm.

Wire B was adjusted so that its length would remain constant, a load weighing 120 N was hung on it, and the length of the wire A was then altered until both wires produced the same note. This length (40.6 cm) was noted.

Without altering either the tensile force in A or the length of B, the experiment was repeated several times with a different load on B each time. The observations are tabulated in Table 3.13.

Table 3.13

Length of A (cm)	Tensile force in B (N)	Frequency of A (Hz)
49.7	80	
44.5	100	
40.6	120	
39.0	130	
37.6	140	
35.2	160	

(*a*) Calculate the frequency of vibration of the wires in each case.

(*b*) Determine either graphically or by calculation the proportionality between the frequency of wire B and the tensile force in the wire.

18 The bulb of a constant volume gas thermometer, originally at a temperature of 0°C, was placed in a water bath and heated gradually. The mercury surface on the closed side was kept at a constant level and the difference in the mercury levels was measured at different temperatures.

The results are set out in Table 3.14. For all temperatures shown in the table, the mercury surface on the open side was the higher of the two surfaces, but at 0°C this surface was lower than that on the closed side.

(*a*) Draw a graph of the difference in levels against the temperature, and use it to determine the difference in the mercury levels at 0°C.

(*b*) Explain, from theoretical considerations, why the graph is a straight line.

(*c*) From the graph find the temperature of the air in the bulb corresponding to a difference in mercury levels of 9.6 cm.

(*d*) Check your answer to (*c*) by calculation of the temperature from the data given.

Table 3.14

Temperature (°C)	20	40	60	80	100
Difference in mercury levels (cm)	1.9	7.3	12.7	18.1	23.5

19 A voltmeter, of unknown resistance X ohm, was connected in series with a variable resistance R ohm and an accumulator of e.m.f. E volt. The maximum reading on the voltmeter scale was less than the e.m.f. of the accumulator.

The reading of the voltmeter, V volt, was noted corresponding to different values of the variable resistance R ohm, the results shown in Table 3.15 being obtained.

Table 3.15

Variable resistance R ohm	Voltmeter reading V volt	Values of $\frac{1}{V}$
1000	5.25	
1500	4.83	
2000	4.48	
2500	4.20	
3000	3.95	
3500	3.73	
4000	3.52	
4500	3.32	
5000	3.17	
5500	3.01	
6000	2.88	

(*a*) Draw a diagram of the circuit and use Ohm's law to show that

$$R = EX\frac{1}{V} - X$$

(*b*) Draw a graph of the values of R against the reciprocals of the values of V and hence obtain an estimate of the resistance of the voltmeter. (On the R axis, allow for values from −6000 to +6000 ohm.)

SET 4 CONCEPTS OF DENSITY AND PRESSURE

Topic	*Problems*
Density as mass per unit volume	1–9
Pressure as force per unit area	10–14
Pressure at a depth in a liquid	15–19

1 If 2 m^3 of water has a mass of 2×10^3 kg, what is the density of water in kg m^{-3}?

2 (*a*) The density of benzene is 0.879 g cm^{-3}. Express this density in SI units, i.e. kg m^{-3}.

(*b*) If the density of a substance is x g cm^{-3}, in terms of x what is the density in kg m^{-3}?

(*c*) If the density of a material is y kg m^{-3}, in terms of y what is its density in g cm^{-3}?

3 20.0 cm^3 of mercury is found to have a mass of 272 g. What is the density of mercury in

(*a*) g cm^{-3}?

(*b*) kg m^{-3}?

4 A packet of self-raising flour measures 0.22 m by 0.12 m by 0.080 m. If the density of self-raising flour is 950 kg m^{-3}, what mass of flour is printed on the packet?

5 Some parts of interstellar space have a density of about 10^{-21} kg m^{-3}. Imagine in such a region a cube of space which has a total mass of 1 kg. For this cube of space what would be

(*a*) the volume?

(*b*) the length of one side?

(By way of comparison the volume of the Earth is about 10^{21} m^3.)

6 What is the mass of

(*a*) a block of brass (density 8.5×10^3 kg m^{-3}) which has a volume of 6.0 cm^3?

(*b*) a beaker of mass 20 g which contains 50 cm^3 of turpentine (density 8.6×10^2 kg m^{-3})?

7 Calculate the volume of a glass stopper (density 2.5×10^3 kg m^{-3}) which has a mass of 35 g.

8 The density of a stone is 2.2×10^3 kg m^{-3} and its mass is 165 g. If it is dropped into a vessel full of water, what volume of water will overflow?

9 A density bottle has a mass of 35 g when empty, 85 g when full of water, and 75 g when full of kerosene. Calculate the density of kerosene.

10 The weight of a stack of bricks exerts a total force of 6×10^4 newton on the 2 m^2 of ground it covers. What is the pressure exerted by the stack on the ground in newton per square metre, or, as this SI unit of pressure is normally known, pascal (abbreviated Pa)?

11 If a Volkswagen campervan exerts a pressure of 3×10^5 pascal on the 0.050 m^2 of ground in contact with the tyres, what is the weight of the vehicle?

12 The pressure quoted for a bicycle tyre is 3.6×10^5 Pa. (This is in fact the amount by which the pressure inside should *exceed* the external atmospheric pressure, which is 1.0×10^5 Pa.) If the total force exerted by the air inside the tube is 6.0×10^4 N, what is the total inner surface area of the tube?

13 A girl exerts an average pressure of 3.2×10^4 Pa on the ground when wearing thongs with a total base area of 364 cm^2. What pressure does she exert when wearing a pair of high-heeled shoes with a total area of 182 cm^2 in contact with the ground?

14 In the year 1654, Guericke performed the famous Magdeburg hemispheres experiment before Emperor Ferdinand III and the Reichstag. He showed that the hemispheres could be pulled apart only when teams of four pairs of horses were hitched to each. If the hemispheres were 36.0 cm in diameter, calculate the pull exerted by each team. Atmospheric pressure = 1.01×10^5 Pa. The force required to pull the hemispheres apart is equivalent to the force exerted by the air on one side of a disc of the same diameter.

15 If in a mercury barometer the pressure of the atmosphere supports a vertical column of mercury 0.75 m in height, what is the pressure of the atmosphere in pascal? Density of mercury = 1.36×10^4 kg m^{-3}.

16 How deep would a scuba diver need to go below the surface of the water before he experienced a pressure double that felt at the surface? Atmospheric pressure at the surface = 1.0×10^5 Pa. Density of sea water = 1.03×10^3 kg m^{-3}.

17 In the South Kensington museum there is a barometer in which the liquid used is glycerine. What would be the reading of this barometer when the mercury barometer reads 75.0 cm? Density of mercury = 1.36×10^4 kg m^{-3}. Density of glycerine = 1.25×10^3 kg m^{-3}.

18 For the dam wall shown in Figure 4.1, determine

(*a*) the average pressure exerted by the water on the wall;

(*b*) the total force exerted by the water on the wall.

Density of water = 1.0×10^3 kg m^{-3}.

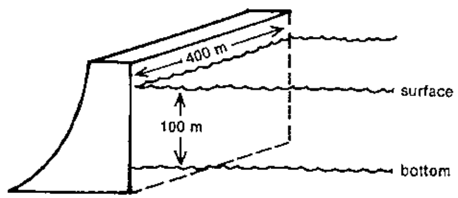


Fig. 4.1

19 Some oil is poured down one arm of a U-tube containing mercury. Measured vertically, the height of the oil column is 22.5 cm, and the difference in levels of the mercury surfaces is 1.5 cm. If the density of the mercury is 1.35×10^4 kg m^{-3}, what is the density of the oil?

PART 2

MOTION

SET 5 MEASUREMENT OF LENGTH, TIME AND FREQUENCY

Topic	*Problems*
Measurement of length by micrometer screw gauge	1–4
Meaurement of length by vernier caliper	5–8
Measurement of small time intervals and frequency by stroboscope	9–14
Measurement of time by multi-image and time-lapse photography	15–18

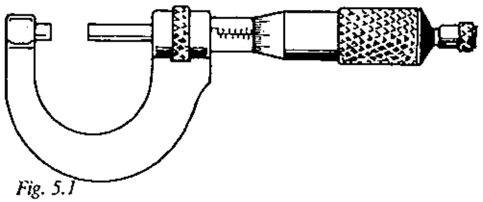


Fig. 5.1

1 Figure 5.1 shows a micrometer screw gauge used for measuring small distances such as the diameter of a wire. What would be the correct reading of a micrometer

(*a*) part of whose scale is shown in Figure 5.2 and which has a pitch of 1.00 mm and 100 graduations on its rotating scale?

(*b*) part of whose scale is shown in Figure 5.3 and which has a pitch of 0.50 mm and 50 graduations on its rotating scale?

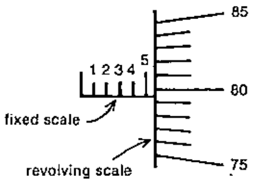


Fig. 5.2

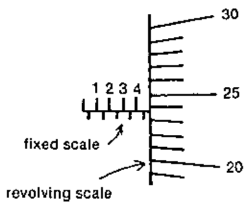


Fig. 5.3

2 A micrometer screw gauge has a pitch of 1.00 mm and there are 100 equal divisions on the circular scale on the screw head. Draw a diagram showing its scales when it reads 4.85 mm.

3 A micrometer screw gauge reads to 0.001 unit. It has 25 equal divisions on its circular scale. What is the pitch of the screw?

4 The pitch of the screw of a micrometer screw gauge is 0.500 mm.

(*a*) How many equal divisions are on its circular scale if it reads to 0.01 mm?

(*b*) What is the reading on the circular scale when the gauge reads 3.87 mm?

5 For the vernier caliper shown in Figure 5.4,

(*a*) what is the length of 10 divisions on the vernier scale?

(*b*) what is the length of 1 division on the vernier scale?

(*c*) what is the difference in length between one division on the vernier scale and one division on the main scale?

(*d*) to what accuracy can this vernier be used?

(*e*) what is the correct reading on the caliper?

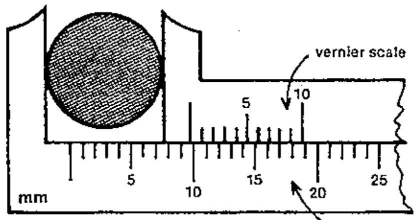


Fig. 5.4

6 A vernier reads to the nearest 0.01 cm and is used in conjunction with a main scale whose divisions are 0.10 cm long.

(*a*) How long is the vernier scale?

(*b*) Into how many equal parts is the vernier scale divided?

(*c*) Which vernier division coincides with a main scale division when the reading is 14.26 cm?

7 Figure 5.5 shows a more accurate vernier than that of Figure 5.4. For this vernier caliper,

(*a*) what is the total length of the vernier scale?

(*b*) what is the length of one division on the vernier scale?

(*c*) to what accuracy can this vernier be used?

(*d*) what is the reading on the caliper?

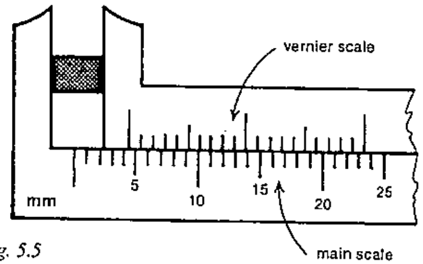


Fig. 5.5

8 A Fortin barometer has a centimetre scale on which each centimetre is divided into 20 equal parts. The vernier used with this scale reads to 0.001 cm.

(*a*) How long is the vernier scale?

(*b*) Into how many equal parts is the vernier scale divided?

9 A motor-driven stroboscope with eight equally spaced slits is used to view a vibrating hacksaw blade which has one end clamped to a bench. Several rates of rotation of the stroboscope produce a stationary picture of the vibrating hacksaw blade. The fastest of these occurs when the stroboscope makes 120 rotations in 20 s. Leave the answers to (*a*), (*c*) and (*d*) in *fractional* form.

(*a*) What is the time taken for one rotation of the stroboscope?

(*b*) How many glimpses of the vibrating blade are gained during one rotation of the stroboscope?

(*c*) What is the time interval between glimpses through the stroboscope?

(*d*) What is the period of vibration of the blade, i.e. the time taken for one complete vibration of the blade?

(*e*) What is the frequency of vibration of the blade? If the frequency turned out to be 20 per second—actually it doesn't—then it would be written as 20 s^{-1} or more usually 20 Hz. Hz is the abbreviation for *hertz*, the SI unit for frequency.

10 A disc stroboscope has four equally spaced slits. Its fastest rate of rotation which "stops" the motion of a vibrating blade is 5 rotations in 1.0 s.

(*a*) What is the interval between glimpses through the slits of the stroboscope?

(*b*) What is the period of vibration of the blade?

(*c*) What is the frequency of vibration of the blade?

(*d*) Other frequencies of vibration could be "stopped" by this stroboscope rotating at the same speed. Name the three of these frequencies that are closest to your answer to (*c*).

11 A white disc marked with a black line along one radius is rotating at a constant rate. The black radial line appears stationary when viewed through a stroboscope with four equally spaced slits and turning at a rate of 3.0 Hz. The next lowest rate of rotation of the stroboscope to make the radial line appear stationary is 1.5 Hz. When the stroboscope is turned at the rate of 6.0 Hz, the disc appears to have two radial lines along the one diameter.

(*a*) What is the frequency of rotation of the disc?

(*b*) Explain briefly why the disc appears stationary when the stroboscope is turned at 1.5 Hz.

12 A mechanically driven ten-slit stroboscope is used to determine the rate of rotation of a flywheel with four identical spokes. The fastest rate of rotation of the stroboscope which gives a stationary picture of the flywheel is 16 Hz.

(*a*) Determine the time elapsing between successive glimpses through the stroboscope.

(*b*) What is the period of rotation of the flywheel?

(*c*) What is the frequency of rotation of the flywheel?

(*d*) What is the fastest rate of rotation of the stroboscope which would give a stationary picture of a flywheel with eight spokes, this wheel rotating at the same speed as the above?

13 A large tuning fork is observed through a stroboscope which has twenty-one equally spaced slits and appears stationary when the stroboscope rotates 100 times per minute. The frequency of the stroboscope is gradually reduced until the tuning fork again appears stationary, the new rotation frequency being 80 rotations per minute. What is the frequency of the fork?

14 A multiflash lamp flashes 20 times per second, and is used to illuminate a spoked wheel with ten spokes. The wheel starts from rest and is made to rotate steadily faster and faster until it reaches a speed such that, for the first time, it appears stationary. When this happens,

(*a*) through what fraction of a turn does the wheel rotate between successive flashes of the lamp?

(*b*) how long is the wheel taking to make one rotation?

15 A cricket coach films a bowler's action with a movie camera which takes 24 frames per second.

(*a*) To calculate the speed of the ball down the pitch, he examines the developed film and finds that the ball reaches the batsman six frames after leaving the bowler's hand. How long did it take to travel from the bowler to the batsman?

(*b*) To illustrate some faults in the bowler's action he wishes to show the film at one-quarter actual speed. At what rate, expressed in frames per second, should the film be projected?

16 A coach wishes to analyse the high-jumping technique of some of his athletes using TV videotaping. If the videotape is replayed at 18 pictures per second and the coach wishes to study the jumping style at one-third normal speed when screened, at what rate should the TV camera record the movement?

17 If a biology student wishes to make a two-minute movie film showing the growth of a plant which would normally occur during 60 days, what time should elapse between photographing successive frames of the film? Assume that the projector shows the film at the rate of 24 frames per second.

18 In order to measure accurately the speed of a squash ball hit horizontally, a multiflash photograph is to be taken in a darkened squash court. If the speed is known to be approximately 40 m s^{-1}, at what frequency should a xenon stroboscope flash in order to photograph the ball about every 20 cm?

SET 6 MOTION IN A STRAIGHT LINE

Graphical treatment

Topic	*Problems*
Deductions from velocity–time graphs	1–6
Deductions from displacement–time graphs	7–15
Further graphical questions	16–19
Analysis of ticker tapes, spark dot tapes and multi-image photographs	20–24

1 The velocity of a skier as a function of time is shown in Figure 6.1.

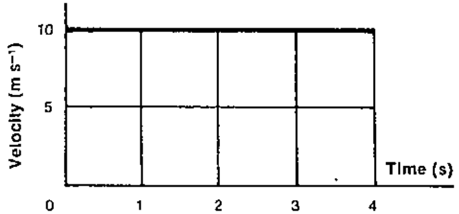


Fig. 6.1

(*a*) What was her acceleration?
(*b*) What distance did she cover in the 4 s interval?
(*c*) What distance did she travel during the *fourth* second?

2 The velocity–time graph of a motor car is given in Figure 6.2.

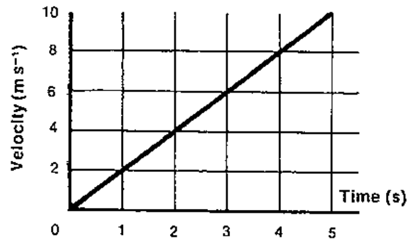


Fig. 6.2

(*a*) By how much did the velocity change in 5 s?
(*b*) What was the acceleration of the motor car?
(*c*) What was the average velocity of the car during the 5 s interval?
(*d*) How far did the car move in
  (*i*) 4 s?
  (*ii*) 5 s?
(*e*) Is the distance travelled proportional to the time taken?

3 Referring to the velocity–time graph in Figure 6.3 determine
(*a*) the distance travelled by the body in the first 2 s;
(*b*) the average velocity of the body in the first 5 s;
(*c*) the instantaneous acceleration of the body after 1, 4 and 7 s respectively.
(*d*) Draw a graph of the acceleration of the body against the time.

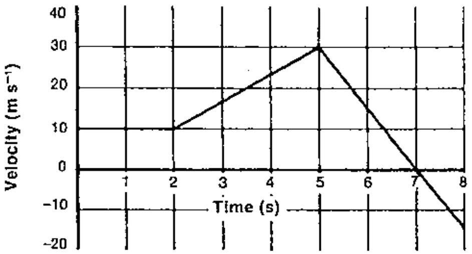


Fig. 6.3

4 Figure 6.4 is the velocity–time graph of a body travelling in a straight line.
(*a*) What was the displacement of the body after 4 s?
(*b*) What was its acceleration during the first second?
(*c*) What was its acceleration 4 s after starting?
(*d*) What was the magnitude of its displacement from the starting point after 10 s?

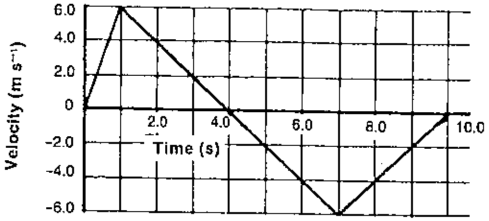


Fig. 6.4

5 Figure 6.5 is the velocity–time graph for a train travelling north along a straight track.

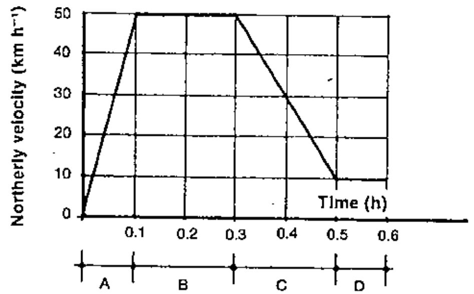


Fig. 6.5

(*a*) In which section(s) (A, B, C, D)
  (*i*) is the velocity a maximum?
  (*ii*) is the acceleration a maximum?
  (*iii*) are the velocity and acceleration constant throughout?
(*b*) What distance is travelled by the train in section C?
(*c*) What is the average speed of the train during the first 0.3 h? Give the answer to two significant figures.
(*d*) Determine the acceleration of the train at 0.4 h. Give the answer to two significant figures.

6 The graphs in Figures 6.6 and 6.7 describe two motions.

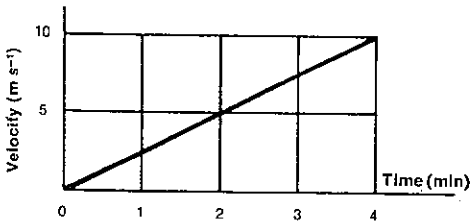


Fig. 6.6

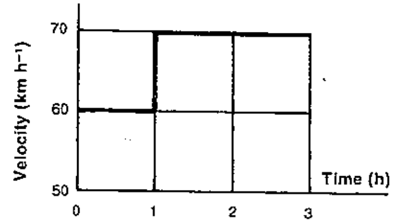


Fig. 6.7

Determine the distance covered
(*a*) in the first 4 min in Figure 6.6;
(*b*) in the first 3 h in Figure 6.7.

7 Figure 6.8 is the distance–time graph for a body.

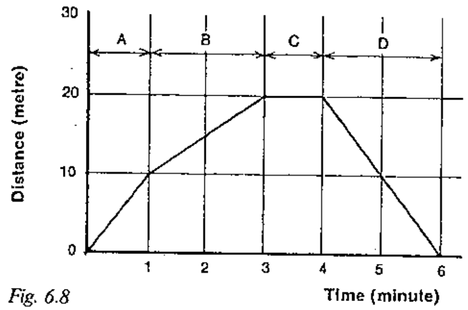


Fig. 6.8

(*a*) In which section of the motion was the body stationary?
(*b*) In which section was it moving with the least speed?
(*c*) What distance did the body travel in the first 4 min, and what was its average speed during that time?
(*d*) What is the maximum speed shown on the graph?

8 The distance–time graph for a moving body is shown in Figure 6.9.
(*a*) During which parts of the motion did the body have a speed of 4 m s^{-1}?
(*b*) When was the body stationary?
(*c*) What was its speed at C?
(*d*) What was its velocity at D? Give two significant figures.

9 An object moves along a straight line. Its displacement from the starting point is shown as a function of time in Figure 6.10.

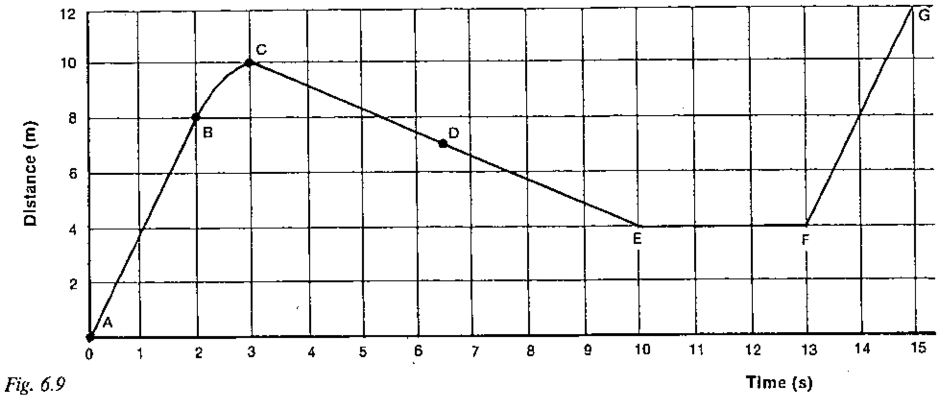


Fig. 6.9

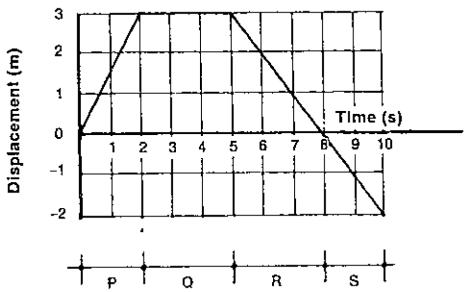


Fig. 6.10

(*a*) What is the displacement of the object after 5 s?
(*b*) Which section of the graph represents a constant velocity of 1.5 m s^{-1}?
(*c*) Which section of the graph represents the object at rest?
(*d*) What was the instantaneous velocity of the object after 7 s?
(*e*) What time elapsed before the object returned to its starting point?

10 Figure 6.11 represents part of a short trip in a car. AB and CD are two straight sections of the graph and section BC is slightly curved in a parabolic shape.

(*a*) What is the speed in the section AB?
(*b*) What is the speed in the section CD?
(*c*) What are the average speeds between B and C and between A and D respectively?
(*d*) What is the instantaneous speed 60 s after the start of the trip?
(*e*) What is the acceleration in the section BC?

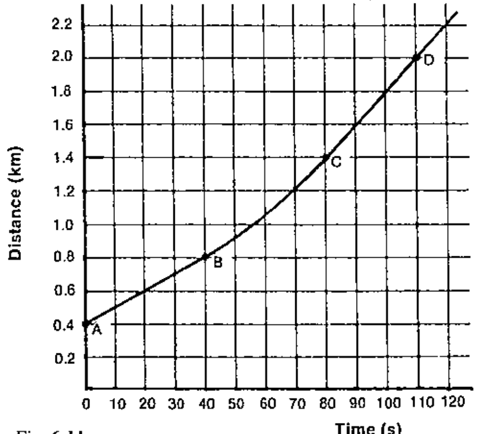


Fig. 6.11

11 In a machine a certain part A moves to and fro horizontally with a period of 0.9 s. The displacement of A to the right of its initial position is shown in Figure 6.12 as a function of time from the beginning of a cycle.

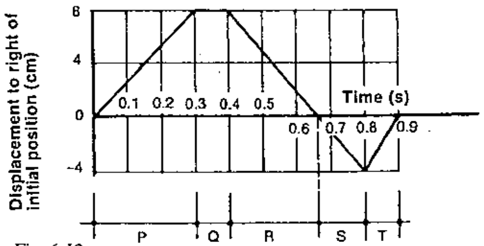


Fig. 6.12

(*a*) During which section(s) of its motion is A moving with constant velocity?
(*b*) During which section is A moving with the greatest speed?
(*c*) During which section is A at rest?
(*d*) What is the displacement of A from its initial position 0.8 s after the beginning of a cycle?
(*e*) What distance does A travel between 0.4 s and 0.9 s?
(*f*) What velocity does A reach 0.6 s after the beginning of each cycle?
(*g*) Calculate the average speed of A during one cycle. Give the answer to two significant figures.

12 The velocity–time graphs for the motions of a train and a car travelling along a road parallel to the railway line are shown in Figure 6.13. Initially the car, travelling at a speed of 30 km h^{-1}, is passing a station where the train has stopped. The graphs illustrate the subsequent motions.

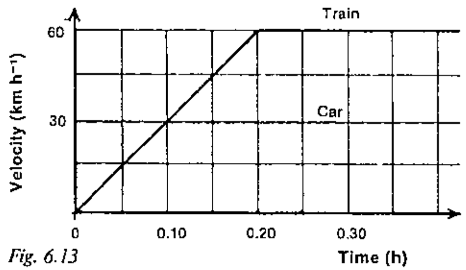


Fig. 6.13

(*a*) Find the acceleration of the train as it pulls out of the station.
(*b*) How far does the train travel during the first 0.30 h?
(*c*) At what time does the train meet the car again?

13 Figure 6.14 is the velocity–time graph for a motor car which crossed an amphometer at an excessive speed. The driver failed to notice the policeman's signal to stop, and his car was chased by another policeman on a motor-cycle for which the graph is also given. At what time, on the time scale given, does the motor-cycle overtake the car? The zero of time is the instant when the car passes the motor-cycle.

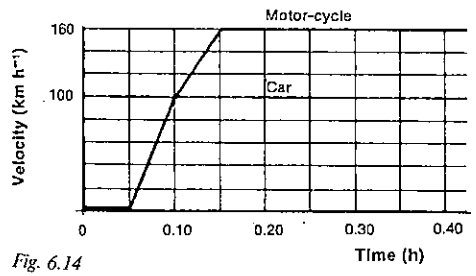


Fig. 6.14

14 Figure 6.15 is the velocity–time graph for a train travelling on a straight track.

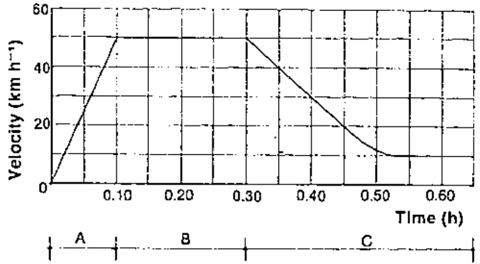


Fig. 6.15

(*a*) In which section of the motion (A, B, C) is
(*i*) the velocity a maximum?
(*ii*) the acceleration a maximum?
(*b*) How far does the train travel in section A?
(*c*) Estimate the distance travelled by the train in section C.
(*d*) What is the average velocity of the train in section C?
(*e*) What is the acceleration of the train after 0.50 h has elapsed?

15 Figure 6.16 is the velocity–time graph of the motion of a motor-cycle. (Give all the answers in this question to three significant figures.)

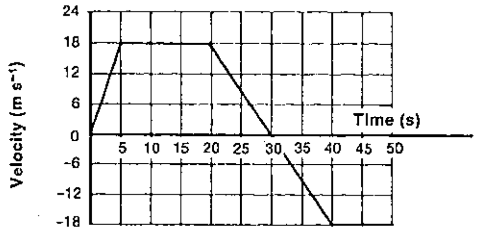


Fig. 6.16

(*a*) The graph shows a positive and a negative acceleration. What is the ratio of the magnitude of the positive acceleration to that of the negative acceleration?
(*b*) What distance is travelled by the motor-cycle in the first 30 s?
(*c*) How far is the motor-cyclist from his starting point after 50 s have elapsed?
(*d*) If another motor-cyclist started from rest from the same starting point with a constant acceleration, what would be the magnitude of this acceleration if
(*i*) each cyclist reached the same speed 20 s after starting?
(*ii*) both motor-cycles travelled the same distance in the first 20 s of their motions?

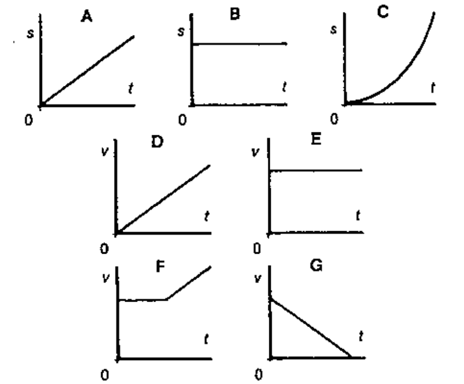


Fig. 6.17

16 From the graphs in Figure 6.17 select the appropriate graph(s) which describe(s) each of the following cases (s = displacement, v = velocity, t = time).

(*a*) A parked car.
(*b*) A car moving at a constant speed along a straight road.
(*c*) A bus accelerating uniformly from a standing start.
(*d*) The brakes of a train bring the train to rest with a uniform retardation.
(*e*) A hiker walks at an even pace.
(*f*) A rocket moving with a uniform velocity fires its second stage and accelerates uniformly.

17 Figure 6.18 is the velocity–time graph of the motion of a car.

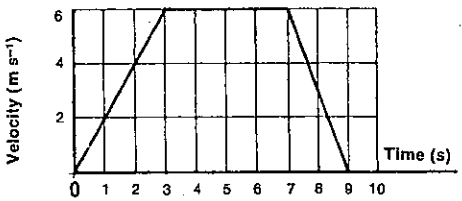


Fig. 6.18

(*a*) Sketch the corresponding acceleration–time graph for this car, showing the scales on the axes.
(*b*) What was the total distance travelled by the car in the 9 s?

18 A car takes 2 min to travel between two sets of traffic lights 2145 m apart. It has a uniform acceleration for 30 s, then uniform velocity, and then uniform retardation for the last 15 s. Without showing a scale on the velocity axis, sketch the velocity–time graph for the car, and use this graph to determine the maximum velocity and the acceleration.

19 Between two stations a train travels the first $1/n$ th of the distance with uniform acceleration, then with uniform speed v, and for the last $1/n$ th of the distance with uniform retardation. Without showing scales on the axes, sketch the velocity–time graph for the train. What is the average speed?

If the acceleration and the retardation each occupied $1/n$th of the total *time*, what would be the average speed in this case?

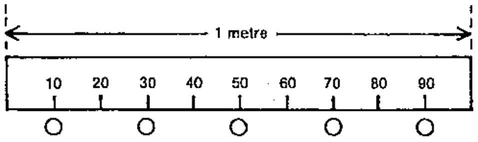


Fig. 6.19

20 Figure 6.19 represents a stroboscopic photograph of a ping-pong ball in flight with the light flashing 1500 times per minute. Determine the speed of the ball in metre per second.

21 A table has one end raised so that a dynamics cart will run down it with a constant acceleration. A tape attached to the cart passes through a ticker timer, and part of the tape obtained is shown in Figure 6.20. The period of the ticker timer is 1/20 s.

(*a*) What was the instantaneous velocity (in mm tick^{-1}) of the cart at the instant that dot B was made on the tape?
(*b*) What was the acceleration of the cart in
 (*i*) mm tick^{-2}?
 (*ii*) mm s^{-2}?
 (*iii*) m s^{-2}?

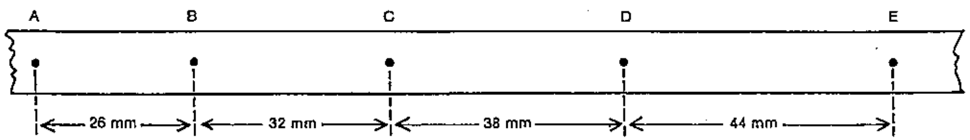


Fig. 6.20

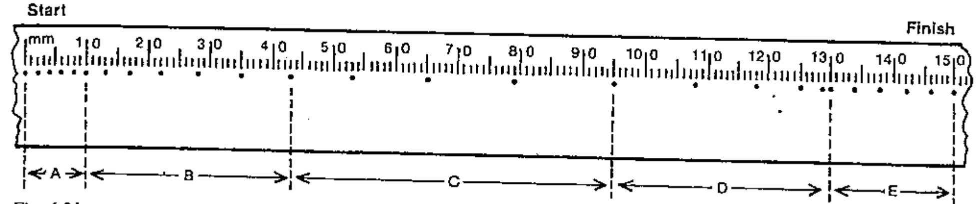


Fig. 6.21

22 On a linear air track a high voltage spark pen attached to a glider produced the trace of dots shown with a millimetre scale in Figure 6.21. The spark frequency was 20 Hz. Sections of the tape are marked A, B, C, D and E.

(*a*) In which section or sections of the tape (A, B, C, D or E)
 (*i*) was the greatest distance covered?
 (*ii*) did the longest time elapse?
 (*iii*) was the glider moving at constant velocity?
 (*iv*) did the glider have a negative acceleration?
 (*v*) did the glider have its highest positive acceleration?
(*b*) What was the glider's average speed (in mm s^{-1}) in section D?
(*c*) Determine the acceleration of the glider (in mm s^{-2}) in section C.

23 A dynamics trolley with ticker tape attached rolled along an undulating track. A trace of dots was produced on the tape by a 50 Hz ticker timer. A student cut a section of the tape into "5-tick" or "pentatick" pieces and attached these to centimetre squared paper (as shown in Fig. 6.22) to represent graphically the motion of the trolley.

(*a*) What is the difference (in cm pentatick^{-1}) between the average velocity of the trolley during the first pentatick and its average velocity during the second pentatick?
(*b*) During the interval $t = 0$ to $t = 2$ pentatick, what is the acceleration of the trolley
 (*i*) in cm pentatick^{-2}?
 (*ii*) in m s^{-2}?
(*c*) What is the total distance covered by the trolley from $t = 0$ to $t = 4$ pentatick?
(*d*) What physical quantity is represented by the shaded area in Figure 6.22? Calculate its value.

24 Suppose that henceforth the metre (the unit of length) and the second (the unit of time) are to be replaced by the *vel* (a unit of speed) and the *tick* (a unit of time). Using secondary standards calibrated by the National Standards Laboratory, a ticker-timer making 1.0 click per tick makes the record shown in Figure 6.23 of a test object moving at 10 microvel, using the well-known method of tape marking.

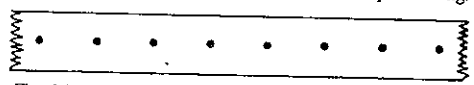


Fig. 6.23

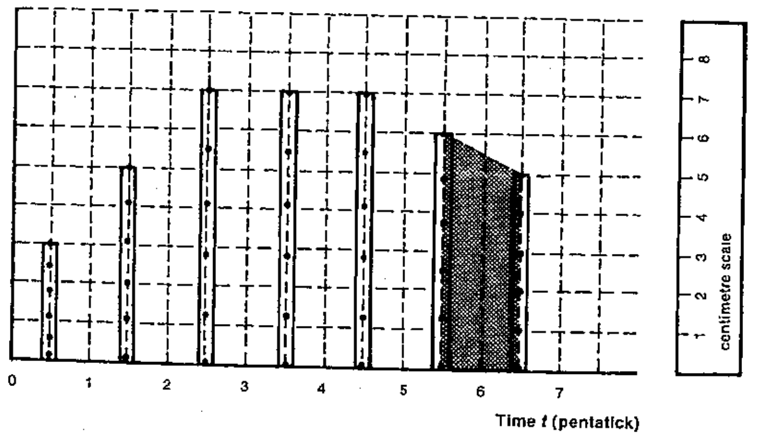


Fig. 6.22

Use the new units only in answering the following questions.
(*a*) What time elapsed between making the first and last dots on this sample of the tape?
(*b*) What is the distance from the first to the last dot on the tape?
(*c*) An astronomer determines that, to a good approximation, 10^8 tick = 1 year. Establish in vel (to one significant figure) a reasonable legal speed limit for motor traffic in a suburban area.

A rocket sled starts from rest and accelerates as shown in Figure 6.24.

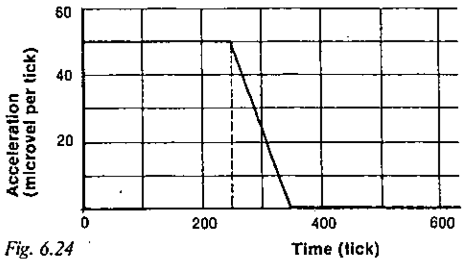


Fig. 6.24

(*d*) How fast is it going after 200 tick?
(*e*) How far has it gone in that time?
(*f*) How fast is it going after 500 tick?
(*g*) How far has it travelled in 500 tick?

SET 7 MOTION IN A STRAIGHT LINE
Algebraic treatment

Topic	*Problems*
Introductory questions using the equations of motion	1–8
More difficult problems using the equations of motion	9–20

1 What is the speed of a stone 3.0 s after it falls from rest with a uniform acceleration of 10 m s^{-2}?

2 Determine the distance covered by a glider on a linear air track, while it accelerates at 4.0 m s^{-2} for 0.50 s from an initial speed of 1.0 m s^{-1}.

3 What is the displacement of a cyclist while he accelerates from 1.5 m s^{-1} to 2.5 m s^{-1} in 2.0 s?

4 Calculate the final speed of a runner who accelerates at 0.5 m s^{-2} from an initial speed of 3.0 m s^{-1} while she covered 16 m.

5 Calculate the uniform acceleration of a sports car which
(*a*) starts from rest and reaches a speed of 15 m s^{-1} in 10 s;
(*b*) changes its speed from 10 m s^{-1} to 32 m s^{-1} in 4.0 s;
(*c*) starts from rest and traverses a distance of 98 m in 7.0 s;
(*d*) starts from rest and travels a distance of 22 m during the sixth second of its motion;
(*e*) slows down from a speed of 66 m s^{-1} and comes to rest in 12 s.

6 A ball rolls from rest down an incline with a uniform acceleration of 4.0 m s^{-2}.
(*a*) What is its speed after 8.0 s?
(*b*) How long will it take to reach a speed of 36 m s^{-1}?
(*c*) How long does it take to travel a distance of 200 m, and what is its speed after that time?
(*d*) How far does it travel during the third second of its motion?

7 A model train moving at 20 cm s^{-1} increases its speed uniformly to 60 cm s^{-1} in 5.0 s. Calculate its acceleration and the distance travelled in that time.

8 A car, accelerating uniformly, reaches a speed of 30 m s^{-1} from rest in 3.0 min. Calculate its acceleration and the distance travelled in this time. It comes to rest in 20 s when the brakes are applied. Find the retardation and the distance traversed while it is pulling up.

9 A car, which is accelerating uniformly from rest, travels a distance of 28.5 m in the tenth second of its motion. Determine its acceleration.

10 A vehicle travelling with a uniform velocity was given a uniform acceleration, and in the sixth and tenth seconds after the acceleration began it was observed to travel 15 m and 23 m respectively. Find the acceleration and the initial uniform velocity of the vehicle.

11 A motor-cycle which is travelling at a speed of 10.0 m s^{-1} undergoes a uniform acceleration of 2.00 m s^{-2} for 3.00 s and of 3.00 m s^{-2} for the nest 4.00 s. Calculate its final velocity and the total distance travelled in the 7.00 s.

12 Convert:
(*a*) a speed of 72 km h^{-1} to m s^{-1};
(*b*) an acceleration of 5.4 km h^{-1} s^{-1} to m s^{-2}.

13 The speedometer of a motor-car registers speeds of 24.0 km h^{-1}, 31.2 km h^{-1} and 38.4 km h^{-1} at intervals of 2.5 s. If the acceleration is uniform, calculate its value in m s^{-2}. After what further interval of time will the speedometer indicate 81.6 km h^{-1}?

14 A goods train travelling at 18.0 m s^{-1} slips the last truck as it approaches a siding. The truck, which is retarded uniformly, travels 117 m before it comes to rest in the siding. If the train does not alter its speed, how far will it be from the siding when the truck comes to rest?

15 A cyclist commences a ride with a uniform acceleration of 1.0 m s^{-2} which he maintains for 8.0 s and then continues with a uniform speed. He pulls up with a uniform retardation over an interval of 13.5 s. If he travels a total distance of 438 m, for what period of time does he travel with a uniform speed?

16 A bus moves off from one of its stops and accelerates uniformly for 20.0 s over a distance of 100 m. It continues with the speed acquired for another 30.0 s, then the driver applies the brakes and the bus is uniformly retarded so that it comes to rest at the next stop after a further 8.00 s. Calculate the distance between the bus stops.

17 Between two stops a tram accelerates uniformly at the rate of 0.750 m s^{-2} for 12.0 s, travels for the next 20.0 s with the speed acquired, then comes to rest with uniform retardation which takes place over a distance of 27.0 m. Calculate the distance between the stops and the time taken for the tram to travel that distance.

18 A moving object covers a distance s while it accelerates uniformly at a rate a for time t, thereby attaining a final speed v_f. Derive an expression for s in terms of v_f, a and t.

19 A sports car travelling at 30 m s^{-1} passes a police van which is cruising at 15 m s^{-1}. The van, accelerating at the rate of 1.2 m s^{-2}, immediately begins to chase the car. Write down expressions for the distance travelled by each vehicle in t second, and hence calculate the time required for the van to overtake the car.

20 A motor-cycle patrolman has his machine stationary with the engine running when a speeding motor-car passes him at a speed of 126 km h^{-1}. As the car passes, he sets off after it with an acceleration of 1.40 m s^{-2} which he maintains until he overtakes it. Calculate the time required for the motor-cycle to overtake the car, and the distance travelled in doing this.

21 Estimate as an order of magnitude the average acceleration of a tennis ball while it is being served by an average tennis player, i.e. while the ball is in contact with the racquet.

SET 8 VECTORS AND SCALARS

Topic	*Problems*
Classifications of physical quantities as vectors or scalars	1–2
Addition of vectors	3–7
Subtraction of vectors	8–11
Resolution of vectors	12–16
Relative vectors	17–28

1 By writing V or S, classify as vectors or scalars any of the following physical quantities you have met in your physics course so far:
(*a*) distance covered;
(*b*) displacement;
(*c*) speed;
(*d*) velocity;
(*e*) acceleration;
(*f*) mass;
(*g*) refractive index;
(*h*) potential energy;
(*i*) momentum;
(*j*) density;
(*k*) power;
(*l*) electric charge;
(*m*) magnetic flux density.

2 For each of the following physical quantities, write V if it is a vector or S if it is a scalar:
(*a*) the time taken to cross the road;
(*b*) the distance covered by a billiard ball after being hit by the cue;
(*c*) the change in velocity of a downhill skier as she makes a turn;
(*d*) the electric current in an overhead power line;
(*e*) the kinetic energy of an arrow in flight;
(*f*) the weight of a weightlifter;
(*g*) the pressure inside a bicycle tube;
(*h*) the potential difference across an electric motor;
(*i*) the gravitational field strength on the surface of the Moon.

3 Two boys are dragging an object along level ground by pulling on a rope held horizontally with forces of 150 newton and 180 newton respectively. What single force on the rope would have the same effect?

4 If the frictional force in the previous problem is 30 newton, what is the resultant horizontal force on the object?

5 Vectors P, Q, R, S and T are shown on a grid in Figure 8.1.

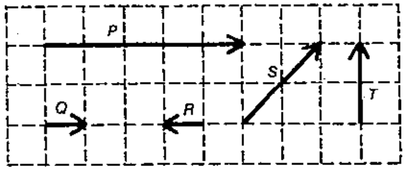


Fig. 8.1

Using the large squares on a sheet of graph paper as your grid, draw and label vectors to represent
(*a*) $P+Q$;
(*b*) $P+R$;
(*c*) $P+T$;
(*d*) $Q+S$;
(*e*) $T+2R$.

6 Two tugs are pulling on a ship, one with a force of 6.0 kN (kilonewton) and the other with a force of 8.0 kN. The cables from the tugs to the ship are at right angles to each other. Find the magnitude of the resultant force on the ship, and the angle that its direction makes with the cable in which the force is 8.0 kN.

7 A bushwalker travels 6.5 km north east, then 4.5 km north. What is his resultant displacement?

8 Refer to the vectors shown in Figure 8.1. Using the large squares on a sheet of graph paper as your grid, draw and label vectors to represent
(*a*) $P-Q$;
(*b*) $P-R$;
(*c*) $P-T$;
(*d*) $Q-S$.

9 In a tennis match at Kooyong a girl receives a serve which is travelling at 30 m s^{-1} south just prior to hitting her racquet. Immediately after leaving her racquet it is travelling at 25 m s^{-1} north. Specify fully the change in velocity she gave the ball.

10 A cyclist travelling at 12 km h^{-1} east makes a right-hand turn at an intersection without changing speed. What is his change in velocity?

11 A student whirls a rubber stopper on the end of a nylon string in a horizontal clockwise circle (as seen from above) at a constant speed of 6.0 m s^{-1}. From an instant when the stopper is moving in a northerly direction, find its change in velocity after moving round
(*a*) one-half of the circumference;
(*b*) one complete turn;
(*c*) one-quarter of a turn;
(*d*) one-ninth of a turn.

12 A crate is pulled along the floor by means of a rope making an angle of 30° with the horizontal, the pull in the rope being 80 N. Calculate the effective part of the pull in moving the crate, and also the force tending to lift it off the floor.

13 A pedestrian walking up High Street in a certain city is moving in a direction 60° east of north.
(*a*) If he walks 0.25 km up High Street, what is the easterly component of his displacement?
(*b*) If he takes 5.0 min for this walk, what is the northerly component of his velocity?

14 The rails of a "big dipper" in a fun park slope at an angle of 30° to the horizontal as shown in Figure 8.2. A car is being pulled up the rails at a constant speed of 4.0 m s^{-1}. A battery of lights above causes a shadow directly below the car on the level ground underneath.

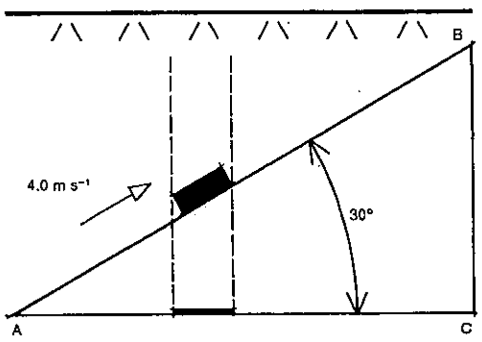


Fig. 8.2

(*a*) At what speed is the shadow of the car moving from A to C?
(*b*) If the distance BC = 26 metre, how long does it take for the car to move from A to B?

15 A ramp rises vertically 3.0 m for every 5.0 m of its length. A body weighing 100 N is placed on the ramp. Find the component of its weight
(*a*) parallel to the ramp;
(*b*) perpendicular to the ramp.

16 A man pushes a garden roller weighing 1000 N over a level lawn, holding the end of a 150 cm long handle 90 cm above the axis of the roller as shown in Figure 8.3. He pushes down the handle with a force of 200 N.

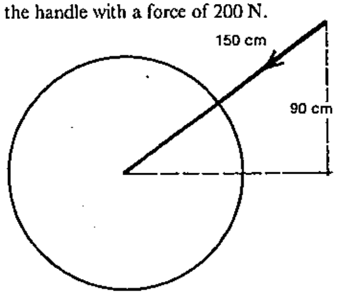


Fig. 8.3

(*a*) What is the size of the effective force pushing the roller along?
(*b*) With what force does the roller press against the lawn while the man is pushing the handle?
(*c*) What force would the roller exert on the lawn if the man pulled it (with the same force) instead of pushing it, the handle remaining in the same position?
(*d*) If the roller lodges in a small rut, would it be easier to get it out by pulling or by pushing?

17 A train is travelling at 50 km h^{-1} towards the north and a passenger is walking along the corridor of a carriage. What is the velocity of the passenger relative to the ground when he walks along the corridor at 4 km h^{-1} (relative to the train) towards
(*a*) the front of the train?
(*b*) the rear of the train?

18 The air-speed of an aeroplane is 240 km h^{-1} and its course is north. It is carried out of its course by a wind blowing at 70.0 km h^{-1} from the west. Find its ground-speed and track.

19 A yacht travelling with a velocity of 10 km h^{-1} north relative to the water is in a tidal stream flowing south-east at 4.0 km h^{-1}. Determine, either by scale drawing or by calculation, the yacht's velocity relative to the ocean floor.

20 An aeroplane travels due east at 360 km h^{-1} relative to the ground while the wind is blowing from the south at 100 km h^{-1}. In what direction does the pilot point the plane, and what is the speed of the plane through the air?

21 A motor-cyclist is travelling due north with a speed of 60.0 km h^{-1} and a wind is blowing from the east at 20.0 km h^{-1}. What is the apparent direction and speed of the wind to the cyclist?

22 An aeroplane A is travelling at 200 km h^{-1} east. Another B is travelling at 300 km h^{-1} south-west. Find the velocity of B relative to A.

23 To a cyclist A travelling at 12 km h^{-1} due north the wind appears to blow directly from the west. To another cyclist B whose speed relative to A is 24 km h^{-1} in a direction 30° south of east, the wind appears to blow directly from the south. What is the speed and direction of the wind?

24 At 1.30 p.m. an aircraft is 400 km north-east of an aerodrome. If there is a wind blowing from the south-east at 100 km h^{-1} and the aircraft is due over the drome at 2.10 p.m., find
(*a*) the direction in which the pilot must steer the aircraft to make a direct track to the drome;
(*b*) the speed of the aircraft relative to the air.

25 A river 2.0 km across has a current of velocity 8.0 km h^{-1}. In what direction must a launch, moving at 16 km h^{-1} relative to the water, steer in order to reach one bank directly opposite its starting point on the other? What will be the time for the crossing?

26 A river is 1.0 km wide and the speed of the stream is 6.0 km h^{-1}. A ferry boat crosses from one bank to a point directly opposite on the other bank in 7.5 min.
(*a*) Find the speed of the boat relative to the water and the direction in which it is actually steered.
(*b*) If the point of landing on the opposite bank does not matter, how much time could the ferry boat save by making the fastest possible crossing?

27 An aeroplane is scheduled to fly between two airports A and B. B is due north of A. The economical cruising air-speed of the aeroplane is 400 km h^{-1}. At 3000 m altitude there is a 50 km h^{-1} wind from a direction 15° south of east, and at 5000 m altitude an 80 km h^{-1} wind is coming from the south-east. At which of these altitudes should the aircraft fly to traverse the distance between A and B in minimum time?

28 A man wishing to cross a river 300 m wide can row at the same speed as the current.
(*a*) What is his quickest way across?
(*b*) If he wishes to reach a point 400 m downstream from his starting point, in what direction must he row?

SET 9 INTRODUCTION TO NEWTON'S LAWS OF MOTION

Topic	*Problems*
Simple direct applications of Newton's second law	1–4
Determining forces from analysis of ticker tapes, spark-dot tapes and multi-image photographs	5–8
Velocity changes produced by net forces	9–13
Momentum—impulse of a force and the change in momentum it produces—area under a force–time graph (further problems in Set 17)	14–17
Accelerating objects with more than one force acting on them	18–24
Bodies in equilibrium with several forces acting on them	25–26
Connected bodies moving horizontally	27–30
Using both dynamics and kinematics	31–34
Forces produced by continuous flow of solids or liquids	35–36

1 A dry-ice puck (object capable of moving with very little friction) is pulled along by a constant force. After 2.0 s, its speed is 0.30 m s^{-1}. What will be its speed after 12 s?

2 Find the acceleration caused by a net force of
(*a*) 20 N east acting on a 5 kg mass;
(*b*) 0.2 N south acting on a mass of 100 kg.

3 What is the net force required to cause an acceleration of
(*a*) 4 m s^{-2} upwards in a mass of 2 kg?
(*b*) 5 cm s^{-2} west in a mass of 400 g?

4 While hitting a cricket ball with a force of 70 N east, a boy gives it an acceleration of 500 m s^{-2} east. What is the mass of the ball?

5 A student made three different experiments in pulling a paper tape through a vibrating ticker-timer. From somewhere about the middle of each tape, he cut off a number of pieces, each produced in the time for 5 ticks (pentatick) and pasted them side by side as shown in diagrams P, Q and R in Figure 9.1.

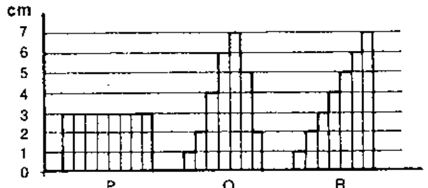


Fig. 9.1

(*a*) Which diagram shows a uniform speed of the tape?
(*b*) Which diagram shows a uniformly accelerated motion?
(*c*) What is the magnitude of the greatest acceleration shown in diagram Q?
(*d*) In which case was the net force on the tape constant?

6 A dynamics cart was pulled along a smooth horizontal surface by a rubber band in which the extension was kept constant. A record of the motion was obtained on a paper tape attached to the cart and passing through a vibrating ticker-timer.

Omitting the first few dots which were very close together, a clearly marked dot was chosen as the origin for measurement and called dot 0. The distance between dots 0 and 5 was 4.80 cm and, counting along in groups of 5 dots, the distance between dots 25 and 30 was 11.3 cm.
(*a*) What was the average velocity (in cm/pentatick) between dots 0 and 5?
(*b*) At what instant was the velocity equal to the average velocity calculated in (*a*)?
(*c*) What was the average velocity between dots 25 and 30?
(*d*) What was the acceleration of the cart?
(*e*) Given that the ticker-timer was making 20.0 ticks per second, express the acceleration in cm s^{-2}.
(*f*) What would be the acceleration if the same cart were pulled by three bands, all similar to the first, all attached to the same point on the cart, and with the extension the same as before?
(*g*) If the mass of the cart was 1.50 kg, what was the magnitude of the resultant force acting on it, when pulled by the three rubber bands?

7 On a linear air track glider, a spark pen whose frequency was 50 Hz produced on the tape shown in Figure 9.2 a record of its motion as it moved eastwards.

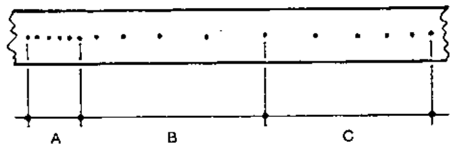


Fig. 9.2

(*a*) In section A, each of the five distances between successive dots is 2.0 mm. State the speed, in cm s^{-1}, in this section.
(*b*) In section B, the distances between successive dots are 3.0, 5.0, 7.0, 9.0 and 11.0 mm respectively. State the acceleration, in cm s^{-2}, in this section.
(*c*) If the mass of the glider is 0.30 kg, what net force was acting on the glider in section B?
(*d*) In section C, the distances between successive dots are 10.5, 9.5, 8.5, 7.5 and 6.5 mm respectively. What type of motion occurred in this section?
(*e*) Calculate the average speed, in cm s^{-1}, over the complete time interval.

8 A loaded dynamics cart carrying 4 bricks was pulled along a special horizontal surface in a straight line by a rubber band kept at a constant extension. The mass of the cart was equal to the mass of 2 bricks. The motion of the cart was recorded on a paper tape by a ticker-timer. The velocity–time graph drawn from the resulting tape is given in Figure 9.3. (1 pentatick = 5 ticks)

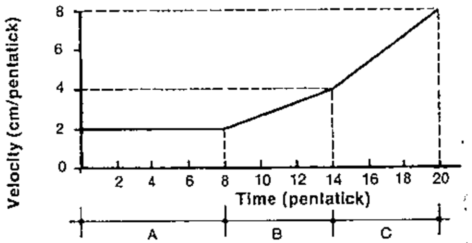


Fig. 9.3

In the first part of the motion, represented by section A of the graph, the surface was covered by carpet; in sections B and C the surface was perfectly smooth. At the end of section B some bricks were removed by a device which did not exert any horizontal force on the loaded cart.

(*a*) What was the resultant force acting on the cart in section A?
(*b*) What was the acceleration of the cart in section B?
(*c*) What was the acceleration of the cart in section C?
(*d*) How many bricks were removed from the cart?

9 A net force of 0.049 N acts on a body at rest and causes it to acquire a speed of 14.7 m s^{-1} in 6.0 s. Calculate the mass of the body.

10 What is the magnitude of the force needed to accelerate an electron of mass 9.1×10^{-31} kg from rest to a speed of 10^5 m s^{-1} in 10 s?

11 A force of 0.784 N west acts on a body for 0.25 s and causes its velocity to change by 0.70 m s^{-1} west. Determine the mass of the body.

12 A cricket ball, of mass 0.14 kg, moving horizontally at 24 m s^{-1} north was hit back with a speed of 16 m s^{-1} south. If the contact with the bat lasted for 4.0×10^{-2} s find the average force exerted by the bat on the ball.

13 A net force of 20.0 N acts on a 5.00 kg mass for 6.00 s. The original velocity of the mass is 100 m s^{-1} east. Calculate the velocity after the 6.00 s, if
(*a*) the force is applied in an easterly direction;
(*b*) the force is applied in a westerly direction.

14 What is the momentum of
(*a*) a 70 kg sprinter running north at 9.0 m s^{-1}?
(*b*) an apple of mass 110 g falling downwards at 5.0 m s^{-1}?

15 (*a*) Find the impulse given to a body by a constant force of (*i*) 6.0 N upwards acting for 0.50 s;
(*ii*) 40 N south-west acting for 10 s.
(*b*) What would be the change in momentum of the bodies in each of the cases (*i*) and (*ii*) above?

16 A net force of 20 N towards the east acts on a mass of 8.0 kg for a period of 5.0 s. Find
(*a*) the impulse of the force;
(*b*) the change in momentum of the body;
(*c*) the final momentum of the body if its momentum when the force began to act was 20 kg m s^{-1} west;
(*d*) the final velocity of the body.

17 A body of mass 3.2 kg and initially at rest is acted on by a resultant westward force which varies with time as shown in Figure 9.4. For the 10 s interval, find
(*a*) the impulse exerted on the body;
(*b*) the change in momentum of the body;
(*c*) the change in the velocity of the body;
(*d*) its final velocity.

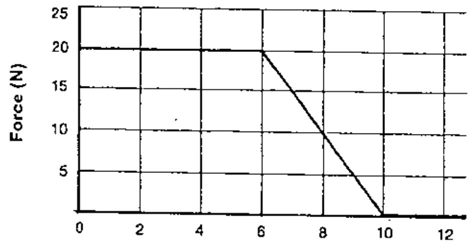


Fig. 9.4

18 For how long must a force of 120 N act on a mass of 50.0 kg to produce a change in velocity from 1.00 to 5.50 m s^{-1} if there is a force of 20.0 N opposing the motion?

19 A 3 kg mass resting on a level table top is pulled along by a horizontal force of 16 N north, and accelerates at 3 m s^{-2} north. Find the force of friction between the block and the table.

20 Forces of 5.0 N and 12 N act at right angles to each other on a mass of 10 kg. Determine the magnitude of the acceleration they produce if there is a frictional force of 2.00 N opposing the motion.

21 A cart, of mass 2 kg, was pulled along a rough level surface by a horizontal force of 6 N. The speed–time graph of the motion of the cart is given in Figure 9.5.

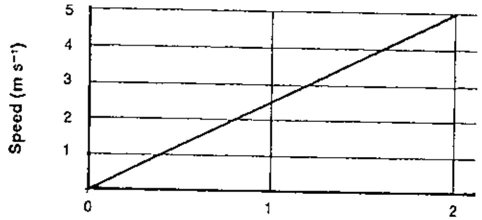


Fig. 9.5

(*a*) What was the magnitude of the net horizontal force on the cart?
(*b*) What was the magnitude of the force of friction?
(*c*) Sketch a graph of the acceleration of the cart for any pulling force from 0 to 6 N.

22 A mass of 3.0 kg is acted on by two forces, each of 10 N, which are at an angle of 60° to each other as shown in Figure 9.6.

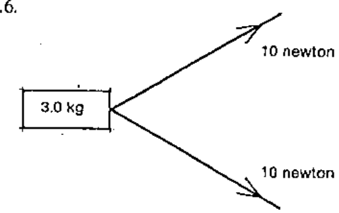


Fig. 9.6

(*a*) Find the resultant force acting on the mass.
(*b*) Find the acceleration produced by the forces.

23 A simple pendulum hangs from the ceiling of a railway carriage. When the carriage is moving with an acceleration of 2.45 m s^{-2}, find the angle that the pendulum makes with the vertical.

24 A mascot in a motor-car hangs vertically when the car is moving with a uniform velocity along a straight level road. The brakes are applied so that the car is stopped with a uniform retardation. The mascot is deflected through an angle of 10.0°. What is the acceleration of the car while it is coming to rest?

25 F_1, F_2 and F_3 are three forces as illustrated in Figure 9.7. They act on an object at A which remains at rest under the action of the three forces. In this problem leave your answers in surd form.

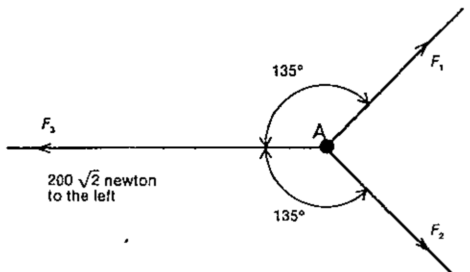


Fig. 9.7

(*a*) What is the resultant of the two forces F_1 and F_2?
(*b*) What is the magnitude of the force F_1?
(*c*) What would be the resultant of the three forces if F_1 and F_2 were both trebled in size? In this case, in which direction would the object accelerate?

26 Three horizontal forces of 4.50 N, 7.50 N and 6.00 N act on a puck on a frictionless air table as indicated in Figure 9.8, where the angles are not correctly drawn to scale.

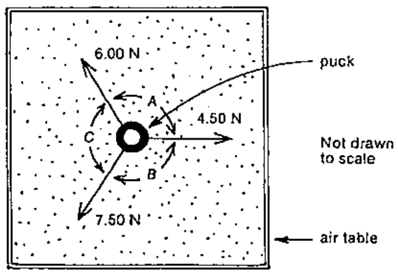


Fig. 9.8

What are angles *A*, *B* and *C*,
(*a*) if the puck is at rest?
(*b*) if the puck moves with a constant velocity of 0.35 m s^{-1} in the direction of the 4.50 N force?

27 Two masses of 2 kg and 3 kg respectively rest on a smooth table and are connected by a taut string of negligible mass. A force of 20 N is applied to the 3 kg mass as shown in Figure 9.9. Calculate
(*a*) the tension in the string;
(*b*) the acceleration of the system.

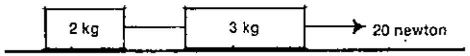


Fig. 9.9

28 A force of 2.4×10^{-3} N acts horizontally on a mass of 50 g which is connected by means of a taut string to a mass of 30 g, both masses resting on a smooth laboratory table. Find the magnitude of the acceleration of the system and the tensile force in the string.

29 An engine provides a steady driving force of 2.56×10^4 N on a train with a total mass of 80 tonne (1 tonne = 10^3 kg). The frictional resistance to the forward motion of the train is 20 N per tonne of mass of the train. Find:
(*a*) the magnitude of the acceleration of the train;
(*b*) the pull in the coupling to the guard's van which has a mass of 1.0 tonne.

30 A tug tows a barge A of mass m_1, which in turn tows another barge B of mass m_2. The acceleration of the system is *f*. Ignoring frictional resistances, find expressions for the tensions in the towing cables.

31 A net force of 450 N acts for 4.00 s on a body of mass 90 kg. Calculate the acceleration, the speed acquired, and the distance travelled in the 4.00 s.

32 A proton, of mass 1.7×10^{-27} kg, moving at a speed of 5.0×10^7 m s^{-1}, is brought to rest by a constant force in a distance of 0.25 m. What is the size of the force?

33 A mass of 50 kg moving with a speed of 20 m s^{-1} is brought to rest by a constant force in a distance of 5.0 m. Calculate the retardation and the size of the force.

34 A 2.0 kg block of wood thrown with a speed of 7.0 m s^{-1} along the floor of a large hall comes to rest in a distance of 100 m. What average frictional force was retarding the block?

35 A stream of sand moving at 4 m s^{-1} is directed horizontally east against a wall at the rate of 120 kg per minute. If the sand falls vertically downwards after impact, find
(*a*) the horizontal force exerted by the wall on the sand;
(*b*) the horizontal force exerted by the sand on the wall.

36 Gravel is thrown horizontally with a velocity of 3.00 m s^{-1} against a vertical screen at the rate of 600 kg per minute. 25 per cent of the gravel passes freely through the screen, but the remainder has its momentum completely destroyed. Calculate the magnitude of the horizontal force exerted on the screen.

SET 10 NEWTON'S LAW AND GRAVITATIONAL EFFECTS NEAR THE EARTH'S SURFACE

Topic	*Problems*
Gravitational field strength and weight	1–5
Newton's law and vertical motion under the influence of weight and other forces	6–13
Vertically moving bodies connected by strings	14–19
Static problems involving weight	20–22

Before commencing problems in this set, refer to the note on the *gravitational field strength* on page x.

1 If a mass of 5.00 kg has a weight of 49.09 N in Stockholm, what is the Earth's gravitational field strength in that city?

2 A mass of 4.00 kg is falling freely. What is the magnitude of the gravitational force acting on it?

3 A man of mass 80 kg travels from the south island of New Zealand, where the Earth's gravitational field is 9.81 N kg^{-1}, to a place on the equator where he is in a gravitational field of 9.78 N kg^{-1}. Find the change in the force of gravity acting on him.

4 Estimate as an order of magnitude the weight of
(*a*) a car;
(*b*) a ballpoint pen;
(*c*) a postage stamp.

5 An astronaut in his space suit has a mass of 133 kg.
(*a*) What is his weight on the Earth?
(*b*) What would he weigh if he landed on the Moon whose gravitational field strength is one-sixth that of the Earth?
(*c*) What would be his mass on the Moon?

6 A bucket of water with a total mass of 20 kg is raised from a well by means of a vertical rope. During the first part of the rise the bucket has an upward acceleration of 0.200 m s^{-2}. During this initial accelerating period,
(*a*) what is the net force acting on the bucket of water?
(*b*) what is the force exerted by the rope on the bucket of water?

7 A parachutist has a mass of 70.0 kg. Find the average force exerted on him by his parachute when his vertical velocity is reduced from 13.0 m s^{-1} to 3.00 m s^{-1} in 2.00 s after it opens.

8 A student of mass 75.0 kg investigates forces exerted on him in a lift by standing on a set of bathroom scales. Taking g = 10 N kg^{-1}, what would the scales read,
(*a*) if the lift is stationary?
(*b*) if the supporting cable broke, and the lift fell freely?
(*c*) if the lift is ascending or descending with uniform velocity?
(*d*) if the lift has an upward acceleration of 2.00 m s^{-2}?
(*e*) if the lift has a downward acceleration of 2.00 m s^{-2}?

9 The mass of a lift and its occupants is 600 kg. Calculate the tension in the cable supporting the lift when it is moving
(*a*) upwards with a uniform velocity;
(*b*) with a uniform acceleration of 0.450 m s^{-2} directed downwards;
(*c*) with a uniform acceleration of 0.450 m s^{-2} directed upwards.
(*d*) Can you tell from the statements in (*b*) and (*c*) above whether the lift is moving upwards or downwards in each case?

10 A mass of 3.00 kg is hung from a spring balance (graduated in newtons) in a lift. Calculate the reading of the spring balance while the lift is slowing to a stop on an upward trip, the magnitude of the acceleration being 2.00 m s^{-2}.

11 A man weighing 980 N slides down a rope that can support a weight of only 755 N.
(*a*) How is this possible?
(*b*) What is the least acceleration he can have without breaking the rope?
(*c*) What will be his minimum speed after sliding down 8.0 m?

12 A tennis ball of mass 50 g is dropped vertically, meeting the ground with a speed of 3.2 m s^{-1} and leaving the ground with a speed of 2.4 m s^{-1}. During the 0.080 s interval during which the ball is in contact with the ground,
(*a*) what is the change in velocity of the tennis ball?
(*b*) what is the average net force acting on the ball?
(*c*) what is the average force exerted on the ball by the ground?

13 A man holds a bucket under a tap from which water issues vertically with a speed of 10 m s^{-1}. It takes 10 s to fill the bucket which holds 10 kg of water. What is the apparent loss in weight of the bucket when the tap is turned off?

14 Bodies of mass 500 g and 200 g respectively are connected together by a light string which passes over a frictionless pulley, as shown in Figure 10.1. Find
(*a*) the magnitude of the acceleration of the bodies;
(*b*) the tensile force in the string.

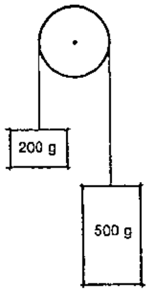


Fig. 10.1

15 Bodies of mass 3.0 kg and 4.0 kg respectively are connected together by a light string which passes over a frictionless pulley. The arrangement is similar to that shown in Figure 10.1. Determine
(*a*) the acceleration of the 3.0 kg mass;
(*b*) the time taken for the masses, starting from rest, to reach a speed of 3.5 m s^{-1};
(*c*) the net force acting on the 3.0 kg mass.

16 Two masses of 1.00 kg and 3.00 kg are at the ends of a string passing over a smooth pulley. If the arrangement were taken to the planet Jupiter and the masses released, they would be found to move a distance of 49.0 m in 2.00 s. What value does this information give for the gravitational field strength on the surface of Jupiter?

17 A string has bodies with masses m_1 and m_2 (m_1 is greater than m_2) attached to its ends and is hung over a peg. If the tension in the string leading down to the mass m_1 is k times greater than the tension on the other side, determine the condition under which the masses will begin to accelerate.

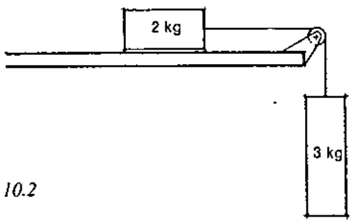


Fig. 10.2

18 A mass of 2 kg resting on a smooth, level table is connected by a light string passing over a frictionless pulley to a suspended 3 kg mass as shown in Figure 10.2. Find
(*a*) the tension in the string;
(*b*) the magnitude of the acceleration of the two masses;
(*c*) the resultant force acting on the 3 kg mass.
(Take g = 10 N kg^{-1})

19 A mass of 1.22 kg rests on a smooth, horizontal table 50 cm from its edge. Another mass of 5 g hangs by a light string which is attached to the 1.22 kg mass and passes over a smooth pulley at the edge of the table. When the system is allowed to move, how long does the 1.22 kg mass take to reach the edge of the table?

20 A boy's finger supports his yo-yo which hangs stationary on the end of its vertical string. If the plastic part of the yo-yo, which normally rotates, has a mass of 0.150 kg, what is the force exerted on the plastic part by the string?

21 A bag of sand weighing 100 N is suspended by a long, light rope, but is pulled aside by a horizontal force so that the rope makes an angle of 25.0° with the vertical. Find the magnitudes of the horizontal force and the tensile force in the rope.

22 A pendulum bob is hung on a string 1.00 m long and is displaced by a horizontal force so that the string makes an angle of 45° with the vertical. If the mass of the bob is 20.0 g, what is the horizontal force?

SET 11 FLOTATION AND GRAVITATIONAL EFFECTS IN FLUIDS

Topic	*Problems*
Direct application of Archimedes' Principle	**1-3**
Archimedes' Principle and flotation in liquids	**4-11**
Archimedes' Principle and flotation in gases	**12-14**
Forces on objects completely immersed in fluids	**15-21**

Where necessary in this set, use

$$\text{density of water} = 1.00 \times 10^3 \text{ kg m}^{-3} = 1.00 \text{ g cm}^{-3}$$

For problems where the question number is highlighted, e.g. Problems **12** and **13**, the buoyancy effect of the surrounding *air* should be taken into account; in all other problems in this set the buoyancy effect of the air is to be regarded as negligible.

1 A piece of aluminium is attached to a spring balance. It reads 54 g when the aluminium is surrounded by air but reads 34 g when the aluminium is immersed in water.

(*a*) What is the true weight of the piece of aluminium (in newton)?
(*b*) What is its apparent weight when immersed in water?
(*c*) What is the net force exerted on the immersed piece of aluminium?
(*d*) What must be the upward buoyancy force exerted by the water on the aluminium?
(*e*) What is the weight of the water displaced by the aluminium?
(*f*) What mass of water was displaced by the aluminium? Give the answer in gram.
(*g*) What is the volume of water displaced? Give the answer in cm^3.
(*h*) What is the volume of the piece of aluminium?
(*i*) What is the density of the aluminium in g cm^{-3} and in kg m^{-3}?

2 A stone of mass 48 g weighs 0.47 N in air and apparently weighs 0.31 N when submerged in water.
(*a*) What is the upward buoyancy force exerted on the stone by the water?
(*b*) What is the mass of water displaced by the stone?
(*c*) What is the volume of the stone?
(*d*) Determine the density of the stone in g cm^{-3} and in kg m^{-3}.
The stone attached to the spring balance is now immersed in glycerine of density 1.25×10^3 kg m^{-3}.
(*e*) What would be the mass of glycerine displaced by the stone?
(*f*) What would be the upward buoyancy force exerted on the stone by the glycerine?
(*g*) What would the apparent weight of the stone read on the spring balance?

3 A piece of metal appears to weigh 0.24 N less in kerosene (density 800 kg m^{-3}) than it does in air.
(*a*) What is the weight of the kerosene displaced by the piece of metal?
(*b*) What is the mass of kerosene displaced?
(*c*) What is the volume of the piece of metal?

4 A block of wood which has a volume of 60 cm^3 floats in water with 36 cm^3 submerged.
(*a*) What mass of water is displaced by the block?
(*b*) What is the weight of the displaced water?
(*c*) What is the upward buoyancy force the water exerts on the block?
(*d*) What is the net force acting on the block?
(*e*) What then is the weight of the block?
(*f*) What is its mass?
(*g*) Determine the density of the wood in g cm^{-3}.

5 A block of wood in the shape of a cube with a 10 cm edge is floating in water. The density of the wood is 850 kg m^{-3}.
(*a*) What is the density of the wood in g cm^{-3}?
(*b*) What is the volume of the block?
(*c*) What is its mass?
(*d*) What is its weight?
(*e*) What is the upward buoyancy force the water exerts on the block?
(*f*) What is the weight of the water displaced by the block?
(*g*) What mass of water is displaced by the block?
(*h*) What volume of water is displaced by the block?
(*i*) What is the volume of the immersed part of the block?
(*j*) What fraction of the block is immersed? Give the answer in decimal form.
(*k*) What is the value of the ratio

$$\frac{\text{density of the floating object}}{\text{density of the supporting liquid}}?$$

Compare your answers to (*j*) and (*k*).
(*l*) If the same block were floating in an oil of density 930 kg m^{-3}, what fraction would be submerged?

6 A raft which is made of wood of density 600 kg m^{-3} just floats with its top level with the water surface when a man of mass 80 kg stands on it. Find the volume of the wood in the raft.

7 A flat aluminium box weighs 540 g and its inside volume is 340 cm^3. If it is sealed and placed in fresh water, it just floats when the lid coincides with the water surface. What is the density of aluminium?

8 What mass of wire (density 8.0×10^3 kg m^{-3}) must be wrapped around a cork (density 250 kg m^{-3}) of mass 7.0 g in order to make it just sink in water?

9 A common hydrometer has the shape shown in Figure 11.1. When immersed in water it sinks to level P and when immersed in a liquid of density 1.50×10^3 kg m^{-3} it sinks to level Q.

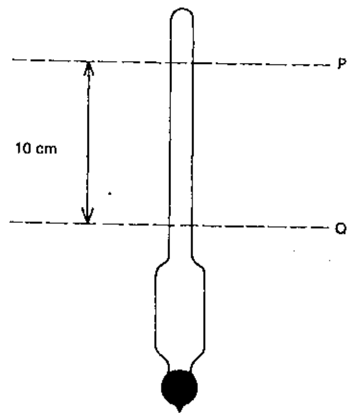


Fig. 11.1

(*a*) If the volume of the bulb and the stem up to the level of P is 12 cm^3, what is the volume of 1.0 cm length of the stem?
(*b*) How far above Q on the stem would the graduation mark corresponding to a density of 1.25×10^3 kg m^{-3} be placed?
(*c*) What density corresponds to a graduation mark half-way between P and Q?

10 A tall jar of water has a layer of petrol, 6 cm deep, above the surface of the water. A pencil which is 12 cm long floats upright in the jar with 3 cm of its length above the petrol surface. If the density of petrol is 700 kg m^{-3}, what is the average density of the pencil?

11 A very light dish with a 1 kg iron weight in it floats in a bowl of water. If the block of iron is removed from the dish and dropped into the water, will the level of the water in the bowl rise, fall or remain unchanged?

12 The envelope and basket of an instrument-carrying balloon has a total mass of 1.6 kg, and its volume when inflated with hydrogen is 3.0 m^3. What is the mass of the instruments which can just be raised by the balloon? Density of hydrogen = 9.0×10^{-2} kg m^{-3}; density of air = 1.29 kg m^{-3}.

13 What load could just be lifted off the ground by a spherical balloon 2.0 m in diameter containing hydrogen if the mass of the envelope and fittings is 200 g? Use the densities of hydrogen and air given in the previous question.

14 The mass of the envelope and basket of a balloon is 2000 kg. What is the volume of the envelope if the balloon is just on the point of rising from the ground? The volume of the basket may be neglected. Density of hydrogen = 9.00×10^{-2} kg m^{-3}; density of air = 1.29 kg m^{-3}.

15 A cork of mass 12 g is attached to a piece of brass (density 8.2×10^3 kg m^{-3}) which weighs 49.2 g in air. The combination immersed in water weighs 7.2 g. Calculate the density of the cork.

16 A badly prepared iron casting contains cavities. It weighs 40.0 kg in air and 34.2 kg in water. Given that the densities of iron and water are 8.00×10^3 and 1.00×10^3 kg m^{-3} respectively, determine the volume of the cavities in the casting.

17 What is the buoyancy correction for a brass weight of 500 g? Density of brass = 8.4×10^3 kg m^{-3}; density of air = 1.25 kg m^{-3}.

18 A gold sphere is weighed in air with brass weights and its mass is found to be 160.20 g. What is the buoyancy correction to be applied to this result? Density of gold = 1.93×10^4 kg m^{-3}; densities of brass and air given in previous question.

19 A block of wood of density 2.0×10^2 kg m^{-3} is placed in a liquid of density 8.00×10^2 kg m^{-3}. What fraction of the volume of the block is above the surface of the liquid?

The block is then completely submerged by means of a thread attached to it and to the bottom of the vessel containing the liquid. Draw a diagram showing the forces acting on the block and, given that the mass of the wood is 50 g, calculate the tension in the thread.

20 In Figure 11.2, AB is a uniform rod free to swing about a fulcrum at its centre of gravity; C and D are identical blocks of iron, each of volume 10.0 cm^3, one being immersed in water and the other in kerosene. E is a mass of 4.16 g. Given that the arrangement is just balanced and that kerosene floats on water, state which mass is immersed in water and calculate the density of kerosene.

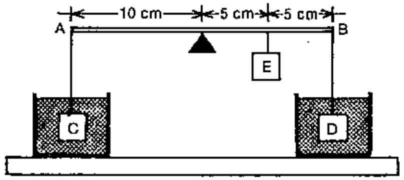


Fig. 11.2

21 A body weighs 86.0 g in air. Its apparent weights in each of two liquids are 72.4 g and 63.9 g respectively. In a completely miscible mixture of the two liquids it appears to weigh 67.1 g. Calculate the proportion by volume in which the two liquids have been mixed.

SET 12 VERTICAL MOTION NEAR THE EARTH'S SURFACE

Topic	*Problems*
Objects dropped, with negligible air resistance	1–3
Objects projected upwards, with negligible air resistance	4–9
More difficult questions, with negligible air resistance	10–11
Vertical flight, with significant air resistance	12–14

Before commencing this set, refer to the note on the *gravitational field strength* on page x.

For problems where the question number is highlighted, i.e. Problems **12** to **14**, the effect of air resistance *should* be taken into account; in all other problems in this set air resistance is to be regarded as negligible.

1 An object falls freely from the top of a high overhanging cliff.
(*a*) What is its speed at the end of 4.00 s?
(*b*) How far did it fall in that time?
(*c*) How far does it fall in the third second of its motion?

2 A stone is dropped from a height of 51.2 m above the ground. (Take g = 10 m s^{-2}.)
(*a*) How long does it take to reach the ground?
(*b*) What is its speed just before it strikes the ground?
(*c*) How far does it fall during the last second of its motion?

3 A balloon is moving vertically downwards with a uniform speed of 5.0 m s^{-1}. When it is 60 m from the ground, a small object is dropped from it. How long before the balloon will the small object reach the ground? (Take g = 10 m s^{-2}.)

4 A ball is thrown vertically upwards with an initial speed of 24.5 m s^{-1}.
(*a*) Calculate the greatest height it will attain.
(*b*) How long does it take to reach its maximum height?
(*c*) What is its speed when it returns to the starting point?
(*d*) What is the total time for its upward and downward movement?
(*e*) What is the ball's acceleration at the highest point of its flight?

5 An arrow which is fired vertically upwards leaves the bow at a speed of 30.0 m s^{-1}. How far will it be above the ground when it has been in flight for 2.00 s?

6 A stone is projected vertically upwards with an initial speed of 29.4 m s^{-1} near a building which is 24.5 m high. For what interval of time is the stone above the level of the top of the building?

7 A cricket ball is thrown vertically upwards with a speed of 19.6 m s^{-1}. For what fraction of its total time of flight is it above a height of 14.7 m?

8 A helicopter is rising vertically with a uniform speed of 14.7 m s^{-1} and a wheel drops off when it is 49.0 m from the ground. Calculate the time taken for the wheel to reach the ground and its velocity at impact.

9 A balloon is 30.0 m above the ground and is rising vertically with a uniform speed when a coin is dropped from it. If the coin reaches the ground in 4.00 s, what is the speed of the balloon?

10 A surveyor's tape is hung vertically from the top of a building and a small ballbearing is dropped from the zero mark. A photograph of the falling ballbearing shows that it falls from the 4.90 m mark to the 4.91 m mark while the shutter of the camera is open. What is the length of the exposure?

11 A stone is projected vertically upwards from the top of a tower. It is found that its speed at a distance x m below the point of projection is twice its speed at the same distance above that point. Show that the stone rises a distance $\frac{5x}{3}$ m above the top of the tower.

12 A stroboscopic photograph is taken of a falling ping-pong ball of mass 2.50 g. Analysis of the photograph showed that at a point X the ball had a downward acceleration of 4.00 m s^{-2}.
(*a*) What is the net force acting on the ball at point X?
(*b*) What is the weight of the ping-pong ball?
(*c*) What is the force of air resistance acting on the ball at point X?

13 Consider again the questions asked in Problem 4 (*a*), (*b*), (*c*) and (*e*) in this set, but this time assume that the ball *is* affected by air resistance. For each question—omit (*d*)—write

H if the answer is higher
L if the answer is lower
S if the answer is the same

as it was in Problem 4.

14 If a tennis ball is hit straight up in the air, its flight will be affected by air resistance. Will the upward part of the flight take a longer, a shorter or the same time as the downward part of the flight?

SET 13 PROJECTILE MOTION NEAR THE EARTH'S SURFACE

Topic	*Problems*
Horizontal projection, without considering air resistance	1–6
Oblique projection, without considering air resistance	7–14
Horizontal and oblique projection, including the effects of air resistance	15–17

Before commencing this set, refer to the note on the *gravitational field strength* on page x.

In *all* questions in this set assume that the effects of *spinning* are negligible.

For problems where the question number is highlighted, i.e. Problems **15** to **17**, the effect of *air resistance* should be taken into account; in all other problems in this set air resistance is to be regarded as negligible.

1 When a projectile is launched horizontally,
(*a*) which component of its velocity remains constant, and

(*b*) which component changes uniformly with time?
(*c*) If the initial horizontal velocity of the projectile is decreased, will the time taken for the projectile to fall to the ground be decreased, increased, or remain the same?

2 When an aeroplane is at an altitude of 490 m and moving horizontally at a speed of 160 m s^{-1}, a small object is dropped from it. The following questions refer to the period from immediately after the instant the object is dropped till just before it hits the ground.
(*a*) What is the vertical component of the object's initial velocity?
(*b*) What is the vertical component of its displacement?
(*c*) What is the vertical component of its acceleration?
(*d*) Deduce the time it is in flight.
(*e*) What is the horizontal component of its initial velocity?
(*f*) What is the horizontal component of its final velocity?
(*g*) What is the horizontal component of its acceleration?
(*h*) Deduce the horizontal component of its displacement.

3 A ball is thrown horizontally with a speed of 14.0 m s^{-1} from a point 6.40 m above the ground. Calculate
(*a*) the time taken by the ball to reach ground level;
(*b*) the horizontal distance travelled in that time;
(*c*) its velocity when it reaches the ground.

4 A stone is projected out to sea from the edge of a vertical cliff 122.5 m high, the velocity of projection being 7.00 m s^{-1} horizontally. Find
(*a*) the time the stone takes to reach the water;
(*b*) the distance from the base of the cliff at which it strikes the water;
(*c*) the velocity at that instant.

5 Two stones are thrown simultaneously in line and horizontally from the top of a cliff 78.4 m high with speeds 5.00 m s^{-1} and 20.0 m s^{-1} respectively. How far apart will they strike the water? Find the velocity of the first stone as it reaches the water.

6 A bomber travelling horizontally with a speed of 576 km h^{-1} at an altitude of 10 000 m drops a bomb.
(*a*) Taking the origin of a Cartesian coordinate system as the point at which the bomb was dropped and the *x*-direction as that in which the plane was travelling, deduce two equations which give the position of the bomb at time *t* after it was released. (*x* and *y* will represent distances in metres, and *t* the time in seconds. The two equations will give *x* and *y* as functions of *t*.)
(*b*) Deduce the equation which describes the path of the bomb while in flight.
(*c*) What is the name of the shape of the path followed by the bomb?

7 A tennis player hits a lob shot at point X in Figure 13.1 with a velocity of 13 m s^{-1} at an angle of 60° to the horizontal.

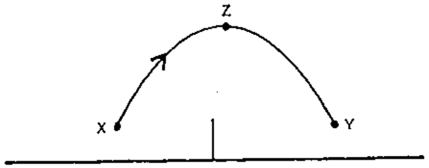


Fig. 13.1

Answer the following questions for the flight of the ball from X to a point Y, the same height above the ground as X.
(*a*) What is the vertical component of the ball's initial velocity?
(*b*) What is the vertical component of the ball's velocity at Z, the highest point of its flight?
(*c*) What is the vertical component of the ball's acceleration as it moves from X to Z?
(*d*) How long does the ball take to travel from X to Z?
(*e*) How long does the ball take to travel from X to Y?
(*f*) What is the horizontal component of the ball's velocity at X?
(*g*) What is the horizontal component of the ball's velocity at Z?
(*h*) What is the horizontal distance from X to Y?

8 A stone is thrown with a velocity of 20 m s^{-1} at an angle of 40° with the horizontal. Determine
(*a*) the time for which it is above the level at which it is thrown;
(*b*) the greatest vertical height reached;
(*c*) its range on a horizontal plane.

9 At the XXI Olympiad in Montreal in the Women's High Jump a new Olympic record was set by Rosemarie Ackermann of East Germany.
(*a*) If 0.44 s elapsed between the instant she left the ground and the time she passed over the bar, what vertical height did she raise her centre of mass?
(*b*) If her centre of mass was 1.10 m above the ground at take-off and 0.12 m above the bar as she cleared it, what was the new Olympic record?

10 A missile is projected with a velocity of 30.0 m s^{-1} at an angle of 60.0° with the horizontal. At what height will it strike a cliff 45.0 m distant?

11 A cricketer hits a six, the ball having travelled 100 m horizontally when it strikes the ground beyond the boundary. The greatest height reached by the ball is 25 m. Calculate the velocity with which the ball left the bat.

12 A boy standing on the tray of a utility travelling at 18 km h^{-1} throws a ball with a velocity of 7.35 m s^{-1} vertically upwards (relative to himself) and catches it again at the same level. What *horizontal* distance does the ball move while it is in the air?

13 Prove that, if the barrel of a pistol is held at an angle to the horizontal so that it is directed towards a particular point, a bullet fired from the pistol will always collide with a second bullet which is released from that point at the instant of firing the pistol. Assume that the distance of the second bullet from the pistol is within the range of the weapon.

14 The greatest horizontal distance through which a person can throw a stone is *d*. What is the greatest vertical height to which he can throw it? Assume that the initial speed of the stone is the same in each case.

15 Consider again the questions asked about the object dropped from the aeroplane in Problem 2 of this set, but this time take into account the effects of air resistance. For (*a*) to (*g*)—omit (*h*)—write
H if the new answer is higher,
L if it is lower, and
S if it remains unchanged.

16 Refer back to the flight of the tennis ball in Problem 7, only this time consider the effect of air resistance. Omit (*e*). Write
H if the new answer is higher,
L if it is lower, and
S if it is the same
as it was in Problem 7.

17 Would a cricket ball thrown in from the boundary cover a longer, a shorter or the same *horizontal* distance during the rising part of its flight as during the falling part? Take into account the frictional effect of the air on the ball.

SET 14 UPHILL AND DOWNHILL MOTION

Topic	*Problems*
Objects at rest on a frictionless inclined plane	1-2
Objects at rest on a rough inclined plane	3
Motion at constant velocity on an inclined plane	4
Accelerated motion on an inclined plane	5-8

1 A small barrel of mass 50.0 kg is held at rest on a smooth ramp inclined at 30.0° to the horizontal by means of a light rope parallel to the ramp. By means of resolution determine
(*a*) the component of the weight of the barrel acting down the ramp;
(*b*) the tensile force in the rope;
(*c*) the force exerted on the barrel by the ramp.

2 Find the horizontal force required to keep a small box of mass 200 g at rest on a smooth plane inclined at 30.0° to the horizontal.

3 A VW delivery van of mass 1.3×10^3 kg is parked on a 5° hill. What is the total uphill frictional force acting on the tyres of the van?

4 In order to set up a friction-compensated ramp for a dynamics experiment, a girl finds that she has to raise one end of the 2.0 m ramp 3.0 cm before a 1.5 kg dynamics trolley will run down the ramp at constant velocity.
(*a*) What is the net force on the moving trolley?
(*b*) What is the effective uphill frictional force acting on the trolley?

5 A 66 kg cyclist pedals his 14 kg bike down a 6° hill accelerating at 1.2 m s^{-2}.
(*a*) What is the total net force acting on the cyclist and his bike?
(*b*) What is the total downhill component of the weight of the cyclist and his cycle?
(*c*) If frictional forces on the bike produce an effective uphill force of 4.0 N, what is the effective downhill driving force provided by the cyclist?

6 A mass of 200 kg is being pulled up a slope inclined at 17.0° to the horizontal by means of a force along a rope parallel to the slope with an acceleration of 2.40 m s^{-2}. If the force of friction between the mass and the slope is 117 N, calculate the size of the force in the rope.

7 A 1.8 tonne (1.8×10^3 kg) railway truck moving at a speed of 3.0 m s^{-1} up a steady incline comes to rest in 45 m. If the frictional forces opposing its motion amount to 80 N, what is the angle of inclination?

8 A mass of 4.0 kg and a mass of 3.0 kg are connected by a light string. The 4.0 kg mass is placed on a smooth plane inclined at 30° to the horizontal and the string passes over a smooth pulley at the top of the plane, the 3.0 kg mass hanging freely. Draw a diagram of the arrangement, show the forces acting on each mass and determine the acceleration of each mass.

SET 15 MOTION IN A CIRCLE

Topic	*Problems*
Motion in a horizontal circle at constant speed	1–8
Motion in a horizontal circle with speed varying	9
Conical pendulum	10–11
Motion on banked curves	12–13
Motion in vertical circles or arcs	14–16

1 A Holden Gemini is driven around a curve of radius 18 m at a constant speed of 12 m s^{-1}.
(*a*) What is the magnitude of the acceleration of the car?
(*b*) If the car has a mass of 923 kg and is being driven by a 77 kg driver, what is the magnitude of the net force on the car and driver?
(*c*) In what direction does this net force act?

2 An engine of mass 4.0×10^4 kg travels at a constant speed of 10 m s^{-1} on a level track in the form of a circular arc of radius 200 m. Calculate the total horizontal thrust exerted by the rails on the flanges of the wheels.

3 Given that the radius of the Earth at the equator is 6.4×10^6 m, calculate the magnitude of the centripetal acceleration of a point on the equator.

4 A flywheel of radius 19 cm on a horizontal rotary engine rotates at 2400 r.p.m. (revolutions per minute).
(*a*) What is the frequency of rotation in hertz (or revolutions per second)?
(*b*) What is the centripetal acceleration of a point on the rim of the flywheel?

5 A mass of 5.0 kg is whirled in a horizontal circle at one end of a string 49 cm long, the other end being fixed. If the string when hanging vertically will just support a load of 200 kg mass without breaking, find the maximum whirling speed in revolutions per second.

6 Without special equipment a pilot in a high speed aircraft "blacks out" if his aircraft describes a path with too small a radius of curvature. Under these conditions he is subject to excessive acceleration. Supposing he cannot withstand an acceleration due to the aircraft's motion in excess of 4 *g*, what should be the minimum permissible radius of a turn (in a horizontal circle) of an 800 km h^{-1} aircraft?

7 A mass of 1.7 kg, on the surface of a smooth horizontal table, revolves in a circular path, to the centre C of which it is attached by a light string of length 2.0 m. See Figure 15.1.

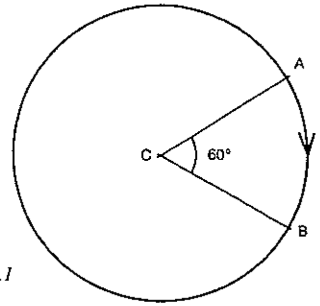


Fig. 15.1

Given that the angular velocity is 3.0 rad s^{-1}, what is
(*a*) the velocity at A?
(*b*) the velocity at B?
(*c*) the time to travel from A to B?
By using a graphical method, or by calculation, determine, for the motion from A to B,
(*d*) the change in velocity (relating its direction to the tangent at B);
(*e*) the magnitude of the average acceleration;
(*f*) the tension in the string.

8 A body of mass 2.0 kg has a velocity of 5.0 m s^{-1}. When it reaches a point P, it is acted on by a force of 10 N which is always at right angles to its direction of motion. Find the time taken for the body to return to the point P.

9 A 70 kg speedway rider on his 90 kg bike is travelling at 16 m s^{-1} north at point X on a section of the track which has a radius of 64 m. See Figure 15.2. At X he is speeding up at a rate of 0.30 m s^{-1} every 0.10 s.

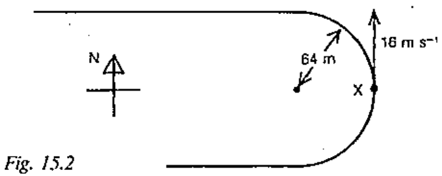


Fig. 15.2

At the point X specify fully
(*a*) his tangential acceleration;
(*b*) his centripetal acceleration;
(*c*) his net acceleration;
(*d*) the net force on him and his bike.

10 In an investigation of uniform circular motion, a student, with his hand above his head, whirls a rubber stopper of mass 50 g in a horizontal circle on the end of a 1.7 m nylon thread. His hand is 0.80 m above the horizontal circle.
(*a*) Draw a diagram, marking in the two forces acting on the rubber stopper.

(*b*) What is the radius of the circular path?
(*c*) What is the net force acting on the stopper?
(*d*) Determine the tension in the nylon thread.
(*e*) What is the period of revolution of the stopper?

11 Two spheres A and B are attached to the ends of light strings XA and XB. They are swung in horizontal circles as indicated in Figure 15.3. The time for one complete revolution is the same for both A and B. The radius of A's path is 0.20 m, and the radius of B's path is 0.30 m.

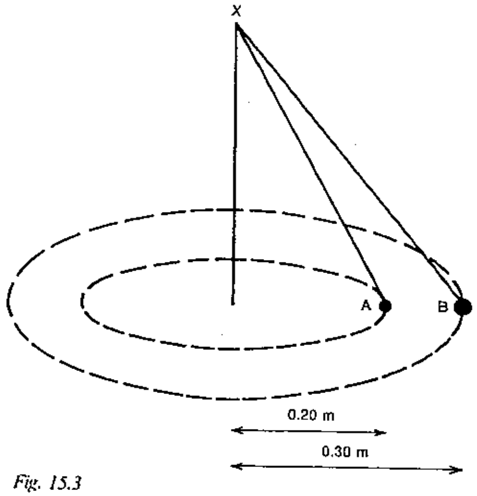


Fig. 15.3

(*a*) What is the value of the ratio
$$\frac{\text{speed of sphere B}}{\text{speed of sphere A}}?$$
(*b*) What is the value of the ratio
$$\frac{\text{centripetal acceleration of B}}{\text{centripetal acceleration of A}}?$$
(*c*) If mass of sphere B = 4 times mass of sphere A, what is the value of
$$\frac{\text{centripetal force on B}}{\text{centripetal force on A}}?$$

12 A cycling velodrome track is so banked at a certain point, where the horizontal radius of curvature is 30 m, that a cyclist may take the bend at 16 m s^{-1} without depending on a sideways frictional force on his tyres. What is the angle between the horizontal and the track?

13 A turn on a railway line has a radius of 351 m and is banked to suit a speed of 22 m s^{-1}. Find the angle of banking.

14 A 0.500 kg stone is whirled on the end of a 1.20 m string in a vertical circle at a constant speed of 6.00 m s^{-1}.
(*a*) Specify fully the net force acting on the stone at
(*i*) the highest, and
(*ii*) the lowest point of its circular path.
(*b*) Specify fully the force exerted on the stone *by the string* at
(*i*) the highest, and
(*ii*) the lowest point of the circular path.

15 A pilot flies his plane in a vertical loop of radius 500 m at such a speed that at the top of the loop he feels no force from either the seat or the seat belt.
(*a*) At what speed is the plane then flying?
(*b*) If his speed rises to 100 m s^{-1} by the time he reaches the bottom of the loop, what force does he experience from the seat? Take his mass as 67 kg.

16 A car of mass 500 kg is travelling on a "big dipper" at Luna Park. Find the force exerted by the track on the car
(*a*) when it passes over the top of a curve of 7.00 m radius at a speed of 8.00 m s^{-1};
(*b*) when it passes over the bottom of a curve of the same radius at 10.0 m s^{-1}.

SET 16 SIMPLE HARMONIC MOTION

Topic	*Problems*
Dependence of net force and acceleration on displacement from the mean position	1–2
Period and frequency of a simple harmonic motion (SHM)	3–5
Velocity of an object moving in SHM	6–8
More complex problems on SHM	9–13
Simple pendulum as an approximation to SHM	14–16
SHM variables in trigonometric form	17–18

Questions relating to kinetic and potential energies of objects moving in SHM will be found in Set 20, Problems 4 to 8.

1 A mass of 0.50 kg executes simple harmonic motion while hanging on the end of a spring which has a force constant of 25 N m^{-1}.
(*a*) Specify fully the net force acting on the mass when it is displaced 0.10 m below its mean position.
(*b*) What is its acceleration when it is 0.18 m above its mean position?

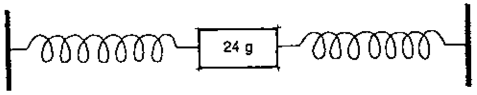


Fig. 16.1

2 A mass of 24 g attached to two springs as shown in Figure 16.1 is caused to execute simple harmonic motion along a frictionless surface with an amplitude of 0.40 m. The force–displacement graph for the body moving under the action of the springs is given in Figure 16.2, with *positive* force indicating force to the *right*.

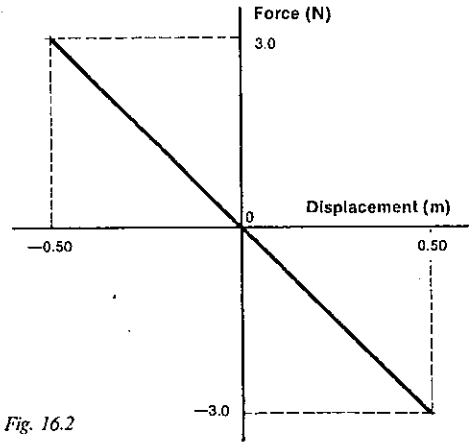


Fig. 16.2

(*a*) What is the force constant of the system of the two springs?
(*b*) What is the net force on the mass when it is 0.30 m to the right of its mean position?
(*c*) What is its displacement when it has its maximum acceleration to the right?
(*d*) What is its maximum acceleration to the right?

3 A 810 kg car has a suspension system with a force constant of 5×10^4 N m^{-1}. If the car goes over a bump, what is the period of the SHM set up? (The fact that the SHM is quickly damped by the shock absorbers does not affect the period.)

4 An ice-puck of mass 0.20 kg is caused to move with SHM between two points A and E on a smooth level surface with a period of $\frac{2\pi}{5}$ s. Its path is illustrated in Figure 16.3 in which the points A, B, C, D and E are evenly spaced.

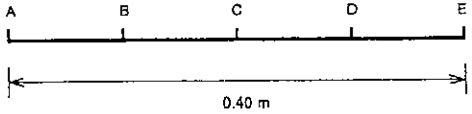


Fig. 16.3

(*a*) What is the force constant of the system?
What is the magnitude and direction of the force acting on the ice puck,
(*b*) when it is in position A?
(*c*) when it is at C?
(*d*) when it is at D?

5 A body of mass 2.00 kg is moving in SHM with a frequency of 1.90 Hz and an amplitude of 40.0 cm. Calculate the magnitude of the restoring force when
(*a*) the body is in its extreme position;
(*b*) its displacement is 20.0 cm from its mean position.

6 Refer back to Problem 4. At which point or points in Figure 16.3 is the speed of the ice-puck
(*a*) a maximum?
(*b*) zero?
Determine the puck's speed
(*c*) at C;
(*d*) at D.

7 Calculate the maximum speed of the tip of the prong of a tuning fork of frequency 256 Hz when its amplitude is 1.00 mm.

8 A particle of mass 0.500 g moves with SHM, the period being $\frac{\pi}{5.00}$ s and the amplitude 8.00 cm. Find the velocity of the particle when it is at a distance of 4.00 cm from its mean position.

9 A body of mass 16.0 g is executing SHM. When its displacement is 4.00 cm, the force acting on it is 0.196 N. If the amplitude is 6.00 cm, find the period and the maximum speed.

10 A particle is executing SHM. Show that the time interval required to undergo a displacement from the mean position equal to half the amplitude is one-twelfth of the period of vibration.

11 A stick of length L mm floats vertically in a liquid, with a length h mm projecting above the surface of the liquid. Show that it executes SHM when given a small vertical displacement, and find the period of the motion.

12 A U-tube which has a uniform bore stands vertically. It contains mercury which measures 40 cm around the tube from surface to surface. Show that, if the mercury is slightly depressed on one side, it oscillates up and down the two sides of the tube with SHM, and calculate the period of the motion.

13 A long glass tube open at both ends is partly immersed in water. Oil of density 8×10^2 kg m^{-3} is poured into it until the top of the oil is h m above the free surface of the water. What is the length of the oil column? The oil column is given a small vertical displacement. Find the period of its oscillation. Density of water $= 10^3$ kg m^{-3}.

14 What is the period of a simple pendulum 0.400 m long in a locality where $g = 9.80$ N kg^{-1}?

15 A "seconds" pendulum (that is, one having a period of 2.000 s) is 99.30 cm long in Melbourne in 99.13 cm long in Madras. What is the difference in the gravitational attraction on a mass of 1.000 kg in Melbourne and in Madras?

16 Find the gain or loss in seconds per day when a pendulum clock which keeps correct time at a place where $g = 9.7998$ N kg^{-1} is taken to another place where $g = 9.8100$ N kg^{-1}.

17 At time t second the displacement y metre of an object of mass m kilogram moving in SHM is given by

$$y = R \sin \omega t,$$

where ω is the angular velocity of the SHM expressed in rad s^{-1}.
Deduce expressions for
(*a*) the velocity v of the object;
(*b*) the acceleration a of the object;
(*c*) the net force F acting on the object
as functions of time t.

18 The motion of the end E of a pendulum in a grandfather clock is described by the expression:

$$x = 0.10 \sin \pi t,$$

where x (in metres) is the displacement of E from its mean position at time t (in seconds).
(*a*) What is the amplitude of the SHM?
(*b*) What is the frequency of the SHM?
(*c*) In terms of t, derive expressions for
(*i*) the velocity v of E;
(*ii*) the acceleration a of E.
(*d*) What is the maximum value of the velocity of E?
(*e*) What is the magnitude of the acceleration of E at $t =$ 1.8 s?

SET 17 MOMENTUM AND ITS CONSERVATION

Topic	*Problems*
Impulse, momentum and Newton's Second Law of Motion	1–5
Conservation of momentum in one dimension	6–14
Newton's Third Law of Motion	15–16
Force and pressure produced by a continuous stream of particles	17–18
Centre of mass	19–23
Conservation of momentum in two dimensions	24–26

Further problems involving conservation of momentum along with conservation of kinetic energy will be found in Set 21.

1 Specify fully the momentum of
(*a*) a 0.16 kg hockey ball moving at 6.0 m s^{-1} east;
(*b*) an oxygen molecule of mass 5.3×10^{-26} kg in the air moving straight upwards at 4.7×10^2 m s^{-1};
(*c*) a 700 kg Suzuki driven by a 50 kg driver at 72 km h^{-1} south.

2 In a game of snooker a player hits a 0.20 kg billiard ball with the cue, exerting an average force of 40 N south on the ball for 12 millisecond.
(*a*) What is the impulse of the force exerted on the ball?
(*b*) What is the change in momentum of the ball?
(*c*) With what velocity does the ball leave the cue?

3 Estimate, as an order of magnitude,
(*a*) the momentum of a football which has just been kicked by a league player;
(*b*) the impulse given by a typist's finger to a typewriter key.

4 Figure 17.1 shows a somewhat idealised graph of the force–time relationship for the interaction between a 50 g golf ball and the club head hitting it.

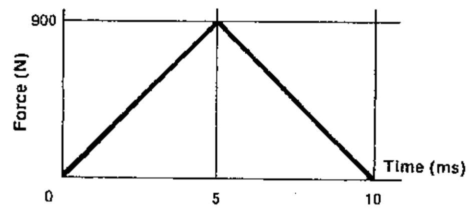


Fig. 17.1

Determine the magnitude of
(*a*) the average force exerted by the club head on the golf ball;
(*b*) the impulse of the force exerted on the ball;
(*c*) the resulting change in momentum of the ball;
(*d*) the velocity of the ball as it leaves the club head;
(*e*) the average rate of change of momentum of the ball while being hit. Be sure you understand the connection between the answers to (*a*) and (*e*).

5 A competitor in a women's shot-put event drops the 4.00 kg shot and it falls vertically. Assume the air resistance is negligible. While it is falling,
(*a*) what is the net force on the shot?
(*b*) at what rate is its momentum changing?

6 A truck of mass 4×10^3 kg moving at 3.5 m s^{-1} collides with a stationary truck to which it becomes automatically coupled. Immediately after the impact the trucks move along together at 2 m s^{-1}. Find the mass of the second truck.

7 A bag of sand of mass 9 kg which is suspended by a rope is used as a target for a pistol bullet of mass 30 g and which strikes the bag with a horizontal velocity of 301 m s^{-1}. If the bullet remains embedded in the sand, determine the velocity of the bag immediately after impact.

8 A man of mass 80 kg stands on a trolley of mass 120 kg and throws a parcel of mass 10 kg with a velocity of 4.0 m s^{-1} horizontally away from the rear of the trolley. What is the speed with which the trolley and the man commence to move?

9 A bullet of mass 20 g is fired from a rifle of mass 4.0 kg with a speed of 500 m s^{-1}. Calculate the initial speed of recoil of the rifle.

10 A tape running through a ticker-timer is attached to a small cart X of mass 0.50 kg. The cart X collides with a stationary cart Y and both move together on a frictionless surface. Figure 17.2 shows portion of the tape, including the instant of collision.

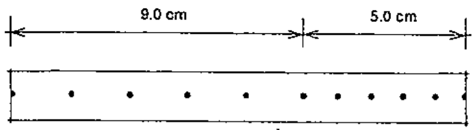


Fig. 17.2

(*a*) What is the mass of cart Y?
(*b*) The two carts are standing together on the frictionless table 2.4 m long with barriers at its ends. Spring bumpers between the carts can be suddenly exploded. If the ends of the carts are 0.60 m apart, how far from the barrier nearest the 0.50 kg cart must the centre of the two-cart system be placed so that, after the explosion, the carts hit the barriers simultaneously?
(*c*) A brick of mass 1.5 kg is dropped vertically from a height on to cart X when it is moving with a speed of 0.20 m s^{-1}. What is the new speed of the loaded cart?

11 A small cart of mass 300 g when empty can move freely on a horizontal surface. The cart contains 100 g of sand and moves initially with a speed of 18 cm s^{-1}.
(*a*) What would be the speed of the cart if a further 200 g of sand were dropped vertically into it from a stationary container?
(*b*) If, instead of adding sand to the cart, the original 100 g of sand were allowed to pour out through a hole in the bottom of the cart, what would be the final speed of the cart?

12 A toy locomotive of mass 420 g moving at 30 cm s^{-1} south collides with a carriage of mass 200 g moving at 32 cm s^{-1} north. If they become coupled together, what is their common velocity after the collision?

13 A ball of mass 50.0 g moving horizontally at a speed of 40.0 cm s^{-1} strikes a suspended plate of mass 1000 g and rebounds from it with a speed of 25.0 cm s^{-1}. Find the speed with which the plate commences to move.

14 Two wooden blocks of mass 6 kg and 2 kg respectively approach each other from opposite directions on a smooth level surface at a relative speed of 15 m s^{-1}. After a head-on collision they separate at a relative speed of 5 m s^{-1}. The initial velocity of the 6 kg block was 9 m s^{-1} east. Find the velocity of
(*a*) the 2 kg block immediately before the impact;
(*b*) the 6 kg block immediately after the impact;
(*c*) the 2 kg block immediately after the impact.

15 On a linear air track a glider A of mass 0.10 kg moves at 4.0 m s^{-1} east towards a stationary glider B of mass 0.40 kg. Both gliders are fitted with spring buffers. After they interact A moves west at 0.80 m s^{-1} while B moves east at 1.2 m s^{-1}.
(*a*) Use these measurements to check if the total momentum of the two gliders is the same before and after the collision.
(*b*) What is the change in momentum of
(*i*) A?
(*ii*) B?
(*c*) How are the changes in momentum related?
(*d*) How then is the impulse that B exerts on A related to the impulse that A exerts on B?
(*e*) Since B acts on A for the same time interval as A acts on B, how is the force that B exerts on A related to the force that A exerts on B?

16 Suppose your hockey stick exerts an average force of 150 N south on the ball as you hit it.
(*a*) What is the average force that the ball exerts on your hockey stick?
(*b*) Does it then follow that the net force on the ball is zero?

17 Water from a fire hose which discharges 960 kg min^{-1} hits an east-west wall horizontally at a velocity of 15.0 m s^{-1} south. Suppose that the cross-section of the jet is 20 cm^2 when it hits the wall, and (unrealistically) that it spreads out evenly over the surface without splashing backwards.
(*a*) What is the change in momentum of 960 kg of water as it strikes the wall?
(*b*) What is the rate of change of momentum of this 960 kg of water as it hits the wall?
(*c*) What is the force on the water from the wall, causing this change in momentum?
(*d*) What is the force on the wall exerted by the water?
(*e*) Determine the pressure exerted on the wall by the water.
(*f*) Actually there would be some splashback from the wall. Would the *real* pressure be higher or lower than your answer to (*e*)?

18 An air blower with a nozzle velocity v and nozzle cross-sectional area A is held very close and perpendicular to a surface to be cleaned. Suppose that, on average, the air molecules have a mass of m and bounce back perpendicularly off the surface with a speed of $\frac{3}{4}v$. If there are N molecules per unit volume in the stream, derive an expression for
(*a*) the force, and
(*b*) the pressure
exerted on the surface by the air stream.

19 Find the position of the centre of mass of
(*a*) a 5.0 kg mass and a 1.0 kg mass which are 2.4 m apart;
(*b*) masses of 4.0 kg and 3.0 kg which are 8.4 m apart.

20 Three small masses are arranged as shown in Figure 17.3. Find the position of the centre of mass of the system.

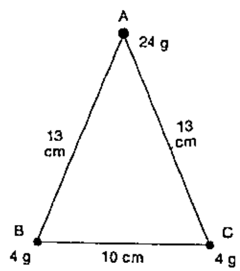


Fig. 17.3

21 A mass of 5.0 kg is moving with a speed of 12 m s^{-1} towards a mass of 3.0 kg at rest on a smooth surface.
(*a*) How far from the 5.0 kg mass is the centre of mass of the system at the instant when this mass is 2.0 m from the stationary mass?
(*b*) What is the speed of the centre of mass?
(*c*) If the two masses move along together after colliding, what is their common speed?
(*d*) What is the change in velocity of the centre of mass?

22 Two masses, of 4.0 kg and 3.0 kg respectively, are travelling to the east along a frictionless surface with respective speeds of 12.0 m s^{-1} and 5.0 m s^{-1} as illustrated in Figure 17.4.

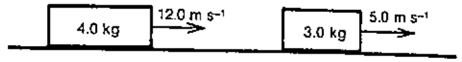


Fig. 17.4

(*a*) What is the velocity of the centre of mass?
(*b*) What is the velocity of the 4.0 kg mass relative to the centre of mass?
(*c*) What is the velocity of the 3.0 kg mass relative to the centre of mass?
(*d*) What is the total momentum of the system relative to the centre of mass?
(*e*) If the 3.0 kg mass continues to move to the east with a speed of 9.0 m s^{-1} after the collision, what is the velocity of the 4.0 kg mass?
(*f*) What is then the velocity of the centre of mass?

23 Masses of 20 kg and 4 kg respectively are approaching each other on a frictionless surface with the speeds indicated in Figure 17.5.

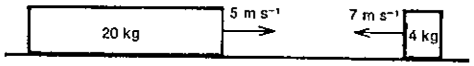


Fig. 17.5

(*a*) Find the velocity of the centre of mass.
(*b*) What is the momentum of the system relative to the centre of mass?
(*c*) If the velocity of the 4 kg mass changes by 10 m s^{-1} to the right as a result of the collision, find the velocity of the 20 kg mass after the collision.

24 A ball A with a mass of 1.0 kg and moving at 4.0 m s^{-1} strikes a glancing blow on a second ball B which is initially at rest. After the collision, ball A is moving at right angles to its original direction at a speed of 3.0 m s^{-1}, as illustrated in Figure 17.6.

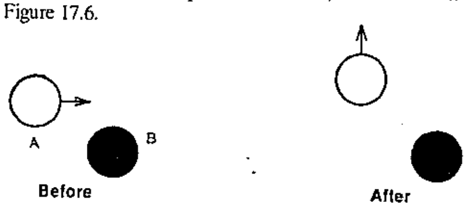


Fig. 17.6

(*a*) What is the magnitude of the momentum of B after the collision?
(*b*) In what direction is B moving after the collision?

(*c*) If B has a mass of 5.0 kg, what is its speed after the collision?
(*d*) If the time of the impact was 0.020 s, what was the magnitude of the average force exerted on B during the collision?

25 A radioactive nucleus decays by emitting an electron and a neutrino at right angles to each other in a horizontal plane. The momentum of the electron is 1.2×10^{-22} kg m s^{-1} north, and that of the neutrino is 6.0×10^{-23} kg m s^{-1} west.
(*a*) Assuming that the original nucleus was at rest, what is the total momentum of the electron, neutrino and product nucleus immediately after the decay?
(*b*) What was the momentum of recoil of the product nucleus?
(*c*) If the mass of the product nucleus is 3.3×10^{-25} kg, with what velocity did it recoil?

26 A body of mass 400 g is moving along a smooth surface at a velocity of 1.25 m s^{-1} towards the east. It strikes a body of mass 600 g, initially at rest, and then the 400 g body moves at a velocity of 1.00 m s^{-1} in a direction 36.9° north of east.
(*a*) What is the *easterly* component of the total momentum of the system before and after the collision?
(*b*) What is the *northerly* component of the total momentum of the system before and after the collision?
(*c*) Determine the final velocity of the 600 g body.

SET 18 WORK, KINETIC ENERGY AND POWER

Topic	*Problems*
Work done by a force	1-8
Kinetic energy in terms of mass and speed	9-10
Equality of work done and consequent increase in kinetic energy	11-17
Relation between kinetic energy, mass and momentum	18-20
Power	21-25

1 What work is done by
(*a*) a force of 12 N west which pushes a box 6.0 m west?
(*b*) a force of 16 N south which moves an object 20 cm south?

2 How much work is done when
(*a*) a bricklayer lifts a brick weighing 40 N a vertical height of 1.5 m?
(*b*) 1000 kg of coal is raised from the bottom of a mine 300 m deep?
(*c*) a box is pushed 10 m north across a floor at a constant speed against a frictional force of 42 N south?

3 How much work is done by a force which causes a mass of 30 kg to have an acceleration of 1.5 m s^{-2} south while it is displaced 0.60 m south?

4 240 J of work done on a body of mass 8.0 kg caused it to move 12 m south-west. If the acceleration over this distance was constant, calculate its value.

5 A toboggan is pulled along for a distance of 15 m by a rope making an angle of 26° with the horizontal. If the pull on the rope is 4.0 N, find the work done.

6 A body B moves in a plane under the action of a force F which is always parallel to a fixed line Ox in the plane. It is restricted by rails to move along the path pqrs as shown in Figure 18.1. If the magnitude of the force F is 8.0 N, how much work is done by this force as the body moves
(*a*) from p to q?
(*b*) from q to r?
(*c*) from r to s?

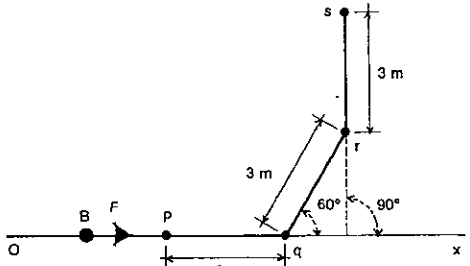


Fig. 18.1

7 A hill rises 1.0 m vertically for every 14 m of its *inclined* length. A truck of mass 3000 kg travels a distance of 40 m up the slope at a constant speed. Find the work done by the motor if frictional resistance is equivalent to 100 N down the hill.

8 Figure 18.2 gives the net force–distance graph for a body which was moved from a rest position along a rough level surface by a horizontal force.

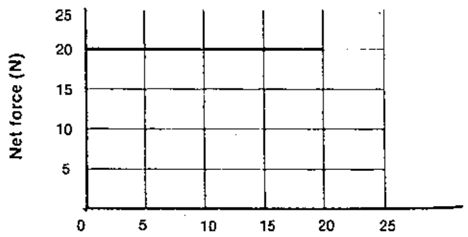


Fig. 18.2

(*a*) Calculate the work done by the net force in moving the body 20 m.
(*b*) If the frictional resistance was 3.0 N, what was the magnitude of the force applied?
(*c*) If the mass of the body was 10 kg, what was the magnitude of the acceleration?

9 Determine the kinetic energy of
(*a*) a 0.2 kg billiard ball moving at 3 m s^{-1};
(*b*) a 50 g golf ball moving at 80 m s^{-1}.

10 Estimate as an order of magnitude the kinetic energy of
(*a*) an adult athlete while running a 100 m sprint;
(*b*) a moving snail.

11 A body of mass 4.0 kg is moving at a constant velocity of 3.0 m s^{-1} west. It is then acted upon by a force of 5.0 N west, while it moves 2.0 m west.
(*a*) What is the work done by the force?
(*b*) What is the resulting increase in the kinetic energy of the body?
(*c*) What was the initial kinetic energy of the body?
(*d*) What is the final kinetic energy of the body?
(*e*) Determine its final speed.

12 How much work is done in changing the speed of a vehicle of mass 5000 kg from 20.0 m s^{-1} to 30.0 m s^{-1}?

13 What would be the kinetic energy of the body in Problem 8 when it had travelled a distance of
(*a*) 10 m?
(*b*) 20 m?

14 The graph in Figure 18.3 shows how a force varies with distance as it acts on a mass of 4.0 kg, initially at rest.

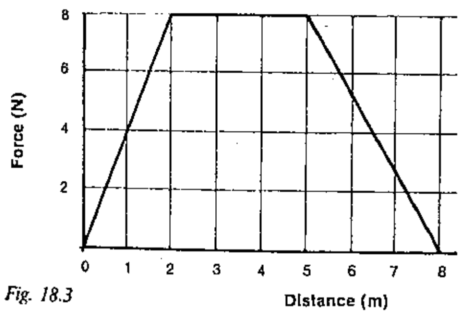


Fig. 18.3

(*a*) How much work was done by the force in moving the body 2.0 m?
(*b*) How fast was the mass moving when it had been pulled a distance of 2.0 m?
(*c*) How much kinetic energy did the mass have by the time it had been moved 5.0 m and what was its speed then?
(*d*) How much kinetic energy did the mass gain between completing the fifth and the eighth metre?

(*e*) How large an impulse did the 4.0 kg mass receive while it was moving over the first 2.0 m?

15 A body of mass 50 kg moving with a speed of 20 m s^{-1} is brought to rest by a constant force in a distance of 5.0 m. Calculate
(*a*) the loss in kinetic energy of the body;
(*b*) the work done by the body against the retarding force;
(*c*) the size of the force.

16 The graph in Figure 18.4 shows the force applied to a mass of 1.5 kg, initially at rest and free to move on a frictionless surface.

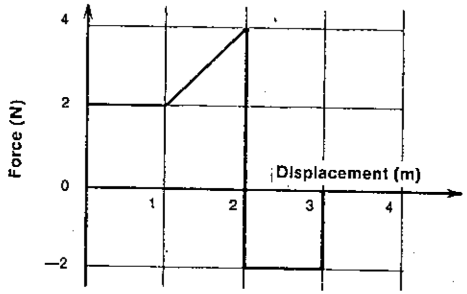


Fig. 18.4

(*a*) What is the magnitude of the acceleration of the body during the first metre of travel?
(*b*) What is its kinetic energy after it has moved 2.0 m?
(*c*) How fast is it travelling 3.0 m from the starting point?

17 On a merry-go-round at the Showground a girl of mass 50 kg is moving at a constant speed of 6.0 m s^{-1} in a horizontal circle of radius 12 m.
(*a*) What is her kinetic energy?
(*b*) What is the net force acting on her?
(*c*) How much work does this net force do on her in one half revolution?

18 (*a*) A 0.16 kg hockey ball has a kinetic energy of 0.50 J just after it is hit. What is the magnitude of its momentum at this instant?
(*b*) Write down an expression for p, the magnitude of the momentum of a body, in terms of its mass m and kinetic energy E_k.
(*c*) Express E_k as a function of p and m.

19 A cannon of mass 1.00×10^5 kg fires a 500 kg shell horizontally, imparting to it a speed of 1.00×10^3 m s^{-1}. Find the work done on the cannon by the expanding gases (assuming that it is free to move).

20 Two bodies, initially at rest, of mass 10 g and 50 g respectively are acted on in turn by the same force. Determine the ratio of the times for which this force must be operative to produce

(*a*) the same kinetic energy;
(*b*) the same momentum.

21 An upward force of 6.0 kN raises a mine cage a vertical distance of 100 m in 40 s at a constant rate.
(*a*) How much work is done by the force in the 40 s?
(*b*) What work is done by the force in one second?
(*c*) At what power is the force working?

22 The frictional forces on a small car moving along a level road are estimated to be 500 N. Calculate the power exerted by the engine to maintain a constant velocity of 14 m s^{-1}.

23 The engine of a jet aeroplane develops 2.4 megawatt of power when in level flight at a constant speed of 200 m s^{-1}. Calculate the air resistance on the plane.

24 Find the power necessary to move a body of mass 1.5 kg up a smooth plane inclined at 30° to the horizontal with a uniform speed of 5.0 m s^{-1}.

25 A car of mass 2000 kg travels up a slope of 30.0° at a uniform speed of 15.0 m s^{-1}. If the frictional resistance is one-tenth of the weight of the car, at what power is the engine working?

SET 19 GRAVITATIONAL POTENTIAL ENERGY IN A UNIFORM FIELD

Topic	*Problems*
Gravitational potential energy in a uniform field	1-4
Conservation of total energy for a projectile motion, frictionless motion and motion with friction	5-12
Gravitational potential, gravitational potential difference and gravitational potential gradient in a uniform field	13-16

1 Taking the gravitational potential energy at ground level as zero, determine the gravitational potential energy of
(*a*) a 1.4 kg book while being carried 0.80 m above the ground;
(*b*) a 50 g golf ball at the top of its flight 9.0 m above the fairway.

2 What is the increase in gravitational potential energy of
(*a*) a 870 kg car which is raised 2.0 m by a garage hoist?
(*b*) a 0.55 kg book taken from a drawer where it is 0.30 m above the floor and placed on a desk top 0.80 m above the floor?
(*c*) a 190 kg barbell dropped a vertical distance of 2.2 m to the ground?

3 A shopper of mass 65 kg goes from the first to the second floor of a store by travelling 10 m on an escalator which moves in a direction at 30° to the horizontal. What is her increase in gravitational potential energy?

4 A Swiss mountain railway car of the rack and pinion type leaves Zermatt (altitude 1608 m) at 6.00 a.m. and completes the 9.0 km journey to the Gornergrat (altitude 3136 m) at 6.45 a.m. If the car and passengers together have a mass of 20 000 kg, calculate the average power in kilowatt required by the electric motors (neglecting friction).

5 A boy on a bridge drops a 5.0 kg rock into the water 10 m below. Taking the water surface as the zero gravitational potential energy level, determine
(*a*) the gravitational potential energy of the rock as it left the boy's hand;
(*b*) its total energy (kinetic plus potential) as it left his hand;
(*c*) the total energy of the rock half-way down;
(*d*) its kinetic energy just before hitting the water;
(*e*) its speed just before hitting the water.

6 A tennis ball of mass m is hit vertically into the air with an initial speed of v_i. Take the point where the ball was hit as the zero gravitational potential energy level.
(*a*) With what kinetic energy did it leave the racquet?
(*b*) What was the sum of the kinetic and potential energies of the ball at any point in the flight?
(*c*) What is the potential energy of the ball at the highest point it reaches?
(*d*) What is this maximum height above the point where it was hit?

7 A pendulum bob is pulled aside until it is 10 cm vertically above its lowest point and then released. With what speed does it pass through its mean position?

8 A body of mass 0.36 kg is projected horizontally, with a speed of 8.0 m s^{-1}, from the top of a tower AD which is 45 m above the level ground DC. It moves on a parabolic path and strikes the ground at C as shown in Figure 19.1. B is a point on the path of the body 32 m above the ground.

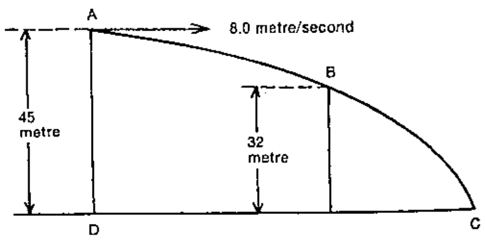


Fig. 19.1

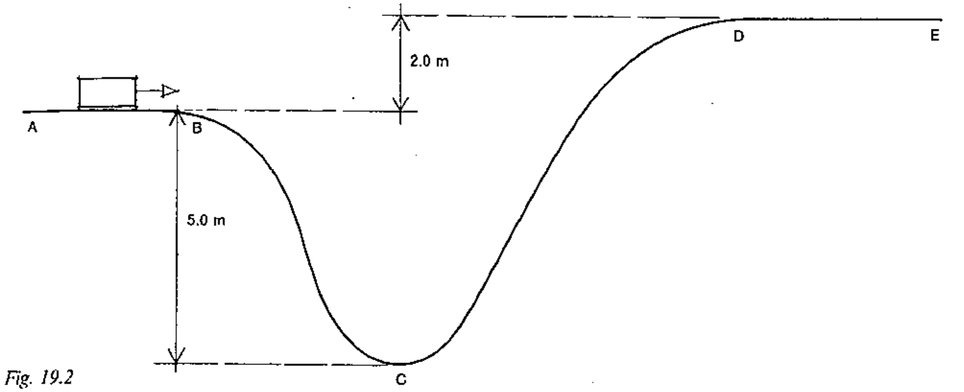


Fig. 19.2

Ground level is taken as the zero level of gravitational potential energy.
(*a*) What is the total energy of the body at A?
(*b*) How much potential energy has it lost in moving from A to B?
(*c*) What is its kinetic energy at B?
(*d*) What is its speed at B?
(*e*) What is its kinetic energy just before it strikes the ground at C?

9 Figure 19.2 represents a track which has been erected to demonstrate certain aspects of the energy of a body. A small cart of mass 4.0 kg moves without friction on the track, with a speed 10 m s^{-1} along AB.
(*a*) What is the kinetic energy of the cart as it moves along AB?
(*b*) At which point on the track is its kinetic energy a maximum?
(*c*) What is the change in the potential energy of the cart between B and D?
(*d*) What is the kinetic energy of the cart as it moves along the horizontal level DE?
(*e*) Calculate the speed of the cart as it passes the point C.

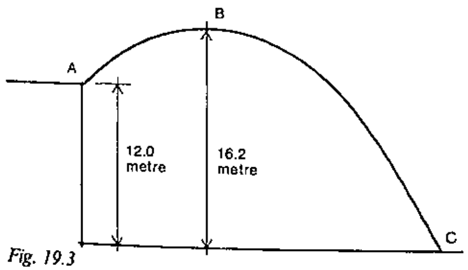


Fig. 19.3

10 In this question, take the intensity of the Earth's gravitational field as 10.0 N kg^{-1} and the zero of potential energy at the ground level shown. Figure 19.3 shows the path of a stone of mass 0.500 kg after being thrown from the point A with a speed of 22.0 m s^{-1}. Find the value of
(*a*) the total energy of the mass at A just as it leaves that point;
(*b*) its potential energy at the highest point B;
(*c*) its speed at B;
(*d*) its total energy at the ground level C, just prior to impact.
(*e*) Given that the horizontal component of the velocity does not alter, find the angle to the horizontal at which the stone was projected.

11 A sack of sand of mass 30 kg swings from rest from a point A on the end of a rope 3.2 m long as in Figure 19.4. It falls to point B, the lowest point of its swing.

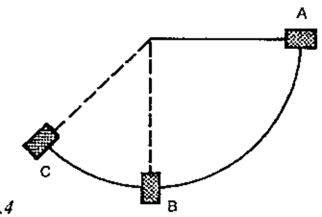


Fig. 19.4

(*a*) What speed had it acquired on reaching point B?
(*b*) At point B, the sack of sand comes into contact with an arresting device to which half its kinetic energy is transferred. If the arresting device acts for a distance of 0.10 m, what was the average force applied by it?

(*c*) The sack continues swinging, reaching its maximum height at the point C as shown in Figure 19.5. What is the vertical height of C above B?

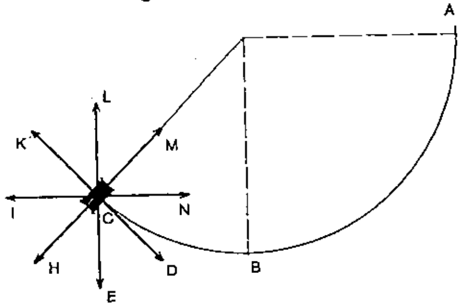


Fig. 19.5

(*d*) In which one or more of the directions D to N in Figure 19.5 are there applied forces acting on the sack at the point C?
(*e*) In which one of the directions D to N does the resultant force act on the sack at the point C?

12 A body of mass 5.0 kg slides from rest down a plane inclined at 30° to the horizontal. After sliding 12 m down the plane it is found to have a speed of 10 m s^{-1}.
(*a*) How much work has been done in overcoming friction along the plane?
(*b*) Calculate the average value of the force of friction.

13 One method of describing gravitational effects at a point in space is to quote the gravitational *force* per unit mass, that is, the *gravitational field strength g*, at that point. Another approach is to give the gravitational *potential energy* per unit mass, known as the *gravitational potential* V_g, at that point.
(*a*) What would be the SI unit of gravitational potential?
(*b*) Taking floor level as the zero gravitational potential energy level, what is the gravitational potential energy of a telephone directory of mass 2.00 kg at a point
 (*i*) X, 1.02 m above the floor?
 (*ii*) Y, 2.02 m above the floor?
 (*iii*) Z, 3.02 m above the floor?
(*c*) What is the gravitational potential V_g at
 (*i*) X?
 (*ii*) Y?
 (*iii*) Z?
(*d*) What is the *increase in gravitational potential per metre rise* above the floor? Is there any relation between this quantity, which is known as the *gravitational potential gradient*, and *g*, the magnitude of the gravitational field strength?

14 Draw a graph of gravitational potential V_g against vertical height *h* above the floor (to be taken as the zero gravitational potential energy level) for values of *h* from 0 to 5 m. From the graph deduce
(*a*) the gravitational potential at a point
 (*i*) P, 3.00 m above the floor;
 (*ii*) Q, 5.00 m above the floor;
(*b*) the gravitational potential difference between P and Q;
(*c*) the work necessary to raise a 10.0 kg box from P to Q;
(*d*) a meaning and value for the gradient of the graph, that is, $\Delta V_g / \Delta h$.

15 Figure 19.6 shows gravitational field lines and gravitational equipotential surfaces near ground level. (Give all answers to three significant figures.)

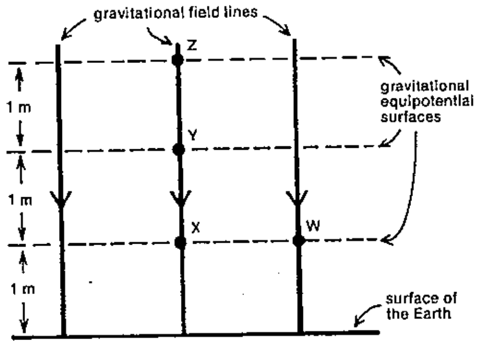


Fig. 19.6

(*a*) What is the gravitational potential at point Y?
(*b*) What is the gravitational potential difference between points
 (*i*) X and W?
 (*ii*) Z and Y?
 (*iii*) Z and X?
 (*iv*) Z and W?
(*c*) How much energy is required to move a mass of 4.00 kg from W to Z?

16 The gravitational potential difference between the surface of the planet Jupiter and a point 3.0 m above it is 75 J kg^{-1}. The gravitational field in this region is uniform.
(*a*) What work would be done in lifting a 2.0 kg mass from the surface of Jupiter to a point 3.0 m above it?
(*b*) Deduce the magnitude of the gravitational field strength near the surface of Jupiter.

SET 20 ELASTIC POTENTIAL ENERGY

Topic	*Problems*
Potential energy stored in an elastic material	1-3
Conservation of energy involving elastic potential and kinetic energies	4-8
Conservation of energy involving elastic potential, gravitational potential and kinetic energies	9-10

1 The graph in Figure 20.1 shows the force exerted by an unusual kind of spring bumper device as the bumper is depressed and as it returns to its previous shape. The device is fixed to a table with a frictionless surface.

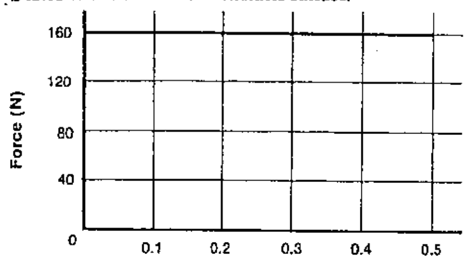


Fig. 20.1

(*a*) What work is done on the bumper if it is compressed 0.30 m?
(*b*) What is the elastic potential energy stored in the bumper when it is compressed 0.30 m?
(*c*) What aspect of the force–compression graph gives the elastic potential energy stored in the bumper?

2 Figure 20.2 shows the force–compression graph for a perfectly elastic spring.
(*a*) How much work is done in compressing the spring 0.20 m?
(*b*) What is the potential energy of the spring when it is compressed this amount?

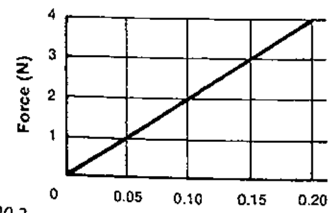


Fig. 20.2

3 If the force constant of a spring in a car suspension is 5.0×10^4 N m^{-1}, what potential energy is stored in the spring when it is compressed 4.0 cm?

4 A 5.0 kg puck moving at 0.20 m s^{-1} on a horizontal frictionless air table strikes the spring described in Problem 2 and Figure 20.2. Assume the mass of the spring is negligible.
(*a*) What is the kinetic energy of the puck before it hits the spring?
(*b*) While the puck is in contact with the spring, what is the sum of the kinetic energy of the puck and the elastic potential energy of the spring?
(*c*) What is the potential energy stored in the spring at its maximum compression?
(*d*) What is the maximum compression of the spring?

5 The force–compression graph of a complex spring is shown in Figure 20.3.

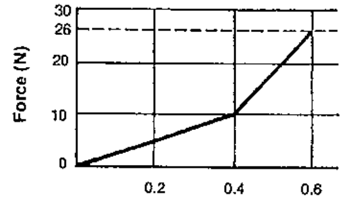


Fig. 20.3

(*a*) How much work is done in compressing the spring 0.40 m?
(*b*) What is the potential energy of the spring when it is compressed 0.60 m?
A 5.0 kg mass is placed next to the end of the spring when it is compressed 0.60 m and the whole thing let go. Assume the mass of the spring is negligible in comparison with 5 kg.
(*c*) How much potential energy has been lost by the mass when it passes the point where the spring is compressed 0.40 m?
(*d*) What is the magnitude of the momentum of the mass at this stage?

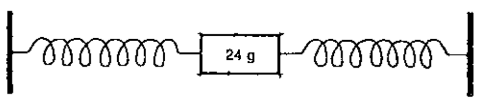


Fig. 20.4

6 A mass of 24 g attached to two very light springs as shown in Figure 20.4 is caused to execute simple harmonic motion along a frictionless surface with an amplitude of 0.40 m. The force–displacement graph for the body moving under the action of the springs is shown in Figure 20.5.

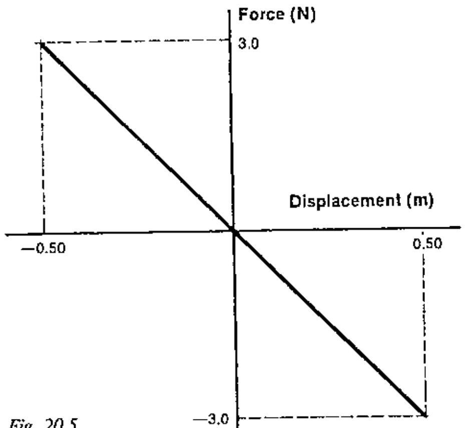


Fig. 20.5

(*a*) How much potential energy is stored in the system when the 24 g mass is in an extreme position, i.e. 0.40 m from its mean position?
(*b*) What will be the kinetic energy of the body when its displacement is 0.10 m?
(*c*) What is its speed at this instant?

7 A body of mass 0.200 kg is vibrating with simple harmonic motion with an amplitude of 5.00 cm. When it is 3.00 cm from its mean position, the restoring force acting on it is 0.294 N.
(*a*) Calculate the maximum value of its potential energy.
(*b*) When it is displaced 4.00 cm from its mean position, what is
 (*i*) the potential energy of the system?
 (*ii*) the kinetic energy of the 0.200 kg mass?

8 Show that the total energy at any instant of a particle executing simple harmonic motion is
(*a*) proportional to the square of the amplitude;
(*b*) inversely proportional to the square of the period.

9 The charging device of a toy gun is pushed back 10 cm and an average force of 51 N is used in doing this. If all the energy stored in the gun is transmitted to a pebble of mass 20 g which is fired straight up into the air, how high will the pebble rise?

10 A projectile of mass 5.0 kg can be launched by either of two types of mortar. One uses a small explosive charge; the other is powered by compressed air. Each has a barrel 0.70 m long. The graphs in Figures 20.6 and 20.7 show the propulsive force exerted on the projectile as it moves along the barrel of each mortar.

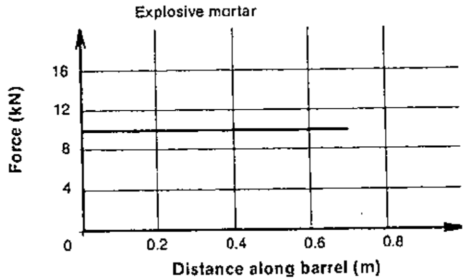


Fig. 20.6

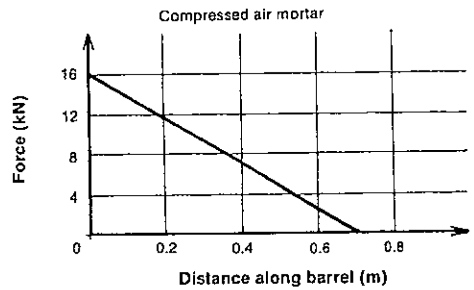


Fig. 20.7

(*a*) In which mortar does the projectile experience the greatest propulsive force?
(*b*) Ignoring the effects of friction in the barrels, find the kinetic energy given to the projectile by
 (*i*) the explosive mortar;
 (*ii*) the compressed air mortar.
(*c*) The explosive mortar fires the projectile along a path such that at the highest point of its trajectory the projectile has lost half of its initial kinetic energy. At this point,
 (*i*) at what angle to the horizontal is its direction of motion?
 (*ii*) what potential energy has it gained since it was launched?
 (*iii*) what is its height above the level of its launching position?

SET 21 ELASTIC AND INELASTIC COLLISIONS

Topic	*Problems*
Qualitative comparison of elastic and inelastic interactions	1–2
Inelastic interactions in one dimension (see also Set 17)	3–4
Inelastic interactions in two dimensions (see also Set 17)	5–6
Elastic interactions in one dimension	7–13
Elastic interactions in two dimensions	14–15

1 This question concerns what takes place when objects interact in various types of collision. Which of the lines (*A*, B, C, . . .) in Figure 21.1 best represents the variation with time of
(*a*) the total momentum of the bodies in
 (*i*) an inelastic interaction?
 (*ii*) an elastic interaction?
(*b*) the total kinetic energy of the bodies in
 (*i*) an inelastic interaction?
 (*ii*) an elastic interaction?
(*c*) the total potential energy of the bodies during an elastic interaction?

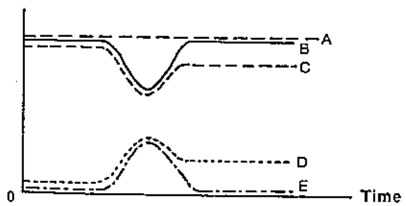


Fig. 21.1

2 By using the letters X, Y and Z, arrange the following interactions in order of *elasticity*, starting with the most inelastic and finishing with the most elastic:
X A golf club hits a golf ball.
Y A positive proton is deflected as it passes close to a positively charged nucleus.
Z A custard pie hits the face of a clown skating on ice.

3 Figure 21.2 represents two trolleys on a horizontal surface, able to move only in the same straight line. Their masses are as shown in the diagram. The questions below refer to three different collisions between them.

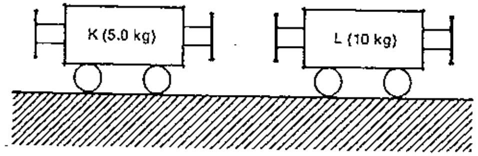


Fig. 21.2

(*a*) L is stationary and K strikes it with a speed of 1.5 m s^{-1}. During the impact, the trolleys lock together. With what speed do they move off after the collision?
(*b*) This time, K and L are both moving in the same direction, with K gaining on L. They collide when the speed of K is 1.5 m s^{-1} and the speed of L is 0.50 m s^{-1}. Again, they lock together. With what speed do they move off after the collision?
(*c*) This time, the initial situation is the same as in (*b*), but the trolleys no longer lock together. They move off, after the collision, with a relative speed which is half of their relative speed before impact. Calculate their separate speeds immediately after the impact.

4 A block of wood of mass 2000 g is suspended from a fixed point by a long light vertical string. A bullet of mass 20 g, travelling horizontally with a speed of 303 m s^{-1}, embeds itself in the block, and the string is deflected so that the block describes an arc of a circle and rises through a vertical height *h* before coming momentarily to rest.
(*a*) What is the kinetic energy of the bullet just before the impact?
(*b*) Determine the speed of the block and the bullet immediately after the impact.
(*c*) What is the total kinetic energy of the block and bullet immediately after the impact?
(*d*) What is the value of *h*?
(*e*) What percentage loss of mechanical energy results from the collision? What happens to this lost energy?
(*f*) Indicate briefly how such an experiment could be used to determine the speed of a bullet.

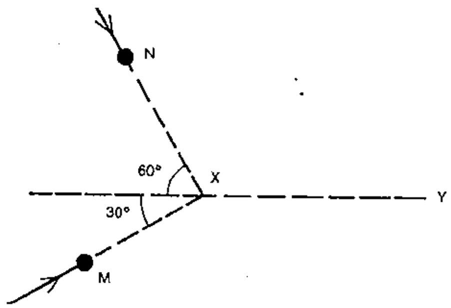


Fig. 21.3

5 Figure 21.3 shows two masses, M and N, which will collide at X and stick together. No outside forces act on the system. M has a mass of 1.0 kg and a speed of 6.0 m s^{-1}; N has a mass of 2.0 kg and a speed of 3.0 m s^{-1}.
(*a*) What is the value (to the nearest degree) of the angle which the path of the masses makes with the direction XY, after collision?
(*b*) What is the value of the ratio:

$$\frac{\text{kinetic energy of the system after collision}}{\text{kinetic energy of the system before collision}}?$$

6 In a game of billiards the white ball moving at 2.0 m s^{-1} hits the stationary black a glancing blow, and the two balls move off in the directions shown in Figure 21.4. The balls have equal mass. Determine the speeds of the two balls immediately after the collision.

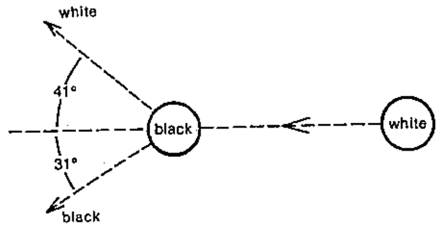


Fig. 21.4

7 (*a*) A proton of mass m moving with a velocity u_1 makes a head-on elastic collision with a second stationary proton. If v_1 and v_2 represent the final velocities of the incident and target protons respectively, show by using the laws of conservation of momentum and kinetic energy that $v_1 = 0$ and $v_2 = u_1$.
(*b*) Now suppose a proton of mass m moving with a velocity u_1 makes a head-on collision with a stationary deuteron of mass $2m$. If the proton came to rest and the deuteron moved off with a velocity of $\frac{1}{2}u_1$,
(*i*) would momentum be conserved?
(*ii*) would kinetic energy be conserved?

8 On a linear air track a 0.3 kg glider moving at 0.5 m s^{-1} east approaches a 0.2 kg glider at rest. Repelling magnets mounted in the near ends of the gliders prevent the gliders from coming in contact and they make an elastic interaction. Determine the final velocity of
(*a*) the 0.3 kg glider;
(*b*) the 0.2 kg glider.

9 Consider a *one-dimensional* collision between two bodies of mass m_1 and m_2, moving with velocities u_1 and u_2 before, and v_1 and v_2 after the collision.
(*a*) If it is possible to say that

$$-(m_1v_1 - m_1u_1) = (m_2v_2 - m_2u_2),$$

what can be said about the system of the two bodies?
(*b*) If it is also possible to say that

$$-(\tfrac{1}{2}m_1v_1^2 - \tfrac{1}{2}m_1u_1^2) = (\tfrac{1}{2}m_2v_2^2 - \tfrac{1}{2}m_2u_2^2),$$

what can be said about the interaction?
(*c*) From the equations in (*a*) and (*b*), deduce that

$$v_1 + u_1 = v_2 + u_2$$

(*d*) The relation in (*c*) could be rewritten as

$$u_1 - u_2 = v_2 - v_1$$

Express this relationship of *relative velocities* in words. This is true for all one-dimensional interactions between two bodies when both momentum and kinetic energy are conserved.

10 A particle of mass m_1 travelling with a velocity u_1 collides head-on with a second particle of mass m_2, which is initially at rest, and imparts to it in an elastic collision a velocity v_2, while mass m_1 acquires a final velocity v_1. Show that:
(*a*) $v_1 = \dfrac{(m_1 - m_2)u_1}{m_1 + m_2}$;
(*b*) $v_2 = \dfrac{2m_1u_1}{m_1 + m_2}$
(The derivations in Problem 9 above could be useful here.)

11 In 1932 Sir James Chadwick first determined a value for the mass of the neutron by passing neutrons with a constant speed in turn through hydrogen and nitrogen gas. In elastic head-on collisions, effectively stationary hydrogen nuclei (protons) were given a velocity of 3.3×10^7 m s^{-1}, while nitrogen nuclei were given a velocity of 4.7×10^6 m s^{-1}. If the mass of a proton is 1.0 atomic mass unit (abbreviated u) and that of a nitrogen nucleus is 14 u, what value did Chadwick obtain for the mass of the neutron? Refer back to Problem 10 (*b*) above.

12 A 0.10 kg glider moving at 0.40 m s^{-1} south interacts elastically on a linear air track with a 0.40 kg glider moving at 0.30 m s^{-1} north. Determine the final velocity of
(*a*) the 0.10 kg glider;
(*b*) the 0.40 kg glider.
The relation deduced in Problem 9 (*d*) above will prove helpful.

13 (*a*) For the special type of collision described in Problem 10 above, show that the fractional transfer of kinetic energy from the incident body to the target body is $4m_1m_2/(m_1 + m_2)^2$.
(*b*) Giving answers to two significant figures, determine the fractional transfer of kinetic energy if
(*i*) $m_1 = \frac{1}{10}m_2$;
(*ii*) $m_1 = m_2$;
(*iii*) $m_1 = 10m_2$.
(*c*) When a particular body X undergoes a collision of this type with a stationary body Y of different mass, 28 per cent of the kinetic energy of X is transferred to Y. If then Y becomes the incoming body and X the stationary target in the same type of collision, what percentage of the kinetic energy of Y would be passed on to X?

14 On an effectively frictionless air table a 0.60 kg puck A, moving at speed x in a westerly direction, interacts with a 0.70 kg puck B moving at 0.50 m s^{-1} in a direction S 30° W. After the interaction A moves at a speed y in a direction S 39° W, while B moves at a speed z in a westerly direction.
(*a*) If the interaction was an *inelastic* one, how many equations could have been written down?
(*b*) Actually, it is an *elastic* interaction. How many equations can be written down?
(*c*) Deduce the values of x, y and z.

15 If one marble collides elastically but not head-on with another marble which is stationary and of equal mass, show that the angle between their velocities immediately after the collision should be a right angle.

SET 22 GRAVITATION IN A NON-UNIFORM FIELD AND SATELLITE MOTION

Topic	*Problems*
Kepler's law of elliptical orbits (I)	1–4
Kepler's law of equal areas (II)	5–6
Kepler's law of periods (III)	7–10
Newton's law of gravitation	11–17
Kepler's third law and Newton's law of gravitation	18–22
Gravitational potential energy of a satellite in a non-uniform field	23–24
Potential, kinetic, total and binding energies of a satellite in a stable orbit	25–29
Potential, kinetic, total and binding energies of an object not in a stable orbit	30–31
Escape energy and escape speed from a planet	32–33
Gravitational potential and gravitational potential gradient in a non-uniform field	34–37

Where necessary in this set, use information from Table 22.1, and gravitation constant $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.

For problems where the question number is highlighted, e.g. Problems **23** to **37**, assume the gravitational potential energy of two bodies is zero when their separation is infinite.

1 Using Table 22.1 and no arithmetic calculations,
(*a*) state what is meant by 1 astronomical unit (AU);
(*b*) select from the table the times (in seconds) which best approximate
(*i*) 1 year;
(*ii*) 1 day;
(*iii*) 1 lunar month.

2 Figure 22.1 shows the elliptical orbit followed by Pluto around the Sun. a is the length of the major axis PA and c is the separation of the two foci F_1 and F_2 of the ellipse. The *eccentricity* e of the orbit is defined by $e = c/a$. The mean distance R_{av} between the centres of Pluto and the Sun (the equivalent *average radius* of its orbit) is the average of the *perihelion* or least distance R_p and the *aphelion* or greatest distance R_a.

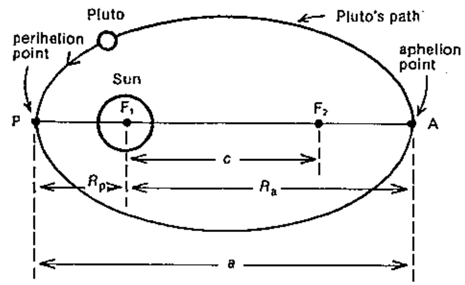


Fig. 22.1

(*a*) Express R_p in terms of
(*i*) a and c;
(*ii*) a and e.
(*b*) Express R_a in terms of
(*i*) a and c;
(*ii*) a and e.
(*c*) Express R_{av} in terms of
(*i*) a;
(*ii*) c and e.
(*d*) If the eccentricity of Pluto's orbit is 0.249 and the length of the major axis is 1.18×10^{13} m, deduce
(*i*) the perihelion distance
(*ii*) the aphelion distance, and
(*iii*) the equivalent average radius
of Pluto's orbit.

3 If the eccentricity of the Earth's elliptical orbit around the Sun is 0.0170 and the centre of the Sun is at one focus of the ellipse,
(*a*) how far apart are the two foci?
(*b*) approximately how far is the second focus outside the surface of the Sun? Give the answer to the nearest whole number of Sun radii.

Table 22.1

Object	Mass (kg)	Radius (m)	Period of rotation (s)	Mean radius of orbit (m)	(AU)*	Period of revolution (s)	(y)*
Sun	1.98×10^{30}	6.95×10^{8}	2.14×10^{6}				
Mercury	3.28×10^{23}	2.57×10^{6}	5.05×10^{6}	5.79×10^{10}	0.387	7.60×10^{6}	0.241
Venus	4.83×10^{24}	6.31×10^{6}	2.10×10^{7}	1.08×10^{11}	0.723	1.94×10^{7}	0.615
Earth	5.98×10^{24}	6.38×10^{6}	8.61×10^{4}	1.49×10^{11}	1.00	3.16×10^{7}	1.00
Mars	6.37×10^{23}	3.43×10^{6}	8.85×10^{4}	2.28×10^{11}	1.52	5.94×10^{7}	1.88
Jupiter	1.90×10^{27}	7.18×10^{7}	3.54×10^{4}	7.78×10^{11}	5.20	3.74×10^{8}	11.86
Saturn	5.67×10^{26}	6.03×10^{7}	3.60×10^{4}	1.43×10^{12}	9.54	9.30×10^{8}	29.46
Uranus	8.80×10^{25}	2.67×10^{7}	3.88×10^{4}	2.87×10^{12}	19.19	2.66×10^{9}	84.01
Neptune	1.03×10^{26}	2.48×10^{7}	5.69×10^{4}	4.50×10^{12}	30.07	5.20×10^{9}	164.78
Pluto	5.37×10^{23}	2.96×10^{6}	5.51×10^{5}	5.91×10^{12}	39.52	7.82×10^{9}	248.42
Moon	7.34×10^{22}	1.74×10^{6}	2.36×10^{6}	3.84×10^{8}		2.36×10^{6}	

*Use these non-SI units, the astronomical unit (AU) and the year (y), only when specifically directed to do so in a question.

4 The orbit of the Moon about the Earth has an eccentricity of 0.0549.
(*a*) Relative to the Earth, what is the Moon's
(*i*) perihelion distance?
(*ii*) aphelion distance?
(*b*) What is the difference between the least and greatest distances from the centre of the Moon to the centre of the Earth?

5 What is the constant rate (in $m^2\,s^{-1}$) at which area is swept out by the radius of the Moon's orbit as the Moon encircles the Earth?

6 (*a*) Using Kepler's law of equal areas, show that the speeds v_p and v_a at the perihelion and aphelion points respectively (see Fig. 22.1) are related by

$$\frac{v_p}{v_a} = \frac{R_a}{R_p}$$

(*b*) Using answers from Problem 2 (*d*) in this set, deduce the ratio of Pluto's fastest to its slowest orbital speed.

7 If R represents the mean orbital radius and T the period of revolution, determine for the following Sun satellites the value of R^3/T^2
(*a*) in SI units for
(*i*) Venus;
(*ii*) Uranus;
(*b*) in AU^3/y^2 for
(*i*) Jupiter;
(*ii*) Earth.

8 Ceres, the largest of the asteroids or minor planets which circle the Sun, has a mean orbital radius of 4.14×10^{11} m. Use Kepler's law relating periods and radii to deduce the period of revolution of Ceres about the Sun
(*a*) in seconds;
(*b*) in years.

9 Raduga 5 is a Russian satellite launched into a circular orbit in April 1979. It is a synchronous communications satellite, that is, one that remains directly above a constant position on the Earth's equator.
(*a*) What is the period of revolution of its orbit?
(*b*) Using data on the Moon's orbit, deduce a value for the orbital radius of Raduga 5.
(*c*) What altitude does the satellite maintain above the surface of the Earth?

10 Halley's Comet completes its elliptical orbit about the Sun once every 76 years.
(*a*) Using Kepler's third law, determine an equivalent average radius for the comet's orbit.
(*b*) If its distance of closest approach to the Sun is 8.9×10^{10} m, does Halley's Comet move farther from the Sun than all of the Sun's known planets?

11 Using Newton's law of gravitation, determine the weight of an 80 kg man on the surface of
(*a*) the Earth;
(*b*) the Moon.

12 Determine the order of magnitude of the gravitational attraction between two teenagers standing talking to one another.

13 A UFO is approaching the Earth at a uniform velocity. When the craft is at a distance from the Earth's surface equal to $2R$, where R is the radius of the Earth, the navigators take a depth sounding. Their device is a sensitive spring from which they suspend their standard mass.

(*a*) Find the ratio of the tension in the spring at this distance to that when the craft is at a distance R above the Earth's surface.
(*b*) What is the tension in the spring when the craft is orbiting the Earth in a stable orbit?

14 How far were the Apollo spacecrafts from the centre of the Earth when they experienced a zero net gravitational force from the Earth and the Moon?

15 (*a*) Show that the magnitude g of a planet's gravitational field strength at a distance R from its centre is given by

$$g = \frac{GM}{R^2},$$

where M is the mass of the planet. (R is equal to or greater than the radius of the planet.)
(*b*) Use your answer to (*a*) to deduce the value of g on the surface of the Earth.
(*c*) What would be the magnitude of the acceleration of a free-falling object near the surface of Venus?

16 A planet which has a diameter twice that of the Earth has a mass six times greater.
(*a*) What is the ratio of the gravitational field at the surface of this planet to that at the surface of the Earth?
(*b*) What is the ratio of the density of the planet to that of the Earth?

17 Assuming that the Moon is a sphere of the same mean density as the Earth, and one-quarter its radius, what will be the length of a seconds pendulum on the Moon? Its length on the Earth's surface is 992 mm.

18 For a satellite of mass m moving in a circular orbit of radius R and period of revolution T about a central body of mass M, two expressions for the force on the satellite can be equated:

$$\frac{m4\pi^2 R}{T^2} = \frac{GMm}{R^2},$$

where G is the gravitation constant.
(*a*) Deduce an expression for the constant of Kepler's third law, R^3/T^2.
(*b*) If R_1 and T_1 are respectively the orbital radius and period of revolution of one of Saturn's moons, write an expression for R_1^3/T_1^2 in terms of M_S, the mass of Saturn.

19 Imagine a planetary system in which the gravitational force varied inversely as the *distance* (R) instead of the *square of the distance*. What proportionality relation between T (period of revolution) and R in this two-dimensional world would correspond to Kepler's third law?

20 Callisto, Jupiter's largest moon, completes an orbit every 16 d 16 h 32 min at a mean distance of 1.88×10^9 m from the centre of Jupiter. Deduce a value for the mass of Jupiter.

21 An artificial satellite of the Earth is to be established in the equatorial plane of the Earth, and *to an observer at the equator* it is required that the satellite will move
(*a*) eastward, completing one round trip per day;
(*b*) westward, completing two round trips per day.
In each case determine the distance of the satellite from the centre of the Earth.

22 Four satellites, A, B, C and D are put into circular orbits about the Earth. Table 22.2 shows certain measured and calculated quantities relating to the satellites. In some cases the units used are arbitrary. Calculate the values of the "missing" quantities, M_C, R_D, F_B and T_C. Leave the answers in fractional or surd form.

Table 22.2

Satellite	Mass (arbitrary units)	Radius of orbit (arbitrary units)	Gravitational force exerted by Earth (arbitrary units)	Period of revolution (days)
A	3	1	3	1/8
B	1	4	F_B	1
C	M_C	25/4	16/25	T_C
D	1024	R_D	4	8

23 If we adopt the convention that the gravitational potential energy of two bodies is zero at infinite separation,
(*a*) what is the gravitational potential energy of
(*i*) the Earth–Moon system?
(*ii*) the Sun and Pluto?
(Since this energy is more significant to the motion of the *smaller* of the two bodies when the masses are not comparable, it is often referred to as the gravitational potential energy of the *satellite* rather than of the *system* composed of the satellite and the central body.)
(*b*) Which of these two systems has the higher gravitational potential energy?

24 If the gravitational potential energy relative to the Earth of the Russian supply vehicle Progress 7 while being launched in June 1979 was -4.377×10^{11} J,
(*a*) what was its mass?
(*b*) what was its gravitational potential energy when it docked with the Salyut 6 space station in an orbit with a perigee of 395 km and an apogee of 405 km? (*Perigee* and *apogee* are respectively the minimum and maximum altitudes of the orbital path above the *surface* of the Earth.)

25 If a satellite of mass m moves in a circular orbit of radius R about a central body of mass M, give an expression for
(*a*) the gravitational force acting on the satellite;
(*b*) the gravitational potential energy of the satellite;

(*c*) its kinetic energy;
(*d*) its total energy (i.e. potential plus kinetic);
(*e*) its binding energy.

26 Considering the Earth as a satellite of the Sun, determine the Earth's
(*a*) gravitational potential energy;
(*b*) kinetic energy;
(*c*) speed relative to the Sun;
(*d*) total energy;
(*e*) binding energy.

27 The total energy (i.e. kinetic plus potential) of the US navigational satellite NavStar 2 is -3.26×10^9 J, and the mean radius of its orbit about the Earth is 2.65×10^7 m. Relative to the Earth, what is
(*a*) the kinetic energy of NavStar 2?
(*b*) its gravitational potential energy?
(*c*) its binding energy?
(*d*) the gravitational force exerted on it by the Earth?

28 A cannon is fired in a horizontal direction from a point a short distance above the Earth's surface. Find an expression in terms of the gravitation constant G, the radius R, and the mass M of the Earth, for the minimum velocity required in order that the cannon ball should continue to circulate around the Earth, assuming the Earth to be spherical and the atmosphere to offer no resistance.

29 Figure 22.2 shows the variation in gravitational potential energy U_g of a body of mass m with distance r from the centre of the Earth.

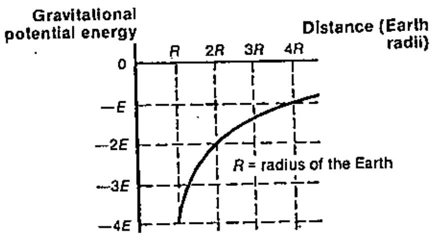


Fig. 22.2

(*a*) If a satellite of mass m was in a stable circular orbit of radius $2R$, in terms of E what would be its total energy?
(*b*) If another satellite of mass m in a stable circular orbit had a binding energy of $\frac{1}{2}E$, in terms of R how far would it be from the Earth's surface?
(*c*) If a spacecraft of mass m returning to Earth fell from a distance $3R$ to R above the Earth's surface, how much kinetic energy would it gain? Assume air resistance is negligible.

30 During the outward journey of the Apollo 11 spacecraft leading to the first manned Moon landing in July 1969, when the 4.40×10^4 kg spacecraft was 5.64×10^7 m from the centre of the Earth, its speed was 3.56×10^3 m s^{-1}. At this point, relative to the Earth, what was
(*a*) the spacecraft's gravitational potential energy?
(*b*) its kinetic energy?
(*c*) its total energy?
(*d*) its binding energy?

31 Refer to Figure 22.2 which shows the gravitational potential energy of a mass m at various distances from the Earth's centre. In (*a*) to (*c*) assume air resistance is negligible.
(*a*) If a rocket of mass m moving outwards from the Earth has a kinetic energy of $\frac{1}{2}E$ and a total energy of $-3E/2$, in terms of R how far is it from the surface of the Earth?
(*b*) If a second rocket of mass m on the launching pad was given a kinetic energy of $3E$ causing it to move radially outwards from the centre of the Earth, in terms of R at what distance from the centre of the Earth would it come to rest?
(*c*) If a nose cone of mass m fell (from rest) from a point R above the Earth's surface, in terms of E and m with what speed would it hit the Earth?
(*d*) If in (*c*) the effect of air resistance had been taken into account, would your answer to (*c*) have been smaller, larger or unchanged?

32 A 4.8×10^3 kg rocket is sitting on a launching pad on the equator.
(*a*) What is the gravitational potential energy of the rocket?
(*b*) What is the kinetic energy of the rocket due to the rotation of the Earth?
(*c*) Which of these two energies—your answers to (*a*) and (*b*)—has the far larger *magnitude*?
(*d*) What is the total energy of the rocket on the pad?
(*e*) What is the escape energy of the rocket from the Earth?
(*f*) What is its escape speed, that is, the minimum speed it must be given to escape completely from the Earth's gravitational field?

33 A body of mass m is on the surface of a planet of radius R and mass M. Assume the kinetic energy of the body due to the rotation of the planet is negligible in comparison with the magnitude of the body's gravitational potential energy.
(*a*) Deduce an expression for
(*i*) the escape energy, and
(*ii*) the escape speed
of the body from the planet.

(*b*) How does the escape speed depend on the mass of the body?
(*c*) What is the escape speed from the Moon?

34 Consider a point P which is a distance R from the centre of a planet of mass M. (R is greater than the radius of the planet.) Write expressions for
(*a*) the magnitude of the gravitational field strength g at point P;
(*b*) the gravitational potential energy U_g of a satellite of mass m at point P;
(*c*) the gravitational potential V_g at P (the *gravitational potential* at a point is the gravitational potential energy per unit mass at that point, i.e. $V_g = U_g/m$);
(*d*) the gravitational potential gradient at P (the *gravitational potential gradient* is the rate of change of gravitational potential per unit distance moved from the centre of the Earth, i.e. dV_g/dR).
(*e*) Is there any connection between your answers to (*a*) and (*d*)?

35 The US oceanographic and meteorological satellite Nimbus 7, launched in October 1978, has a mass of 715 kg and moves in a circular orbit of radius 7.33×10^6 m.
(*a*) State the SI unit of gravitational potential.
(*b*) What is the gravitational potential
(*i*) at a point S on the surface of the Earth?
(*ii*) at a point N on the orbital path of Nimbus 7?
(*c*) What is the gravitational potential difference between S and N?
(*d*) How much energy is required to raise from S to N
(*i*) a mass of 1.00 kg?
(*ii*) Nimbus 7?
(*e*) Apart from your answer to (*d*) (*ii*) what additional energy was needed to place Nimbus 7 in orbit?

36 Refer back to Figure 22.2 which shows the gravitational potential energy U_g of a body of mass m as a function of distance r from the centre of the Earth.
(*a*) What is the gravitational potential at a point A, distant $4R$ from the centre of the Earth?
(*b*) What is the gravitational potential difference between A and a point B situated $2R$ from the centre of the Earth?
(*c*) How much potential energy would be converted to kinetic energy while a 120 kg meteor fell from A to B? Give the answer in terms of E and m, and ignore the effects of air resistance.
(*d*) (*i*) From Figure 22.2 determine a general expression for U_g in terms of r.
(*ii*) Deduce an expression for the gravitational potential V_g as a function of r.
(*iii*) Then deduce a general expression for the gravitational potential gradient, dV_g/dr, in terms of r.
(*e*) (*i*) Compare your answer to (*d*) (*i*) with $U_g = -GMm/r$. Write down an expression for the gravitational force F_g on mass m in terms of r, E and R.
(*ii*) Then deduce an expression for the magnitude of the gravitational field strength g as a function of r, E, m and R.
(*f*) Compare and comment on your answers to (*d*) (*iii*) and (*e*) (*ii*).

37 Table 22.3 shows data from the Apollo 11 mission in 1969. The spacecraft of mass 4.4×10^4 kg was *coasting* from point X to point Y during this early part of its outward journey from the Earth to the Moon.

Table 22.3

Position	Time from launch (h:min:s)	Distance from Earth's centre (10^7 m)	Speed (10^3 m s^{-1})
Point X	5:58:00	5.44	3.63
Point Y	6:08:00	5.64	3.56

(*a*) While moving from point X to point Y, what was
(*i*) the decrease in the kinetic energy of the spacecraft?
(*ii*) the increase in its gravitational potential energy?
(*iii*) the increase in gravitational potential energy per unit mass of the spacecraft?
(*b*) What is the gravitational potential difference between points X and Y?
(*c*) What then would be the average gravitational potential gradient in the region between X and Y?
(*d*) Using the relation $g = GM/R^2$, determine the gravitational field strength at a point midway between X and Y.
(*e*) Explain why answers (*c*) and (*d*) are similar.
(*f*) What would have been
(*i*) the average acceleration of the rocket, and
(*ii*) the average net force on the rocket
as it moved from X to Y?

SET 23 ROTATIONAL KINEMATICS

Topic	*Problems*
Using equations of uniformly accelerated rotational motion	1–7
Graphs of uniformly accelerated rotational motion	8–9
Linking rotational and translational kinematics	10–12

In this simple introduction to rotational motion, the *direction* of the vectors angular displacement, angular velocity and angular acceleration will be ignored.

1 Suppose a spinning top undergoes a constant angular acceleration of α, thereby raising its initial angular velocity ω_i to a final angular velocity ω_f, while it rotates through an angular displacement of θ in a time interval t. Write down expressions for:
(*a*) ω_f in terms of ω_i, α and t;
(*b*) θ in terms of ω_i, α and t;
(*c*) θ in terms of ω_i, ω_f and t;
(*d*) ω_f in terms of ω_i, α and θ.

2 An electric fan blade, which is originally at rest, undergoes an angular acceleration of 22 radian per second squared (22 rad s^{-2}) for 3.0 s, after which it maintains a constant angular velocity.
(*a*) What is the fan's final angular velocity after 3.0 s?
(*b*) What is the total angular displacement of the fan during the 3.0 s interval?

3 Convert the rate of rotation of
(*a*) a shaft of a motor doing 1900 r.p.m. (rotations or revolutions per minute) into rad s^{-1};
(*b*) a twisting springboard diver, turning at 16 rad s^{-1} into rotations per second.

4 A flywheel which is rotating at 6000 r.p.m. slows down at a uniform rate to 1000 r.p.m. in 40 s. While slowing down,
(*a*) what is the magnitude of the angular acceleration of the flywheel?
(*b*) what is the angular displacement of the flywheel?
(*c*) how many rotations does it make?

5 A flywheel, initially at rest, is given a constant angular acceleration. In the first 9.00 s of its motion it rotates through an angle of 450 rad. Determine
(*a*) its average angular velocity;
(*b*) its angular velocity after 9.00 s;
(*c*) its angular acceleration.

6 A wheel rotating at 3.14 rad s^{-1} is accelerated uniformly to an angular velocity of 10.0 rad s^{-1} while making 100 rotations. What was the angular acceleration of the wheel?

7 Quoting your answers to four significant figures, determine the angular velocity of
(*a*) the sweep second hand on a wristwatch;
(*b*) the hour hand on a clock.

8 What physical quantity would be represented by
(*a*) the gradient of an angular displacement versus time graph?
(*b*) the gradient of an angular velocity versus time graph?
(*c*) the area between the graphed line and the time axis in an angular velocity versus time graph?
(*d*) the area between the graphed line and the time axis in an angular acceleration versus time graph?

9 Figure 23.1 shows—slightly idealised—how the angular velocity of a car wheel varies with time, while the car is backed out of a supermarket parking bay and driven off.
(*a*) For how long is the car accelerating in first gear?
(*b*) What is the angular velocity of the wheel while the driver is changing from second to third gear?
(*c*) Is the car slowing down, speeding up or at constant velocity at the time 9 s?
(*d*) What is the angular displacement of the wheel while the car is being reversed?
(*e*) How many rotations has the wheel made by the time 13 s?
(*f*) What is the angular acceleration of the wheel at the following times?
(*i*) 15.5 s;
(*ii*) 19 s.

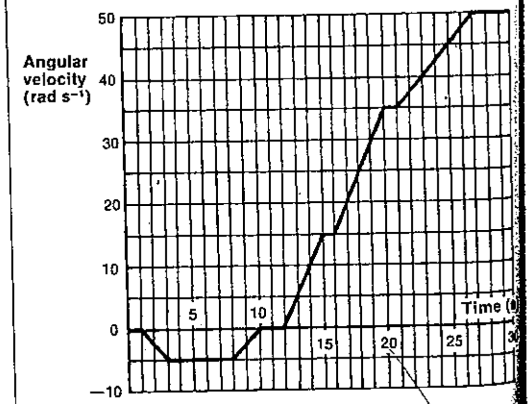


Fig. 23.1

10 The grindstone shown in Figure 23.2 is rotating at 7π rad s^{-1}. Its radius is $1/\pi$ m. Q is mid-way between the centre O and a point P on the edge of the grindstone.

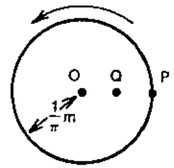


Fig. 23.2

Give the answers to the following questions to two significant figures.
(*a*) What is the angular velocity of point Q?
(*b*) What is the speed of point P?
The grindstone now slows down at a uniform rate and comes to rest in 20 s. During this 20 s period,
(*c*) what is the angular displacement of the wheel?
(*d*) what distance does the point P cover?
(*e*) what is the angular acceleration of the wheel?
(*f*) what is the tangential acceleration of point Q?

11 Shortly after a motor is turned on, a point E on the edge of its flywheel of radius 0.50 m has a tangential acceleration of 1.5 m s^{-2}. At the instant when E is moving at 15 m s^{-1}, what is the flywheel's
(*a*) angular velocity?
(*b*) angular acceleration?

12 Estimate as an order of magnitude the angular speed of a wheel of a car which is travelling at about 60 km h^{-1}.

SET 24 ROTATIONAL STATICS AND DYNAMICS

Topic	*Problems*
Rotational equilibrium (zero net torque)	1–5
Rotational dynamics (non-zero net torque), moment of inertia, radius of gyration, relation between net torque, moment of inertia and angular acceleration	6–10
Angular impulse and angular momentum	11–12
Conservation of angular momentum	13–14
Work done by a torque, rotational kinetic energy and power	15–19

1 A man weighing 840 N sits on a see-saw 1.5 m from its centre. He is balanced by a boy and a girl sitting on the other side. The girl, who weighs 420 N, is 0.90 m from the centre and the boy is 1.8 m from the centre. What is the weight of the boy?

2 A metre ruler of mass 80 g has masses of 105 g and 180 g hung from its ends. At what point will it balance?

3 A garden roller has a radius of 300 mm, weighs 500 N and stands on level ground against a step 60.0 mm high. Assume the frictional force between the roller and its axle is insignificant.
(*a*) If only a *horizontal* force is applied to the handle, what minimum force must be exerted on it to make the roller begin to mount the step?
(*b*) If, instead, the force may be applied to the handle in any direction, what then would be the magnitude of the *least* force on it necessary to cause the roller to begin to mount the step?

4 An oregon beam, 5.00 m long and weighing 700 N, is carried horizontally on the shoulders of Cedric and Edna who support it at points 1.00 m and 0.500 m respectively from its ends. Find the load carried by each.

5 A painter stands on a horizontal platform suspended at its ends by vertical ropes A and B 3.60 m apart. The platform is uniform and weighs 200 N. The tension in rope A is 600 N and in rope B is 350 N. What is the weight of the painter and where is he standing?

6 For the three-body system shown in Figure 24.1 find the moment of inertia about
(*a*) the axis AB;
(*b*) an axis through A and perpendicular to the plane containing A, B and C.

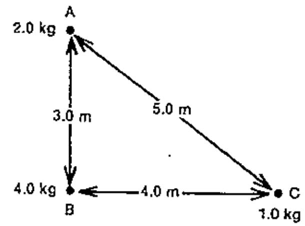


Fig. 24.1

7 A rod has a mass of 10 kg. It is 2.0 m long and 4.0 cm in diameter. Find the moment of inertia of this rod about an axis which
(*a*) is perpendicular to the rod and passes through its centre;
(*b*) is perpendicular to the rod and passes through one end;
(*c*) passes longitudinally through the centre of the rod.

8 (*a*) What is the moment of inertia of
(*i*) a uniform disc D of radius r and mass m about an axis perpendicular to and through the centre of the disc?
(*ii*) a solid sphere S of uniform density, of radius R and mass M about any diameter?

(*b*) Relative to the axes used in (*a*), what is the radius of gyration of
(*i*) the disc D in terms of r?
(*ii*) the sphere S in terms of R?

9 What would be the order of magnitude of the moment of inertia of the Earth about its axis of rotation? (Refer to Table 22.1 on page 52.)

10 A wheel has a mass of 10 kg, a radius of 0.25 m and a radius of gyration of 0.20 m. A string is wound around its circumference and pulled off by a constant tangential force of 16 N. Determine the magnitude of
(*a*) the moment of inertia of the wheel;
(*b*) the torque applied to the wheel;
(*c*) its angular acceleration.

11 A wheel is set in rotation by means of a cord wound around its axle. The tension in the cord is 9.8 N and the radius of the axle is 1.5 cm. When the cord has been pulled for 5.0 s, the wheel is turning at the rate of 2π rad s^{-1}. Find
(*a*) the magnitude of the angular momentum given to the wheel;
(*b*) the moment of inertia of the wheel.

12 If the torque acting on the wheel described in Problem 10 was applied for 5.0 s, what is the magnitude of
(*a*) the angular impulse provided by the torque?
(*b*) the angular momentum given to the wheel?
(*c*) the rate of increase in the angular momentum of the wheel?

13 To demonstrate the effect of radius of rotation on moment of inertia, a boy is asked to stand on a small frictionless platform which can be set in rotation. He stretches out his arms and holds a weight in each hand and the platform is set in rotation with a speed of 2π rad s^{-1}. By drawing in his hands, he reduces the total moment of inertia of the rotating system from 6.0 kg m^2 to 2.0 kg m^2. Determine the resulting angular speed of the platform.

14 The turntable disc of a record player has a mass of 0.30 kg, a radius of 10 cm and is rotating at $33\frac{1}{3}$ r.p.m. The belt driving the turntable breaks at the very instant an LP record is dropped vertically on to the turntable. The LP has a mass of 0.11 kg and a radius of 15 cm. What is the angular speed of the turntable just after the record lands on it? (Assume the turntable is a disc of uniform density.)

15 A flywheel of mass 20 kg and radius 12 cm has a radius of gyration of 10 cm. It is rotating at 20π rad s^{-1}. If it is brought to rest while executing 40 rotations,
(*a*) what was the flywheel's rotational kinetic energy before slowing down?
(*b*) how much work was done by the flywheel while coming to rest?
(*c*) what average torque was acting on the flywheel while it was slowing down?
(*d*) what average tangential force was acting on the rim of the wheel?

16 A design constraint on the flywheel of a small machine is that it requires 570 J of energy to increase its angular velocity from 18π rad s^{-1} to 20π rad s^{-1}. What must be the moment of inertia of the flywheel?

17 Estimate as an order of magnitude the rotational kinetic energy of an ordinary plastic "frisbee" in flight.

18 A grindstone whose moment of inertia is 150 kg m^2 comes to rest in 5.0 min from an angular velocity of 20π rad s^{-1}. Assuming that the frictional torque in the bearings remains constant, how much power must be expended to keep the grindstone rotating at a constant angular speed of 20π rad s^{-1}?

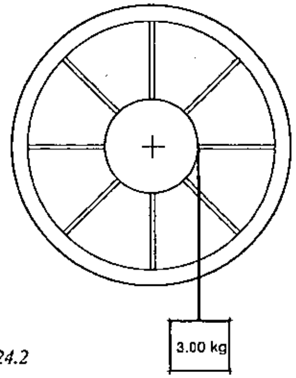


Fig. 24.2

19 A wheel with a heavy rim, as illustrated in Figure 24.2, is driven by a falling mass at the end of a thin rope wrapped around its hub. The moment of inertia of the wheel and hub is 11.5 kg m^2, the suspended mass is 3.00 kg and the radius of the hub is 0.400 m. The suspended mass falls from rest. At the instant the mass has fallen 5.10 m, what is
(*a*) the loss of gravitational potential energy by the falling mass?
(*b*) the speed of the falling mass?
(*c*) the speed of a point on the rim of the hub?
(*d*) the translational kinetic energy of the falling mass?
(*e*) the rotational kinetic energy of the wheel and hub?

SET 25 SIMPLE MACHINES

Topic	*Problems*
Mechanical advantage, velocity ratio, efficiency of a machine	1–5
Lever	6–7
Wheel and axle	8–10
Pulleys	11–13
Screw	14–15
Inclined plane	16

1 What is the mechanical advantage of a machine which raises a load of 2000 N when an effort of 250 N is applied?

2 An effort exerted on the handle of a lifting jack over a distance of 48 cm raises a load through 1.0 cm.
(*a*) What is the velocity ratio of this machine?
(*b*) If the efficiency of the machine was 100 per cent, what effort would be required to lift a load of 1248 N?

3 In a machine which has a velocity ratio of 10, an effort of 60 N is required to move a load of 420 N. Find the efficiency of the machine.

4 A machine has an efficiency of 75 per cent and a mechanical advantage of 9.0. What is its velocity ratio?

5 In a certain machine, which has a velocity ratio of 30, it is found the forces of 150 N and 250 N will just raise weights of 2.50 kN and 4.50 kN respectively.
(*a*) If the relation between the force exerted F and the weight lifted W is of the form $F = pW + q$, determine the values of p and q.
(*b*) What is the efficiency of the machine for a load of 6.00 kN?

6 A lever 60 cm long has a force of 30 N exerted on one end, 45 cm from the fulcrum. What is the greatest load that it will support at the other end?

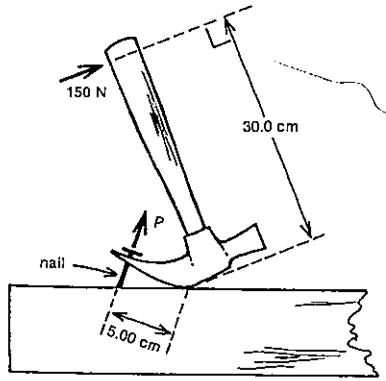


Fig. 25.1

7 A hammer is used to pull a nail out of a plank as shown in Figure 25.1. Assuming 100 per cent efficiency, what is the force P exerted on the nail when the 150 N force is applied to the handle?

8 What effort is required to raise a load of 210 N by means of a wheel and axle, if the radius of the axle is 2.50 cm and that of the wheel 30.0 cm?

9 The crank of a bicycle is 22.5 cm long and a man exerts a force of 300 N on the pedal, the force being at right angles to the direction of the crank. What is the pull on the chain if the diameter of the chain wheel is 25.0 cm and the efficiency is 80.0 per cent?

10 The radii of a wheel and axle are 25 cm and 5.0 cm respectively. It is found that an effort of 40 N is required to raise slowly a load of 160 N. Find
(*a*) the mechanical advantage;
(*b*) the efficiency;
(*c*) the part of the effort used in overcoming friction.

11 A block and tackle contains two sheaves in the upper block and one in the lower block. An effort of 75 N is required to support a load of 200 N. Calculate
(*a*) the velocity ratio;
(*b*) the efficiency.

12 A block and tackle has two sheaves in each block. If the lower blocks weighs 30 N and 10 per cent of the work done is lost in friction, find
(*a*) the effort required to raise a load of 1.5 kN;
(*b*) the efficiency of the machine at this load.

13 An endless chain passes round a fixed pulley of diameter 30 cm, round a movable pulley supporting a load of 2.4 kN, and finally around a pulley 25 cm in diameter attached to the side of the 30 cm pulley. Find the effort required to raise the load if the efficiency is 80 per cent.

14 In a screw jack the pitch of the screw is 1.00 cm and the length of the handle is 28.0 cm. A force of 100 N is required to lift a load of 2.8 kN. Find
(*a*) the mechanical advantage;
(*b*) the velocity ratio;
(*c*) the efficiency.

15 The screw of a jack has a pitch of 0.050 cm and the handle is 42 cm long. What effort must be applied to the handle to raise a load of 6.6 kN if the efficiency is 50 per cent?

16 A body of mass 50 kg rests on an incline which rises 2.0 m vertically for every 7.0 m of its (inclined) length. Frictional resistance to the motion of the body is 210 N. Find
(*a*) the force acting parallel to the plane that would draw the body up the plane at constant velocity;
(*b*) the mechanical advantage of this system for raising the 50 kg body.

SET 26 FLUID MECHANICS

Topic	*Problems*
Pascal's principle, the hydraulic press	1-2
Rate of fluid flow in tubes	3-4
Equation of continuity, Bernoulli's equation Torricelli's theorem	5-9
Viscosity, Reynold's number, Poiseuille's equation, Stoke's equation	10-16
Surface tension	17-19
Capillarity	20-23

Where necessary in this set, use
density of water $= 1.00 \times 10^3$ kg m^{-3}

1 A boy compresses the air in his bicycle pump by pushing with a force of 60 N on the handle while blocking the outlet with his finger. See Figure 26.1.

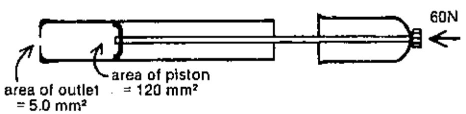


Fig. 26.1

(*a*) What pressure is he exerting on the air to the left of the piston?
(*b*) What pressure is transmitted to his finger blocking the outlet?
(*c*) What force is the compressed air exerting on the part of his finger that is blocking the outlet?

2 A Volkswagen Kombi of mass 1.3 tonne is being raised for servicing on a hydraulic hoist in a service station. The diameter of the large piston in the hydraulic system below the vehicle is 24 cm and that of the small piston in the driving cylinder is 2.1 cm.
(*a*) With no load on the hoist the pressure in the oil in the hydraulic system is approximately atmospheric. When the VW is being supported, what is the resulting increase in the oil pressure?
(*b*) What force is being exerted by the small piston on the oil in the driving cylinder?
(*c*) How far does the vehicle rise when the driving piston moves in 17 cm?

3 The nozzle speed of water from a fire hose is 12 m s^{-1}. If the diameter of the nozzle is 40 mm, what is the discharge rate of the water in
(*a*) m^3 s^{-1}?
(*b*) m^3 min^{-1}?

4 A swimming pool filtration system pumps 7800 litre per hour. At what speed is the water moving to and from the pool through the pipes, which have an internal diameter of 50 mm? (1 litre $= 10^3$ cm^3 $= 10^3 \times 10^{-6}$ m^3 $= 10^{-3}$ m^3)

5 An air-conditioning duct with a circular cross-section narrows at one point from a 0.45 m to a 0.35 m diameter. What is the ratio of the speed of the air in the wider section to that in the narrower section?

6 Water flows past a point X in a horizontal pipe which then narrows down passing a point Y at 0.82 m s^{-1} in a narrower section. The pressures at X and Y are 103.90 kPa and 103.70 kPa respectively. At what speed is the water flowing at X?

7 Oil of density 800 kg m^{-3} flows along a pipeline from a lower horizontal section A to another horizontal section B, which is 5.0 m higher than A. See Figure 26.2. Sections A and B have internal radii of 0.30 m and 0.15 m respectively. The pressure and speed in section A are 120 kPa and 0.40 m s^{-1} respectively. *In section B,*
(*a*) at what speed does the oil flow?
(*b*) what is the pressure in the oil?

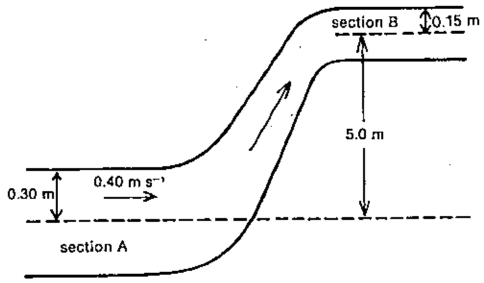


Fig. 26.2

8 Using either conservation of energy or Bernoulli's equation, show that the speed v with which water leaves a small outlet E of a wide tank is given by

$$v = \sqrt{2gh},$$

where g is the gravitational field strength, and h is the vertical height above E of the water surface in the tank. (This relation is known as *Torricelli's theorem.*)

9 A large yacht strikes a submerged rock at a point 0.8 m below the water line. A hole about 5 cm by 5 cm results.
(*a*) At what speed does water pass through this hole into the boat?
(*b*) At what rate (in m^3 min^{-1}) does the boat begin to fill?

10 The question of whether the flow of a fluid through a tube will be *laminar* (also called *streamline*) or *turbulent* depends on the speed of bulk flow (or average speed) v, the internal radius r of the tube, and the density ρ and (dynamic) viscosity η of the fluid. These factors are combined in the Reynold's number

$$N_R = \frac{2rv\rho}{\eta}.$$

For the flow to be laminar in tubes of circular cross-section, N_R must be less than about 2000. If it is greater than about 3000, the flow is turbulent. Between these values it is unstable and may vary from one type of flow to the other.

Derive a second expression for the Reynold's number in terms of the volume V of fluid, the time t that volume takes to flow past, the internal radius r of the tube, and the density ρ and viscosity η of the fluid.

11 Using information from Problem 10 above, determine whether each of the following would represent laminar or turbulent flow.
(*a*) Water passes through an ordinary hose while watering the garden. Viscosity of water at 20°C is 10^{-3} Pa s. You will need to make your own estimates of some of the quantities.
(*b*) In a bottling machine in a pharmaceutical factory castor oil runs through a tube of internal diameter 6 mm into 200 cm^3 bottles which fill in 3 s. Density and viscosity of castor oil at 20°C are 962 kg m^{-3} and 0.986 Pa s respectively.

12 A layer of glycerol 0.20 mm thick separates two microscope slides of area 72 mm by 17 mm. When tilted slightly from the horizontal position, the top slide moves over the lower one at 1.2 mm s^{-1}. If the movement within the glycerol is laminar, what is the magnitude of the shearing force exerted on the slides by the glycerol? Viscosity of glycerol is 1.5 Pa s.

13 A cooling process in a factory involves air at room temperature being directed on to a passing warm plastic strip. The air is forced through a tube of length 3.4 m and internal diameter 5.0 mm at a speed of 0.61 m s^{-1}. The density and viscosity of air at this temperature are 1.2 kg m^{-3} and 1.8×10^{-5} Pa s.
(*a*) Is the flow turbulent or laminar?
(*b*) What is the rate of volume flow (in m^3 s^{-1}) through the tube?
(*c*) Use Poiseuille's equation to determine the pressure difference between the ends of the tube.

14 Data furnished from Robert Millikan's famous experiment which established the quantised nature of electric charge:

An uncharged oildrop of density 800 kg m^{-3} and diameter 6.5×10^{-3} mm fell at constant velocity with no electric forces acting on it. Viscosity and density of air at room temperature are 1.8×10^{-5} Pa s and 1.2 kg m^{-3} respectively.
(*a*) What was the downward gravitational force acting on the oildrop?
(*b*) What was the upward *buoyancy* force the air exerted on the oildrop?
(*c*) What was the net force acting on the drop?
(*d*) What was the upward *viscous* force exerted on the oildrop by the air?
(*e*) Use Stoke's equation to determine the downward velocity of the oildrop.

15 If steel ballbearings of diameter 1.0 mm are released in a tall cylinder of olive oil, they reach a terminal downward velocity of 45 mm s^{-1}. Determine the viscosity of olive oil. The densities of steel and olive oil are 7.8×10^3 kg m^{-3} and 915 kg m^{-3} respectively.

16 Use the following experimental results to determine the approximate diameter of the smallest particles in a small sample of clay of density 2.2×10^3 kg m^{-3}.

The sample was shaken vigorously in a measuring cylinder of water, and then allowed to settle. The water finally became clear after $6\frac{1}{2}$ h. The depth of the water was 0.25 m. The viscosity of water is 10^{-3} Pa s.

17 Calculate the magnitude of the force required to pull a circular flat plate of radius 50 mm away from a water surface, the surface tension of the water being 7.3×10^{-2} N m^{-1}.

18 If the surface tension of a soap solution is 3.00×10^{-2} N m^{-1}, calculate
(*a*) the work done in blowing a bubble 200 mm in diameter;
(*b*) the excess pressure inside the bubble.

19 Air is bubbled into water through a vertical capillary tube, which has an internal diameter of 1.0 mm, and the lower end of which is 0.12 m below the surface of the water. Assume that each bubble breaks away when its diameter equals the internal diameter of the tube. By how much does the pressure required to blow the bubbles exceed that of the atmosphere? The surface tension of water $= 7.3 \times 10^{-2}$ N m^{-1}.

20 When a capillary tube stands upright in a beaker of water (surface tension $= 7.3 \times 10^{-2}$ N m^{-1}), the water rises 35 mm in the tube. Calculate the internal diameter of the tube.

21 Two vertical parallel plates 0.20 mm apart are immersed in water for which the surface tension is 7.3×10^{-2} N m^{-1}, the angle of contact being zero. To what maximum height does the water rise at some distance from the edge of the plates?

22 When a glass capillary tube of internal diameter 0.200 mm stands vertically in mercury, the difference in the mercury levels inside and outside the tube is 62.8 mm. If the angle of contact of mercury at an air–glass surface = 140°, what is the surface tension of mercury?

23 A U-tube which is standing vertically contains water. If the internal diameters of the two arms of the tube are 5.0 mm and 1.0 mm respectively, what will be the difference of the levels of the water in these arms? The surface tension of the water is 7.3×10^{-2} N m^{-1}.

SET 27 ELASTICITY

Topic	*Problems*
Longitudinal deformations: Hooke's law, force constant, normal stress and strain, Young modulus, elastic limit, ultimate strength	1–10
Shearing deformations: shearing stress and strain, shear modulus, ultimate shearing stress	11–13
Bulk deformations: volume stress and strain, bulk modulus	14–15

Other problems relating to elasticity can be found in Sets 16 and 20.

1 A mass of 100 g is attached to the end of an open-coiled spring hanging vertically, and causes its length to increase by 49 mm.
(*a*) What is the force constant of the spring?
(*b*) What force would give the spring a compression of 5.0 mm (assuming that adjacent coils are still not in contact with one another)?

2 A load of 16.3 kg causes an extension of 3.84 mm in a wire of cross-sectional area 1.0 mm^2 and 6.0 m long.
(*a*) What is the normal (or linear) stress on the wire?
(*b*) What is the normal (or linear) strain?
(*c*) Determine the Young modulus for the material of which the wire is made.

3 The Young modulus for steel is 2.0×10^{11} Pa.
(*a*) Calculate the stress produced when a steel wire increases in length by 0.10 per cent of its original length.
(*b*) If the diameter of the wire is 0.50 mm, what load (in kg) would produce that stress?

4 Figure 27.1 shows graphically the result of an experiment in which an aluminium wire of diameter 0.50 mm and length 2.5 m was stretched.
(*a*) What is the Young modulus for aluminium?
(*b*) What is the elastic limit for the aluminium wire?
(*c*) As the wire reached its elastic limit,
(*i*) what weight was acting on it?
(*ii*) what was its extension?

5 A load of 2.00 kg stretches a certain wire 1.00 mm. How much work is required to stretch it 5.00 mm?

6 A brass rod 2.0 cm in diameter and 20 cm long is held in position vertically on a plate and a load of 5.0 kg placed on its upper end. Find
(*a*) the shortening of the rod;
(*b*) the energy stored in it.
The Young modulus for brass is 1.0×10^{11} Pa.

7 The *ultimate strength* (also known as the *ultimate tensile stress* or *breaking stress*) of a material is—not surprisingly—the maximum stress it can withstand before breaking. The ultimate strength of the steel used for a particular car towing cable is 7.0×10^8 Pa and the Young

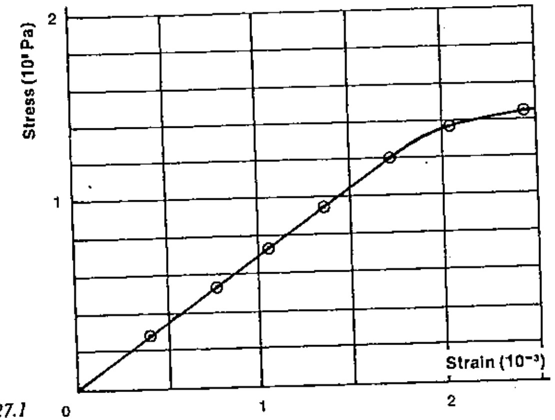


Fig. 27.1

modulus for the material is 2.0×10^{11} Pa. If the cable has a length of 4.0 m and a total cross-sectional area of 0.20 cm^2,
(*a*) what maximum tension can it withstand?
(*b*) what is the maximum extension possible without breaking, if we assume for simplicity that elastic behaviour continues right up to the instant when fracture occurs?

8 The ultimate strength of aluminium is 7.5×10^7 Pa and its density is 2.7×10^3 kg m^{-3}.
(*a*) Find the greatest length of aluminium which could hang vertically without breaking.
(*b*) Theoretically, where should the break occur?

9 A load of 6.0 kg causes an extension of 5.6 mm in a certain wire. What would be the extension if the wire, still supporting the load, were raised vertically with
(*a*) a uniform velocity of 2.45 m s^{-1}?
(*b*) a uniform acceleration of 2.45 m s^{-2}?

10 A load of 8 kg extends a given wire by 6 mm. Given that the density of the material in the load is 6×10^3 kg m^{-3}, find the extension that would be caused if the load were completely submerged in water. Density of water is 10^3 kg m^{-3}.

11 A cube of material, of edge 7.0 cm, has one face fixed, and a tangential shearing force equal to the weight of 200 kg is applied to the opposite face. Calculate
(*a*) the shearing strain produced;
(*b*) the distance through which one face moves relative to the other.
The *shear modulus* (or *modulus of rigidity*) of the material is 2.0×10^6 Pa.

12 A bracket is fastened to a wall by three horizontal steel bolts 4.76 mm in diameter. If the bracket is meant to support 600 kg and the ultimate shearing stress of the steel is 2.4×10^8 Pa,
(*a*) will the bolts shear off?
(*b*) what is the maximum mass the bracket can support?

13 Estimate as an order of magnitude the maximum shearing strain on a rubber eraser when a student uses it to rub out a mistake.

14 Mercury will contract in volume by 0.010 per cent when subjected to a pressure of 2.6×10^6 Pa. Calculate the bulk modulus of mercury.

15 If a cube of aluminium, of edge 0.10 m, is placed in a vacuum, by how much will its volume increase? Bulk modulus of aluminium is 7.0×10^{10} Pa; atmospheric pressure = 1.0×10^5 Pa.

SET 28 INTRODUCTORY RELATIVITY

Topic	*Problems*
The speed of light in different frames of reference	1
Variation of mass with speed	2–4
Relativistic kinetic energy	5–6
Energy–mass equivalence	7–9

Where necessary in this set,
use speed of light (or any other form or electromagnetic radiation) in a vacuum or air,
$c = 3.00 \times 10^8$ m s^{-1}

1 A spacecraft X moves at a constant (but very high) velocity of 2×10^8 m s^{-1} along a straight line from Earth to Mars. A second spacecraft Y follows at a constant distance behind and therefore at the same velocity as X. Both space vehicles have external lights. What is the speed of light observed by
(*a*) R_2, an occupant of X, as he looks back at the lights from Y?
(*b*) D_2, an occupant of Y, as he looks forward at the lights from X?
(*c*) Uncle Martin on Mars, as he observes X and Y approaching?

2 What is the relativistic mass of an electron (rest mass = 9.1×10^{-31} kg), if it is moving at
(*a*) 30 m s^{-1}?
(*b*) 4.4×10^7 m s^{-1}?
(*c*) 2.985×10^8 m s^{-1}?

3 Determine the relativistic *increase* in mass of a cricket ball of mass 0.14 kg (giving all answers to one significant figure),
(*a*) when it is bowled by a boy at 15 m s^{-1};
(*b*) when bowled by Jeff Thomson at 147.9 km h^{-1} or 41 m s^{-1};
(*c*) if delivered at 2.1×10^8 m s^{-1} by an advanced bowling machine developed by the Brytons.

4 In a colour TV tube electrons reach a speed of about 7.3×10^7 m s^{-1} as they approach the screen. At this point what is the ratio of the electron's relativistic mass to its rest mass?

5 One result of Einstein's special theory of relativity is the more general expression for the kinetic energy of a body of mass m,

$$E_k = mc^2 - m_0c^2,$$

rather than

$$E_k = \tfrac{1}{2}m_0v^2,$$

given by classical mechanics. m_0 is the rest mass and v the speed of the body.

From the relativistic expression, deduce that

$$E_k \approx \tfrac{1}{2}m_0v^2,$$

if $v \ll c$, which of course represents the experience of all of us except perhaps those scientists who work with very high speed elementary particles.

6 What is the kinetic energy of an electron of mass 9.1×10^{-31} kg and travelling at 50 per cent of the speed of light, if calculated by
(*a*) classical mechanics?
(*b*) relativistic mechanics?

7 One tonne of Yallourn brown coal yields 6.75 gigajoule (GJ) of energy for generating electricity. (1 GJ = 10^9 J)
(*a*) How much mass must be changed into energy in a nuclear fission or fusion process to provide the same energy output as 13 000 tonne of brown coal? Give your answer in gram.
(*b*) If an average home requires about 16 GJ of electric energy per year, about how many households would be supplied with electricity for one year by converting this mass—your answer to (*a*)—into energy?

8 If a positron, which has the same mass as but opposite charge to an electron, encounters an electron, and both are at low speed, they can annihilate one another to produce two identical photons or bundles of electromagnetic radiation. What is the total energy of one of these photons? Mass of an electron or positron is 9.1×10^{-31} kg.

9 One sequence of nuclear fusion reactions that occur in the Sun amounts to the fact that four hydrogen nuclei (or protons) fuse to produce one helium nucleus and two positrons.

$$4\,^1_1\mathrm{H} \rightarrow \,^4_2\mathrm{He} + 2\,^0_1\mathrm{e}$$

Given the masses in Table 28.1, determine for one such fusion reaction
(*a*) the loss of mass (or *mass defect*) in atomic mass units (u);
(*b*) the amount of energy produced. (1.00 kg = 6.02×10^{26} u.)

Table 28.1

Particle	Mass (u)
Hydrogen nucleus ($^1_1$H)	1.007 276
Helium nucleus ($^4_2$He)	4.002 604
Positron ($^0_1$e)	0.000 549

SET 29 REVISION PROBLEMS FOR PARTS 1 AND 2

Set	Topic	Problems
1	Mathematical conventions in science	1
2	Using dimensional analysis	2
4	Concepts of density and pressure	3
5	Measurement of length, time and frequency	4
6	Motion in a straight line—graphical treatment	5-7
7	Motion in a straight line—algebraic treatment	8
8	Vectors and scalars	9-11
9	Introduction to Newton's laws of motion	12-13
11	Flotation and gravitational effects in fluids	14
12	Vertical motion near the Earth's surface	15
13	Projectile motion near the Earth's surface	16
14	Uphill and downhill motion	17
15	Motion in a circle	18
16	Simple harmonic motion	19-20
17	Momentum and its conservation	21-22
18	Work, kinetic energy and power	23-24
19	Gravitational potential energy in a uniform field	25-26
21	Elastic and inelastic collisions	27-28
22	Gravitation in a non-uniform field and satellite motion	29
26	Fluid mechanics	30-31
27	Elasticity	32

1 (*a*) A metal rod, of length 7.6039 cm, is heated and expands to 7.6052 cm. What was its change in length in *metre*? Express your answer in *standard form*, to an appropriate number of significant figures.
(*b*) In a Young's double-slit experiment, a student takes the following measurements:
separation of slits,

$$d = 0.21 \text{ mm}$$

distance between slits and screen,

$$L = 1.34 \text{ m}$$

distance between successive black bars on screen,

$$\Delta x = 3.2 \text{ mm}$$

Substituting in the relation,

$$\lambda = 10^9\, d\Delta x / L,$$

which gives directly the wavelength of the light in nanometre (nm), the student's calculator display reads 501.492 53. The "10^9" in the formula comes from the definition of 1 nm as exactly 10^9 m. Express this result in *metre*, in *standard form*, to the appropriate number of *significant figures*.

2 Imagine that the first man on the Moon discovers rational beings living in underground caves, and that they have developed a system of units based on a unit of velocity, the "luna", a unit of time, the "tic", and a unit of force, the "fran". In this system, what is the unit of
(*a*) displacement?
(*b*) acceleration?
(*c*) mass?
(*d*) energy?
(*e*) power?

3 A square piece of gold foil, of the quality used for gold lettering, has sides of length 0.14 m. It has a mass of 8.0×10^{-5} kg, and its density is assumed to be uniform, with a value of 2.0×10^4 kg m^{-3}.
(*a*) Calculate the area of the gold foil.
(*b*) Calculate the volume of the gold foil.
(*c*) Calculate the thickness of the gold foil.
(*d*) Assuming a gold atom to take up the same space as a cube of side 4.0×10^{-10} m, how many atoms thick is the foil?
(*e*) If the foil could be spread out into a monolayer, i.e. a layer one atom thick, what would be its area?

4 A square plate is capable of being rotated about its centre O. It has a small black disc near one corner and, at rest, appears as in Figure 29.1 (*a*). It is illuminated by a flashing lamp which flashes 24 times per second and is set in rotation.

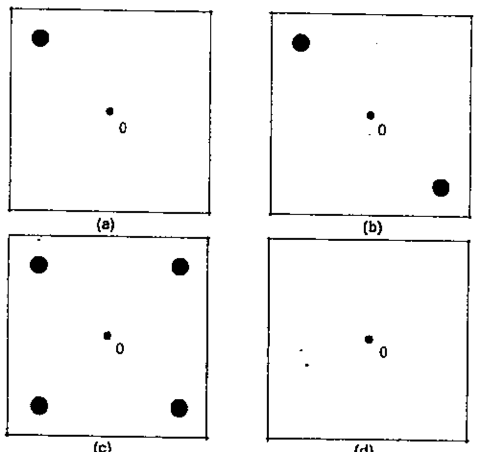


Fig. 29.1

(*a*) What is the least rate of rotation of the plate at which it would appear as in Figure 29.1 (*b*)?
(*b*) What is the largest period of rotation of the plate which would give it the appearance shown in Figure 29.1 (*c*)?
(*c*) With the plate rotating at 100 Hz, the lamp is set to flash 80 times per second. On a copy of Figure 29.1 (*d*), illustrate the appearance of the plate.

5 A trolley, with a ticker-tape attached, rolled along a track without any frictional losses of energy. The ticker-timer marked the tape at the rate of 50 ticks per second. A student cut the tape into "5-tick" pieces, and fixed the "pentatick" strips on squared paper as shown in Figure 29.2

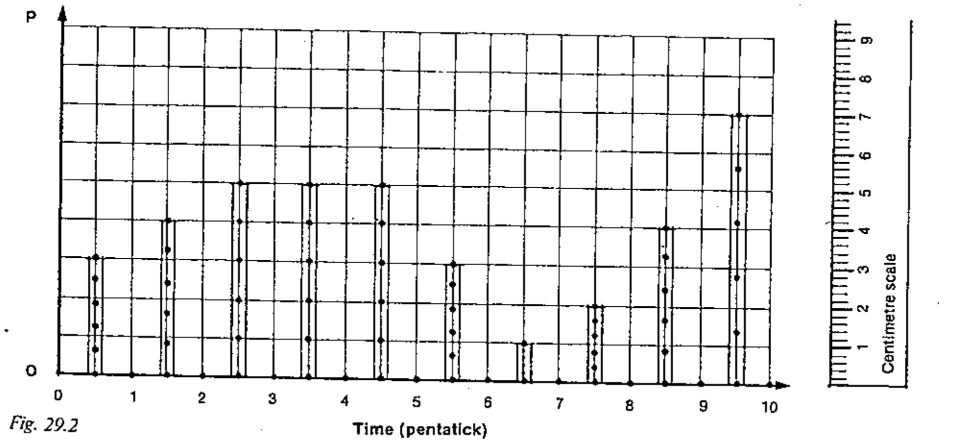


Fig. 29.2

in order to represent the motion of the trolley graphically. A centimetre scale is included in the right-hand side of the diagram.

(*a*) What is the difference (in cm/pentatick) between the average velocity of the trolley in the first pentatick and the average velocity of the trolley in the second pentatick?

(*b*) Express the acceleration of the trolley during the second pentatick in
  (*i*) cm pentatick^{-2};
  (*ii*) m s^{-2}.

(*c*) What physical quantity is represented along the axis OP?

(*d*) What is the total distance travelled by the trolley in the first five pentaticks?

6 The velocity–time graph in Figure 29.3 represents a journey along a straight road in which a man started from home in his car and drove to a mail box, and then travelled back along the same road, finally stopping at a shop. He stopped at traffic lights on his way to and from the mail box.

(*a*) How far did the man drive from his home to the mail box?

(*b*) How far from his home was the shop?

(*c*) How far apart were the two intersections where he encountered traffic lights?

7 In the laboratory, a trolley is being pulled using a rubber band kept extended to a constant length. A tape which passes through a ticker-timer is attached to the trolley and from the dots on the tape, the acceleration is calculated. This procedure is repeated with one, three and then finally four identical bricks added to the trolley. Figure 29.4 shows graphically the reciprocal of acceleration versus the number of bricks.

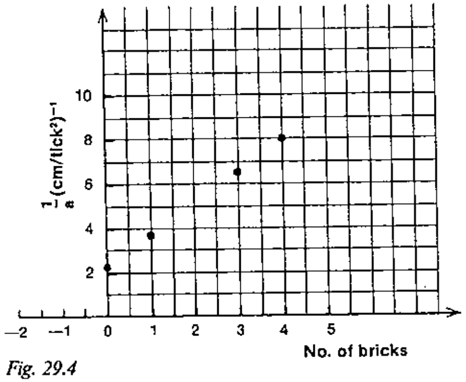


Fig. 29.4

(*a*) What would be the acceleration if the load on the trolley were two bricks?

(*b*) Using the graph, estimate the mass of the trolley (in terms of number of bricks). Express the answer to two significant figures.

(*c*) The original rubber band is now replaced by a number of rubber bands each identical with the first, side by side, and extended by the same distance as the original band. If the acceleration is now found to be 0.37 cm tick^{-2} with four bricks on the trolley, how many bands are being used?

8 An observer on the Moon would find that a body allowed to fall freely would travel a distance of 4.0 m in the third second of its motion. Calculate the acceleration due to gravity on the Moon.

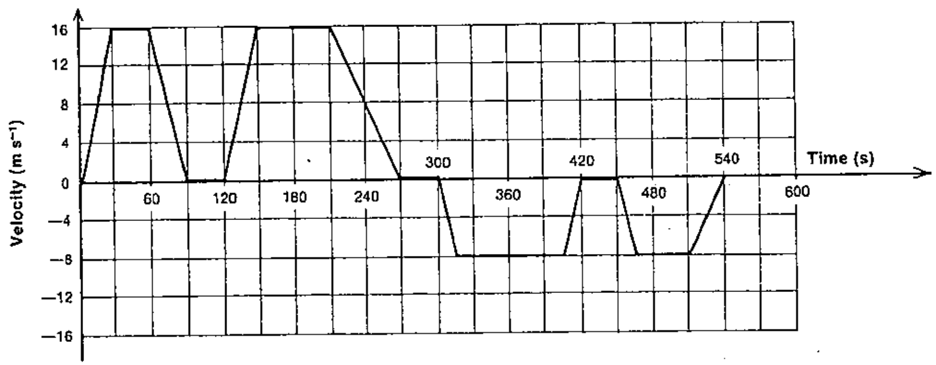


Fig. 29.3

9 An aeroplane flying east at 200 km h^{-1} in still air commences at 2.30 p.m. to turn right to follow a circular path of circumference 40.0 km without changing its speed. See Figure 29.5.

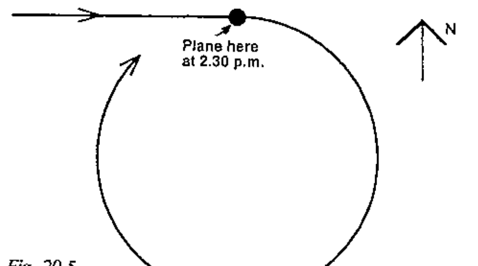


Fig. 29.5

(*a*) How many minutes will it take the plane to complete the circle?

(*b*) At what time will the plane be travelling in a westerly direction?

(*c*) What will be its change in velocity during the period in which its direction of motion changes from east to south?

(*d*) What is the plane's displacement between 2.30 p.m. and 2.36 p.m.?

10 An aeroplane pilot sets a course due west and maintains an air speed of 240 km h^{-1}. After flying for half an hour he finds himself over a town which is 150 km west and 40 km south of his starting point. Find the wind velocity in magnitude and direction.

11 A river 1.0 km wide flows due north at 8.0 km h^{-1}. A motor launch travels at 6.0 km h^{-1} relative to the water. A man starts from the west bank in the launch and wishes to reach the point directly opposite on the east bank. The launch is incapable of reaching this point because it cannot travel as fast as the water. It is necessary for the man to land downstream and walk back along the bank. If he walks at 3.0 km h^{-1}, find the direction in which he should head the launch to reach his objective in minimum time.

12 A car of mass 500 kg is being towed along a straight road so that its velocity changes uniformly from 20.0 m s^{-1} to 30.0 m s^{-1} in a distance of 100 m. During this time, the frictional resistance is constant and equal to 500 N.

(*a*) Calculate the acceleration of the car.

(*b*) What is the size of the net force being exerted on the car during this 100 m?

(*c*) What is the size of the force being exerted on the car by the towing vehicle?

After the speed of 30.0 m s^{-1} is reached, the towing force is reduced so that the car now moves with a constant velocity.

(*d*) What is now the size of the net force on the car?

(*e*) What is now the size of the towing force?

13 A stream of water from a fireman's hose strikes a wall at right angles with a speed of 25 m s^{-1}. If the water is projected from the hose at the rate of 20 litre per second, find the force exerted on the wall

(*a*) when the water does not rebound;

(*b*) when it rebounds with a velocity of 2.0 m s^{-1} perpendicular to the wall.

14 A stone of mass 3.0 kg has a density of 2.0×10^3 kg m^{-3}. With what acceleration will it commence to sink in water in which it is initially completely immersed?

15 Two boys are experimenting to find the depth of an old mine shaft, dropping a stone down the shaft and measuring the time taken for the stone to make a splash which they can observe in the water at the bottom. They find that 4.0 s is a good average time taken for the stone to fall from the top into the water.

(*a*) Find the depth of the mine.

(*b*) With what speed did the stone strike the water?

(*c*) What is the average speed of the stone over the whole distance?

(*d*) What was the speed of the stone when it was half-way down the mine?

(*e*) If another stone were thrown down the shaft with an initial speed of 30 m s^{-1}, how long would it take to reach the bottom?

16 A stone of mass 8.0 kg is dropped from a cliff in a high wind blowing parallel to the cliff face and horizontally. The wind exerts a constant horizontal force of 49 N on the stone as it falls.

(*a*) Write down equations giving the vertically downward displacement y and the horizontal displacement x after the time interval t has elapsed.

(*b*) Is the path of the stone a straight line, a parabola, or some more complicated path? Give your reasoning.

17 A block of mass 2.0 kg is pulled by a cord in the various ways described below. Calculate the magnitude of the tension in the cord in each case.

(*a*) Along a smooth horizontal table with a constant acceleration of 0.75 m s^{-2}.

(*b*) Along a horizontal table with a constant velocity of 0.75 m s^{-1}, the force of friction between the block and the table being 1.2 N.

(*c*) Vertically upwards with a constant acceleration of 1.0 m s^{-2}.

(*d*) With a constant speed up a slope making an angle of 30° with the horizontal against a frictional force of 3.0 N. See Figure 29.6.

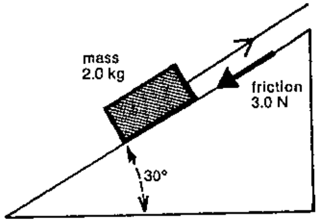


Fig. 29.6

18 A motor-cyclist of mass 70.0 kg rides over a culvert in the shape of an arc of a circle of 10.0 m radius.

(*a*) What is the force exerted by the seat on the cyclist when he passes over the top of the curve at a speed of 5.00 m s^{-1}?

(*b*) What is the greatest speed at which he can ride over the top without the machine leaving the ground?

19 A man carrying a suitcase of mass 10.0 kg is walking at a rate of 5.00 km h^{-1}. His steps are 75.0 cm long and as he makes each step the case rises and falls 2.50 cm. If the vertical motion of the case is simple harmonic, find

(*a*) its period;

(*b*) the maximum acceleration of the case;

(*c*) the maximum load on the man's arm.

20 A pan of mass 30.0 g is suspended from one end of a light spiral spring hanging vertically. A mass of 20.0 g is placed on the pan, and this causes the spring to lengthen by 5.00 cm. The pan, with the mass resting on it, is then pulled downwards and released, and it oscillates with an amplitude of 10.0 cm.

(*a*) Calculate the period of oscillation.

(*b*) Determine the reaction between the pan and the mass at the top of the oscillation.

(*c*) For what amplitude would the mass be just on the point of separating from the pan at the top of the oscillation?

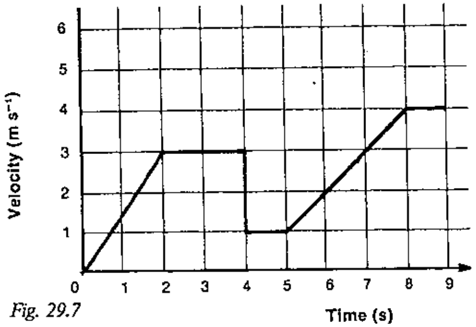


Fig. 29.7

21 The graph in Figure 29.7 represents a plot of velocity against time of a cart of initial mass 1.0 kg as it travels in a straight line across a level surface. At some time during the 9.0 s, an additional mass was dropped vertically on to the cart.

(*a*) At what time did this occur?

(*b*) What was the size of the additional mass dropped vertically on to the cart?

(*c*) What was the net force acting on the cart at $t = 1.0$ s?

(*d*) What was the kinetic energy of the cart at $t = 3.0$ s?

(*e*) How far did the cart travel between $t = 0$ and $t = 4.0$ s?

(*f*) Draw the position–time graph for the first 5.0 s of the cart's motion.

22 A mass at rest on a frictionless horizontal surface explodes into three pieces. A 1.0 kg piece travels due north with a speed of 12 m s^{-1} and a 2.0 kg piece travels due west with a speed of 8.0 m s^{-1}. The third piece has a mass of 4.0 kg.

(*a*) What is the speed of the third piece?

(*b*) In what direction does the third piece move?

(*c*) What is the magnitude of the impulse delivered to the 1.0 kg mass during the explosion?

23 A box slides down a smooth incline with an acceleration of 2.45 m s^{-2}. Determine

(*a*) the angle that the incline makes with the horizontal;

(*b*) the kinetic energy of the box after sliding from rest a distance of 3.00 m down the slope, the mass of the box being 4.00 kg.

24 A simple pendulum consists of a bob of mass 100 g supported by a light inelastic string of length 500 mm. The bob is pulled aside until the string makes an angle of 60.0° with the vertical, and is then released (see Fig. 29.8). Consider the subsequent motion of the pendulum and neglect air resistance.

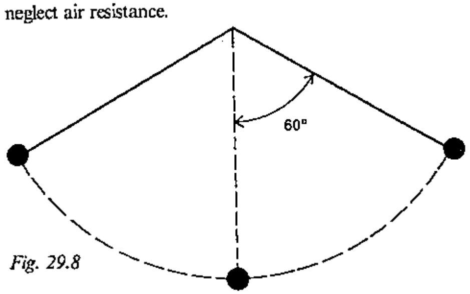


Fig. 29.8

(*a*) When the bob is at its highest point, find the tension in the string and the resultant force on the bob.

(*b*) When the bob is at its lowest point, find the kinetic energy of the bob and its momentum.

(*c*) How much work has been done by the tension force in the string during the time that the bob moves between the lowest point and the highest point in its path?

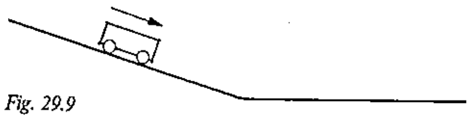


Fig. 29.9

25 A small trolley is allowed to roll from rest down a straight ramp and on to a horizontal surface, as illustrated in Figure 29.9. Assume frictional forces in the moving parts of the trolley are negligible. The trolley runs down the ramp for 3.0 s before reaching the horizontal surface. The gravitational potential energy of the trolley relative to the horizontal surface has the values given in Table 29.1.

Table 29.1

Time elapsed (s)	0	1.0	2.0	3.0
Potential energy (J)	9.0	8.0	5.0	0

(*a*) Draw a graph of the potential energy of the trolley against time elapsed for the first 5.0 s of its motion.

(*b*) From your graph estimate the potential energy of the trolley 1.5 s after the start.

(*c*) What is the kinetic energy of the trolley at the time 2.0 s?

(*d*) On the same graph sheet draw a graph of the kinetic energy of the trolley for the first 5.0 s of its journey.

(*e*) From your graph estimate the kinetic energy of the trolley after 2.5 s and 4.0 s have elapsed.

(*f*) Giving reasons, explain which of the two estimates in (*e*) is the more reliable.

26 A model rocket of total mass 110 g is fired vertically. Assume that the charge lasts for 10.0 s and during that time the rocket moves with a uniform acceleration to a height of 100 m, its mass having reduced to 100 g. Calculate, neglecting air resistance,

(*a*) the maximum upward velocity attained;

(*b*) the maximum height reached by the rocket in its flight;

(*c*) its kinetic energy when it reaches the ground.

27 Two spherical balls whose masses are in the ratio 3:1 are suspended side by side so that their centres are in a horizontal line and the balls hang just touching. The smaller ball is drawn aside and let go. It strikes the larger ball with a velocity of 20 cm s^{-1} and the collision is perfectly elastic. Find the velocities of the two balls

(*a*) after the first collision;

(*b*) after the second collision.

28 1.00 kg of matter is split into two parts, X and Y, of equal mass by an explosion which releases 1.00 J of energy. X flies off in a westerly direction. Assuming complete conversion into kinetic energy, what is the velocity of Y relative to X immediately after the explosion?

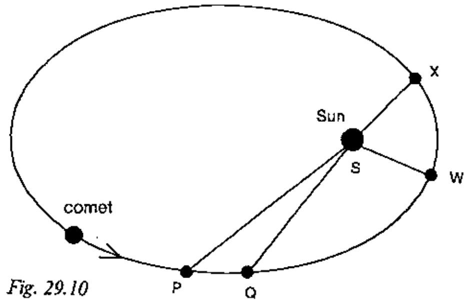


Fig. 29.10

29 A comet is travelling in an elliptical path with respect to the Sun. The mass of the comet is constant during its orbit. See Figure 29.10 which is not drawn to scale.

Information:

$\frac{PS}{WS} = 4$ Area of sector QSP = area of sector WSX

$\frac{QS}{WS} = 3$ $\frac{\text{Arc length WX}}{\text{Arc length PQ}} = \frac{3}{2}$

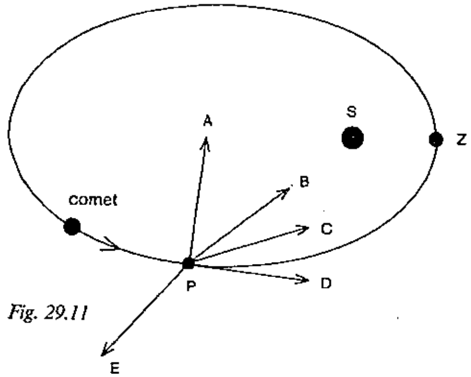


Fig. 29.11

Give the answers to (*a*), (*b*) and (*c*) in *fractional* rather than decimal form.

(*a*) Taking the gravitational potential energy of two masses to be zero when they are infinitely separated, what is the value of the ratio,

$$\frac{\text{gravitational potential energy of the system when the comet is at P}}{\text{gravitational potential energy of the system when the comet is at Q}}?$$

(*b*) What is the value of the ratio,

$$\frac{\text{magnitude of the gravitational force of the Sun on the comet when it is at P}}{\text{magnitude of the gravitational force of the Sun on the comet when it is at Q}}?$$

(c) What is the value of the ratio,

$$\frac{\text{time taken for the comet to travel from P to Q}}{\text{time taken for the comet to travel from W to X}}?$$

(d) Which arrow in Figure 29.11 best describes the *direction* of the net force on the comet when it is at P?

(e) Which arrow best describes the direction of the acceleration of the comet when it is at P?

(f) The kinetic energy of the comet at Z (the point of closest approach) is

A momentarily zero.
B less than at any other point on its path, but not zero.
C greater than at any other point on its path.
D the same as at every other point on its path, because energy is conserved.

(g) Gravitation constant $= G$
Mass of Sun $= M$
Mass of comet $= N$
Distance SP $= p$
Distance SQ $= q$

Using only these symbols, what is the change in potential energy of the system as the comet moves from P to Q?

30 A soap bubble 20.0 mm in diameter is formed at the end of a tube connected to a water manometer, and an excess pressure in the bubble of 1.20 mm of water is observed. Calculate the surface tension of the soap solution.

31 A U-tube which stands vertically has arms whose diameters are 1.20 mm and 0.600 mm respectively. A liquid of surface tension 3.00×10^{-2} N m^{-1} and zero angle of contact is introduced into the tube, the density of the liquid being 800 kg m^{-3}. Calculate the difference in levels of the liquid surfaces.

32 Calculate the period of vertical vibration of a mass of 1.0 kg hanging by a steel wire 3.0 m long and 0.50 mm in diameter, given that the Young modulus for steel is 2.0×10^{11} Pa.

PART 3
WAVES

SET 30 WAVES IN ONE DIMENSION

1 Figure 30.1 illustrates a pulse which is moving along a stretched spring at a speed of 0.50 m s^{-1}.

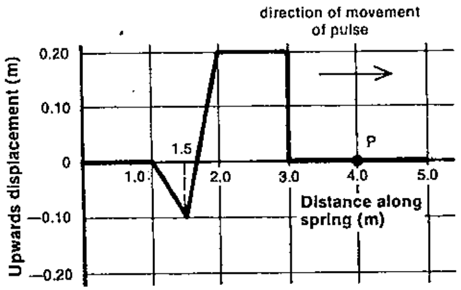


Fig. 30.1

(*a*) Draw a displacement–time graph for the particle P (4.0 m from the origin), taking zero time at the instant for which the graph in Figure 30.1 was drawn.
(*b*) What will be the velocity of the particle P after 3 s, 4.5 s and 5.5 s respectively?
(*c*) You will appreciate that the graph in Figure 30.1 is an idealisation. Give one reason why an actual pulse could not have the shape shown.

2 The graph in Figure 30.2 illustrates a pulse which is moving to the right along a stretched spring at a speed of 10 m s^{-1}.

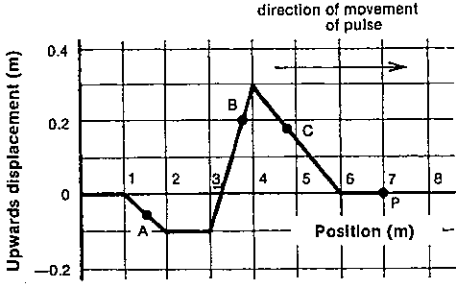


Fig. 30.2

(*a*) What is the direction of motion (up or down, right or left), of the particles marked A, B and C at the instant for which the pulse is shown?
(*b*) Which of the points A, B or C has the greatest speed at the instant shown?
(*c*) Draw the displacement–time and the velocity–time graphs for the particle marked P, taking zero time as the instant for which the graph in Figure 30.2 was drawn.
(*d*) What is the instantaneous velocity of the particle marked B in the graph?

3 A guitar string is plucked, sending 50 progressive waves along it in 0.200 s. What is
(*a*) the frequency, and
(*b*) the period
of the waves?

4 A series of pulses flicked along an extended skipping rope have a period of 0.40 s. What is their frequency?

5 An electromagnetic vibrator attached to a stretched string sends transverse waves of wavelength 0.48 m at a frequency of 100 hZ along the string. What is the speed of the waves?

6 Longitudinal pulses of frequency 2.4 Hz are sent along a slinky spring on the floor at a speed of 3.6 m s^{-1}. What is their wavelength?

7 If the sideways displacement y (in metre) of *a particular point* P on a progressive transverse wave at time t second is given by

$$y = 0.15 \sin 4\pi t,$$

and the wave speed is 6 m s^{-1}, determine
(*a*) the amplitude of the wave;
(*b*) its frequency;
(*c*) its period;
(*d*) its wavelength.

8 A stretched string vibrates transversely in the x–y plane while travelling in the positive x-direction. *At a particular instant* the sideways displacement of y metre of a point a distance x metre from the end of the string is given by

$$y = 0.002 \sin 8\pi x.$$

Determine
(*a*) the amplitude of the waves;
(*b*) their wavelength.

9 A transverse wave in a vibrating string is described by

$$z = 0.003 \sin 5\pi(80t + x),$$

where x and z are expressed in metres and t is the time (in seconds) elapsed since the waves were first generated at the origin of the Cartesian coordinate system.
(*a*) In which plane (x-y, y-z or x-z) is the transverse vibration occurring?
(*b*) In which direction is the wave travelling?
(*c*) What is the wavelength of the waves?
(*d*) What is their period?
(*e*) At what speed are the waves travelling?
(*f*) What is their amplitude?
(*g*) What is the transverse displacement of the wave at a point 0.75 m from the origin in the direction in which the waves are moving, 10 ms after the waves were first generated?

10 Write an equation in trigonometric form to express the sideways displacement of a transverse wave in terms of distance travelled from the origin of the waves and time t (in seconds) elapsed since they were first generated. The transverse vibration is in the y-z plane, and the waves are travelling in the positive z-direction. They have a frequency of 40 Hz, an amplitude of 0.12 m and a wavelength of 0.20 m. y and z are to be expressed in metres. At $t = 0$ and $z = 0$, $y = 0$. Immediately after $t = 0$, at $z = 0$ transverse movement occurs initially in the *positive* (not negative) y-direction.

11 A light rope is attached at J to a heavier one which is fixed at its other end as illustrated in Figure 30.3. The ropes are stretched and a pulse is sent along the light rope in the direction indicated.
(*a*) Will the pulse transmitted in the heavy rope be the same way up or inverted?
(*b*) Is the pulse reflected at J the same way up or inverted?
(*c*) Which of the pulses, the transmitted or reflected, will move away from J with the greater speed?

12 Three rubber cords, A, B and C, each with a different mass per unit length, are set up as shown in Figure 30.4. The cords are stretched and a pulse is sent along cord A towards the right.
(*a*) In which of the cords does the pulse travel with the least speed?
(*b*) Is the pulse reflected at the junction of A and B the same way up or inverted? Sketch the shape of this reflected pulse.
(*c*) When the pulse reflected at the junction of B and C passes back into A, is it the same way up or inverted?

13 A man stands at the centre of a very long coil spring resting on a smooth bench. Each half-second he gives the spring a sideways flick which sends pulses travelling away from him in both directions.
(*a*) If V is the velocity of the pulses in the spring, what is the separation x of successive crests on the spring?
(*b*) Now the man walks beside the spring with a speed $S(<V)$ and again flicks the spring every half-second. What is the distance d travelled by the man in the time interval between successive flicks?
(*c*) In terms of x and d only give an expression for the distance between successive pulses travelling along the spring ahead of the man.
(*d*) In terms of x and d only give an expression for the distance between successive pulses travelling along the spring behind the man.
(*e*) The man wishes to produce a set of pulses whose separation is $\frac{1}{3}x$. He could do this by flicking the spring every half-second while moving forward with a speed S given by $S = kV$. What is the numerical value of K?

14 A stretched string vibrates in three segments, the period of the vibration being T s. Assuming that the string passes through its equilibrium position at the instant $t = 0$, draw a series of diagrams to show the shape of the string at the subsequent instants $t = \frac{T}{4}, \frac{T}{2}, \frac{3T}{4}$ and T s.

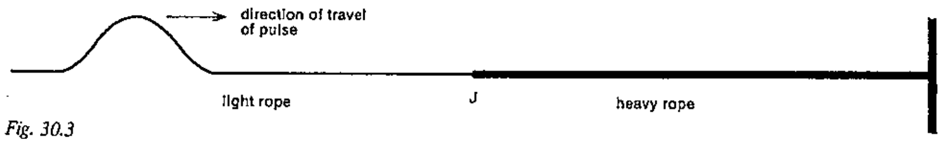


Fig. 30.3

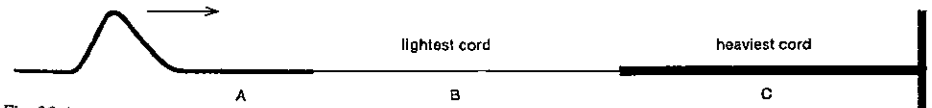


Fig. 30.4

15 Progressive transverse waves of wavelength 60 cm, amplitude 5.0 cm and speed 15 m s^{-1} interact as they travel in opposite directions. What is the
(*a*) amplitude,
(*b*) wavelength and
(*c*) frequency
of the stationary waves so produced?

16 A rope 3.0 m long is tied to a hook on a wall. Find the frequency at which the free end should be shaken to make the rope form three equal loops. The speed of the transverse wave in the rope is 7.0 m s^{-1}.

17 (*a*) What is the longest wavelength of progressive waves that can produce a stationary wave in a stretched string 5.0 m long?
(*b*) The distance between two adjacent nodes in a stationary wave is 0.18 m. What is the wavelength of the component progressive waves?

18 A string is stretched between two fixed points 40 cm apart and when vibrating in its fundamental mode the displacement x of its middle point M from its mean position is given by $x = 3 \sin 60\pi t$ cm. Working to two significant figures, determine
(*a*) the nature of the motion of M;
(*b*) the amplitude;
(*c*) the period;
(*d*) the acceleration of M when its displacement is greatest;
(*e*) the velocity of M when its displacement is zero;
(*f*) the wavelength of the transverse disturbance in the string.

SET 31 WAVES IN TWO DIMENSIONS

Topic	*Problems*
Relations between (*a*) period and frequency, and (*b*) speed, frequency and wavelength for waves in two dimensions	1
Reflection of water waves by a straight barrier	2
Standing or stationary waves produced by reflection	3
Partial reflection and transmission of water waves at an "interface" between deep and shallow water; refraction, relative refractive index, critical angle, total internal reflection	4–6
Diffraction of water waves	7
Interference of water waves from two sources in phase, based on the path difference from the sources	8–9
Other interference problems for two sources in phase	10–11
Interference of water waves from two sources not in phase	12
Single-slit diffraction pattern with water waves, Huygens' principle	13–14
Standing waves on a vibrating surface	15

1 A stone is thrown into a pond and after 5.0 s fifty ripples have spread out from the spot where the stone entered the water. The radius of the outermost ripple is 2.0 m. Calculate
(*a*) the period,
(*b*) the frequency,
(*c*) the wavelength, and
(*d*) the speed
of these water waves.

2 In Figure 31.1 CO represents a wavefront approaching a fixed wooden barrier AB in a ripple tank.

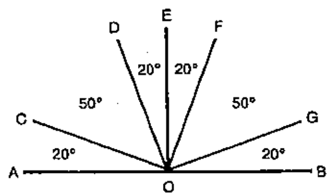


Fig. 31.1

(*a*) which line in the diagram represents the reflected wave?
(*b*) Of the possible directions CO, OC, DO, OD, EO, OE, etc., which represents the direction of travel of
(*i*) the incident wave?
(*ii*) the reflected wave?
(*c*) What is the size of
(*i*) the angle of incidence?
(*ii*) the angle of reflection?

3 In a ripple tank a 4.0 Hz straight wave generator AB sends waves towards a fixed barrier CD as shown in Figure 31.2. A light source 0.60 m above the water surface enables wave effects to be observed on a sheet of white paper 0.60 m below the surface.

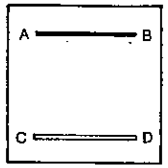


Fig. 31.2

A standing wave pattern is set up near CD. On the paper below the tank bright flickering lines are observed 12 mm apart below the region near CD.
(*a*) *As seen on the paper below the tank*, what is the distance between
(*i*) adjacent nodal lines near CD?
(*ii*) adjacent anti-nodal lines near CD?
(*iii*) travelling progressive waves near AB?
(*b*) As seen on the paper, what is the wavelength of the waves?
(*c*) *In the ripple tank* (not on the paper) what is the
(*i*) wavelength?
(*ii*) speed of the waves?

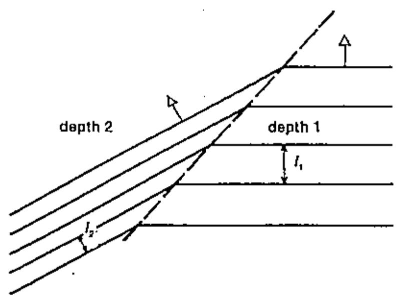


Fig. 31.3

4 Straight waves in a ripple tank pass from one depth of water to another as shown in Figure 31.3. The distances between crests in depth 1 and depth 2 are l_1 and l_2 respectively. In terms of l_1 and l_2, what is the ratio

(*a*) $\dfrac{\text{wavelength in depth 1}}{\text{wavelength in depth 2}}$?
(*b*) $\dfrac{\text{wave frequency in depth 1}}{\text{wave frequency in depth 2}}$?
(*c*) $\dfrac{\text{wave speed in depth 1}}{\text{wave speed in depth 2}}$?

5 A ripple tank containing deep water has a triangular region of shallow water BAC. The boundary BA makes an angle of 45° with a train of straight waves whose frequency is 12 Hz. It is desired to study the refraction of this wave train as it meets the boundary BA. The diagram in Figure 31.4 was drawn from a photograph of the wave pattern. A centimetre scale has been added which enables wavelengths of waves in the ripple tank to be measured directly.

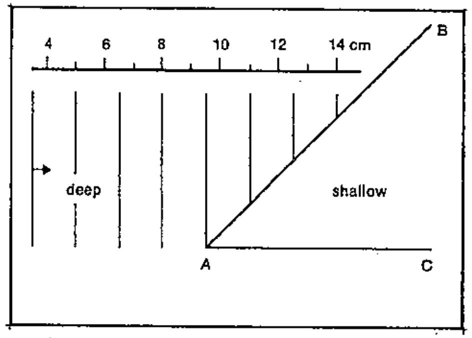


Fig. 31.4

(*a*) What is the wavelength (in cm) of the waves in the deep region of the tank?
(*b*) Calculate the speed of the waves in the deep region.
(*c*) In the shallow region BAC of the tank, the wavelength is measured as 1.0 cm. Draw the wave pattern in the shallow region.
(*d*) What is the frequency of the waves in the region BAC?
(*e*) Calculate the speed of the waves in the shallow region BAC.
(*f*) Calculate the "relative refractive index" for waves travelling from the deep into the shallow section.
(*g*) If the wave pattern is viewed through a six-slit stroboscope, what is the highest frequency of rotation of the stroboscope that would give a stationary pattern with the crests in the deep region spaced as shown in Figure 31.4?
(*h*) What pattern would you observe in the ripple tank if you viewed it through the same stroboscope rotating at 4.0 Hz?

6 Figure 31.5 shows a ripple tank with two different depths of water separated by a boundary XY. A wave generator is producing straight waves as indicated. The separation of the wave fronts is not drawn to scale, but the

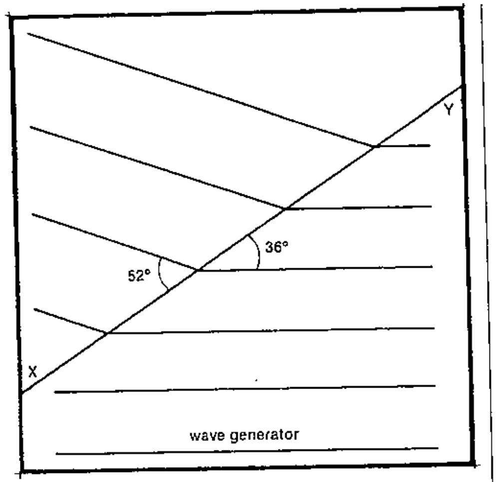


Fig. 31.5

angles between the wave fronts and the barrier XY are as shown.

(*a*) What is the ratio of the speed of the waves in deep water to their speed in shallow water?

(*b*) If this ripple tank set-up is taken to represent light going from one medium to another, one of the media being air, what is the refractive index of the other medium?

(*c*) The waves are viewed through a disc stroboscope with two radial slits at opposite ends of a diameter. The waves appear stationary when the stroboscope is rotated at $1\frac{1}{2}$ Hz and, as the frequency of rotation is gradually increased, the next frequency which "stops" the motion is $2\frac{1}{4}$ Hz. What is the frequency of the waves, and what is the highest rate of rotation of the stroboscope which would give the same stationary pattern?

(*d*) The waves partially reflected at the boundary XY have been omitted from Figure 31.5. What would be their angle of reflection?

(*e*) What would be the critical angle for waves travelling between these two depths?

(*f*) By varying the angle of incidence but not the location of the wave generator in Figure 31.5, would it be possible to observe total internal reflection?

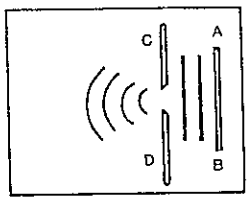


Fig. 31.6

7 In Figure 31.6 AB represents a periodic wave generator in a ripple tank sending out straight waves of wavelength λ. These waves move towards two straight barriers C and D separated by a gap of width w. The waves passing through the gap undergo diffraction. Will the degree of diffraction (or sideways spreading) be increased or decreased, if

(*a*) λ is increased, while w is kept constant?

(*b*) w is increased, while λ is kept constant?

8 Using a 25 m swimming pool, two students decide to set up an interference pattern by leaning over the edge and moving two mops vertically up and down in phase at F and G, each 1.0 m away from one wall as shown in Figure 31.7. The pool has an overflow gutter all round at surface level, thus preventing any reflection of waves from the sides. (FG = 2.0 m.) They find they produce waves with a wavelength of 0.40 m when they execute six complete cycles with each mop in 4.0 s.

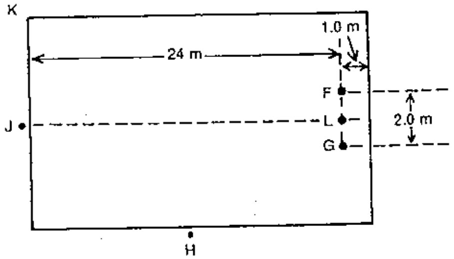


Fig. 31.7

(*a*) What is the frequency of the waves produced by each mop?

(*b*) At what speed do the waves generated at G move out radially from that point?

(*c*) If there is a point P in the pool such that the distance PF is 1.2 m longer than the distance PG, is P on a point of maximum or minimum disturbance?

(*d*) A point R in the half of the pool nearer the point H is on the second nodal line from the centre of the pool. If the distance RF is 18.5 m, what is the distance RG?

9 This question concerns the *surface of the water between F and G* in the set-up described in Problem 8 and Figure 31.7. L is the mid-point between F and G.

(*a*) How far apart are adjacent nodal points along the line FG?

(*b*) Is there a node or an antinode at L?

(*c*) How far from L along the line LF is the node nearest to L?

(*d*) How many nodal points are there on the line FG between F and G?

(*e*) How far from F along the line FG is the nodal point closest to F?

10 Refer again to the situation described in Problem 8 and Figure 31.7.

(*a*) A third student at H observes a series of "double crests" and "double troughs" travelling down the centre of the pool towards J, and measures the time taken for 12 "double crests" and 12 "double troughs" to go past. What time did he obtain?

(*b*) A fourth student at J moves towards the corner K until he is in line with the nodal line nearest the centre of the pool. How far did he move from J?

11 Two point sources in a ripple tank are generating waves with the same wavelength λ. They are placed a distance apart $d = 5.5\lambda$ as in Figure 31.8.

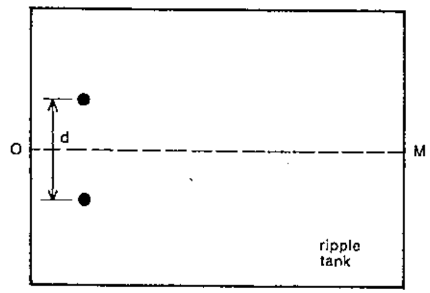


Fig. 31.8

(*a*) If the sources are in phase, what is the sine of the angle which the straight portion of the first nodal line makes with the central line OM?

(*b*) How many nodal lines will be produced in the tank on each side of OM?

12 If the sources described in Problem 11 and Figure 31.8 have a phase difference of one-half, what is the sine of the angle between the central nodal line and either of the nodal lines nearest to it?

13 By reducing the frequency of the wave generator AB shown in Figure 31.6 a diffraction pattern is produced to the left of barriers C and D. The wavelength of the water waves is 1.1 cm and the width of the opening between C and D is 2.6 cm.

(*a*) How many nodal lines could be observed?

(*b*) At a large distance from the opening what is the angle between the perpendicular bisector of the opening and one of the nodal lines closest to it?

14 In order to test Huygens' theory for a single-slit diffraction pattern, a 20 mm opening between two barriers (similar to that in Fig. 31.6) is replaced by a series of point sources from a piece of comb 20 mm in length. The vibrating comb, generating waves with a wavelength of 5.0 mm, produces an interference pattern.

(*a*) In a region well away from the comb, what is the path difference between the two edges of the piece of comb and a point on one of the nodal lines closest to the centre of the pattern?

(*b*) P, a point on the second nodal line from the centre of the pattern, is 0.40 m from the centre of the piece of comb. Approximately how far is P from the line of symmetry of the pattern?

15 Fine powder is sprinkled on a metal plate which is clamped at the centre. An edge of the plate is stroked with a violin bow. This sets up standing waves and causes the powder to form the lines shown in Figure 31.9.

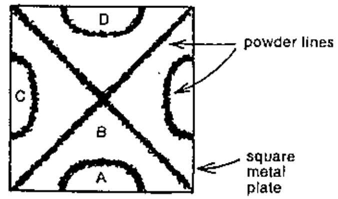


Fig. 31.9

(*a*) What do the powder lines locate?

(*b*) At the instant that region A (in Fig. 31.9) is vibrating upwards, are the following regions vibrating upwards or downwards?

(*i*) B;

(*ii*) C;

(*iii*) D.

SET 32 REVISION PROBLEMS

Set	*Topic*	*Problems*
30	Waves in one dimension	1–3
31	Waves in two dimensions	4–7

1 Two water beetles are sitting 40 mm apart on the surface of the water in a ripple tank in which straight waves are being generated. When one of the beetles is on a crest, he sees the other beetle also on a crest at the same time and with one crest between them. The beetles bob *up and down again* ten times in 2 s.

(*a*) What is the wavelength of the water waves?

(*b*) What is the period of the waves?

(*c*) How far does a wave travel in one second?

2 Two similar wave trains are travelling in opposite directions in a stretched string, the velocity of each being 1 cm s^{-1}. Figure 32.1 represents each of the wave trains at the instant $t = 0$. Reproduce the diagram in pencil on graph paper and below it draw two other pencil diagrams showing the wave trains at the subsequent instants $t = \frac{1}{2}$ s, and $t = 1$ s. In each of the three diagrams draw a line to depict the shape of the string at the specified instant.

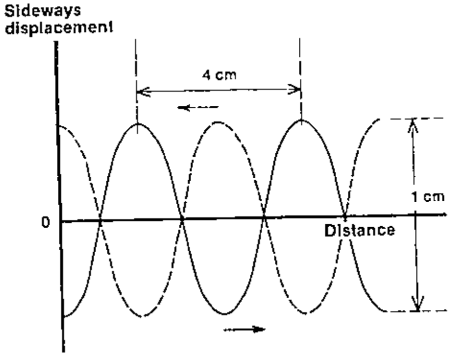


Fig. 32.1

3 A student vigorously shakes the end of a slinky spring stretched to 12 m, the other end of which is firmly fastened to a table leg, as shown in Figure 32.2. He begins slowly and then increases the rate of shake. A stationary wave pattern, with one node between the student and table leg, appears when he is shaking the end (which is a point of *maximum* displacement) three times per second.

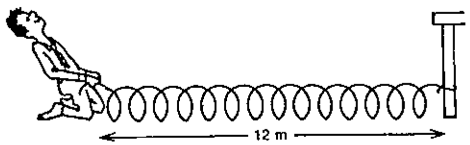


Fig. 32.2

(*a*) What is the wavelength of the wave produced?
(*b*) What is the speed of the wave propagation along the spring?
(*c*) If a point P on the spring undergoes exactly the same motion at exactly the same time as a point 3.0 m from the table leg, how far is P from the table leg?
(*d*) As the student gradually speeds up his shaking, at what frequency will be obtain the *next* stationary pattern, still with an antinode at the end he is holding?

4 A ripple tank is illuminated by a point source 30 cm above it, and a straight wave generator is producing waves whose crests show up as bright bands moving across a screen 50 cm below the tank. When the bright bands are viewed through an electrically driven stroboscope with four equally spaced slits, there are found to be several rates of rotation which produce Pattern 1 in Figure 32.3. Doubling the fastest of these to 12 rotations per second produces Pattern 2. The lines in these diagrams represent bright bands.

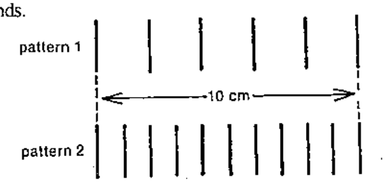


Fig. 32.3

(*a*) What is the highest frequency of rotation of the stroboscope that produces Pattern 1?
(*b*) What is the frequency of the water waves?
(*c*) What is the wavelength of the water waves in the tank?
(*d*) Determine the speed of the water waves in the tank.
(*e*) Suppose the speed of the waves is altered by draining some water from the tank. In order to "stop" the waves one wavelength apart now, should the stroboscope be rotated at a higher, lower or the same frequency as before?

5 Waves are travelling across a boundary between two media, as illustrated in Figure 32.4.

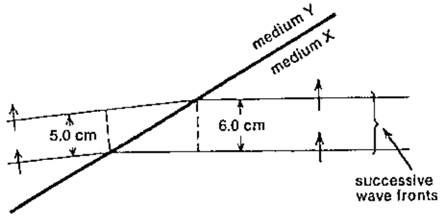


Fig. 32.4

(*a*) What is the value of the "relative refractive index" for waves travelling from X to Y?
(*b*) What is the value of the ratio $\dfrac{\text{frequency of waves in medium X}}{\text{frequency of waves in medium Y}}$?
(*c*) What is the value of the ratio $\dfrac{\text{speed of waves in medium X}}{\text{speed of waves in medium Y}}$?

6 In a ripple tank two regions X and Y contain water of different depths. In Figure 32.5 six wavefronts are shown as waves cross the interface between the two regions. The wavelengths in the regions X and Y are 5.0 cm and 4.0 cm respectively.

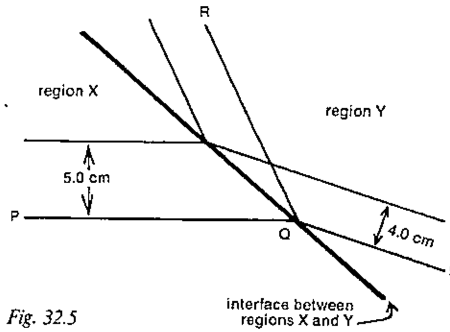


Fig. 32.5

(*a*) Which of the wavefronts, PQ, RQ and SQ, is an *incident* (rather than a *reflected* or *refracted*) wavefront?
(*b*) What is the value of the ratio $\dfrac{\text{period of the waves in region X}}{\text{period of the waves in region Y}}$?
(*c*) What is the value of the ratio $\dfrac{\text{speed of the waves in the shallower region}}{\text{speed of the waves in the deeper region}}$?

7 Figure 32.6 shows a small artificial harbour with an entrance 40 m wide running east-west. Also running east-west is a wharf 200 m south of the opening. On a particular day the waves in the sea are 4.0 m apart and parallel to the opening. Neglect any effects due to change in depth. On the wharf a fisherman at O directly opposite the centre of the opening notices that in the harbour there are a number of lines along which there is little disturbance of the surface.

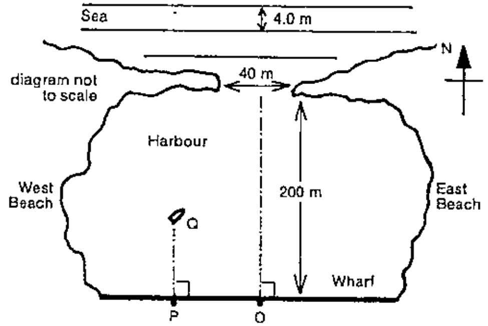


Fig. 32.6

The fisherman walks west from O until he reaches a point P where the nearest of these lines on the west side strikes the wharf.

(*a*) How far did he walk?
(*b*) If he looks straight along this line of calm water, in what direction is he facing? Express the answer to the nearest degree east or west of north.
(*c*) While at P he notices a small boat Q anchored directly north of P and on the *next* line of calm water. How far is he from the boat?

PART 4
SOUND

SET 33 TRANSMISSION OF SOUND

Topic	Problems
Determination of the speed of sound in air	1–2
Relation between period and frequency	3–5
Relation between speed, frequency and wavelength	6–8
Effect of temperature on the speed of sound in air	9–10
Refraction of sound	11–12
Inverse square law of sound intensity	13–14
Sound intensity and sound level or loudness in decibels	15–18
Doppler effect with sound	19–23

1 Thunder is heard 5.0 s after a flash of lightning is seen. How many kilometres is the storm centre from the observer? Speed of sound in air = 340 m s^{-1}.

2 In order to make a rough measure of the speed of sound in air a girl claps her hands at 1.0 s intervals as she walks away from the front of a building across a playing field. When she is about 175 m from the building, she hears the echo of each clap at the same instant she makes the next clap. What value does she deduce for the speed of sound in air?

3 If the frequency of the E above Middle C is 329.6 Hz, what is the period of this note?

4 If it takes 1.911 ms for the vibrating string of C above Middle C on a piano to send out one wave, what is the frequency of the note?

5 Estimate as an order of magnitude the total number of sound waves produced by a 40-piece orchestra playing a symphony which lasted for 20 min.

6 If a string of a guitar vibrates with a frequency of 400 Hz, what is the wavelength of the progressive wave produced in the air? Speed of sound in air = 336 m s^{-1}.

7 A note with a wavelength of 0.600 m in air, where its speed is 345 m s^{-1}, is transmitted into water where it travels at 1.49 km s^{-1}.
(*a*) What is the frequency of the note in air?
(*b*) What is its frequency in the water?
(*c*) What is its wavelength in the water?

8 Draw a diagram on graph paper to represent a transverse wave of amplitude 2.0 cm and wavelength 8.5 cm. If this wave is propagated with a velocity of 340 m s^{-1}, what is the frequency?

Assuming that the curve you have drawn represents a sound wave, draw curves that represent
(*a*) a louder sound of the same pitch;
(*b*) a sound of higher pitch.

9 If the speed of sound in air at a temperature of 273 K is 331 m s^{-1}, what is it at 295 K?

10 At 24°C sound travels through carbon dioxide at 270 m s^{-1}. At what Celsius temperature will its speed in carbon dioxide be 285 m s^{-1}?

11 (*a*) Sound waves strike an air-to-water interface at an angle of incidence of 5.0°. If the speed of sound in air is 346 m s^{-1}, and the angle of refraction is 22°, what is its speed in water?
(*b*) What is the critical angle (to the nearest degree) for sound travelling from air to water?

12 Under certain atmospheric conditions sound intensity does not decrease appreciably with distance from the source. For instance, when occasionally the temperature of the air increases with distance above ground level (see Fig. 33.1), it is possible for an observer at O to hear a sound from source S more clearly than would be the case under other atmospheric conditions. Figure 33.2 shows a satisfactory wave model explanation, the continuous lines representing wavefronts moving from S to O.

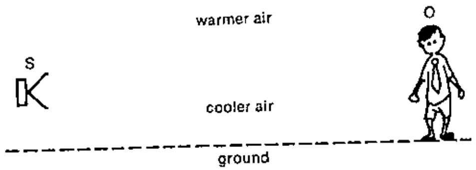


Fig. 33.1

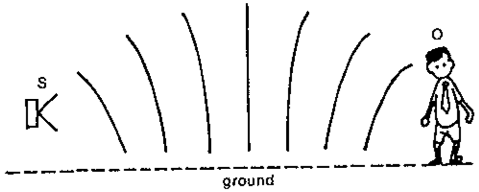


Fig. 33.2

Using the information given in Figures 33.1 and 33.2, decide whether each of the following ratios are greater than one, equal to one or less than one:
(*a*) $\dfrac{\text{speed of sound in warm air}}{\text{speed of sound in cold air}}$;
(*b*) $\dfrac{\text{frequency of sound in warm air}}{\text{frequency of sound in cold air}}$;
(*c*) $\dfrac{\text{wavelength of sound in warm air}}{\text{wavelength of sound in cold air}}$.

13 If the intensity of sound 3.0 m from a small source is 1.2 × 10^{-6} W m^{-2}, what will it be 6.0 m from the source? Assume there is no reflection of sound in this region.

14 If the sound intensity 1.1 m from a small source is 2.3 × 10^{-4} W m^{-2},
(*a*) how far from the source will it be 2.9 × 10^{-4} W m^{-2}?
(*b*) what is the power of the source?

15 Although the *intensity of sound* in the SI system is expressed in W m^{-2}, *loudness* or *sound level* is often measured on a logarithmic scale relative to an arbitrary intensity I_o of 1.0 pW m^{-2} (picowatt per square metre), which is about the minimum intensity detectable by the human ear. For sound of intensity I,

$$\text{loudness or sound level in } bel = \log \frac{I}{I_o}, \text{ and}$$

$$\text{loudness or sound level in } decibel\ (\text{db}) = 10 \log \frac{I}{I_o}$$

(*a*) Determine the sound level in decibel of a loud party of sound intensity 1 μW m^{-2}.
(*b*) What is the intensity of the sound in W m^{-2} in an excessively noisy factory where the loudness is 95 db?

16 Prior to entering a tunnel the sound intensity from a train is 2.0 × 10^{-4} W m^{-2}. As it enters, the sound level changes by +10 db.
(*a*) Before entering the tunnel what is the sound level in decibel?
(*b*) In the tunnel what is the sound intensity in W m^{-2}?

17 What is the ratio of the rate at which energy is received by the ear in a normal conversation (43 db) to that received when listening to a whisper (20 db)?

18 Suppose the sound level for a listener 1.7 m from a radio is 40 db. What minimum distance would the person need to be from the radio in order to be unable to hear it at all, i.e. for the sound level to become 0 db?

19 The sound from a police siren is transmitted through the air as a series of compressions and rarefactions of the air. The siren produces compressions which are separated in time by 1.0 × 10^{-3} s and the speed of the sound is 3.4 × 10^2 m s^{-1}.
(*a*) What is the wavelength of the sound wave?
(*b*) If the siren is on a car moving at 40 m s^{-1}, how far does the car move between sending out successive compressions?
(*c*) How far apart are the compressions as heard by a stationary observer directly ahead of the moving car?
(*d*) What is the frequency of the sound heard by a stationary observer directly ahead of the car?
(*e*) What is the frequency of the sound heard by a stationary observer after the car has passed?

20 A boy in a sports car moving at 108 km h^{-1} is blowing a whistle which emits a frequency of 512 Hz. Speed of sound = 340 m s^{-1}. What is the frequency of the note heard by a pedestrian standing close to the road, when the car
(*a*) is approaching him?
(*b*) is moving away from him?

21 Refer back to the police siren in Problem 19. Suppose the police car was stationary with the siren left on.
(*a*) How many sound waves would an observer nearby hear in 1.0 s, if she was stationary?
(*b*) How many *additional* waves would she hear in 1.0 s, if she was driving towards the police car at a constant speed of 34 m s^{-1}?
(*c*) What frequency would she observe as she approached the police car?
(*d*) What frequency would she hear after she had passed the police car?

22 The frequency of the sound from a warning bell at a railway crossing is 400 Hz. The speed of sound in air = 340 m s^{-1}. What frequency will be heard by the driver of a train travelling at 30.0 m s^{-1}
(*a*) while approaching the crossing?
(*b*) while moving away from the crossing?

23 The whistle of a train has a frequency of 330 Hz, but it appears to be 360 Hz to a stationary observer. What is the velocity of approach of the train to the observer? Speed of sound in air = 340 m s^{-1}.

SET 34 SOUND FROM VIBRATING STRINGS

Topic	Problems
Relation between wave speed, tension and linear density for a stretched string	1–2
Relation between frequency, length, tension and linear density for the fundamental mode of vibration of a stretched string	3–6
Overtones and harmonics for a stretched string	7–9

1 If a guitar string with a linear density of 4.0 g m^{-1} is under a tension of 58 N, at what speed will progressive waves travel along the string when it is plucked?

2 A piece of cotton stretched to a tension of 49 N between two points 36 cm apart is plucked, causing transverse waves to travel along the cotton at 420 m s^{-1}. What is the mass of the piece of cotton?

3 What is the frequency of vibration of a tuning fork which gives the same note as a wire, 1.20 m long, of mass 1.50 g and stretched by a force of 49 N?

4 If the mass of the wire between P and Q in Figure 34.1 is 3.20 g, and the wire PQ is plucked, what is the frequency of the fundamental note produced?

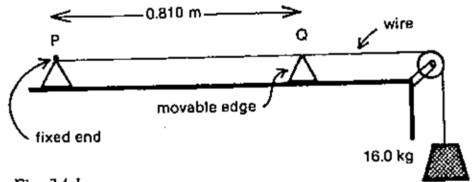


Fig. 34.1

5 Two wires equal in mass and dimensions are stretched over a sounding board by forces in the ratio 9:16. Determine the ratio of the frequencies of the notes produced by their fundamental transverse vibrations.

6 Two wires, equal in length and made of the same material, are subject to tensile forces in the ratio 1:3, the one under the lesser tension being the thinner. The thinner wire emits a fundamental note of double the frequency of the other. If the diameter of the thinner wire is 0.13 mm, what is the diameter of the thicker one?

7 A wire 50.0 cm long and having a mass per unit length of 2.00 g m^{-1} is stretched by a tensile force of 157 N and plucked. Find the frequency of
(*a*) the fundamental mode of vibration;
(*b*) the second mode of vibration, i.e. the first overtone;
(*c*) the third overtone;
(*d*) the fifth harmonic.

8 Deduce an expression for the frequency f of the nth simplest mode of vibration of a stretched wire of length L and mass m, in terms of n, L, m and T, the tension in the wire.

9 What is the frequency of the second harmonic of a string of length 60.0 cm and of mass 4.00 g if it is stretched by hanging a load of mass 16.0 kg on it?

SET 35 SOUND FROM VIBRATING AIR COLUMNS

Topic	*Problems*
Vibrating air columns closed at one end	1–6
Vibrating air columns open at both ends	7–9
Situations involving both types of vibrating air columns	10–12

Unless indicated otherwise, ignore the *end corrections* of vibrating gas columns.

1 An organ pipe 1.20 m long, which is closed at one end, is sounded on a day when the speed of sound in air is 366 m s^{-1}.
(*a*) What is the wavelength of the sound for the fundamental mode of vibration?
(*b*) Calculate the frequency of
  (*i*) the fundamental mode of vibration;
  (*ii*) the first overtone;
  (*iii*) the second overtone.
(*c*) What harmonic is
  (*i*) the first overtone?
  (*ii*) the second overtone?
  (*iii*) the sixth overtone?

2 The shortest length of a tube, closed at one end, which resonates to a tuning fork of frequency 326 Hz is 0.260 m.
(*a*) What is the wavelength of the note emitted by the fork?
(*b*) What is the speed of sound in air in this case?

3 What is the length of the air column in a long, narrow sound box, closed at one end, if its fundamental note is in resonance with a 384 Hz fork? Speed of sound in air = 332 m s^{-1}.

4 The level of water in a glass tube (diameter 1.8 cm) is adjusted until the air column above the water resounds strongly the note of a tuning fork marked 253. The length of the air column is 33.5 cm.
(*a*) Including the end correction, what is the effective length of the vibrating air column?
(*b*) What is the wavelength of the note?
(*c*) Deduce a value for the speed of sound in air.

5 A brass tube stands vertically in a tall jar of water so that its top is level with the water surface. The tube is raised vertically and a continuously vibrating tuning fork is held above the open end. A second resonance is observed when the tube is raised 0.293 m from the position of the first resonance. If the speed of sound in air is 340 m s^{-1}, find the frequency of vibration of the fork.

6 A closed organ pipe blown with air emits a note of frequency 145 Hz. The pipe is then blown with coal gas, and the note emitted is 160 Hz.
(*a*) If the speed of sound in air is 334 m s^{-1}, find the speed in coal gas.
(*b*) Find the frequency of the first overtone of the pipe when blown with air.

7 A pipe open at both ends is 0.400 m long. Take the speed of sound in air as 344 m s^{-1}.
(*a*) What is the wavelength of the sound in air for the fundamental mode of vibration?
(*b*) What is the frequency of the pipe's
  (*i*) fundamental mode of vibration?
  (*ii*) first overtone?
  (*iii*) second overtone?
  (*iv*) fifth overtone?
(*c*) What harmonic is
  (*i*) the first overtone?
  (*ii*) the second overtone?
  (*iii*) the nth overtone?

8 The notes most easily produced on a bugle have frequencies of 264, 352 and 440 Hz.
(*a*) What is the frequency of the bugle's fundamental note?
(*b*) Given that the speed of sound in air is 330 m s^{-1}, what is the length of the air column in the bugle?

9 The shortest length of an open resonance tube which gives resonance with a fork of frequency 256 Hz is 64.0 cm. For a fork of frequency 384 Hz it is 41.0 cm. Find the velocity of sound in air and the end correction for the tube (assumed to be independent of frequency).

10 Two organ pipes, one open and the other closed, are giving the same note. What is the ratio of the length of the open pipe to that of the closed pipe?

11 The third overtone of an open pipe resonates in unison with the second overtone of a closed pipe. What is the ratio of the length of the open pipe to that of the closed one?

12 An organ pipe when blown softly emits a note of frequency 192 Hz. When blown more strongly it emits a note of frequency 384 Hz. Is it a closed or an open pipe, and what is its length? Speed of sound = 340 m s^{-1}.

SET 36 INTERFERENCE EFFECTS WITH SOUND

Topic	*Problems*
Interference caused by reflection of sound at a plane surface	1–2
Interference problems based on path difference	3–4
More difficult problems on interference from two sources	5–6
Beats	7–10

Fig. 36.1

1 A student sets up at S a loudspeaker, facing a wall W and emitting a note of frequency 210 Hz. See Figure 36.1. As she walks from the wall to the speaker, she observes regions alternating between loud and soft sound. If the distance between adjacent points where the sound is loud is 0.800 m,
(*a*) what is the wavelength of the sound emitted by S?
(*b*) what is the speed of sound in air?

2 A source of sound emits continuously a tone of constant frequency. Interference between sound from the source and sound reflected by a wall 7.5 m distant from the source produces a standing wave pattern. Investigation shows that successive nodes are 0.17 m apart and that the time taken for the sound to travel from source to wall and back to source is 0.044 s. What is the frequency of the tone?

3 Two loudspeakers X and Y, joined to a signal generator, emit a sound of wavelength 0.60 m. See Figure 36.2. A girl walks from point A to point I, and observes that the note she hears alternates between loud (at points A, C, E, G and I) and very soft (at points B, D, F and H). OE is the perpendicular bisector of the line XY.

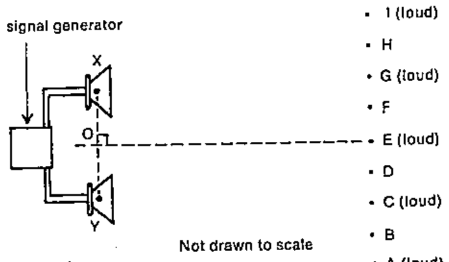


Fig. 36.2

(*a*) What is the path difference from the two speakers to the point F, i.e. (YF − XF)?
(*b*) If XA = 12.9 m, what is the distance YA?

4 The device in Figure 36.3 is a sound interferometer. The sound waves enter at A and travel to C via paths ABC and ADC. The length of the path ADC can be changed by a sliding U-tube as shown in the diagram. The source emits a sound of a single frequency. The wavelength in air of this sound is λ.

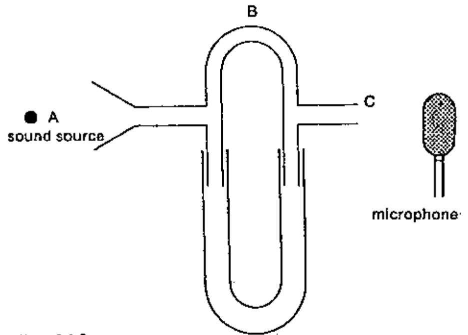


Fig. 36.3

(*a*) Initially ABC and ADC are equal. Then ADC is increased until the microphone records the first minimum in intensity. In terms of λ what is the change in the pathlength ADC?
(*b*) The frequency is fixed at a certain value and the tube is filled with oxygen gas. Successive minima are detected each time the distance ADC is increased by 0.300 m. When the experiment is repeated using hydrogen gas the corresponding change in ADC is 1.20 m. If the speed of sound in oxygen is 320 m s^{-1}, what is the speed of sound in hydrogen?

5 Refer back to Problem 3 and Figure 36.2. XY = 3.60 m and OB = 12.4 m.
(*a*) How far is B from E?
(*b*) What is the size (to the nearest degree) of angle EOH?
(*c*) What is the distance DF?

6 A teacher wishing to demonstrate the interference of sound waves sets up two loudspeakers S and T, 4.3 m apart and on top of two stepladders out on a playing field. S and T are connected to a signal generator set at 220 Hz. The speed of sound in air is 330 m s^{-1}.
(*a*) What is the wavelength of the sound waves generated?
(*b*) The teacher asks the students to move about and find how many nodal lines there are in the interference pattern. How many are there?
(*c*) He then asks them to locate and stand at any point on the *first* nodal line from the centre of the pattern on the side nearer to S, but not to come too close to the speakers. They all finish up standing in a straight line. What angle does this line make with the perpendicular bisector of the line ST?

7 An unmarked tuning fork when sounded with a fork of frequency 250.0 Hz produces six beats in 5.000 s both before and after being loaded by having elastic bands wound round the end of each prong. Find its frequency before and after being loaded.

8 The movable bridge of a monochord is placed so that three beats per second are heard when the string, which is 50.0 cm long, is sounded simultaneously with a tuning fork. When the bridge is moved so as to lengthen the string by 1.0 cm, three beats per second are again observed. What is the frequency of the fork?

9 A siren in which air is blown through fifteen holes in a rotating disc gives one beat per second with a tuning fork when the frequency of rotation of the disc is 2000 rotations per minute, and three beats per second when it is 2016 rotations per minute. Determine the frequency of the tuning fork.

10 An open organ pipe at the front of a town hall is tuned to 256 Hz at 20°C and an open organ pipe in the echo organ above the rear balcony is tuned to 512 Hz at 20°C. What will be the beat frequency caused by the fundamental of one pipe and the first overtone of the other on a night when the front of the hall is at 20°C and the rear is at 26°C? The speed of sound in a gas is proportional to the square root of the absolute temperature.

SET 37 REVISION PROBLEMS

Set	*Topic*	*Problems*
33	Transmission of sound	1–3
34	Sound from vibrating strings	4–7
35	Sound from vibrating air columns	8–11
36	Interference effects with sound	12–14

1 To test the depth of the water a ship sends out a sound which travels through the water at 1.5 km s^{-1}. If the echo is received by the ship after 20 ms, how deep is the water?

2 A cog-wheel having 25 teeth is rotated 3.00 times per second and a thin strip of metal is fixed so that it is struck by the cogs. If the speed of sound is 330 m s^{-1}, what is the wavelength of the note emitted?

3 A motorist drives at 14.0 m s^{-1} (about 50 km h^{-1}) on a still day along a road adjoining two factories. The knock-off sirens of the two factories sound simultaneously at a frequency of 500 Hz. If the speed of sound in air is 350 m s^{-1}, what is the apparent frequency he hears from the factory he is driving
(*a*) towards?
(*b*) away from?

4 A cord with a mass per unit length of 0.050 kg m^{-1} is stretched between two points 30 m apart by a tension of 250 N. The cord is then plucked at a point 1 m from one end. How long will it take for the pulse to reach the other end?

5 A monochord wire 1.00 m long has a fundamental frequency of 288 Hz when plucked. A bridge is placed underneath the wire at a point 0.60 m from one end.
(*a*) What are the fundamental frequencies generated by the two segments?
(*b*) The bridge is then removed and the tension altered so that a note of frequency 864 Hz is emitted. What is the ratio of the new to the previous tension?

6 A string stretched with a tension of 60 N produces the note Middle C. What tension must be applied to the string to produce the C one octave below Middle C?

7 A brass wire and a steel wire of the same dimensions are mounted on a sounding board and set in transverse vibration. It is found that the second harmonic of the brass wire is in unison with the first harmonic of the steel wire. Given that the densities of brass and steel are 8.4×10^3 kg m^{-3} and 7.8×10^3 kg m^{-3} respectively, find the ratio of the tension in the brass wire to that in the steel wire.

8 A vertical cylindrical glass tube contains water the level of which can be varied. A vibrating tuning fork of frequency 520 Hz is held over the open end of the tube, and the sound is greatly intensified when the water level is 15.2 cm below the open end and also when it is 47.5 cm below the open end. On repeating the experiment further values obtained for the first level are 15.0, 14.8 and 15.3 cm, and for the second level are 47.9, 47.4 and 47.7 cm. What value do you obtain for the speed of sound in air from this experiment?

9 The length of an open organ pipe is 0.50 m. In this question ignore end effect corrections, and take the speed of sound in air to be 340 m s^{-1}.
(*a*) What is the frequency of
(*i*) its fundamental?
(*ii*) the first overtone?
(*b*) If the organ pipe had been closed at one end, what would have been the frequency of
(*i*) the fundamental?
(*ii*) the first overtone?

10 An open organ pipe is 25 cm long. Find the change in frequency of its fundamental note when the air temperature changes from 15°C to 25°C, given that the speed of sound in a gas is directly proportional to the square root of the absolute temperature. Assume that the speed of sound in air at 15°C = 340 m s^{-1}.

11 A closed organ pipe of length 1.1 m is filled with a gas, and it is found to give the same note as an open pipe of length 1.3 m filled with air. Find the ratio of the speed of sound in the gas to the speed of sound in air. Ignore the end corrections of the vibrating gas columns.

12 Two loudspeakers L_1 and L_2 are placed 2.5 m apart on a plank on top of two stepladders. The speakers are emitting sound waves of wavelength 0.60 m. Figure 37.1 shows the view from above. The sound sources are in phase.

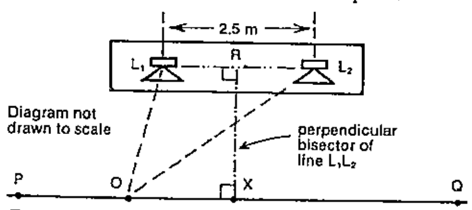


Fig. 37.1

(*a*) If at point O the intensity of the sound is a maximum rather than a minimum, which one or more of the following are possible values for the path difference, $L_2O - L_1O$?
0.15 m, 0.30 m, 0.60 m, 0.90 m, 1.2 m, 1.5 m
(*b*) Suppose the point P is on the *n*th nodal line from the centre of the interference pattern. Express the sine of the angle PRX as a function of *n*.

13 A bridge is placed under the string of a monochord at a point near the middle, and it is found on plucking the two parts of the string that 3.0 beats per second are produced. The force stretching the string is then increased by 44 per cent. Determine the new rate of beating of the two parts of the string.

14 A monochord wire is stretched tightly over two bridges, the stretching force being applied by a mass of 10.0 kg of copper hanging from the free end of the wire. The positions of the bridges are adjusted until the wire between them vibrates in unison with a 96.0 Hz tuning fork. The mass of copper is totally immersed in water, and it is then observed that the tuning fork and wire, when vibrating simultaneously, produce 5.6 beats per second. Calculate the density of the copper. Density of water = 1.00×10^3 kg m^{-3}.

PART 5
MAINLY LIGHT

SET 38 TRANSMISSION OF LIGHT

Topic	Problems
Speed of light	1-2
Shadows	3
Reflection at a plane surface, law of reflection	4-8
Inverse square law of illumination, source intensity, luminous flux, photometry	9-14

Where necessary in this set, use
speed of light in air or a vacuum,
$c = 3.00 \times 10^8$ m s^{-1}

1 What would be the minimum time required for a laser beam to travel from the surface of the Earth to the Moon and back again at a time when their centre-to-centre separation is 3.84×10^8 m? Radius of Earth $= 6.38 \times 10^6$ m and radius of Moon $= 1.74 \times 10^6$ m.

2 Albert Michelson, an American physicist, once used an arrangement somewhat like that in Figure 38.1 to measure the speed of light. P is an eight-sided prism with reflecting faces and can be rotated by a motor. Light from a lamp, after reflection at face Q, travels 1.0 km to a mirror M, back to face R and then forms a spot on the scale at 2 when the mirror is stationary. The motor is started up and its speed is gradually increased until the spot is again at 2 on the scale.

(*a*) What is the frequency of rotation of the mirror when this happens?

(*b*) If this frequency, your answer to (*a*), is denoted by N, at which one or more of the following rates of rotation would the spot of light be again at 2 on the scale?
$N/8$, $N/2$, $2N$, $9N/8$, $3N$, $7N$

3 In Melbourne at midday on 22 December the angle of elevation of the Sun above the horizontal is 75°. At that time in Melbourne how long would the shadow of a 9.7 m street light pole be?

4 In Figure 38.2 a particular ray of light from a small light source S passes through the point X *after* reflection by a small mirror MN.

(*a*) Through which one or more of the points, A, B, C, D, E and F, will this ray also pass before, during or after reflection by the mirror?

(*b*) What are the angles of incidence and reflection for this same ray (to the nearest half of a degree)?

(*c*) If the separation of adjacent grid lines in Figure 38.2 is 10 cm, what is the distance from the source S to its image produced by the mirror?

5 A man whose eye is 1.6 m above the ground is facing a plane mirror which is standing vertically with its lower edge on the ground. If the man can just see the whole of his shoes in the mirror, find the approximate height of the mirror.

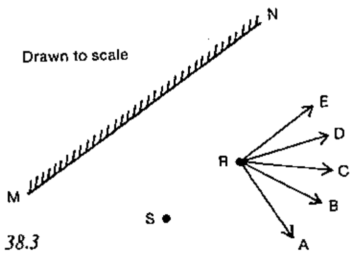


Fig. 38.3

6 In Figure 38.3 light is given out by a small light source S. Of the rays, A, B, C, D and E, which could represent part of a ray of light from S *after* it has been reflected by the plane mirror MN?

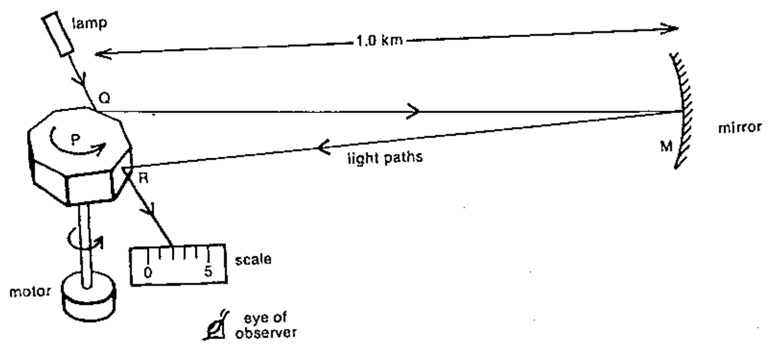


Fig. 38.1

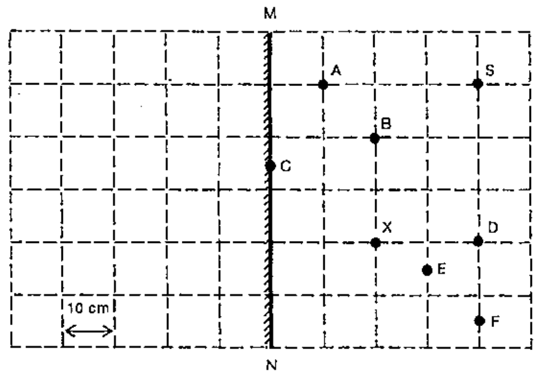


Fig. 38.2

7 In a ballistic galvanometer the deflection of the coil is displayed by means of a small mirror which rotates with the coil and to which a fixed light beam is directed. When the coil rotates through 4°, through what angle does the reflected light beam move?

8 Two plane mirrors are inclined at an angle θ. Show that a ray of light incident on one mirror in a plane normal to the line of intersection is deviated by an angle 2θ after reflection from the other mirror.

9 A light bulb of luminous intensity 70 candela (abbreviated cd) is 2.7 m directly above a 22 cm by 18 cm page of an open book on a table.

(*a*) What is the luminous flux in lumen (abbreviated lm) given out by the bulb?

(*b*) What is the illumination or illuminance in lux (abbreviated lx) on the open page?

(*c*) What is the luminous flux falling on the open page?

10 If a theatre spotlight concentrates all the light from a 120 cd source in a circle of radius 0.94 m on the back wall of the stage

(*a*) what luminous flux is falling on the wall?

(*b*) what is the illumination produced?

11 An illumination of 220 lx is appropriate for reading.

(*a*) At what height directly above an opened book should a 110 cd source be to provide the correct illumination?

(*b*) If the book is moved 1.0 m horizontally to the side so that the light source is no longer directly above it, what is the new average illumination?

(*c*) Strictly, in answering (*a*), did you take 220 lx as the average, maximum or minimum illumination on the opened pages?

12 Two identical small light bulbs are placed 56 cm apart. A screen is placed between them in such a position that the illumination on one side of it is 9.0 times the illumination on the other side. How far from the screen is the nearer light bulb?

13 A lamp is placed at a distance r from a non-reflecting screen in a dark room as in Figure 38.4. The lamp is switched on. A perfectly reflecting mirror is brought to the

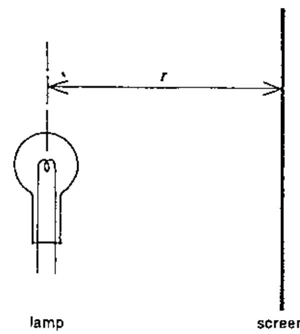


Fig. 38.4

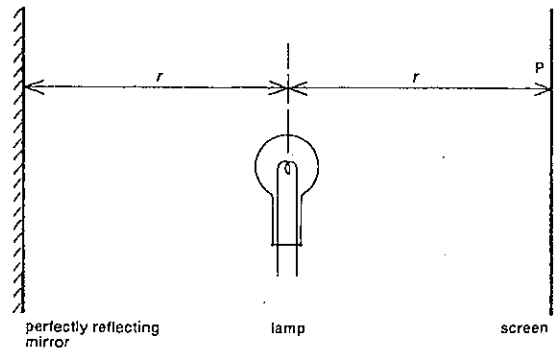


Fig. 38.5

position shown in Figure 38.5. Does the effect of the mirror increase the intensity of light in the vicinity of P by about 100, or 50, or 25, or 10 per cent?

14 If a photographic print can be made in 12 s when the printing frame is held 60 cm from a lamp, what is the correct exposure time when it is held 90 cm away?

SET 39 CURVED MIRRORS

Topic	*Problems*
Concave mirrors	1–9
Convex mirrors	10–12
More difficult problems with curved mirrors	13–15

1 A small object is placed at a distance of 60 cm from a concave mirror of focal length 20 cm.
(*a*) Is the image formed on the same side of the mirror as the object or on the opposite side?
(*b*) Determine, either by ray tracing or by calculation, the distance from the mirror to the image.
(*c*) Is the image real or virtual?
(*d*) Is it upright or inverted?
(*e*) What is its magnification?

2 Determine the size, nature and position of the image formed when an object 2.0 cm high is placed 15 cm from a concave mirror of focal length 10 cm.

3 When an object is placed 40 cm from a concave spherical mirror, the image formed is also 40 cm from the mirror.
(*a*) Is the image inverted or erect?
(*b*) Is it real or virtual?
(*c*) What is its magnification?
(*d*) What is the focal length of the mirror?
(*e*) Is the radius of curvature of the mirror about 10, 20, 40 or 80 cm?

4 While shaving, a man uses a concave mirror of focal length 36 cm.
(*a*) If a whisker is 24 cm from the mirror, will its image be real or virtual, upright or inverted, diminished or enlarged, on the same side of the mirror as the whisker or on the opposite side?
(*b*) How far will the image of this whisker be from the mirror?
(*c*) What will be its magnification?

5 In Figure 39.1, M represents a concave mirror with its principal focus at F. Its focal length is *f*. The points A, B, C, D and E all lie on the principal axis.
At which of the points, A, B, C, D, E and F, should an object be placed, if
(*a*) an enlarged image is to be cast on a screen?
(*b*) an image identical in size to the object is to be produced?
(*c*) an enlarged upright image is to be seen?

6 Figure 39.2 shows a ray of light XY meeting a concave mirror at the point Y. F and C represent the mirror's principal focus and centre of curvature respectively. Which of the paths, YP, YQ, YR, YS, YT and YU, best represent the reflected ray?

7 A concave spherical mirror has a focal length 8.0 cm. An object is placed in such a position that there is formed
(*a*) a real image of half the height of the object;
(*b*) a virtual image of twice the height of the object.
Find the distances of the object and image from the mirror in each case.

8 A dentist uses a concave mirror for examining patients' teeth. Design a mirror suitable for this purpose, assuming that the image is to be three times the size of the object, erect, and that the mirror is not to be held nearer to the tooth than 1.0 cm.

9 The Sun has a diameter of 1.38×10^6 km and it is at a distance of 1.49×10^8 km. An image of the Sun is formed by a concave mirror of focal length 15.0 m. What is the position of the image and what is its diameter?

10 A match is placed upright 60 cm from a convex mirror with a focal length of 20 cm.
(*a*) Determine, either by ray tracing or by calculation, the distance from the image to the mirror.

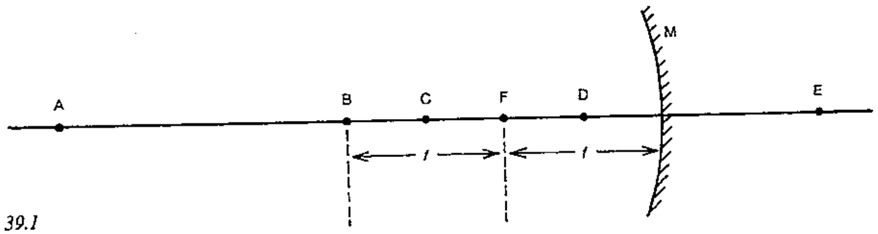


Fig. 39.1

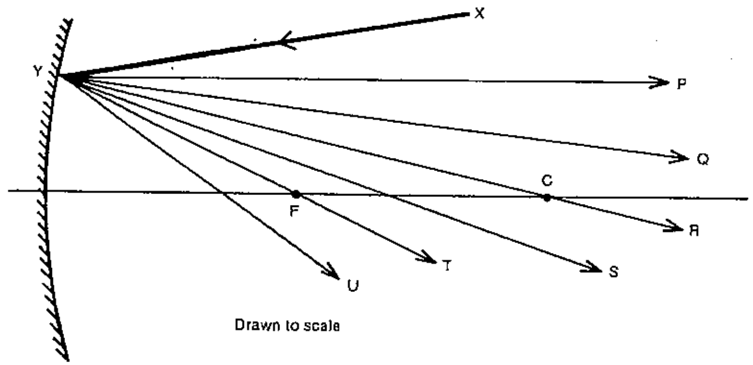


Fig. 39.2

(*b*) Is the image formed on the same side of the mirror as the object or on the opposite side?
(*c*) Is the image real or virtual?
(*d*) Is it upright or inverted?
(*e*) What is its magnification?

11 Determine the size, nature and position of the image formed when an object 3.0 cm high is placed 20 cm from a convex mirror of radius of curvature 60 cm.

12 The radius of curvature of a convex mirror is 25 cm. An object 5.0 cm high placed in front of the mirror yields an image 1.0 cm high. By ray tracing or otherwise, determine
(*a*) whether the image is real or virtual, upright or inverted;
(*b*) how far the object is from the mirror;
(*c*) the position of the image.

13 A convex mirror and a plane mirror are placed facing each other and 28 cm apart. Midway between them and on the principal axis of the convex mirror a small luminous object is placed. If the distance between the two images observed on looking in the plane mirror is 24 cm, calculate the radius of curvature of the convex mirror.

14 A real image is formed 40.0 cm from a spherical mirror, the image being twice the size of the object. What kind of a mirror is it, and what is its radius of curvature?

15 A pin is set up 30 cm in front of a convex mirror. It is found that if a plane mirror is placed between the pin and the convex mirror 20 cm from the pin (so that it covers only half the convex mirror) the images of the pin in both convex and plane mirrors coincide. Find the focal length of the convex mirror.

SET 40 REFRACTION AT A PLANE INTERFACE

Topic	*Problems*
Partial reflection and transmission at an interface	1
Law of refraction, relative refractive index	2–3
(Absolute) refractive index	4–9
Apparent thickness of a transparent medium	10–11
Refraction and dispersion by prisms	12–17
Critical angle, total internal reflection	18–25

Where necessary in this set, use
absolute refractive index of air = 1.00

1 A ray of light is directed down on to the surface of the water in a fish tank, the ray making an angle of 30° with the surface. The light that is transmitted into the water makes an angle of 49° with the surface.
(*a*) What is the angle of incidence at the air–water interface?
(*b*) What would be the angle of reflection for the partially reflected light?
(*c*) What is the angle of refraction for the light transmitted into the water?
(*d*) What is the angle of deviation for this transmitted light?

2 A ray of light in air with an angle of incidence of 35.0° enters a block of polystyrene. The angle of refraction is 21.0°.
(*a*) If the direction of the incident ray is altered, what is the ratio of the sine of the new angle of incidence to the sine of the new angle of refraction?
(*b*) What is the relative refractive index for light passing from air into polystyrene?

(*c*) What would be the relative refractive index for light passing out of the polystyrene into air?

3 A ray of light enters the side wall of a fish tank. The angle of refraction is 47° as the light passes from the glass into the water. If the relative refractive index for light passing from glass to water is 0.88, what is the angle of incidence?

4 Figure 40.1 shows a ray of light travelling from medium 1 (absolute refractive index 1.20) to medium 2 (absolute refractive index 1.50).

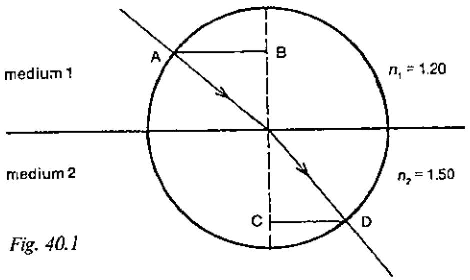


Fig. 40.1

(*a*) What is the relative refractive index for light travelling from medium 1 to medium 2?
(*b*) If CD = 7.20 cm, what is the length of the semichord AB?
(*c*) What is the angle of refraction?
(*d*) Determine the angle between the reflected and refracted rays, given that the angle of incidence in medium 1 is 50.0°.

5 A ray of light in water meets a flat-sided block of perspex with an angle of incidence of 55.0°. The angle of refraction in the perspex is 47.0°. If the absolute refractive index (more commonly known as the *refractive index*) of water is 1.33, determine the refractive index of perspex.

6 A ray of light is incident on a plane glass surface at an angle of 30.0°. Given that the refractive index of the glass is 1.50, calculate the angle of refraction and the angle of deviation.

7 A ray of light makes an angle of incidence of 42.8° at the boundary of glycerine and flint glass. If the absolute indices of these materials are 1.47 and 1.72 respectively, through what angle is the ray deviated?

8 Calculate the lateral displacement of a ray of light incident at an angle of 30° on a parallel-sided block of glass of thickness 40 mm and refractive index 1.6.

9 A narrow parallel beam of light strikes a glass block so that it makes an angle of 60.0° with the normal to the glass surface. Part of the beam is reflected and part is refracted, the angle between the reflected and refracted portions being 90.0°. Find, either by scale drawing or by calculation, the refractive index of the glass.

10 A slab of glass 8.0 cm thick is placed upon a printed page. If the refractive index of the glass is 1.5, how far from the surface would the letters appear to be when viewed from directly above?

11 A coin is viewed directly from above through liquid 8.0 cm deep and appears to be brought 3.0 cm nearer the surface. Calculate the refractive index of the liquid.

12 A ray of white light travelling through the air strikes the surface of a sheet of crown glass at an angle of incidence of 38.0°. The refractive indices for crown glass for red and violet light are 1.513 and 1.532 respectively.
(*a*) Which is refracted more, the red or the violet light?
(*b*) What is the angle of refraction for the red light? Give your answer to the nearest tenth of a degree.
(*c*) What is the angle of refraction for the violet light?
(*d*) What is the angle of dispersion, that is, the angle between the directions in which the red and violet light emerge into the water?

13 Three rays of yellow, orange and blue light meet a liquid-to-air interface at an angle of incidence of 10°.
(*a*) Of the yellow, orange and blue light, which will have the largest angle of refraction?
(*b*) Will all these angles of refraction be greater than or less than 10°?
(*c*) Which colour light will be deviated the most?
(*d*) Which colour will be deviated the least?
(*e*) The refractive indices of the liquid for yellow, orange and blue light, n_y, n_o and n_b respectively, differ slightly. Arrange n_y, n_o and n_b in order of size, with the largest first.

14 A ray of light PQ meets the face AC of a triangular glass prism with an angle of incidence of 25° as shown in Figure 40.2. The refractive index of the glass is 1.53.

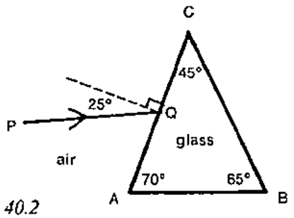


Fig. 40.2

(*a*) Which one of the three angles A, B or C will affect the direction in which the light travels after emerging from the surface BC? This angle is referred to as the *refracting angle of the prism*.

(*b*) For the passage of the light through the prism, what is
(*i*) the angle of refraction as it passes through the surface AC?
(*ii*) the angle of incidence as it passes through the surface BC?
(*iii*) the angle of refraction as it passes through the surface BC?
(*iv*) the angle of deviation produced by the prism?

15 A ray of light strikes the surface of a 60° glass prism at an angle of incidence of 40°. Given that the refractive index of the glass is 1.5, calculate the angular deviation caused by the prism.

16 Calculate the angle of minimum deviation of the prism described in Problem 15 above.

17 A prism of small refracting angle θ is made of glass for which the refractive index is n, and a ray of light falls normally on one of the faces that include the angle θ. Show that the deviation experienced by the ray is $(n - 1)\theta$.

18 The refractive index of water is 1.33. Determine the greatest angle of incidence for which a ray of light, coming from the bottom of the pool and striking the water surface, can emerge into the air above.

19 Water (refractive index 1.33) is placed upon a block of glass of refractive index 1.50. What is the critical angle for light passing from the glass to the water?

20 The critical angle for light travelling from a particular type of glass to air is 42.0°. What is the refractive index of this glass?

21 A narrow beam of white light falls normally on the surface XZ of a crown glass prism, as in Figure 40.3. It subsequently makes an angle of incidence of 41°15′ at the face XY of the prism, this being the critical angle for crown glass for yellow light.

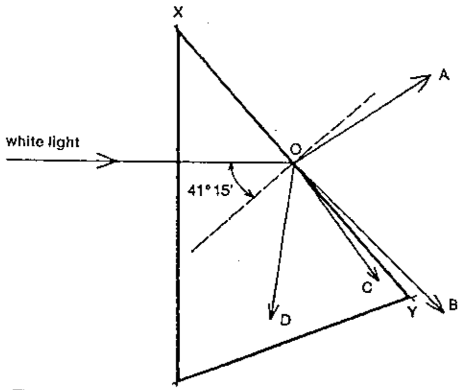


Fig. 40.3

The red and blue components of the white light travel along one or more of the paths OA, OB, OC, OD.
(*a*) Along which of these paths will the red light travel?
(*b*) Along which of the paths will the blue light travel?
(*c*) Find the refractive index of the glass for yellow light.

22 A slab of glass WXYZ has a refractive index of 1.50 for yellow light. See Figure 40.4. A ray of light AO is a mixture of red and violet light, and is incident on the boundary between the top of the glass and the air at an angle of incidence i, which is the critical angle for glass–air for yellow light.

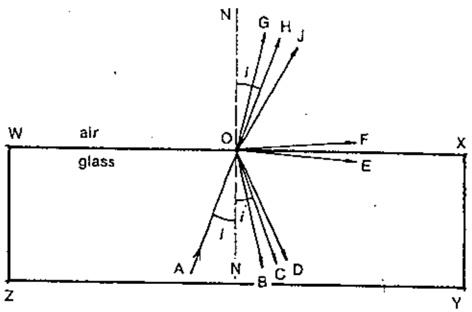


Fig. 40.4

In answering these questions, use the letters B, C, D, E, X, F, J, G and H to denote the directions of the rays which emanate from O. Note that OC also makes an angle i with the normal to the surface.

(*a*) Along what path or paths will the red light travel from O?

(*b*) Along what path or paths will the violet light travel from O?

(*c*) The space above the glass is now filled with a clear oil whose refractive index for yellow light is 1.42. Along what path or paths would you expect the red light to travel?

(*d*) With this liquid on top of the glass, for what value of i would yellow light travel along the direction OX?

23 With the improved communication provided by fibre optics, light beams carrying stored information are passed along glass fibres by being totally internally reflected as shown in Figure 40.5. Suppose the refractive index of the glass used is 1.5.

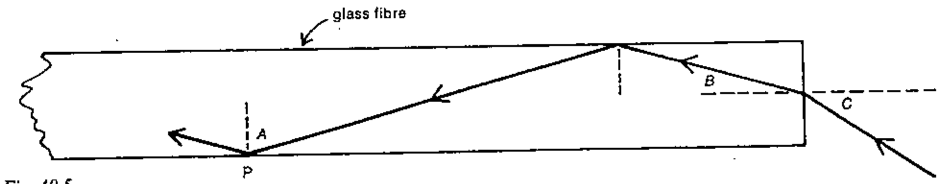


Fig. 40.5

(*a*) For the light beam not to pass out of the fibre at P,
  (*i*) what is the minimum value of the angle of incidence *A*?
  (*ii*) what is the maximum value of angle *B*?
  (*iii*) is there a maximum value for angle *C*?

(*b*) Figure 40.5 shows a straight fibre. If the fibre was bent into a smooth curve, would it be more likely that light might escape from the fibre,
  (*i*) through the inside or outside wall on the curve?
  (*ii*) if the fibre was thicker or thinner?
  (*iii*) if the curve in the fibre was sharper or less sharp?

24 Rays of light are emitted upwards in all directions from a point at the bottom of a bowl containing liquid to a depth of 7.5 cm, the refractive index of the liquid being 1.25. Find the radius of the circle lying on the surface of the liquid, with its centre vertically above the point, outside which all rays will be totally reflected.

25 A ray of light is incident at an angle of 60° on a face of an isosceles prism of refracting angle 75° and refractive index 1.65. Is it refracted or totally reflected at the next face?

SET 41 LENSES

Topic	*Problems*
Situations involving a single convex lens	1–12
Situations involving a single concave lens	13–15
Power of a lens	16–17
Lensmaker's formula	18–20
Chromatic aberration	21
More difficult problems involving single lenses	22–27

This set covers situations involving only *one* lens. More complex optical systems are considered in Set 42.

Where necessary in this set, use
refractive index of air = 1.00

1 A match is placed upright and directly opposite one face of a convex lens of focal length 20 cm. They are 50 cm apart.

(*a*) Is the image formed on the same side of the lens as the match or on the opposite side?

(*b*) What is the nature of the image, or, more specifically,
  (*i*) is it upright or inverted (relative to the object)?
  (*ii*) is it diminished, enlarged or the same size as the object?
  (*iii*) is it real or virtual?

2 Draw rough sketches to determine the nature of the images formed by each of the following convex lenses:

(*a*) one of focal length 10 cm situated 15 cm from an object;
(*b*) one of focal length 25 cm situated 75 cm from an object;
(*c*) one of focal length 30 cm placed 60 cm from an object.

3 A lighted torch bulb is placed 30 cm from a convex lens of focal length 10 cm. By means of a scale drawing on graph paper or by calculation, answer the following questions.

(*a*) What is the nature of the image of the torch bulb?
(*b*) How far from the lens is an image formed?
(*c*) Is the image formed on the same side of the lens as the object or on the opposite side?
(*d*) What is the linear magnification of the image?

4 In a slide projector an illuminated colour slide is 26.0 cm behind a convex lens of focal length 25.0 cm.

(*a*) Is the image real or virtual?
(*b*) Is it upright or inverted relative to the picture on the slide in the projector?
(*c*) If the image is well focused on the screen, how far is the screen from the projector lens?
(*d*) If the colour slide is 35.0 mm high, how high is the illuminated part of the screen?

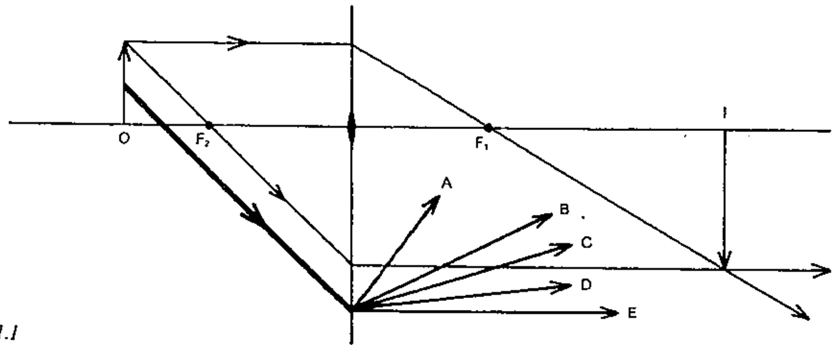


Fig. 41.1

5 Figure 41.1 shows two standard rays traced correctly to show the formation of the image I of an object O. There is also an incomplete third ray drawn with a heavier line. Which of A, B, C, D or E is the correct continuation of the incomplete ray?

6 To photograph the detail on a leaf a student uses extension tubes to increase the distance between the camera lens and the film at the back of his camera. The lens has a focal length of 40 mm, and is 80 mm from the leaf in order to produce a focused image on the film.

(*a*) How far is the film from the lens?
(*b*) What is the magnification of the image?

7 Figure 41.2 shows five rays of light, which come from the top T of a luminous source ST and pass through a thin convex lens. Which one or more of the rays A, B, C, D and E are *incorrect*?

8 A convex lens of focal length 15.0 cm is used to form an image 10.0 times the height of an object on a screen. Find the distances from the screen of

(*a*) the object;
(*b*) the image.

9 At night a girl uses a convex lens of focal length 20 cm to cast a well-focused image of a full moon on a white card. The Moon has a diameter of 3.5×10^6 m and is 3.8×10^8 m from the Earth.

(*a*) How far is the card from the lens?
(*b*) What is the diameter of the image of the Moon?
(*c*) What is the magnification of this image? An order of magnitude answer will do.

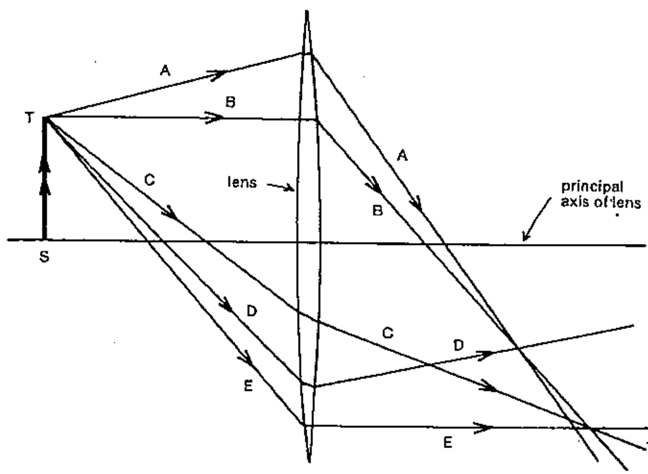


Fig. 41.2

10 A pin AB of length 2.0 cm is perpendicular to the principal axis of a convex lens of focal length 20 cm with B on the axis and distant 40.0 cm from the centre of the lens.
(*a*) Determine the height of the image.
The pin is now rotated through a right angle about B, so that AB is on the axis with A distant 38.0 cm from the centre of the lens.
(*b*) What is the length of its new image?

11 In an experiment with a lens of unknown focal length the following readings were obtained for D_i, the distance from the image to the lens, and M, the magnification.

D_i (cm)	6.0	8.0	10.0	12.0	14.0
M	0.2	0.6	1.0	1.4	1.8

Draw a graph of M against D_i, and use the graph to deduce the focal length of the lens.

12 A student holds a lens near the laboratory wall and moves it back and forth until she sees on the wall a clear image of the window opposite. The lens is then 10 cm from the wall.
(*a*) Is the focal length of the lens approximately 5, 10, 20 or 40 cm?
(*b*) Strictly, is the focal length slightly greater than or less than your answer to (*a*)?

13 An illuminated ballpoint pen is held upright 15 cm from a concave lens of focal length 10 cm.
(*a*) Determine, either by ray tracing or calculation, the distance of the image from the lens.
(*b*) Is the image formed on the same side of the lens as the pen or on the opposite side?
(*c*) Is the image of the pen real or virtual?
(*d*) Is it upright or inverted?
(*e*) What is its magnification?

14 A person holds a pair of concave spectacle lenses 24 cm from the newspaper and notices that the print appears to be one-third its real size. (The questions below refer to the images formed by the spectacle lenses and not that formed by the combination of the lenses and the observer's eyes.)
(*a*) What is the nature of the image formed by each lens, i.e. is it real or virtual, and upright or inverted?
(*b*) Is the image of the newsprint formed on the same side of the lenses as the paper or on the opposite side?
(*c*) What is the focal length of the lenses?
(*d*) How far from each lens is its image formed?

15 Select from the letters below those which correspond to correct statements about images formed by *concave* lenses.
A They are real.
B They are virtual.
C They are closer to the lens than the object.
D They are further from the lens than the object.
E Their magnification is greater than one.
F Their magnification is less than one.
G They are formed on the opposite side of the lens to the object.
H They are formed on the same side of the lens as the object.
I They are upright (relative to the object).
J They are inverted (relative to the object).
K When the object is a very long distance from the lens, the image is formed in the principal focal plane.

16 The smaller the focal length of a lens, the more powerful it is said to be. The *power* of a lens is given as the reciprocal of its focal length in metre. The SI unit of power of a lens is metre^{-1} (formerly known as a dioptre). The power of a convex lens is expressed as a positive value, while that of a concave lens is given as negative. What is the power of
(*a*) a convex spectacle lens of focal length 0.25 m?
(*b*) a concave lens of focal length 20 cm?

17 Suppose you are given four lenses with powers of $-25\ \text{m}^{-1}$, $+20\ \text{m}^{-1}$, $-2.5\ \text{m}^{-1}$ and $+2.0\ \text{m}^{-1}$.
(*a*) Which would be the most suitable as a magnifying glass for a philatelist to examine stamps?
(*b*) What is the focal length of the less powerful concave lens?

18 A biconvex glass lens has surfaces with radii of curvature of 20 and 25 cm. If the refractive index of the glass is 1.53, what is the focal length of the lens when used normally, i.e. surrounded by air?

19 What would be the focal length of the lens described in Problem 18 above, if the lens were immersed in water? Refractive index of water = 1.33.

20 A plano-convex lens is made of glass for which the refractive index is n, and the radius of the spherical surface is R. By considering the refraction of a ray which falls normally on the plane surface and which is parallel and close to the axis of the lens, derive an expression for the focal length of the lens in terms of n and R.

21 The refractive index of heavy crown glass for red light is 1.606 and for violet light is 1.624. It follows that the focal length of a lens of heavy crown glass will vary with the type of light used.
(*a*) Will red or violet light give rise to a larger focal length for the lens?
(*b*) What is the ratio of the focal length of the lens for red light to its focal length for violet light?

22 A lamp and a screen are 2.00 m apart. What is the focal length of a lens that could be used to form an image of the lamp on the screen so that the image is 7.00 times as high as the object?

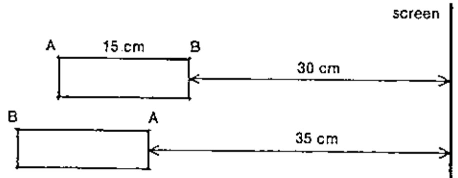


Fig. 41.3

23 AB is a cardboard tube 15 cm long. It is closed at each end with pieces of flat glass and somewhere inside is a single converging lens fitted across the tube. When A faces a bright window at one end of a very long room, the image of the window is well focused on a screen on the opposite wall when B is 30 cm from the screen. If the tube is turned end for end (as in Fig. 41.3), the image of the window is again in focus on the screen when A is 35 cm from the screen. Find the focal length of the lens and its distance from A in the tube.

24 A convex lens forms an image of length 8.96 mm on a screen. Keeping the screen and the object fixed, the lens is moved through a distance of 15.4 cm and a sharp image of the object is again observed on the screen. If the length of the second image is 3.54 cm, calculate the focal length of the lens.

25 A lens forms a thrice magnified image on a screen. The screen is moved 20 cm and the object is then moved until the image is again in focus. The magnification is now found to be 2.5. Determine the focal length of the lens.

26 A bright point is situated on the wall of a room 2.7 m wide. A convex lens of focal length 30 cm and 5.0 cm in diameter is placed 90 cm from the wall. What is the diameter of the circle of light thrown by the lens on the opposite wall?

27 Show that the least distance between a small object and a real image of it formed by a thin convex lens is four times the focal length of the lens.

SET 42 SIMPLE OPTICAL INSTRUMENTS

Topic	*Problems*
Combination of thin lenses (not in contact), virtual objects	1-4
Telephoto lens	5-6
Effective focal length of two thin lenses in contact; the eye and spectacles	7-13
Combinations of mirrors and lenses	14-16
Refracting astronomical telescopes using two convex lenses	17-19
Terrestrial telescopes, prism binoculars	20-21
Galilean telescopes, opera glasses	22-23
Newtonian reflecting telescopes	24-26
Compound microscopes	27-29

1 An illuminated model of Miss Piggy is set up 30 cm in front of a convex lens X, which is in turn 39 cm in front of another convex lens Y. The three are in line. X and Y have focal lengths of 10 cm and 15 cm respectively.
(*a*) Where is the image of Miss Piggy formed by *X alone*?
(*b*) What is the magnification of the image formed by X alone?
(*c*) Where is the image of Miss Piggy formed by the light passing through *both* lenses?
(*d*) What is the ratio of the height of this final image to that of the model?
(*e*) Is her final image right way up or upside down?

2 A lighted bulb is arranged upright and in line with two convex lenses P and Q. P is situated between the bulb and Q, and is 25 cm from the bulb and 15 cm from Q. P has a focal length of 20 cm and Q one of 30 cm.
(*a*) What is the position of the virtual object for lens Q, i.e. the image formed if Q were not present?
(*b*) Where is the image formed by both P and Q?
(*c*) Is it real or virtual, upright or inverted?

3 A convex lens of focal length 30 cm is placed 20 cm away from a concave lens of focal length 5.0 cm. An object is placed 6.0 m distant from the convex lens (which is nearer to it) and on the common axis of the lens system. Determine the position and magnification of the image formed.

4 Light is converging towards a point 12 cm behind a concave lens of focal length 15 cm. Find the position of the point to which the light will actually come to a focus.

5 A telephoto lens system in a particular camera consists of a convex lens of focal length 20 cm situated 15 cm in front of a concave lens of focal length 10 cm. In order to correctly focus and photograph a kangaroo about 100 m away in the bush, how far should the concave lens be from the film at the back of the camera?

6 Investigate the effect of the telephoto lens system described in Problem 5 above by determining the height of the kangaroo's image on the film if the kangaroo itself is 1.2 m tall, using

(*a*) the *convex lens* (of focal length 20 cm) *alone*;
(*b*) the telephoto lens system.

7 Two thin lenses are in contact. What would be the approximate effective focal length of the pair, if they were

(*a*) two convex lenses with focal lengths of 60 cm and 20 cm?
(*b*) two concave lenses with focal lengths of 60 cm and 20 cm?
(*c*) a convex lens with focal length of 60 cm and a concave lens with focal length of 20 cm? Indicate in this last case whether the combination would act as a convex or concave lens.

8 A thin convex lens of focal length 12 cm is attached to another thin lens X. When placed 60 cm from a fluorescent tube the combination forms a real image of the tube on a white card 150 cm from the tube.

(*a*) What is the focal length of the combination?
(*b*) What is the focal length of lens X?
(*c*) Is lens X convex or concave?

9 The lens system of the human eye can change its focal length so that objects at different distances can be seen clearly. The image formed by the lens system of the eye falls on the "screen" or retina at the back of the eye. See Figure 42.1. A normal eye can clearly see objects as far away as infinity and as close as 25 cm. The distance between the lens system and the retina is 3.0 cm.

For the lens system of a normal eye determine
(*a*) the maximum value, and
(*b*) the minimum value
of its focal length.

10 A short-sighted person cannot see objects distinctly at a distance greater than 50 cm.

(*a*) Does he need spectacles with convex or concave lenses in order to see objects a long way off?
(*b*) Calculate the focal length of the lenses he should use. The spectacle lens and the lens system of the eye may be treated as thin lenses in contact.

11 A person who is long-sighted wears glasses containing convex lenses of focal length 30 cm, and with them his least distance of distinct vision is 25 cm. What is the shortest distance at which he can read without using his glasses?

12 A person can focus objects only when they lie between 75 cm and 1.8 m from his eyes. What spectacles should he use

(*a*) to see distant objects?
(*b*) to reduce his least distance of distinct vision to 30 cm?

13 A certain person who is short-sighted has spectacle lenses with a power of $-1.3\ \mathrm{m}^{-1}$. What is the maximum distance beyond which he cannot focus objects clearly without his glasses?

14 A ballpoint pen was moved up and down above a thin convex lens sitting on a plane mirror until parallax showed that its real image was the same distance, 21 cm, above the lens as the pen itself. See Figure 42.2. What is the focal length of the lens?

15 A beam of light diverges from a point P on the axis of a convex lens, and after passing through the lens is reflected from the surface of a convex mirror. The reflected beam is brought to a focus by the lens at a point coinciding with P. Find the radius of curvature of the mirror, given that the distance of the lens from the mirror is 10 cm, the distance of P from the mirror is 30 cm, and the focal length of the lens is 12 cm.

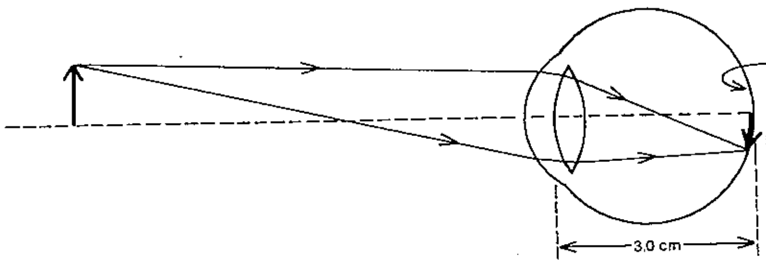


Fig. 42.1

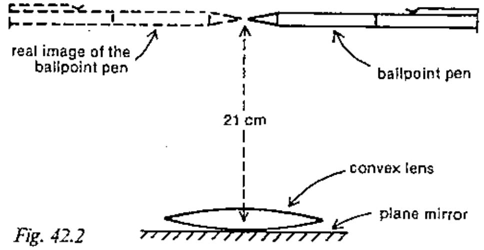


Fig. 42.2

16 A convex lens of focal length 20 cm is placed 30 cm from a plane mirror and an object is placed on the principal axis 10 cm from the mirror, which is also perpendicular to the axis. Where will the two images of the object be formed?

17 A boy views the Moon through an astronomical telescope consisting of two convex lenses arranged coaxially. The focal length of the objective lens is 1.00 m, while that of the eyepiece is 10 cm. The lenses are at a separation which places the final image at infinity.

(*a*) How far from the objective is the image of the Moon formed by the objective only?
(*b*) How far is the eyepiece from this image formed by the objective only?
(*c*) What is the approximate length of the telescope?
(*d*) What is the angular magnification?

18 A refracting astronomical telescope consists of two thin convex lenses placed coaxially 91 cm apart and has a magnifying power of 12 in normal adjustment, i.e. with the final image at infinity. What is the focal length of

(*a*) the objective?
(*b*) the eyepiece?

19 Suppose the boy referred to in Problem 17 above adjusts his telescope to produce the final image *at his near point*, 25 cm from his eye, which is directly behind the eyepiece.

(*a*) How far is the eyepiece from the image formed by the objective only?
(*b*) How far is the objective from the image formed by the objective alone?
(*c*) What is the new angular magnification?

20 A student constructs a model refracting telescope from a long cardboard tube and two convex lenses of focal length 100 cm and 20 cm. He observes the inverted image of a building 50 m high and distant 1.0 km with the telescope in normal adjustment, i.e. with the final image at infinity.

(*a*) What is the angular magnification or magnifying power of the telescope?

He decides to convert the instrument into a *terrestrial* telescope by righting the image of the building using a convex lens of focal length 8.0 cm between the objective and the eyepiece, yet he wishes to keep the length of the instrument as short as possible.

(*b*) What is the minimum distance between the image formed by the objective and that formed by the erecting lens?
(*c*) What is the total length of the new instrument?
(*d*) What is the angular magnification of the terrestrial telescope?

21 The awkward length of a terrestrial telescope or spyglass is overcome in prismatic binoculars by means of two right-angled isosceles glass prisms, arranged both to fold the long light path between the objective and the eyepiece and also to produce an upright image without the insertion of an erecting lens. Consider a pair of binoculars with objective and eyepiece systems equivalent to convex lenses of focal length 2.00 m and 10 cm respectively.

(*a*) What is the magnifying power of the binoculars in normal adjustment?
(*b*) What would be the approximate total length of an equivalent terrestrial telescope using the objective and eyepiece from the binoculars and an erecting lens of focal length 10 cm?
(*c*) In order to bring the final image from infinity to the observer's near point, are the eyepieces moved towards or away from the objective?

22 In a particular Galilean telescope the objective is a convex lens of focal length 84 cm and the eyepiece or ocular is a concave lens of focal length 14 cm.

(*a*) At normal adjustment what is the angular magnification of the telescope?
(*b*) What is the approximate length of the telescope?
(*c*) What would have been the length of the telescope if the concave eyepiece lens had been replaced by a convex one of the same focal length?

23 A small set of opera glasses, which give a magnifying power of 2.5, consist of a convex objective lens of focal length 6.0 cm and a concave eyepiece lens.

(*a*) What is the focal length of the eyepiece?
(*b*) At normal adjustment what is the approximate distance between the objective and eyepiece?
(*c*) Apart from compactness, what other advantage have the opera glasses over an equivalent telescope using a *convex* eyepiece lens?

24 A Newtonian telescope consists of a concave mirror of focal length 100.0 cm, a secondary plane mirror and a convex eyepiece lens of focal length 2.5 cm as shown in Figure 42.3.

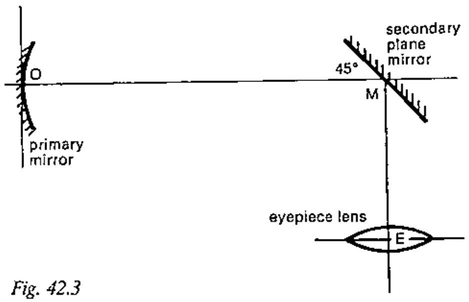


Fig. 42.3

If the telescope is focused on the Moon so that the final image is formed at infinity,
(*a*) what is the magnifying power of the telescope?
(*b*) what is the distance from the primary mirror to the eyepiece *along the optical axis* (OME)?

25 The Moon has a diameter of 3.5×10^6 m and is 3.8×10^8 m from the Earth.
(*a*) What is the *angular diameter* of the Moon as seen from the Earth, i.e. the angle subtended at the Earth by the diameter of the Moon? Give your answer in radians and in degrees.
Suppose the Moon is being observed by the Hale reflecting telescope at Mt Palomar in California, using an eyepiece of focal length 5.6 cm. The focal length of the primary mirror is 16.8 m. At normal adjustment,
(*b*) what would be the diameter of the real image of the Moon produced by the primary reflector?
(*c*) what would be the angular magnification provided by the telescope?
(*d*) what would be the angular diameter in radians of the final image of the Moon seen through the eyepiece?

26 The Kitt Peak reflecting telescope in Arizona has a primary mirror of focal length 5.6 m.
(*a*) If it was used to photograph Mars when it was 8.0×10^{10} m from the Earth, how far from the primary reflector would the film be placed?
Suppose the film is removed and the image is viewed by means of an eyepiece of focal length 3.5 cm. The astronomer examines the image of Mars at his near point, 25 cm from his eye, and finds the image has a diameter of 3.9 mm.
(*b*) What do these measurements enable him to deduce for the diameter of Mars?

27 The objective of a compound microscope has a focal length of 50 mm and is 62.5 mm above a fly's wing that is being examined by a biology student.
(*a*) How far above the objective is the image *formed by the objective alone*?
(*b*) What is the magnification of this intermediate image?
277 mm above the objective is a convex eyepiece of focal length 30 mm and through which a biology student observes the image formed by the objective.
(*c*) How far from the eyepiece is the final virtual image?
(*d*) By what factor has the *eyepiece alone* increased any linear dimension of the image?
(*e*) What is the overall magnification of the microscope?

28 A compound microscope is made of two convex lenses placed 15 cm apart, the focal lengths of the objective and eyepiece being 2.5 cm and 5.0 cm respectively. If the final image is to be at the least distance of distinct vision, i.e. 25 cm from the eyepiece,
(*a*) how far from the objective must the object be placed?
(*b*) what is the magnifying power of the microscope?

29 A microscope is focused so that the image coincides with the object at a distance of 28.0 cm from the eye. If the distance of the object from the objective is 4.00 cm and the magnification produced is 14.0, find the focal lengths of the two lenses.

SET 43 SIMPLE WAVE PROPERTIES OF LIGHT AND OTHER FORMS OF ELECTROMAGNETIC RADIATION

Topic	*Problems*
Relation between speed, wavelength, frequency and period	1–2
Order of frequency or wavelength in the electromagnetic spectrum	3–5
Effect of refractive index on speed, frequency, wavelength and period	6–8
Doppler effect (other problems in Sets 30 and 33)	9–11
Diffraction as a phenomenon (diffraction patterns are treated in Set 44)	12
Comparing wave and Newtonian particle models for light	13–14
Polarisation (see Set 59)	

1 Sydney FM radio station 2JJJ has a listed frequency of 105.7 MHz (megahertz) and wavelength of 2.8 m. What does this indicate for the speed of radio waves in air?

2 Red light from a neon light has a wavelength of 640 nm in air. If the speed of light in air is 3.00×10^8 m s^{-1}, what is
(*a*) the frequency, and
(*b*) the period
of this red light in air?

3 Arrange the following colours of the visible spectrum in order of increasing frequency, commencing with the one having the smallest frequency:
green, orange, blue, red, violet, yellow.

4 The various forms of electromagnetic radiation are referred to as infrared light, radio waves, x-rays, visible light, gamma rays, ultraviolet light and microwaves.
Arrange them in order of increasing wavelength, starting with the one with the smallest wavelength.

5 Of the various forms of electromagnetic radiation referred to in Problem 4 above, which type or types would
(*a*) have a wavelength of 500 nm?
(*b*) have a wavelength of 1 cm?
(*c*) have a frequency of 10^{18} Hz?
(*d*) have a frequency of 10^6 Hz?
(*e*) travel in air at 3.0×10^8 m s^{-1}?
(*f*) be produced by accelerating particles carrying a charge?

6 One type of yellow light from a mercury street lamp has a wavelength of 577 nm in air. The cover of the lamp is made of a glass which has a refractive index of 1.62 for yellow light. The speed of all electromagnetic radiation in air is 3.00×10^8 m s^{-1}.
(*a*) When this yellow light has left the lamp and is in the air, what is its frequency?
(*b*) While the yellow light is travelling through the glass cover, what is its
  (*i*) frequency?
  (*ii*) wavelength?
  (*iii*) speed?

7 In this problem assume you are able to pick up Geelong radio station 3GL, which transmits radio waves of frequency 1.35 MHz (megahertz). The speed of radio waves in air is 3.00×10^8 m s^{-1}.
(*a*) How long does it take 3GL to send out one complete cycle of a radio wave?
(*b*) What is the wavelength of the radio waves transmitted by 3GL as they travel to your home?
(*c*) With what speed do these waves travel through the glass of a window into your home, if the glass has a refractive index of 1.5?

8 The three angles θ_1, θ_2 and θ_3 in Figure 43.1 were measured for rays of yellow light from a sodium vapour lamp. Boundary 1 is parallel to boundary 2. The measurements for one such ray are shown in Table 43.1.

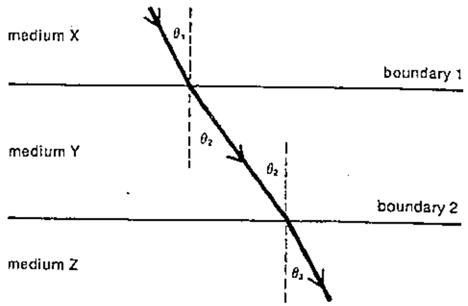


Fig. 43.1

Table 43.1

θ_1 (deg)	θ_2 (deg)	θ_3 (deg)
30.0	48.5	34.0

(*a*) In which of the three media, X, Y and Z, would the yellow light have the lowest
  (*i*) frequency?
  (*ii*) period?
  (*iii*) wavelength?
  (*iv*) speed?
  Write S if the property is the same in all three media.
(*b*) If medium Y is air and the speed of yellow light in air is 3.00×10^8 m s^{-1}, what is the refractive index of medium Z for yellow light?

9 The K-line in the emission spectrum of calcium is normally 393.37 nanometre (nm). However, in the light reaching us from a cluster nebula in the constellation Ursa Major the observed wavelength for this line is displaced 18.36 nm towards the red end of the spectrum.
(*a*) Is this group of stars moving towards or away from us?
(*b*) At what speed relative to us is it moving?

10 What would be the expected "red shift" in wavelength for the D-line of the sodium spectrum in the light from a star which is receding from us at 1.2×10^6 m s^{-1}? The "D-line" consists of two very close wavelengths with a mean of 589.3 nm.

11 One method used by police for detecting suspected speeding drivers uses the Doppler shift in the frequency of radar (microwaves) reflected back to the stationary police transmitter by the moving vehicle. If the radar frequency is 10^{10} Hz, what change in frequency must the police equipment be able to detect in order to differentiate between speeds of 60 and 61 km h^{-1}?

12 Electromagnetic waves exhibit diffraction, i.e. they tend to spread sideways as they pass through openings or go past obstacles. With each pair that follows select the one in which more diffraction would occur:
(*a*) light waves passing through a keyhole *or* light waves passing between the threads of a handkerchief;
(*b*) red light *or* violet light passing through the aperture of a microscope;
(*c*) radio waves *or* light waves travelling past the corner of a building.

13 Answer key:
B Both the wave and Newtonian particle models of light give satisfactory accounts for this behaviour.
W The wave model of light gives the more satisfactory account for this behaviour.
P The Newtonian particle model of light gives the more satisfactory account for this behaviour.
Select from the answer key above a statement appropriate to each of the following properties of light:
(*a*) Light travels through water in a straight line.
(*b*) When light passes through very small openings, it undergoes diffraction.
(*c*) When light is refracted at the interface between two media, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant.
(*d*) A thin film of oil on a puddle of water gives rise to coloured regions.
(*e*) If light crosses an interface from one medium where its speed is higher to a second medium where its speed is lower, the light is refracted towards the normal to the interface.

14 Suppose that the refractive index of an artificial diamond is 2.40. If the speed of light in a vacuum is 3.00×10^8 m s^{-1}, what is the speed of light in the artificial diamond, as predicted by
(*a*) the wave model of light?
(*b*) the Newtonian particle model of light?

SET 44 INTERFERENCE AND DIFFRACTION WITH ELECTROMAGNETIC RADIATION

Topic	*Problems*
Interference from two coherent sources producing standing or stationary waves in one dimension	1-2
Interference from two coherent sources, based on path difference only	3-5
More advanced problems involving interference from two sources in phase (Young's experiment)	6-10
Interference from two sources not in phase	11-12
Lloyd's mirror	13
Fresnel's biprism	14-15
Diffraction patterns with a single slit	16-19
Resolution	20-24
Diffraction gratings	25-29
Interference in thin films and wedges	30-35
Newton's rings	36-39
Bragg diffraction (or reflection)	40-42
Michelson interferometer	43-44

Where necessary in this set, use
speed of electromagnetic radiation in a vacuum or in air,
$c = 3.00 \times 10^8$ m s^{-1}

1 A transmitter T sends out radio waves of wavelength 2 km towards a reflector R which returns the waves to T. A ship travels between T and R, as shown in Figure 44.1. What is the distance between successive points of maximum intensity recorded by the ship's receiver when tuned to this wavelength?

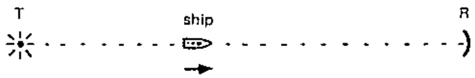


Fig. 44.1

2 Two microwave transmitters P and Q, which generate the same frequency, are set up facing each other. A microwave receiver is moved to a point on the line PQ where a maximum is detected. The receiver is then moved 3.0 cm towards Q in order to locate the next maximum but one. What is the
(*a*) wavelength of the microwaves?
(*b*) frequency of the microwaves?

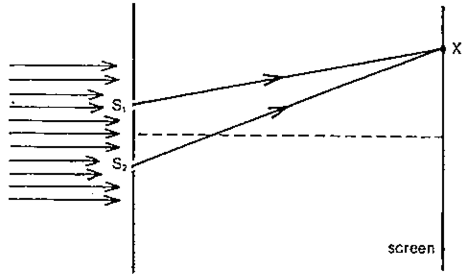


Fig. 44.2

3 Figure 44.2 shows the paths of light rays which leave two narrow slits S_1 and S_2 in phase, and travel to a point X on a screen. The distance S_2X is greater than S_1X by 1.2 μm. Of the following wavelengths for light, which one or more would produce a bright illumination at X?
0.40 μm, 0.60 μm, 0.80 μm, 2.4 μm

4 A source M of microwaves is set up centrally behind a metal barrier with two openings P and Q as shown in Figure 44.3. A microwave detector R is placed at X directly opposite the openings and connected to a loudspeaker. The sound output of the loudspeaker increases with the intensity of the microwaves detected. The wavelength of the microwaves is 3.2 cm. With the detector R at X a loud note is heard from the loudspeaker. As R is moved from X to Y, the sound drops to a minimum and then rises to a maximum at Y, which is 42.1 cm from Q.
(*a*) How far is P from Y?
(*b*) Select any one or more of the following statements that are correct.
A The distance XY must equal one wavelength.
B Y is on a nodal line.
C If the distance PQ had been larger with the wavelength unchanged, there would have been more nodal lines in the microwave interference pattern.
D If the microwave wavelength was gradually decreased to 2.0 cm, the detector R would have to be moved from Y towards X to keep the sound loud.
E If the detector R were moved slowly from X towards the source M, the loudness of the sound would alternate between high and low.

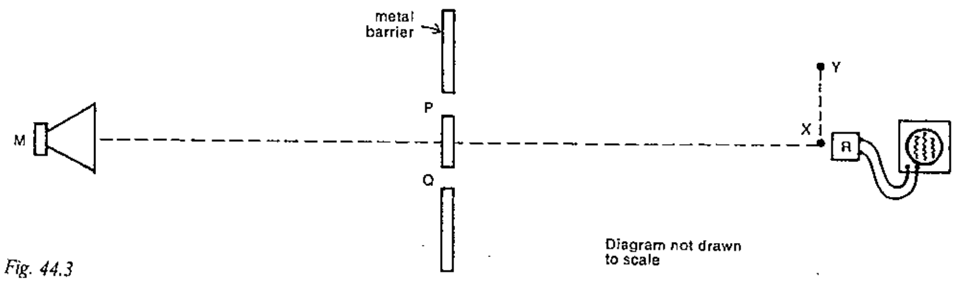


Fig. 44.3

5 A student sets up Young's experiment using a pair of slits C and D, and obtains interference patterns on a screen for two colours P and Q, whose wavelengths are p and q respectively. Figure 44.4 shows the experimental set-up, which is identical for the two colours. As shown in Figure 44.4, he notices that only two of the nodes for one colour line up exactly with nodes produced by the other colour.

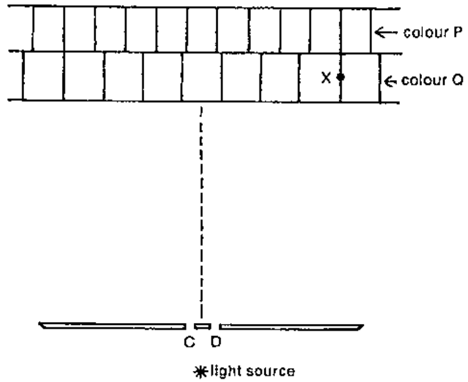


Fig. 44.4

(*a*) A point X is shown on the screen in Figure 44.4. What is the light path difference, CX − DX, in terms of q?
(*b*) What is the value of the ratio p/q?

6 Young's experiment is performed with light of wavelength λ using the apparatus shown in Figure 44.5. G is equidistant from slits M and N.

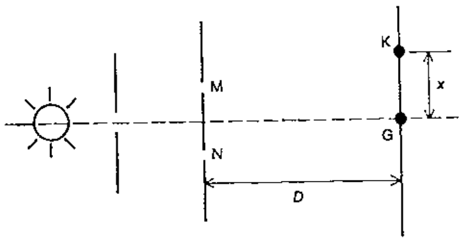


Fig. 44.5

(*a*) What would you expect to see on the screen in the vicinity of G?

(*b*) K is a point such that $NK - MK = \lambda/2$. What would you expect to see in the vicinity of K?

(*c*) If the distance between slits M and N is 1.0×10^{-3} m, $D = 2.0$ m, and $\lambda = 6.0 \times 10^{-7}$ m, find the value in metre of x, the distance of K from G.

7 The two slits in a Young's experiment apparatus are 0.200 mm apart. The interference fringes for light of wavelength 600 nm are formed on a screen 800 mm away from the slits.

(*a*) How far from the centre of the central bright band is the second dark line from the centre?

(*b*) How far apart are the centres of the central bright band and the second bright band from the centre?

8 Sodium light of wavelength 589.3 nm falls on a double slit of separation 0.400 mm. What will be the distance between successive interference fringes on a screen which is 150 cm from the slits?

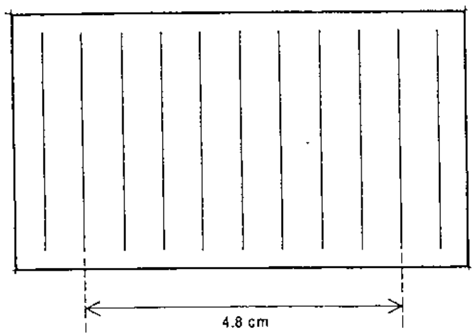


Fig. 44.6

9 Two razor blades of thickness 0.105 mm are held parallel and in contact, and are used to scratch slits on an opaque glass slide. The slide is then held 1.00 m away from a screen and illuminated by a helium–neon laser. A pattern similar to Figure 44.6 was observed on the screen behind the slide. What is the wavelength of the laser light?

10 A double slit arrangement is illuminated by light that consists of two wavelengths, one of which is known to be 600 nm. The interference pattern on a screen shows that the fourth dark fringe for the known wavelength coincides with the fifth bright fringe for the unknown wavelength. What is the unknown wavelength?

11 An airport is equipped with radio antennas X and X′, situated symmetrically about the centre line of a runway and spaced 30 m apart, as shown in Figure 44.7. X and X′ emit radio waves of wavelength 10 m.

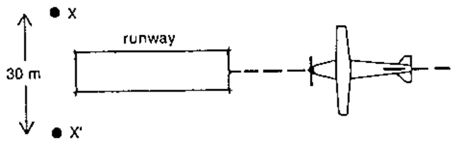


Fig. 44.7

(*a*) If the direction of the runway is to be marked by a line along which a radio receiver situated in an aircraft shows a minimum signal, what should be the phase delay of X′ relative to X?

(*b*) If the phase delay between X and X′ is such that there is a minimum signal along the runway, then it is still possible for an aircraft a long way away to be flying in another straight line along which the radio signal is also a minimum. What is the approximate size of the smallest possible angle between such a line and the direction of the runway?

12 Figure 44.8 shows two receiving aerials P and Q located 20 m apart along an east-west line at the equator so that as the Earth rotates some radiostars pass from horizon to horizon in a vertical plane through PQ. The aerials are connected to a receiver through cables of equal length.

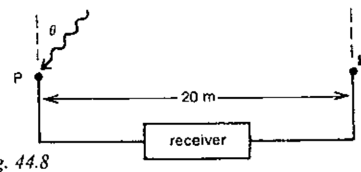


Fig. 44.8

(*a*) If the receiver is tuned to accept a signal of wavelength 6.0 m, how many maxima in the signal will occur from a particular radiostar emitting this wavelength during the 12 h it is above the horizon?

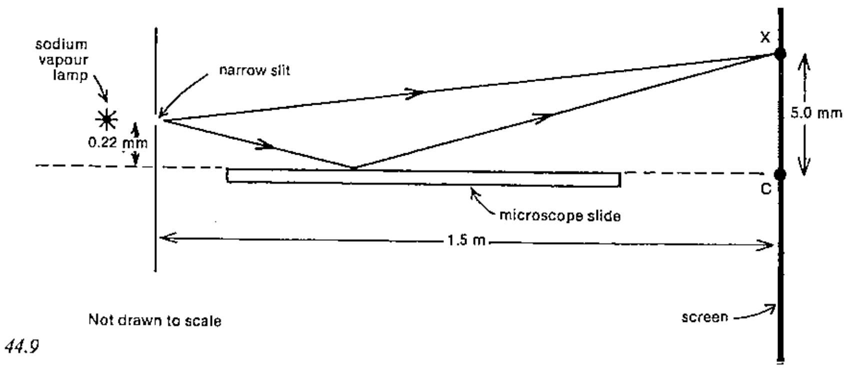


Fig. 44.9

(*b*) To "look" at an angle θ to the vertical (i.e. to obtain a maximum signal when the direction of a star is at an angle θ to the vertical), what phase delay must be introduced between Q and the receiver when $\sin\theta = 3/40$?

13 In the arrangement known as Lloyd's mirror and shown in Figure 44.9, an illuminated *single* narrow slit produces on the screen an interference pattern similar to that produced by *two* slits.

(*a*) What is the "separation" of the two effective sources?

At C, the point where the perpendicular bisector of the line joining the two effective sources meets the screen, a *dark* line rather than a *bright* band is seen, in contrast to Young's experiment. The wavelength of the light used is 589 nm.

(*b*) What phase delay must the light undergo as it is reflected?

(*c*) At X on the screen is there a bright band or a dark line? What number bright band or dark line (counting from C) is the point X on?

14 Figure 44.10 shows a device known as Fresnel's biprism used for producing two virtual light sources S_1 and S_2 in phase from a single monochromatic yellow source situated behind a slit S. The axis of the biprism lies on the line SR drawn from S perpendicular to the screen.

(*a*) If point P on the screen is such that $PS_1 - PS_2 = 7\lambda/2$, where λ is the wavelength of the yellow light, how many dark lines would there be between P and R? Include the dark lines, if any, that occur at P and/or R.

(*b*) In terms of λ, what is the separation of adjacent dark lines on the screen?

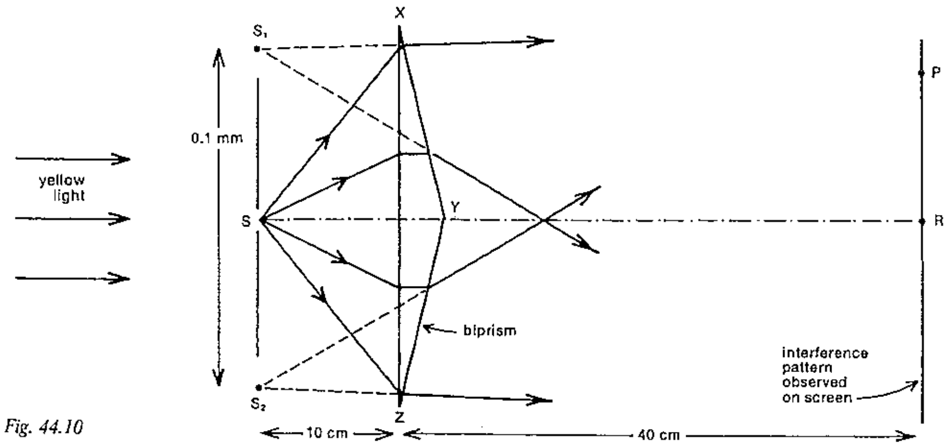


Fig. 44.10

15 A ray of light falls normally on a triangular glass prism with a small refracting angle A radian.

(*a*) Derive an expression for the angle of deviation D (in rad) in terms of A and n, the refractive index of the glass. Assume that the angles are small enough to warrant the approximation $\sin\theta \approx \theta$ (in rad).

Two prisms, each of vertical angle 1.5° and of refractive index 1.5, are placed base to base. A narrow slit is placed 20 cm from them and parallel to their common base. A screen is situated 80 cm from the prisms, and the slit is illuminated with light of wavelength 633 nm.

(*b*) What is the fringe spacing (in mm) seen through a travelling microscope focused on the screen?

16 Monochromatic light of wavelength λ falls normally on a narrow slit of width w, as shown in Figure 44.11. For light diffracted in direction Y the path difference from the two edges of the slit is y.

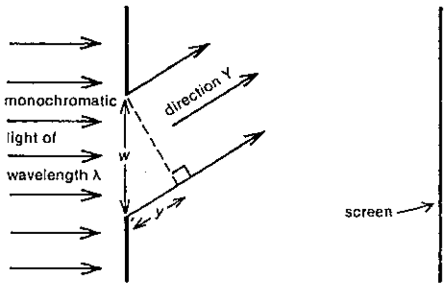


Fig. 44.11

For light falling on the screen in direction Y, would one expect bright light or a dark line, if y was

(*a*) zero?

(*b*) $\lambda/2$?

(*c*) λ?

(*d*) $3\lambda/2$?

(*e*) 2λ?

17 Consider the diffraction pattern produced on the screen in Figure 44.11. Figure 44.12 shows the intensity of the light plotted as a function of distance down the screen in the region of the pattern.

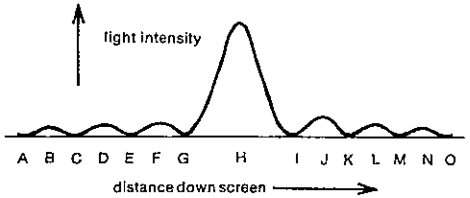


Fig. 44.12

(*a*) In terms of λ, the wavelength of the light, what is the path difference from the two edges of the slit to

(*i*) point G in Figure 44.12?

(*ii*) point L?

(*iii*) point C?

(*b*) If the distance CE on the screen is 1.2 mm, how far is H from N?

18 Light from a monochromatic red source of wavelength 700 nm emerges from a single vertical slit and travels 4.5 m before falling on a screen placed at right angles to the beam. Three dark lines either side of the centre of the diffraction pattern are labelled from left to right in order U, V, W, X, Y and Z, the centre of the pattern being between W and X. The distance from W to X is 30 mm.

(*a*) What is the path difference from the two edges of the slit to V?

(*b*) What is the approximate distance between U and Z?

(*c*) What is the width of the slit?

19 A diffraction pattern is produced on a screen when light of wavelength 6.0×10^{-7} m falls on a single slit of width 8.0×10^{-5} m. See Figure 44.13. The screen is 2.4 m from the slit.

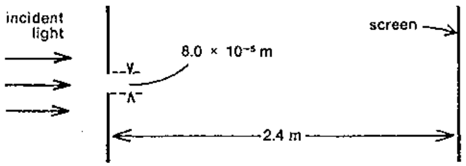


Fig. 44.13

(*a*) What is the separation on the screen of the two dark lines nearest to the centre of the diffraction pattern? Give the answer in millimetre.

(*b*) If R represents your answer to (*a*) above, express *in terms of R* the separation on the screen of the adjacent third and fourth dark lines from the centre of the pattern.

20 If light from two sources very close together passes through a single slit, two diffraction patterns will be formed. It will not be possible to *resolve* two separate images on a screen if the patterns overlap excessively. The two images could just be resolved if the centre of the central maximum of one falls no nearer than the first minimum of the other. Show that the minimum angle θ subtended by two such sources at the slit is given by

$$\sin\theta = \lambda/w,$$

where λ is the wavelength of the light used and w is the slit width.

21 In Problem 20 the minimum conditions for resolving two sources through a *straight slit* were considered. The corresponding relation for *circular openings*, known as *Rayleigh's criterion*, is

$$\sin\theta = 1.22\lambda/d,$$

where λ is the wavelength of the light used, d is the diameter of the aperture, and θ is the minimum angle subtended by two points that can just be resolved.

If a camera with a lens diameter of 2.8 cm is being used to photograph an intricate piece of embroidery 1.5 m away, what is the minimum separation of two points on the embroidery that will be resolved in the photograph? Use 550 nm as an average wavelength for white light.

22 Refer to the Rayleigh criterion explained in Problem 21.

(*a*) If two stars 12 light year from the Earth can just be resolved by the Mt Palomar telescope, which has an objective with an aperture of 5.1 m, how far apart are the stars? Use 550 nm for an average wavelength of visible light.

(*b*) The Jodrell Bank radio telescope in England is just able to resolve radiostars emitting a wavelength of 0.21 m and having an angular separation of 3.7 mrad. What is the diameter of the receiving parabolic dish?

23 Making suitable estimates, determine the maximum distance a car could be from you at night if you can resolve the two headlights. To which of the following is your answer closest?

60 m, 600 m, 6 km, 60 km

24 An optical microscope is used to view a small object, and its image is recorded on film. Which one or more of the following procedures would give an image with improved resolution?

A Use infrared radiation rather than visible light.

B Use ultraviolet radiation rather than visible light.

C Use a larger diameter objective lens in the microscope.

D Use a smaller diameter objective lens in the microscope.

E Move the object closer to the objective lens.

F Move the object further away from the objective lens.

25 A transmission diffraction grating ruled with 200 lines to the millimetre is employed to determine the wavelength of a monochromatic beam of light. For normal incidence it is found that the angular separation of the first bright line from the central one is 4°30′. What is the wavelength of the incident light?

26 Light of wavelength 662.4 nm falls normally on a diffraction grating with 400 lines per millimetre. Find the order of the image which is diffracted 32.0° from the normal.

27 When a parallel beam of light is incident normally on a certain transmission grating, a diffraction pattern is produced in which the first order maximum for light of wavelength 515 nm appears at an angle of 18.0° away from the central image. How many lines per millimetre are ruled on the grating?

28 Using light of wavelength 480 nm, what is the highest order image that can be produced by a diffraction grating having 600 lines per millimetre?

29 When light from a helium gas discharge tube is directed normally on to a diffraction grating, it is found that the second order yellow line corresponding to a wavelength of 590 nm coincides with a third order line in the deep violet. Determine the wavelength of the violet light.

30 Blue light of wavelength 480 nm falls normally on the surface of a horizontal soap film. The wavelength of the light *in the soap film* is λ'. In Figure 44.14 (in which, for clarity, the incident ray is shown as being not quite normal to the surface) rays 1 and 2 are those reflected from the two surfaces of the soap film. The soap film is $\lambda'/4$ thick and has a refractive index of 4/3.

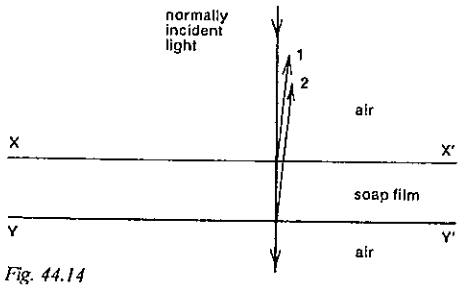


Fig. 44.14

(*a*) Does ray 1 undergo a phase change as it is reflected by the interface XX′?

(*b*) Does ray 2 undergo a phase change as it is

(*i*) transmitted through XX′?

(*ii*) reflected by YY′?

(*c*) Will the soap film appear blue or black when viewed directly from

(*i*) above?

(*ii*) below?

(*d*) What is the thickness of the film?

(*e*) In terms of λ' what is the next thickest film (after $\lambda'/4$) which would produce the same effect on transmitted and reflected light?

31 Monochromatic light, whose wavelength in oil is λ_o, is incident very near the normal (exaggerated for clarity in Fig. 44.15) on to an oil film. The film, which rests on water, is exactly $3\lambda_o/2$ thick. The refractive indices together with the reflected and transmitted rays are as shown in Figure 44.15.

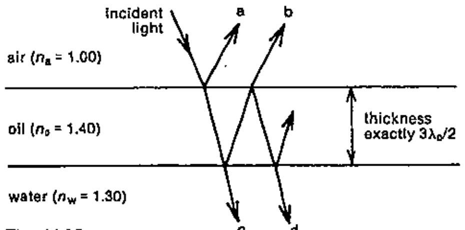


Fig. 44.15

(*a*) Are the following pairs of rays in phase or out of phase with each other?
(*i*) a and b;
(*ii*) c and d.

(*b*) Does destructive interference occur for either the light reflected by the oil film or that transmitted through the film?

(*c*) If the film is 632 nm thick, what is the wavelength *in air* of the incident light?

(*d*) If the oil shown in Figure 44.15 is replaced by an oil of refractive index 1.1, which one or more of the rays a, b, c and d will have its phase relationship to the incident ray changed *as a result of this replacement*?

32 The "blooming" of a camera lens involves coating its front surface with a thin layer of magnesium fluoride of refractive index 1.38 to increase the amount of light transmitted into the camera. Take 550 nm as an average wavelength of sunlight in air, and 1.61 as the refractive index of the glass lens.

(*a*) What is the corresponding average wavelength for sunlight in the magnesium fluoride coating?

(*b*) What is the appropriate minimum thickness for this coating? Give your answer to the nearest nanometre.

33 In order to find out the diameter of a human hair, two optically flat glass plates P and Q, 120 mm long, were placed in contact at one end with the hair holding the other ends apart as shown in Figure 44.16. The air wedge so formed was illuminated from above with yellow light of wavelength 590 nm from a sodium vapour lamp. Viewed from above, parallel interference fringes were observed, with 20 fringes occupying a distance of 26 mm along the plates.

(*a*) Would a bright yellow line or a dark line be observed at the end where the glass plates are touching?

(*b*) As one moves from one dark band to the next, by how much does the thickness of the air wedge increase?

(*c*) Approximately how many dark bands would be observed in all?

(*d*) What is the diameter of the hair?

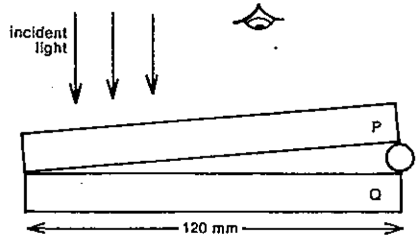


Fig. 44.16

34 Two squares of plane plate glass are placed edge to edge with a piece of tinfoil between the two at the opposite edge. When viewed at normal incidence with light of wavelength 5.9×10^{-7} m, 8 interference fringes per cm are observed. Find the angle in milliradian and in degree between the plates.

35 A vertical draining soap film is examined by reflected sodium light and shows a series of horizontal fringes 5.00 mm apart. If the wavelength of the sodium light is 589 nm and the refractive index of the soap solution is 1.33, find the angle (in μrad) between the surfaces of the film.

36 A plano-convex lens is placed on a flat slab of glass and illuminated from above with monochromatic light of wavelength λ, as indicated in Figure 44.17. Viewed from above, a pattern caused by interference between light reflected from the upper and lower surfaces of the air film can be observed. It consists of a black dot at the centre surrounded by alternating bright and dark circles, known as *Newton's rings*.

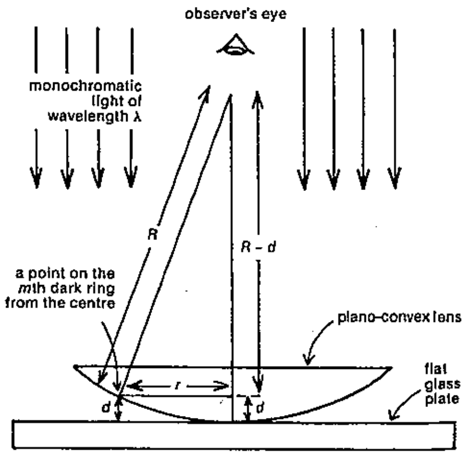


Fig. 44.17

Give answers to (*a*), (*b*) and (*c*) in terms of λ.

(*a*) What is the thickness of the air film at
(*i*) the first bright ring from the centre?
(*ii*) the second bright ring?
(*iii*) the first dark ring?
(*iv*) the second dark ring?

(*b*) How much thicker is the air film at the sixth dark circle than it is at the fifth dark circle?

(*c*) How thick is the film at the 14th dark ring?

d is the thickness of the air film at the mth dark ring which has a radius r. R is the radius of curvature of the lower surface of the lens.

(*d*) Express d in terms of m and λ.

(*e*) Express R in terms of r and d. (Remember that $d \ll r$, and so d^2 is negligible in comparison with r^2.)

(*f*) Express R in terms of r, m and λ.

37 A convex lens is placed on a plane glass surface and illuminated with sodium light of wavelength 589.3 nm. The diameter of the tenth dark ring is found to be 4.06 mm.

(*a*) What is the radius of curvature of the lower face of the lens?

(*b*) Using another light source, the diameter of the twelfth dark ring was found to be 47.4 mm. Calculate the wavelength of the light from this source.

38 Newton's rings are formed with neon light of wavelength 640 nm. The radii of two successive dark rings are 1.00 mm and 1.118 mm. What is the radius of curvature of the lens surface?

39 Newton's rings are formed with reflected light of wavelength 590 nm using a plano-convex lens and a plane glass plate with liquid between them. The diameter of the third dark ring is 2.00 mm. If the radius of curvature of the curved surface of the lens is 80 cm, find the refractive index of the liquid.

40 In a microwave analogue of the constructive interference caused in Bragg reflection (or diffraction) by the superposition of waves reflected off parallel planes of atoms in a crystal, a large block of polystyrene was impregnated with ballbearings arranged in a regular structure with "layers" distance d apart as indicated in Figure 44.18.

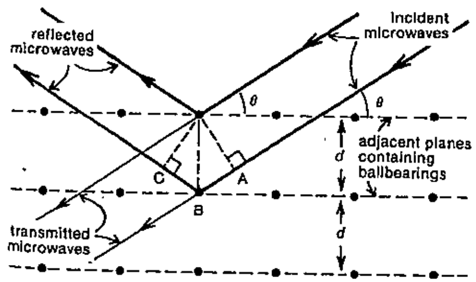


Fig. 44.18

(*a*) Derive an expression for the path difference ABC in Figure 44.18 in terms of d and θ, the glancing angle.

(*b*) Deduce an expression relating d, θ and λ, the microwave wavelength, for constructive interference to occur for the reflected waves. Your answer, if correct, is known as the *Bragg relation* or *Bragg's law*.

(*c*) If d = 43 mm and λ = 32 mm, determine the possible glancing angles for which there could be reinforced waves reflected off the parallel planes shown in Figure 44.18.

41 In a sodium chloride crystal one particular spacing between parallel planes of ions is 0.252 nm. If a crystal of sodium chloride was irradiated by a beam of x-rays of wavelength 0.154 nm, at what possible glancing angles would you expect strong reflected beams?

42 When x-rays of wavelength 0.124 nm are directed on to a potassium chloride crystal, reflected maxima result when the glancing angle is 11.4°. What does this suggest as a possible separation for parallel planes of ions in a potassium chloride crystal?

43 In a Michelson interferometer using sodium light of wavelength 589 nm, 254 fringes were counted as one of the mirrors was moved backwards. How far in μm was the mirror moved?

44 When a thin shaving of perspex is introduced into one of the beams of a Michelson interferometer, while using green light of wavelength 546 nm from a mercury source, 158 fringes were seen to sweep across the field of view. If the refractive index of perspex for green light is 1.49, what is the thickness of the shaving in micrometre?

SET 45 WAVE-PARTICLE DUALITY OF ELECTROMAGNETIC RADIATION

Topic	*Problems*
Photoelectric effect	1–7
Photons and photon energy	8–11
Relation between wavelength (wave property) and energy (particle property)	12–13
Photon momentum	14–16
Compton effect (without considering the relativistic increase in mass with speed)	17–18
Compton effect (considering the relativistic increase in mass with speed)	19–20
Pressure exerted by electromagnetic radiation	21–24
Pair production	25–26
Interpretation of wave-particle duality, and predominance of wave or particle properties	27–30

Where necessary in this set, use
speed of electromagnetic radiation in a vacuum or air, $c = 3.00 \times 10^8$ m s^{-1}
Planck's constant, $h = 6.63 \times 10^{-34}$ J s
$= 4.14 \times 10^{-15}$ eV s
$1\text{ e} = 1.60 \times 10^{-19}$ C
$1\text{ eV} = 1.60 \times 10^{-19}$ J
mass of an electron $= 9.11 \times 10^{-31}$ kg

1 An experiment is carried out with a photoelectric cell using the circuit shown in Figure 45.1. By varying the intensity and frequency of the light used, and by varying the voltage and polarity of the d.c. power supply, the graphs shown in Figure 45.2 were obtained.

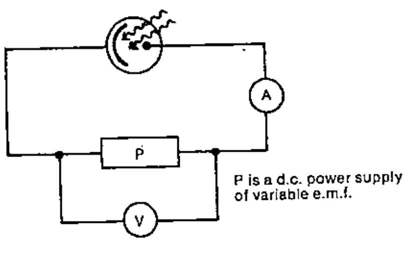


Fig. 45.1

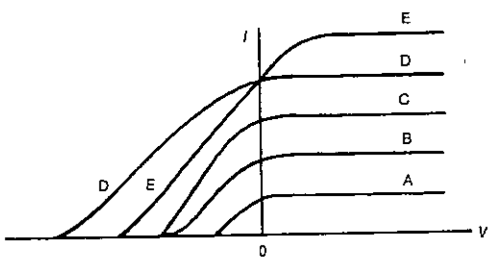


Fig. 45.2

(*a*) Which *two* of the five curves (A, B, C, D, E) in Figure 45.2 correspond to incident light of the same frequency but of different intensities?
(*b*) Which of the curves corresponds to the highest frequency of light used?
(*c*) Which of the curves corresponds to the situation where the highest energy electrons are ejected from the photoelectric surface?

2 The maximum kinetic energy of electrons ejected from a particular metal surface by monochromatic light of frequency ν is measured for several different values of ν. The line of best fit to the experimental points is shown in Figure 45.3.

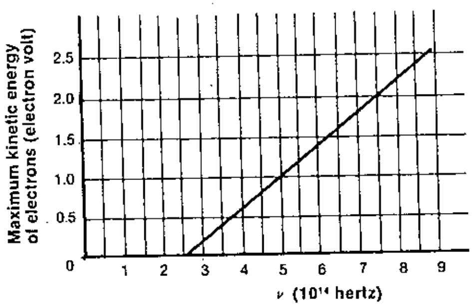


Fig. 45.3

(*a*) What is the value of Planck's constant (in eV s) as calculated from the graph?
(*b*) What is the minimum energy (in eV) required to extract an electron from this metal surface?
(*c*) If deep violet light of frequency 7.5×10^{15} Hz is used, what will be
(*i*) the maximum kinetic energy (in eV) of the ejected photoelectrons?
(*ii*) the minimum stopping potential difference that would have to be applied across the photocell to prevent the photoelectrons from reaching the collector electrode?

3 An experiment on the photoelectric effect was carried out, in which the maximum kinetic energy of electrons emitted from the surface of a particular metal was measured and then plotted against the frequency of the incident electromagnetic radiation. In Figure 45.4 the graph of E_{max} versus ν was obtained using electromagnetic radiation of a fixed intensity I_0, and shows the line of best fit to the experimental points.

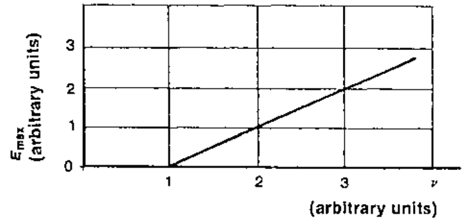


Fig. 45.4

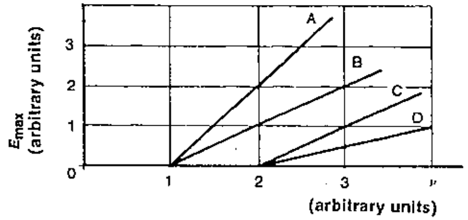


Fig. 45.5

(*a*) Which of the graphs A–D drawn to the same scale in Figure 45.5 would be obtained for the same metal, using electromagnetic radiation of intensity $2I_0$?
(*b*) Which of the graphs A–D in Figure 45.5 would be obtained using intensity I_0 for another metal for which the minimum energy required to remove an electron is twice that for Figure 45.4?

4 Electrons are ejected from a silver surface by radiation of wavelength 253.6 nm. The cut-off potential difference required to stop the photoelectrons is 0.12 V. Determine (in eV)
(*a*) the maximum kinetic energy of the ejected electrons;
(*b*) the work function of the silver surface.

5 The threshold frequency for a rubidium surface is 5.0×10^{14} Hz. Find
(*a*) the work function for the metal surface in eV;
(*b*) the maximum speed of electrons ejected by light of frequency 8.0×10^{14} Hz.

6 Radiation of wavelength 375 nm is incident upon a surface which has a work function of 2.0 eV. Calculate
(*a*) the energy (in eV) of each of the incident photons;
(*b*) the maximum kinetic energy (in eV) of the ejected electrons;
(*c*) the maximum speed of the electrons.

7 The threshold frequency for a particular material is 3.0×10^{14} Hz. If blue light of frequency 7.0×10^{14} Hz falls on this surface, what is
(*a*) the energy (in joule) of the incident photons?
(*b*) the work function (in joule) of the material?
(*c*) the maximum kinetic energy of the ejected electrons?
(*d*) the maximum speed of the ejected electrons?

8 What is the energy (in joule) of a photon of yellow light of frequency 5.1×10^{14} Hz from a sodium vapour lamp?

9 Radio photons from station 2CA in Canberra have an energy of 6.95×10^{-28} J. On what radio frequency does 2CA transmit?

10 A typical microwave apparatus used in schools uses a 32 mm wavelength.
(*a*) What is the frequency of the microwaves?
(*b*) What is the energy of one of the microwave photons?

11 Visible light ranges in wavelength from about 400 nm in the deep violet to about 750 nm for red light. Estimate as orders of magnitude
(*a*) the energy of a photon coming from a 100 W household light globe;
(*b*) the number of photons leaving the filament of the globe in 1 s.

12 Like many other questions in this set this problem relates *particle* and *wave* properties of electromagnetic radiation.
(*a*) Write down an expression for photon energy E (a particle property) in terms of radiation frequency ν (a wave property).
(*b*) Deduce an expression for photon energy E (a particle property) in terms of radiation wavelength λ (a wave property).
(*c*) Your answer to (*b*) should be in the form $E = k/\lambda$, where k is a constant. If SI units are used, i.e. E is in joule and λ is in metre, what is the value of k?
(*d*) Often it is useful to be able to relate photon energy in electron volt to radiation wavelength in nanometre. Deduce the relationship for E (in eV) in terms of λ (in nm), including the numerical value of the constant. This relation is possibly worth memorising.

13 (*a*) What are the approximate maximum and minimum energies of the photons that enable you to see? Give your answers in eV. The relation deduced in Problem 12 (*d*) should prove useful. The wavelength of visible light ranges from about 400 to 750 nm.
(*b*) What is the wavelength (in nm) corresponding to an x-ray photon of energy 1.24 keV?

14 What is the momentum of each photon in a helium–neon laser beam of wavelength 633 nm aimed across a laboratory in a westerly direction?

15 If a gamma ray photon has a momentum of magnitude 2.21×10^{-21} kg m s^{-1}, what is the wavelength of the gamma rays?

16 (*a*) Deduce an expression for the momentum p of a photon in terms of the corresponding wave frequency ν, the speed c of the photon in a vacuum and Planck's constant h.

(*b*) A mercury vapour lamp emits ultraviolet radiation of frequency 8.22×10^{14} Hz. What is the magnitude of the momentum of a photon of this ultraviolet light?

(*c*) Using your answer to (*a*) deduce an expression for p in terms of c and the photon energy E.

17 An x-ray photon travelling east and of wavelength 0.100 nm hits an electron essentially "at rest". As a result the photon is deflected in a northerly direction. For this collision the fractional loss in energy—and thus in the magnitude of the momentum—of the photon is small: $\nu' \approx \nu$ and $p'_x \approx p_x$. (The *primed* quantities ν' and p'_x represent the wave frequency and photon momentum *after* the interaction, while the *unprimed* ν and p_x represent the values *before* the interaction.) Work to three significant figures and use the joule as the energy unit.

(*a*) Draw a vector diagram showing the photon momentum before the collision and the photon and electron momenta after the collision.

(*b*) State the initial and final momenta of the photon.

(*c*) Determine the momentum of the electron after the collision.

(*d*) What was the kinetic energy of the electron after the interaction?

(*e*) What was the energy of the incident photon?

(*f*) Deduce the final photon energy.

(*g*) By calculating the ratio ν'/ν, find the percentage error in the original assumption that $\nu' = \nu$.

(*h*) Determine the Compton shift $\Delta\lambda$, i.e. the increase in wavelength caused by the scattering. Give the answer in picometre (1 pm = 10^{-12} m).

18 Figure 45.6 shows a Compton scattering of a 2.00 MeV gamma ray photon, which collides with an electron effectively at rest. The Compton recoil electron moves off with a momentum of 9.27×10^{-22} kg m s^{-1} N 30° E.

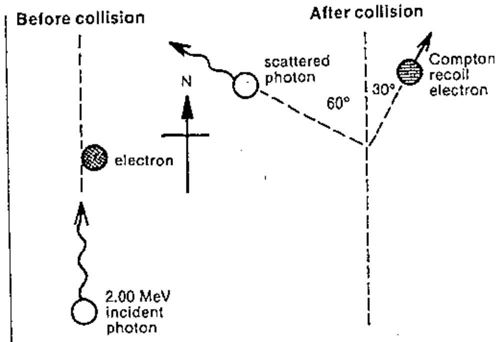


Fig. 45.6

(*a*) What is the wavelength of the incident photon?

(*b*) What is the momentum of the incident photon?

(*c*) Determine the momentum of the scattered photon.

(*d*) What is the wavelength of the scattered photon?

(*e*) What is the Compton shift, i.e. the increase in electromagnetic wavelength?

19 In this question on the Compton effect the relativistic increase in mass with speed will be considered. Suppose a high energy photon corresponding to a frequency of ν makes a Compton collision with an effectively stationary electron of mass m (and rest mass m_0). As a result the electron with a speed v and a photon of lower frequency ν' move off in the directions indicated in Figure 45.7.

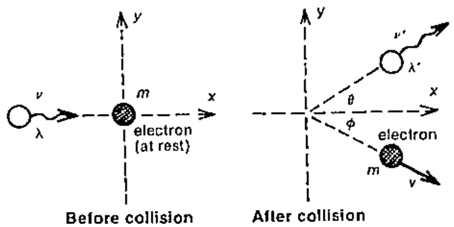


Fig. 45.7

(*a*) On what basis can one claim that

(*i*) $m = m_0/\sqrt{1 - v^2/c^2}$?

(*ii*) $h\nu = h\nu' + (m - m_0)c^2$?

(*iii*) $h/\lambda = (h/\lambda')\cos\theta + mv\cos\phi$?

(*iv*) $0 = (h/\lambda')\sin\theta - mv\sin\phi$?

(*b*) From these equations deduce that the Compton shift,

$$\Delta\lambda = \lambda' - \lambda = h(1 - \cos\theta)/m_0c.$$

20 A 221 keV x-ray photon undergoes a Compton scattering through 70.0°. What is the energy of the x-ray photon after scattering?

21 A small plane mirror of area A is hung vertically in a uniform beam of monochromatic light of frequency ν directed horizontally to the east. The surface of the mirror is hit by N photons per unit area per unit time.

(*a*) What is the momentum of a photon as it approaches the mirror?

(*b*) What is the change in momentum of a photon as it is reflected by the mirror?

(*c*) What is the *total* change in momentum of all the photons that hit the mirror per unit time?

(*d*) What total force does the mirror exert on these photons hitting it per unit time?

(*e*) What total force do these photons hitting the mirror per unit time exert on the mirror?

(*f*) What is the pressure of the light on the mirror?

22 Refer to Problem 21 (*f*) above and its answer.

(*a*) If E_T is the *total* energy falling on the mirror per unit area per unit time, express E_T in terms of N, h and ν.

(*b*) Deduce an expression for the pressure on the mirror in terms of E_T. Would this relation still hold if the light, instead of being monochromatic, were spread over a whole range of frequencies?

(*c*) Determine the order of magnitude of the pressure of sunlight at noon, if at that time $E_T \approx 10^3$ J m^{-2} s^{-1}.

(*d*) Approximately how much greater is the pressure due to the atmosphere (10^5 Pa) than the pressure due to sunlight?

23 In order to determine the pressure of light on an A4 sheet of black paper of area 6.2×10^{-2} m^2 facing a 60 W light bulb 3.0 m from the paper, consider the following.

(*a*) If the average wavelength of visible light is about 550 nm, what is the magnitude of the average momentum of one of the photons from the light bulb?

(*b*) What is the magnitude of the change in momentum of one of these photons as it is absorbed by the black paper?

(*c*) What is the average energy of the photons?

(*d*) How many photons are leaving the light bulb per second?

(*e*) How many photons per second are passing through an imaginary spherical surface of radius 3.0 m and centred on the bulb?

(*f*) What fraction of the photons passing through this spherical surface strike the sheet of paper?

(*g*) How many photons per second hit the paper?

(*h*) What is the magnitude of the total change in momentum of the photons hitting the paper in 1 s?

(*i*) What is the magnitude of the total force exerted by the paper on these photons hitting the paper in 1 s?

(*j*) What is the magnitude of the total force exerted on the paper by these photons hitting the paper in 1 s?

(*k*) What then will be the pressure exerted by the photons on the paper?

24 By considering photon momentum, estimate the order of magnitude of the power of a light source in your bedroom so strong that, when you turn the light on, you are almost knocked over.

25 If a gamma ray photon of energy x MeV passes very close to a nucleus, the photon may disappear to be replaced by a negative electron and a positron (with identical mass to the electron but exactly opposite charge). This process is known as *pair production*. What is the minimum value of x for this to occur?

26 In an observed electron–positron pair production the kinetic energy of the electron was found to be 0.29 MeV.

(*a*) What is the total kinetic energy of the electron–positron pair?

(*b*) What was the energy of the original gamma ray photon?

27 When referring to electromagnetic radiation, which one or more of the following include both particle and wave properties in the one equation? (Do not take c into account here, since it is both a particle and a wave property.)

A $c = \nu\lambda$
B $E = h\nu$
C $p = h/\lambda$
D $\lambda E = hc$
E $\nu = 1/T$ (T = period)
F $p = h\nu/c$

28 (*a*) For which one of the following forms of electromagnetic radiation would you expect *particle* rather than *wave* properties to be the most obvious?

A microwaves
B x-rays
C orange light
D radio waves
E ultraviolet light
F gamma rays

(*b*) Using the answer key from (*a*), which type of electromagnetic radiation would most obviously exhibit *wave* rather than *particle* properties?

29 Would you expect

(*a*) *higher* or *lower* frequency electromagnetic radiation to exhibit *interference* more obviously?

(*b*) *longer* or *shorter* wavelength radiation to be more likely to give rise to *interactions between individual photons and particles of matter*, e.g. the photoelectric effect and the Compton effect?

30 Accepting the dual nature of electromagnetic radiation, which one or more of the following statements would you agree with?

A Only high energy photons land at a maximum of an interference pattern on a screen.
B Visible light can illustrate both wave and particle properties.

C In a two-slit interference pattern maxima, predicted by the wave model, are regions where there is a high probability of a photon landing.
D It can be predicted where a particular photon will land in an interference pattern.
E The particle rather than the wave model predicts the location of regions of minimum disturbance in a diffraction pattern.
F If only a few photons are present, wave properties are difficult to demonstrate.

SET 46 REVISION PROBLEMS

Set	*Topic*	*Problems*
38	Transmission of light	1–3
39	Curved mirrors	4–6
40	Refraction at a plane interface	7–10
41	Lenses	11–14
42	Simple optical instruments	15–18
43	Simple wave properties of light and other forms of electromagnetic radiation	19–20
44	Interference and diffraction with electromagnetic radiation	21–27
45	Wave-particle duality of electromagnetic radiation	28–29

Where necessary in this set, use
refractive index of air = 1.00
speed of electromagnetic radiation in a vacuum or air,
$c = 3.00 \times 10^8$ m s^{-1}
Planck's constant, $h = 6.63 \times 10^{-34}$ J s
$= 4.14 \times 10^{-15}$ eV s
1 e $= 1.60 \times 10^{-19}$ C
1 eV $= 1.60 \times 10^{-19}$ J
mass of an electron $= 9.11 \times 10^{-31}$ kg

1 A laser beam is reflected from the middle of one face of a 16-sided rotating mirror to a distant station 35.0 km away and is then reflected back along its incident path. At what frequency must the mirror be rotated if the beam is received back on the middle of the next face?

2 Two lamps on opposite sides of a screen produce the same illumination at the screen. Lamp X is 60.0 cm away on one side and the other lamp Y is a 60.0 watt lamp 90.0 cm from the screen.
(*a*) What is the power of lamp X?
(*b*) A piece of glass which transmits 64.0 per cent of the light falling on it is placed between X and the screen. How far from the screen must Y now be placed in order to keep the illuminations equal?
(*c*) If, as shown in Figure 46.1, a plane mirror is placed, facing X and 30 cm from it, by what fraction will the intensity on that side of the screen increase?

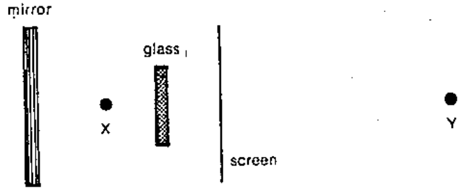


Fig. 46.1

3 A transmitter is used to send radiation vertically upwards, as though from a point source, to passive satellites when they are overhead. The satellites are thin spherical reflecting shells which reflect all the radiation falling on them as though from a point source. Reflected radiation is picked up by a receiver on the Earth's surface. The distance between the transmitter and receiver is negligible by comparison with their distances from the satellite.

The first satellite of a series is 1.0 m in outside diameter. Its distance from the Earth's surface varies from time to time over a ratio of 1 to 4.
(*a*) Over what ratio will the intensity of radiation reaching the satellite vary?
(*b*) Over what ratio will the radiation received back at the Earth vary?
(*c*) Certain scientists suggest that the signal strength of the first satellite is weak and that it should be replaced in the same orbit by a satellite designed to reflect eight times as much radiation as the first. What should be the diameter of the second satellite?

4 Answer key:
A Real
B Virtual
C Upright relative to the object
D Inverted relative to the object
E On the opposite side of the mirror (or lens) to the object
F On the same side of the mirror (or lens) as the object
G $M > 1$, where M is the linear magnification of the image

H $M = 1$
I $M < 1$
J $D_i < f$, where D_i is the distance from the image to the mirror (or lens) and f is the focal length
K $D_i = f$
L $2f > D_i > f$
M $D_i = 2f$
N $D_i > 2f$
O $D_i = D_o$, where D_o is the distance from the object to the mirror (or lens)

From the answer key above select all the responses that *always* apply to the image formed by
(*a*) a plane wall mirror;
(*b*) a concave mirror when used correctly as a shaving mirror;
(*c*) a concave mirror used in a reflecting astronomical telescope.

5 When an object is placed 40 cm from a concave spherical mirror, the image formed is also 40 cm from the mirror.
(*a*) Is the image upright or inverted?
(*b*) What is the focal length of the mirror?

6 A concave mirror has a focal length of 15 cm.
(*a*) As an object is moved from a point 40 cm from the mirror to a point 20 cm from it, does the image become larger or smaller?
(*b*) If the object, 4.8 cm high, is now placed 60 cm from the mirror, the image formed is 40 cm away from the object. What is the height of this image?

7 A ray of light is directed from medium X into medium Y and then medium Z along the path PQRS as shown in Figure 46.2.

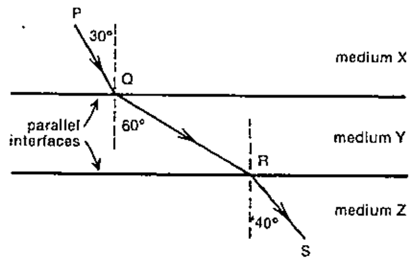


Fig. 46.2

(*a*) Which of the three media has the highest refractive index?
(*b*) What is the relative refractive index for light travelling from medium X to medium Y?
(*c*) If the refractive index of medium Y is 1.2, what would be the refractive index of medium Z?
(*d*) For light travelling downwards in medium X, as shown in Figure 46.2, could total internal reflection occur at the first interface, the second interface, neither or both interfaces?

8 A rectangular block of glass 5.0 cm thick and of refractive index 1.5 is held just in front of a concave mirror of radius of curvature 35 cm, the faces of the block being perpendicular to the principal axis of the mirror. How far from the mirror must a pin be placed for its reflected image to coincide with the pin itself?

9 A parallel beam of light from a hydrogen discharge tube is passed through a prism of refracting angle 60°0′, the setting being such that the C line of the hydrogen spectrum experiences minimum deviation, and the transmitted beam is focused on a photographic plate by a converging lens. Given that the focal length of the lens is 40.0 cm for both the C line and the F line of the hydrogen spectrum, calculate the distance between these lines on the photographic plate. You may assume that the deviation of the F line is also approximately minimum. The refractive indices of the glass for the C and F lines are 1.604 and 1.620 respectively.

10 A ray of light is incident normally on the face AB of a glass prism of refractive index 1.52 as in Figure 46.3. If the prism is immersed in water of refractive index 1.34, find the largest value for the angle marked β so that the ray is totally reflected at the face AC.

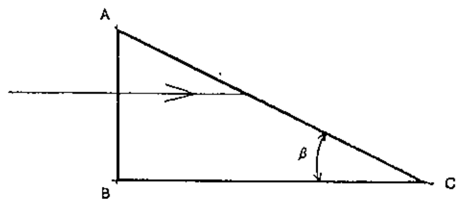


Fig. 46.3

11 Using the answer key given in Problem 4 of this set, select all the responses that *always* apply to the image formed by
(*a*) a convex lens being used in a camera to photograph a whole page of a book;
(*b*) a convex lens used as a magnifying glass.

12 An object which is 30 cm from a lens forms an image 6.0 cm from the lens on the same side. Is the lens converging or diverging and what is its focal length?

13 A convex lens 10 cm in focal length is held in a horizontal position just above the surface of a liquid 20 cm deep in a tank. The image of a point 38 cm above the centre of the lens is brought to a focus on the bottom of the tank.

Draw a diagram showing the path of the rays and calculate the refractive index of the liquid.

14 In the camera shown in Figure 46.4 the convex lens of focal length 6.40 cm is situated 6.60 cm from the film at the back of the camera.

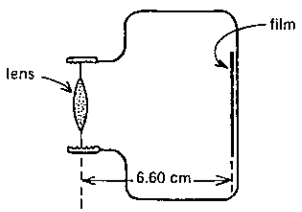


Fig. 46.4

(*a*) With this 6.60 cm separation of the lens and film, how far away from the lens would an object be in order to be most clearly photographed? Give the answer in metre.
(*b*) Determine the magnification in this case.
(*c*) To then photograph a distant mountain range, write down
  (*i*) "in" or "out", depending on whether the lens should be moved *in* towards the film or *out* away from the film, and
  (*ii*) how far it should be moved.

15 A convex lens of focal length 24 cm is placed 12 cm in front of a convex mirror. It is found that when a pin is placed 36 cm in front of the lens it coincides with its own inverted image formed by the lens and the mirror. Find the focal length of the mirror.

16 (*a*) An observer looking down on a pool of water (refractive index 1.34) judges the depth to be 1.00 m. Accepting this figure, calculate the real depth if the observer's line of sight is normal to the surface.
(*b*) The observer wishes to photograph a flat object on the bottom of the pool, using a camera with a lens of focal length 10.0 cm placed 1.00 m above the surface and directly over the object. Find the required distance between the lens and the film to give a sharp image.

17 If the angular diameter of the Sun is 32 minute of arc, what will be the size of its image formed at the focus of a telescope objective having a focal length of 18 m?

18 Three convex lenses of focal lengths 1.0 cm, 3.0 cm and 30 cm are supplied.
(*a*) Which pair of lenses would you select to construct each of the following instruments giving the greatest magnifying power?
  (*i*) an astronomical telescope;
  (*ii*) a compound microscope.
(*b*) Estimate the magnifying power for
  (*i*) the telescope;
  (*ii*) the microscope. Assume that the microscope tube, in the ends of which the two lenses are mounted, is 20 cm long, and that the least distance of distinct vision is 25 cm.

19 Camera lenses are sometimes coated with a thin film of magnesium fluoride to reduce loss of light by reflection. The refractive indices of magnesium fluoride and glass are 1.4 and 1.5 respectively. Consider monochromatic yellow light from a sodium lamp passing from the *air*, through the *coating* and into the *glass*. In which of the three media would
(*a*) the speed of the light be the lowest? (Write S if the property indicated is the same in all three media.)
(*b*) the frequency of the light be the lowest?
(*c*) the wavelength of the light be the lowest?
(*d*) the period of the light be the lowest?

20 For which of the following phenomena do you think the *wave* model for light clearly gives a more acceptable picture than the *Newtonian particle* model?
A The resolution provided by a microscope improves with the size of the aperture.
B Light travels in straight lines.
C As sunlight passes through the surface of the water in a swimming pool, the refraction of light towards the normal is associated with its reduction in speed.
D If you look through a narrow gap between two of your fingers held in front of one of your eyes, you can see dark lines.
E Light can be shown to exert a pressure on any surface on which it falls.

21 In Figure 46.5, S_1 and S_2 are point sources of an electromagnetic radiation. They are connected to the same oscillator and remain in phase. Placed 4.0 m apart, they emit energy at equal rates in waves of 1.0 m wavelength.

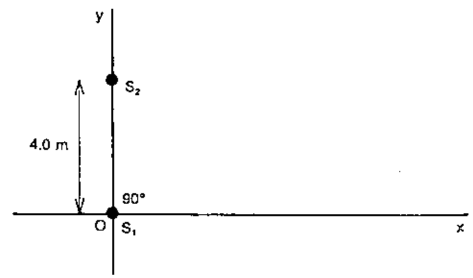


Fig. 46.5

(*a*) If a detector of this radiation is moved from O along Oy, how far from O will the first minimum be detected?

(*b*) Explain why the detector would register a maximum when 3.0 m from O along Ox.

22 Figure 46.6 shows parallel light rays incident normally on a barrier in which there are two narrow openings M and N which are 15 wavelengths of light apart. The point O is midway between M and N. Light passing through M and N falls on a screen situated a long way from the barrier. OP is the line perpendicular to the barrier and screen. When the wavelength of the light is 6.0×10^{-7} m, X is the position of the second dark band on the screen below P.

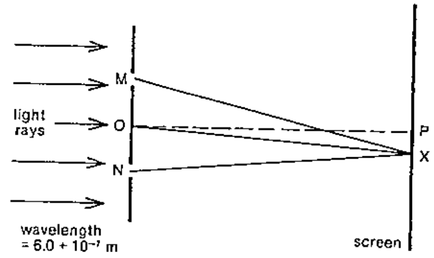


Fig. 46.6

(*a*) State the sine of the angle XOP.
(*b*) What thickness of transparent material of refractive index 2.0 would be required in front of opening M in order to delay the light through that opening and make the second dark band which was at X now fall on P?
(*c*) If the plate of transparent material is removed, what is the wavelength of the incident light which would make the third dark band appear at X?
(*d*) With incident light of the original wavelength, the barrier between M and N is removed, leaving a gap of width 16 times the wavelength of the light. Determine the sine of the angle which the direction of the first minimum would make with the line OP.

23 The inclined faces of a glass biprism make angles of 2° with the base of the prism, the refractive index of the glass being 1.5. A slit which is 10 cm from the prism is illuminated with sodium light of wavelength 590 nm. Calculate the distance between successive fringes observed at a distance of 1.0 m from the prism.

24 A screen is placed 5.0 m from a graphite coated glass plate on which a slit of width 50 μm has been scratched. The slit is illuminated by light of wavelength 460 nm, and a pattern of light and dark bands is observed on the screen.
(*a*) Calculate the distance (in mm) between the two dark bands adjacent to the centre of the pattern.
(*b*) A second slit, also of width 50 μm, is now scratched on the coated glass plate so that the centre of the second slit is 100 μm from the centre of the first slit. Light of wavelength 460 nm is again used to illuminate the slits, but now a new pattern is seen on the screen. Calculate the distance (in mm) between the two dark bands adjacent to the centre of this new pattern.

25 A parallel beam of light falls normally on a diffraction grating in which the spacing is 10.0 μm. If the beam contains violet light of wavelength 400 nm and red light of wavelength 750 nm, calculate the angular deviations of these two wavelengths in the second order image.

26 Aluminium is "anodised" by coating a polished sheet of it with a thin transparent film of aluminium oxide of refractive index 1.8.
(*a*) What is the wavelength of green light *in the aluminium oxide*, if it has a wavelength in air of 5.4×10^{-7} m?
(*b*) If the anodised aluminium is viewed normally (i.e. perpendicularly to the surface) and the green light referred to in (*a*) is allowed to fall normally on to the anodised surface, what would be the minimum thickness of the aluminium oxide coating which would cause the material to appear black?

27 A parallel beam of sodium light of wavelength 589.3 nm falls normally on the plane surface of a plano-convex lens whose convex surface rests on a piece of plate glass. The radius of curvature of the convex surface is 500 mm. Calculate the radius of the fiftieth dark ring seen by reflected light.

28 The photoelectric threshold of tungsten is 230 nm. Determine the maximum energy, in electron volt, of the electrons ejected from the surface by ultraviolet light of wavelength 180 nm.

29 The SI unit of length, the metre, is defined as 1 650 763.73 wavelengths in a vacuum of the radiation corresponding to the transition between the levels $2p_{10}$ and $5d_5$ of the krypton-86 atom.
(*a*) What is the wavelength (to the nearest nanometre) of this orange light from krypton?
(*b*) What is the energy (in eV) of a photon of this light?
(*c*) What is the magnitude of the momentum of one of these photons?

PART 6
HEAT

SET 47 THERMAL EXPANSION OF SOLIDS AND LIQUIDS

Topic	*Problems*
Temperature scales	1–4
Linear expansion of solids	5–11
Superficial expansion of solids	12–13
Cubic expansion of solids and liquids	14–18
Expansion of liquids in solid containers; real and apparent expansion	19–21

1 If the triple point of water, the temperature at which water can exist in equilibrium with ice and its own vapour, is defined as 273.16 K and is also 0.01°C,

(*a*) what is the temperature

(*i*) of the ice point on the thermodynamic (or Kelvin) temperature scale?

(*ii*) at which oxygen boils (90.19 K) on the Celsius scale?

(*b*) Deduce an expression for the thermodynamic temperature T in terms of its equivalent Celsius temperature θ.

2 Convert

(*a*) the melting point of zinc 692.73 K to its temperature on the Celsius scale;

(*b*) the boiling point of water to its temperature on the thermodynamic temperature scale.

3 Would you be less comfortable playing tennis at 264 K or 301 K?

4 If the length of the mercury column in an uncalibrated mercury-in-glass thermometer was 12 mm at the ice point and 237 mm at the boiling point of water, what would be the Celsius temperature when the column was 57 mm long?

5 A brass rule which is 300 mm long at 21°C is found to expand by 0.45 mm when it is heated to 100°C. Determine the linear expansivity (formerly known as the coefficient of linear expansion) of brass.

6 The Golden Gate Bridge in San Francisco is 1.25 km long and made of steel which has a linear expansivity of $1.2 \times 10^{-5}\ K^{-1}$. What total range for expansion must be allowed for, if temperatures might vary between −5°C and 45°C?

7 An aluminium ring has a diameter of 400.0 mm at 20°C. What is its diameter at a temperature of 240°C? The average linear expansivity of aluminium in this temperature range is $2.4 \times 10^{-5}\ K^{-1}$.

8 If steel rails are being laid in 30 m lengths for a new train line on a day when the temperature is 20°C, what is the appropriate width of the gap where one length meets the next, if the rail temperatures can go as high as 56°C? The linear expansivity of steel is $1.2 \times 10^{-5}\ K^{-1}$.

9 A steel measuring tape which is correct at 20°C is used to measure out a 2000 m cross-country skiing course, while the temperature is −0.8°C. What is the correct length of the course? Is the discrepancy likely to worry the organisers? Linear expansivity of steel is $1.2 \times 10^{-5}\ K^{-1}$.

10 A pendulum clock is controlled by a light rod of linear expansivity α supporting a relatively massive bob and oscillating with a period T_1 when its length is l_1. A temperature increase of $\Delta\theta$ will raise the period and length to T_2 and l_2 respectively. (For a pendulum, $T = 2\pi\sqrt{l/g}$, where g is the local gravitational field strength.)

(*a*) Deduce an expression for T_2/T_1 in terms of

(*i*) l_1 and l_2;

(*ii*) α and $\Delta\theta$.

(*b*) If the changes in l and T are small, show that

$$(T_2 - T_1)/T_1 \approx \tfrac{1}{2}\alpha\Delta\theta.$$

11 At 20°C the brass rod pendulum of a particular grandfather clock correctly beats a second each swing (its period of swinging *to and fro* is therefore 2 s). During a sudden hot spell the temperature jumps from 20°C to 30°C and remains there for 24 h. Linear expansivity of brass = $1.9 \times 10^{-5}\ K^{-1}$.

(*a*) At the end of the 24 h period, by how much will the clock be wrong?

(*b*) Will it be fast or slow?

(*c*) See if you can find out at least one way in which this effect is reduced or corrected in a grandfather clock.

12 A rectangular sheet of metal has a length L and width W as shown in Figure 47.1. The temperature of the sheet is raised by $\Delta\theta$. The linear expansivity of the material is α.

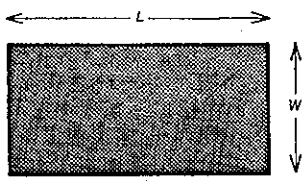


Fig. 47.1

(*a*) What is the new

(*i*) length?

(*ii*) width?

(*iii*) area?

(*b*) If the new area A_2 is related to the previous area A_1 as follows

$$A_2 = A_1(1 + \beta\Delta\theta),$$

where β is the superficial expansivity of the material, show that

$$\beta \approx 2\alpha$$

13 When the engine of a 1934 MG NA sports car warms up, the temperature in the cylinders may rise from, say, 20°C to 300°C. The piston heads have a diameter of 57 mm and are made of steel of linear expansivity $1.2 \times 10^{-5}\ K^{-1}$.

(*a*) What is the increase in the

(*i*) diameter, and

(*ii*) surface area

of the piston heads?

(*b*) What is the percentage change in pressure exerted on the piston heads, if the average force exerted by the exploding mixture has not changed while the engine has warmed up? Indicate whether the change is positive or negative.

14 Consider a cube of metal being heated, and show that the cubic expansivity of the metal is approximately three times its linear expansivity.

15 The cubic expansivity of solids lies roughly in the range $2 \times 10^{-5}\ K^{-1}$ to $2 \times 10^{-4}\ K^{-1}$ while that of liquids is roughly ten times greater.

Suppose in a moment of frustration at home you burst into an absolute rage, and roar, scream and yell at the whole family for a minute or so. Estimate as orders of magnitude, as a result of this outburst,

(*a*) your increase in temperature;

(*b*) your cubic expansivity;

(*c*) your linear expansivity;

(*d*) your increase in height;

(*e*) your increase in mass;

(*f*) your increase in volume;

(*g*) your decrease in density.

16 Copper has a cubic expansivity of $5.1 \times 10^{-5}\ K^{-1}$. If the temperature is raised by 10 K, by what percentage does

(*a*) the length of a 40 cm copper rod increase?

(*b*) the length of a 4 cm copper rod increase?

(*c*) the mean separation of two adjacent copper ions in either rod increase?

17 The density of mercury is $1.360 \times 10^4\ kg\ m^{-3}$ at 0°C. What is it at 25°C? Cubic expansivity of mercury = $1.82 \times 10^{-4}\ K^{-1}$.

18 A mercury barometer has a brass scale which is correct at 20°C. What will be the corrected height of the barometer when the reading is 750.0 mm and the temperature is 30°C? Linear expansivity of brass = $2.0 \times 10^{-5}\ K^{-1}$; cubic expansivity of mercury = $1.8 \times 10^{-4}\ K^{-1}$.

19 A partially constructed mercury thermometer consists of a mercury filled glass bulb of internal volume $6.0 \times 10^{-7}\ m^3$ at the lower end of a capillary tube of cross-sectional area $5.2 \times 10^{-8}\ m^2$ and containing a 30 mm mercury column. These measurements are all correct at 20°C. The tube is then immersed in a beaker of boiling water as indicated in Figure 47.2.

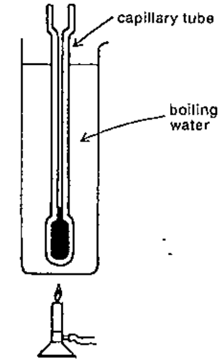


Fig. 47.2

(*a*) Determine to two significant figures the total volume of mercury in the bulb and capillary tube at 20°C.

To assist our understanding of what takes place, let us suppose (unrealistically) that

Step I the glass heats up to 100°C *while the mercury remains at 20°C*, and

Step II the mercury then heads up to 100°C.

The linear expansivity of glass is $9.0 \times 10^{-6}\ K^{-1}$ and the cubic expansivity of mercury is $1.8 \times 10^{-4}\ K^{-1}$.

(*b*) *As a result of step I,*

(*i*) what is the increase in the internal volume of the bulb?

(*ii*) what volume of mercury would have moved from the capillary tube into the bulb?

(*iii*) if the *increase* in the internal cross-sectional area of the capillary can be regarded as negligible, by what distance has the top of the mercury column fallen?

(*iv*) what is the new length of the mercury column?

(*c*) *As a result of step II,*

(*i*) what is the increase in the total volume of the mercury?

(*ii*) by what distance (expressed to three significant figures) will the top of the mercury column move up the tube since the completion of step I?

(*iii*) what is the final length of the mercury column?

20 With the threat of a petrol strike looming, the manager of a self-serve petrol station fills—right to the brim—the 40 litre tank of his Mazda 1300, when he arrives at work with the temperature at 5°C. By the time he knocks off, the temperature has risen to 18°C. How much petrol has he lost through the tank's overflow pipe during the day? The tank is made of steel which has a cubic expansivity of 3.5×10^{-5} K^{-1}, while that of petrol is 9.0×10^{-4} K^{-1}.

21 In the poisons cupboard in a school chemistry department mercury is to be stored in a 250 cm^3 glass reagent bottle. It is desired that there be at least a 5 cm^3 safety gap below the stopper at all times. What is the maximum volume of mercury that should be put into the bottle on a day when the temperature is 15°C, if temperatures could rise as high as 38°C? Glass has a linear expansivity of 9×10^{-6} K^{-1}, while mercury has a cubic expansivity of 1.8×10^{-4} K^{-1}. (Assume that the volume of the reagent bottle is 250 cm^3 at 15°C.)

SET 48 THERMAL EXPANSION OF GASES AND THE IDEAL GAS LAWS

Topic	*Problems*
Variation of volume with pressure of a gas at constant temperature (Boyle's law)	1–7
Variation of volume with temperature of a gas at constant pressure (Charles' law or Gay-Lussac's law)	8–11
Variation of pressure with temperature of a gas at constant volume (Amonton's law)	12–15
The mole and the Avogadro constant	16–21
Ideal gas law	22–30
Dalton's law of partial pressures, vapour pressure, relative humidity, dew point	31–37
Kinetic molecular theory of gases: average translational kinetic energy of molecules, root mean square speed, average speed, collisions with container walls	38–47
Kinetic molecular theory of gases: average separation of molecules, mean free path, collision frequency, effect of density on molecular speed, escape of gas through a hole	48–54

Where necessary in this set, use
pressure due to a vertical column of mercury mm 1 mm high,
$1\ \text{mmHg} = 0.133\ \text{kPa}$
$0°\text{C} = 273\ \text{K}$ (rather than the more exact 273.15 K)
Avogadro constant, $L = 6.02 \times 10^{23}\ \text{mol}^{-1}$
$1\ \text{g} = 6.02 \times 10^{23}\ \text{u}$ (unified atomic mass unit)
molar gas constant, $R = 8.31\ \text{J mol}^{-1}\ \text{K}^{-1}$
Boltzmann constant, $k = 1.38 \times 10^{-23}\ \text{J molecule}^{-1}\ \text{K}^{-1}$
STP (standard temperature and pressure) $= 273$ K and 101 kPa

1 A certain mass of gas, occupying a volume of 350 cm^3 at a pressure of 120 kPa, is subjected to a change of pressure without any change in temperature. If the new volume is 400 cm^3, what is the new pressure?

2 It has been common practice to use the vertical height of mercury columns to measure the pressure of gases in enclosed equipment and the pressure of the atmosphere in mercury barometers. The density of mercury is 1.36×10^4 kg m^{-3}.
(*a*) What is the pressure (in kilopascal) of the atmosphere, if on a particular day it can support a vertical column of mercury 760 mm high in a mercury barometer?
(*b*) Deduce the pressure in kPa equivalent to that of a vertical column of mercury 1 mm high (written 1 mmHg). This relation should prove useful in any situation where it is necessary to convert a pressure in mmHg to the SI unit of pressure, the pascal.

3 In a Boyle's law apparatus shown in Figure 48.1 the mercury surface in the open tube is 60 mm higher than the surface in the closed tube. The atmospheric pressure is 750 mmHg.

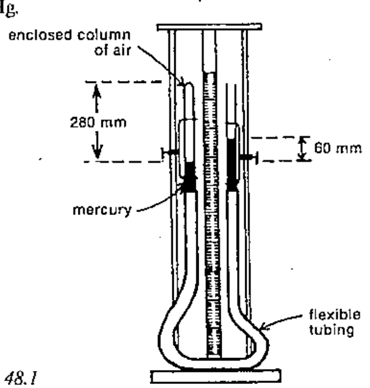


Fig. 48.1

(*a*) What is the pressure on the 280 mm column of air in the left-hand tube? Give the answer in mmHg.
The right-hand tube is now raised and the vertical distance between the two mercury surfaces is now 90 mm.
(*b*) What is the new pressure on the trapped air in mmHg?
(*c*) What is the new length of the enclosed column of air?

4 A glass tube with a uniform bore is closed at one end and contains a bead of mercury 9 cm long. The length of the air column between the closed end and the bead is
15.4 cm when the tube is held horizontally;
13.75 cm when held vertically, open end up;
17.5 cm when held vertically, open end down.
Given that the barometer reads 75.00 cm of mercury, show that these results are consistent with the Boyle's law statement that pressure is inversely proportional to volume when the temperature of a gas is constant.

5 A faulty barometer has some air trapped above the mercury. The height of the mercury column is 732 mm when the air space is 100 mm long. When the tube is pushed down into the bowl so that the air space is 50.0 mm long, the height of the mercury column is 716 mm. What is the correct atmospheric pressure
(*a*) in mmHg?
(*b*) in kPa?

6 A car tyre contains 0.028 m^3 of air after being pumped up to a gauge pressure of 250 kPa. The pressure read on a tyre gauge is the internal tyre pressure *in excess of* the pressure of the air outside the tyre. If the atmospheric pressure is 100 kPa,
(*a*) what is the pressure of the air in the tyre?
(*b*) to what volume will this air expand if a blowout occurs?

7 A 120 m^3 balloon to carry cosmic ray experiments to the upper atmosphere is to be filled with hydrogen at atmospheric pressure (100 kPa). If the hydrogen comes in gas cylinders with an internal volume of 0.042 m^3 and an internal pressure of 20 MPa, how many such cylinders will be needed? Assume the volume of the balloon remains effectively constant as it rises.

8 A quantity of gas has a volume of 120 cm^3 at a temperature of 7°C. What would be its volume at 77°C, the pressure remaining constant?

9 At what temperature (in °C) will the volume of a given mass of gas be
(*a*) twice, and
(*b*) one-third
of its volume at 0°C, if its pressure does not alter?

10 Suppose the mercury surfaces were in the positions shown in Figure 48.1 at a temperature of 15°C early in the day. Later in the afternoon direct sunlight through a laboratory window raises the temperature of the apparatus. The right-hand tube is moved to bring the vertical distance between the two surfaces back to 60 mm. The new length of the air column is 297.5 mm. The pressure of the atmosphere has remained constant during the day.
(*a*) Was the right-hand tube moved up or down?
(*b*) What is the new temperature of the apparatus?
(*c*) If the atmospheric pressure was 101 kPa, what was the constant pressure (in kPa) on the enclosed column of air when the two measurements were taken?

11 Suppose you live in an average sized house and the air temperature inside has dropped to about 12°C overnight. In the morning you turn on heaters to raise the temperature to about 18°C. What is the order of magnitude of the volume of air you expel from the house while warming it up?

12 An aeroplane carries a supply of oxygen in cylinders at an internal pressure of 1.0 MPa measured at the ground temperature of 27°C. When the aeroplane reaches its cruising altitude, the temperature of the cylinders falls to −3°C. Assuming that the change in the volume of the cylinders is negligible, what is the pressure inside the cylinders when the cruising altitude is reached?

13 The pressure of the air in a constant volume gas thermometer is 804 mmHg at 0°C. What is the temperature of a beaker of water in which the thermometer bulb is immersed if the pressure on the enclosed air is 881 mmHg when constant volume is restored?

14 The tyres on a car are inflated until the tyre gauge reads 250 kPa. The air temperature is 24°C and atmospheric pressure is 100 kPa. After a long trip the driver tests her tyres again and the gauge reads 276 kPa. Assume the expansion of the tyres is negligible.
(*a*) What is the internal pressure in her tyres at the beginning of the trip? (The answer is not 250 kPa.)
(*b*) What is it at the end of her trip?
(*c*) What is the temperature of the air in the tyres at the end of the journey?

15 To determine the maximum internal pressure in glass tubes filled with argon gas at 27°C and 100 kPa, a number of them are slowly heated. If they burst at an average temperature of 177°C, what internal pressure can they withstand? Assume that the expansion of the glass is insignificant.

16 When dealing with the *mass* of atomic and subatomic particles, scientists often find it convenient to use the small (*unified*) *atomic mass unit* (abbreviated u) rather than the inconveniently large SI unit of mass. 1 u is defined as exactly one-twelfth of the mass of the atom of carbon which contains 6 protons and 6 neutrons. The *gram* is related to the *atomic mass unit* as follows:

$$1\ \text{g} = 6.02 \times 10^{23}\ \text{u}$$

It is handy to be able to refer to a particular number of atoms, molecules, etc., which indicate a measurable quantity of the substance. The quantity we have selected is called the *mole* and is 6.02×10^{23}. Just as we have 12 per *dozen* and 2 per *pair*, here we use 6.02×10^{23} per *mole*, written $6.02 \times 10^{23}\ mol^{-1}$ and known as the *Avogadro constant*. (It should be noted that mol is the accepted abbreviation for mole, not molecule.)

A periodic table of the elements shows that the *relative* atomic mass of helium is 4.0. So the mass of 1 helium atom = 4.0 u.

(*a*) What is the mass (in u) of 20 helium atoms?
(*b*) What is the mass (in u) of 1 mole or 6.02×10^{23} atoms of helium?
(*c*) Remembering that there are 6.02×10^{23} u in one gram, what is the mass (in g) of 1 mole of helium atoms?
(*d*) If the relative atomic mass of aluminium is 27, what is the mass (in g) of 1 mole of aluminium atoms?

17 Neon has a relative atomic mass of 20. It is a *monatomic* gas, i.e. each independent molecule consists of *one* atom.
(*a*) What is the mass (in u) of
 (*i*) 1 atom of neon?
 (*ii*) 1 molecule of neon?
 (*iii*) 4 molecules of neon?
(*b*) What is the mass (in g) of
 (*i*) 1 mole of neon molecules?
 (*ii*) 4.5 mole of neon molecules?

18 Oxygen is a diatomic gas, i.e. each molecule consists of two atoms. Its relative atomic mass is 16.
(*a*) What is its relative molecular mass?
(*b*) What is the mass (in u) of 3 molecules of oxygen?
(*c*) What is the mass (in g) of
 (*i*) 1 mole of oxygen atoms?
 (*ii*) 6.02×10^{23} oxygen molecules?
 (*iii*) 0.25 mole of oxygen molecules?

19 How many mole of students are there in your school? Give your answer as an order of magnitude.

20 Each molecule of water vapour consists of two atoms of hydrogen and one of oxygen. Hydrogen and oxygen have relative atomic masses of 1.0 and 16 respectively.
(*a*) What is the relative molecular mass of water vapour?
(*b*) How many moles of molecules are there in
 (*i*) 3.01×10^{24} molecules?
 (*ii*) 36 g of water vapour?
(*c*) What is the total number of atoms in 5.0 mole of water vapour molecules?

21 The main constituents of air are nitrogen (approximately 80 per cent) and oxygen (approximately 20 per cent). The relative atomic masses of nitrogen and oxygen are 14 and 16 respectively, and both are diatomic gases.
(*a*) What is an approximate "relative molecular mass" for air?

The density of air at normal temperatures and pressures is about $1.2\ kg\ m^{-3}$. Give the following answers as orders of magnitude.
(*b*) What would be the volume of air in your bedroom?
(*c*) What is the mass of air in your bedroom?
(*d*) How many moles of "air molecules" does the room contain?
(*e*) How many molecules of "air" are there in the room?

22 A sample of a certain gas occupies $1.47 \times 10^{-3}\ m^3$ at a temperature of 18°C and a pressure of 102 kPa.
(*a*) How many moles of gas molecules are there in the sample?
(*b*) If the mass of this sample is 1.24 g, what is the relative molecular mass of the gas?
(*c*) Use a periodic table to see if you can identify the gas.

23 Commencing with the ideal gas equation, $PV = nRT$, deduce expressions giving
(*a*) the pressure–volume product PV of a gas sample in terms of
 m, the mass of the gas sample,
 M, the mass of 1 mole of the gas,
 T, the thermodynamic temperature of the gas;
(*b*) the pressure P of a gas sample in terms of
 ρ, the density of the gas at pressure P and thermodynamic temperature T,
 M,
 R, and
 T.

24 A carbon dioxide cylinder has an internal volume of $0.018\ m^3$. If the internal pressure is 12 MPa and the temperature is 20°C, what mass of carbon dioxide does the cylinder contain? Relative molecular mass of carbon dioxide is 44.

25 1.0 g of hydrogen occupies $1.12 \times 10^{-2}\ m^3$ at STP (standard temperature and pressure of 0°C and 101 kPa). What would be the volume of 3.0 g of this gas at a temperature of 52°C and a pressure of 126 kPa?

26 If the density of helium at STP is $0.178\ kg\ m^{-3}$, what is its density in a party balloon at 26°C and 202 kPa?

27 If the temperature and pressure inside an argon filled light globe of internal volume $114\ cm^3$ is 130°C and 96 kPa, what mass of argon is in the globe? The density of argon at STP is $1.78\ kg\ m^{-3}$.

28 An exhaust pump extracts one-tenth of the air in a bell jar at each stroke. If the original pressure in the jar is 100 kPa, determine the pressure after three strokes of the pump.

29 A gas cylinder A contains oxygen at a pressure of 600 kPa. An identical cylinder B contains the same gas at a pressure of 200 kPa, and both cylinders are at room temperature, 17°C.
(*a*) Find the values of the ratios
 (*i*) $\dfrac{\text{number of oxygen molecules in A}}{\text{number of oxygen molecules in B}}$;
 (*ii*) $\dfrac{\text{density of the gas in A}}{\text{density of the gas in B}}$.
(*b*) Cylinder A is heated to a temperature of 127°C, the volume change in the cylinder being negligible. Find the value of the ratio
 $\dfrac{\text{pressure of the gas in A at 127°C}}{\text{pressure of the gas in A at 17°C}}$.
(*c*) If the oxygen in cylinder A, again at 17°C, were allowed to escape into the air at 100 kPa pressure, what would be the value of the ratio
 $\dfrac{\text{volume of the gas at atmospheric pressure}}{\text{volume of cylinder}}$?
(*d*) The cylinder A is connected to a third cylinder C which has half the volume of A and contains oxygen at a pressure of 200 kPa. Both cylinders are at room temperature. Find the value of the ratio
 $\dfrac{\text{pressure in the cylinders A and C}}{\text{original pressure in A}}$.

30 Two identical glass bulbs are joined by a narrow tube of negligible volume, and the whole is initially filled with gas at STP and sealed. What will be the pressure of the gas when one of the bulbs is maintained at 0°C and the other at 100°C?

31 A closed flask contains a mixture of gaseous nitrogen and carbon dioxide with partial pressures of 70 kPa and 35 kPa respectively.
(*a*) What is the total pressure inside the flask?
(*b*) If the carbon dioxide were extracted without any other changes taking place, what would the pressure then be in the flask?

32 Two flasks having equal volumes are connected by a narrow tube with a tap which is closed. The pressure of the air in one flask is double that in the other. After the tap is opened the common pressure is 96.0 kPa. Find the original pressures in the flasks.

33 A closed vessel contains a mixture of a gas and water vapour in contact with excess water. The pressures in the vessel at 27°C and 60°C are respectively 102.7 kPa and 130 kPa. Assuming that the gas is not soluble in water, and given that the saturated vapour pressure of water at 27°C is 3.6 kPa, calculate the saturated vapour pressure of water at 60°C.

34 A graduated jar is filled with water and inverted in a trough of water. Oxygen is collected in the jar until the level of the water inside the jar is the same as that outside. The volume of oxygen collected is $250\ cm^3$, the temperature is 30°C, and the atmospheric pressure is 100 kPa. Calculate the mass of oxygen in the jar.
The density of oxygen at 0°C and 101 kPa pressure is $1.43\ kg\ m^{-3}$; the saturation vapour pressure of water at 30°C is 4.2 kPa.

35 On a particular day the air temperature is 15°C and the dew point is 11°C. The saturated vapour pressures of water at 15°C and 11°C are 1.71 kPa and 1.31 kPa.
(*a*) What is the relative humidity?
(*b*) What is the absolute humidity, i.e. the mass of water vapour per unit volume of the air? The relative molecular mass of water is 18.0.

36 In a laboratory the dew point is found to be 12°C when the air temperature is 16.5°C. Using the saturation vapour pressures of water given in Table 48.1, find the relative humidity of the air.

Table 48.1

Temperature (°C)	Saturation vapour pressure (kPa)
12	1.40
16	1.82
17	1.94

37 A cubic metre of air, with a relative humidity of 70.0 per cent, is dried without changing its pressure and temperature which are 100.0 kPa and 20°C respectively. Calculate the change in volume. The saturation vapour pressure of water at 20°C is 2.3 kPa.

38 If seven very slow molecules had the following speeds (in $m\ s^{-1}$): 2.0, 3.0, 4.0, 5.0, 8.0, 8.0, 12.0, what would be
(*a*) the mean or average speed?
(*b*) the most probable speed?
(*c*) the median speed?
(*d*) the root mean square speed?

39 Determine the average translational kinetic energy of a gas molecule at a temperature of 300 K.

40 Air has an approximate composition by volume of 79 per cent nitrogen, 20 per cent oxygen and 1 per cent argon. Nitrogen and oxygen are diatomic, while argon is monatomic.
(*a*) Determine the average translational kinetic energy of
 (*i*) an argon molecule, if the air temperature and pressure are 17°C and 101 kPa respectively;
 (*ii*) an oxygen molecule at the same temperature and pressure;
 (*iii*) an argon molecule, if the temperature and pressure are 17°C and 202 kPa;
 (*iv*) an argon molecule, if the temperature is 97°C?

(*b*) If oxygen has a relative atomic mass of 16.0, what is the mass of an oxygen molecule *in kg*?
(*c*) Determine the root mean square speed of an oxygen molecule in the air when the temperature is 17°C.
(*d*) A nitrogen molecule has a greater average energy than an argon molecule at the same temperature, even though they have the same average translational kinetic energy. What additional forms of energy does the nitrogen molecule possess over the argon molecule?

41 The pressure of a sample of a gas is doubled while the temperature remains constant,

$$\text{i.e. } \frac{\text{pressure in final state}}{\text{pressure in original state}} = 2.$$

For each of the following quantities, what is the ratio of the value of the quantity in the final state of the gas to its value in the original state?
(*a*) volume;
(*b*) density;
(*c*) number of molecules per unit volume;
(*d*) root mean square speed of the molecules;
(*e*) number of impacts per second per unit area of the wall of the container.

42 The pressure of a sample of gas is held constant and the thermodynamic temperature is doubled,

$$\text{i.e. } \frac{\text{temperature in final state}}{\text{temperature in original state}} = 2.$$

For each of the following quantities, what is the ratio of the value of the quantity in the final state of the gas to its value in the original state?
(*a*) number of molecules per unit volume;
(*b*) momentum transfer per second per unit area of the wall of the container;
(*c*) root mean square speed of the molecules.

43 Two relations predicted by the kinetic molecular theory of gases are

(1) $\frac{1}{2}m\overline{v^2} = \frac{3}{2}kT$,
where m = the mass of a molecule,
$\overline{v^2}$ = the mean of the squares of the speeds of all the molecules,
k = the Boltzmann constant,
T = the thermodynamic temperature, and

(2) $\bar{v} = \sqrt{8kT/\pi m}$,
where $\bar{v}$ = the mean or average speed of the molecules.

(*a*) Deduce an expression for the mean speed $\bar{v}$ in terms of the root mean square speed $\sqrt{\overline{v^2}}$.
In Problem 40 it was found that the average translational kinetic energy of an argon molecule in the air at 17°C is 6.00 $\times 10^{-21}$ J.
(*b*) What is the mass of an argon molecule in *kilogram*?
(*c*) Determine the mean square speed of the argon molecules.
(*d*) What is their root mean square speed?
(*e*) What is their mean speed?

44 Figure 48.2 shows two containers both filled with helium gas. The pressures, volumes and temperatures are indicated.

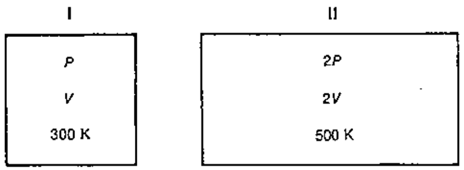


Fig. 48.2

Giving answers to two significant figures, determine the values of the following ratios:

(*a*) $\dfrac{\text{number of helium molecules in I}}{\text{number of helium molecules in II}}$;

(*b*) $\dfrac{\text{mean translational kinetic energy of helium molecules in I}}{\text{mean translational kinetic energy of helium molecules in II}}$;

(*c*) $\dfrac{\text{root mean square speed of helium molecules in I}}{\text{root mean square speed of helium molecules in II}}$;

(*d*) $\dfrac{\text{average magnitude of momentum of helium molecules in I}}{\text{average magnitude of momentum of helium molecules in II}}$;

(*e*) $\dfrac{\text{rate of momentum transfer to unit wall area in I}}{\text{rate of momentum transfer to unit wall area in II}}$.

45 The molecules of a monatomic gas at thermodynamic temperature T have root mean square speed v and the relative atomic mass of the element is M. Another gaseous element, which is diatomic, has molecules with a root mean square speed of $\frac{1}{2}v$ and a relative atomic mass of $2M$. What is the thermodynamic temperature of the second gas in terms of T?

46 The molecules of two well-known gases, which we will denote as A and B, have an average translational kinetic energy of 6.21 $\times 10^{-21}$ J at 27°C, as of course do the molecules of all gases at this temperature. The molecular masses of A and B are 7.31 $\times 10^{-26}$ kg and 6.65 $\times 10^{-27}$ kg respectively.

(*a*) Determine the root mean square speed of the molecules of A and B, giving your answers in km s^{-1}.
(*b*) The escape speed from the Earth is 11.2 km s^{-1}. Which of the gases A and B is likely to be found in lower concentration in the Earth's atmosphere? Why is it so?

47 The *average* translational kinetic energy of the molecules of a gas equals $\frac{3}{2}kT$, where k is the Boltzmann constant and T is the thermodynamic temperature.
(*a*) Deduce an expression for the *total* translational kinetic energy of the molecules in a gas sample in terms of only the pressure P and volume V of the gas.
(*b*) What is the total translational kinetic energy of the air molecules inside a basket ball with an internal diameter of 0.23 m and internal pressure of 300 kPa?

48 By imagining a sample of gas divided up into little cubes, each containing one molecule, determine the average separation of the molecules in the air around you. Give your answer as an order of magnitude.

49 Consider a very large sphere, say, one which would just contain Australia; its diameter would have to be about 4200 km.
(*a*) What would be the volume of this sphere in m^3?
(*b*) If one molecule of nitrogen (relative molecular mass = 28) were inserted in each cubic metre of this sphere, what total mass of nitrogen would be required? The answer might surprise you.

50 The *mean free path* or average distance between successive collisions for a gas molecule is given by

$$\lambda = V/\sqrt{2}\pi N d^2,$$

where λ = the mean free path,
N = the number of molecules in the gas sample,
V = the volume of the sample, and
d = the molecular diameter.

(*a*) In order to determine the mean free path in the air around you, determine first the number of molecules per unit volume, N/V, for air at a typical pressure of 101 kPa and temperature of 20°C.
(*b*) Using your answer to (*a*) find the mean free path for the molecules in the air about you. Take 3 $\times 10^{-10}$ m as the approximate diameter of an "air molecule".
(*c*) In order to determine a value for the collision frequency of the air molecules, i.e. the number of times a molecule in the air collides with other molecules per unit time, find first the mean speed of an air molecule at the temperature and pressure we are considering. Refer to Problem 43 above. Take the approximate "relative molecular mass of air" as 29.
(*d*) Then deduce a value for the collision frequency of an air molecule. It is probably higher than you expected.

51 One prediction of the kinetic molecular theory is

$$P = \tfrac{2}{3}E_k N/V,$$

where P = the pressure of the gas sample,
E_k = the average translational kinetic energy of the molecules,
N = the number of molecules in the sample, and
V = the volume of the gas.

(*a*) Deduce an expression giving the pressure P of a gas in terms of its density ρ and the mean square speed $\overline{v^2}$ of its molecules.
(*b*) You know—or you should know—that the mean square speed $\overline{v^2}$ of the molecules is proportional to the thermodynamic temperature of the gas. How is it that T does not appear in your answer to (*a*)?

52 Given that the density of oxygen at STP is 1.428 kg m^{-3}, determine the root mean square speed of the oxygen molecules under these conditions.

53 Graham found that the rate at which gases escape from a small orifice is inversely proportional to the square root of the density of the gas. Show that this "law", which is based on experimental evidence, can be deduced from the relations obtained from the kinetic molecular theory of gases. (Assume that the rate of escape varies as the average speed of the molecules.)

54 An container holds an equal number of hydrogen and oxygen molecules. If a small hole is made in the side of the container, what is the ratio of the rate at which hydrogen molecules diffuse through the hole to the rate at which oxygen molecules do? The relative molecular masses of hydrogen and oxygen are 2.0 and 32 respectively.

SET 49 CHANGE OF TEMPERATURE AND CHANGE OF PHASE

Topic	*Problems*
Change in temperature, heat capacity and specific heat capacity of solids and liquids	1–11
Mixtures of substances (other than two gases) involving changes in temperature but not in phase, water equivalent	12–17
Change in phase, specific latent heat of fusion and vaporisation	18–22
Change in temperature accompanied by a change in phase	23–25
Mixtures of substances (other than two gases) involving changes in temperature and phase	26–30
Newton's law of cooling, "cooling correction"	31–35
Dulong and Petit law, molar heat capacity	36–37

Where necessary in this set, use
specific heat capacity of water = 4.19 kJ kg^{-1} K^{-1}
specific latent heat of vaporisation of water = 2.26 MJ kg^{-1}
specific latent heat of fusion of ice = 334 kJ kg^{-1}

1 If a cup of coffee loses 10 kJ of heat while its temperature drops 20 degree on the Celsius scale, what is its heat capacity?

2 If the heat capacity of an electric kettle is 400 J K^{-1}, how much heat does it absorb as its temperature rises from 20°C to 95°C while heating some water for a cup of tea?

3 In a family room with an initial temperature of 10°C a wood fire is being started on a steel grate of heat capacity 1.50 kJ K^{-1}. What is the temperature of the grate by the time it has absorbed 540 kJ of heat from the fire?

4 How much heat is absorbed by a 2.8 kg brick sitting in the sun while its temperature rises from 18°C to 28°C? The specific heat capacity of a brick is 7.5×10^2 J kg^{-1} K^{-1}.

5 In an experiment to determine a value for the specific heat capacity of water two students found that the temperature of 0.15 kg of water rose from 17°C to 35°C as the water absorbed 11 kJ of heat energy. What value did they obtain?

6 If you had a suitable source which could provide you with 5 MJ of heat energy, as well as an unlimited supply of water (at room temperature), teabags, cups, teaspoons, etc., how many cups of tea could you make? Give your answer as an order of magnitude.

7 A solar flat plate collector for a domestic hot water system on a Sydney home has a collector area of 3.2 m^2. At one stage during the day when the collector happens to be at right angles to the Sun's rays, water is moving through it at the rate of 12 g s^{-1}. Radiant energy is falling on the collector at the rate of 210 W m^{-2}, and 82 per cent of the incident energy is absorbed by the water. By how much does the temperature of the water rise as it passes through the collector?

8 In the continuous flow calorimeter shown in Figure 49.1 with the 200 W electric heater turned on, water flows through at the rate of 0.119 kg min^{-1}. When the two thermometer readings become steady, the various parts of the apparatus will have attained different but steady temperatures, and so no part of the apparatus is being heated up. If the inlet temperature is 16°C, what is the outlet temperature?

9 As a cloud cools from 24°C to 4°C, 72 kg of rain is formed. How much heat is released into the atmosphere during this process?

10 At what minimum speed would a steel bullet at a temperature of 21°C have to be travelling in order to reach a temperature of 1000°C on hitting a sheet of armourplate steel? Assume the bullet retains 70 per cent of the heat energy it generates. Specific heat capacity of steel = 4.8×10^2 J kg^{-1} K^{-1}.

11 On his honeymoon Joule measured the temperature at the bottom of a waterfall to see if it was higher than the temperature at the top.
(*a*) What increase in temperature would you expect at the Tully Falls near Ravenshoe in Queensland? They are 270 m high.
(*b*) In fact would you expect the temperature rise to be greater or less than your answer to (*a*)? Give a reasoned response.

12 The specific heat capacity of copper is 3.8×10^2 J kg^{-1} K^{-1}. If a copper calorimeter has a mass of 55 g, determine its
(*a*) heat capacity;
(*b*) water equivalent, i.e. the mass of water, which has the same heat capacity as the calorimeter.

13 54 g of water at 60°C is poured into a calorimeter which has a heat capacity of 25 J K^{-1} (or water equivalent 6.0 g) and which contains 120 g of water at 10°C. If the heat exchange with the surroundings is negligible, what is the highest temperature reached by the mixture?

14 A calorimeter of heat capacity 33 J K^{-1} (or water equivalent 8.0 g) contains 123.4 g of water at 16.2°C. To this is added some water at 37.3°C, and, after stirring, the highest temperature reached in the calorimeter is 21.0°C.

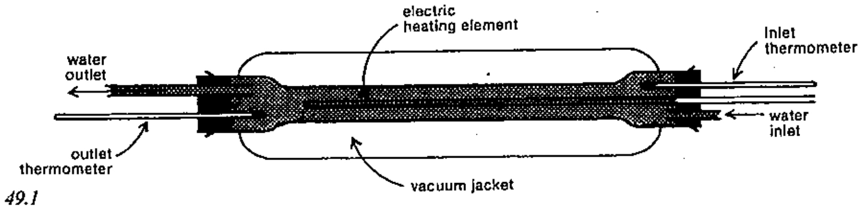


Fig. 49.1

(*a*) Assuming the heat losses to the surroundings are negligible, determine the mass of water added.
(*b*) Actually there would have been some heat exchange with the surroundings. Would the actual mass of water added be greater or less than your answer to (*a*)?

15 The single outlet connected to both taps over a kitchen sink is supplied with cold water at 8.0°C and hot water at 64°C. The cold water tap is giving a flow of 8.0 litre min^{-1} and the hot water tap is then adjusted to give a total flow rate of 14.0 litre min^{-1} through the outlet. What is the resulting temperature of the warm water?

16 In an experiment to find the specific heat capacity of iron, 30.0 g of iron tacks at 100°C are dropped into a calorimeter of heat capacity 25 J K^{-1} (or water equivalent 6.0 g) containing 54.0 g of water at 13.2°C, and the final temperature is found to be 18.0°C. What value does this give for the specific heat capacity of iron? Assume heat losses to the surroundings are insignificant.

17 You pour yourself a beer (or perhaps an orange drink) from the refrigerator into a glass which is at room temperature. Making sensible estimates, determine by how much the temperature of your drink will rise as a result of using an unchilled glass. Give your answer as an order of magnitude. Specific heat capacity of glass $\approx 8 \times 10^2$ J kg^{-1} K^{-1}.

18 If 4.2×10^3 J of heat energy are required to vaporise 5.0 g of ethanol at its boiling point, what is the specific latent heat of vaporisation of ethanol?

19 A 2100 W electric kettle containing 1.00 litre (i.e. 1.00 kg) of water has just come to the boil.
(*a*) How much energy is required to boil all the water away?
(*b*) How long (to the nearest minute) will it take for the kettle to boil dry?

20 Two adjacent units at a ski resort are identical apart from the fact that the owner of one has installed insulation in the ceiling of his unit. Just 4 h after the snow stops falling, the 80 m^2 roof of the insulated unit has 10 cm of snow on it as the last of the snow melts on the roof of the other unit.
(*a*) How much heat has been wasted by the owner of the uninsulated unit? Take the density of snow as 94 kg m^{-3}.
(*b*) If heat energy costs 0.65 cent MJ^{-1}, what approximate cost saving for the three month skiing period results from insulating the roof? Give your answer to the nearest $10.

21 A cross-country skier descends a 25° slope at a constant speed of 12 m s^{-1}. If all the work done by friction on the lower side of her skis goes into producing heat, what mass of snow does she melt in the 50 s she is descending the slope? Her total mass including skis and pack is 80 kg.

22 Liquid in a vacuum flask is boiled steadily by passing an electric current through a coil of wire immersed in the liquid. When energy is supplied to the coil at the rate of 806.4 J min^{-1}, liquid is changed into vapour at the rate of 0.34 g min^{-1}, and when energy is supplied at the rate of 302.4 J min^{-1}, liquid is changed into vapour at the rate of 0.12 g min^{-1}. Assuming that the rate of loss of heat from the liquid in the flask to the surroundings is H J min^{-1} in each case, write down two equations connecting L, the specific latent heat of vaporisation of the liquid, and H; hence calculate the value of L.

23 Heat is passed at a uniform rate into a vessel containing 50.0 g of ice at 0°C, and the whole of the ice is just melted in 10.0 min.
(*a*) Neglecting the heat gained by the vessel and any heat losses to the surroundings, calculate
(*i*) the energy being supplied per minute;
(*ii*) the additional time required to boil all the water away if heat continues to be supplied at the same rate.
(*b*) Actually heat exchanges with the vessel and the surroundings would occur. If these were taken into account, would the real answer to (*a*) (*i*) be higher or lower?

24 An ice-block tray holding 0.25 kg of water at 18°C is placed in the freezing compartment of a refrigerator. If it takes $1\frac{1}{2}$ h to form ice-blocks, at what rate was the refrigerator extracting heat from the water?

25 In Figure 49.2 a hypothetical temperature–time curve has been drawn for 0.60 kg of a solid which is warmed in a well-insulated container by a 100 W heater.

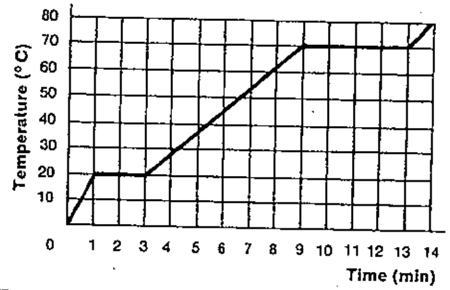


Fig. 49.2

(*a*) What is the melting point of the solid?
(*b*) Which is greater: the substance's specific latent heat of fusion or vaporisation?
(*c*) Which is greatest: the substance's specific heat capacity as a solid, liquid or gas?
(*d*) Determine its specific latent heat of fusion.
(*e*) What is its specific heat capacity as a liquid?

26 A copper calorimeter has a mass of 50 g and contains 85 g of water at 16°C. 6.0 g of dried ice is then added and the contents stirred until all the ice has melted. What is the temperature of the mixture? Specific heat capacity of copper $= 3.8 \times 10^2$ J kg^{-1} K^{-1}.

27 In an expresso coffee machine steam is bubbled through 200 g of coffee-flavoured milk taken from a refrigerator at 2°C. If the final temperature of the coffee is 90°C, what mass of steam condensed? Take the specific heat capacity of the coffee as 4.2 kJ kg^{-1} K^{-1}.

28 A bottle of lemonade is taken from a cupboard (not a refrigerator) at 20°C and 200 g is poured into a glass. Two 25 g ice-blocks are added. The glass has a mass of 120 g and a specific heat capacity of 8.4×10^2 J kg^{-1} K^{-1}. Take the specific heat capacity of lemonade as that of water.
(*a*) If we make the simplification that there is no heat exchange with the surroundings, what would be the temperature of the drink when half the ice has melted?
(*b*) In reality would its temperature be higher or lower than this?

29 A calorimeter which has a mass of 1.00 kg and a specific heat capacity of 8.4×10^2 J kg^{-1} K^{-1} is cooled to 0°C and 1.40 kg of ice at 0°C is placed in it. 1.00 kg of water at 100°C is then poured into the calorimeter. Find the resulting temperature.

30 A lump of copper is heated and placed on a large block of ice at 0°C, into which it sinks until it is three-quarters buried. What was its original temperature? The densities of copper and ice are 8.9×10^3 kg m^{-3} and 9.2×10^2 kg m^{-3}. The specific heat capacity of copper is 3.8×10^2 J kg^{-1} K^{-1}.

31 During dinner the room temperature is 18°C. If your soup has a temperature of 70°C, while your brother's which was served earlier is at 44°C, what is the ratio of the rate at which your soup is cooling down to that at which his is cooling?

32 The temperature of a body falls from 30.0°C to 20.0°C in 5.00 min, the air temperature being 13.0°C. Find the temperature after an equal time interval.

33 A kettle of water previously boiled cools from 62.0°C to 50.0°C in 10.0 min and to 42.0°C in the next 10 min interval.
(*a*) What is the temperature of the surroundings?
(*b*) Calculate the temperature of the kettle after another 10 min interval has elapsed.

34 A copper calorimeter of heat capacity 25 J K^{-1} contains 0.13 kg of glycerol and a 15 W immersion heater. With the room temperature at 20°C the heater is turned on for 4.0 min and then turned off. Temperature readings in the calorimeter are shown in Table 49.1.

Table 49.1

Time (min)	Temperature (°C)
0	20.0
1	22.6
2	25.1
3	27.5
4	30.0
5	29.7
6	29.5
7	29.3
8	29.0

(*a*) Draw a graph of temperature against time for the calorimeter and glycerol.
(*b*) What is the maximum temperature reached?
(*c*) What is the cooling rate (in °C min^{-1}) after the heater is turned off?
(*d*) What is the average "cooling rate" during the first 4 min due to heat loss to the surroundings?
(*e*) What "cooling correction" should be made to the maximum temperature observed to allow for these heat losses?
(*f*) Determine the specific heat capacity of glycerol.

35 In an experiment to determine specific heat capacity of a solid by the method of mixtures, the calorimeter and its contents were initially at 15.0°C, the room temperature being 20.0°C. The heated solid was immersed, and 0.750 min later the maximum temperature was reached at 30.0°C. The temperature then fell at a constant rate of 0.50°C min^{-1}. Find the "cooling correction" to be applied to give the corrected maximum temperature.

36 Table 49.2 shows the specific heat capacity at constant pressure c_P at 273 K and molar mass M for five metals. The column for the molar heat capacity c_M is yet to be filled in.

Table 49.2

Element	c_P (J kg^{-1} K^{-1})	M (kg mol^{-1})	c_M ( )
chromium	464	0.0520	
copper	381	0.0635	
gold	127	0.197	
iron	438	0.0558	
zinc	384	0.0654	

(*a*) What is meant by the *molar heat capacity* of a substance?
(*b*) Express c_M in terms of c_P and M.
(*c*) What is the missing unit for molar heat capacity in Table 49.2?
(*d*) Write down in order the missing molar heat capacities.
(*e*) Use the table to suggest approximately how much heat would be required to raise the temperature of an aluminium saucepan (about 20 mol of aluminium) from about 20°C to 100°C when boiling potatoes.

37 The relative atomic masses of cobalt and tungsten are 58.9 and 184 respectively. If the specific heat capacity of cobalt is 431 J kg^{-1} K^{-1}, find an approximate value for the specific heat capacity of tungsten.

SET 50 THERMODYNAMICS

Topic	*Problems*
Zeroth law of thermodynamics	1–2
Work done by a fluid expanding at constant pressure	3–7
First law of thermodynamics	8–9
Enthalpy	10–13
Molar heat capacities of a gas at constant pressure and constant volume, equipartition of energy	14–18
Relations between pressure, volume and temperature for the adiabatic expansion of a gas	19–21
Applications of the first law to isochoric, isobaric, isothermal and adiabatic processes	22–26
Mixtures of gases involving changes in pressure, volume and temperature but not phase	27
Carnot cycle, Carnot cycle efficiency, second law of thermodynamics	28–31
Entropy, heat engines, steam, internal combustion and diesel engines	32–41
Refrigerators, heat pumps, air conditioners, coefficient of performance	42–46
Third law of thermodynamics	47

Where necessary in this set use
molar gas constant,
$R = 8.31$ J mol^{-1} K^{-1}
Boltzmann constant,
$k = 1.38 \times 10^{-23}$ J molecule^{-1} K^{-1}
Avogadro constant,
$L = 6.02 \times 10^{23}$ mol^{-1}

1 For two bodies to be in thermal equilibrium what physical property must be identical for them?

2 A thermistor (an electronic temperature sensor) is in thermal equilibrium with a large tank of water. The thermistor is transferred quickly to a glass of milk and found to be at the same temperature as the glass of milk. If the glass of milk was then placed in contact with the tank, would there be a net flow of heat from the water to the milk, from the milk to the water, or neither?

3 If as a result of being heated a sample of air is allowed to expand from a volume of 1.20×10^{-3} m^3 to 1.70×10^{-3} m^3 at a constant pressure of 1.1×10^5 Pa, how much work has been done by the expanding air?

4 A sample of ideal gas is contained in a cylinder with a freely moving piston. The gas was initially at a temperature of 300 K, and at the pressure of the air in the laboratory, 10^5 Pa. The volume of the gas was 10^{-3} m^3. The sample of gas was heated, and allowed to expand so that its pressure remained constant until its volume was 2×10^{-3} m^3. This process is represented on the pressure–volume graph in Figure 50.1.

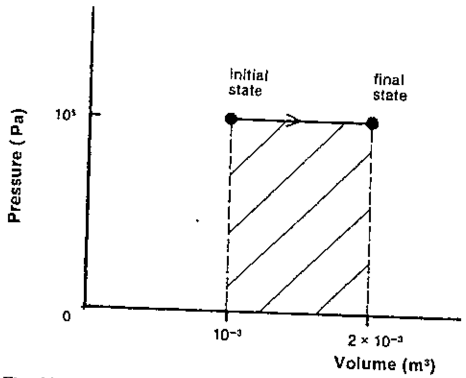


Fig. 50.1

(*a*) What was the final kelvin temperature of the gas?
(*b*) What physical quantity is represented by the area shaded on the graph?
(*c*) How much work was done during the expansion by the gas against the force exerted on the piston by the surrounding air?

5 Taking your average blood pressure as about 13 kPa and making suitable estimates of other quantities, determine as an order of magnitude how much work your heart does in a day.

6 A gas syringe with the outlet closed is held upright and filled with carbon dioxide. The plunger of mass 92 g and diameter 24 mm is fitted and let go. As it moves down the tube a short distance, the plunger does 3.6 mJ of work on the carbon dioxide gas.
(*a*) What pressure is exerted on the carbon dioxide by the weight of the plunger?
(*b*) How far did the plunger drop before coming to rest?
(*c*) In order to answer (*b*) which item of information given in the question was strictly not required?

7 Four balloons I, II, III and IV, each initially containing $0.040\,m^3$ of air at 1.5×10^5 Pa, are put up at various spots in a room in preparation for a party. The room heater is turned on. The four balloons undergo the changes (somewhat idealised) indicated in Figure 50.2.

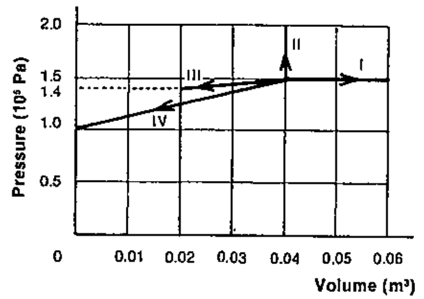


Fig. 50.2

(*a*) Which of the balloons (I, II, III or IV)
 (*i*) was not well sealed and went down?
 (*ii*) was made of very strong and inflexible rubber?
 (*iii*) was in contact with a window pane (it was very cold outside)?
 (*iv*) contained air which expanded at constant pressure (a very unusual balloon!)?
(*b*) Calculate the work done by the gas in
 (*i*) balloon I;
 (*ii*) balloon II;
 (*iii*) balloon III.

8 The first law of thermodynamics states that for a system

$$\Delta U = \Delta Q - \Delta W,$$

where ΔU = the change in the internal energy of the system,
ΔQ = the heat absorbed by the system, and
ΔW = the work done by the system.
If a gas syringe containing oxygen is thrust into a large container of hot water, by how much will the internal energy of the oxygen have increased if it absorbs 365 mJ of heat and does 25 mJ of work while expanding?

9 When a teaspoon of water (5.00 g or $5.00 \times 10^{-6}\,m^3$) is boiled at 100°C and 100 kPa pressure, $8355 \times 10^{-6}\,m^3$ of steam is formed. The specific latent heat of vaporisation of water is 2.26 MJ kg^{-1}.
(*a*) How much work is done by the expanding water?
(*b*) What heat energy is absorbed by the water while it boils?
(*c*) What is the increase in the internal energy of the water molecules?

10 The *enthalpy* of a system is defined by the equation

$$H = U + PV,$$

where H = the enthalpy of the system,
U = its internal energy,
P = its pressure, and
V = its volume.
(*a*) What would be the SI unit of enthalpy?
(*b*) For a system that has undergone some change, deduce the following expression for the change in enthalpy:

$$\Delta H = \Delta Q - \Delta W + \Delta(PV),$$

where ΔQ = the heat absorbed by the system, and
ΔW = the work done by the system.
(*c*) Chemists often carry out reactions at constant pressure (the pressure of the atmosphere). In this case show that

$$\Delta H = \Delta Q$$

11 Using the definitions of symbols given in Problem 10 above, show that, for changes in systems involving only solids and liquids and not gases,

$$\Delta H \approx \Delta U$$

12 When an 18 g ice-block melts at 0°C and 101 kPa, the volume changes from 19.6 cm^3 to 18.0 cm^3. The specific latent heat of fusion of ice is 334 kJ kg^{-1}. Determine
(*a*) the heat absorbed during the melting;
(*b*) the change in enthalpy;
(*c*) the change in the pressure–volume product;
(*d*) the change in internal energy. (Do your answers agree with the expression derived in Problem 11?)

13 Read Problem 9 in this set. Would the answer to (*a*), (*b*) or (*c*) be equal to the change in enthalpy for the boiling process?

14 Suppose that two identical samples of n mole of an ideal gas are raised from temperature T to temperature $T + \Delta T$, one change being carried out at constant pressure and the other one at constant volume, as shown in Figure 50.3, which shows the pressure–volume isothermals for the two temperatures.

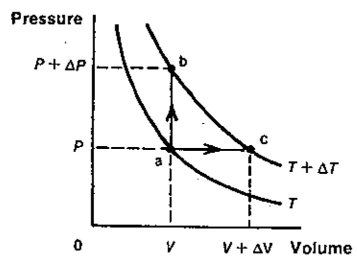


Fig. 50.3

(*a*) Is the final internal energy U_b greater than, equal to or less than U_c?

Use c_V and c_P to represent the molar heat capacities of the gas at constant volume and constant pressure respectively.
(*b*) Determine expressions for the change in internal energy ΔU for
 (*i*) the constant volume process, and
 (*ii*) the constant pressure process,
 in terms of symbols given above and in Figure 50.3.
(*c*) Deduce an expression for c_P in terms of c_V and R, the molar gas constant. Your answer (if correct!) applies not only to ideal gases, but reasonably well to real monatomic and polyatomic gases.
(*d*) The kinetic theory of gases gives the following expression for the internal energy of an ideal gas

$$U = \tfrac{3}{2}nRT$$

Deduce expressions in terms of R for
 (*i*) c_V;
 (*ii*) c_P.
The relations derived here apply to an ideal gas and reasonably well to monatomic gases but not to polyatomic gases.

15 3.33 kJ of heat is required to raise the temperature of 2.00 mol of helium from 20.0°C to 100°C at constant pressure.
(*a*) Determine the molar heat capacity of helium at constant pressure, c_P.
(*b*) Suggest an approximate value for the molar heat capacity of helium at constant volume, i.e. c_V.
(*c*) Deduce the value of the ratio c_P/c_V for helium. This ratio is often written as γ.
(*d*) Refer back to Problem 14 (*d*) and determine a value for γ by an alternate method. This result is typical of an ideal gas and monatomic gases such as helium, but not of polyatomic gases.

16 Figure 50.4 shows the variation of the molar heat capacity at constant volume c_V for hydrogen with thermodynamic temperature T.

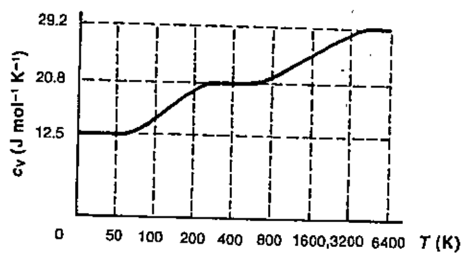


Fig. 50.4

(*a*) At 50 K, c_V the heat capacity *per mole* = 12.5 J mol^{-1} K^{-1} $\approx 3R/2$, where R is the molar gas constant. The heat capacity *per molecule* at this temperature would be $3R/2L$, where L (the Avogadro constant) is the number of molecules per mole of molecules, or $3k/2$, since k (the Boltzmann constant) = R/L. It follows that the average internal energy per molecule of hydrogen at about 50 K is therefore $3kT/2$, where T is the thermodynamic temperature.
For hydrogen at 400 K, express
 (*i*) c_V in terms of R;
 (*ii*) the average internal energy of a hydrogen molecule in terms of kT.
(*b*) For hydrogen at 6000 K, express
 (*i*) c_V in terms of R;
 (*ii*) the average internal energy of a hydrogen molecule in terms of kT.
(*c*) How many degrees of freedom does a hydrogen molecule have at 50 K? What types of energy do they represent?
(*d*) If the temperature of hydrogen is raised from 50 K to 400 K, how many additional degrees of freedom does a hydrogen molecule now have? What additional types of energy does the molecule now possess?
(*e*) If the temperature is now raised from 400 K to 6000 K, how many further degrees of freedom does the hydrogen molecule gain? What additional types of energy does the molecule now have?

17 If γ, the ratio of the molar heat capacities, for nitrogen is about 1.4 at normal temperature and pressure, estimate
(*a*) the molar heat capacity of nitrogen at constant volume;
(*b*) the molar heat capacity of nitrogen at constant pressure;
(*c*) the number of degrees of freedom possessed by a nitrogen molecule at normal temperature and pressure.

18 Argon is a monatomic gas. What would you expect to be the value of
(*a*) the molar heat capacity of argon at constant volume?
(*b*) the molar heat capacity of argon at constant pressure?
(*c*) the number of degrees of freedom of an argon molecule?
(*d*) the amount of heat required to raise the temperature of 0.020 mol of argon from 20°C to 100°C in a bomb calorimeter where the volume is held constant?

19 An *adiabatic* process, in which no heat enters or leaves the system, can be carried out by completely insulating the system or by carrying out the process very rapidly. The pressure P and volume V of a gas before and after an adiabatic expansion or contraction are related as follows:

$$P_1V_1^{\gamma} = P_2V_2^{\gamma},$$

where γ is the ratio of the molar heat capacities.
(*a*) From the relation given deduce equivalent expressions relating
 (*i*) V and thermodynamic temperature T, and
 (*ii*) P and T
of a gas before and after an adiabatic change.

(*b*) Does the temperature of a gas increase, decrease or remain the same during
(*i*) an adiabatic compression?
(*ii*) an adiabatic expansion?

20 A car tyre has a blowout. The actual internal pressure (tyre gauge pressure plus atmospheric pressure) was 350 kPa. The atmospheric pressure is 100 kPa. If the expansion is sufficiently rapid to be regarded as adiabatic, by what factor did the volume of the air in the tyre increase? Take γ, the ratio of the molar heat capacities for air, as 1.4.

21 If the compression stroke in a diesel engine reduces the gases to 1/15 of their original volume, and the rapid change can be regarded as adiabatic, estimate
(*a*) the pressure (to the nearest 100 kPa), and
(*b*) the temperature (to the nearest 100 K)
of the gases resulting from the compression stroke. Take the value of γ as 1.4.

22 Reread the first few lines of Problem 8 above.
(*a*) What does the relation $\Delta U = \Delta Q - \Delta W$ become in the case of
(*i*) an *isochoric* process, i.e one which takes place at constant volume?
(*ii*) an *isobaric* process, i.e. one which takes place at constant pressure?
(*iii*) an *isothermal* process, i.e. one taking place at constant temperature?
(*iv*) an *adiabatic* process, i.e. one in which there is no exchange of heat with the surroundings?
(*b*) If a sample of gas expands at a constant pressure P by an amount ΔV, write down a relation for ΔU (using only ΔQ, P and ΔV), if the process is
(*i*) an isobaric change;
(*ii*) an adiabatic change.

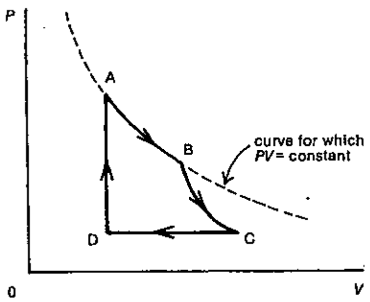


Fig. 50.5

23 A cyclic series of changes for a sample of a gas is shown on a P–V diagram in Figure 50.5. Which section (AB, BC, CD or DA) involves
(*a*) an isobaric change?
(*b*) an isothermal change?
(*c*) an isochoric change?
(*d*) an adiabatic change?

24 When 0.0500 mol of argon at 21°C and a pressure of 100 kPa is heated in a bomb calorimeter, the pressure rises to 150 kPa. Assume the expansion of the calorimeter is negligible. Molar heat capacity of argon at constant volume is 12.6 J mol^{-1} K^{-1}.
(*a*) Is this an isobaric, isothermal, isochoric or adiabatic change?
(*b*) What is the final temperature of the argon?
(*c*) How much heat has been absorbed by the gas?
(*d*) How much work has been done by the gas?
(*e*) What is the change in the internal energy of the argon?

25 A student draws 60 cm^3 of carbon dioxide into a gas syringe and seals the inlet. The plunger of the syringe is allowed to find its equilibrium position with the room temperature and pressure at 23°C and 100 kPa respectively. With the piston free to move, the syringe is heated until the volume of the gas reaches 75 cm^3. Molar heat capacity of carbon dioxide at constant volume is 28.6 J mol^{-1} K^{-1}.
(*a*) Is this change an isobaric, isothermal, isochoric or adiabatic process?
(*b*) What is the final temperature of the carbon dioxide?
(*c*) How many mole of carbon dioxide molecules are there in the syringe?
(*d*) Determine the change in the internal energy of the enclosed gas.
(*e*) How much work has been done by the gas as it expands?
(*f*) How much heat has the carbon dioxide absorbed?

26 Refer back to the changes to a sample of gas shown in Figure 50.5. Suppose the area between the curve AB and the V axis represents an energy of 14 J. In the expansion of the gas from A to B,
(*a*) what is the change in its internal energy?
(*b*) how much work is done by the gas?
(*c*) how much heat does it absorb?

27 1.00 litre of helium (41 millimole) at 23°C and 101 kPa is allowed to expand adiabatically into an evacuated 2.0 litre well-insulated container. For helium $c_V = 12.6$ J mol^{-1} K^{-1} and $\gamma = 1.30$.
(*a*) What is the final temperature of the helium?
(*b*) What is the change in its internal energy?
(*c*) How much work was done by the expanding gas?
3.00 litre of nitrogen (123 mmol) at 23°C and 101 kPa is then compressed into the vessel containing the helium. For nitrogen $c_V = 20.5$ J mol^{-1} K^{-1} and $\gamma = 1.42$.
(*d*) What temperature for the nitrogen results from this adiabatic compression?

(*e*) After the two gases have reached thermal equilibrium, what is their temperature?

28 Figure 50.6 shows the pressure–volume diagram for a rather inefficient reversible cycle for a heat engine, where an ideal gas is
heated at constant pressure from e to f,
cooled at constant volume from f to g,
cooled at constant pressure from g to h, and
heated at constant volume from h back to e.

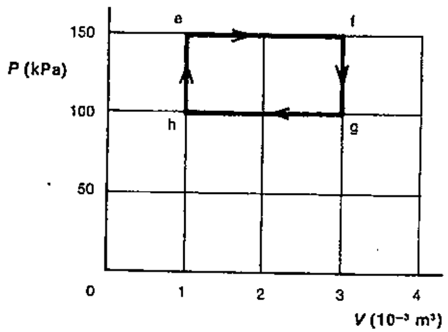


Fig. 50.6

(*a*) What is the work done by the gas in the change
(*i*) from e to f?
(*ii*) from f to g?
(*iii*) from g to h?
(*iv*) from h to e?
(*b*) What is the net work done by the gas during the full cycle?

29 A *Carnot cycle* for a heat engine using an ideal gas is shown in Figure 50.7. During each cycle the gas undergoes
an isothermal expansion from a to b, while absorbing heat Q_{ab} from the surroundings,
an adiabatic expansion from b to c,
an isothermal compression from c to d while losing heat Q_{cd} to the surroundings, and
an adiabatic compression from d to a.
The shaded area under the section of the graph cd represents an energy of 35 J. Similar areas beneath the sections ab, bc and da represent energies of 190 J, 85 J and 65 J respectively. $Q_{ab} = 875$ J.
(*a*) What total work is done by the gas while *expanding*?
(*b*) What is the net work done by the gas during one cycle?

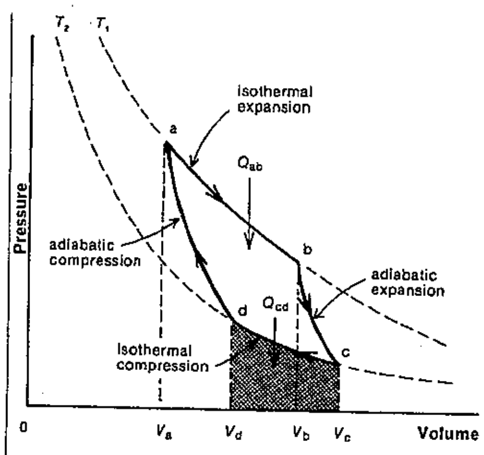


Fig. 50.7

(*c*) What is the net heat change for the gas during each cycle?
(*d*) What is the heat Q_{cd} lost to the surroundings during each isothermal compression cd?
(*e*) Determine the Carnot efficiency of this heat engine as a percentage.

30 A Carnot engine absorbs 440 J each cycle from a high temperature reservoir at 227°C. If it loses 330 J per cycle to its low temperature reservoir,
(*a*) what is the temperature of the low temperature reservoir?
(*b*) what is the (percentage) Carnot efficiency of this engine?

31 As an incorrigible optimist you decide to design a 95 per cent efficient Carnot engine using the atmosphere at normal temperatures as your hot reservoir. Estimate (to the nearest 5 K) the temperature of the cold reservoir you will require.

32 In each of the following cases select the item which has the greater "entropy":
(*a*) a cup of coffee before *or* after the sugar has been stirred in;
(*b*) uncombed hair *or* hair that has just been permed;
(*c*) two glasses of water, one at 19°C and one at 21°C, *or* two glasses of water both at 20°C;
(*d*) a diver leaving the 3 m board *or* the diver entering the water at the end of her dive;
(*e*) the Solar System in 1997 *or* in 1987;
(*f*) playing cards collected by the dealer immediately after a hand of poker *or* the pack a few moments later when it has been shuffled.

33 The specific latent heat of fusion of ice is 334 kJ kg^{-1} and the specific latent heat of vaporisation of water is 2.26 MJ kg^{-1}.

(*a*) What is the change in entropy of a 10 g ice-block which melts reversibly at 0°C?

(*b*) Would you expect the increase in entropy of 10 g of water changing reversibly into steam at 100°C to be greater or less than your answer to (*a*)? Why?

(*c*) To test your prediction determine the change in entropy for this boiling process.

34 What is the change in entropy of 50 mmol of argon which is heated reversibly from 20°C to 21°C at constant volume? The molar heat capacity of helium at constant volume is 12.6 J mol^{-1} K^{-1}.

35 The zero for the entropy of a substance is arbitrarily taken as its entropy at 0 K. Given only that the specific heat capacity of lead is less than that of aluminium, select from the values 3.5, 0.97 and 0.30 kJ K^{-1} the entropy of

(*a*) 1.0 kg of aluminium at 373 K;

(*b*) 1.0 kg of water at 373 K;

(*c*) 1.0 kg of lead at 373 K.

36 This problem considers the entropy changes of an object and its surroundings during a typical *irreversible* process. We will consider the melting of a 10 g ice-block in a kitchen sink where the temperature is 20°C. Specific latent heat of fusion of ice = 334 kJ kg^{-1}.

(*a*) What energy is absorbed by the ice while it melts at 0°C?

(*b*) What is the change in entropy of the ice as it melts at 0°C?

(*c*) The surroundings provided this energy at a temperature of 20°C. What is the change in entropy of the surroundings while this energy was supplied to the melting ice?

(*d*) What then is the total change in entropy of the melting ice and its surroundings?

(*e*) What aspect of your answer to (*d*) is typical of all normal irreversible processes?

37 Refer to Figure 50.7 and Problem 29 above.

(*a*) Write expressions in terms of Q_{ab}, Q_{cd}, T_1 and T_2 for the entropy change of the gas for

(*i*) the isothermal expansion ab;

(*ii*) the adiabatic expansion bc;

(*iii*) the isothermal compression cd;

(*iv*) the adiabatic compression da;

(*v*) a complete cycle.

(*b*) Since for a Carnot cycle $Q_{ab}/Q_{cd} = T_1/T_2$, deduce a simpler expression than your answer to (*a*) (*v*) for the total change in entropy for a complete cycle.

38 A modern steam turbine in a power station might use superheated steam at about 600°C and exhaust condensed water at about 30°C.

(*a*) Express as a percentage its Carnot cycle efficiency.

(*b*) Which of the following do you think would approximate the actual efficiency of a modern steam turbine: 38 per cent, 65 per cent or 78 per cent?

39 The cycle in the internal combustion or petrol engine approximates the *Otto cycle* shown in Figure 50.8.

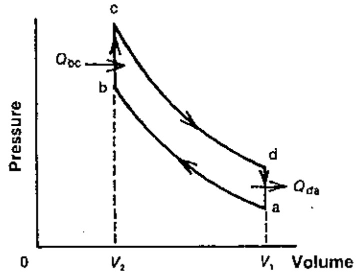


Fig. 50.8

(*a*) Which change in the diagram (ab, bc, cd or da) represents

(*i*) the compression stroke?

(*ii*) the period when the exhaust valve is open?

(*iii*) the working or power stroke?

(*b*) Using symbols in Figure 50.8, write an expression which equals the quantity represented by the area enclosed by the lines ab, bc, cd and da.

40 Refer to Figure 50.8 showing the Otto cycle, which is an idealised form of the petrol engine cycle.

(*a*) If T_1 and T_2 represent the initial and final thermodynamic temperatures for the adiabatic compression ab, show that the efficiency *e* is given by

$$e = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

(*b*) If the compression ratio V_1/V_2 in a Mazda 1300 is 9.2 and the ratio of heat capacities γ is 1.4, which of the following would you expect to be closest to the actual efficiency of the engine?
25 per cent, 59 per cent or 67 per cent

(*c*) If in each of the four cylinders of this Mazda engine Q_{bc} = 430 J and Q_{da} = 30 J and the engine while idling is executing 600 r.p.m. (revolutions per minute), what is the power of the idling engine? (During each revolution of the engine, the petrol–air mixture is exploded in only two of the four cylinders.)

41 A slightly idealised cycle for the diesel engine is shown in Figure 50.9.

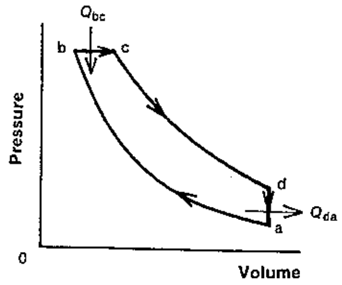


Fig. 50.9

(*a*) At which point (a, b, c or d) in the cycle

(*i*) is the diesel fuel injected into the cylinders?

(*ii*) does the exhaust valve open?

(*b*) If the thermodynamic temperatures at points a, b, c and d are represented by T_a, T_b, T_c and T_d respectively, show that the ideal efficiency of a diesel engine is given by

$$e = 1 - \frac{T_d - T_a}{\gamma(T_c - T_b)}$$

42 Refrigerators, heat pumps and air conditioners are really heat engines in reverse, with work *W* being done to transfer heat Q_2 from a low temperature reservoir at thermodynamic temperature T_2 to a reservoir at a higher temperature T_1 where heat Q_1 is released.

(*a*) Write an expression for *W* in terms of Q_1 and Q_2.

The coefficient of performance of a refrigerator is defined as follows

$$\eta = \frac{\text{extracted heat}}{\text{input work}}$$

(*b*) Deduce expressions for η in terms of

(*i*) Q_1 and Q_2 for a real refrigerator;

(*ii*) T_1 and T_2 for an ideal, i.e. a reverse cycle Carnot, refrigerator.

43 The freezing compartment in a refrigerator is at −9°C and the condenser unit outside the fridge is at 31°C.

(*a*) What would be the ideal coefficient of performance for this fridge?

(*b*) If it has an efficiency one-third that of an ideal fridge operating at the same temperatures, at what rate is it extracting heat from the freezing compartment, while the 180 W motor is running?

44 An ice-block tray containing 0.61 kg of water at 20°C is placed in a freezer with an internal temperature of −22°C and external condenser temperature of 34°C. The freezer is 30 per cent as efficient as an equivalent reverse cycle Carnot engine. It is driven by a 230 W motor. The specific heat capacity of water is 4.2 kJ kg^{-1} K^{-1}. The specific latent heat of fusion of ice is 330 kJ kg^{-1} K^{-1}.

(*a*) What would be the coefficient of performance of this freezer if it performed ideally?

(*b*) What is the actual coefficient of performance of the freezer?

(*c*) In order to change the water into ice-blocks at 0°C, how much energy must be

(*i*) extracted from the freezer?

(*ii*) supplied by the motor?

(*d*) How long do the ice-blocks take to form? Give your answer to the nearest minute.

45 During winter a householder used a heat pump to transfer heat from his inground swimming pool to the inside of his house. The expansion coils in the pool were at a temperature of 10°C, while the condenser coils in the house were at 32°C. The heat pump was driven by a 200 W motor. The pump's efficiency is 31 per cent of that of an ideal reverse cycle heat engine.

(*a*) Calculate the actual coefficient of performance of the heat pump.

(*b*) At what rate is heat energy being generated inside the house by the heat pump?

46 An air conditioner is required to operate with the following average temperatures:

evaporating coils (inside the family room window)	15°C
condensing coils (outside the window)	33°C

(*a*) What would be the coefficient of performance for a reverse cycle Carnot engine operating at these temperatures?

(*b*) The air conditioner cycle is 25 per cent as efficient as the ideal Carnot cycle. If it is required that heat be removed from the house at a rate of 720 W, what is the power of the required motor?

47 Consider a Carnot engine operating at very low temperatures.

(*a*) Theoretically, at what temperature could a system undergo a reversible isothermal process without a transfer of heat?

(*b*) If it were possible to attain 0 K for the lower temperature reservoir for a Carnot engine, what would be the efficiency of such an engine?

SET 51 TRANSFER OF HEAT

Topic	*Problems*
Conduction through a single material, Fourier's law, thermal conductivity, temperature gradient, Searle's apparatus	1-6
Conduction through two or more materials in contact, Lee's apparatus	7-12
R-value of building materials	13-15
Convection	16-19
Newton's law of cooling (see Set 49)	
Radiation, Stefan's law (or Stefan–Boltzmann law)	20-25
Wien's law (or Wien's displacement law)	26-27

Where necessary in this set, use
Stefan–Boltzmann constant (or Stefan constant), $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}
Wien constant $= 2.9 \times 10^{-3}$ m K

1 The temperatures on the two sides of an 8.0 mm thick plaster wall are 20°C and 16°C. If the area of the wall is 17 m^2 and plaster has a thermal conductivity of 0.13 W m^{-1} K^{-1}, determine
(*a*) the temperature gradient in the plaster;
(*b*) the heat flow rate through the plaster.

2 Figure 51.1 shows one version of a Searle's apparatus used for measuring the thermal conductivity of aluminium. It consists of a well-insulated rod of aluminium of cross-sectional area 8.0×10^{-4} m^2 with an electric heater embedded in one end. Thermistors at points A and B, 64 mm apart, register temperatures of 67.9°C and 58.0°C respectively. Water flowing through the metallic tubing at the left-hand end at a rate of 1.2 g s^{-1} has its temperature raised from 18.0°C to 23.9°C. The specific heat capacity of water is 4.2 kJ kg^{-1} K^{-1}.

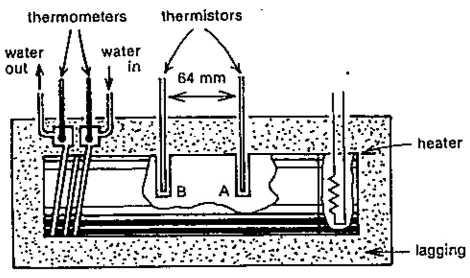


Fig. 51.1

(*a*) Determine the thermal conductivity of aluminium.
(*b*) At what rate is heat being absorbed by the circulating water?
(*c*) Would the power of the electric heater be actually equal to, less than or greater than your answer to (*b*)?

3 A rod of copper of which the thermal conductivity is 3.8×10^2 W m^{-1} K^{-1} is maintained at 500°C at one end while the other is in contact with a block of ice. The rod is well insulated along its sides and is 1.0 m long and 2.0 cm^2 in cross-sectional area.
(*a*) How much heat energy is conducted along the rod in 1.0 min?
(*b*) As a result what mass of ice melts per minute, given that the specific latent heat of fusion of ice is 334 kJ kg^{-1}?

4 It is necessary to maintain a laboratory at a constant temperature of 20°C. The laboratory is 10 m long, 9.0 m wide and 6.0 m high and has concrete walls and ceiling all 0.30 m thick. The temperature is maintained by means of electric radiators. The loss of heat through the floor can be regarded as negligible. The outside temperature is 10°C, and the thermal conductivity of concrete is 1.05 W m^{-1} K^{-1}.
(*a*) Calculate the total rate of heat flow through the walls and ceiling.
(*b*) What is the electrical power used in maintaining the laboratory temperature?

5 A storage tank, a cube of 1.0 m edge, is filled with water at 98°C and is lined on all sides with felt insulation 10 mm thick. Thermal conductivity of felt = 0.038 W m^{-1} K^{-1}.
(*a*) At what rate is heat energy lost by conduction if the outer surface of the felt is at a temperature of 18°C?
(*b*) How many minutes would it take for the temperature of the water to drop to 97°C? Density of water = 1.0×10^3 kg m^{-3}; specific heat capacity of water = 4.2 kJ kg^{-1} K^{-1}.

6 Your internal body temperature is about 37°C while your skin temperature is about 33.5°C. If these two regions are separated by a layer of fat about 2 cm thick, estimate as an order of magnitude the amount of energy you lose per day by conduction. Thermal conductivity of fat = 0.21 W m^{-1} K^{-1}.

7 Figure 51.2 shows a sectional view through one form of Lee's apparatus for determining the thermal conductivity of a poor conductor. A 40.0 W electric heater in a cavity is sandwiched between two thin circular discs of polypropylene each 0.550 mm thick. These are in turn sandwiched between two copper discs 22.6 mm thick. All the discs have a diameter of 120 mm. Three thermocouples enable temperatures θ_1 and θ_3 to be measured. θ_1 is 69.5°C. The thermal conductivities of polypropylene and copper are 0.120 and 400 W m^{-1} K^{-1} respectively.

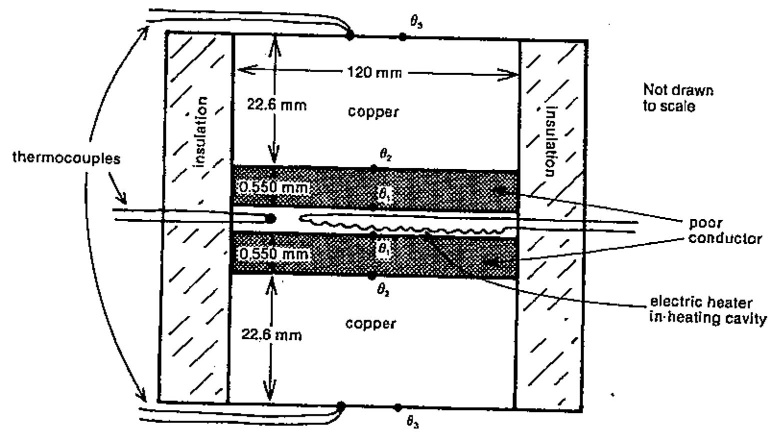


Fig. 51.2

(*a*) At what rate is heat flowing through the upper polypropylene and copper discs?
(*b*) Determine temperature θ_2.
(*c*) Determine temperature θ_3.
(*d*) In which of the two materials can the temperature gradient be regarded as negligible? (You should now realise why Lee's apparatus does not require thermocouples at the *lower temperature surface* of the poor conductor.)

8 In a Lee's apparatus similar to that shown in Figure 51.2, the poor conductor consists of 0.66 mm thick discs of cork of radius 60 mm. When thermal equilibrium is reached using a 30 W heater, $\theta_1 = 84$°C and $\theta_3 = 61$°C.
(*a*) At what rate is heat energy transferred through each disc of cork?
(*b*) Determine the thermal conductivity of cork.

9 A copper pipe 7.3 m long with an internal diameter of 12 mm and copper thickness of 1.0 mm carries hot water from your hot water system to the bathroom. It is lagged with a 10 mm thickness of insulation with a thermal conductivity of 0.040 W m^{-1} K^{-1}. How much heat energy is lost through the insulation if hot water at 74°C is running through the pipe while you are having a 5.0 min shower? The ambient temperature is 18°C. (Since the thermal conductivity of the copper is much larger than that of the insulation, the temperature gradient within the copper can be regarded as negligible.)

10 A plate of material of thermal conductivity 42 W m^{-1} K^{-1} and thickness 40 mm is placed in contact with another flat plate of thermal conductivity 84 W m^{-1} K^{-1} and thickness 20 mm. The outer surface of the first plate is maintained at a temperature of 120°C and the outer surface of the second plate at 40°C. Find the temperature of the interface when conditions are steady, assuming that there is no loss of heat from the edges.

11 In order to test the effectiveness of fibreglass insulation in the roof of a house, two 4.0 m by 3.2 m bedroom ceilings were compared. Mark's ceiling was of plasterboard (thermal conductivity 0.13 W m^{-1} K^{-1}) 10 mm thick, while Joanne's had in addition fibreglass batts (thermal conductivity 0.040 W m^{-1} K^{-1}) 50 mm thick laid on top of the plasterboard. The temperature below the ceilings was found to be 22°C while in the roof cavity above it was 10°C. At what rate is heat passing through
(*a*) Mark's ceiling?
(*b*) Joanne's ceiling?

12 To consider the effectiveness of using double-glazed windows in very cold or very hot climates, compare a conventional 2.0 m by 1.0 m window 3.0 mm thick with a double-glazed one of the same area having a 5.0 mm air gap between two 3.0 mm sheets of glass. If the temperature difference between inside and outside is 15 K, what is the heat flow rate for
(*a*) the conventional window?
(*b*) the double-glazed window?
The thermal conductivities of glass and air are 1.1 W m^{-1} K^{-1} and 0.026 W m^{-1} K^{-1} respectively. (Actually the real rates will be lower than your answers. The temperatures of the inner and outer surfaces of the windows differ by a few degrees from the nearby air since air is a poor conductor.)

13 The *thermal resistance* or *R-value*, a term used in the building industry, indicates how well a given form of insulation restricts the transmission of heat. The R-value of a material is the reciprocal of the rate of heat flow through one square metre per unit temperature difference between the outer surfaces. Would a 10 mm thick sheet of copper or a 10 mm thick sheet of rubber have the higher R-value?

14 (*a*) Deduce an expression for the R-value of a material in terms of its thermal conductivity k and its thickness l.
(*b*) What would be the SI unit of R-value?
(*c*) *Without doing any calculations*, suggest which of the following you would regard as the better insulating material: an ordinary brick 110 mm thick or a pine weatherboard 20 mm thick. Now *calculate* their R-values. The thermal conductivities of brick and pine are 1.2 W m^{-1} K^{-1} and 0.11 W m^{-1} K^{-1} respectively. Did the result surprise you?

15 What is the R-value of a 50 mm thick fibreglass insulation batt measuring 1.8 m by 0.45 m, if it conducts 4.9 J of heat per second when the temperatures on its two sides differ by 12 K?

16 In a domestic hot water storage tank is the *top* or the *bottom* of the tank the region where
(*a*) the density of the water is greater?
(*b*) the temperature of the water is greater?
(*c*) the electric or gas heater is located?
(*d*) the outlet pipe leading to the hot water tap for your shower is located?

17 As the temperature of a liquid drops, the liquid ordinarily contracts thereby increasing its density. However, water is an anomaly in that it *expands* for the temperature fall from 4°C to 0°C. Suppose a shallow lake is at a temperature of about 6°C, when there is a sudden cold snap with the air temperature dropping to −10°C.
(*a*) Select any of the following which indicate possible temperatures at the *surface* (S) and at the *bottom* (B) during the subsequent cooling of the lake:
A S: 6°C and B: 4°C
B S: 4°C and B: 6°C
C S: 4°C and B: 2°C
D S: 2°C and B: 4°C
E S: 6°C and B: 2°C
F S: 2°C and B: 6°C
(*b*) Will the rate at which the lake is cooling increase or decrease when the water temperature reaches 4°C? Why?

18 Figure 51.3 shows a passive solar hot water system which operates on natural convection. The solar collector is connected by pipes to the hot water storage tank in the roof cavity. Other inlet and outlet pipes to the storage tank are not shown.

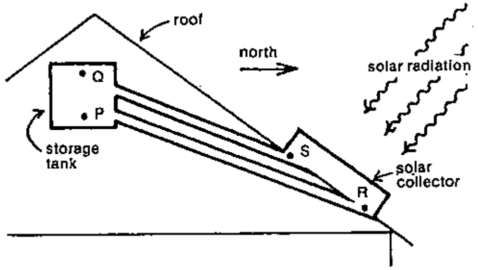


Fig. 51.3

(*a*) Would the water be denser at R or S?
(*b*) In Figure 51.3 would the water be circulating in a clockwise or anticlockwise sense?
(*c*) If temperatures at P, Q, R and S are all different and the water is circulating, arrange the points P, Q, R and S in decreasing order of temperature, i.e. starting with the hottest.

19 One method of heating and cooling a house by natural convection is to incorporate a greenhouse on the north side (in the southern hemisphere) of the house. One form is indicated in Figure 51.4. Vents A, B, C, D and E can be set in an open or closed position.

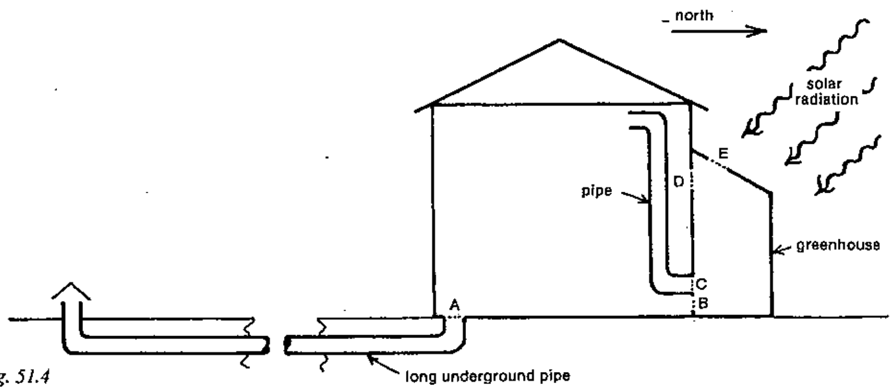


Fig. 51.4

How should they be set
(*a*) to heat the house in winter?
(*b*) to cool the house in summer?

20 If the temperature of a filament in a light bulb is raised from 1000 K to 2000 K, by what factor does the total radiant power emitted increase?

21 A small hole in a furnace may be taken to approximate a black body in so far as emitted radiation is concerned. If the temperature is 1700°C, calculate the energy per square metre per second radiated through the hole. The Stefan–Boltzmann constant is given at the beginning of this set.

22 A light globe was broken, and the tungsten filament extracted and carefully measured. Its diameter and length were 0.12 mm and 0.21 m respectively. The emissivity of tungsten is 0.32, and the normal operating temperature of the filament was 2700 K.
(*a*) Calculate the total surface area of the filament.
(*b*) Determine by calculation whether the globe was marked 40 W, 60 W, 75 W or 100 W.

23 In this problem use any information you require from Table 22.1. The Sun's surface temperature is 5.8×10^3 K.
(*a*) Assuming the Sun acts as a black body, at what rate (in watt) is it emitting energy?
(*b*) Imagine a spherical surface, which is centred on the Sun and passes through the Earth. At what rate is the Sun's radiant energy passing through this spherical surface?
(*c*) What fraction of the energy passing through this spherical surface would hit the Earth?
(*d*) At what rate is this energy reaching the Earth?
(*e*) At what rate is this energy impinging on the Earth per square metre (at right angles to the direction of the Sun's rays)? This answer is known as the *solar constant*.
(*f*) Actually the average amount of energy passing through one square metre (normal to the Sun's rays) at the Earth's surface is about 750 W m^{-2}. What do you think is the major reason why this is less than your answer to (*e*)?

24 Read Problem 23 (*a*). The answer is 3.9×10^{26} W. Einstein's equation, $E = mc^2$, relates the energy E produced when mass m is converted into energy. c, the speed of electromagnetic radiation in a vacuum, is 3.0×10^8 m s^{-1}. If the Sun continued to radiate energy at the present rate, how many more years will it last, assuming the mass–energy conversion proceeds to completion? Mass of Sun = 2.0×10^{30} kg.

25 You are standing in a shower recess about to take a shower. By making suitable estimates determine as an order of magnitude the net rate at which you are losing radiant energy. Don't forget that the walls around you are radiating energy too.

26 The relative intensity of the radiation emitted by black bodies at different temperatures is shown in Figure 51.5 as a function of wavelength.
(*a*) If the most intense wavelength at 5000 K is 0.58 μm, what would be the most common wavelength coming from the Southern Cross star Alpha Crucis, which has a surface temperature of 23 000 K?
(*b*) Would Alpha Crucis appear to be predominantly red, white or bluish?

27 The temperature of a furnace is determined by optical means and it is found that more energy is emitted for a wavelength of 784 nm than for any other wavelength. What is the temperature of the furnace?

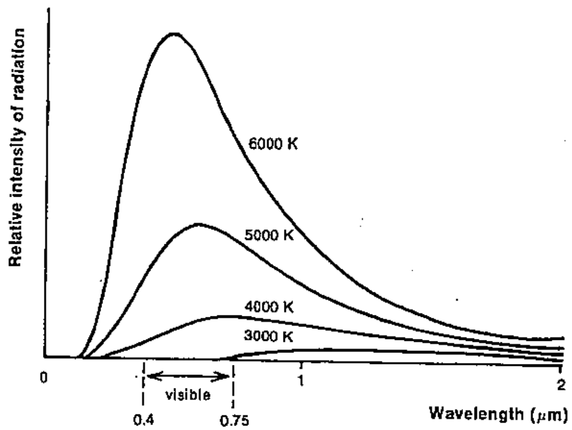


Fig. 51.5

SET 52 ENERGY TRANSFORMATIONS
(Mainly involving heat)

Topic	*Problems*
Transformation of work into heat energy	1–2
Transformation of rotational kinetic energy into heat and electrical energy, wind generators	3–7
Transformation of translational kinetic energy into heat and electrical energy, wave power generators	8–11
Transformation of gravitational potential energy into heat and electrical energy, tidal power	12–16
Transformation of solar energy into heat and electrical energy	17–20
Transformation of electrical energy into heat and other forms of energy (see Set 54)	
Transformation of chemical energy into mechanical and heat energy, energy content of foods and fuels	21–25
Transformation of mass energy into heat and other forms of energy (see Set 28)	
Energy use and conservation on the Earth	26–29

Where necessary in this set, use
density of water $= 1.0 \times 10^3$ kg m^{-3}
specific heat capacity of water $= 4.2$ kJ kg^{-1} K^{-1}

1 When a boy has to make a sudden stop on his BMX bike and applies the back pedal brake, his back wheel locks and his back tyre exerts a forward force of 400 N on the road as he skids to a stop in 3.0 m.
(*a*) How much work does the skidding tyre do on the road?
(*b*) Ideally how much heat energy is developed in the parts of the skidding tyre and road that are in contact?
(*c*) If he was on a loose gravel road, why strictly would the answer to (*b*) be less than the answer to (*a*)?

2 In filing the metal edges of her downhill skis with a steel file of mass 0.12 kg, a girl generates an average frictional force of 11 N between the file and the ski edges. If she does 125 strokes of average length 0.20 m with the file, and 40 per cent of the work done ends up as heat in the file,
(*a*) how much heat is generated in the file?
(*b*) by how much does the temperature of the file rise? Specific heat capacity of steel $= 4.8 \times 10^2$ J kg^{-1} K^{-1}.

3 A 400 W electric power drill is used to drill a hole in a 0.43 kg block of aluminium. In the 40 s taken to drill the hole the temperature of the block rises from 20°C to 44°C. What percentage of the energy used was converted into internal energy of the aluminium? Specific heat capacity of aluminium $= 9.0 \times 10^2$ J kg^{-1} K^{-1}.

4 A chisel to be sharpened is held against the edge of the wheel of an electric grinder rotating at 300 r.p.m. (rotations per minute). The tangential force exerted by the chisel on the grinding wheel is 4.0 N and the wheel has a radius of 0.10 m. If 85 per cent of the energy input is converted into heat, how much heat is developed in the 45 s required to grind the chisel?

5 In order to measure the power of a motor a *band brake* shown in Figure 52.1 can be used. The two spring balances X and Y attached to a band wound around the flywheel read 450 N and 220 N respectively. The flywheel of diameter 0.81 m rotates at 65 Hz.

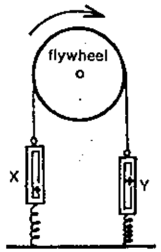


Fig. 52.1

(*a*) What is the tangential frictional force between the band and the flywheel?
(*b*) What is the power output of the motor?

6 Assume a *wind generator* with a two blade propeller of diameter d extracts all the kinetic energy of the air which meets it at speed v. Represent the density of the air by ρ. Write down expressions for
(*a*) the volume of the "cylinder" of air that "hits" the blades per unit time;
(*b*) the mass of this air hitting the blades;
(*c*) the kinetic energy of this air hitting the blades per unit time;
(*d*) the power input to the generator.
Obviously no wind generator can extract *all* the kinetic energy of the air approaching its blades, and so the power expression you derived in (*d*) must be multiplied by C_p, the *power coefficient*, which of course must be less than one.
(*e*) What would be the power output of the US Windcharger generator with a power coefficient of 0.14 in a 36 km h^{-1} wind, if the propeller diameter is 1.83 m? Density of air $= 1.2$ kg m^{-3}.

7 For a wind generator the power developed is proportional to the cube of the wind speed. The Soma 500, a New Zealand wind generator, develops 62.5 W in a 20 km h^{-1} wind. What power would it develop if the wind were blowing at 40 km h^{-1}?

8 An 850 kg car travelling at 72 km h^{-1} brakes and stops in a short distance.
(*a*) Before braking what was the car's
(*i*) speed in SI units?
(*ii*) kinetic energy?
(*b*) While braking, what total heat energy is developed in the brakes, tyres and road in contact with the tyres?

9 At a rifle range a 24 g bullet travelling at 316 m s^{-1} strikes a mound after missing the target.
(*a*) What is the resulting increase in internal energy of the bullet and mound?
(*b*) If 70 per cent of this energy is absorbed by the bullet, by how much does its temperature rise as a result of the impact? Specific heat capacity of lead $= 1.3 \times 10^2$ J kg^{-1} K^{-1}.

10 A fireman directs a fire hose at a burning wall. Making suitable estimates, determine the order of magnitude of the temperature rise resulting from some of the kinetic energy of the jet of water being converted into internal energy of the water.

11 The energy per metre width of a water wave is given by
$$E = \tfrac{1}{8}\lambda\rho g A^2(1 - 4.93A^2/\lambda^2)$$
where λ = wavelength of the wave,
ρ = density of the water,
g = gravitational field strength, and
A = amplitude of the wave.
Consider a *sea wave power generator*, which absorbs 70 per cent of the wave energy impinging on it, and which is set up with its 20 m wave-absorbing side parallel to the waves. The waves have a wavelength of 25 m, an amplitude of 1.8 m and a speed of 3.2 m s^{-1}. Density of sea water $= 1.02 \times 10^3$ kg m^{-3}.
(*a*) What is the total energy in the 20 m section of a wave which meets the generator?
(*b*) What total energy is absorbed by the generator during the time taken for
(*i*) one wave to strike it?
(*ii*) 10 waves to strike it?
(*c*) How long does it take for 10 waves to strike it?
(*d*) What is the power input to the wave power generator?

12 After sliding from rest down a vertical pole a distance of 10.0 m, a fireman of mass 70.0 kg has a speed of 2.00 m s^{-1}. For the 10.0 m slide
(*a*) how much gravitational potential energy did he lose?
(*b*) how much kinetic energy did he gain?
(*c*) how much heat was generated by friction?

13 Suppose an ordinary glass marble drops out of your hand on to a concrete path. Estimate as an order of magnitude the rise in temperature caused by its first bounce. Specific heat capacity of glass $= 7 \times 10^2$ J kg^{-1} K^{-1}.

14 A hydroelectric scheme relies on a reservoir of area 5.1×10^6 m^2 and average depth 40 m. If the reservoir were allowed to completely empty through the power station turbines situated 34 m below the bottom of the reservoir, how much electrical energy would be produced?

15 The demand for electric power by consumers varies considerably during the 24 h cycle. However, since it is more efficient to operate power stations continuously, it is desirable to have some means of storing energy during periods of low demand. One such method used at Loch Ness in Scotland is the Foyers *pumped storage system*, where water is pumped from Loch Ness to Loch Mhor, a vertical rise of 176 m.
(*a*) This is carried out by a 336 MW pumping station. If the efficiency of this process is 86 per cent, at what rate (in kg s^{-1}) is the water moved up to Loch Mhor?
(*b*) During peak periods the process is reversed with water running down at a rate of 2.0×10^5 kg s^{-1}, generating electrical energy at 300 MW. What is the efficiency of this generating process?

16 One source of renewable energy is *tidal power*. At the Rance tidal power plant in France, tidal water flowing into the estuary is blocked and then allowed to run through hydro generators. A similar process is used when the tide goes out, thus enabling power to be generated during both phases of the tidal movement. If 2.6×10^3 m^3 s^{-1} pass through the generators when the vertical difference between the two levels is 11 m, what is the power input to the plant?

17 One of the world's first photovoltaic solar generators to convert the Sun's energy directly into electrical energy was set up in desert conditions in Chile in 1960. It consisted of 144 panels each comprising thirty-six 19 mm diameter solar cells. With a solar energy flux of 1.0 kW m^{-2} the generator produced an electrical power output of 87 W.
(*a*) What is the total front surface area of the solar cells?
(*b*) At what rate is solar energy being received by the whole array of solar cells?
(*c*) With what percentage efficiency is solar energy being converted into electrical energy? (Solar cell efficiency has more than tripled since 1960, though there is a maximum theoretical efficiency of 24 per cent.)

18 Consider a thermosiphon solar hot water system similar to that shown in Figure 51.3, representing 4.0 m^2 of collecting panels installed on a Melbourne home with a roof inclined at 40° to the horizontal. The average daily insolation received by a surface with this orientation in Melbourne is 15.7 MJ m^{-2}. If the collector retains 78 per cent of the incident energy,

(*a*) what is the daily average energy it absorbs?
(*b*) what will be the average daily increase in temperature for the water, if the system has a 300 litre capacity. 1 litre $= 10^{-3}$ m^3.

19 One method of solar heating for a home is a *rock bed storage* system. An example would be an underground insulated tank containing 4.4 tonne of rocks about 20 mm in diameter. Hot air from solar collectors is circulated downwards through the bed, losing 93 per cent of its heat to the rocks which have a relatively high specific heat capacity of 8.0×10^2 J kg^{-1} K^{-1}. At night when the temperature has fallen, air is circulated upwards through the rock pile and through radiators in the house and back to the base of the rock bed. On an average day 96 MJ of solar radiation falls on the solar hot air collectors, which are 75 per cent efficient.
(*a*) What is the average daily absorption of energy by
(*i*) the hot air collectors?
(*ii*) the rock bed?
(*b*) What is the average daily increase in the temperature of the rock bed?

20 The 10 MW solar electric power plant at Barstow, California, is one of the world's largest. It consists of an array of 1818 heliostats (mirrors which track the Sun) each of area 39.3 m^2, focusing the Sun's radiation on to a water–steam collector on top of an 80 m high tower, which the heliostats surround. If the solar insolation incident on the heliostats is about 700 W m^{-2}, what is the approximate efficiency of the plant?

21 How many slices of buttered bread would a 53 kg girl need to eat to enable her to climb a 2000 m high mountain, if about 80 per cent of the energy is used up as heat? The energy value of a slice of buttered bread is about 4×10^5 J.

22 You are planning for your first marathon (42.195 km), which you estimate you can complete in about 4 h without the usual "carbohydrate debt" you have seen on the faces of so many lesser mortals at the finishing line. You intend to "carbohydrate load" sufficiently before the race to provide your body with sufficient energy to complete the event without the usual discomfort. At your suggested pace you will be metabolising carbohydrates at about 420 W, providing mechanical and heat energy.
(*a*) What total energy will you metabolise from carbohydrates during the marathon?
(*b*) How many 0.45 kg cans of cooked spaghetti (energy value 1.5 MJ kg^{-1}) would you need the night before the race to provide this energy?

23 Suppose you could control your metabolism to the extent that having eaten a plain doughnut (energy value 8×10^5 J) you could use its energy exclusively for a single high jump on your school's annual athletics sports day. What new high jump record would you create? You will need to know your mass.

24 Use the following information gathered on a long car trip to determine the efficiency of the vehicle:
average speed of the car = 90 km h^{-1}
average power output = 30 kW
petrol consumption = 10.7 litre per 100 km
1 litre = 10^{-3} m^3
density of petrol = 7.0×10^2 kg m^{-3}
heat of combustion of petrol = 4.7×10^7 J kg^{-1}

25 An engineering student decides to use a novel—but probably not the best—method of determining the total air and road resistance opposing the motion of his car. He drives the car at a constant speed of 80 km h^{-1} on a straight level deserted road for 6.0 min and finds he has used 1.21 litre of petrol. The motor has an efficiency of 24 per cent. You will need to use the last three items in Problem 24.
(*a*) For the 6.0 min trial determine
(*i*) the volume (in m^3) of petrol used;
(*ii*) the mass of petrol used;
(*iii*) the energy provided by the petrol;
(*iv*) the energy output from the motor.
(*b*) How far did the car travel in the 6.0 min?
(*c*) What forward driving force did the engine provide?
(*d*) What was the total backward force of the air and road opposing the motion of the car?

26 At the beginning of 1984 the total oil reserves in the world were estimated at about 1.7×10^{12} barrels. We derive about 6×10^9 J from a barrel of oil. In 1983 oil supplied about 1.6×10^{20} J of energy.
(*a*) If oil continued to be used at the 1983 rate, by about what year would the world's reserves run out?
(*b*) Why is it likely to run out before your predicted date?
(*c*) Optional: If the growth in the use of oil can be restricted to 2.7 per cent per annum, till about what year would the world's reserves last?

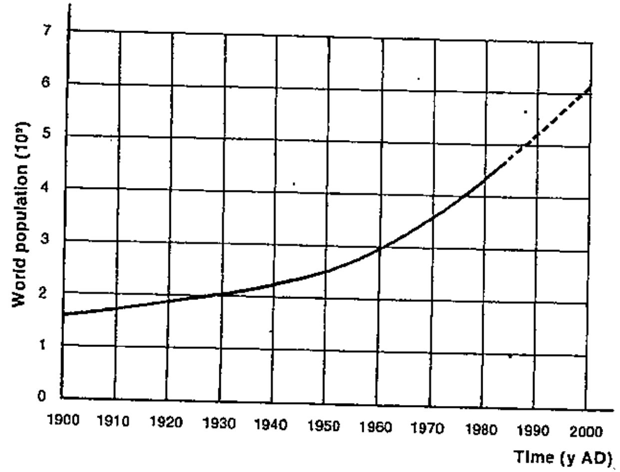


Fig. 52.2

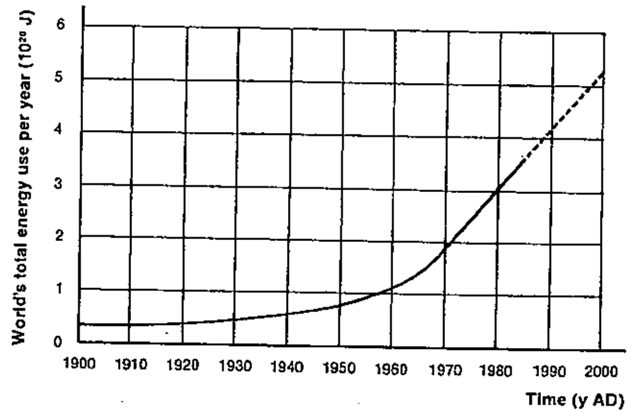


Fig. 52.3

27 Figures 52.2 and 52.3 show past and probable future growth in world population and energy use. Use the graphs to answer the following:
(*a*) Determine the annual energy consumption per person for the year
(*i*) 1900;
(*ii*) 1950;
(*iii*) 2000 (predicted).
(*b*) In the first 45 y of this century (1900–45) the world energy use doubled. In the next 45 y (1945–90) how many times will it probably have doubled? Use only whole numbers in your answer.

28 Refer to Figure 52.3.
(*a*) What is represented by the total area between the graphed line and the time axis?
(*b*) At the beginning of 1984 the world's reserves of fossil fuels amounted to about 2×10^{23} J. If the predicted energy use shown in Figure 52.3 continued beyond 2000, and the world used fossil fuels as the only source of energy, roughly when would the supply be exhausted?

29 Much energy is wasted by modern society's acceptance of inbuilt obsolescence. Milk has a food energy value of 2 MJ per 600 cm^3. Actually it takes about three times that energy to manufacture either the milk bottle or cardboard carton to hold the milk. With a 1985 Australian population of 18 million, estimate as an order of magnitude
(*a*) the amount of energy wasted that year, if all Australians used milk cartons rather than bottles and discarded them afterwards (not of course taking into account the energy required to reuse milk bottles);
(*b*) the number of homes that could be lit and heated for a full year by the same amount of energy. An average Australian home uses about 10^{11} J per year on heating and lighting.

SET 53 REVISION PROBLEMS

Set	*Topic*	*Problems*
47	Thermal expansion of solids and liquids	1–3
48	Thermal expansion of gases and the ideal gas laws	4–11
49	Change of temperature and change of phase	12–15
50	Thermodynamics	16–19
51	Transfer of heat	20–26
52	Energy transformations (mainly involving heat)	27–31

Where necessary in this set, use
$0°C = 273$ K (rather than the more exact 273.15 K)
specific heat capacity of water $= 4.19$ kJ kg^{-1} K^{-1}
1 litre $= 10^{-3}$ m^3
density of water $= 1.0 \times 10^3$ kg m^{-3}

1 A wire which has an area of cross-section of 0.50 mm^2 is 5.0 m long at 16°C. To what temperature must this wire be heated to cause an extension equal to that produced by hanging a load of 8.0 kg mass on it?
Young modulus of the material $= 1.6 \times 10^{11}$ Pa
Linear expansivity $= 1.4 \times 10^{-5}$ K^{-1}

2 A brass rod is measured with a steel scale which is correct at 0°C, and a reading of 413.0 mm is obtained. If the measurement is made at a temperature of 20°C, what is the true length of the brass rod at 0°C?
The linear expansivities of steel and brass are 1.2×10^{-5} K^{-1} and 1.9×10^{-5} K^{-1} respectively.

3 The bob of a mercury pendulum consists of a light steel cylinder, linear expansivity α K^{-1}, filled to a height h m with mercury, cubic expansion γ K^{-1}. Show that the change in h m for an increase of temperature of 1°C is $(\gamma - 2\alpha)h$ m. If the bob is supported by a light rod of the same steel, so that the total length to the base of the cylinder is l m, find the ratio of h to l if the centre of mass of the mercury is not to alter in height with change of temperature. Take $\alpha = 1.1 \times 10^{-5}$ K^{-1} and $\gamma = 1.82 \times 10^{-4}$ K^{-1}.

4 You are pumping up a tyre on your bike and have the piston drawn out with the pump full of air at a pressure of 100 kPa. For a full stroke the piston travels 0.40 m. If prior to the present stroke the *gauge* pressure for your tyre reads 300 kPa, how far will the piston move in before air starts to enter the tyre from the pump? Assume the compression is isothermal rather than adiabatic.

5 Two glass flasks, one of which has twice the volume of the other, are connected together by a capillary tube of negligible volume. They contain dry air at a temperature of 20.0°C and a pressure of 100 kPa. The larger flask is then immersed in steam at 100°C and the smaller in melting ice at 0°C. Neglecting any change in volume of the flasks, find the resulting pressure in them.

6 A glass bulb contains helium gas at a temperature of 27°C. Its pressure is 106 kPa. Determine the pressure of the gas in the bulb if
(*a*) some helium gas containing twice as many molecules is added without changing the temperature;
(*b*) instead, the temperature of the gas is lowered to -123°C without changing the volume.

7 A capillary tube closed at one end contains air enclosed by a bead of water. When the tube is held horizontally, and with the barometer reading 760.0 mmHg, the length of the air column is 156 mm at 20°C and 257 mm at 70°C. Given that the saturation vapour pressure of water is 17.0 mmHg at 20°C, find its value at 70°C.

8 Helium, of mass M kg, is contained in a vessel of volume V litre at a pressure of P Pa. Another vessel of volume $2V$ litre contains m kg of argon at a pressure of $3P$ Pa. While the temperature is kept constant, the vessels are connected so that the gases can mix freely. The mass of an atom of helium is 4 u and that of an argon atom is 40 u.
(*a*) What is the ratio of the average kinetic energy of the helium molecules to that of the argon molecules?
(*b*) What is the pressure of the mixture of the two gases?

(*c*) When molecules collide normally with the walls of the container, will the change in momentum of a helium molecule be less than, equal to or greater than that of an argon molecule?

9 Container Y contains N molecules of a certain gas at atmospheric pressure. Container Z, whose volume is twice that of Y, contains $5N$ molecules of the same gas. Both containers are at the same temperature, 0°C, and are connected by a tube in which a stopper separates the gas in the containers. What is the value of the ratio
(*a*) $\dfrac{\text{density of gas in Z}}{\text{density of gas in Y}}$?
(*b*) $\dfrac{\text{pressure of gas in Z}}{\text{pressure of gas in Y}}$?
(*c*) $\dfrac{\text{average molecular speed in Z}}{\text{average molecular speed in Y}}$?
(*d*) $\dfrac{\text{average number of collisions per unit time per unit area with the walls in Z}}{\text{average number of collisions per unit time per unit area with the walls in Y}}$?
With the temperature remaining at 0°C, the stopper is now turned so that the containers Y and Z are connected.
(*e*) After a short time, what is the value of the ratio $\dfrac{\text{pressure of gas}}{\text{original pressure of gas in Y}}$?
(*f*) What is now the value of the ratio $\dfrac{\text{average molecular speed of the gas}}{\text{original average molecular speed in Z}}$?
(*g*) If the temperature of the gas is now decreased from 0°C to -91°C, what is now the value of the ratio $\dfrac{\text{pressure of gas}}{\text{original pressure of gas in Y}}$?

10 Refer to Figure 53.1, which shows two containers X and Y of equal volume. X contains 1.0 mole of oxygen molecules at 1000 K, while Y contains 1.0 mole of helium molecules at 250 K. Oxygen is a diatomic gas with a relative atomic mass of 16, while helium is monatomic with a relative atomic mass of 4.0.
Determine the value of the following ratios:
(*a*) $\dfrac{\text{number of molecules in X}}{\text{number of molecules in Y}}$;
(*b*) $\dfrac{\text{number of atoms in X}}{\text{number of atoms in Y}}$;
(*c*) $\dfrac{\text{mass of gas in X}}{\text{mass of gas in Y}}$;
(*d*) $\dfrac{\text{pressure of gas in X}}{\text{pressure of gas in Y}}$;
(*e*) $\dfrac{\text{average translational kinetic energy of a molecule in X}}{\text{average translational kinetic energy of a molecule in Y}}$;
(*f*) $\dfrac{\text{root mean square speed of a molecule in X}}{\text{root mean square speed of a molecule in Y}}$.

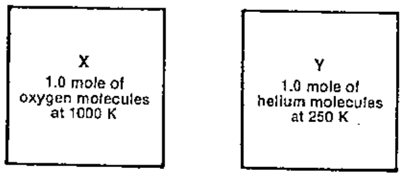


Fig. 53.1

11 The monatomic gas krypton has a density 3.0 times that of diatomic nitrogen gas at the same temperature and pressure. A flask is filled with krypton and an identical flask is filled with nitrogen at the same temperature and pressure. Concerning the gases in the flasks, what is the value of each of the following ratios?
(*a*) number of molecules of krypton to the number of molecules of nitrogen;
(*b*) mass of krypton to mass of nitrogen;
(*c*) the average translational kinetic energy of the krypton molecules to that of the nitrogen molecules.

12 200 g of tin (specific heat capacity 235 J kg^{-1} K^{-1}) at 100.0°C is dropped into a calorimeter of heat capacity 29 J K^{-1} (or water equivalent 7.0 g) containing 43.0 g of water at 30.0°C.
(*a*) What will be the final temperature?
(*b*) In practice there will be some heat exchange with the surroundings. Would the experimental result be above or below your answer to (*a*)?

13 10.0 g of ice at a temperature of -78.5°C is added to 120 g of water at a temperature of 30.0°C. Assuming that heat exchanges with the surroundings are prevented, what will be the final temperature? Specific heat capacity of ice $= 2.10 \times 10^3$ J kg^{-1} K^{-1}; specific latent heat of fusion of ice $=$ 334 kJ kg^{-1}.

14 A 25.0 g block of brass (specific heat capacity 368 J kg^{-1} K^{-1}) is placed in a thermos of liquid air for about one minute. It is then quickly transferred to another thermos of water at 0°C. A 6.25 g coating of ice forms on it. What is the temperature of the liquid air? Specific latent heat of fusion of ice $= 334$ kJ kg^{-1}.

15 A cooked frankfurt initially at a temperature of 80°C is cooling. Its temperature after one 5.0 min interval is 64°C and is 52°C after the next 5.0 min interval. Determine
(*a*) the temperature of the surroundings;
(*b*) the temperature of the frankfurt after a third 5.0 min interval.

16 Answer key:
A the zeroth law of thermodynamics
B the first law of thermodynamics
C the second law of thermodynamics
D the third law of thermodynamics

Using the answer key above, select the law which effectively is being stated in each of the following:

(*a*) Heat cannot flow spontaneously from a lower to a higher temperature level.

(*b*) The change in internal energy of a system equals the heat absorbed by the system less the work done by the system.

(*c*) If two bodies X and Y are in thermal equilibrium with a third body Z, then X and Y are in thermal equilibrium with each other.

(*d*) It is impossible by any procedure to reduce any system to the absolute zero of temperature by a finite number of operations.

(*e*) The entropy of the universe is always increasing.

(*f*) It is impossible to construct an engine which will convert a given amount of heat energy completely to mechanical work.

(*g*) There is a scalar quantity called temperature which is a property of all thermodynamic systems such that temperature equality is the necessary and sufficient condition for thermal equilibrium.

17 If you have a blowout while driving a car, the expansion of the air would take place so rapidly as to be regarded as adiabatic. Determine as an order of magnitude the resulting fall in the temperature of the air from your tyre. Take the ratio of the principal heat capacities of air as 1.4.

18 A heat engine absorbs 500 J from its high temperature reservoir and loses 410 J to its low temperature reservoir each cycle.

(*a*) How many kilojoule of work is done during each 50 cycles of the engine?

(*b*) Determine the maximum possible efficiency of the engine.

19 Write down the letters (A, B, . . .) corresponding to any of the following statements that are correct:

A No refrigerator or heat pump can transfer internal energy from a cold reservoir to warmer surroundings without some external agent doing some work.

B If a gas expands very suddenly its temperature increases.

C It is impossible to convert heat energy entirely into work; some must be wasted as heat.

D The change in entropy for a completely reversible process is zero.

E When a gas expands isothermally no heat enters the gas from the surroundings.

F In an isobaric expansion of a gas the gas does external work.

G The total entropy of the universe is continually decreasing.

20 (*a*) The tank of a hot water system is a cube of 0.70 m edge, and it holds 300 kg of water. It is enclosed by a layer of insulating material 0.10 m thick with a thermal conductivity 4.2×10^{-2} W m^{-1} K^{-1}. The heating current is switched off at 6 a.m. when the temperature of the water is 90°C. Assuming that the temperature of the outer surface of the covering is 15°C throughout the day and that no hot water is drawn off, estimate the temperature of the water at 11 p.m. on the same day. (The effect at the corners may be neglected.)

(*b*) If half the water is run off at 6 a.m. and replaced by water at 10°C, how long would it take to raise the temperature to 75°C with a 1.94 kW heater. (For this part of the question neglect heat loss.)

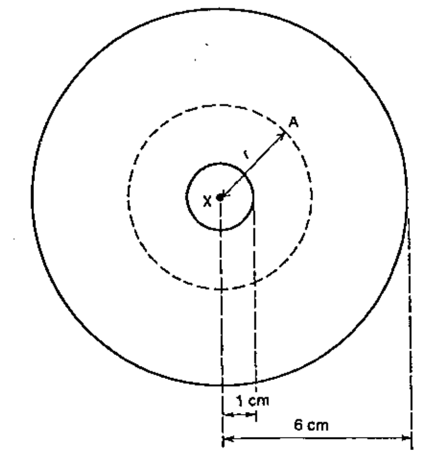


Fig. 53.2

21 Figure 53.2 shows a cross-section of a cylindrical tube 1.00 m long. The interior of the tube is heated electrically by a wire X which coincides with the long axis of the tube and when conditions are steady it is found that the energy dissipated inside the tube is 40.0 W. Given that the thermal conductivity of the material is 8.40×10^{-2} W m^{-1} K^{-1}, calculate the temperature gradient S at points such as A, distant r cm from the axis of the tube, and well removed from the ends of the tube. Heat losses through the ends of the tube are to be neglected. Draw a graph of S against r and use it to make an estimate of the difference in temperature between the inner and outer surfaces.

22 Calculate the heat energy conducted per hour through the side walls of a cylindrical steel boiler of 1.00 m diameter and 3.0 m long if the internal and external temperatures of the walls are 140°C and 40°C respectively and the thickness

of the walls is 6.0 mm. Thermal conductivity of steel = 42 W m^{-1} K^{-1}.

23 A wall of a building consists of 0.120 m thickness of brickwork and 0.240 m thickness of concrete. The temperature of the outside face of the brick is 40.0°C, and that of the outside face of the concrete is 20.0°C. Find the temperature of the interface between the brick and the concrete. Thermal conductivity of brickwork = 0.126 W m^{-1} K^{-1}; thermal conductivity of concrete = 0.294 W m^{-1} K^{-1}.

24 A solar hot water system uses a collector of area 1.4 m by 2.5 m. The collector has a tracking device which enables it to remain perpendicular to the Sun's rays. The average irradiation normal to the collector is 750 W m^{-2}. By how many degrees does the temperature of the water rise in one pass through the collector, if it flows at a rate of 2.5 litre min^{-1}?

25 A copper sphere, whose surface is blackened, has a radius of 20.0 mm and a mass of 300 g. The sphere is suspended by a thin thread in bright sunlight on a still day when the air temperature is 20.0°C. The temperature of the sphere is found to vary with time as shown in Table 53.1.

Table 53.1

Time (min)	Temperature (°C)
0	20.00
1.0	20.93
2.0	21.83
3.0	22.70

Assuming that the sphere absorbs all the radiation which falls upon it and assuming that none of the Sun's radiation is absorbed by the atmosphere, determine

(*a*) the rate (in watt) at which the Sun is radiating energy;

(*b*) the rate at which matter is being annihilated in the Sun.

Specific heat capacity of copper = 378 J kg^{-1} K^{-1}; distance of the Sun from the Earth = 1.50×10^{11} m; speed of electromagnetic radiation in a vacuum = 3.0×10^{8} m s^{-1}.

26 In a laboratory analysis of the light coming from the Sun it was found that the wavelength with the highest intensity was 491 nm. Deduce a value for the surface temperature of the Sun. Wien constant = 2.9×10^{-3} m K.

27 In the crime-buster film "Even Stevens" two identical lead bullets moving with equal kinetic energies make a head-on collision. They stop and just reach their melting point, 335°C, under the heat generated at the impact. The temperature of the two bullets before colliding was 35°C. What was their common speed before impact? Specific heat capacity of lead = 126 J kg^{-1} K^{-1}.

28 The bearing of an engine is cooled by a stream of oil which flows through it. The temperature of the oil on entering is 20°C and on leaving is 28°C. The specific heat capacity of the oil is 2.1 kJ kg^{-1} K^{-1}. If oil is supplied at the rate of 50 g s^{-1}, at what rate is mechanical energy dissipated in the bearing?

29 A stream flowing at the rate of 10 m^{3} s^{-1} falls over a waterfall 20 m high.

(*a*) What is the difference in temperature of the stream below and above the falls, assuming that all the kinetic energy of the falling water is converted into heat?

(*b*) If now the stream is caused to run smoothly down a steep incline of the same vertical height as the falls, and at the bottom is made to rotate a turbine coupled to an electrical generator, find the power output of the station, assuming it is 66 per cent efficient.

30 A small electric motor supplies energy at the rate of 300 W to turn a paddle in a calorimeter which contains 0.445 kg of water. To raise its temperature by 1°C, the calorimeter requires the same amount of heat as does 5 g of water. If 63 per cent of the energy supplied is used in heating the calorimeter and its contents, what will be the rate of rise in temperature?

31 While jogging for 1.0 h a girl sweats (or, should we say, glows) 1.2 litre of water. If the specific latent heat of evaporation at her skin temperature is 2.4 MJ kg^{-1}, at what rate is she using energy to keep her body temperature constant? Assume the humidity is not high.

PART 7

ELECTRICITY

SET 54 STATIONARY CHARGED PARTICLES IN ELECTRIC FIELDS

Topic	*Problems*
Electrostatic induction, gold leaf electroscope, testing for charge, charge distribution on conductors	1–4
Coulomb's law	5–12
Electric field strength	13–15
Electric field strength near a single point charge	16–18
Electric field strength between parallel charged plates	19
Electric field strength due to more than one point charge	20–23
Electric potential energy of two point charges	24
Electric potential at a point	25–26
Electric potential difference between two points	27–28
Electric potential near a single point charge	29–30
Electric potential due to more than one point charge	31–32
Problems involving both electric field strength and electric potential due to more than one point charge	33–35
Electric potential between parallel charged plates (further problems in Set 55)	36
Electric potential gradient and its relation to electric field strength (see Set 55)	
Equipotential surfaces	37

In this set assume
all electrical interactions are taking place in air or a vacuum, and
charges at an infinite separation have zero electrical potential energy.

Where necessary in this set, use
Coulomb's law constant in a vacuum or air,
$k = 1/4\pi\epsilon_0$
$= 9.0 \times 10^9\ \mathrm{N\ m^2\ C^{-2}}$
permittivity of a vacuum or air,
$\epsilon_0 = 8.85 \times 10^{-12}\ \mathrm{C^2\ N^{-1}\ m^{-2}}$

1 In each of the following cases indicate whether the object has a net *negative* charge, a net *positive* charge, or is *uncharged*. A plastic ruler when rubbed on your jeans becomes negatively charged.
(*a*) What is the charge on the part of your jeans you rubbed?
(*b*) The charged ruler is brought close to two isolated conducting spheres A and B, in contact, as shown in Figure 54.1. What is the charge on A?

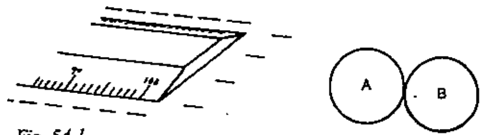


Fig. 54.1

(*c*) While the ruler is kept in the position shown in Figure 54.1, B is removed. What is the charge on B?
(*d*) If, instead, the ruler is removed, then spheres A and B are separated. What is the charge on A?

2 The sketches A to G in Figure 54.2 represent various charge distributions on the cap and leaves of a gold leaf electroscope (GLE). For simplicity, the case of the GLE has been omitted. For an initially uncharged GLE, which of the sketches best represents each of the following situations?
(*a*) A positively charged glass rod is brought near the cap of the GLE and held there.
(*b*) The GLE is placed inside a hollow metal sphere which carries a positive charge.
(*c*) While a negatively charged strip of plastic is held near the cap of the GLE, the cap is connected to earth.
(*d*) *Following* the procedure in (*c*), the earth lead is removed, and then the piece of plastic is removed.

3 Of the following, which is the best way to determine whether a conducting body on an insulating handle is positively charged?
A The body is touched on to the plate of an uncharged gold leaf electroscope (GLE).
B The body is brought close to an uncharged GLE.
C The body is brought close to a positively charged GLE.
D The body is brought close to a negatively charged GLE.
E Methods C and D above are equally satisfactory.

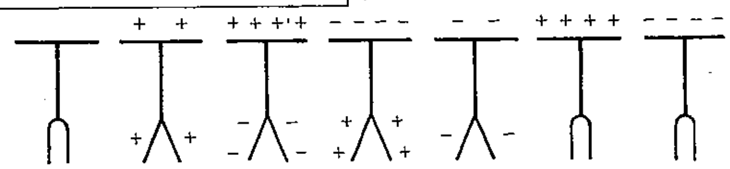


Fig. 54.2

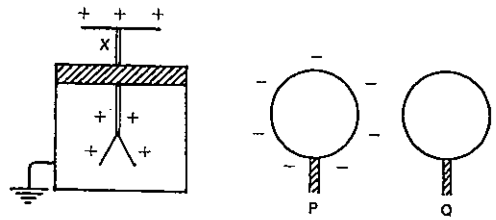


Fig. 54.3

4 In Figure 54.3, X is a positively charged electroscope, and P and Q are identical metal spheres on insulating handles. P carries a negative charge equal in magnitude to the positive charge on X. Q is uncharged. For each of the questions below assume the *initial* conditions of P, Q and X are as stated above, and that they are all well separated from one another.
Answer key:
A The leaves will partly collapse.
B The leaves will collapse and then diverge again.
C The leaves will not move.
D The leaves will collapse completely.
E The leaves will diverge further.
Use the key above to describe what will happen to X,
(*a*) if P is touched to the plate of X;
(*b*) if, instead, Q is touched to the plate;
(*c*) if, instead, Q is touched to P and then to the plate.

5 What is the force of repulsion between two point charges of $+6\ \mu$C (microcoulomb) and $+5\ \mu$C, situated 0.3 m apart in a vacuum?

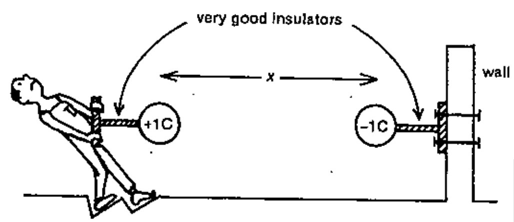


Fig. 54.4

6 In order to appreciate the force of attraction between two objects carrying charges of $+1$ C and -1 C, imagine the situation in Figure 54.4. With your feet well dug in, you could probably exert a horizontal force of, say, 560 N. *Guess* an answer for the minimum value of x, and then actually calculate it. You might be surprised.

7 Two small objects, 0.15 m apart and carrying identical charges, experience a force of repulsion of 32.4 N. What is the magnitude of the charge on each?

8 In Figure 54.5, A and B are insulated conducting spheres situated distance x apart and carrying equal charges of $+q$. The force that each exerts on the other is 4.0×10^{-5} N.

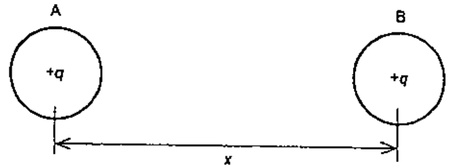


Fig. 54.5

What would be the magnitude of the force between A and B in each of the following cases?
(*a*) An additional charge of $+3q$ is added to the charge already on A.
(*b*) Instead, the distance between A and B is increased to $2x$.
(*c*) Instead, a charge of $-2q$ is added to the original charge on B.
(*d*) Instead, the distance between A and B is decreased to $\frac{x}{3}$, and the charges on A and B are changed to $+0.2q$ and $+5q$ respectively.

9 Two small conducting spheres on insulating stands carry charges $+Q$ and $+q$ respectively and are situated at B and C as in Figure 54.6. ABC is a right-angled triangle in which the lengths BC and AB can be considered as 1 unit and 2 units respectively. The force of repulsion between $+Q$ and $+q$ is 9.0×10^{-1} N.

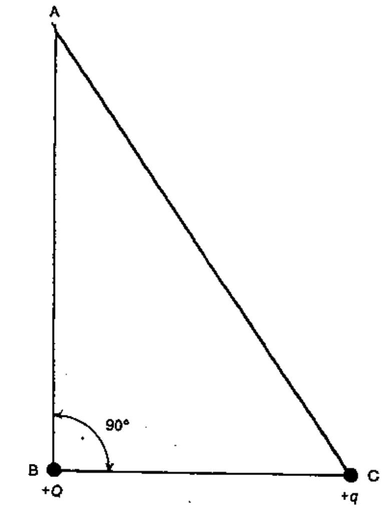


Fig. 54.6

(*a*) If a small positive charge were placed at A, would the net force acting on it have an upward or downward component?
(*b*) What is the magnitude of the force of repulsion between the charges if $+Q$ is moved from B to A?
(*c*) If, with the charges at B and C as originally, one-half of the charge $+Q$ and one-third of the charge $+q$ leak away through poor insulation, what will then be the magnitude of the force of repulsion between the remaining charges?

10 A, B and C are three small, insulated, conducting spheres with equal charges on them, and are placed as shown in Figure 54.7. A and B are positively charged and C is negatively charged. A exerts a force of 3.0 μN on B.

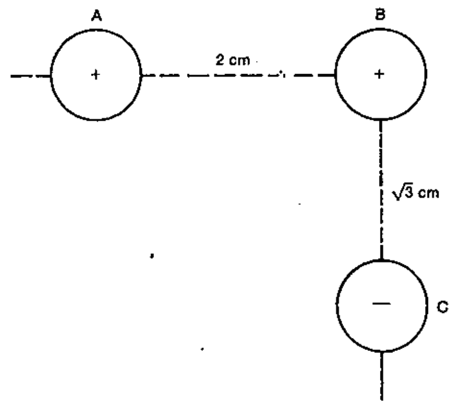


Fig. 54.7

Give all answers to two significant figures.
(*a*) Find the magnitude of the force that C exerts on B.
(*b*) Find the magnitude of the resultant force on B.
(*c*) What angle does the direction of the resultant force on B make with AB produced?
(*d*) What would be the size of the resultant force if all the charges were three times as great?

11 In Figure 54.8, A, B and C are three small insulated conductors all carrying positive charges of equal size. A exerts a force of 2.5 μN on B.
Give answers to two significant figures.
(*a*) What is the magnitude of the force exerted by B on C?
(*b*) What is the direction of the resultant force exerted by A and B on C?
(*c*) What is the magnitude of this resultant force?

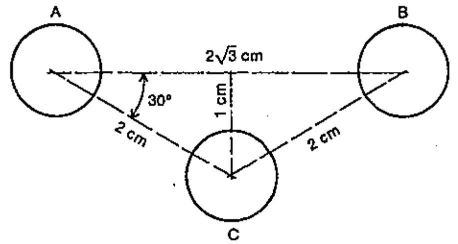


Fig. 54.8

12 Two small spheres, each of mass 5.0 g, are each supported from the same point by a light insulating thread 500 mm long. When they are given equal charges they repel each other so that the distance between them is 80 mm. Find the charge on each.

13 At a point X near your charged polyester shirt a charge of 2 μC would experience an electric force of 6 μN west. Specify fully the electric field strength at X.

14 After brushing your hair there is a point Z near it where the electric field strength is 75 N C^{-1} east. If a dust particle carrying -8.0×10^{-18} C arrives at Z, what is the
(*a*) magnitude, and
(*b*) direction
of the electric force acting on the dust particle?

15 At a point P in a fluorescent tube at a given instant the electric field strength is 25 N C^{-1} south. A mercury ion at P at this instant experiences an electric force of 8.0×10^{-18} N south. What is the charge on the ion?

16 A small graphite-coated polystyrene sphere suspended on a piece of thin nylon fishing line carries a charge of $+2.0 \times 10^{-9}$ C. In order to determine the electric field strength at a point X, 0.10 m to the west of the sphere's centre, you place a minute test charge of $+4.0 \times 10^{-18}$ C at X.
(*a*) What will be the electric force on the test charge?
Now you throw away the test charge.
(*b*) Deduce a value for the electric force per coulomb at X.
(*c*) What is the electric field strength at X?

17 What is the electric field strength at a point 0.15 m directly above a point charge of -12 pC in air?

18 You may have thought the Earth was electrically neutral. Actually it carries a net charge of -6.0×10^5 C. The electric field strength at a distance *d* from the centre of the Earth has a magnitude of 128 N C^{-1}. (*d* is greater than the radius of the Earth.)

(*a*) Specify fully the electric field strength at a distance 2*d* from the Earth's centre.
(*b*) Determine the value of *d*.

19 Figure 54.9 shows two horizontal conducting plates connected to a cell. The top plate becomes positively charged. X, A and D are on the same horizontal level.

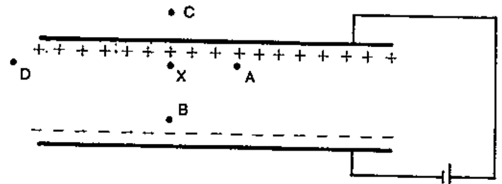


Fig. 54.9

(*a*) What is the direction of the electric field at X?
(*b*) For each of the following pairs of points select the one where the magnitude of the electric field strength is greater (write E if they are equal):
(*i*) X and A;
(*ii*) X and B;
(*iii*) X and C;
(*iv*) X and D.

20 In Figure 54.10 A and B are two small insulated spheres 100 mm apart which are capable of retaining electric charge. P is a point 75 mm from A and 25 mm from B.

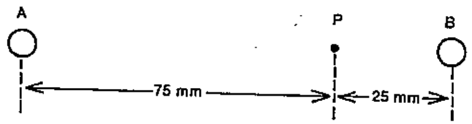


Fig. 54.10

What is the direction of the electric field at P when
(*a*) A and B carry positive charges of equal magnitude?
(*b*) A carries a positive charge and B a negative charge of equal magnitude?

21 (*a*) Find the electric field strength at a point midway between two charges of +400 pC and −500 pC situated 100 mm apart in air.
(*b*) What would be your answer to (*a*) if both charges had been positive?

22 Two small spheres, carrying charges of $+Q$ and $+4Q$, are 12 cm apart. At what distance from the charge of $+4Q$ will the net electric field be zero?

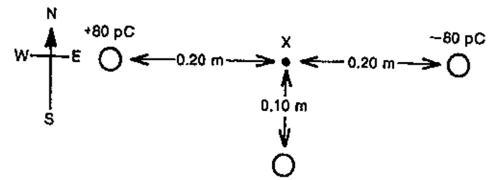


Fig. 54.11

23 Three charges are distributed in a horizontal plane as shown in Figure 54.11. What will be the
(*a*) magnitude, and
(*b*) direction
of the electric field strength at point X?

24 If in Figure 54.10 A and B each carry charges of +500 μC, 100 mm apart, find their electrical potential energy.

25 When placed at a point P near a charged Van de Graaff generator, a test charge of $+4.8 \times 10^{-19}$ C is found to have 7.2×10^{-18} J of electrical potential energy. What is the electric potential at P?

26 If the cap and leaves of an electroscope is at an electric potential of 500 V, what electric potential energy is possessed by a charge of + 13 pC on the electroscope?

27 On the positive terminal of a calculator battery 2 nC of charge has 19 nJ of electrical potential energy, while on the negative terminal it has 1 nJ. What is the electric potential difference between the two terminals?

28 You charge one side of a ballpoint pen negatively by rubbing it on your sleeve, thereby setting up a potential difference of 360 V between the two sides of the pen.
(*a*) Would a small positive charge have more electrical potential energy on the charged or the uncharged side?
(*b*) How much more electrical potential energy would a charge of −5.0 pC have on the charged side than on the uncharged side?

29 Determine the electric potential at a point which is 0.20 m from the centre of an insulated small conducting sphere carrying a charge of 8.0 nC.

30 An electron, of charge $-x$, moves around a nucleus in a circular orbit of radius *R*. The electric potential energy of the nucleus–electron system is $-U$. Using some or all of the given symbols *x*, *U* and *R*, which are all positive, write an expression for the electric potential at a distance *R* from the centre of the nucleus.

31 Refer to Figure 54.10. What is the electric potential at point P,
(*a*) if A and B both carry charges of +20 pC?
(*b*) if A and B carry charges of +20 pC and −20 pC respectively?

(c) if A and B carry charges of +60 pC and −20 pC respectively?

32 Two point charges $+q$ and $-q$ are a distance $2r$ apart, as illustrated in Figure 54.12. The points A, B, C and D are a distance r from $+q$. The points A, E and F are a distance r from $-q$.

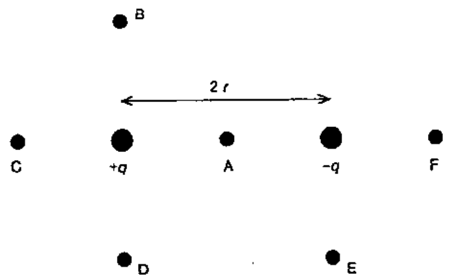


Fig. 54.12

(a) Which of the points A to F is at zero electric potential?
(b) At which one point is there the largest positive potential?
(c) Which two points are at the same potential?
(d) What is the value of the ratio

$$\frac{\text{potential at D} - \text{potential at F}}{\text{potential at E} - \text{potential at C}}?$$

33 Two small spheres U and V, carrying charges $+Q$ and $-Q$ respectively, are separated in air by a distance $2r$ as shown in Figure 54.13. XY is a line perpendicular to the line joining the charges.

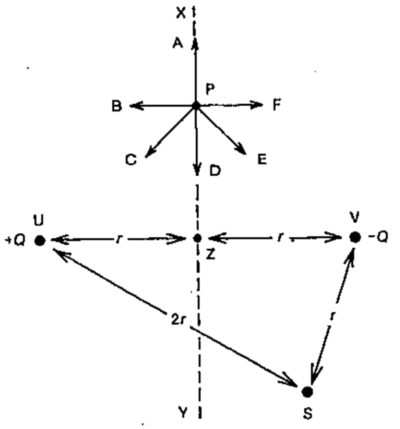


Fig. 54.13

(a) Which of the directions, A, B, C, . . . F in Figure 54.13 represents the direction of the electric field at point P?
(b) What is the electric potential at point Z?
(c) In terms of Q, r and ϵ_0 (or k), write expressions for
 (i) the electric potential at point S;
 (ii) the magnitude of the electric field strength at point Z. Also state its direction.

34 Consider the region along the line joining the charged spheres U and V in Figure 54.13. Which of the sketch graphs (Fig. 54.14) best represents the variation of
(a) electric potential with distance?
(b) electric field strength to the right with distance?
(c) electric potential with distance, if the charge on V is changed to $+Q$?
(d) electric field strength to the right with distance, if the charge on V is changed to $+Q$?

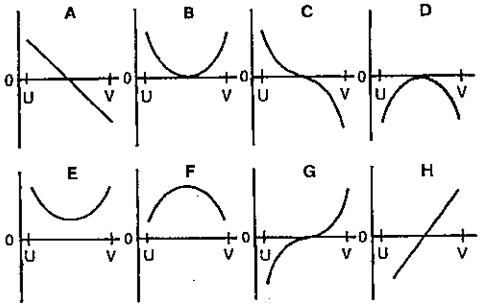


Fig. 54.14

35 Points A, B and C are situated in a horizontal plane as shown in Figure 54.15. There is a charge of +45 pC at A and one of −96 pC at B.

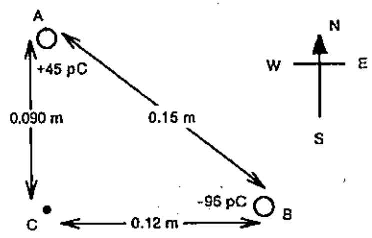


Fig. 54.15

For C determine the resulting
(a) electric potential;
(b) electric field strength.

36 Consider the arrangement of parallel charged plates shown in Figure 54.9.
(a) For each of the following pairs of points select the one where the electric potential is greater (write E if they are equal):
 (i) X and A;
 (ii) X and B;
 (iii) D and a point on the negative plate.
(b) The separation of the plates is 8.0 mm and the potential difference between them is 12 V. If A and B are 2.0 mm from the positive and negative plates respectively, what is the potential difference between
 (i) the positive plate and A?
 (ii) A and B?

37 An *equipotential surface* (or *equipotential line*) in an electric field is a surface (or line) joining all points of the same potential.
(a) What would be the shape of an equipotential surface surrounding a positively charged sphere?
(b) If in Figure 54.9 you were asked to work down the diagram and mark in three equipotential lines L_1, L_2 and L_3, with a constant potential difference between adjacent lines, would the spacing between L_1 and L_2 be less than, equal to or greater than the spacing between L_2 and L_3?
(c) Consider (b) again, this time for lines L_1, L_2 and L_3 drawn around a positively charged sphere, L_1 being closest to the sphere.
(d) At what angle do electric field lines cross equipotential surfaces?

SET 55 MOVING CHARGED PARTICLES IN ELECTRIC FIELDS

Topic	*Problems*
Electric field strength in a uniform field (between parallel charged plates) and its relation to electric potential gradient	1-5
Millikan experiment with stationary charged particles	6-7
Millikan experiment with moving charged particles	8-9

Topic	*Problems*
Conservation of charge	10-11
Charged particles moving between points with a known potential difference, electric forces on the particles, work done, changes in potential and kinetic energies	12-19
The electron volt	20-22
Electric field strength in a non-uniform field (near a point charge) and its relation to electric potential gradient	23-24
Electric current as the rate of movement of charge	25-29
E.m.f. and power of a source (of negligible resistance)	30-34
E.m.f. and internal resistance of a source (see Set 57)	
Moving charged particles in electrolytes	35-39
Deflection of charged particles in uniform electric fields, cathode ray oscilloscope	40-42
Deflection of charged particles in non-uniform electric fields (see Set 63)	
Potential, kinetic, total and binding energies of a charged particle in a stable circular orbit	43-45

In this set, assume
 all electrical interactions are taking place in air or in a vacuum,
 charges at an infinite separation have zero electrical potential energy.
Energy answers are to be given in *joule*. If the energy is required in *electron volt*, this will be indicated in the question.

Where necessary in this set, use
 Coulomb's law constant in a vacuum or in air,
 $k = 1/4\pi\epsilon_0$
 $= 9.0 \times 10^9\ \text{N m}^2\ \text{C}^{-2}$
 permittivity of a vacuum or air,
 $\epsilon_0 = 8.85 \times 10^{-12}\ \text{C}^2\ \text{N}^{-1}\ \text{m}^{-2}$
 $1\ \text{e} = 1.60 \times 10^{-19}\ \text{C}$
 $1\ \text{eV} = 1.60 \times 10^{-19}\ \text{J}$
 Avogadro constant,
 $L = 6.02 \times 10^{23}\ \text{mol}^{-1}$
 $1\ \text{g} = 6.02 \times 10^{23}\ \text{u}$ (unified atomic mass unit)

1 A uniform electric field is set up between two parallel charged plates P and N as shown in Figure 55.1. There is a potential difference V between P and N which are distance d apart. The electric field provides a force F on a small plastic sphere, which is carrying a positive charge q and is on the point of moving from P to N. Assume that gravitational effects on the sphere are negligible.

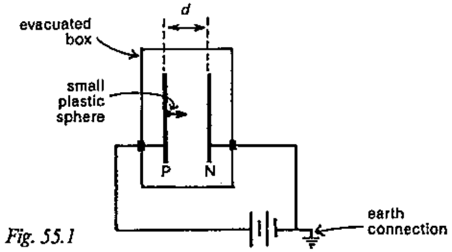


Fig. 55.1

(*a*) In terms of V and q
- (*i*) how much more electrical potential energy has the sphere on P than it has on N?
- (*ii*) what must be the increase in the kinetic energy of the sphere as it moves from P to N?
- (*iii*) how much work is done by the electric field in moving the sphere from P to N?

(*b*) Answer (*a*) (*iii*) in terms of F and d.
(*c*) Using your last two answers, deduce an expression (other than $E = F/q$) for the electric field strength E between the plates.
(*d*) What then would be an alternative to *newton per coulomb* for the SI unit for E? Your answer (if correct) is the more commonly used unit for E.

2 The *electric potential gradient* at a point is the rate at that point at which the electric potential is increasing per unit distance along the electric field line, i.e. $\Delta V/\Delta d$.
(*a*) Which *two* of the following are equal in magnitude to the electric potential gradient at a point?
- A The electric potential energy per coulomb at that point.
- B The electric field strength at that point.
- C The kinetic energy of a particle carrying one coulomb at that point.
- D The electric force per coulomb at that point.

(*b*) As you move along an electric field line in the direction of the field, is the electric potential gradient positive or negative?

3 If in Figure 55.1 the potential difference between the plates is 80 V and $d = 20$ mm, what is
(*a*) the magnitude, and
(*b*) the direction
of the electric field strength between the plates?

4 Consider the region between the charged plates in Figure 55.1.

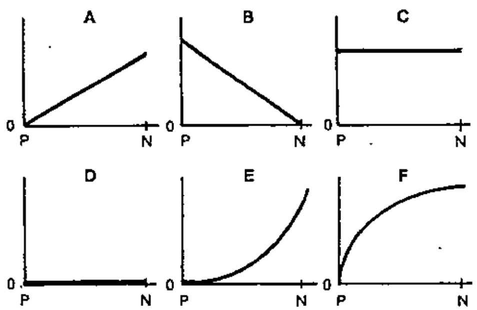


Fig. 55.2

Which of the graphs in Figure 55.2 best represents the variation of
(*a*) electric potential with position from P to N?
(*b*) electric field strength with position from P to N?

5 Two insulated brass plates, P and Q, stand vertically 12 cm apart, and are connected as shown in Figure 55.3 to a 720 V power supply. When a small positively charged sphere at the end of an insulating rod is placed as illustrated at the point X, 6.0 cm from the plate P, the electrical force acting on it is 12 mN to the left.

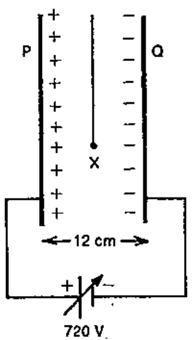


Fig. 55.3

(*a*) Specify fully the electric field strength between the plates.
(*b*) What is the charge on the sphere?
(*c*) The separation of the plates is increased to 16 cm. What is now the magnitude of the electrical force on the charged sphere?
(*d*) With the plates again 12 cm apart, the voltage selector on the power supply is adjusted. If the new force on the sphere is 1.6 mN, what now is the potential difference across the plates?

6 The potential difference between two large horizontal aluminium plates 20 mm apart is 1.0 kV. An oil drop, carrying five excess electrons and introduced into the space between the plates, remains stationary. Charge on an electron $= -1.6 \times 10^{-19}$ C.
(*a*) Is the top plate positive or negative?
(*b*) What is the magnitude of the electric field strength between the plates?
(*c*) Specify fully the electric force acting on the oil drop.
(*d*) What is the net force acting on the drop?
(*e*) Specify fully the gravitational force acting on it.
(*f*) What is the mass of the drop?

7 In an experiment similar to Millikan's, a small plastic sphere of radius 9.03×10^{-7} m and density 941 kg m^{-3} is held stationary between two horizontal metal plates, which are 4.91 mm apart and have a potential difference of 285 V between them. Other tests on this sphere indicated that it must be carrying three elementary charges. Determine
(*a*) the volume of the sphere;
(*b*) its mass;
(*c*) its weight;
(*d*) the upward electric force acting on it;
(*e*) the magnitude of the electric field strength between the plates;
(*f*) the charge on the sphere;
(*g*) a value for the elementary charge expressed in coulomb.

8 In a Millikan-type experiment, a small charged plastic sphere moved downwards under gravity alone with a terminal speed of v, as illustrated in Figure 55.4. With a 90 V power supply connected to the plates as shown in Figure 55.5, the sphere remained in balance.

Fig. 55.4

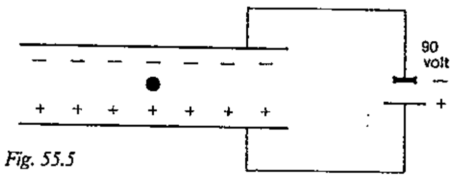


Fig. 55.5

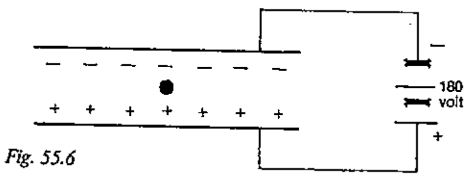


Fig. 55.6

(*a*) State the sign of charge on the sphere.
(*b*) State the direction of the electric field between the plates.
(*c*) The power supply setting in Figure 55.5 is doubled to 180 V as indicated in Figure 55.6. In which direction will the charged sphere now move?
(*d*) In terms of v, state the speed of the sphere in Figure 55.6.
(*e*) If the power supply connections are reversed in the arrangement shown in Figure 55.6, what will then be the *velocity* of the sphere?

9 In a type of Millikan experiment performed with tiny charged plastic spheres, a certain sphere fell between the plates under gravity alone at a steady speed of 8.0×10^{-5} m s^{-1}. The sphere remained stationary between the plates when a 90 V power supply was connected across them.
(*a*) What potential difference applied to the plates would cause the sphere to move upwards at 3.2×10^{-4} m s^{-1}?
(*b*) If the power supply connections are reversed, what potential difference would cause the sphere to move downwards at a speed of 3.2×10^{-4} m s^{-1}?
(*c*) With 360 V across the plates, connected so that the electric force is upwards, with what speed would the sphere move?
(*d*) The original 90 V source was connected as at first and, by means of a beam of x-rays, the charge on the sphere was altered. Determine the ratio of the final to the initial charge on the sphere, if it finally moved with a constant velocity of 2.4×10^{-4} m s^{-1}
- (*i*) upwards;
- (*ii*) downwards.

10 Three identical conducting spheres X, Y and Z, carrying +3.0 nC, −5 nC and +4 nC of charge respectively, are supported on insulating nylon fishing line as shown in Figure 55.7. X and Y are then allowed to touch and separate. Y and Z are then touched together and separated. What are the final charges on the three spheres?

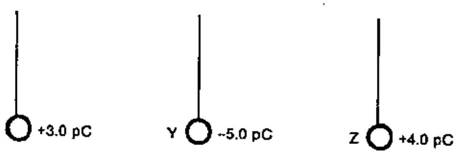


Fig. 55.7

11 If in the arrangement and procedure described in Problem 10 and Figure 55.7, the initial charges on spheres X, Y and Z had been +3 e, −5 e and +4 e respectively, what would have been the final charge on Z? (e is the abbreviation for elementary charge.) Think! Don't disappoint Millikan.

12 A and B, two points 2 m apart, are at potentials of 110 V and 100 V respectively. They are situated in a uniform electric field directed to the right as shown in Figure 55.8. A charge of +4 μC is moved from A to B.

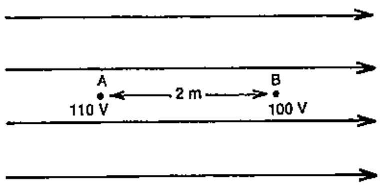


Fig. 55.8

(*a*) In moving the +4 μC charge from A to B what work is done?
(*b*) What is the magnitude of the electric force on the charged particle?
(*c*) What is the magnitude of the electric field strength?
(*d*) What is the electric potential gradient?

13 What is the potential difference between two points, if
(*a*) 24 J of work is done to move a total charge of 3 C from one point to the other?
(*b*) an alpha particle (charge 3.2×10^{-19} C) gains 1.6×10^{-15} J of energy in moving freely between two points?

14 How much work is done by an electric field which moves a total charge of 5.0 C from one end of the heating element in an electric jug to the other, if the potential difference between the two ends is 240 V?

15 In a computer disc-drive circuit an electron (charge -1.6×10^{-19} C) uses an energy of 8.0×10^{-19} J in moving from a point A at a potential of 16 V to a point B. What is the potential at B?

16 An electric charge of 4.0 μC is placed in a uniform electric field of intensity 500 V m^{-1}. Find
(*a*) the force acting on the charge;
(*b*) the work done to move the charge between two points 10 cm apart in the field;
(*c*) the potential difference between these two points.

17 In Figure 55.9 electrons emitted by a hot filament F are accelerated between metal plates which are located in an evacuated enclosure (not shown) and connected to a 200 V power supply.

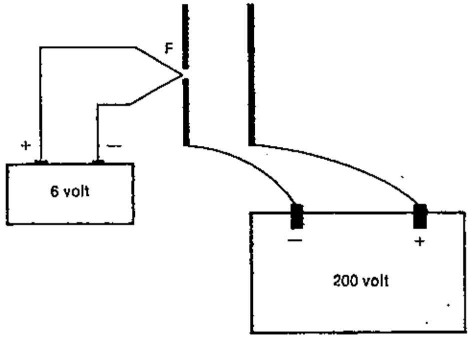


Fig. 55.9

(*a*) Which d.c. source—the 6 V or the 200 V—supplies the kinetic energy which the electron gains while in the region between the plates?
(*b*) How much kinetic energy does an electron gain while between the plates
(*i*) in the situation illustrated?
(*ii*) if the separation of the plates is doubled?
(*c*) How do you justify your answer to (*b*) (*ii*), when you consider that the work done equals the product of force and displacement, and doubling the separation has doubled the displacement?

18 In the electron gun in a typical colour TV tube the potential difference between the cathode and the anode is 15 kV. Assuming that the electrons leave the cathode with a negligible speed and that relativistic effects are to be ignored, find
(*a*) the kinetic energy, and
(*b*) the speed
of an electron as it passes through the opening in the anode. Mass of an electron = 9.1×10^{-31} kg.
(*c*) Travelling at this speed, how long would it take an electron to circle the Earth once? Radius of the Earth = 6.4×10^6 m.

19 Singly charged hydrogen ions are produced inside the container shown in Figure 55.10. They drift through a small hole in the first plate and are accelerated by the electric field. After passing through the hole in the second plate, they enter a metal cylinder 50.0 cm long. A cathode ray oscilloscope registers their time of entry into this cylinder and their time of arrival at the far end C.

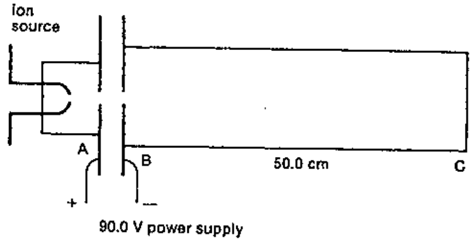


Fig. 55.10

(*a*) What is the kinetic energy of each ion as it traverses the cylinder from B to C?
(*b*) If the ions take 3.80 μs to travel from B to C, what is their speed?
(*c*) Calculate the mass of the hydrogen ion.

20 Although it is not the SI unit of energy, the *electron volt* (eV) is in some circumstances very convenient.
(*a*) When five electrons travel through the filament of your torch globe, across which there is a potential difference of 3.0 V, what electrical energy in
(*i*) joule, and
(*ii*) electron volt (eV)
is transferred into light and heat energy?
(*b*) Deduce how many joule of energy are equivalent to 1 eV.

21 What potential difference is required to give an alpha particle (charge +2 e) an increase in kinetic energy of 80 eV?

22 What is the speed of a 3.0 MeV proton? Proton mass = 1.67×10^{-27} kg. (1 MeV = 10^6 eV.)

23 Two points X and Y are 5.1×10^{-11} m and 5.3×10^{-11} m respectively from the centre of a helium nucleus, which carries a charge of +2 e, i.e. $+3.2 \times 10^{-19}$ C.
(*a*) What is the electric potential at
(*i*) X?
(*ii*) Y?
(*b*) What is the potential difference between X and Y?
(*c*) What is the distance between X and Y?
(*d*) What is the average electrical potential gradient in the region between X and Y?
(*e*) Determine the magnitude of the electric field strength midway between X and Y, i.e. 5.2×10^{-11} m from the centre of the nucleus. Compare your answers to (*d*) and (*e*).

24 Consider a point P which is distance R from a point charge Q. Use ϵ_0 or the Coulomb law constant k wherever necessary in this problem. Write expressions for
(*a*) the magnitude of the electric field strength E at point P;
(*b*) the electrical potential energy U_e of a small test charge q at point P;
(*c*) the *electrical potential* V_e at P (the electrical potential at a point is the electrical potential energy per unit positive charge at that point);
(*d*) the *electrical potential gradient* at P (the electrical potential gradient is the rate of change of electrical potential per unit distance moved from the point charge, i.e. dV_e/dR).
(*e*) Is there any connection between your answers to (*a*) and (*d*)?

25 During one start of a VW Kombi 220 C passes through the starter motor in 5.5 s. What was the electric current in the motor?

26 There is a current of 5.0 A in the heating element of a portable refrigerator. How long does it take for 60 C of charge to flow through the heating element?

27 In a particular computer monitor the current in the electron beam between the cathode and the anode is 2.5 mA. In 32 s
(*a*) what charge, and
(*b*) how many electrons
leave the cathode?

28 The current at a point X in a beam of electrons moving at 4.00×10^6 m s^{-1} is 1.00 mA. Assuming the electrons in the beam are equally spaced,
(*a*) how many electrons per second pass the point X?
(*b*) how many electrons are there in a 1.0 mm length of the beam?

29 Free electrons exist in a conductor. When these electrons drift in the same direction along the conductor a current is set up. Figure 55.11 shows a section of such a conductor PR, made of copper, in which there is a current of 1.6 A. Copper contains 1.0×10^{29} free electrons m^{-3}.

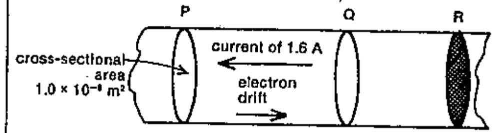


Fig. 55.11

(*a*) *Guess* (without any calculations) a value for the average drift speed of the free electrons along the wire.
(*b*) Now calculate how many electrons pass through the shaded section R in 1.0 s.
(*c*) Consider the piece of wire from Q to R which contains the free electrons that will pass through the cross-section R in the next 1.0 s.
(*i*) What is the volume of the piece of wire from Q to R?
(*ii*) What is the length of the piece of wire from Q to R?
(*d*) What is the average drift speed of the free electrons along the wire?

30 Determine the e.m.f. of
(*a*) a cell that supplies 24 J of energy to every 16 C of charge passing through it;
(*b*) a calculator battery which provides each electron passing through it with 1.44×10^{-18} J of electrical potential energy.

31 How much energy is supplied to each electron passing through a car battery of e.m.f. 12 V? Give your answer in
(*a*) joule;
(*b*) electron volt.

32 A d.c. power pack with an e.m.f. of 8.0 V is delivering a current of 2.5 A.
(*a*) What is the power output of the power pack?
(*b*) If it is used for 5.0 min, how much electrical energy does it supply?

33 If the button battery in an LCD calculator has a power rating of 0.7 mW and an e.m.f. of 1.5 V, what current is it delivering when the calculator is being used?

34 A particular 12 V car battery has a rating of "64 A h" (ampere hour) when fully charged.
(*a*) Is the ampere hour a unit of time, energy, power, current or charge?
(*b*) How much charge has passed through the battery by the time it is discharged?
(*c*) What total energy was stored in the battery when charged?

35 It is known that a given aqueous solution of a soluble salt contains 4×10^{20} doubly charged positive ions.
(*a*) How many negative elementary charges are carried by negative ions in this same solution?
(*b*) Does your answer to (*a*) depend on whether the negative ions are singly, doubly or trebly charged?

36 Three electrolytic cells are connected in series with a battery and an ammeter. The positive ions in the solutions are silver (Ag^+), hydrogen (H^+) and copper (Cu^{2+}) respectively. A current of 1.00 A is indicated on the ammeter.
(*a*) How many hydrogen atoms will be released in 1.00 s?
(*b*) What is the ratio of the number of silver atoms deposited to the number of copper atoms deposited in the same time?
(*c*) What is the ratio of the number of copper atoms deposited to the number of hydrogen molecules released in the same time? Hydrogen is a diatomic gas.

37 Two electrolytic cells containing aqueous solutions of sulphuric acid and copper sulphate are connected in series as in Figure 55.12. The ions are charged as shown in the diagram. After the current has been flowing for some time, the negative electrode in the copper sulphate cell is found to have increased in mass by 2.12 g. The atomic masses of copper and hydrogen are 63.6 and 1.00 u respectively. Hydrogen is a diatomic gas.

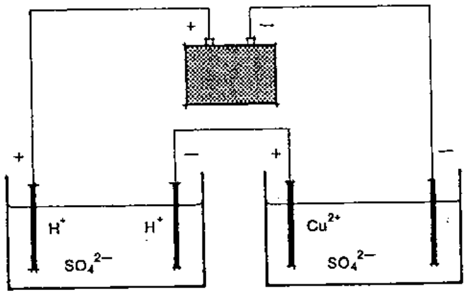


Fig. 55.12

(*a*) How many copper atoms have been deposited on the cathode?
(*b*) How many electrons have passed through the battery?
(*c*) How many hydrogen molecules have been deposited on the cathode of the sulphuric acid cell?
(*d*) What mass of hydrogen (in mg) was evolved at this electrode?

38 An electrolytic cell containing silver nitrate solution, a power pack and an ammeter are connected in series. After 10 min it is found that 0.80 g of silver has been deposited on the negative electrode. What was the average reading on the ammeter? Silver has a singly charged ion and an atomic mass of 108 u.

39 In the electrolysis of water 3.8×10^{-5} m^3 of a gas at a temperature of 23°C and a pressure of 104 kPa was evolved in 2 min 30 s. The current reading on a poorly calibrated ammeter in series with the electrolytic cell was 2.4 A. What correction should be applied to the ammeter reading? Both hydrogen and oxygen are diatomic gases. The molar volume of a gas at STP (0°C and 101 kPa) is 2.24×10^{-2} m^3.

40 An electron of mass 9.1×10^{-31} kg is projected with a horizontal velocity of 6.0×10^6 m s^{-1} into a uniform electric field of strength of 1.5×10^4 V m^{-1} vertically downwards. The field ends abruptly after the electron has traversed it for a period of 5.0×10^{-9} s. Given that the whole process takes place in an evacuated container and that the weight of the electron can be neglected, determine
(*a*) the vertical force on the electron (don't forget its direction);
(*b*) the horizontal force on the electron;
(*c*) the vertical acceleration of the electron;
(*d*) the vertical displacement of the electron;
(*e*) the vertical component of the velocity with which the electron leaves the field;
(*f*) the magnitude of the velocity with which it leaves the field;
(*g*) the direction of this final velocity.

41 Figure 55.13 shows a simplified picture of the evacuated interior of a cathode ray tube. Electrons removed from the filament by heating are accelerated through the electron gun before passing between the plates YY'. The electron mass = 9.1×10^{-31} kg.

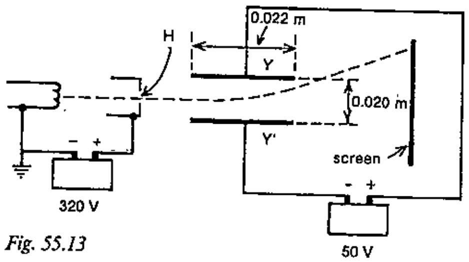


Fig. 55.13

(*a*) What is the energy of each electron as it reaches the hole H in the electron gun, assuming that it leaves the filament with a negligible speed?
(*b*) What is the electron's speed at H?
(*c*) State fully the electric field strength between Y and Y'.
(*d*) How long is the electron in the region between the plates Y and Y'?
(*e*) What is the vertical displacement of the electron while it is between the plates Y and Y'?
(*f*) What is the horizontal component of the electron's velocity when it finally reaches the screen at the end of the tube?

42 Use the following data gathered with a cathode ray tube similar to that shown in Figure 55.13 to determine a value for the mass of the electron:
speed of the electron leaving the electron gun = 1.52×10^7 m s^{-1}
potential difference between the deflecting plates = 234 V
vertical displacement of the electron beam while between the deflecting plates (determined from the position of the spot on the screen and tube geometry) = 2.1 mm upwards
length and separation of the deflecting plates were as in Figure 55.13.

43 A small particle S carrying charge $-q$ moves in a stable circular orbit of radius r about the centre of a much more massive particle L carrying charge $+Q$. In terms of Q, q, r and ε_0 (or k the Coulomb law constant), write expressions for
(*a*) the electric force exerted on S by L;
(*b*) the electric potential energy of S (strictly this potential energy is the combined potential energy of S and L, but since it has very little influence on the much larger mass L, it is often ascribed to the smaller mass S);
(*c*) the kinetic energy of S;
(*d*) the total energy, i.e. potential plus kinetic, of S;
(*e*) the binding energy of S to L.

44 In the Bohr model of the hydrogen atom in its lowest energy state, a negative electron moved in a stable circular orbit of radius 5.29×10^{-11} m about a proton carrying a charge of +1 e.
(*a*) Giving your answers to three significant figures, determine the electron's
(*i*) electric potential energy;
(*ii*) kinetic energy;
(*iii*) total energy;
(*iv*) binding energy.
(*b*) Before doing any calculations have a *guess* at an approximate value for the ratio of the *electrical* force of attraction between the electron and proton to their *gravitational* force of attraction. Then calculate it, giving your answer as an order of magnitude. Electron mass = 9.1×10^{-31} kg; proton mass = 1.7×10^{-27} kg; gravitation constant = 6.7×10^{-11} N m^2 kg^{-2}.

45 An oversimplified picture of a singly charged helium ion consists of an electron moving in a stable circular orbit about a nucleus consisting of two protons, each carrying a charge of +1 e, and two uncharged neutrons. If the total energy of the orbiting electron is -8.5×10^{-18} J, what is the radius of its orbit?

SET 56 SIMPLE D.C. CIRCUITS

Topic	*Problems*
Variation of electric potential and field strength with distance around a circuit	1–6
Resistance, Ohm's law	7–11
Colour coding and international number–letter coding for values of resistors	12–13
Resistivity	14–16
Conductance and conductivity (electrical)	17–18
Transformation of electrical energy into other forms of energy, power developed in a component, cost of electrical energy	19–25
The kilowatt hour	26–27
Circuit components in series	28–31
Resistors in series	32
Circuit components in parallel	33–35
Resistors in parallel	36–38
Combinations of series and parallel resistors	39–41
Simpler general d.c. circuit problems	42–48

1 A uniform wire 2.4 m long is connected across the terminals of a 12 V car battery. What is the magnitude of the electric field strength in the wire?

2 A 0.50 m length of nichrome wire is connected across the terminals of a d.c. power supply. If the magnitude of the electric field strength in the wire is 3.0 V m^{-1}, what is the potential difference between the ends of the wire?

3 A wire 1.0 m long and uniform in area of cross-section is connected to the terminals of a 3.0 V battery. What is the potential difference between two points in the wire 0.40 m apart?

4 In the circuit shown in Figure 56.1 AB and BC are two uniform resistance wires of the same material but of different thickness. AB and BC are 0.60 m and 0.20 m long respectively. Readings on the voltmeter were used to plot the graph (Fig. 56.2) of potential difference V between A and a variable point X on the wires against the distance AX along the wires.

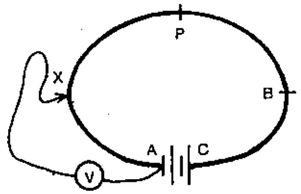


Fig. 56.1

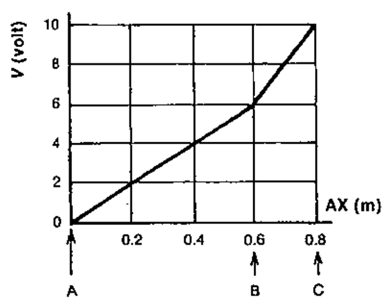


Fig. 56.2

Give your answers to two significant figures.

(*a*) What is the potential difference across

(*i*) AB?

(*ii*) BC?

(*b*) What is the e.m.f. of the d.c. power supply?

(*c*) Draw a graph of the magnitude of the electric field strength E inside the wire at X against the distance AX.

(*d*) Which is the thicker wire, AB or BC?

(*e*) If the wire was cut at the points P and B shown in Figure 56.1 and this 0.20 m length (PB) was removed from the circuit, which of the graphs in Figure 56.3 correctly shows the new variation of

(*i*) V with distance AX?

(*ii*) E with distance AX?

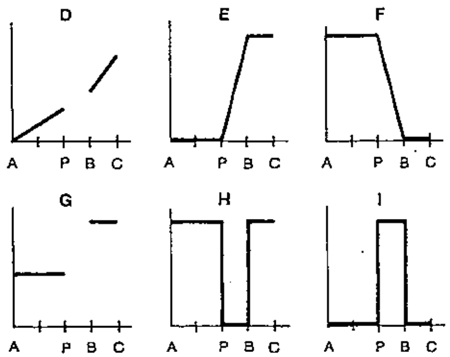


Fig. 56.3

5 Two uniform resistance wires A and B, each of length 2.0 m, are connected in series with a battery as shown in Figure 56.4. Figure 56.5 shows a comparison of the electric fields in the two resistors, and Figure 56.6 shows the relation between potential and distance along the resistor A.

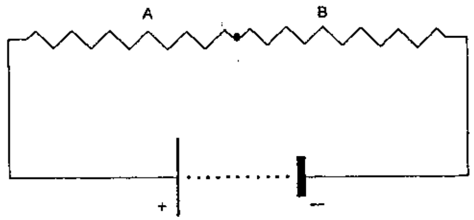


Fig. 56.4

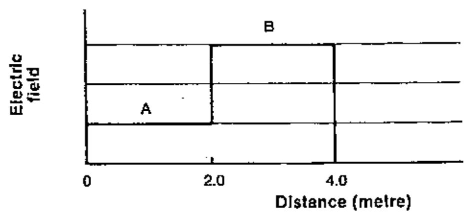


Fig. 56.5

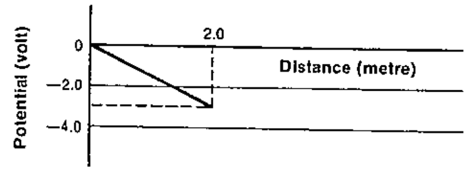


Fig. 56.6

(*a*) What is the magnitude of the electric field strength in wire B? What is its direction—to the left or the right—in Figure 56.4?

(*b*) What is the e.m.f. of the battery?

(*c*) One of the terminals of the d.c. source is earthed. Which one?

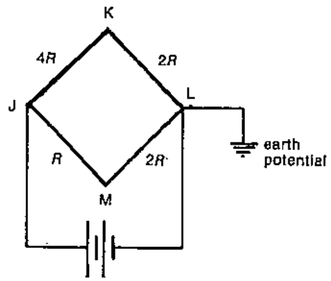


Fig. 56.7

6 Four wires, JK, KL, JM and LM, are joined as shown in Figure 56.7 and a battery is connected between J and L. The resistance of each wire, in terms of R, is marked on the diagram.

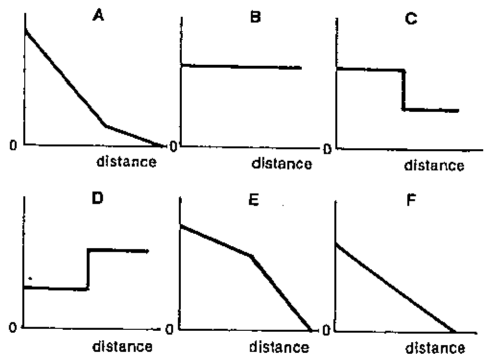


Fig. 56.8

Which of the graphs in Figure 56.8 best represents

(*a*) the electric potential as a function of distance from J along the path JML?

(*b*) the magnitude of the electric field strength as a function of distance from J along the path JKL?

(*c*) the current as a function of distance from J along the path JKL?

7 Determine the resistance of the filament in a commonly used light globe, if the current in it is 0.25 A when the potential difference between its ends is 240 V?

8 What is the current in a 1.2 kΩ carbon resistor connected across a potential difference of 6.0 V?

9 What is the potential difference across a 48 Ω heating element when there is a 5.0 A current in it?

10 Does Ohm's law hold for

(*a*) the coil C whose current–potential difference characteristics are given in Figure 56.12?

(*b*) the semiconducting device S whose current–potential difference characteristics are given in Figure 56.13?

11 What is the resistance of the coil C partly described in Figure 56.12?

12 Use the resistor colour code to answer the following (colours are quoted in order from the end nearer to which the colours are marked on the resistor):

(*a*) What is the resistance of a carbon resistor marked red, violet, black, silver? What is the tolerance of the resistor?

(*b*) What resistance and tolerance corresponds to yellow, violet, orange, with no fourth band?

(*c*) If you are searching in a box of mixed resistors for one which must be within 5 per cent of 3.9 kΩ, what colour order should you be looking for?

13 Use the international number–letter code to answer the following:

(*a*) What resistance and tolerance is indicated by

(*i*) 5R6J?

(*ii*) K15M?

(*iii*) 4M3G?

(*b*) What marking would you be looking for if you required a resistance within 10 per cent of 6.8 kΩ?

14 A common type of copper wire has a cross-sectional area of 1.00×10^{-8} m^2. If a 6.03 m length of it has a resistance of 10.2 Ω, what is the resistivity of copper?

15 If a standard resistor of 6.00 Ω is to be made of constantan wire of diameter 0.250 mm, what length is required? Resistivity of constantan = 4.9×10^{-7} Ω m.

16 Estimate as an order of magnitude the resistance between your right foot and the ground if you are wearing rubber thongs. Resistivity of rubber = 10^{12} Ω m.

17 The present SI unit of *electrical conductance* is the siemens (S).
(*a*) How is the siemens related to the ohm?
(*b*) Some years back the unit of conductance was called the *mho*. Can you suggest why?
(*c*) In terms of the siemens what is the SI unit of *electrical conductivity*?

18 Determine
(*a*) the conductance of a 0.2 Ω resistor;
(*b*) the resistance of a 37 mS resistor (or conductor!);
(*c*) the conductance of a 1.6 m length of copper wire of diameter 1.25 mm and conductivity 6.4×10^7 S m^{-1}.

19 The heating element H of an electric toaster carries a current of 5.0 A when the potential difference across it is 240 V.
(*a*) How many coulomb of charge move through H in one second?
(*b*) How much energy does each of these coulombs of charge use while moving through H?
(*c*) How much energy is therefore used up in H in one second?
(*d*) What is the power rating marked on the toaster? This is the rate at which it transforms electrical energy into heat and light energy.

20 An electric jug is marked "240 V 1.7 kW".
(*a*) When in normal use, what is the current in its heating element?
(*b*) How much electrical energy does it transform into heat energy if it takes 2 min 21 s to boil some water?

21 If V is the potential difference across an electrical component, such as a heater, light or motor, of resistance R, and I is the current in it, deduce expressions for the power P developed in it in terms of
(*a*) I and R;
(*b*) V and R.

22 If the quartz halogen underwater light in a swimming pool is rated at 150 W and draws a current of 4.7 A, what is the resistance of the filament?

23 Two electric light bulbs X and Y are marked "240 V 100 W" and "240 V 40 W" respectively.
(*a*) Without doing any calculations *guess* which bulb has the higher resistance. Now determine their resistances.
(*b*) How much electrical energy is used if X is left on for 5.0 min?

24 A student calculates that the 240 V electric radiator she runs in her bedroom while studying has an operating resistance of 28.8 Ω.
(*a*) What power rating is marked on the back of the radiator?
(*b*) What is the total cost (to the nearest dollar) to run it during the winter months of June, July and August, if she has it on Sunday to Friday evenings from 8.00 p.m. to 10.30 p.m.? Assume electric energy costs 3.1 cent MJ^{-1}.

25 A well-insulated calorimeter contains 0.20 kg of water at 15°C. A heating coil R is placed in the water and connected to a power source as indicated in Figure 56.9. The temperature of the water rises to 25°C in 7.0 min while the ammeter reads 2.5 A. Assume the heat lost to the calorimeter and the surroundings is negligible. Specific heat capacity of water = 4.2 kJ kg^{-1} K^{-1}.

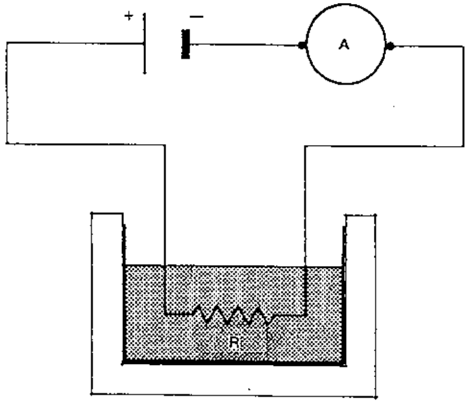


Fig. 56.9

Determine
(*a*) the amount of electrical energy transformed into heat energy in the 7.0 min;
(*b*) the power developed in R;
(*c*) the potential difference across R.

26 A non-SI unit still used by some electricity authorities is the kilowatt hour (kW h). Is it a unit of power, time, charge or energy?

27 (*a*) How many megajoule are equivalent to 1 kW h?
(*b*) If a 690 W electric motor is used to run a filter pump for a swimming pool, how much energy is used per week, if the pump runs for 6.0 h per day? Give your answer in
(*i*) kW h;
(*ii*) MJ.

28 If two circuit components A and B are connected in series to a battery, and V_A and V_B represent the potential differences across and I_A and I_B the currents in A and B respectively, write down expressions relating
(*a*) I_T, I_A and I_B, where I_T represents the current in the series group regarded as *one* component;
(*b*) V_T, V_A and V_B, where V_T represents the potential difference across the series group regarded as a *single* component.

29 A d.c. power supply is connected in series, as in Figure 56.10, with a 5.0×10^4 Ω resistor, a milliammeter of negligible resistance and a diode valve which requires a potential difference of 140 V to produce a current of 3.2 mA. This is the current reading on the meter.

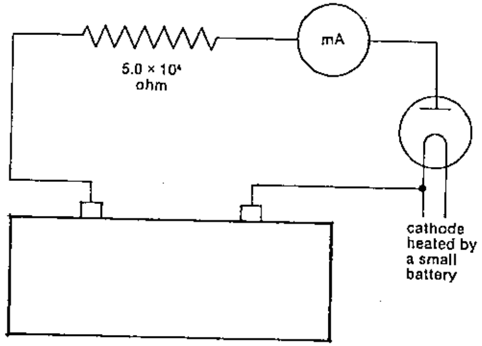


Fig. 56.10

Determine
(*a*) the current in the resistor;
(*b*) the potential difference across the resistor;
(*c*) the potential difference between the power supply terminals.

30 Figure 56.11 shows the current–potential difference relationships for two components X and Y.

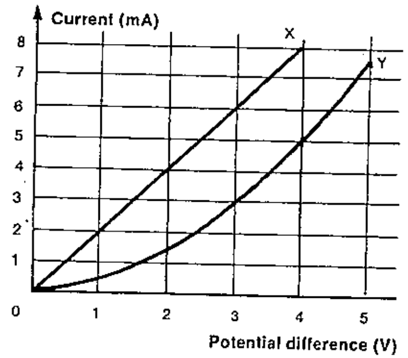


Fig. 56.11

(*a*) If X and Y were connected in series in a circuit and the current in X was 5.0 mA, what would be
(*i*) the current in Y?
(*ii*) the potential difference across X?
(*iii*) the potential difference across Y?
(*iv*) the potential difference across X and Y taken as one component?
(*b*) When X and Y are connected in series into a *new* circuit, the potential difference across Y is 3.0 V. What is
(*i*) the current in Y?
(*ii*) the current in X?
(*iii*) the potential difference across X?
(*iv*) the potential difference across X and Y taken as a single component?

31 An ohmic coil C and a semiconducting device S are found to behave according to the current–potential difference graphs shown in Figures 56.12 and 56.13 respectively.

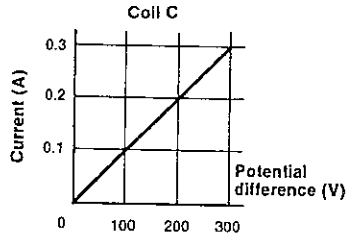


Fig. 56.12

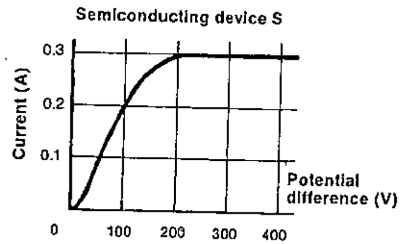


Fig. 56.13

(*a*) C and S are connected in series with a d.c. power supply. If a voltmeter connected across S reads 200 V, what is
(*i*) the current in S?
(*ii*) the potential difference across C?
(*iii*) the potential difference between the power supply terminals?
(*b*) The e.m.f. of the power supply is altered to raise the potential difference between its terminals to 600 V. What now is
(*i*) the current in S?
(*ii*) the potential difference across S?

32 What is the total resistance of
(*a*) a 10 Ω resistor in series with a 12 Ω resistor?
(*b*) two 18 Ω resistors in series?
(*c*) twenty 18 Ω resistors in series?
(*d*) a 2.2 Ω resistor in series with a 47 MΩ resistor?

33 Suppose two circuit components A and B are connected in parallel with one another and then to a battery, and V_A and V_B represent the potential differences across, and I_A and I_B the currents in A and B respectively. Write down expressions relating
(*a*) I_T, I_A and I_B, where I_T represents the current in the parallel group regarded as *one* component;
(*b*) V_T, V_A and V_B, where V_T represents the potential difference across the parallel group regarded as a *single* component.

34 Refer back to Figure 56.11 which shows the current–potential difference relationships for two components X and Y.
(*a*) If X and Y are connected in parallel and then to a power supply, and the potential difference across X is 2.0 V, what is
(*i*) the current in X?
(*ii*) the potential difference across Y?
(*iii*) the current in Y?
(*iv*) the current in the power supply?
(*b*) In a *new* circuit X and Y are again in parallel, and the current in Y is 0.5 mA. What is the current in X?

35 The current–potential difference characteristics for an ohmic coil C and a semiconducting device S are given in Figures 56.12 and 56.13. C and S are connected in parallel in a circuit.
(*a*) If the current in C is 0.10 A, what is
(*i*) the potential difference across C?
(*ii*) the potential difference across S?
(*iii*) the current in S?
(*iv*) the total current in C and S taken as one component?
(*b*) With C and S connected in parallel in a different circuit, it was found that the total current in the parallel group was 0.70 A. What is the potential difference across the parallel group?

36 If two resistors R_1 and R_2 are connected in parallel, show that the total resistance R_T of the parallel group is given by

$$R_T = \frac{R_1 R_2}{R_1 + R_2}.$$

37 What is the effective total resistance of each of the following parallel groups?
(*a*) a 24 Ω and a 48 Ω resistor;
(*b*) two 36 Ω resistors;
(*c*) three 36 Ω resistors;
(*d*) twenty 36 Ω resistors;
(*e*) a 2.2 Ω resistor and a 47 MΩ resistor.

38 In a circuit it becomes necessary to reduce the 10 Ω resistance of a component C to 6 Ω without removing C from the circuit. What is the resistance of the resistor that should be connected in parallel with C?

39 X and Y are two resistors with resistances R_X and R_Y. $R_X < R_Y$.
(*a*) If X and Y are connected in *series*, the total resistance of the combination must be
A less than R_X.
B greater than R_X but less than R_Y.
C greater than R_Y.
(*b*) Using the answer key from (*a*), what can be said about the total resistance of X and Y if they are connected in *parallel*?

40 What is the effective total resistance between the points P and Q in
(*a*) Figure 56.14?
(*b*) Figure 56.15?

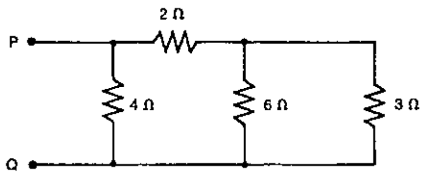


Fig. 56.14

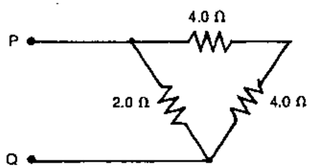


Fig. 56.15

41 Assume you have only five 12 Ω resistors. How could you connect some or all of them to produce an effective total resistance of
(*a*) 48 Ω?
(*b*) 4 Ω?
(*c*) 15 Ω?
(*d*) 28 Ω?

42 Two resistors of 500 Ω and 1000 Ω are joined in parallel and carry a *total* current of 60 mA. Find
(*a*) the potential difference across the combination;
(*b*) the current in the 500 Ω resistor.

43 Figure 56.16 shows an electrical circuit in which the potential difference between B and C is 6 V.

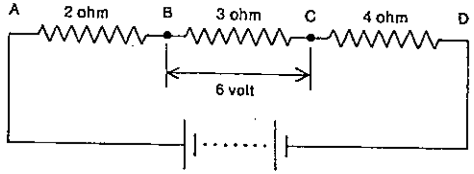


Fig. 56.16

Calculate
(*a*) the current in the 2 Ω resistor;
(*b*) the quantity of charge that passes through the 4 Ω resistor in 5 min;
(*c*) the potential difference between the terminals of the battery (to two significant figures).

44 A projector lamp is designed to take a current of 4.0 A when used with a 32 V home lighting unit. What resistance ought to be placed in series with the lamp if it is to be used on a 240 V supply?

45 A circuit is shown in Figure 56.17, in which the straight wires represent conductors of negligible resistance.

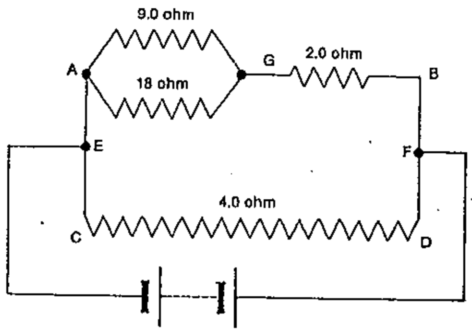


Fig. 56.17

If the current in the 18 Ω resistor is 0.50 A, what is
(*a*) the potential difference between A and G?
(*b*) the current in the 9.0 Ω resistor?
(*c*) the current in the 2.0 Ω resistor?
(*d*) the potential difference between G and B?
(*e*) the potential difference between A and B?
(*f*) the potential difference between C and D?
(*g*) the current in the 4.0 Ω resistor?
(*h*) the current in the battery?

46 Three 1.5 V 0.50 A lamps are to be operated normally, but the only battery available provides a potential difference of 12 V. The lamps are connected as shown in Figure 56.18 with a resistor R joined in series. What must be the resistance of R?

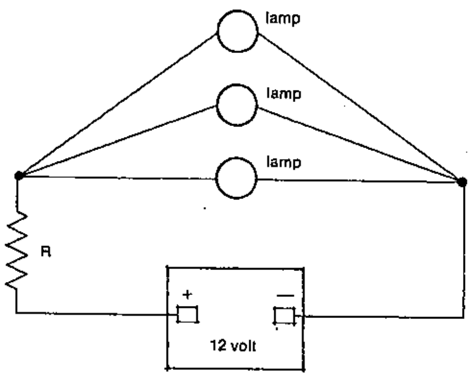


Fig. 56.18

47 In the circuit shown in Figure 56.19 the point A is maintained at a potential of +300 V relative to B.

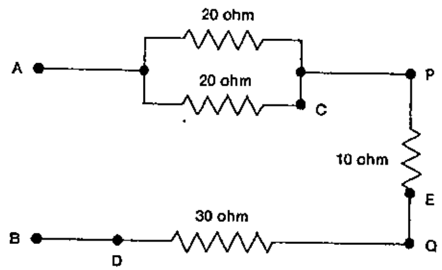


Fig. 56.19

What is the potential difference between P and Q
(*a*) in the circuit shown?
(*b*) if a break occurs at C?
(*c*) if, instead, a break occurs at D?
(*d*) if, instead, a break occurs at E?

48 Three uniform wires, P, Q and R, are identical in composition, length (0.50 m) and diameter. The ends of P, Q and R are joined as in Figure 56.20. The unit is then connected by wires of negligible resistance to a 24 V power supply. The current in P is found to be 4.0 A.

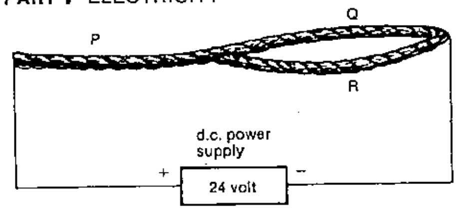


Fig. 56.20

Determine
(*a*) the current in wire Q;
(*b*) the magnitude of the electric field strength in wire R;
(*c*) the resistance of wire R;
(*d*) the current in the d.c. power supply.

SET 57 LESS SIMPLE D.C. CIRCUITS

Topic	*Problems*
E.m.f. and internal resistance of a d.c. power source	1–6
Identical d.c. power sources in series and in parallel	7–9
Kirchhoff's laws	10–13
Voltage dividers	14–16
Potentiometer	17–20
Wheatstone bridge	21–24
D.c. meter conversion, multimeters	25–30
Less simple general d.c. circuit problems	31–40

1 A battery of e.m.f. 4.0 V and internal resistance 0.4 Ω is connected to a 7.6 Ω resistor. Determine
(*a*) the current in the circuit;
(*b*) the potential difference across the 7.6 Ω resistor.

2 A lead-acid battery of e.m.f. 12 V and internal resistance 1.0 Ω is connected across a series of resistors one at a time.
(*a*) What would be the current in the circuit if the external resistor has a resistance of
(*i*) 0.20 Ω?
(*ii*) 1.0 Ω?
(*iii*) 11 Ω?
(*b*) What would be the potential difference across the battery terminals in each of the three cases (*i*), (*ii*) and (*iii*) above?
(*c*) For the potential difference between the terminals of a d.c. source to approach the e.m.f. of the source, should the resistance of the external circuit be *much less than, equal to* or *much greater than* the internal resistance of the source?

3 Refer to Problem 2 above.
(*a*) What is the rate at which energy is being *supplied* by the source in cases (*i*), (*ii*) and (*iii*)?
(*b*) What is the rate at which energy is *received* by the external resistor in each case?
(*c*) Where else has energy been used?
(*d*) From the limited information in your answers to (*b*), would you expect that a maximum amount of energy is transferred from the source to the external circuit when the resistance of the external circuit is *much less than, equal to* or *much greater than* the internal resistance of the source?

4 In the circuit shown in Figure 57.1, V_1 and V_2 are very high resistance voltmeters. The battery of e.m.f. 12 V has internal resistance of 1.0 Ω. Connecting wires have negligible resistance.

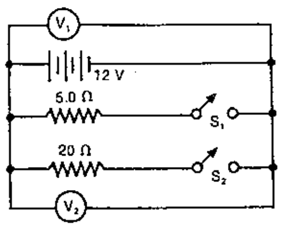


Fig. 57.1

(*a*) With S_1 closed, what is the reading on
(*i*) V_1?
(*ii*) V_2?
(*b*) With both S_1 and S_2 closed, what does V_1 read?

5 A battery with an internal resistance of 0.300 Ω is connected up to a resistance Z. The potential difference between its terminals is 1.48 V while the current through the circuit is 0.200 A.
(*a*) What is the e.m.f. of the battery?
(*b*) What is the resistance of Z?

6 The e.m.f. of a copper–constantan thermocouple is known to be 4.00 mV, but when a millivoltmeter of resistance 260 Ω is connected to it, the meter reads 3.25 mV. What is the internal resistance of the thermocouple?

7 Four Daniel cells, each of e.m.f. 1.08 V and internal resistance 0.20 Ω, are connected together.
(*a*) If they are connected in series, what is the effective total
(*i*) e.m.f.?
(*ii*) internal resistance?
(*b*) If they were connected in parallel, what would be their effective total
(*i*) e.m.f.?
(*ii*) internal resistance?

8 What is the internal resistance of each of two cells (e.m.f. 2.1 V), which when connected in parallel produce a current of 10 A in a load resistor of 0.20 Ω?

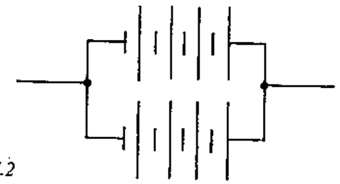


Fig. 57.2

9 Figure 57.2 shows a battery of eight cells. If each cell has an e.m.f. of 2.0 V and an internal resistance of 0.50 Ω, what is
(*a*) the e.m.f., and
(*b*) the internal resistance
of the battery?

10 Suppose with the circuit in Figure 57.3 you were attempting to use Kirchhoff's laws to find the currents in the two cells and the current in and potential difference across the 8.0 Ω resistor.
(*a*) In using Kirchhoff's current or junction law,
(*i*) how many junctions are there?
(*ii*) how many independent equations do these junctions provide?
(*b*) In using Kirchhoff's voltage or loop law,
(*i*) how many loops or closed paths are there? (Think!)
(*ii*) how many independent equations do these loops provide?

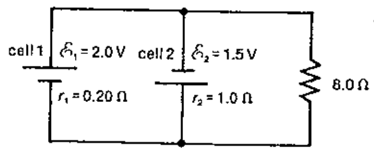


Fig. 57.3

11 Using Kirchhoff's laws (or otherwise) determine for the circuit in Figure 57.3
(*a*) the current in cell 1;
(*b*) the current in cell 2;
(*c*) the potential difference across the 8.0 Ω resistor.

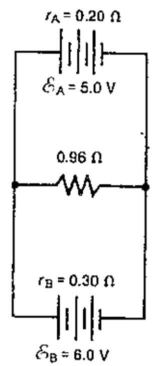


Fig. 57.4

12 For the circuit shown in Figure 57.4 determine by means of Kirchhoff's laws (or otherwise)
(*a*) the current in, and
(*b*) the potential difference across
the 0.96 Ω resistor.

13 In the circuit in Figure 57.5, A is a cell of e.m.f. 6.0 V and internal resistance 1.0 Ω, and B is another cell of e.m.f. 4.0 V and internal resistance 0.50 Ω.

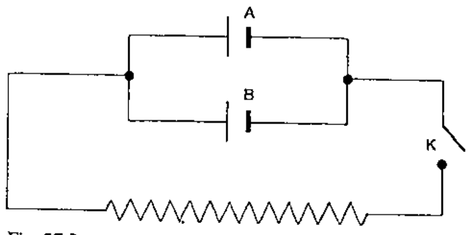


Fig. 57.5

(*a*) Find the potential difference between the contacts of the key K when it is open.
(*b*) What is the current in the 4.0 Ω resistor when K is closed?

14 In Figure 57.6 XY represents a 24 Ω resistor made from uniform wire which is connected to a 12 V battery. The length XP is one-quarter of the length XY.

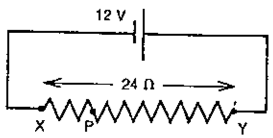


Fig. 57.6

(*a*) What is the potential difference across XP?
(*b*) If a 3.0 Ω resistor is connected in parallel with the section XP, what now is the potential difference across XP?

15 In Figure 57.7 AB is a resistor of 1000 Ω, across which there is a potential difference of 20.0 V. C is a sliding contact so that the resistance of the part AC is 800 Ω. X is an unknown resistance and K is a key.

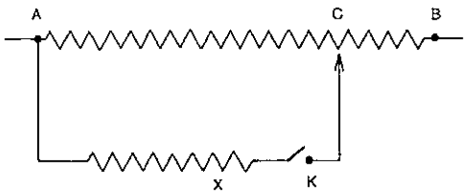


Fig. 57.7

(*a*) With K open what is the potential difference between A and C?
(*b*) When K is closed, the potential difference between A and C is found to be 14.4 V. Determine
  (*i*) the current in X;
  (*ii*) the resistance of X.

16 In Figure 57.8 PQ is a resistor of 12 kΩ and R is a resistor which is connected in parallel with two-thirds of PQ at P and S. If a potential difference of 20 V is maintained between P and Q, calculate the potential difference between P and S when there is a current of 2.0 mA in R.

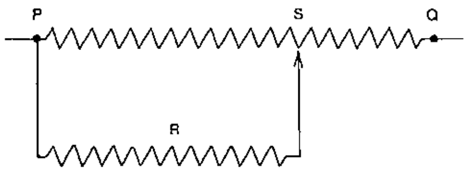


Fig. 57.8

17 A potentiometer wire XY of length 1.000 m is connected to a car battery and ammeter as shown in Figure 57.9. A cell C of unknown e.m.f., a galvanometer G and key K are connected as shown, and the sliding contact S is moved along XY until there is zero current in the galvanometer when K is closed. The length XS is found to be 0.133 m. At the same time a voltmeter (not shown in Fig. 57.9) connected across XY reads 11.5 V. Calculate the e.m.f. of cell C.

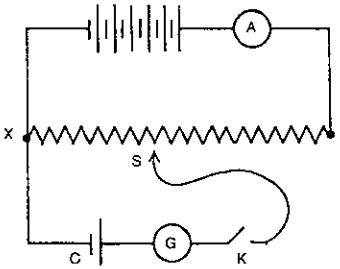


Fig. 57.9

18 With a potentiometer circuit similar to that shown in Figure 57.9, it is known that the linear resistance of the 500 mm long potentiometer wire XY is 5.18 Ω m^{-1}. When the galvanometer current is zero with K closed, XS = 89 mm and the ammeter reads 4.62 A. Determine the e.m.f. of cell C.

19 The wire of a potentiometer is 200 cm long and has a resistance of 5.0 Ω. When a dry cell of e.m.f. 1.5 V in series with a galvanometer is being tested on the potentiometer wire, no current flows through the galvanometer when the points tapped are 120 cm apart. Find the current flowing through the potentiometer wire.

20 In the circuit shown in Figure 57.10 the 6.0 V battery has negligible internal resistance and the ammeters M_1 and M_2 both read zero.

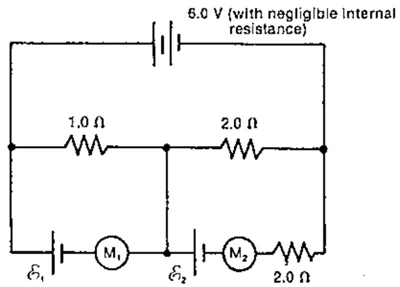


Fig. 57.10

Determine
(*a*) the e.m.f. $\mathscr{E}_1$;
(*b*) the e.m.f. $\mathscr{E}_2$;

21 In the Wheatstone bridge circuit shown in Figure 57.11, G represents a galvanometer, *R* is the resistance of the arm CD, and I_1 and I_2 are the currents in the arms AB and AC respectively. The galvanometer reads zero.

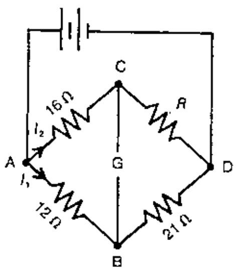


Fig. 57.11

Where necessary, give answers in terms of I_1, I_2 and *R*.
(*a*) What is the potential difference between B and C?
(*b*) Write down a statement equating two expressions for the potential difference between
  (*i*) A and G;
  (*ii*) G and D.
(*c*) Using your answers to (*b*), deduce a value for the unknown resistance *R*.

22 Resistors having 2, 3, 4 and *R* Ω resistance are arranged as in Figure 57.12, there being a potential difference of 2 V applied between A and C.

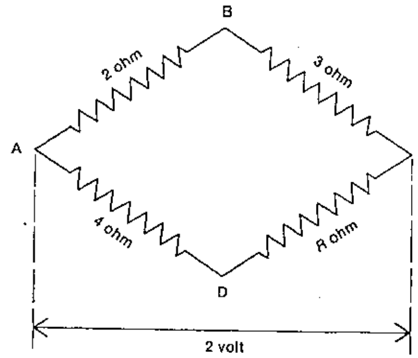


Fig. 57.12

Answers to (*a*) and (*b*) may contain *R*.
(*a*) What is the current in each of the arms AB, BC, AD and DC?
(*b*) What is the potential difference between
  (*i*) A and B?
  (*ii*) A and D?
(*c*) Use your answers to (*b*) to determine a value for *R* for which a galvanometer connected between B and D would read zero.

23 One use of the Wheatstone bridge circuit is in digital clinical thermometers, where one arm of the bridge is a thermistor. The resistance of a thermistor decreases with temperature. Figure 57.13 shows the thermistor in the arm WX. The resistances of the other three arms are chosen to give a balanced bridge, i.e. zero current in the meter, at 0°C.

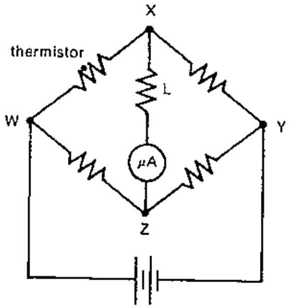


Fig. 57.13

(*a*) For a patient thermometer reading of 36.8°C, would current be flowing from X to Z or Z to X?
(*b*) Would the current in the microammeter be greater for a patient temperature of 36 or 37°C?

24 A balanced Wheatstone bridge ABCDA includes resistances AB of 100 Ω, BC of 10.0 Ω, CD of 14.5 Ω, with a battery of e.m.f. 1.50 V and internal resistance 1.00 Ω connected between A and C. The galvanometer resistance is 5.00 Ω. If there is a break in the arm CD, what would be the current in the galvanometer?

25 A galvanometer has a resistance of 50.0 Ω and gives a full-scale deflection for a current of 100 mA.
(*a*) In order to convert it into an ammeter reading up to 10.0 A,
  (*i*) would you add a resistor in series or parallel with the meter?
  (*ii*) what value resistance would you require?
(*b*) If, instead, you are asked to convert the original galvanometer to a voltmeter, reading up to 20.0 V,
  (*i*) would you add a resistor in series or parallel with the meter?
  (*ii*) what value resistance would you need?

26 A galvanometer has a resistance of 50.0 Ω and deflects five scale divisions for a current of 1.00 mA. What series or parallel resistor must be connected to the galvanometer to convert it to an ammeter on which one scale division corresponds to a current of 0.500 A?

27 A voltmeter has "fsd 100 mV" and "2000 Ω/V" marked on it. This means that it gives a *full-scale deflection* when the potential difference across the meter is 100 mV, and that the resistance between the meter terminals is 200 Ω. What series or parallel resistor should be connected to the meter to convert it to a voltmeter with an fsd of 5.00 V?

28 Figure 57.14 shows part of the circuitry of a multimeter. 50.0 μA is the fsd (full-scale deflection) current for the moving coil which has a resistance of 2000 Ω.

(*a*) Using any range on the multimeter, at fsd what is the potential difference
  (*i*) across the moving coil?
  (*ii*) between the point P and the common negative terminal?

(*b*) Now consider the meter with the selector switch on the 100 μA setting, as in Figure 57.14.
  (*i*) What must be the common current in the two resistors W and X at fsd?
  (*ii*) What is the total resistance of W and X in series, i.e. $(R_W + R_X)$?

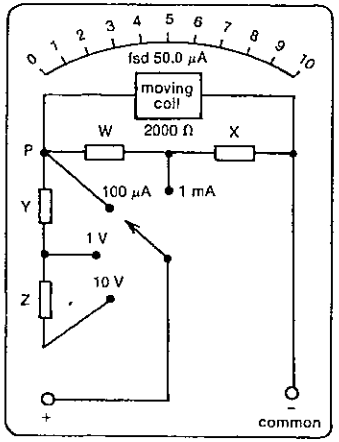


Fig. 57.14

29 Consider again the multimeter described in Problem 28 and Figure 57.14.

(*a*) On the 100 μA *current* setting the multimeter could be used as a *voltmeter*. What would be the fsd potential difference for this setting?

(*b*) Now consider the meter with the selector switch moved to the 1 V setting. At fsd what is
  (*i*) the potential difference across Y?
  (*ii*) the current in Y?
  (*iii*) the resistance of Y?

(*c*) Now the selector switch is moved to the 10 V setting. At fsd what is
  (*i*) the potential difference across Z?
  (*ii*) the resistance of Z?

30 Refer again to the multimeter described in Problem 28 and Figure 57.14, but with the selector switch this time moved to the 1 mA setting.

(*a*) What is the current in W at fsd?

(*b*) What then must be the current in X at fsd?

(*c*) Using R_W and R_X (the resistances of W and X), equate two expressions for the potential difference between the 1 mA selector point and the common negative terminal of the meter. Reduce this to an expression for R_W in terms of R_X.

(*d*) In Problem 28 (*b*) (*ii*) you found (or you should have) that

$$R_W + R_X = 2000$$

Determine values for
  (*i*) R_W;
  (*ii*) R_X.

31 A 4.0 Ω resistor is connected in parallel with a 12 Ω resistor, and this combination is then connected in series with a 5.0 Ω resistor, a 6.0 Ω resistor, and a d.c. power supply of e.m.f. 28 V and negligible internal resistance. Draw a diagram of this circuit and calculate

(*a*) the effective resistance of the parallel group;
(*b*) the current in the battery;
(*c*) the current in the 12 Ω resistor;
(*d*) the potential difference across the 6.0 Ω resistor;
(*e*) the charge that passes through the 4.0 Ω resistor in 1.0 min;
(*f*) the power supplied by the battery.

32 A moving coil voltmeter (resistance 1500 Ω and full-scale deflection for 1.50 V) is placed in series with a 1.50 V cell of negligible internal resistance and an unknown resistance. If the voltmeter reads 1.00 V, find the value of the unknown resistance.

33 A tramway substation maintains (at the station) a potential difference of 600 V between the overhead wire and the rail of an electric tramway. The resistance of the wire is 0.45 Ω km^{-1} and that of the rail is 0.03 Ω km^{-1}. If a tram motor requires 25 A of current at 540 V, how far from the power station can a tram travel? Assume that there is only one tram on the track.

34 In the circuit shown in Figure 57.15, $R_1 = R_2 = R_3 =$ 40 Ω, M_1 and M_2 are ammeters, and S is a switch.

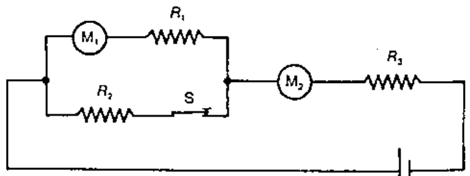


Fig. 57.15

Indicate whether each of the readings on M_1 and M_2 will *increase* or *decrease*, if

(*a*) switch S is opened;
(*b*) the resistance R_3 is halved while S remains closed.

35 The current drawn from a 2 V accumulator of negligible internal resistance is 1.25 A when connected across two resistors in parallel, and 0.2 A when connected across the same two resistors arranged in series. What is the resistance of each resistor?

36 A circuit supplied by a constant e.m.f. of 500 V has a total resistance of 1.00 MΩ. This resistance is tapped at two points across which there is found to be, on measurement with a potentiometer, a potential difference of 500 mV. What would be the reading of an accurate millivoltmeter, with an internal resistance of 10 MΩ, connected across these two points?

37 A cell of e.m.f. 2.0 V and negligible internal resistance is used to run an electric motor in series with a 0.20 Ω resistor, as shown in Figure 57.16. With the key K open, the ammeter, which has a negligible resistance, reads 2.5 A.

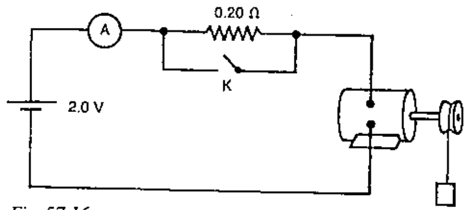


Fig. 57.16

(*a*) What is the potential difference across the motor when K is open?
(*b*) What is the power input to the motor when K is open?
(*c*) If K were closed, what would the ammeter register?

38 An unknown resistor is to be measured by the "ammeter–voltmeter" method using a voltmeter of resistance 3000 Ω and an ammeter of resistance 0.250 Ω. The voltmeter is connected in parallel with the unknown resistor and this combination is then connected in series with the ammeter and a battery. The ammeter reads 42.8 mA and the voltmeter reads 8.40 V. Calculate the resistance of the unknown resistor, taking into account any errors introduced by the use of the meters.

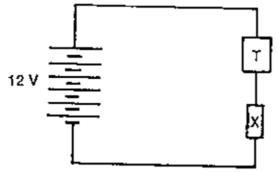


Fig. 57.17

39 In Figure 57.17 T is a 6.0 V car radio, which is known to draw 2.4 A when operating normally. To use the radio in a car with a 12 V battery, a component X is to be fitted in series with the radio as shown. Three suggestions are put forward for X:

A Six resistors, each marked "15 Ω 1 W", joined in parallel.
B One light globe marked "6 V 15 W".
C One 2.5 A fuse.

Which of the above suggestions would be the best choice for X?

40 A circuit consists of resistances of 10.0 kΩ and 50.0 kΩ in series with a d.c. source of 75.0 V. If a voltmeter of resistance 100 kΩ is placed across each resistance in turn, what reading will it give in each case?

SET 58 MAGNETIC EFFECTS OF MOVING CHARGED PARTICLES

Topic	*Problems*
Magnetic force on a current-carrying conductor in a magnetic field, magnetic flux density	1–5
Current balance	6–7
D.c. meters and motors (qualitative questions)	8–9
Torque in d.c. meters and motors	10–11
Direction of the magnetic flux density due to moving charges within or outside a straight conductor	12–13
Magnitude of the magnetic flux density due to moving charges within or outside a straight conductor	14–15
Direction of the magnetic flux density due to current-carrying loops, coils and solenoids	16
Magnitude of the magnetic flux density due to current-carrying loops, coils and solenoids	17–19
Earth's magnetic field, inclination or angle of dip, declination	20–21
Vector addition of magnetic fields	22–27
Magnetic force between two current-carrying conductors, definition of the ampere	28–32
Charged particles moving in a region containing a magnetic field	33–38
Charged particles moving in a region containing both a magnetic and an electric field	39–41
Mass spectrometers	42–45
Ampere's circuital law or circulation of the magnetic flux density	46–48

Where necessary in this set, use
permeability of a vacuum or air,
$\mu_0 = 4\pi \times 10^{-7}$ N A^{-2}
$1\ e = 1.60 \times 10^{-19}$ C
mass of an electron $= 9.11 \times 10^{-31}$ kg

1 The small circle in Figure 58.1 represents a conductor carrying a current of 5.0 A in a direction into the paper. If the 15 mm long section of the wire, which is in between the poles of the magnet, experiences a magnetic force of 3.0×10^{-4} N downwards, what is the magnitude and direction of the magnetic flux density at the central position between the poles?

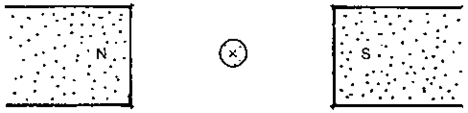


Fig. 58.1

2 A horizontal wire, carrying a current of 2.0 A in a northerly direction, has a portion 80 mm long in a magnetic field where the magnetic flux density is 25 mT (millitesla) vertically upwards. What is the
(*a*) magnitude, and
(*b*) direction
of the magnetic force on this portion of the wire?

3 A horizontal strip of aluminium foil, 25 mm long carries a current of 3.0 A in an easterly direction. What is the magnetic force on this length of foil, if the magnetic flux density in the region is
(*a*) 40 mT north?
(*b*) 40 mT N 30° E?

4 A current-carrying loop of wire VWXYZ is placed in a magnetic field which is directed to the left, as indicated in Figure 58.2.

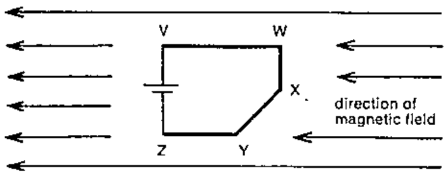


Fig. 58.2

What is the direction of the magnetic force acting on section
(*a*) WX?
(*b*) XY?
(*c*) YZ?

5 The voice coil of a moving coil loudspeaker is situated in a radial field in which the magnetic flux density is 0.301 T. It is a circular coil containing 40 turns each of radius 10.0 mm. With a current X in the coil, it experiences a magnetic force of 0.151 N.
(*a*) On what total length of wire is this force acting?
(*b*) What is the size of current X?

6 A conducting rectangular loop is positioned horizontally inside a solenoid with its long sides parallel to the axis of the solenoid as shown in Figure 58.3. The short side PQ is at right angles to the axis and is 25 mm long. The current in the loop is 5.0 A, while the current in the solenoid is such as to produce within the solenoid a magnetic flux density of 40 mT to the right.

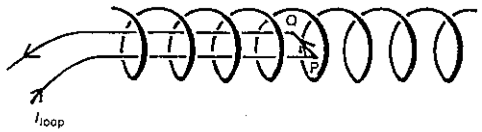


Fig. 58.3

(*a*) What is the direction of the magnetic force on the side PQ?
(*b*) What is the magnitude of this force?

7 A spring with a pointer attached is set up alongside a scale marked in centimetres. With *nothing* further attached to the spring, the pointer reads zero. A coil of 10 turns and mass 0.10 kg is then attached to the end of the spring, as in Figure 58.4. The lower edge of the coil is perpendicular to a uniform magnetic field of magnetic flux density 1.5 T. With no current flowing around the coil, the pointer reads 5.0 cm on the scale as shown in Figure 58.4.

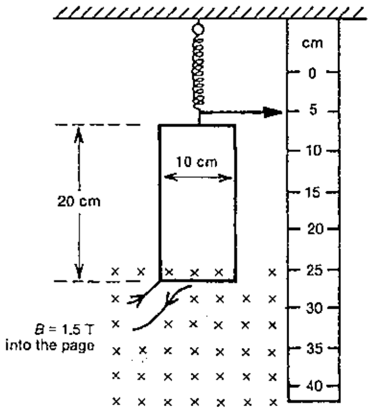


Fig. 58.4

(*a*) What would the pointer read if there was a clockwise current of 2.0 A in the coil?
(*b*) With this current in the coil, what would be the resultant force acting on the coil?

8 Figure 58.5 shows a square loop of wire PQRS of side length 0.15 m carrying a current of 4.0 A in a clockwise sense. The loop is suspended within a uniform magnetic field of flux density 0.20 T directed to the right.

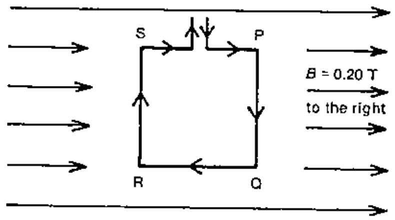


Fig. 58.5

What is the magnetic force on the side
(*a*) PQ?
(*b*) QR?

9 A view from above of a rotating coil in a simplified d.c. motor is shown in Figure 58.6. The direction of the current is shown for the two sides of the coil, which is free to rotate about an axis through O perpendicular to the page.

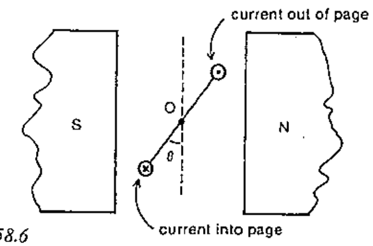


Fig. 58.6

(*a*) Is the coil of the motor rotating in a clockwise or anticlockwise sense?
(*b*) What is the value of θ at the instant when the greatest turning effect is experienced by the coil? There are a number of possible answers; give the lowest.
(*c*) What is the value of θ at the instant when the commutator reverses the direction of the current through the coil? Give the lowest of the possible answers.
(*d*) If Figure 58.6 represented the position of a coil in an oversimplified d.c. *meter* for a zero meter current, would the scale divisions on the meter be evenly spaced, more spread out near the middle of the scale, or more spread out near the ends of the scale?

10 Refer to the information given in Problem 8 and Figure 58.5.
(*a*) What is the magnitude of the net torque on the loop in the position shown in Figure 58.5?
(*b*) After the loop has rotated through 90° from the position shown in Figure 58.5, what would be the magnitude of the net torque on the loop?

11 For the coil described in Problem 9 and Figure 58.6, determine the magnitude of the net torque acting on the coil when $\theta = 30°$. The coil is square, of side 20 mm, has 80 turns and a current of 0.20 A. The magnitude of the magnetic flux density between the poles is 0.60 T.

12 An electric cable carrying a direct current passes along a duct which lies within a factory wall running north–south. It is found that a horizontal compass needle on the east side of the wall points south instead of north.
(*a*) Within the wall does the cable run vertically or horizontally?
(*b*) What is the direction of the current in the cable?

13 A cathode ray tube is set up in a north–south direction with the cathode at the south end. While the tube is producing cathode rays, a horizontal compass needle is placed directly above it. In what direction will the north pole of the needle be deflected?

14 Calculate the magnitude of the magnetic flux density 40 mm from a long straight wire carrying a current of 12 A.

15 Suppose your wrist watch remains unaffected by any magnetic field if its flux density is less than 8.0 T. If you walk under an overhead tram wire carrying a current of 100 A, is your watch likely to be damaged?

16 (*a*) In Figure 58.5, at the centre of the loop PQRS what is the direction of the magnetic field produced by the *current in the loop*?
(*b*) At O in Figure 58.6, is the direction of the magnetic field due to the *current in the coil* roughly towards top left or bottom right?
(*c*) Refer back to Figure 58.3 where the magnetic flux density inside the solenoid (and produced by the *solenoid current*) is directed to the right. Is the current entering the solenoid at the left- or right-hand end?

17 Suppose that in Figure 58.10 the vertical loop carries a current of 2.5 A and has a radius of 4.3 cm. Determine the magnitude of the magnetic flux density at O due to the current in this vertical loop.

18 In order to produce a strong magnetic field you wind a tightly wound circular coil of 600 turns. The coil lying horizontally on a table carries a current of 2 A clockwise (as seen from above), and has an average diameter of 5 cm. Specify fully the magnetic flux density at the centre of the coil.

19 The magnetic flux density within the solenoid in Figure 58.3 has a magnitude of 40 mT. If the solenoid has 995 turns and a length of 0.15 m, what is the current in it?

20 In Brisbane the Earth's magnetic field is specified by three elements:

magnitude of the magnetic flux density = 54 μT
inclination or angle of dip = $-55°$
declination = 10°E

(*a*) If you were holding a pocket compass horizontally in Brisbane, it would point to "magnetic north". How many degrees east or west of this direction is true geographic north?
(*b*) What is the horizontal component of the Earth's magnetic flux density in Brisbane?

21 (*a*) At the south magnetic pole, what would be
(*i*) the angle of dip?
(*ii*) the declination?
(*b*) Arrange Auckland, London, San Francisco and Hobart in the order of their angles of dip, starting with the most negative and ending with the most positive.

22 Two very long straight parallel wires carry currents of 15 A and 5.0 A as shown in Figure 58.7. The wires are 10 cm apart and P is a point 5.0 cm from the wire carrying the 5.0 A current and in the same plane as the two wires.

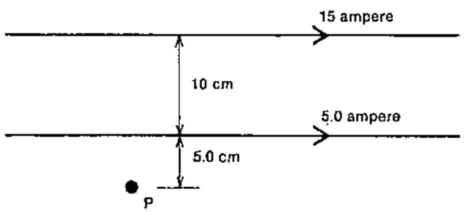


Fig. 58.7

(*a*) Determine at P the magnitude and direction of the magnetic flux density due to
(*i*) the 5.0 A current;
(*ii*) the 15 A current.
(*b*) What is the net magnetic flux density at P?

23 Two parallel horizontal wires I and II carrying currents of 5.0 A and 10 A respectively in opposite directions are 15 cm apart, as shown in Figure 58.8. The points P, Q and R are in the plane of the two wires.

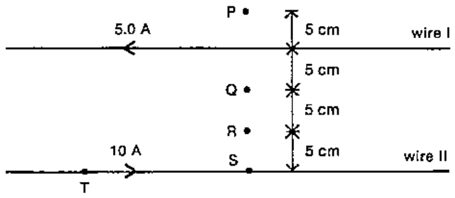


Fig. 58.8

(*a*) What is the direction of the net magnetic flux density at
(*i*) P?
(*ii*) Q?
(*iii*) R?
(*b*) At which of the points P, Q and R is the magnitude of the net magnetic flux density greatest?

24 Figure 58.9 shows two coplanar insulated wires P and Q carrying the currents indicated and crossing each other at right angles.

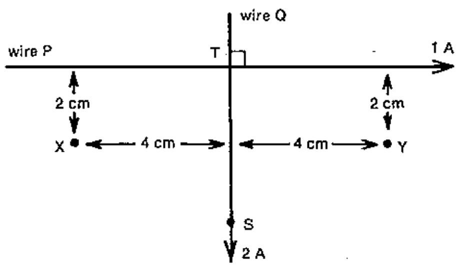


Fig. 58.9

(*a*) What is the net magnetic flux density at
(*i*) point X?
(*ii*) point Y?
(*b*) If wire P was fixed and wire Q was free to rotate about the point T, in what direction would the point S move initially?

25 Two circular loop conductors are mounted at right angles to each other, as in Figure 58.10. The larger loop is horizontal and has a radius 3 times that of the smaller loop and carries a current which is 4 times that in the smaller loop. The directions of the currents are shown in Figure 58.10.

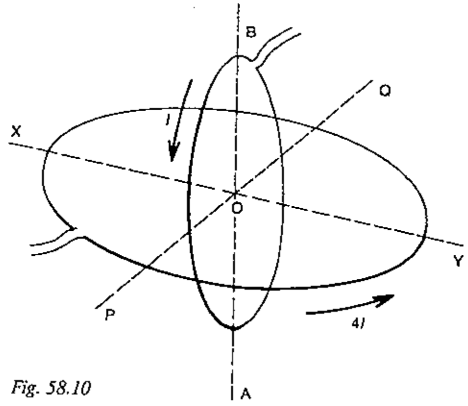


Fig. 58.10

(*a*) What is the direction of the magnetic field at O due to the current in
(*i*) the vertical loop?
(*ii*) the horizontal loop?
(*b*) Determine the direction of the net magnetic flux density at O due to the currents in both loops. Specify to the nearest degree the angle that this direction makes with the direction from A to B.

26 A flat circular coil can rotate about a vertical axis. A small compass needle is placed at the centre of the coil as in Figure 58.11. The coil carries an electric current which produces a magnetic field at the centre of the coil of magnitude equal to that of the horizontal component of the Earth's magnetic field.

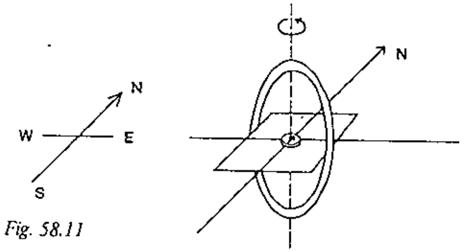


Fig. 58.11

(*a*) The coil is turned so that the direction of the magnetic field at its centre due to the current is towards the south. Looking from the south, is the current in the coil in a clockwise or anticlockwise sense?
(*b*) The coil is now turned so that the direction of its field is towards the east. In what direction does the compass needle now point?
(*c*) With the coil in the same position as in (*b*), the size of the current is trebled. In what direction will the needle point now?

27 A student erects a vertical wire on the edge of a table and establishes a current of 3.0 A in the wire in an upward direction as illustrated in Figure 58.12.

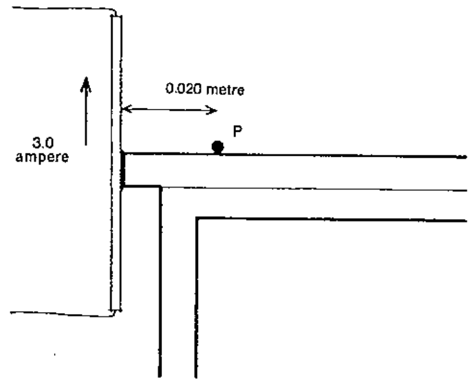


Fig. 58.12

(*a*) What is the magnitude of the magnetic flux density produced by this current at a point P which is on the table at a distance 0.020 m from the wire?
(*b*) If P is east of the wire and the horizontal component of the Earth's magnetic flux density is 23 μT, in what direction would the north pole of a compass needle point at P?
(*c*) In what direction would the north pole of the needle be deflected if P were north of the wire?

28 Refer back to Problem 23 and Figure 58.8.
(*a*) What is the magnitude of B_I, the magnetic flux density produced at S by the current in wire I?
(*b*) In terms of B_I what is the magnitude of the magnetic flux density produced at T by the current in wire I?
(*c*) What is the direction of B_I?
(*d*) Wire II then carries a current in the magnetic field produced by wire I. What is the magnitude of the magnetic force exerted on 2.0 m of wire II by this magnetic field
(*i*) in terms of B_I?
(*ii*) as a numerical answer?
(*e*) What is the direction of this magnetic force on wire II?
(*f*) Deduce the magnitude and direction of the magnetic force produced on 2.0 m of *wire I* by the current in *wire II*.

29 Two overhead trolley-bus wires carry currents of 200 A in opposite directions. If they are 0.40 m apart,
(*a*) what is the magnitude of the magnetic force per metre length on each?
(*b*) is the force between the wires one of attraction or repulsion?

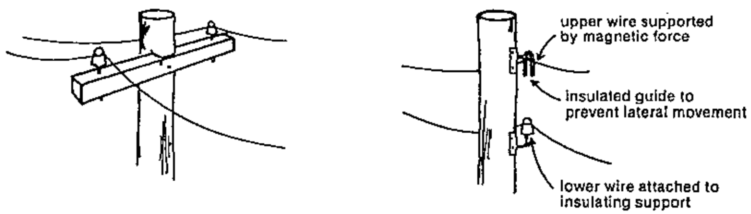


Fig. 58.13

30 The force per unit length between two parallel wires 1 cm apart is 10^{-4} N m^{-1}. If one wire carries a current of 5 A, what is the current in the other wire?

31 You have just thought up a brilliant way of cutting capital costs for your town's local d.c. electricity authority. Rather than have the wires on separate insulators on horizontal crossbars, the upper wire is supported by the magnetic force of repulsion between the currents, as shown in Figure 58.13. If a typical current going to and from the last house in your street was 15 A and the top wire weighs 1.2 N m^{-1}, how far apart would the wires be? You might think of one or two further difficulties with your idea!

32 Three very long conductors, PQ, RS and TU, are suspended vertically with the spacing shown in Figure 58.14. When there is a downward current I in each of RS and TU (but not in PQ) the force per unit length experienced by RS is of magnitude F. Now current I is fed down all *three* conductors.

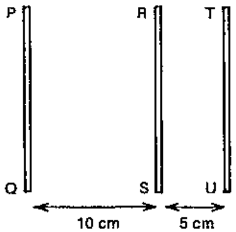


Fig. 58.14

(*a*) In terms of F what is the magnitude of the net force now acting on unit length of RS?
(*b*) What is the direction of this new net force on RS?

33 An electron is projected into a uniform magnetic field of flux density 10 T with a velocity of 3.0×10^7 m s^{-1} in a direction at right angles to the field.
(*a*) What is the magnitude of the magnetic force on the electron?
(*b*) What is the ratio of the magnitude of this force to the weight of the electron?

34 A stream of electrons is projected in front of an observer towards his right. A powerful permanent magnet with the north pole pointing downwards is held above the electron stream.
(*a*) In which direction will the stream of electrons be deflected?
(*b*) If the observer now holds the magnet horizontally so that the south pole of the magnet is towards the electron stream, what will be the direction of the deflection?

35 An electron is moving at a speed of 2.0×10^7 m s^{-1} in an evacuated space. It enters a uniform magnetic field of flux density 1.125 mT, the direction of the field being perpendicular to the plane in which the electron moves. Calculate
(*a*) the radius of the circular path described by the electron;
(*b*) the period of revolution in this path.

36 A beam of electrons is accelerated in a 2.4 kV electron gun and then allowed to pass into a magnetic field with a flux density of 9.7 mT. As the electrons enter the field their velocity is at right angles to the direction of the magnetic field. All this takes place in an evacuated container. (For simplicity, ignore relativistic effects.)
(*a*) What is the speed of the electrons as they leave the electron gun?
(*b*) What is the magnitude of the magnetic force on the electron while in the magnetic field?
(*c*) What is the radius of the path of the electron in the magnetic field?
(*d*) How many loops of its circular path does it traverse in one second?

37 A proton is observed in a cloud chamber to have a circular path of 104 mm radius in a uniform magnetic field of flux density 0.10 T. Calculate the energy of the proton. Proton mass = 1.67×10^{-27} kg.

38 A stream of protons and deuterons in an evacuated enclosure enters a uniform magnetic field which is perpendicular to the stream. Both particles carry a charge of +1 e, and a deuteron has twice the mass of a proton. Both particles have been subject to the same accelerating potential difference and therefore have equal kinetic energies. If the protons move in a circular path of radius 200 mm, what is the radius of the path of the deuterons?

39 In an early form of mass spectrograph designed by J. J. Thomson, a beam of positively charged ions was accelerated and then passed through parallel electric and magnetic fields before striking a screen, as shown in Figure 58.15.

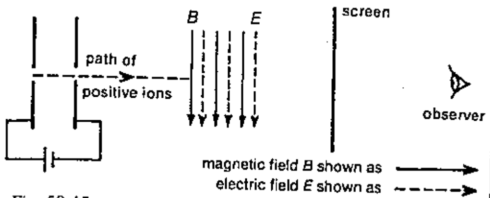


Fig. 58.15

To an observer viewing the screen, in what direction would the spot be deflected by
(*a*) the electric field alone?
(*b*) the magnetic field alone?

40 A stream of positive ions each carrying charge q travels vertically downwards between two charged, vertical plates between which the electric field strength is 2.0×10^4 V m^{-1} towards the west. A magnetic field at right angles to the electric field and of flux density 4.0 mT between the plates keeps the beam travelling in a straight line.
(*a*) What is the net force on each ion due to the electric and magnetic fields?
(*b*) What must be the direction of the magnetic field?
(*c*) In terms of q, specify fully the electric force acting on each ion.
(*d*) In terms of q and v, the speed of the ions, specify fully the magnetic force on each ion.
(*e*) Deduce the value for the speed v.

41 A beam of electrons travelling at 1.7×10^7 m s^{-1} enters a region in which uniform electric and magnetic fields exist at right angles to each other and to the direction of the beam. If the beam of electrons is undeflected and the magnitude of the magnetic flux density is 2.0 mT, what must be the magnitude of the electric field strength?

42 Figure 58.16 represents, schematically, a Dempster mass spectrograph. A spark discharge in a gas produces ions of mass m and carrying charge q. These drift with negligible kinetic energy through a hole T and are accelerated by a potential difference V set up between T and S. The ions pass through a small hole with speed v and enter a region of uniform magnetic field of flux density B at right angles to the velocity of the ions. The ions move in a semicircular path of radius R before impinging on a photographic plate. The whole apparatus is evacuated.

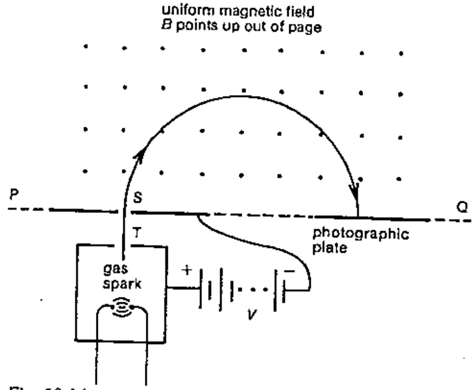


Fig. 58.16

(*a*) Equate two expressions for the energy gained by each ion as it moves from T to S.
(*b*) Equate two expressions for the magnetic force on each ion while in the magnetic field.
(*c*) From (*a*) and (*b*) deduce an expression for m in terms of q, B, R and V.
(*d*) For a Dempster mass spectrograph, indicate as a proportionality statement how m depends on R.
(*e*) If in this apparatus singly charged ions of helium are accelerated by a potential difference of 500 V, and move in a circular path of radius 40.2 mm in a magnetic field of flux density 0.16 T, determine the mass of a helium ion.

43 Two ions, each singly charged, enter the magnetic field of a Dempster type mass spectrograph (similar to Fig. 58.16) at the same point with the same energy. If their masses are in the ratio 5:3 and the path of the lighter particle in the magnetic field has a radius of 150 mm, what is the distance apart of the points where the heavier ion enters and leaves the magnetic field?

44 The essentials of a Bainbridge mass spectrometer are shown in Figure 58.17. Positive ions with a variety of charge and mass are accelerated by a potential difference between slits S_1 and S_2. Ions with a range of speeds pass through S_2 into the velocity selector, where crossed magnetic and electric fields (B and E respectively) ensure that ions with only one velocity v reach slit S_3 undeflected. They are then deflected by the magnetic field alone before being detected by a movable electrometer which can be adjusted for paths of different radius R.

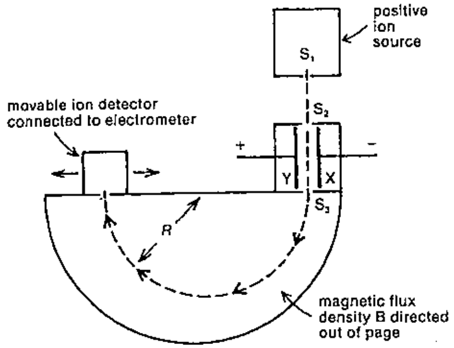


Fig. 58.17

(*a*) Deduce an expression for the speed v of the ions passing through S_3 in terms of E and B.
(*b*) Taking m as the mass of and q as the charge on an ion which passes through the velocity selector, equate two expressions for the centripetal force on the ion after leaving S_3.
(*c*) From (*a*) and (*b*) deduce an expression for m in terms of q, B, E and R.
(*d*) For a Bainbridge mass spectrometer, indicate as a proportionality statement how m depends on R.

45 Chlorine has two isotopes whose respective masses are 34.98 and 36.98 u. If the radius of the path described by the lighter isotope in the Bainbridge mass spectrometer is 50.0 mm, find the separation between the traces produced by the two isotopes on the photographic plate, after traversing a semi-circular path in the magnetic field.

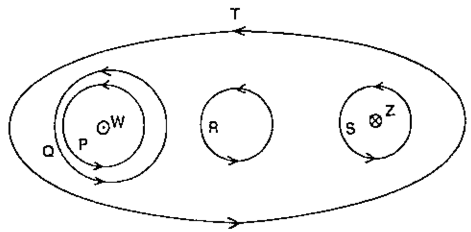


Fig. 58.18

46 In Figure 58.18, W and Z are two parallel conductors carrying currents equal in magnitude. The current in W is directed out of the page, that in Z into the page. Five closed paths, P–T, are drawn in the diagram, and in each case an arrow indicates the direction of the path.
(*a*) For which one or more of the closed paths is the circulation of the magnetic flux density
  (*i*) zero?
  (*ii*) negative?
(*b*) For which one or more of the closed paths is the magnitude of the circulation of the magnetic flux density equal to that around path P?
(*c*) Calculate the value of the circulation of the magnetic flux density for path Q when the current in wire W is 25 mA.

47 The current in the long solenoid shown in Figure 58.19 is 3.5 A. The wire passes through a card WXYZ five times. The length XY is 40 mm. Assume the magnetic flux density in the region WZ is negligible.
(*a*) Calculate the circulation of the magnetic flux density around the path WXYZW in an anticlockwise sense.
(*b*) Deduce a value for the magnitude of the magnetic flux density within the coil. Give its direction also.

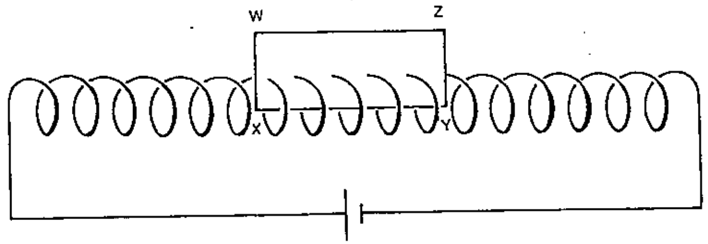


Fig. 58.19

48 A solenoid of spring wire has 48 turns. By placing the ends together it is converted into a toroid as shown in Figure 58.20. The current in the wire is 2.0 A.

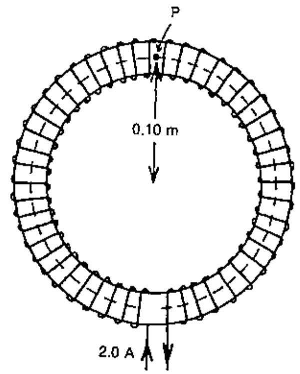


Fig. 58.20

(*a*) What is the magnetic circulation around the circle indicated by the broken line?
(*b*) If the radius of the toroid is 0.10 m, what is the magnitude and direction of the magnetic flux density at point P on the broken line?

SET 59 ELECTROMAGNETIC INDUCTION AND ELECTROMAGNETIC WAVES

Topic	*Problems*
Magnetic flux through an area	1–2
E.m.f. induced in a straight conductor cutting magnetic flux	3–9
Hall effect	10–13
E.m.f. induced in a loop, coil or solenoid by a changing magnetic flux, Faraday's law	14–18
Direction of the induced current, Lenz's law	19–25
Alternating e.m.f. induced in a coil rotating in a magnetic field, a.c. generator (see Set 60)	
Self-induction (qualitative questions)	26–28
Self-inductance or inductance, energy stored in an inductor (quantitative questions)	29–32
Transformers, mutual induction between two coils (see Set 60)	
Circulation of electric field strength and magnetic flux density (see also Set 58)	33
Plane polarisation of electromagnetic radiation	34–37
Other wave properties of electromagnetic radiation (see Sets 43 and 44)	
Wave-particle duality of electromagnetic radiation (see Set 45)	

Where necessary in this set, use
permeability of a vacuum or air,
$$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{N\ A^{-2}}$$
$$1\ e = 1.60 \times 10^{-19}\ \mathrm{C}$$
permittivity of a vacuum or air,
$$\epsilon_0 = 8.85 \times 10^{-12}\ \mathrm{C^2\ N^{-1}\ m^{-2}}$$
speed of electromagnetic radiation in a vacuum or air,
$$c = 3.00 \times 10^8\ \mathrm{m\ s^{-1}}$$

1 In a cyclotron, used for accelerating subatomic particles, a typical magnetic flux density is 1.8 T downwards. Consider a loop of wire 0.25 m² in area in this magnetic field.
(*a*) What would be the magnetic flux threading this loop in a downward sense, if the plane of the loop makes an angle of
  (*i*) 90°, and
  (*ii*) 78°
with the direction of the magnetic field?
(*b*) If the plane of the loop is perpendicular to the magnetic field, what magnetic flux threads it from *below*?

2 If the magnetic flux threading an electromagnet coil of area 4.0×10^{-4} m^2 is 3.2×10^{-5} weber (Wb), what is the magnitude of the magnetic flux density through the coil?

3 XY is a horizontal conductor of length 0.12 m, lying at right angles to a horizontal magnetic field of flux density 0.050 T, as shown in Figure 59.1.

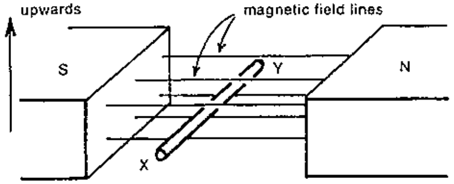


Fig. 59.1

While XY is moving upwards through the magnetic field at a speed of 0.60 m s^{-1},
(*a*) what is the e.m.f. induced in XY?
(*b*) which end of XY gains an excess of positive charge?

4 The axle of a railway carriage is 1.5 m long and is moving with a velocity of 20 m s^{-1} north on a level track. The vertically upward component of the Earth's magnetic flux density is 50 μT.
(*a*) Calculate (in mV) the e.m.f. induced between the ends of the axle.
(*b*) Does the east or the west end of the axle gain an excess of electrons?

5 When in the swept-back position, the wings of the RAAF F111C fighter-bomber have a span of 21 m. With the wings in this position while flying horizontally due magnetic north in a region where the Earth's magnetic flux density is 50 μT and the angle of dip is $-60°$, the F111C develops a potential difference of 0.77 V between its wing tips. At what speed is the plane flying?

6 In Figure 59.2, a conducting rod AB makes contact with the metal rails AD and BC which are 50 cm apart in a uniform magnetic field of flux density 1.0 T perpendicular to and into the plane of the paper as shown. The only resistance in the circuit ABCD is 4.0 Ω in the arm DC. An

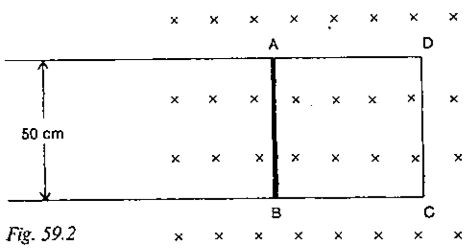


Fig. 59.2

applied force F_a is moving the rod to the left at a constant velocity of 8.0 m s^{-1}.
(*a*) What is the e.m.f. induced between the ends of AB?
(*b*) In which direction does the induced current flow in DC?
(*c*) What is the current in DC?
(*d*) Specify fully the magnetic force F_m acting on AB.
(*e*) What is the net force acting on AB?
(*f*) If the frictional force between the rod and the rails is negligible, specify fully F_a.

7 A magnetic field of uniform flux density B is established in a horizontal direction above a certain horizontal line XY. Below the line XY the field is zero. A square wire frame of side L, mass m and resistance R, is placed part inside and part outside the field with the sides perpendicular to the field as in Figure 59.3. It is projected downwards with an initial velocity v.

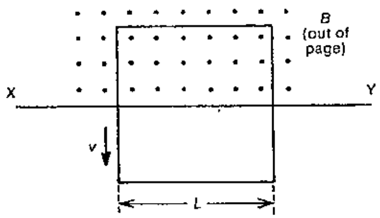


Fig. 59.3

Calculate
(*a*) the initial current in the frame;
(*b*) the value for v for which there would be no subsequent acceleration of the frame while portion of it is still above the line XY.

8 Suppose you lift the front wheel of your bicycle off the ground, orient its plane at right angles to the Earth's magnetic flux density of 55 μT, and spin it at 7.8 rotations per second. What e.m.f. will be induced between the central hub and the rim? The radius of the wheel is 0.31 m.

'**9** A circular copper disc of radius 14 cm rotates about an axis through its centre and normal to its plane. The axis is parallel to a uniform magnetic field. When the frequency of rotation of the disc is 45 Hz, the induced e.m.f. between its centre and the circumference is 6.0 mV. Calculate the flux density of the magnetic field.

10 The current I is passed through a slab of copper of width w and thickness t as shown in Figure 59.4. There is a magnetic field of flux density B at right angles to the front of the slab as shown.

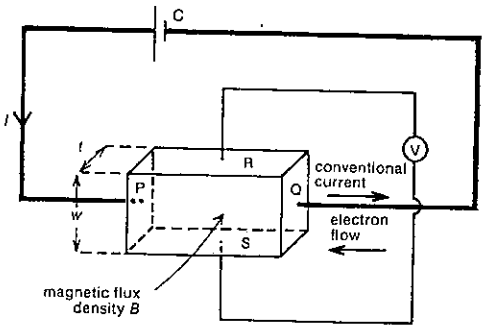


Fig. 59.4

(*a*) As the negative charge carriers, electrons, move from face Q to face P with an average speed v, there will be a magnetic force F_m acting on them. In what direction?
(*b*) On which face of the slab will this cause electrons to accumulate?
(*c*) This accumulation of electrons will generate between faces R and S an electric field E_H and a potential difference V_H (called the *Hall voltage*). What will be the direction of
(*i*) E_H?
(*ii*) the electric force F_H exerted by E_H on the conduction electrons?
(*d*) When the lateral movement of electrons between R and S ceases, how will F_m and F_H be related?
(*e*) Write an expression for F_m in terms of q, the charge on an electron, v and B.
(*f*) Write an expression for F_H in terms of
(*i*) E_H and q;
(*ii*) V_H, w and q.
(*g*) Deduce an expression for V_H in terms of B, V and w.
(*h*) It can be shown for a conductor that

$$I = nAvq,$$

where n = the number of charge carriers per unit volume, and
A = the area of cross-section of the conductor.
Deduce a second expression for the Hall voltage V_H in terms of B, I, n, q and t.

11 Suppose in your bedroom that the copper leads in the wall cavity carrying charge to the electric light happen to be at right angles to the Earth's magnetic flux density (approximately 10^{-4} T). Making suitable estimates of any necessary quantities, determine as an order of magnitude the Hall voltage between the two sides of the wire. The number of conduction electrons per unit volume for copper $\approx 10^{29}$ m^{-3}.

12 A surgical application of the Hall effect is in measuring the rate of blood flow during an operation. A doughnut shaped probe fits snugly around an artery of diameter 5.0 mm, and provides a magnetic flux density through the artery of 3.2 mT. Using our knowledge of the Hall effect we can measure the speed of the charge carriers, positive ions, in the blood and therefore of the blood itself. If in this case the Hall voltage is 4.0 μV, determine
(*a*) the speed of the ions in m s^{-1};
(*b*) the rate of flow of blood in millilitre per minute. (1 millilitre = 10^{-6} m^3.)

13 Consider the arrangement described in Problem 10 and Figure 59.4, only this time the slab is a *semiconductor*.
(*a*) Which of the faces R and S would be at the higher potential, if the semiconductor was
(*i*) p-type germanium?
(*ii*) n-type germanium?
(*b*) With a 0.30 mm thick wafer of n-type germanium in the set-up shown in Figure 59.4, a Hall voltage of 15 mV was generated when the current through the wafer was 1.8 mA and the magnetic flux density had a magnitude of 0.12 T. Determine the number of conduction electrons per metre cubed for this piece of n-type germanium.

14 Three coils A, B and C are placed in the vicinity of a long straight wire which carries a changing current. Which coil, in the configurations shown in Figure 59.5 develops an e.m.f. across its ends?

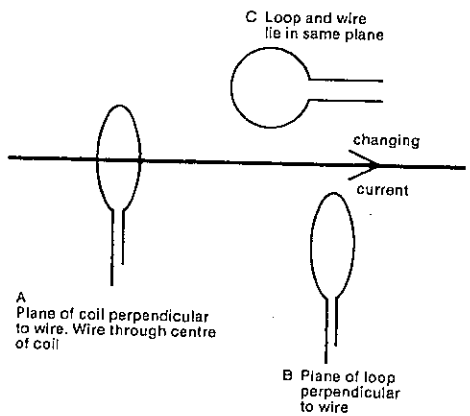


Fig. 59.5

15 A coil, containing 50 turns and of area 4.0×10^{-4} m^2, is situated between the poles of an electromagnet with its plane at right angles to the magnetic field of flux density 0.15 T.

When the current in the electromagnet is turned off, let us suppose it falls to zero at a constant rate (an over-simplification) in 0.12 s.
(*a*) What was the initial magnetic flux threading the coil?
(*b*) While the current was falling to zero,
  (*i*) what was the rate of change of magnetic flux threading each loop of the coil?
  (*ii*) what was the e.m.f. induced in each loop?
  (*iii*) what was the e.m.f. generated between the ends of the wire forming the coil?

16 A coil of 40 turns, each of area 10 cm^2, is mounted in a uniform magnetic field of flux density 0.01 T so that the field is normal to the coil's plane. What e.m.f. is induced in the coil if it is removed completely from the field in 0.01 s?

17 A coil of 500 turns of mean area 20 cm^2 is placed in air with its plane perpendicular to a field of flux density 0.5 T. Calculate the induced e.m.f. in the coil if the direction of the field is reversed at a uniform rate in 2.5 s.

18 A circular coil of 200 turns and mean diameter 200 mm, and having a resistance of 4.00 Ω, is mounted with its plane perpendicular to a uniform magnetic field of flux density 2.0×10^{-2} T. Calculate the quantity of charge which flows around the coil when it is rotated through 180° about a diametral axis.

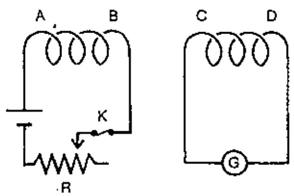


Fig. 59.6

19 A primary coil AB and a secondary coil CD are arranged as shown in Figure 59.6. Looking at the end C of the secondary coil from the left, is the sense of the induced current clockwise or anticlockwise when
(*a*) the sliding contact of the rheostat R is moved to the left?
(*b*) the primary circuit is broken at the key K?

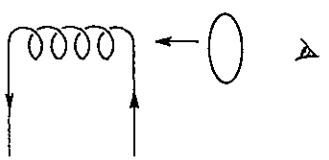


Fig. 59.7

20 A solenoid carrying a current as shown in Figure 59.7 is moved away from a copper ring. When observed from the right as in the diagram, is the induced current in a clockwise or anticlockwise sense?

21 A conducting ring is situated between the poles of a strong horseshoe magnet with its plane perpendicular to the field lines. Figure 59.8 is a diagram of the arrangement showing a side-on view of the ring.

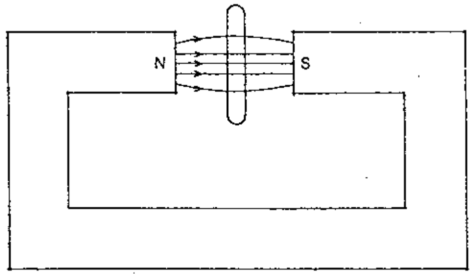


Fig. 59.8

Looking at the ring from the left, is the induced current in it clockwise or anticlockwise
(*a*) while it is being lifted vertically upwards out of the field?
(*b*) while it is being rotated through 10° about a vertical diametral axis in a clockwise sense as seen from above?

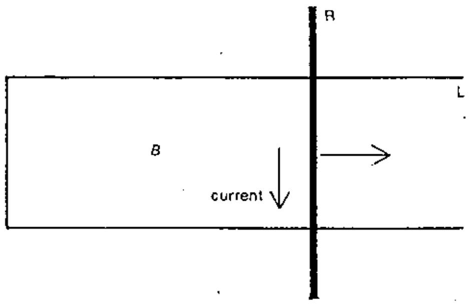


Fig. 59.9

22 Figure 59.9 shows a conducting rod R resting on a U-shaped loop L of heavy wire. The plane containing the loop and the rod is perpendicular to a magnetic field *B* which extends over the whole of the loop L. When the rod is pulled sharply to the right, as indicated in Figure 59.9, a current flows in the rod in the direction shown. What is the direction of the field *B*?

23 In Figure 59.10, C represents a long straight conductor carrying a current. AB is a short rod near the conductor and at right angles to it. When the rod AB is pulled rapidly to the right, the end B becomes positive with respect to the end A.

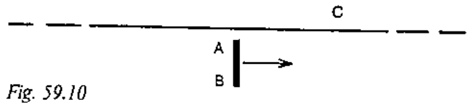


Fig. 59.10

(*a*) What is the direction of the current in C?
(*b*) What is the direction of the force exerted by the magnetic field on the rod when its movement commences?

24 Figure 59.11 shows a conducting loop connected to a resistor R. The plane of the loop is at right angles to the direction of a magnetic field directed into the paper.

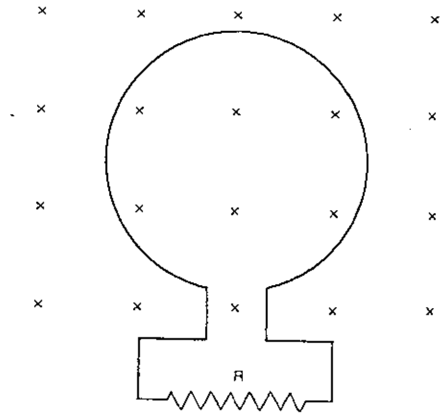


Fig. 59.11

(*a*) What is the direction of the induced current through R while the magnetic flux density is increasing?
(*b*) Given that the area of the loop is 1.0×10^{-2} m^2 and that the flux density increases from 0.20 to 0.50 T in 2.0 s, find the magnitude of the average induced e.m.f.

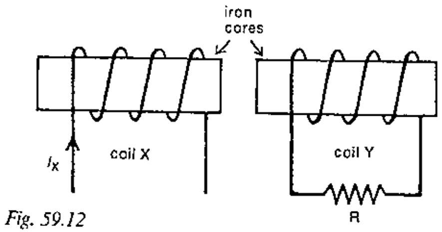


Fig. 59.12

25 Two coils X and Y are placed close together with their axes in line as in Figure 59.12. The current in coil X is varied with time *t* as shown in Figure 59.13.

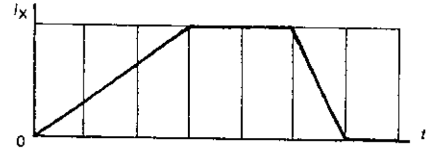


Fig. 59.13

Which of the graphs in Figure 59.14 best represents
(*a*) the magnetic flux density (to the left) threading coil Y as a function of *t*?
(*b*) the induced current (to the left) through the resistor R as a function of *t*?

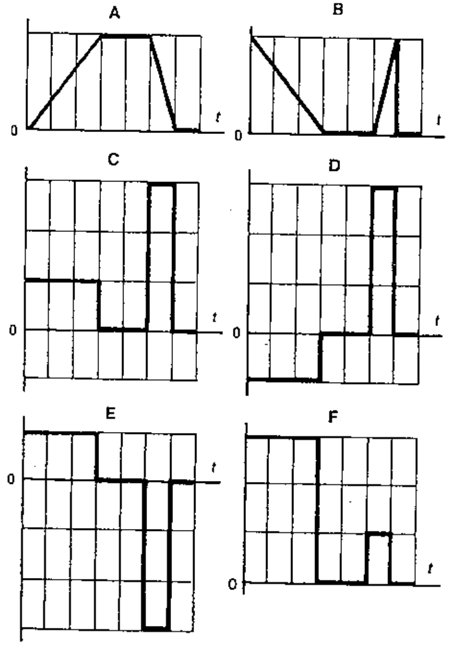


Fig. 59.14

26 In this problem consider the period immediately after the key K in Figure 59.15 is closed and the current has started to rise towards some constant value.

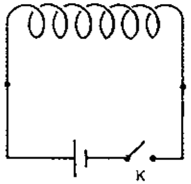


Fig. 59.15

(*a*) In which direction within the coil is the magnetic flux density increasing?
(*b*) Since a changing magnetic flux is threading the coil, an e.m.f. is induced. Is this induced e.m.f. tending to drive current in the same direction as the cell or the opposite direction?

27 Consider again the circuit in Figure 59.15, this time with K closed and a resistor connected between points P and Q. If K was opened and the current began to drop to zero, a self-induced e.m.f. would be set up.
(*a*) Would the resulting current through the resistor be from P to Q or from Q to P?
(*b*) Could this induced e.m.f. be greater than the e.m.f. of the cell?

28 The coils wound on the spools of resistance boxes are made of bifilar wires, i.e. a wire doubled back upon itself. Make a sketch of such a resistance coil, and explain why this arrangement gives coils which are practically non-inductive.

29 The SI unit of self-inductance or inductance is the henry (H), named after Joseph Henry, the American electrical engineer who pioneered the study of self-induction in the 1830s. Express a henry in terms of
(*a*) weber and ampere;
(*b*) volt, second and ampere.

30 If an air-cored coil of 200 turns is threaded by a magnetic flux of 24 μWb when the current in it is 1.2 A, what is the inductance of the coil?

31 What is the inductance of an air-cored 950-turn solenoid of diameter 40 mm and length 150 mm?

32 The current in an electromagnet of inductance 1.6 H is 8.8 A.
(*a*) What energy is stored in its magnetic field at this current?
(*b*) If the current was turned off and fell to zero in 25 ms, what average back e.m.f. would be produced?

33 Suppose a uniform magnetic field of flux density B directed to the south is travelling east at speed v. In Figure 59.16 it is shown just passing the side WX of the vertical loop WXYZ. This will generate an electric field of strength E along the side WX.

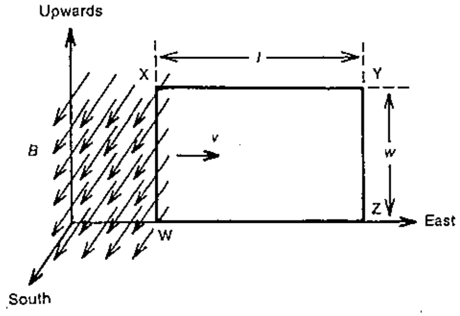


Fig. 59.16

(*a*) Equate two expressions for the clockwise circulation of the electric field strength around the loop WXYZ.
(*b*) Deduce an expression for B in terms of E and v.
Since the electric field strength E will be generated in any region entered by this magnetic field, the upward electric field is effectively moving east also. Consider this upward field strength E moving east at speed v just passing the side PQ of the loop PQRS as in Figure 59.17. It will generate a magnetic flux density B' along the side PQ.

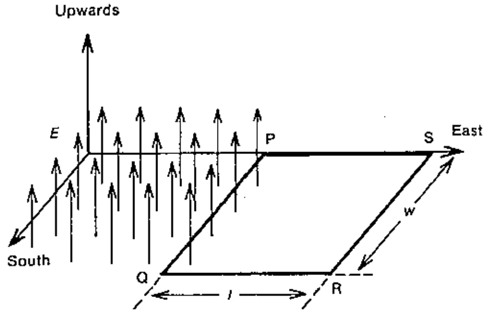


Fig. 59.17

(*c*) Equate two expressions for the anticlockwise circulation of the magnetic flux density around the loop PQRS.
(*d*) From your answer to (*c*) deduce an expression for B'.
(*e*) For these travelling electric and magnetic fields to be self-sustaining, B' must equal B. Deduce the value of v at which this is possible.

34 A plane-polarised electromagnetic wave is travelling towards you.
(*a*) If the magnetic field in the wave at your position is towards the right, what is the electric field at your position?

(*b*) At an instant $T/4$ later, where T is the period of the wave, what will be the direction of the magnetic field at your position?

35 Melbourne radio station 3XY has a frequency of 1422 kHz and you are situated west of 3XY's transmitter.
(*a*) What is the wavelength of 3XY's radio waves?
At one instant the electric field in 3XY's wave at your position is at its maximum value and directed to the south. Assume the radio wave is plane-polarised.
(*b*) At that same instant and position, what is the direction of the magnetic field in 3XY's wave?
(*c*) At this same instant, what would be the direction of the electric field 105.5 m east of your position?
(*d*) What would be the direction of the electric field in 3XY's wave at your position 0.25 μs after the instant we have been considering?

36 A simple periodic wave of electromagnetic radiation is moving in the direction shown in Figure 59.18. P, Q, R, S and T are equally spaced points in the path of the wave and the distance PQ = d. The velocity of the wave is c and its wavelength is $4d$. An observer at T, looking towards P, would see H and D as vertically upwards, while E, A and B_1 would be horizontal and to his left. At a given instant, the magnetic field at P is a maximum, and is represented by the vector B_1.

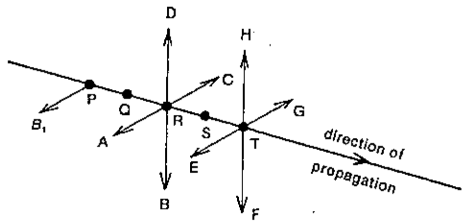


Fig. 59.18

(*a*) Which of A, B, C or D can represent the magnetic field at R at the same instant?
(*b*) Which of E, F, G or H can represent the electric field at T at the same instant?
(*c*) At which of points P, Q, R, S or T is the electric field zero at a time $t = 2d/c$ later than this instant? If at none of these, write N.

37 An electron is ejected from matter by an electromagnetic radiation. Assuming that the electron, before ejection, is moving with a speed v at right angles to the magnetic component of the wave, what is the ratio of the magnitude of the force on the electron due to the magnetic field to that due to the electric field?

SET 60 SIMPLE A.C. CIRCUITS

Topic	*Problems*
Alternating current and potential difference; average, peak, peak-to-peak and r.m.s. values; frequency and period of a.c.	1–4
Trigonometric expressions for alternating current and potential difference	5–6
A.c. generator or alternator	7–10
Transformers	11–14
Capacitors, capacitance, charging and discharging in an RC circuit, time constant, energy stored	15–21
Capacitors in series and parallel	22–27
Reactance of a capacitor, simple low and high pass RC filters	28–30
Reactance of an inductor, simple low and high pass RL filters	31–33
Use of diodes for half-wave and full-wave rectification of a.c., simple smoothing circuits	34–36
RC series circuits, phase relationships between potential difference and current, impedance	37–38
RL series circuits, phase relationships, impedance	39–40
RLC series circuits, phase relationships, impedance	41–42
Series resonance circuit	43–44

Where necessary in this set, use permittivity of a vacuum or air, $\epsilon_0 = 8.85 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$

1 For the alternating potential difference shown in Figure 60.1, what is the
(*a*) peak voltage?
(*b*) peak-to-peak voltage?
(*c*) average voltage?
(*d*) r.m.s. voltage?
(*e*) period?
(*f*) frequency?

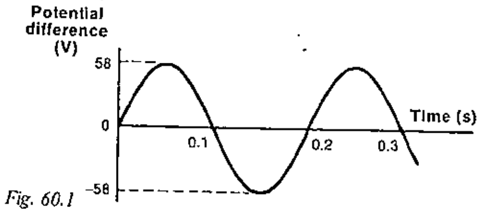


Fig. 60.1

2 The most common a.c. supply in Australia has an r.m.s. voltage of 240 V and a frequency of 50 Hz. Determine its
(*a*) peak voltage;
(*b*) period (in ms).

3 An electric jug with a 48 Ω heating element is connected to the 240 V (r.m.s.) 50 Hz household electricity supply. What would be the
(*a*) r.m.s. current, and
(*b*) peak current
in the heating element?

4 Two identical light globes are lit with equal brightness, one in a d.c. circuit and the other in an a.c. circuit. If the potential difference across the globe in the d.c. circuit is 9.9 V, what must be the
(*a*) r.m.s. value, and
(*b*) peak value
of the potential difference across the globe in the a.c. circuit?

5 If the alternating current I (ampere) as a function of time t (second) in a heating coil is given by

$$I = 4.8 \sin 100\pi t,$$

what is the
(*a*) peak current?
(*b*) r.m.s. current?
(*c*) frequency of the a.c.?

6 If V represents the instantaneous value of the potential difference shown in Figure 60.1 as a function of time t, write down a trigonometric expression for V in terms of t. V and t are in volt and second respectively.

7 In Figure 60.2, QRST is a plane rectangular single turn coil with terminals P and V. The coil is placed in a uniform magnetic field of flux density B directed down the page in the plane of the coil. The length of the side QR is 20 cm, side RS is 10 cm long, and the magnitude of B is 0.01 T. A high resistance is connected between the terminals P and V of the coil in Figure 60.2, and the coil is rotated about the axis XY at a constant frequency of 10 Hz; the sense of rotation is clockwise when viewed along the axis from X to Y. Figure 60.3 shows the coil at a time one-quarter of a period later than shown in Figure 60.2.
(*a*) What is the magnetic flux threading the coil in
 (*i*) Figure 60.2?
 (*ii*) Figure 60.3?
(*b*) During the rotation between the positions shown in

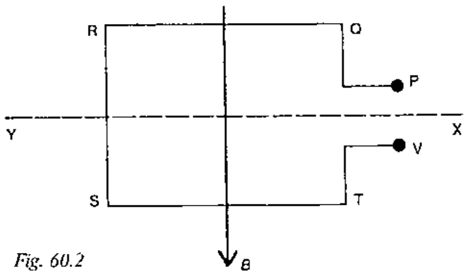


Fig. 60.2

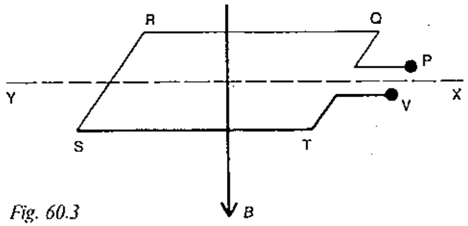


Fig. 60.3

Figures 60.2 and 60.3, does the current flow through the external resistor from P to V or from V to P?
(*c*) Determine the average e.m.f. which is induced in the coil during the time of rotation between positions shown in Figures 60.2 and 60.3.

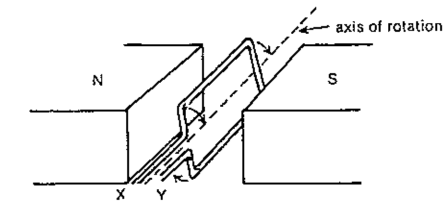


Fig. 60.4

8 A conducting loop is rotating in a clockwise sense at a constant rate in a uniform magnetic field as shown in Figure 60.4. Loop ends X and Y are observed to move successively through positions P, Q, R, S and T, shown in Figure 60.5.

Which one or more of the graphs in Figure 60.6 could represent the variation with coil position of
(*a*) magnetic flux threading the loop?
(*b*) e.m.f. generated between X and Y?

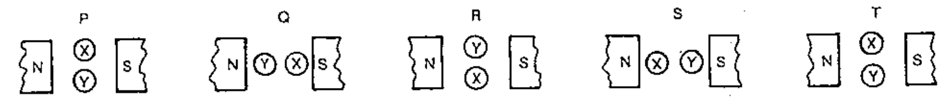


Fig. 60.5

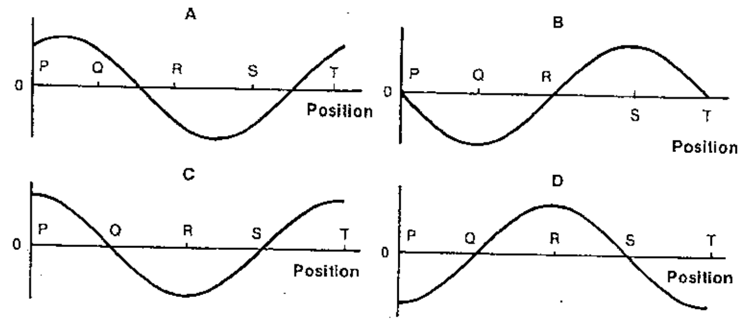


Fig. 60.6

9 Figure 60.7 shows a side view of a rectangular coil of 27 turns, 10 cm long and 5.0 cm wide, rotating about a horizontal axis through O at 50 Hz in a horizontal magnetic flux density of 0.33 T.

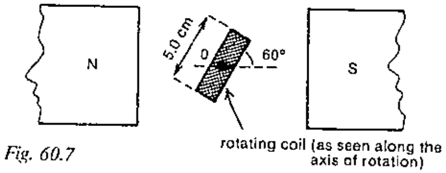


Fig. 60.7

(*a*) At the instant shown, the plane of the loop makes an angle of 60° with the horizontal. What is the e.m.f. induced in the coil at this instant?
(*b*) What is the maximum value of the e.m.f. generated by this alternator?
(*c*) What is the r.m.s. value of the output voltage of the alternator?

10 In order to generate 222 V (r.m.s.) 50 Hz a.c., a 100-turn rectangular coil, 50 cm by 20 cm, is to be rotated in a uniform magnetic field.
(*a*) What should be the frequency of rotation of the coil?
(*b*) What is the magnitude of the necessary magnetic flux density?

11 A step-down transformer with 1000 turns in its primary coil is being used to produce 12 V a.c. from the 240 V a.c. household supply. Assuming the transformer is 100 per cent efficient,
(*a*) how many turns are there in the secondary coil?
(*b*) what is the primary r.m.s. current, if the secondary r.m.s. current is 4.0 A?

12 A transformer has 600 primary turns and 75 secondary turns. An ammeter shows that the r.m.s. current in the secondary coil is 9.6 A.
(*a*) If one assumed there were no energy losses in the transformer, what would be the predicted primary r.m.s. current?
(*b*) Real transformers are not 100 per cent efficient. Would the actual primary r.m.s. current be greater than or less than your answer to (*a*)?

13 Twenty street lights for a toy model village are to be run from the secondary of a transformer, as indicated in Figure 60.8. The 1000-turn primary coil is connected to the 240 V 50 Hz mains supply. Each globe is marked "6.0 V 0.50 A". Assume the globes are to operate as designed.
(*a*) What is the number of turns in the secondary coil of the transformer?
(*b*) Determine the r.m.s. current in the secondary coil.
(*c*) What would be the peak voltage across globe 20?

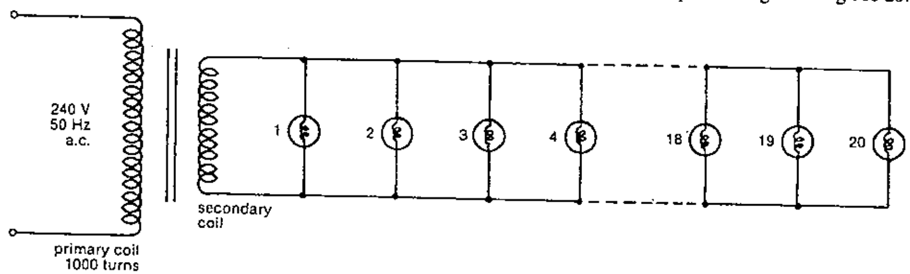


Fig. 60.8

14 Figure 60.9 represents an ideal transformer with a 1000-turn primary coil and two secondary coils, one with 500 turns and the other with 200 turns. X is an electric heater. The primary coil is connected through a 2 A fuse to the 240 V a.c. mains.

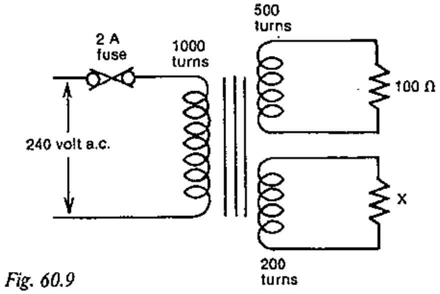


Fig. 60.9

(*a*) What is the maximum power heater that can be used for X without blowing the 2 A fuse?
(*b*) What current would there be in such a heater?

15 At a certain instant the charge on one plate of a polyester capacitor is $-20\ \mu C$ when the potential difference between the two plates is 4 V.
(*a*) What is the charge on the other plate at this instant?
(*b*) Calculate the capacitance of the capacitor.

16 In a 200 pF mica capacitor the separation of the plates is 59 μm. The relative permittivity (or dielectric constant) of mica is 7.0. What is the total area of each of the two plates?

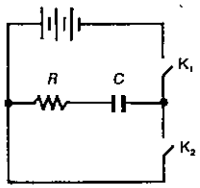


Fig. 60.10

17 The capacitor in the circuit in Figure 60.10 is charged by closing key K_1 at time t_c. K_2 remains open. Figure 60.11 shows the variation of current to the right in the resistor with time.

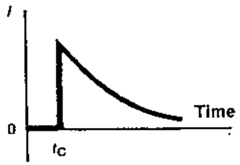


Fig. 60.11

From the graphs shown in Figure 60.12, select the one which best shows the variation with time of
(*a*) the potential difference across the resistor, when K_1 is closed;
(*b*) the potential difference across the capacitor, when K_1 is closed;
(*c*) the charge on each plate of the capacitor, when K_1 is closed;
(*d*) the current to the right in the resistor, when K_1 is opened and then K_2 is closed to allow the capacitor to discharge;
(*e*) the charge on each plate of the capacitor, when K_1 is opened and then K_2 is closed to allow the capacitor to discharge.

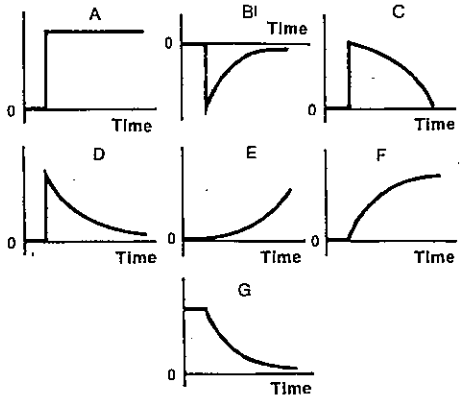


Fig. 60.12

18 In the circuit shown in Figure 60.10, $R = 2.0\ k\Omega$ and $C = 12\ \mu F$.
(*a*) What is the time constant of this circuit?
(*b*) If the 12 μF capacitor was replaced by one of 1.2 μF, and no other change was made to the circuit, would the new capacitor approach its equilibrium charge more or less rapidly than the 12 μF in the original circuit?

19 A student sets up a series circuit containing a 12 V d.c. source, a capacitor, a variable resistor, an ammeter and a switch S. From the instant S is first closed the student manages to keep the current constant at 16 μA by continuously varying the variable resistor, until the capacitor is fully charged. This takes 15 s.
(*a*) During the charging process, is the student decreasing or increasing the resistance of the circuit?
(*b*) How much charge flowed on to each plate during charging?
(*c*) What is the capacitance of the capacitor?

20 A faulty capacitor, the dielectric of which has broken down, is charged to 10 kV. If its capacitance is 40 μF, and it discharges in 50 s, what is its average leakage current?

21 If a potential difference of 19 V is connected across a 10 nF ceramic capacitor, what energy is stored in the capacitor when fully charged?

22 What is the effective total capacitance of the following capacitors connected in *parallel*?
(*a*) a 2 μF and a 4 μF capacitor;
(*b*) two 12 pF capacitors;
(*c*) three 12 pF capacitors;
(*d*) ten 12 pF capacitors;
(*e*) a 2 pF and a 500 μF capacitor.

23 Determine the effective total capacitance of the following capacitors connected in *series*:
(*a*) a 6.0 μF and a 4.0 μF capacitor;
(*b*) two 24 μF capacitors;
(*c*) eight 24 μF capacitors;
(*d*) a 2 pF and a 500 μF capacitor?

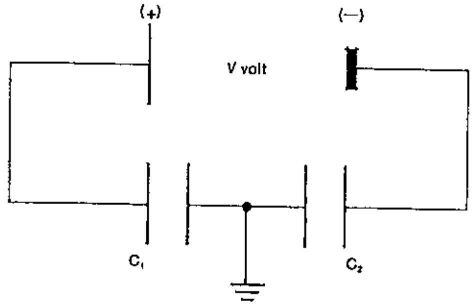


Fig. 60.13

24 Two parallel plate capacitors, capacitances C_1 and C_2, have one plate of each electrically connected to earth. The other two plates are connected to the terminals of a source of potential difference V as shown in Figure 60.13. Taking the earth potential as zero, determine the potential of the four plates, stating them in order from left to right.

25 You are provided with a number of 1 μF capacitors for which the maximum potential difference is 100 V. Draw a diagram to show how you would connect them to give a capacitor of 1 μF able to withstand 200 V.

26 Capacitors with the following working specifications are available—capacitance 2 μF, maximum voltage 500 V. How could a capacitor of capacitance 4 μF be assembled capable of withstanding a maximum voltage of 1000 V?

27 A capacitor of unknown capacitance is charged so that the potential difference between its terminals is 1000 V. It is then disconnected from the charging circuit and connected in parallel with a second capacitor whose capacitance is 1.0 μF. The potential difference across the plates is then found to be 400 V. What is the capacitance of the first capacitor?

28 If a 50 Hz a.c. supply is connected across a 2.0 μF capacitor, what is the capacitative reactance of the capacitor at this frequency?

29 Suppose the input to the circuit shown in Figure 60.14 is an alternating potential difference of frequency $10^5/\pi$ Hz.

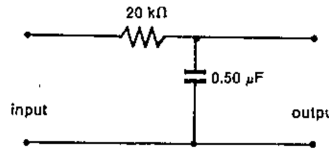


Fig. 60.14

(*a*) What is the reactance of the 0.50 μF capacitor with this input?
(*b*) What is the ratio of the r.m.s. potential difference across the capacitor to that across the 20 kΩ resistor?
(*c*) *With the stated input*, which one of the following functions is this circuit performing?
  A Allowing a low frequency signal to pass from input to output.
  B Allowing a high frequency signal to pass from input to output.
  C Diminishing a low frequency signal passing from input to output.
  D Diminishing a high frequency signal passing from input to output.

30 A signal generator set to an a.c. of frequency $100/\pi$ Hz provides the input to the circuit shown in Figure 60.15.

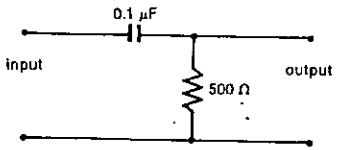


Fig. 60.15

(*a*) What is the reactance of the capacitor at this frequency?
(*b*) Determine the ratio of the peak potential difference across the resistor to that across the capacitor
  (*i*) using the $100/\pi$ Hz input;
  (*ii*) if, instead, the frequency of the input was changed to $10^6/\pi$ Hz.
(*c*) Would the circuit in Figure 60.15 be classed as a high or a low pass filter?

31 Determine the inductive reactance of a coil with an inductance of 0.10 H when an a.c. of frequency 70 Hz is applied to it.

32 Consider the circuit shown in Figure 60.16.

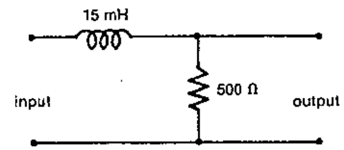


Fig. 60.16

(*a*) What would be the reactance of the inductor if the input signal had a frequency of 70 Hz?
(*b*) *With the stated input*, which one of the following functions is this circuit performing?
A Allowing a low frequency signal to pass from input to output.
B Allowing a high frequency signal to pass from input to output.
C Diminishing a low frequency signal passing from input to output.
D Diminishing a high frequency signal passing from input to output.

33 A signal generator is connected to the input in Figure 60.17.

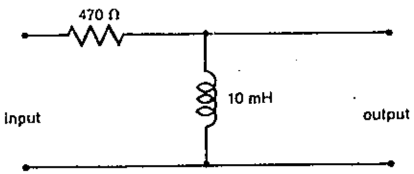


Fig. 60.17

(*a*) Determine the inductive reactance of the coil when the frequency of the signal generator is
(*i*) 140 Hz;
(*ii*) 1.4 MHz.
(*b*) Is the arrangement in Figure 60.17 a high or low pass filter?

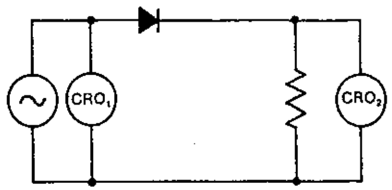


Fig. 60.18

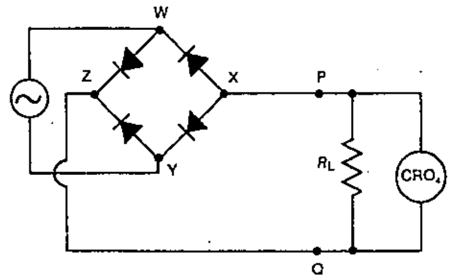


Fig. 60.20

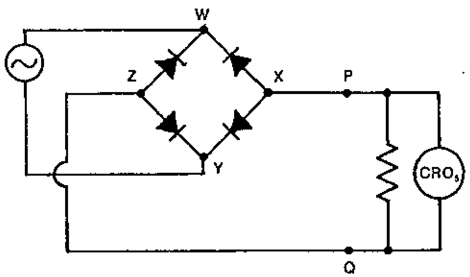


Fig. 60.21

34 A power pack with a 50 Hz a.c. output, of the form shown in the oscilloscope sketch in Figure 60.22A, is connected in turn to the circuits shown in Figures 60.18, 60.19, 60.20 and 60.21. Which of the sketches given in Figure 60.22 would best represent the trace on the screen of oscilloscope

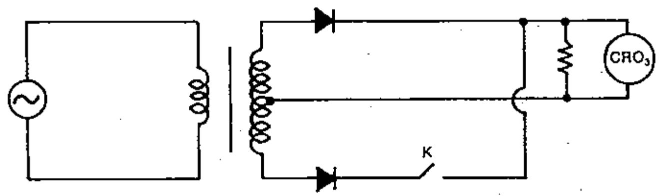


Fig. 60.19

(*a*) CRO_1 in Figure 60.18?
(*b*) CRO_2 in Figure 60.18?
(*c*) CRO_3 in Figure 60.19 when key K is
(*i*) open?
(*ii*) closed?
(*d*) CRO_4 in Figure 60.20?
(*e*) CRO_5 in Figure 60.21?

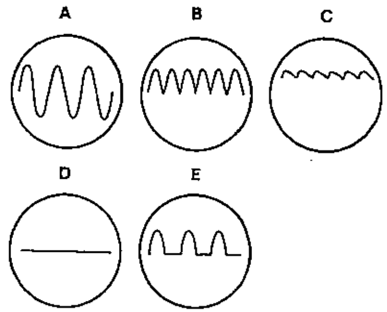


Fig. 60.22

35 With the varying d.c. produced by the circuit shown in Figure 60.20, which of the points W, X, Y and Z on the diode network would you regard as the positive terminal of this d.c. source?

36 Consider the rectifier circuit shown in Figure 60.20.
(*a*) If a capacitor of capacitance C was connected between P and Q, which of the traces in Figure 60.22 would best represent the appearance of the oscilloscope screen?
(*b*) With a 50 Hz a.c. input which of the following combinations would give the better smoothing circuit?
(*i*) $R_L = 50\ \Omega$, $C = 220\ \mu F$ *or* $R_L = 50\ \Omega$, $C = 680\ \mu F$;
(*ii*) $R_L = 50\Omega$, $C = 1.5\ mF$ *or* $R_L = 300\ \Omega$, $C = 680\ \mu F$.

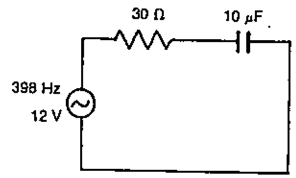


Fig. 60.23

37 Consider the RC circuit shown in Figure 60.23. If the variation of current with time in the 30 Ω resistor is represented in Figure 60.24 by graph A, which one or more of the graphs, A, B, C and D, best represents the variation with time of
(*a*) potential difference across the 30 Ω resistor?
(*b*) current in the 10 μF capacitor?
(*c*) potential difference across the capacitor?

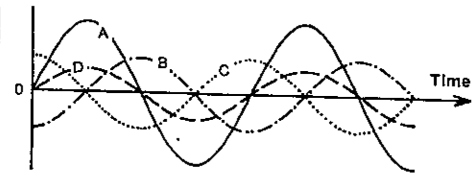


Fig. 60.24

38 Consider the circuit shown in Figure 60.23, powered by a 398 Hz 12 V (r.m.s.) source.
(*a*) What is the capacitative reactance of the 10 μF capacitor?
(*b*) What is the total impedance of the circuit?
(*c*) Determine the r.m.s. current.

39 Figure 60.25 shows an RL series circuit connected to a signal generator. Assume the inductor has a negligible resistance. If graph A in Figure 60.24 represents the variation with time of current in the 50 Ω resistor, which one or more of the graphs, A, B, C and D, best represents the variation with time of
(*a*) potential difference across the 50 Ω resistor?
(*b*) current in the 22 mH coil?
(*c*) potential difference across the coil?

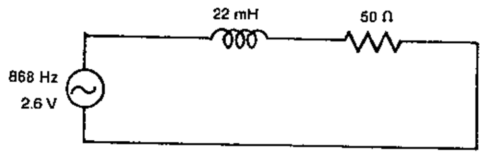


Fig. 60.25

40 The series circuit in Figure 60.25 is driven by a signal generator with a 2.6 V (r.m.s.) 868 Hz output.
(*a*) What is the inductive reactance of the 22 mH coil?
(*b*) What is the impedance of the external circuit?
(*c*) Determine the r.m.s. current (in mA) in the external circuit.
(*d*) What is the r.m.s. potential difference across
(*i*) the 50 Ω resistor?
(*ii*) the coil?
(*e*) Why do your answers in (*d*) (I hope) *not* add up to the output potential difference of the source, 2.6 V?

41 In the series circuit shown in Figure 60.26, C represents the capacitance of the variable capacitor. If graph A in Figure 60.24 represents the variation with time of the current in the 15 Ω resistor, which of the graphs, A, B, C and D, could represent the variation with time of potential difference between
(*a*) W and X?
(*b*) W and Y? (Give both possible answers.)

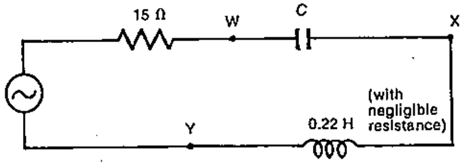


Fig. 60.26

42 The source in the circuit in Figure 60.26 is a 50 Hz 5.0 V (r.m.s.) a.c. power supply. $C = 65\ \mu F$.
(*a*) What is the reactance of the capacitor?
(*b*) What is the reactance of the inductor?
(*c*) What is the total reactance between points W and Y?
(*d*) Determine the total impedance of the circuit, assuming the internal resistance of the source is negligible.
(*e*) Calculate the r.m.s. value of the current in this series circuit.
(*f*) Determine the r.m.s. potential difference between points W and Y.

43 Consider the circuit in Figure 60.26, where C represents a capacitance of $100\ \mu F$ and the source is a variable frequency signal generator.
(*a*) For the impedance of the circuit to be a minimum, what relation exists between the reactances of the capacitor and the inductor?
(*b*) What would be this minimum impedance?
(*c*) If f is the frequency at which the impedance is a minimum, write an expression in terms of f (and π) for
(*i*) the reactance of the inductor;
(*ii*) the reactance of the capacitor.
(*d*) Deduce a value for f. This is known as the resonant frequency of the series circuit.
(*e*) Without doing any further calculations, state whether the r.m.s. current in the circuit is greater at the frequency f or $f + 10$?

44 Suppose you are asked to set up a series resonance circuit using that of Figure 60.26.
(*a*) If the circuit is to resonate at 297 Hz, what must be the value of C?
(*b*) At this resonant frequency the r.m.s. current is 2.0 A. What is the output potential difference of the source?

SET 61 HOUSEHOLD A.C. SUPPLY

Topic	*Problems*
Electrical energy distribution from power station to home, use of high voltage and a.c.	1-6
Electrical energy distribution within the home; active, neutral and earth; fuses	7-15
The kilowatt hour as a unit of energy	16-17

1 Use the following information concerning Yallourn W power station in Victoria to determine the efficiency of the complete process of generating electrical energy from coal:
1 tonne of Yallourn brown coal releases 8.72 gigajoule (GJ) of heat, of which 1.97 GJ is used in drying the coal. ($1\ GJ = 10^9\ J$.) The remaining 6.75 GJ is used in generating steam which drives the generators to produce 2.29 GJ of electrical energy.

2 Table 61.1 shows approximate transformer voltage changes that occur for electrical energy transmitted from Hazelwood power station in Victoria to households in Melbourne.

Table 61.1

Location of transformer		a.c. voltage change
Hazelwood power station	H	18.5 kV to 220 kV
Rowville terminal station	R	220 kV to 66 kV
A suburban substation	S	66 kV to 11 kV
A poletype transformer	P	11 kV to 240 V

(*a*) What is the ratio of the number of secondary turns to the number of primary turns in
(*i*) a transformer at Hazelwood?
(*ii*) a poletype transformer?
(*b*) If a typical current in a single-phase line between Hazelwood and Rowville is 780 A, what is the output current from the Rowville terminal station?
(*c*) Your calculation in (*b*) probably assumed the transformer was ideal. *In practice*, would the current be slightly higher or slightly lower than your answer to (*b*)?
(*d*) By inspecting the voltages in Table 61.1, decide which of the distances, HR, RS and SP, is the longest, giving a reason for your answer.

3 If one considers in Victoria the power transfer from Hazelwood power station (H) to the Rowville terminal station (R) near Melbourne, it can be seen that
the power output from H
= the power losses in a transmission line from H to R
+ the power input to R

These three *power* terms vary during each a.c. cycle. *At one particular instant* when the power output from Hazelwood into one transmission line is 100 MW, the current in that line is about 780 A. The resistance of the line is about 5 Ω.
(*a*) *At this same instant*, what is
(*i*) the power loss in the transmission line?
(*ii*) the percentage loss of the output energy from Hazelwood due to the heating of the transmission line?
(*iii*) the power input to the Rowville terminal station from this transmission line?
(*b*) Name two modifications to the transmission lines which would reduce the transmission line losses. Are there any reasons why your suggestions have not been implemented?

4 A farmer wishes to operate a 960 W appliance at a distance of about one kilometre from his power supply. He has available a suitably long cable with an overall resistance of 20 Ω, i.e. 10 Ω in each of the two leads. See Figure 61.1.

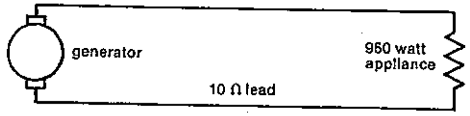


Fig. 61.1

The farmer considers two possibilities:
Plan 1: Buying a 32 V 960 W appliance
Plan 2: Buying a 240 V 960 W appliance
Comparing the two possibilities, determine the following ratios:
(*a*) $\dfrac{\text{power consumed by the appliance in plan 1}}{\text{power consumed by the appliance in plan 2}}$;
(*b*) $\dfrac{\text{energy lost in the leads in plan 1}}{\text{energy lost in the leads in plan 2}}$;
(*c*) $\dfrac{\text{power supplied by the generator in plan 1}}{\text{power supplied by the generator in plan 2}}$.

5 When electrical energy is to be transmitted over long distances through electric cables, it is usually done at *high* voltages (e.g. 220 kV or 500 kV). For which one or more of the following reasons is *high* voltage used?
A Most industrial machinery is designed to operate at voltages like 220 kV or 500 kV.
B It minimises heat losses in the cables.
C It permits the use of smaller diameter cable.
D It allows the use of smaller and therefore cheaper insulators to separate the cables from the steel towers supporting the cables.
E Since lighter cables can be used, fewer supporting towers are needed over a long transmitting distance.

6 Most electricity authorities distribute electrical energy using *a.c.* rather than *d.c.* Which of the following is the main reason for doing this?
A While transmitting the same power, a.c. requires a lower current than d.c. in the transmission cables.
B It is easier to oscillate electrons back and forth about a mean position rather than send them from the power station to the consumer.
C Voltages can be stepped up and down by means of transformers which work with a.c. but not d.c.
D Since the peak voltage of the a.c. used is different to the equivalent constant d.c. voltage, smaller insulators are required between the transmission lines and the supporting towers when using a.c.

7 A common electricity supply in Australian households is "240 V 50 Hz a.c.".
(*a*) What is the average voltage?
(*b*) What is the r.m.s. voltage?
(*c*) What is the peak voltage?
(*d*) What is the period of the a.c. cycle?

8 Which one or more of the following disadvantages would result if all the lights in a house were rewired so that they were in *series* rather than in *parallel*?
A The lights would not be as bright as usual.
B The lights would be brighter than usual.
C There would be a greater likelihood of the filaments burning out.
D The lights would have to be all on or all off.
E A normal 10 A fuse would burn out more easily, if the two leads to an electric light came in contact.
F If the filament in a light globe burnt out, all the lights would go off.

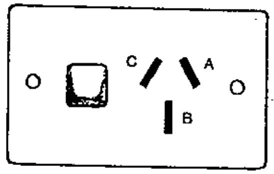


Fig. 61.2

9 For the wall power point shown in Figure 61.2, is the *active, neutral, earth* or *none of these leads* electrically connected to the sockets A, B and C, when the power point switch is
(*a*) turned off?
(*b*) turned on?

10 Two colour codes are used on leads and cables for identifying active, neutral and earth wires. Identify the unnamed colours (*a*), (*b*), (*c*) and (*d*) in Table 61.2.

Table 61.2

Wire	Code 1	Code 2
Active	red	(*c*)
Neutral	(*a*)	(*d*)
Earth	(*b*)	yellow/green

11 In Figure 61.3 the points A, N and E represent respectively the active, neutral and earth contacts of the normal 240 V a.c. power point. The circuit inside the toaster is *incomplete*. PQ is the heating element and F is a contact on the metal casing of the toaster.

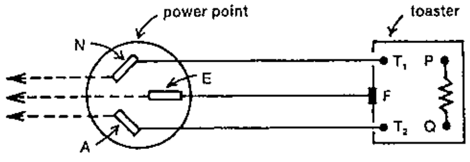


Fig. 61.3

(*a*) Would the toaster switch be connected between T_1 and P, T_2 and Q, T_1 and F, or Q and F?
(*b*) With the toaster switch correctly placed, what other two points would be connected electrically inside the toaster: T_1 to P, T_2 to Q, or F to P?
(*c*) Which of the wires (the one from the active, neutral or earth), shown as broken lines in Figure 61.3, should lead to
(*i*) the power point switch?
(*ii*) a fuse in the house fuse box?

12 Figure 61.4 represents a double-insulated power drill plugged into a three-pin power point. Which of the locations (1, 2, 3, 4, 5 and 6) marked in Figure 61.4 is the best place for
(*a*) the switch on the power drill?
(*b*) the earthing point?
(*c*) the power point switch?

13 In a house the following lights and appliances happen to be operating on one evening at the same time:
2 75 W light bulbs
1 90 W colour TV
1 60 W light bulb
2 30 W desk lamps
1 80 W fan heater for a gas space heater
2 100 W light bulbs
2 40 W fluorescent tubes
1 2400 W electric fan heater
(*a*) If the r.m.s. supply voltage is 240 V, what is the total current in the mains active cable entering the fuse box?
(*b*) In one hour how much electrical energy is used in the house? Give the answer in joule.
(*c*) What percentage of the total energy consumed is used by the electric fan heater?

14 A selection of domestic appliances designed to operate at 240 V had power ratings marked on them as follows:
carving knife 36 W; deep fryer 1450 W; clock 2.0 W; fan heater 2400 W; can opener 175 W; toaster 1150 W.
(*a*) Which appliance has the highest electrical resistance? What is its value?
(*b*) What current does the carving knife draw?
(*c*) Assuming a power point is connected to a 15 A fuse in the fuse box, could any *two* of these appliances plugged into a double adaptor in such a power point cause the fuse to blow?

15 Suppose that the earth wire in the lead to a 240 V a.c. electric toaster has become detached from the earthed pin and that the active wire inside the toaster is touching the metal case. The toaster case is live, but the toaster still operates. A wet swimmer touches the case when the toaster is on and receives a shock, consisting of an electric current travelling through the right hand across the chest to the left hip which is touching the stainless steel kitchen sink. If the total resistance of this path is 1.1 kΩ, is ventricular fibrillation possible? Ventricular fibrillation, in which the heart muscle becomes desynchronised and loses its rhythm, can be caused by a current of about 0.2 A.

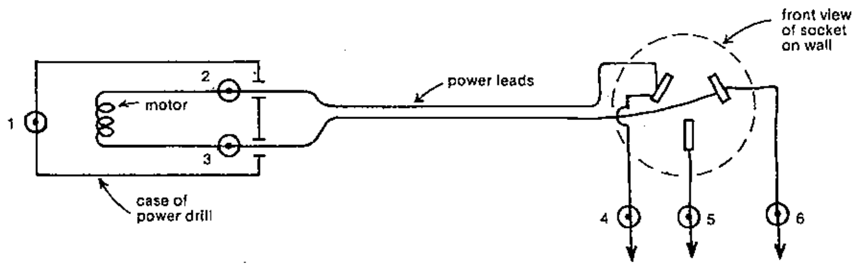


Fig. 61.4

16 Some electricity authorities determine the amount of electrical energy used by consumers in kilowatt hour (kW h) rather than joule. How many megajoule are equivalent to one kilowatt hour?

17 You run a 30 W fluorescent desk lamp in your bedroom for $2\frac{1}{2}$ h each night of the week from Monday to Thursday for a ten week school term.
(*a*) How much electrical energy (in kW h) do you use?
(*b*) How much does it cost at 11 cents per kW h?

SET 62 INTRODUCTORY ELECTRONICS

Topic	*Problems*
p–n junction diode	1
Zener diodes and their use as voltage regulators	2–3
Light-dependent resistors, thermistors	4–5
Inverter or NOT gate	6–7
Electronic switch	8–12
Square wave generator or waveform modifier	13–15
Transistorised amplifier, characteristic curves	16–19
Feedback	20–21
Operational amplifier	22–24
NOT, NOR, OR, AND, NAND and other logic gates	25–33
Bistable or flip-flop, triggered bistable, and more feedback	34–36
Pulse producer	37
Astable, multivibrator, and more feedback	38–40
RC circuits as differentiating and integrating circuits	41–43

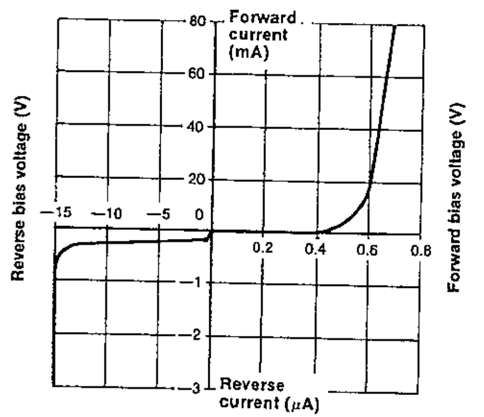


Fig. 62.1

1 Figure 62.1 shows a typical characteristic curve for a p–n junction diode. Note that the current and voltage scales change as you cross the origin. The diode is connected into the circuit shown in Figure 62.2. The voltmeter has a very high resistance.

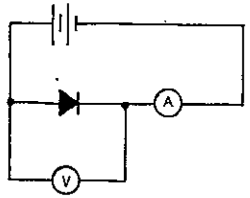


Fig. 62.2

(*a*) In Figure 62.2 is the diode forward or reversed biased?
(*b*) If the voltmeter reads 0.6 V, what is the reading on the ammeter?
(*c*) What is the breakdown voltage of this diode?
(*d*) If the connections to the diode are reversed, which of the following would *not* be possible meter readings?
2 μA, 4 μA, 10 V, 20 V.

2 The Zener diode, acting as a voltage regulator in Figure 62.3, has the characteristics shown in Figure 62.1. It has a maximum power rating of 0.60 W. The applied voltage varies between 21 and 23 V.

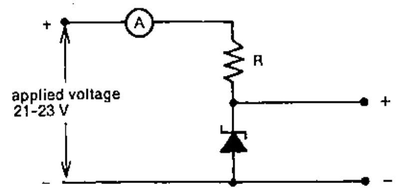


Fig. 62.3

(*a*) Is the Zener diode in Figure 62.3 in forward or reverse bias?
(*b*) What is its Zener voltage?
(*c*) What is the maximum current that should not be exceeded for this diode?
(*d*) When the applied voltage is 23 V, what is
 (*i*) the voltage across the limiting resistor R?
 (*ii*) the maximum desirable current in R?
 (*iii*) the minimum resistance of R?
(*e*) With this resistance for R—your answer to (*d*) (*iii*)—and an applied voltage of 20 V, what is
 (*i*) the voltage across R?
 (*ii*) the current in R?

3 If, in a circuit similar to that in Figure 62.3, the diode had a Zener voltage of 3.9 V and a power rating of 0.78 W, what should be the resistance of R to provide a regulated output voltage of 3.9 V from an applied input varying between 4.2 and 4.5 V?

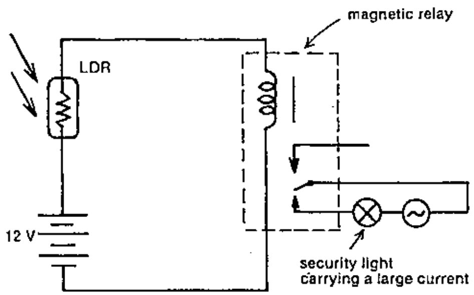


Fig. 62.4

4 A *light-dependent resistor* (LDR), also known as a *photoconductive cell* or *cadmium sulphide cell,* has a resistance which decreases with the amount of light falling on it. One use of an LDR is shown in Figure 62.4, where it can control another circuit carrying a large current by means of a magnetic relay in series with the LDR. Suppose it requires 10 mA in the magnetic relay to open the security light circuit, and that the resistance of the LDR is 1.6 kΩ at 6.00 a.m. and 1.0 kΩ at 6.30 a.m. as the Sun rises. At what approximate time will the security light turn off? The resistance of the coil in the relay would be negligible in comparison with that of the LDR.

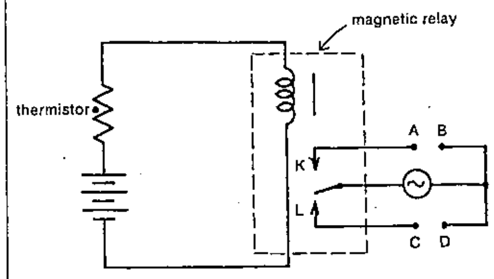


Fig. 62.5

5 A *thermistor* is a semi-conducting device, the resistance of which decreases with increasing temperature. In Figure 62.5 a thermistor is connected in series with a magnetic relay in order to control a secondary circuit which can ring an alarm bell to warn of a rising temperature caused by a fire. With a low current in the relay coil, the moving contact is at L, but with a higher current it is attracted across to K. Should the bell be inserted between A and B or C and D?

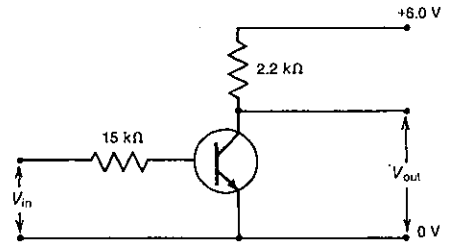


Fig. 62.6

6 For the transistor circuit shown in Figure 62.6 the output-input voltage characteristics are shown in Figure 62.7.

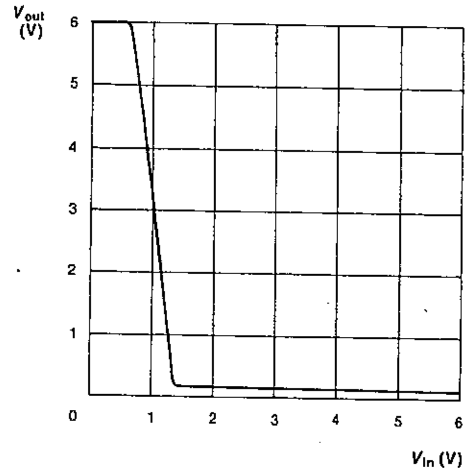


Fig. 62.7

(*a*) What is the output voltage (to the nearest volt) when the input voltage is
 (*i*) 0 V?
 (*ii*) 6 V?

From your answers to (*a*) you should see why this circuit is often referred to as an *inverter* or *NOT gate*. Two simpler symbols used for the circuit in Figure 62.6 or any equivalent one *acting as an inverter* are given in Figure 62.8.

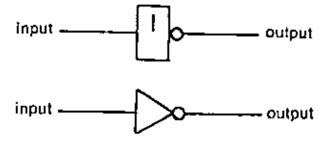


Fig. 62.8

(*b*) To explain why this circuit is called a NOT gate, write a sentence describing its effect, commencing "This circuit gives a *high* output when its input . . ." and using the word "NOT".
(*c*) Tables 62.1 and 62.2, known as *truth tables*, summarise the inverting properties of this circuit. Write down the missing element in
 (*i*) Table 62.1;
 (*ii*) Table 62.2.

Table 62.1

Input	Output
Lo	Hi
Hi	

Table 62.2

Input	Output
0	
1	0

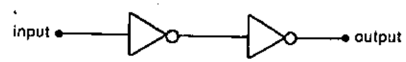


Fig. 62.9

7 How would the logical output in Figure 62.9 be related to the logical input after having passed through two inverters?

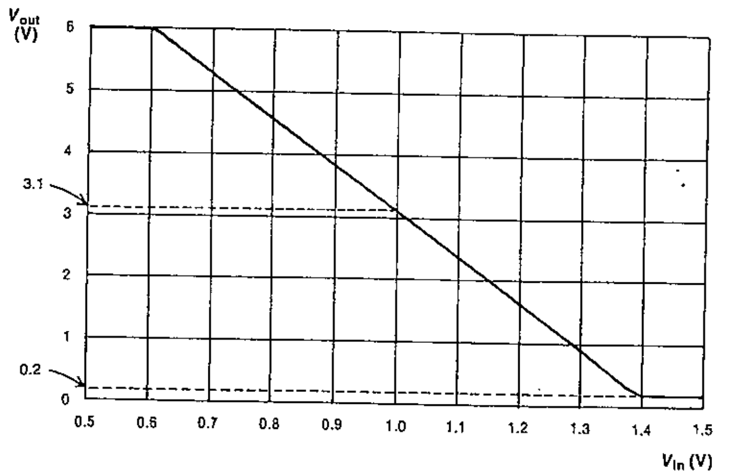


Fig. 62.10

8 Consider again the circuit described in the first few lines of Problem 6 and Figure 62.6. Its output–input voltage characteristics (Fig. 62.7) are shown in greater detail for a range of lower input voltages in Figure 62.10. This circuit is now modified by the inclusion of a potential divider as shown in detail in Figure 62.11, or more simply in Figure 62.12. The movable contact is set at B such that the potential difference across BC is 1.0 V.

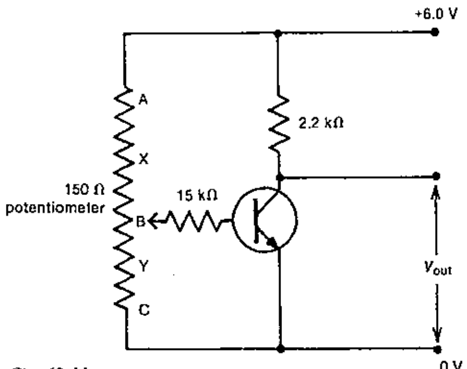


Fig. 62.11

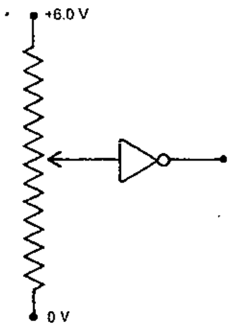


Fig. 62.12

(*a*) What must be the potential difference across AB?
(*b*) What will be the output voltage of the circuit?
(*c*) Assuming the input current in the 15 kΩ resistor is negligible in comparison with the current in AB, determine
 (*i*) the ratio of the voltage across AB, V_{AB}, to V_{BC};
 (*ii*) the ratio of the resistance of AB, R_{AB}, to R_{BC};
 (*iii*) R_{BC};
 (*iv*) R_{AB}.
(*d*) If the potentiometer sliding contact was moved to Y to reduce the input voltage *by* 0.40 V, determine
 (*i*) the new input voltage V_{YC};
 (*ii*) the new output voltage;
 (*iii*) V_{AY};
 (*iv*) the ratio R_{AY}/R_{YC}.
(*e*) If the sliding contact was now moved up slowly from Y until at X the output voltage first reached 0.20 V, what would now be
 (*i*) the input voltage V_{XC}?
 (*ii*) V_{AX}?
 (*iii*) the ratio R_{AX}/R_{XC}?

9 The circuit of Figures 62.11 and 62.12 is known as a switching circuit, and is shown in a slightly modified form in Figure 62.13.

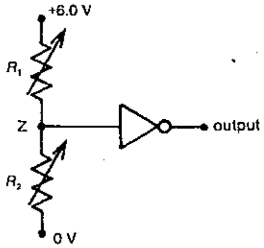


Fig. 62.13

Initially, the potential at the point Z is set at +3.1 V (see Fig. 62.10). Assuming all input changes proceed far enough to effect any possible output changes, would this electronic switch turn *on* or *off*, i.e. would its output go *high* or *low*,
(*a*) if R_1 was increased?
(*b*) if, instead, R_1 was decreased?
(*c*) if, instead, R_2 was increased?
(*d*) if, instead, R_2 was decreased?

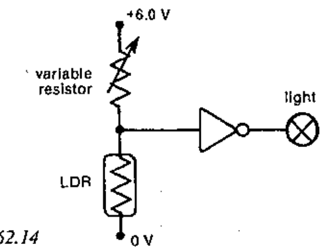


Fig. 62.14

10 As an electronics student you are asked to design a circuit which will automatically turn on a light when it gets dark after sunset. Using a circuit based on Figures 62.11 and 62.12, you come up with Figure 62.14, using an LDR, the resistance of which decreases with an increase in the amount of light falling on it. Actually your circuit will not do the job. Suggest two alternative modifications to rectify this.

11 The circuit in Figure 62.15 is a student's suggestion for a temperature-warning device to turn on a 6.0 V dashboard light if a car engine is overheating. She uses a thermistor, the resistance of which decreases with increasing temperature.

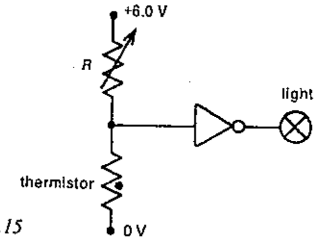


Fig. 62.15

(*a*) Assuming *R* is adjusted appropriately, would her circuit work? If so, go on to (*b*). If not, suggest modifications to rectify it.
(*b*) Using Figure 62.10, which shows the output–input voltage characteristics of this circuit, decide whether there is an upper or lower limit for the potential difference across the thermistor when the engine is overheating. What is this limiting value? For simplicity, assume that the light requires a potential difference of 6.0 V to light up satisfactorily.
(*c*) If the device was not sufficiently sensitive, i.e. the engine was excessively hot by the time the light came on, what modification would you make to *R*, the resistance of the variable resistor?

12 Suppose you have developed for the pest control industry a new, very sensitive electronic device known as a *ringtailistor*, which has an electrical resistance which increases when it comes into contact with the characteristic odour of a possum. Which one or more of the circuits in Figure 62.16 could be used to frighten off possums? The oscillator O, when turned on by a high input, gives out an ultrasonic frequency not detected by humans but very unpleasant for possums. The ringtailistor is shown as a shaded rectangle.

13 Refer again to the circuit shown in Figure 62.6 with its output–input voltage characteristics shown in Figures 62.7 and 62.10.

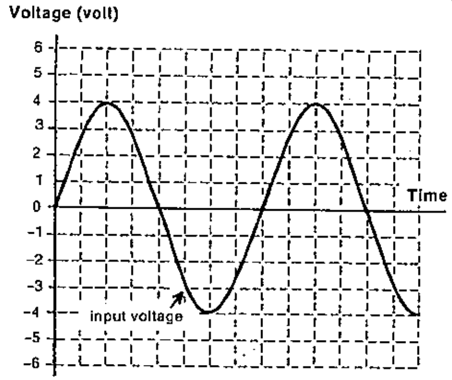


Fig. 62.17

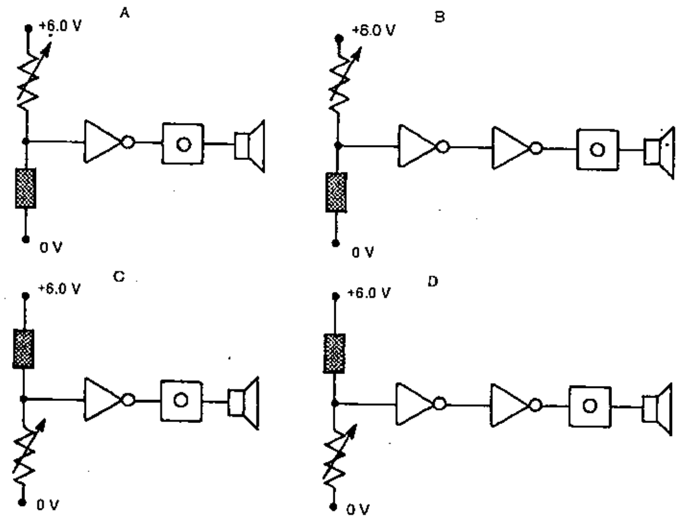


Fig. 62.16

(*a*) This circuit is now given a sinusoidally varying input voltage as shown in Figure 62.17. On a copy of this graph draw a predicted corresponding output voltage against time using Figure 62.7. Incidentally, moderate *negative* input voltages give rise to a 6.0 V output voltage.
(*b*) The square wave you should have produced in (*a*) is not perfectly symmetrical. Is it "on", i.e. at 6.0 V, for a longer or shorter period than it is "off"?
(*c*) Are the "on" periods of the square wave in phase or out of phase with the peaks of the a.c. input?
(*d*) How is the frequency of the square wave related to the frequency of the a.c. input?

14 Repeat Problem 13 for an electronic circuit which has the output-input voltage characteristics shown in Figure 62.18.

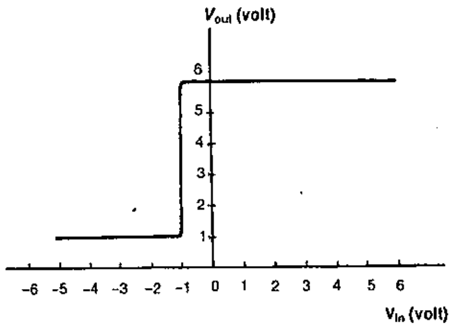


Fig. 62.18

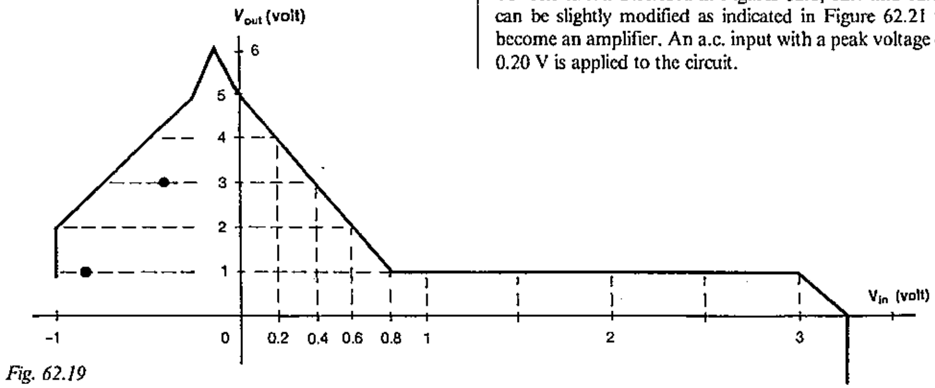


Fig. 62.19

15 A recently developed IC (integrated circuit) used in the totalisator computer at the races has the output–input voltage characteristics shown in Figure 62.19. This IC receives an input voltage which varies with time as shown in Figure 62.20.

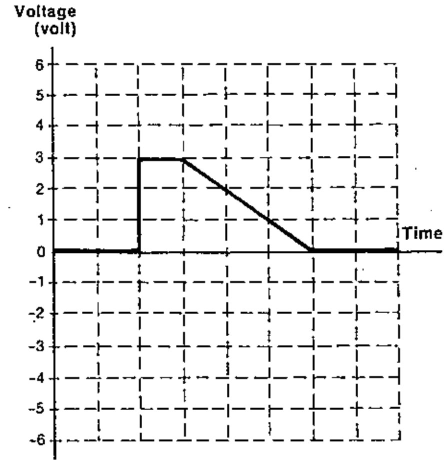


Fig. 62.20

(*a*) Copy Figure 62.20 on to graph paper and draw in the corresponding output voltage.
(*b*) For the output voltage state the
(*i*) maximum value;
(*ii*) minimum value.
(*c*) The input and output voltages are roughly in phase or out of phase. Which?

16 The circuit described in Figures 62.6, 62.7 and 62.10 can be slightly modified as indicated in Figure 62.21 to become an amplifier. An a.c. input with a peak voltage of 0.20 V is applied to the circuit.

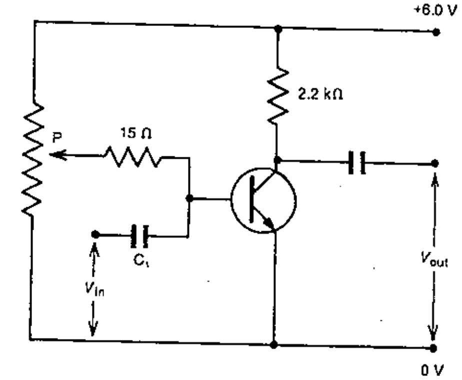


Fig. 62.21

(*a*) If this circuit is to be used as an amplifier, what is the optimum potential at the point P? Refer to Figure 62.10.
(*b*) With P set at this optimum potential, what is the
(*i*) output voltage when the input voltage is a maximum?
(*ii*) output voltage when the input voltage is a minimum?
(*iii*) peak-to-peak input voltage?
(*iv*) peak-to-peak output voltage?
(*v*) voltage amplification provided by the amplifier?
(*c*) Would any distortion of the a.c. signal occur,
(*i*) if, with the same input bias, i.e. your answer to (*a*), a peak voltage of 0.35 V was used?
(*ii*) if, still with the same input bias, a peak voltage of 0.45 V was used?
(*iii*) if the potential at P was changed to 1.10 V, and the original peak voltage of 0.20 V was used?
(*iv*) if, with a 1.10 V potential at P, a peak input voltage of 0.35 V was used?

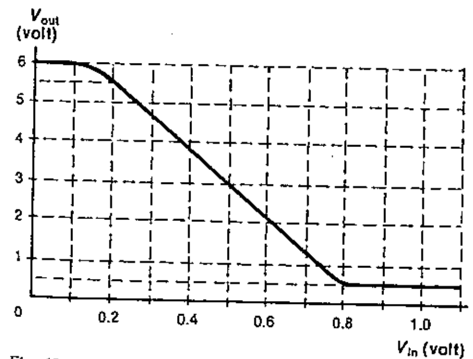


Fig. 62.22

17 Figure 62.22 represents the output–input voltage characteristics for an amplifier. Before any signal is introduced the amplifier is given an input biasing voltage of +0.5 V. Then the sawtooth voltage shown in Figure 62.23 is applied to the input of the amplifier.

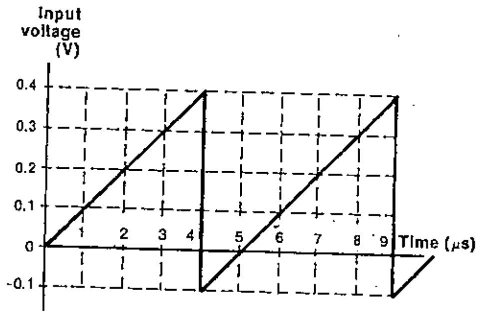


Fig. 62.23

(*a*) Plot a graph of the output voltage as a function of time for the first 6 μs.
(*b*) For the output voltage write down its
(*i*) maximum value;
(*ii*) minimum value.
(*c*) If the intention is to amplify this input signal without distorting it, suggest a more suitable input biasing voltage than +0.5 V.

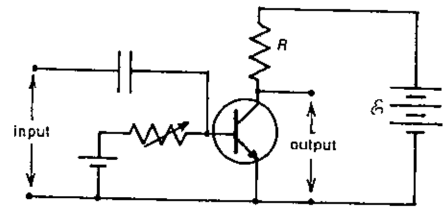


Fig. 62.24

18 The amplifier circuit in Figure 62.24 uses a transistor with the collector characteristics shown in Figure 62.25. Suppose you are told you can use either
Arrangement I : $\mathcal{E} = 9$ V and $R = 1.5$ kΩ, or
Arrangement II: $\mathcal{E} = 12$ V *and* $R = 3$ kΩ

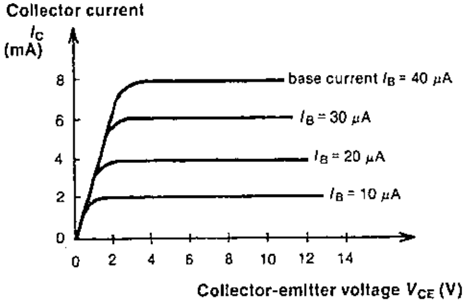


Fig. 62.25

(*a*) On a copy of Figure 62.25 mark in the two loadlines.
(*b*) For the same operating point for each amplifier, which arrangement will give the higher voltage gain?

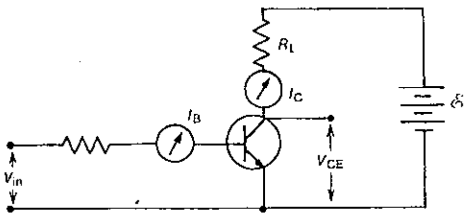


Fig. 62.26

19 In the circuit shown in Figure 62.26 the supply voltage $\mathscr{E}$ and the resistance R_L of the load resistor are unknown. The data in Table 62.3 was obtained using the circuit and a variable input voltage V_{in}.

Table 62.3

V_{in} (V)	I_B (μA)	V_{CE} (V)	I_C (mA)
0	0	15	0
0.5	5	12	1
0.6	10	9	2
0.7	15	6	3
0.8	20	3	4
0.9	25	0.5	4.8
1.0	30	0.5	4.8

Using these measurements, deduce values for
(*a*) $\mathscr{E}$;
(*b*) R_L.

20 Figure 62.27 shows a single-stage amplifier with a feedback link.

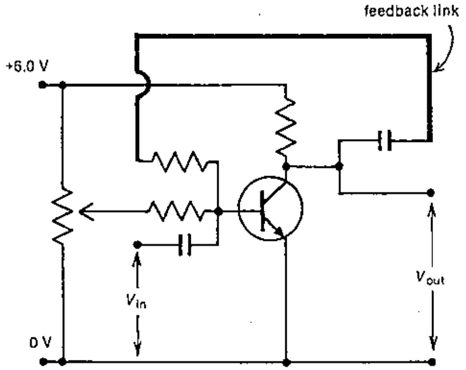


Fig. 62.27

(*a*) Is the feedback positive or negative?
(*b*) Does this feedback reduce or increase
(*i*) the gain of the amplifier?
(*ii*) distortion of the signal being amplified?

21 Decide whether the following are examples of positive or negative feedback.
(*a*) You pour yourself a cup of tea and the phone rings. On returning from the phone you taste the tea and then add some hot water from the kettle.
(*b*) During a petrol-drivers' strike, people expected that petrol supplies would soon be exhausted, and this belief gave rise to panic buying.

22 An amplifier circuit is shown in Figure 62.28. For the operational amplifier, which it incorporates, $V_{out} = 100\,000(V_+ - V_-)$. Determine the gain of the amplifier.

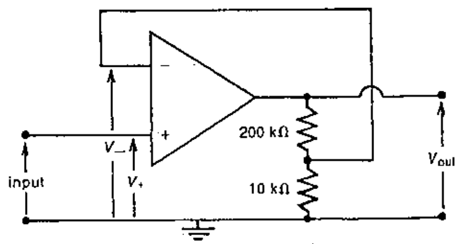


Fig. 62.28

23 An oscillator with an r.m.s. output voltage of 50 mV is connected to the input of the operational amplifier circuit shown in Figure 62.29.

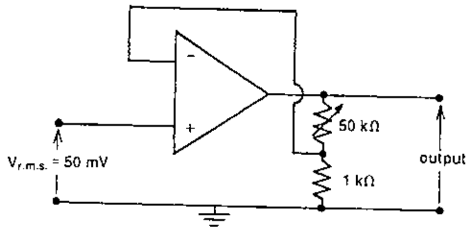


Fig. 62.29

Calculate the r.m.s. values of the
(*a*) maximum, and
(*b*) minimum
output voltages from the amplifier. The gain of the operational amplifier is 10^5.

24 The circuit in Figure 62.30 is used to amplify a 100 mV source sufficiently to operate a 0.20 W 20 Ω buzzer. If R_1 = 10 kΩ, what is the appropriate value for R_2?

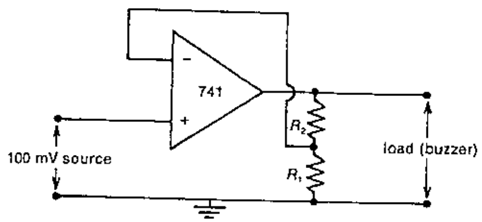


Fig. 62.30

25 Refer again to the circuit shown in Figure 62.6. If it is modified by including a second input resistor to give it two inputs, then we have the circuit in Figure 62.31, which can be represented more simply as indicated in Figure 62.32.

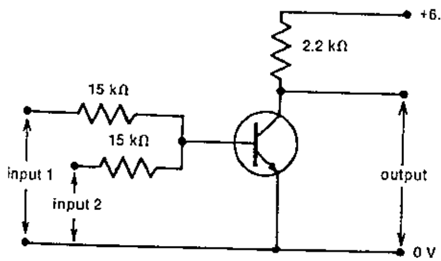


Fig. 62.31

Fig. 62.32

(*a*) Remembering that the input side of the transistor in Figure 62.31 must be low, i.e. close to 0 V for the output to be high, determine whether the output is high or low when
(*i*) input 1 is high and input 2 is high;
(*ii*) input 1 is high and input 2 is low;
(*iii*) input 1 is low and input 2 is high;
(*iv*) input 1 is low and input 2 is low.
(*b*) Using "1" to represent "high" and "0" for "low", copy and complete the truth table, shown here as Table 62.4.

Table 62.4

Input 1	Input 2	Output
1	1	
1	0	
0	1	
0	0	

(*c*) To explain what this circuit does, write a statement commencing, "This circuit gives a *high* output when. . ." without using the word "low".
(*d*) This circuit could have been called a NEITHER gate, but it has a simpler name. What do you think it is? Three letters.

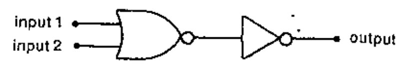
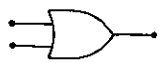


Fig. 62.33

26 Figure 62.33 shows the output of a NOR gate being fed into an inverter (or NOT gate).
(*a*) Draw a truth table for the circuit.
(*b*) Describe the action of this circuit with a statement commencing, "This circuit gives a *high* output when. . ." without using the word "low".
(*c*) If the circuit in Figure 62.32 is a NOR gate, what name do you think is given to the circuit in Figure 62.33?
(*d*) Suggest a possible practical use for this circuit. It is usually shown by the single symbol shown in Figure 62.34.

Fig. 62.34



27 On a production line in a factory you require a gate circuit which will turn on a motor (given a high output) to drive a conveyor belt at a point X, only when both of two different components to be assembled have arrived (and given high inputs) at X.
(*a*) Draw up a truth table for the gate required.
(*b*) Supposing you have access to NOT and NOR gates only, design a suitable gate. Of what does it consist?

(*c*) This circuit is known as an AND gate, normally shown as in Figure 62.35. Using the word "and", describe its function, commencing with the words, "This circuit gives a *high* output when . . .".

Fig. 62.35

28 You are asked to design a circuit which will *not* give a high output when both one *and* the other of two inputs are high. This is known as a NAND gate.
(*a*) Draw up a truth table for a NAND gate.
(*b*) How would you arrange one or more NOT and AND gates to make such a gate? The single symbol for a NAND gate is given in Figure 62.36.

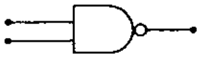


Fig. 62.36

(*c*) Think out a practical use for a NAND gate.

29 What single gate circuit would be equivalent to
(*a*) a NOR gate followed by a NOT gate followed by a second NOT gate?
(*b*) a NAND gate followed by a NOT gate?

30 A NOR gate and a NAND gate are not quite the same.
(*a*) For what inputs (1–1, 1–0, 0–1 or 0–0) do a NOR gate and a NAND gate give a different output?
(*b*) Which of the two gates would you use to turn on an outside security light only when the lights had been turned off in both of two research laboratories?

31 What single logic gate gives
(*a*) an output always opposite to the input?
(*b*) an output of 1 only when both inputs are 1?
(*c*) an output of 1 only when both inputs are 0?
(*d*) an output of 1 unless both inputs are 1?
(*e*) an output of 1 unless both inputs are 0?

32 Computers and calculators carry out their addition using *binary* rather than *decimal* numbers. Table 62.5, showing the sum of two numbers A and B in both decimal and binary notation, is incomplete.
(*a*) Copy and complete Table 62.5.

Table 62.5

A	B	A + B in decimal notation The answer	A + B in binary notation The answer	"Put down" digit	"Carry" digit
0	0	0	0	0	0
1	0	1	1	1	0
0	1	1			
1	1	2	10		

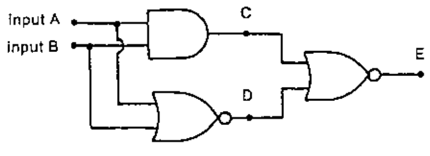


Fig. 62.37

Figure 62.37 represents a binary adding circuit in a computer where a high input represents a "1" and a low input a "0".
(*b*) Copy and complete the truth table shown for this circuit in Table 62.6, which includes the intermediate outputs at C and D as well as the final output at E.

Table 62.6

Input A	Input B	Output C	Output D	Output E
0	0	0	1	0
1	0	0		
0	1			
1	1			

(*c*) In Figure 62.37, which of the outputs C, D and E would represent
(*i*) the "put down" digit, and
(*ii*) the "carry" digit
for the binary addition of the two inputs?

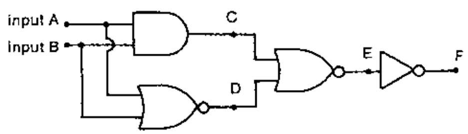


Fig. 62.38

33 Consider the circuit shown in Figure 62.38.
(*a*) Copy and complete the truth table for it, given as Table 62.7.

Table 62.7

A	B	C	D	E	F
0	0				
1	0				
0	1				
1	1				

(*b*) Suggest a name for this circuit, considered as a logic gate.
(*c*) Suggest a possible practical use for this circuit.

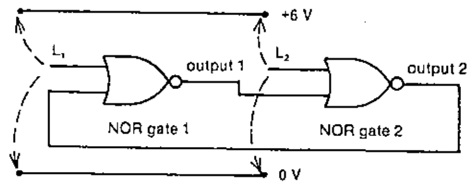


Fig. 62.39

34 In the circuit in Figure 62.39, two NOR gates are linked as shown, while their second inputs are flying leads, L_1 and L_2, which can be
connected to +6 V (a "high" or "1"),
connected to 0 V (a "low" or "0"), or
left *loose* (connected to neither the +6 V nor 0 V lines).
(*a*) A sequence of positions for the flying leads is shown in Table 62.8. Copy the table and add the missing outputs.

Table 62.8

Location of flying leads L_1	L_2	Output 1	Output 2
+6 V	0 V	Lo	Hi
loose	0 V		
loose	loose		
loose	+6 V		
0 V	+6 V		
0 V	loose		
loose	loose		
+6 V	loose		

(*b*) Can you suggest why this circuit is called a *bistable*? It is also called a *flip-flop* for obvious reasons.
(*c*) Is the bistable using positive or negative feedback?
(*d*) Is the output line from the NOR gate on the right or the left the feedback link, or are they equally feedback links? Explain.

35 The bistable or flip-flop shown in Figure 62.39 can be slightly modified so that successive input pulses, applied to an input called the *trigger*, can be fed successively to one and then the other NOR gate of the bistable. Such a circuit is known as a *triggered bistable* or *T-type flip-flop*.

Suppose that the state of a triggered bistable is as shown in the first row of Table 62.9 prior to a series of pulses being fed into the trigger input.

Table 62.9

Number of input pulses to trigger	State of output 1	State of output 2	Progressive number of pulses from output 1	Progressive number of pulses from output 2
0	0	1	0	0
1				
2				
3				
4				
12				

(*a*) Copy and complete Table 62.9.
(*b*) What simple *mathematical* process is this circuit performing?

36 Two triggered bistables are connected as indicated in Figure 62.40, with their outputs monitored by indicator lamps which come on when the output is high. When the trigger input of one of these bistables goes from high to low, the outputs interchange their conditions. The initial state of the circuit is that output 1 of both bistables A and B is high. A series of square pulses is fed into the trigger input of bistable A. What is the condition of the two indicators as the following points on the input train of pulses reach the trigger input of bistable A?
(*a*) P;
(*b*) Q;
(*c*) R.

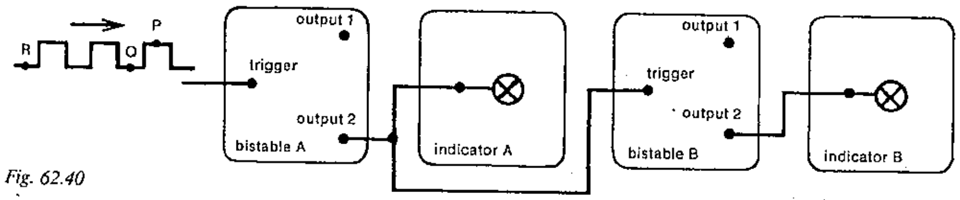


Fig. 62.40

37 If the inverter circuit of Figure 62.6 is slightly modified by connecting the resistive input to +6 V and adding a 25 μF input capacitor as shown in Figure 62.41, then we have a *pulse producer*. When the input goes *from +6 V to 0 V*, the output goes from 0 V to 6 V and stays there before returning to 0 V after an interval which depends on the time constant of the input circuit.

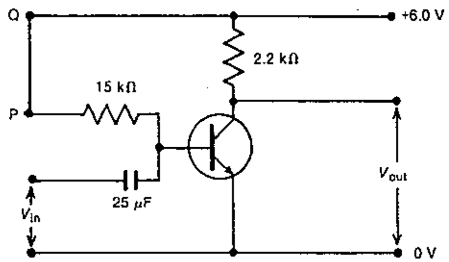


Fig. 62.41

What happens to the length of the output pulse,
(*a*) if the lead between P and Q in Figure 62.41 is replaced by an 18 kΩ resistor?
(*b*) if, instead, an 18 kΩ resistor is connected in parallel with the 15 kΩ resistor?
(*c*) if, instead, a capacitor of 22 μF is connected in series with the 25 μF capacitor?
(*d*) if, instead, a 22 μF capacitor is connected in parallel with the 25 μF capacitor?

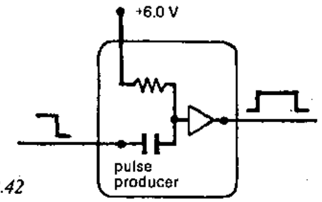


Fig. 62.42

38 Suppose we represented the pulse-producing circuit of Figure 62.41 more simply as shown in Figure 62.42. Two such pulse producers are shown linked together in Figure 62.43.

(*a*) If pulse producer A generates the pulse shown at output A, at what stage will this pulse in turn *begin* to generate a pulse at output B, i.e. as output pulse A goes from low to high or from high to low? (If necessary, refer back to Problem 37.)
(*b*) This circuit, where two pulse producers continually send pulses to each other, is known as an *astable*. Which of the following statements best describes the feedback used in this circuit?
A Immediate positive feedback.
B Immediate negative feedback.
C Positive feedback with a delay.
D Negative feedback with a delay.

39 Consider the astable circuit in Figure 62.43.
(*a*) What would you have to do to the time constant of the input circuit of either pulse producer A or B in order to
(*i*) lengthen the pulse at output A?
(*ii*) shorten the pulse at output B?
(*b*) Which one or more of the following would lengthen the pulse at output A?
A Insert a resistor in series with R_A.
B Insert a resistor in parallel with R_A.
C Insert a capacitor in series with C_A.
D Insert a capacitor in parallel with C_A.

40 Consider the astable or astable multivibrator, as it is often called, shown in Figure 62.43. Suppose for this circuit that each output pulse remained on for 1 s, and $C_A = C_B =$ 25 μF.
(*a*) How many output pulses would be produced at output B in 1 min?
(*b*) What is the frequency at which pulses are produced at either output A or B?
(*c*) If you wished to produce output pulses at an audio frequency of something like 500 hZ, which one of the following capacitors would you connect in series with both C_1 and C_2?
25 mF, 25 μF, 25 nF.

41 A square wave of frequency 100 Hz is fed into the circuit shown in Figure 62.44.

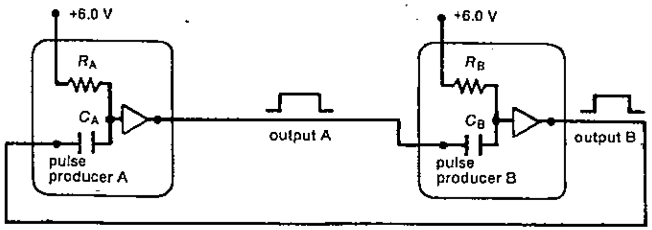


Fig. 62.43

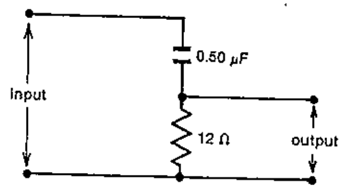


Fig. 62.44

(*a*) Determine the time constant of the circuit.
(*b*) What is the period of the square wave?
(*c*) Which of the sketches in Figure 62.45 best represents the variation of the output voltage with time?
(*d*) Is this circuit acting as an *integrating* or *differentiating* circuit?

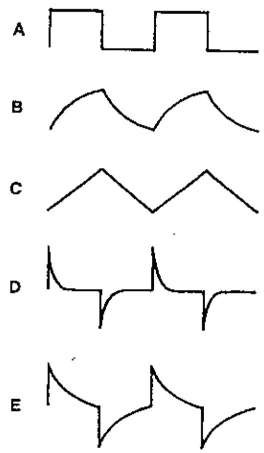


Fig. 62.45

42 A square wave is fed into the circuit shown in Figure 62.46.

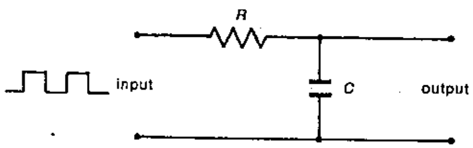


Fig. 62.46

(*a*) In Figure 62.46 R = 500 kΩ, C = 2 μF and the frequency of the square wave is 50 Hz. Compare the time constant of the circuit with the period of the square wave. Are they similar or is one much greater than the other?
(*b*) Of the sketches shown in Figure 62.45 which best represents the variation of the output voltage with time?
(*c*) Is this circuit acting as a differentiator or integrator?

43 If in the circuit in Figure 62.46 R = 100 Ω and C = 0.25 μF, how well would the circuit perform as an integrator of a square wave input of frequency 500 Hz?

SET 63 REVISION PROBLEMS

Set	Topic	Problems
54	Stationary charged particles in electric fields	1–4
55	Moving charged particles in electric fields	5–9
56	Simple d.c. circuits	10–13
57	Less simple d.c. circuits	14–17
58	Magnetic effects of moving charged particles	18–22
59	Electromagnetic induction and electromagnetic radiation	23–27
60	Simple a.c. circuits	28–29
61	Household a.c. supply	30–32
62	Introductory electronics	33–36

Where necessary in this set, use
1 e = 1.60×10^{-19} C
1 eV = 1.60×10^{-19} J
Permeability of a vacuum or air,
$\mu_0 = 4\pi \times 10^{-7}$ N A^{-2}
mass of an electron = 9.11×10^{-31} kg

1 Figure 63.1 shows a positively charged hollow metal capsule, constructed of three sections, hemisphere A, cylinder B and cone C. Points X and Z are at the ends of the capsule and Y is on the surface of the cylinder. In the questions that follow, if all three possibilities are equal, write "equal".
(*a*) Which point, X, Y or Z, is at the highest electrical potential?

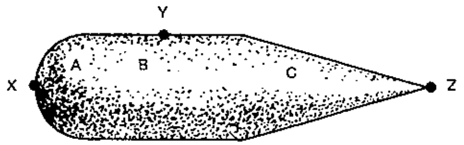


Fig. 63.1

(*b*) At which point, X, Y or Z, is the surface charge density (in C m^{-2}) highest?
(*c*) Within which region, A, B or C, is the electric field strength highest?

2 In Figure 63.2, A and B are two negatively charged hollow spheres. The distance between their centres is 10 cm. It may be assumed that, at this separation, the charge on each sphere is evenly distributed over its surface. The force of repulsion between the two spheres is 9.0×10^{-9} N.

Fig. 63.2

(*a*) What is the value of the electric field strength inside sphere A?
(*b*) If the distance between the centres is increased to 30 cm, what is now the force of repulsion between the spheres?
(*c*) If, at the original separation of 10 cm, the charge on sphere A was halved, what then would be the force of repulsion between the spheres?

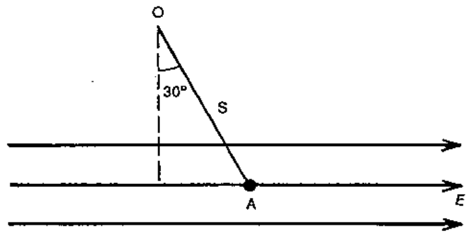


Fig. 63.3

3 A small charged pith ball A, attached to a point of support O by a fine insulating thread S, is at rest in a uniform horizontal electric field of strength 4.7×10^{4} V m^{-1}, as illustrated in Figure 63.3, with the thread displaced through an angle of 30° from the vertical. If the mass of the pith ball is 0.010 g,
(*a*) what is the magnitude of the charge on it?
(*b*) what is the sign of its charge?

4 Consider a point P equidistant from two charges $+q$ and $-q$ a shown in Figure 63.4. Assume the zero of potential to be at infinity.

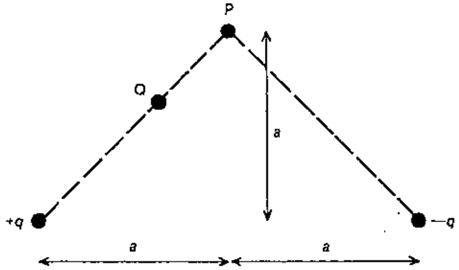


Fig. 63.4

(*a*) What is the electrical potential at P?
(*b*) Is the potential at Q less than, equal to or greater than the potential at P?
(*c*) What is the direction of the electric field strength at P?

5 In a Millikan type experiment with a plastic sphere, the tiny sphere remains balanced between the plates when the upper plate is connected to a power supply of e.m.f. 270 V, the lower plate being connected to the negative terminal of the supply. The magnitude of the charge on each plate is 6.0 nC.
(*a*) What kind of charge is on the sphere?
(*b*) How much work does the power supply do in charging the plates of the apparatus?
Under the action of gravity alone, the sphere was observed to fall with a steady speed of 8.4×10^{-5} m s^{-1}.
(*c*) What potential difference applied to the plates would cause the sphere to move upwards at 2.8×10^{-5} m s^{-1}?
(*d*) With what steady speed will the sphere move if the power supply is set at 540 V and the lower plate is made positive?
(*e*) The distance between the plates is doubled while the original 270 V power supply is connected as before. What will now be the terminal velocity of the sphere? (Give both magnitude and direction.)

6 Electrons are released from a heated filament with an energy of 2 eV. As shown in Figure 63.5, a 180 V power supply accelerates them towards the metal can C, which has a small hole in it at A. The whole apparatus is evacuated.
(*a*) What is the energy (in eV) of each electron as it passes through the hole A?

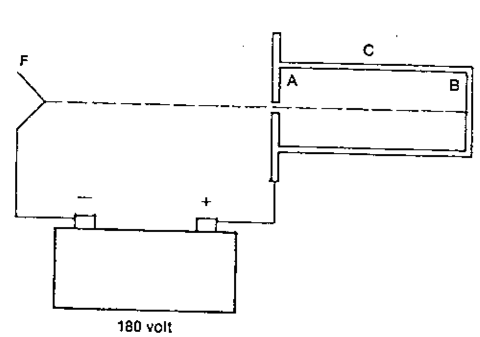


Fig. 63.5

(*b*) With what speed does each electron strike the can at B?

7 A diode valve has a potential difference of 200 V across its terminals, when 2.5×10^{14} electrons reach the positive plate in 10 ms. At what power is the valve operating?

8 Figure 63.6 is a square grid with intersections lettered for reference. At the intersection *b* there is a fixed negative charge X of -10×10^{-11} C, and there is another negative charge Z of -5×10^{-11} C at *g*. The force on Z is found to be *F*.

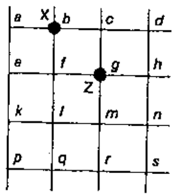


Fig. 63.6

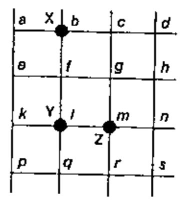


Fig. 63.7

(*a*) What will be the magnitude of the force on Z if it is moved to *m*?
(*b*) Whilst Z is at *m*, a second fixed charge Y of $+3 \times 10^{-11}$ C is placed at *l* as illustrated in Figure 63.7. Under the electric field due to X and Y, towards which lettered intersection will Z begin to move if free to do so? (Choose the nearest intersection if necessary.)
(*c*) In the field due to X and Y it is found necessary to supply 2.5×10^{-10} J of energy to move Z from *d* to *h*. If the electric potential at *d* is −20 V; what is it at *h*?
(*d*) If Z is now removed, X and Y remaining as before, at which other intersection will the electric potential be the same as it is at *m*?

9 Electrons of mass *m* and charge $-e$ from a heated filament are accelerated from J to K by a potential difference V_1, as indicated in Figure 63.8. They then travel a distance *x* from K to L between two oppositely charged, parallel plates, which have a separation *d*. At L they hit a screen. The whole apparatus is evacuated.
Initially, $V_2 = 0$ and the electrons hit the centre of the screen.
(*a*) Write an expression for the speed with which the electrons hit the screen in terms of *m*, *e* and V_1.
(*b*) Write an expression for the time *T* taken for an electron to travel from K to L in terms of *m*, *e*, V_1 and *x*.
(*c*) The potential difference V_2 is now gradually increased from its zero value until the now deflected electrons *just* fail to hit the screen. Write an expression, in terms of *m*, *d* and *T*, for the magnitude of the force experienced by an electron between K and L under this new condition.

10 Three uniform wires, JM, LP and NQ, each of length *d*, are connected as shown in the circuit diagram in Figure 63.9. The resistances of JM, LP and NQ are *R*, 2*R* and 2*R*/3 respectively. Which of the graphs in Figure 63.10 shows

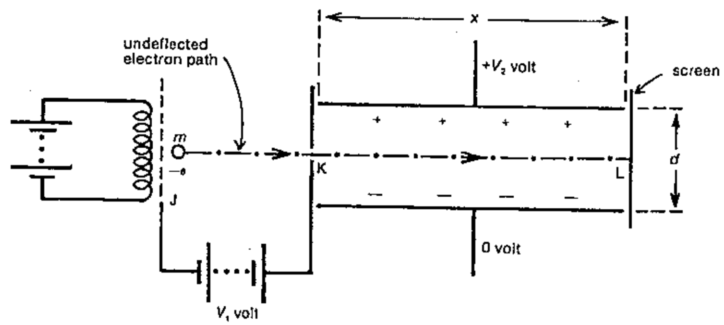


Fig. 63.8

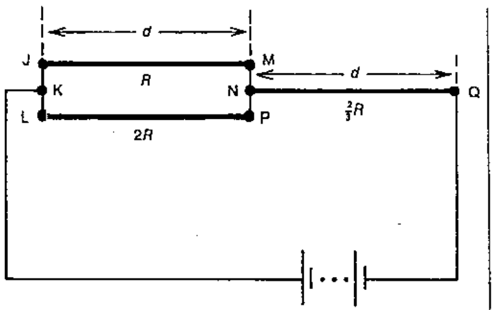


Fig. 63.9

(*a*) the current/distance relationships for the three wires?
(*b*) the electric field strength/distance relationships for the wires?

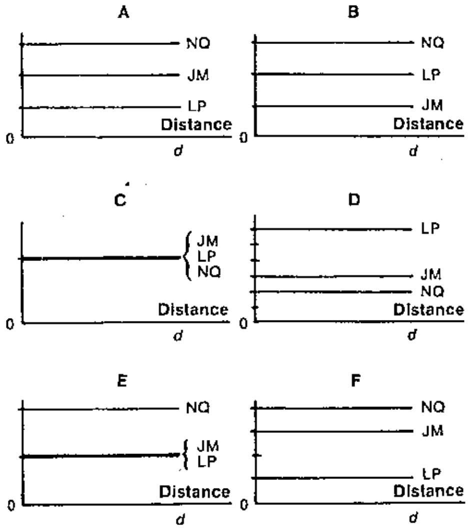


Fig. 63.10

11 It is desired to use nichrome wire of diameter 0.200 mm to produce heat at the rate of 2100 J min^{-1} when there is a potential difference of 12.0 V across it. Find the length of wire required. The resistivity of nichrome = $1.10 \times 10^{-6}\ \Omega$ m.

12 A plastic ice-cream bucket of negligible heat capacity contains 5.0 kg of water at 10°C, and there is a heating coil of 70 Ω resistance in the bucket. If a current of 5.0 A is passed through the coil, in what time will the temperature be raised to 90°C, assuming that 20 per cent of the heat produced is lost to the surroundings? Specific heat capacity of water = 4.2 kJ kg^{-1} K^{-1}.

13 Two coils have an effective resistance of 15 Ω when connected in series, and 2.4 Ω when connected in parallel. Find the resistance of each coil.

14 A cell of e.m.f. 1.50 V and internal resistance 0.100 Ω is connected in series with a coil of 3.00 Ω resistance and an ammeter which reads 0.400 A. A voltmeter is connected so that it will record the potential difference between the terminals of the cell.
(*a*) Draw a diagram of this arrangement.
(*b*) Calculate the resistance of the ammeter.
(*c*) Find the reading of the voltmeter.

15 Two identical cells are connected first in parallel and then in series to a 5.00 Ω resistor. A potentiometer used to measure the potential difference across the resistor gave a value of 1.45 V in the first case and 2.60 V in the second. Calculate
(*a*) the e.m.f., and
(*b*) the internal resistance
of the cells.

16 A galvanometer, internal resistance 100 Ω, gives a full-scale deflection for a current of 10 mA. Calculate the values of the resistances necessary to convert it into
(*a*) a voltmeter to read to 5 V;
(*b*) an ammeter to read to 10 A.

17 Two coils AB and BC are connected in series and a cell of e.m.f. 1.53 V and negligible internal resistance is joined across AC. The potential difference across coil AB is measured with a voltmeter of resistance 500 Ω and the reading is 0.810 V. The same meter then measures the potential difference across coil BC and reads 0.510 V. The resistance of coil AB is known to be 200 Ω.
(*a*) Find the resistance of the coil BC.
(*b*) What would be the reading of the voltmeter when it is joined between the points A and C?
(*c*) Explain why the sum of the potential differences measured across AB and BC separately is less than the potential difference across AC.

18 Two long straight wires X and Y are parallel and 0.150 m apart, with X directly above Y. X carries a current of 5.25 A in an easterly direction, while Y carries a current of 10.0 A west. At a point 0.100 m directly above X, what is the
(*a*) magnitude, and
(*b*) direction
of the magnetic flux density?

19 In Figure 63.11, P and Q are two circular conducting loops with a common centre O and their axes, YY′ and XX′ respectively, are mutually perpendicular. *B* represents a uniform magnetic field of flux density 0.22 mT, directed upwards and extending well beyond the loops.
The current in loop Q is half that in loop P, the radii of the loops are equal, and the magnetic flux density at O due to the current in loop P alone has a magnitude of 0.10 mT.

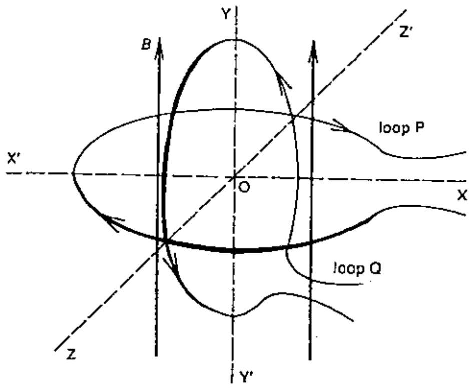


Fig. 63.11

(*a*) Determine the vertical component of the magnetic flux density at O.
(*b*) Determine the magnitude of the resultant magnetic flux density at O.

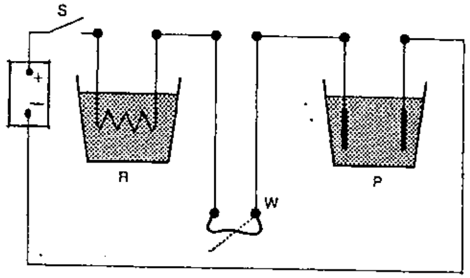


Fig. 63.12

20 The apparatus shown in Figure 63.12 was set up in order to demonstrate the thermal, mechanical and chemical effects of an electric current.
R represents a resistance wire immersed in a beaker of water.
W represents two long, parallel wires.
P represents two copper plates immersed in a copper sulphate solution.
After closing switch S it is observed that
heat is supplied to the water at the rate *H* watt;
the force between the parallel wires is *F* newton;
copper is deposited in P at the rate *M* kg s^{-1}.
Express as a proportionality
(*a*) *H* in terms of *M*;
(*b*) *F* in terms of *M*;
(*c*) *H* in terms of *F*.

21 Ions are produced by sparking, and then accelerated across a vacuum, as shown in Figure 63.13. A constant magnetic field is applied into the tube as indicated, causing the ions to move in the arc of a circle. The masses and charges of the ions used are shown in Table 63.1.

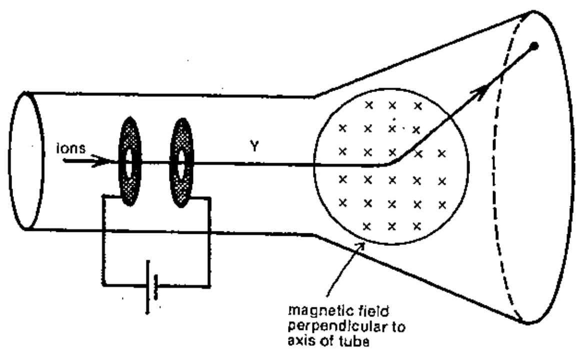


Fig. 63.13

Table 63.1

Ion	Mass (10^{-27} kg)	Charge (10^{-19} C)
proton	1.6	+1.6
deuteron	3.2	+1.6
singly charged helium ion	6.4	+1.6
doubly charged helium ion	6.4	+3.2

(*a*) At the point Y, what is the ratio
$\dfrac{\text{deuteron speed}}{\text{proton speed}}$?

(*b*) What is the ratio
$\dfrac{\text{radius of arc for singly charged helium ions}}{\text{radius of arc for deuterons}}$?

(*c*) If the accelerating potential difference is V when doubly charged helium ions are used, what potential difference (in terms of V) will ensure that a deuteron will move through an arc of equal radius?

22 Figure 63.14 represents a cross-section through a coil of wire consisting of 20 uniformly wound turns. A current flows through the coil in the direction indicated in the diagram. Six paths, lettered A–H, are also shown.

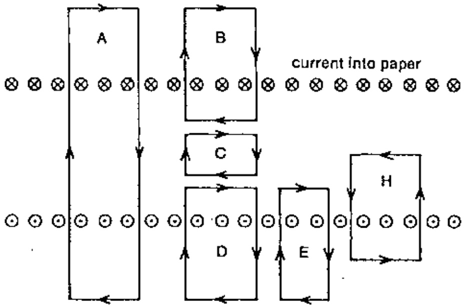


Fig. 63.14

(*a*) For which two of the paths are the circulation of the magnetic flux density around the closed paths equal and non-zero?

(*b*) For which one or more of the paths is the circulation of the magnetic flux density around the closed path(s) equal to zero?

23 A length of copper rod XY is held in a strong magnetic field and is connected to two vertical metal plates A and B as illustrated in Figure 63.15. A light positively charged sphere made of conducting material is suspended between the plates on an insulating thread.

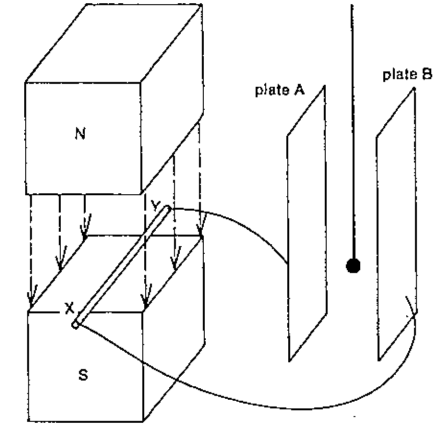


Fig. 63.15

(*a*) If the rod is moved rapidly to the right, what will be the direction of the electric force exerted on the charged sphere?

(*b*) If, instead, the sphere is swung to the left so as to touch plate A, what will be the direction of the magnetic force acting on the rod?

24 A brass rod of length 0.15 m lies at right angles to a magnetic field of flux density 0.50 T. Find the potential difference (in mV) between its ends, if

(*a*) it is moved at right angles to the field and to its own length at a speed of 0.24 m s^{-1};

(*b*) it is moved at right angles to the field and at an angle of 45° to its own length at a speed of 0.24 m s^{-1}.

25 A boy rides a bicycle with handlebars made from a straight metal rod 0.5 m long, as indicated in Figure 63.16. His velocity is 4 m s^{-1} north.

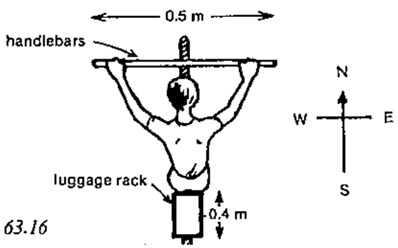


Fig. 63.16

(*a*) Calculate the e.m.f. (in mV) generated between the ends of his handlebars. The vertical component of the Earth's magnetic flux density is 50 μT.

(*b*) Do electrons tend to move to the west or east of his handlebars, assuming he is in
(*i*) Adelaide? (*ii*) Edinburgh?

(*c*) What is the e.m.f. generated between the ends of one of the 0.4 m sides of his luggage rack?

26 ABCD is a square conducting loop, 2 cm by 2 cm, and of resistance 0.1 Ω. As shown in Figure 63.17, it is moving horizontally at 0.4 m s^{-1} towards a block of uniform magnetic field of flux density 0.05 T directed vertically downwards. The magnetic flux is confined to a cube of side 5 cm.

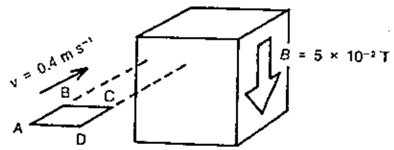


Fig. 63.17

(*a*) What is the induced e.m.f. in the loop when it is exactly halfway through the block?

(*b*) What is the current in the loop at this point?

(*c*) What e.m.f. is induced in the loop 0.05 s after it is exactly halfway through the block?

(*d*) What is the retarding magnetic force on the loop at the instant described in (*c*)?

27 A circular loop of diameter 0.10 m is placed in a magnetic field which is directed perpendicular to the plane of the loop as shown in Figure 63.18. The magnetic flux density is 1.2 T. The loop is pulled at the points Y and Z so that a loop of zero area is formed in 0.20 s. The points C and D are very close, and no part of the circuit to the right of CD is affected by the motion.

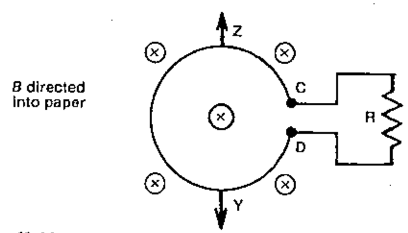


Fig. 63.18

(*a*) What is the average e.m.f. induced across CD?

(*b*) What is the direction of flow of induced current in R?

28 A battery of e.m.f. 2.0 V and negligible internal resistance is connected in series with a 100 Ω resistor and a 4.0 μF capacitor. The capacitor has a 300 Ω resistor connected in parallel with it. The battery is suddenly disconnected from the circuit. While the capacitor is discharging, what is

(*a*) the quantity of charge (in μC) that flows through the 300 Ω resistor?

(*b*) the maximum current (in mA) in the 300 Ω resistor?

29 Capacitors of 5.0 pF and 20 pF respectively are charged so that the respective potential differences between their plates are 200 V and 300 V. They are then connected in parallel, the plates carrying like charges being joined. Calculate

(*a*) the new charge on each capacitor;

(*b*) the common potential difference between their plates;

(*c*) the energy dissipated during the rearrangement of the charges.

30 A radar station has its 50 kW power plant located outside the station and is connected to it by a cable of resistance 0.20 Ω. What is the power loss in the cable if the energy is supplied at

(*a*) 10 kV?

(*b*) 250 V?

31 A pole-type transformer with an output potential difference of 240 V and negligible resistance in its secondary windings, is connected to a nearby farmhouse by a long cable. When three similar radiators in the house are connected in parallel to this supply it is found that the potential difference across the radiators is 210 V and the total current supplied by the source is 15.0 A. What will be the potential difference at the house when one of the radiators is switched off? All quantities quoted are r.m.s. values.

32 Figure 63.19 represents a household electric heater controlled by a switch, which is not shown. At which point, A, B, C, D or E, would the switch be found?

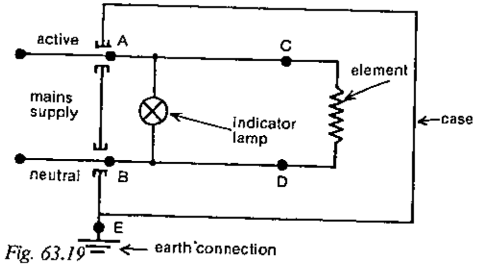


Fig. 63.19

33 Each circuit in Figure 63.20 shows a NOR gate, a light sensitive resistor S of resistance R_S and an ordinary resistor X of resistance R_X.

When S is illuminated (indicated by a squiggly arrow), $R_S < R_X/100$.

When S is not illuminated (no squiggly arrow), $R_S > 100R_X$.

Which one or more of the circuits in Figure 63.20 would give a high output?

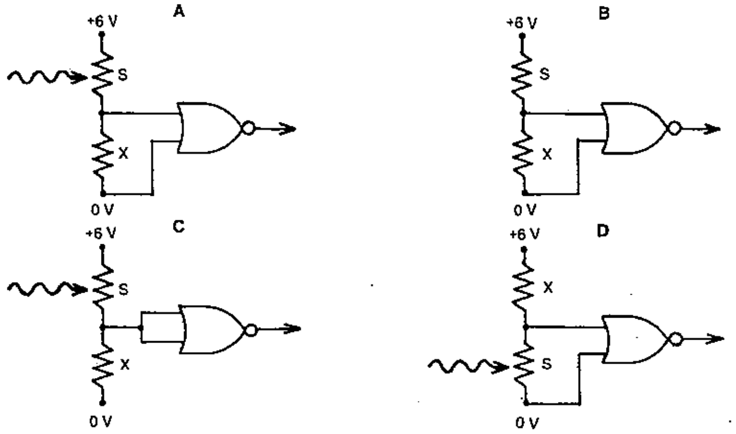


Fig. 63.20

34 Of the logic gates, NOT, NOR, OR, AND and NAND, what single gate is equivalent to
(*a*) three NOT gates in succession?
(*b*) a NAND gate with its two inputs joined?
(*c*) two separate NOT gates with their outputs fed into a NAND gate?
(*d*) an OR gate followed by a NOT gate?

35 Figure 63.21 shows two pulse-producing circuits connected together to make an astable. It is required that indicator 1 should stay on longer than indicator 2. Which one or more of the sets of values for R_1, C_1, R_2 and C_2 given in Table 63.2 would achieve this purpose?

Table 63.2

Set	R_1 (kΩ)	C_1 (μF)	R_2 (kΩ)	C_2 (μF)
A	0	10	10	0
B	10	10	10	10
C	10	0	10	0
D	10	0	0	10
E	10	0	0	0

36 If the input to the circuit in Figure 63.22 is a square wave of frequency f, the circuit will act as
A a differentiating circuit, if $RC \gg 1/f$.
B a differentiating circuit, if $RC \ll 1/f$.
C an integrating circuit, if $RC \gg 1/f$.
D an integrating circuit, if $RC \ll 1/f$.

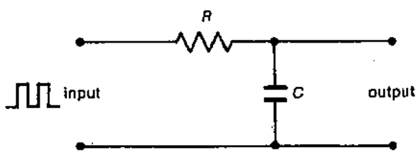


Fig. 63.22

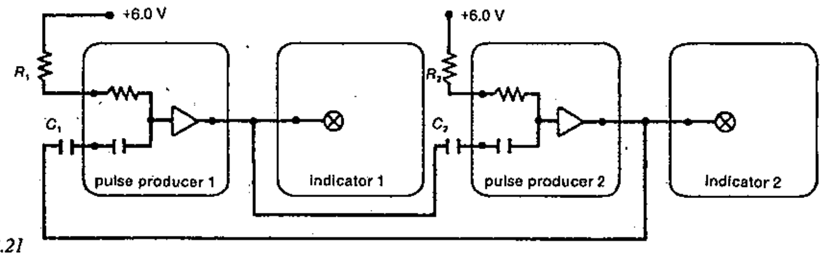


Fig. 63.21

PART 8

ATOMIC AND NUCLEAR PHYSICS

SET 64 STRUCTURE OF THE ATOM

Topic	*Problems*
Estimation of the size of molecules and atoms	1-4
Atomic size and x-ray crystallography (see Set 44, Bragg diffraction)	
Rutherford scattering experiments	5-15
Bohr model of the hydrogen atom, atomic energy levels and spectra, wave mechanical model of the hydrogen atom (see Set 66)	

Where necessary in this set, use

$1\ g = 6.02 \times 10^{23}\ u$ (unified atomic mass unit)

Avogadro constant, $L = 6.02 \times 10^{23}\ mol^{-1}$

mass of an alpha particle $= 6.65 \times 10^{-27}\ kg$

Coulomb's law constant in a vacuum or air,

$k = 1/4\pi\epsilon_0$
$= 9.0 \times 10^{9}\ N\ m^2\ C^{-2}$

permittivity of a vacuum or air,

$1\ e\ \epsilon_0 = 8.85 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$
$= 1.60 \times 10^{-19}\ C$

$1\ eV = 1.60 \times 10^{-19}\ J$

1 Your friend maintains that there are about 200 molecules in a teaspoonful of water. 1 teaspoonful $\approx$ 5400 mm^3. Based on his assertion, what would be
(*a*) the volume of each molecule?
(*b*) the thickness of a molecule if it was regarded as a cube?
(*c*) the area it would cover if it was spread out as a monolayer, i.e. a layer one molecule thick?
(*d*) What evidence have you that your friend has underestimated the number of molecules?

2 100 mm^3 of gold can be beaten into gold leaf so fine that it covers an area 5 m by 1 m.
(*a*) What is the average thickness of the foil in metre?
(*b*) What information about a gold atom can be deduced from your answer to (*a*)?

3 Suppose you are provided with a 0.50 per cent solution of oleic acid in alcohol.
(*a*) What volume of oleic acid is present in 1.0 cm^3 of the solution?
You calibrate a dropper and find that 50 drops occupy 1.0 cm^3. Using the dropper you find that one drop of the solution on warm water forms an almost circular layer with an average diameter of 30 cm. (All the alcohol is either absorbed or evaporated by the warm water.)
(*b*) What is the volume of the layer of oleic acid on the surface of the water?
(*c*) What is the area of the layer?
(*d*) What is the thickness of the layer?
(*e*) Does this thickness represent a maximum or minimum possible dimension for an oleic acid molecule?

4 Consider a large cube of aluminium of side 1.0 m. Aluminium has a density of $2.7 \times 10^3\ kg\ m^{-3}$ and an atomic mass of 27 u.
(*a*) What is the mass of the cube in kg?
(*b*) What is its mass in u?
(*c*) How many aluminium atoms are there in the cube?
(*d*) What is the effective volume of each atom (in m^3)?
(*e*) If each is regarded as a small cube, what is the length of the side of an atom of aluminium?

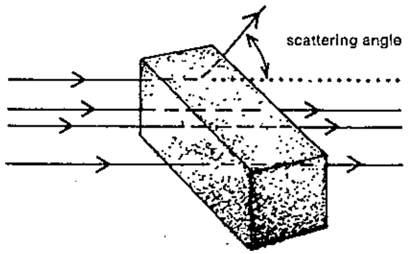


Fig. 64.1

5 Suppose that you have two bales of hay A and B. In one a number of hard steel *spheres* have been inserted while in the other hard steel *cubes* of about the same size have been inserted. Twenty bullets were fired, as indicated in Figure 64.1, through each bale from random positions but in the same direction and the angles through which they were deflected is recorded in Table 64.1.

Table 64.1

Bale	Measured scattering angles (deg)									
A	0	114	49	0	0	139	0	0	0	142
	0	0	87	0	103	0	0	93	0	0
B	0	0	0	0	47	0	0	0	132	0
	0	48	0	0	0	133	0	0	0	47

(*a*) Would you say bale A or B contains the spheres?
(*b*) On the basis of the results in Table 64.1 which bale contains the larger number of steel inserts (spheres or cubes)?
(*c*) Is there any reason for lacking confidence in your answer to (*b*)?
(*d*) Assuming no insert is behind any other, approximately what percentage of the front surface of bale A is "hiding" a sphere?

6 Figure 64.2 shows a path of an alpha particle in a Rutherford-type scattering experiment. *b* represents the *aiming error* or distance of closest approach to the centre of the nucleus if the alpha particle was undeflected, and θ the *scattering angle* or the angle between the directions of the alpha particle path before and after deflection.

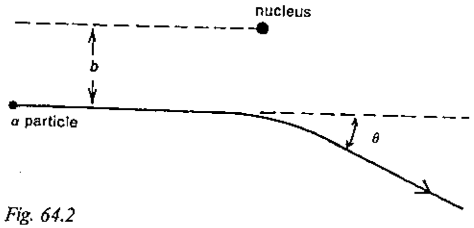


Fig. 64.2

(*a*) When $b = 0$, what is the value of θ?
(*b*) When *b* is very large, what value does θ approach?
(*c*) As *b* increases, what happens to the value of θ?

7 Consider the alpha particle scattering shown in Figure 64.2. Suppose that for a nucleus carrying a charge of $+Z_1$ and alpha particle kinetic energy E_1, θ_1 is the scattering angle corresponding to an aiming error of b_1. Is the new value of θ greater than, equal to or less than θ_1,
(*a*) if we use an aiming error b_2, which is smaller than b_1, without changing the nucleus or the alpha particle energy?
(*b*) if, *instead*, we use a new alpha particle energy E_2, which is greater than E_1, without changing the nucleus or aiming error?
(*c*) if, *instead*, we substitute a new nucleus carrying a charge of $+Z_2$, which is greater than $+Z_1$, without changing the aiming error or alpha particle energy?

8 It was found in a certain scattering experiment using gold foil that 10 alpha particles in 10^4 were scattered through more than 4°.
(*a*) If we *doubled* the thickness of the foil,
(*i*) approximately what number of alpha particles in 10^4 would you predict to be deflected by more than 4°?
(*ii*) would you be safe in assuming that each of these deflections was caused by a single interaction?
(*b*) Could you safely predict the approximate number that would be scattered by more than 4°, if we used a foil
(*i*) three times as thick as the original foil?
(*ii*) ten thousand times as thick as the original foil?

9 For a science exhibition in your school hall you propose to set up a scaled model of a piece of gold foil using tennis balls hanging from the ceiling above head height to represent the nuclei. Approximately how far apart would the tennis balls have to be placed? Take the radii of a gold nucleus and a gold atom to be about 7×10^{-15} m and 1×10^{-10} m respectively.

10 Gold foil of thickness 100 nm contains about 400 layers of atoms. The nuclear radius of gold is 7×10^{-15} m. Give your answers to the following questions as orders of magnitude.
(*a*) Determine for incoming alpha particles approaching the foil at normal incidence the fraction of the *area* of the foil actually blocked out by gold nuclei. Assume that no nucleus is screened by any other.
(*b*) What fraction of the *volume* of the foil is occupied by nuclei?

11 In an experiment typical of those carried out under Rutherford by Geiger and Marsden in 1908, an alpha particle carrying a charge of +2 elementary charges and with a speed of $1.6 \times 10^7\ m\ s^{-1}$ makes a head-on approach to a gold nucleus, which has a charge of +79 elementary charges.
(*a*) What is the kinetic energy of the alpha particle while still a relatively long way off?
(*b*) What is the sum of the potential and kinetic energies of the alpha particle and gold nucleus when they are at their minimum separation *d*?
(*c*) What is the instantaneous speed of the alpha particle at separation *d*?
(*d*) If we assume the gold nucleus does not move as a result of the interaction, what is the total electrical potential energy of the alpha particle and gold nucleus at separation *d*?
(*e*) Determine the value of *d*.

12 Suppose an alpha particle making a head-on collision with a chromium nucleus achieves a minimum separation of 4×10^{-14} m. What would be the minimum separation if an alpha particle of the same energy was making a head-on approach to a cadmium nucleus? The chromium and cadmium nuclei carry charges of +24 and +48 elementary charges respectively.

13 A positively charged particle H of charge *q* and mass *m*, approaches another charged particle K, of charge *nq*, from a large distance. K may be considered to be fixed in space during the interaction. The zero of potential energy of the system is taken to be the potential energy when the particles are at effectively infinite separation, at which stage the kinetic energy of H is E_0. S, T and U are successive points on the trajectory of H. S and U are both distant *r* from K. See Figure 64.3.

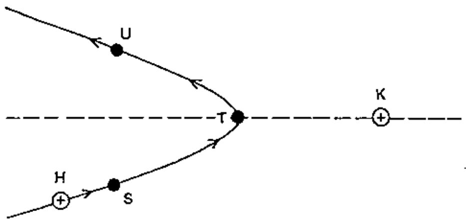


Fig. 64.3

(*a*) In terms of the given quantities q, n, r, E_0 and ϵ_0 (or the Coulomb law constant k), write an expression for the potential energy of particle H at point S.
(*b*) Write an expression for the kinetic energy of particle H at point S.
(*c*) The interaction is perfectly elastic. How does the potential energy of the particle H at the point U compare with its potential energy at the point S?
(*d*) It can be seen that momentum is not conserved in this interaction, so that some external force must be acting on the system. What is this force?
(*e*) A special case of this interaction is that of a head-on collision between the two particles. For such a collision, and with K still held stationary, write an expression for r_0, the least distance apart of H and K during the interaction. Express r_0 in terms of q, E_0, n and ϵ_0 or k.

14 A radioactive isotope of polonium gives off alpha particles with an energy of 5.3 MeV.
(*a*) What is the kinetic energy of an alpha particle in joule?
In passing through a thin gold foil one of these alpha particles is deflected slightly as it passes a gold nucleus, which carries a charge of +97 elementary charges. Their minimum separation is 6.2×10^{-14} m. At this minimum separation
(*b*) what is the electrical potential energy of the alpha particle and the nucleus?
(*c*) what is the kinetic energy of the alpha particle?
(*d*) what is its speed?

15 Consider an alpha particle 2×10^{-13} m from a gold nucleus carrying a charge of +79 elementary charges.
(*a*) Determine the ratio of the electrical to the gravitational force between the two particles. They differ. You might be surprised by how much. Mass of a gold nucleus = 3.3×10^{-25} kg; gravitation constant, $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.
(*b*) Would the ratio have been larger, smaller or the same, if the separation had been 100 times greater?

SET 65 STRUCTURE OF THE NUCLEUS AND RADIOACTIVITY

Topic	*Problems*
Composition of the nucleus, atomic number, mass number, neutron number, notation for representing nuclides	1–5
Isotopes and relative atomic mass	6–7
Nuclear reactions, conservation of mass number and atomic number (or charge)	8–11
Binding energy of the nucleus, binding energy per nucleon, mass deficit or defect	12–19
Radioactive nuclei, alpha, beta$^-$ and gamma decay processes, relative number of protons and neutrons in stable nuclei	20–25
Beta$^+$ decay process	26–27
The neutrino	28
Activity of a radioactive isotope, rate of radioactive decay, simple problems on half-life	29–33
More difficult questions on radioactive source activity, half-life and decay constant	34–37
Radioactive decay series	38–39
Nuclear fission, reactors	40–43
Nuclear fusion	44–45
Radiation and health, radiation absorbed dose, dose equivalent, quality factor, specific gamma ray constant, maximum exposure limits	46–50

Where necessary in this set, use

$1\ e = 1.60 \times 10^{-19}$ C
$1\ eV = 1.60 \times 10^{-19}$ J
Avogadro constant, $L = 6.02 \times 10^{23}$ mol^{-1}
$1\ g = 6.02 \times 10^{23}$ u
(unified atomic mass unit)
$1\ u = 931$ MeV
mass of a proton = 1.0073 u
mass of a neutron = 1.0087 u
speed of electromagnetic radiation in a vacuum or air,
$c = 3.0 \times 10^8$ m s^{-1}

1 One form or isotope of nitrogen has a nucleus represented as $^{15}_{7}N$. Write down
(*a*) its atomic or proton number Z;
(*b*) its mass or nucleon number A;
(*c*) its neutron number.

2 For the isotope of carbon designated by $^{14}_{6}C$, what is
(*a*) the total number of protons and neutrons in the nucleus of an atom of this isotope?
(*b*) the number of protons in such a nucleus?
(*c*) the number of neutrons in the nucleus?
(*d*) the number of positive elementary charges on the nucleus?
(*e*) the number of negative extra-nuclear electrons in an atom of this isotope?

3 The nuclide of carbon referred to in Problem 2 was represented $^{14}_{6}C$. How would you represent the nuclide of lead (symbol Pb) which contains 125 neutrons and carries a charge of 82 positive elementary charges?

4 Nickel has a nuclide represented as $^{64}_{28}Ni$.
(*a*) Which two of the three items, *atomic number*, *mass number* and *chemical symbol*, are redundant?
(*b*) How then could this nuclide be represented more simply than $^{64}_{28}Ni$?

5 Nuclei and subatomic particles can be represented in the form $^{A}_{Z}X$. By replacing Z and A in the following, write down the representation used in nuclear reactions for
(*a*) a neutron $^{A}_{Z}n$;
(*b*) a proton $^{A}_{Z}H$;
(*c*) an alpha particle $^{A}_{Z}He$;
(*d*) a negative electron (also known as a β^- particle) $^{A}_{Z}e$;
(*e*) a positive electron (also known as a positron or β^+ particle) $^{A}_{Z}e$.

6 The element gallium has two naturally occurring isotopes. It consists of 60.40 per cent $^{69}_{31}Ga$ and 39.60 per cent $^{71}_{31}Ga$ with relative isotopic masses of 68.93 and 70.92 respectively. Determine the relative atomic mass of gallium.

7 Copper consists of two naturally occurring isotopes, $^{63}_{29}Cu$ and $^{65}_{29}Cu$, which have relative isotopic masses of 62.9296 and 64.9278 respectively. If the relative atomic mass of copper is 63.545, what percentage of naturally occurring copper is the lighter isotope?

8 The first artificial radioactive substance was produced by Irene Joliot-Curie (daughter of Marie Curie, the discoverer of radium) in 1934. She bombarded an isotope of aluminium with alpha particles producing an isotope of phosphorus and neutrons:

$$_{13}Al + ^{4}_{2}He \rightarrow ^{30}P + ^{1}_{0}n$$

(*a*) What is the missing mass number of the aluminium isotope?
(*b*) What must be the atomic number of phosphorus?

9 The nuclear reaction referred to in Problem 8 can be represented symbolically

$$^{27}_{13}Al(\alpha, n)^{30}_{15}P$$

(*a*) The reaction in the atmosphere, which makes radiocarbon dating of former living organisms possible, is represented symbolically

$$^{14}_{7}N(n, p)^{14}_{6}C$$

Write out this reaction as a full nuclear equation.
(*b*) A nuclear reaction that takes place in some neutron detectors is

$$^{10}_{5}B + ^{1}_{0}n \rightarrow ^{4}_{2}He + ^{7}_{3}Li$$

Write this reaction in the shorter symbolic form.

10 Rewrite and complete the following nuclear equations:
(*a*) The first observed splitting of a nucleus, detected by Ernest Rutherford in 1919:

$$^{14}_{7}N + ^{4}_{2}He \rightarrow ^{17}_{8}O +$$

(*b*) The first nuclear reaction using artificially accelerated particles, achieved by John Cockroft and Ernest Walton in 1932:

$$^{7}_{3}Li + ^{1}_{1}H \rightarrow ^{4}_{2}He +$$

(*c*) A reaction carried out by James Chadwick in 1932 and leading to the discovery of the neutron:

$$^{9}_{4}Be + \qquad \rightarrow ^{12}_{6}C + ^{1}_{0}n$$

11 What else would be emitted when a slow neutron enters a $^{235}_{92}U$ nucleus and causes a fission reaction in which a $^{90}_{36}Kr$ nucleus and a $^{144}_{56}Ba$ nucleus are produced?

12 If, in a series of nuclear reactions, 1 g of mass was converted into energy,
(*a*) how much energy (in joule) would be produced?
(*b*) for how many years could this energy keep a 100 W light globe lit, assuming the globe did not burn out?

13 Mass is a form of energy. Using each answer when working the next part of the question, deduce the equivalent of a mass of 1.00 u in
(*a*) g;
(*b*) kg;
(*c*) J;
(*d*) eV;
(*e*) MeV.

14 Is the mass of a nucleus *less than*, *equal to* or *greater than* the sum of the masses of its constituent nucleons? Why is it so?

15 Deduce an expression for the *binding energy* B of a nucleus in terms of
Z, its atomic number,
A, its mass number
m_p, the mass of a proton,
m_n, the mass of a neutron,
M, the mass of the nucleus, and
c, the speed of electromagnetic radiation in a vacuum.

16 The mass of the helium nucleus $^{4}_{2}He$ is 4.0015 u. Using the proton and neutron masses given at the beginning of this set, determine for this nucleus
(*a*) its mass defect or deficit in u;
(*b*) its binding energy in MeV;
(*c*) its binding energy per nucleon in MeV.

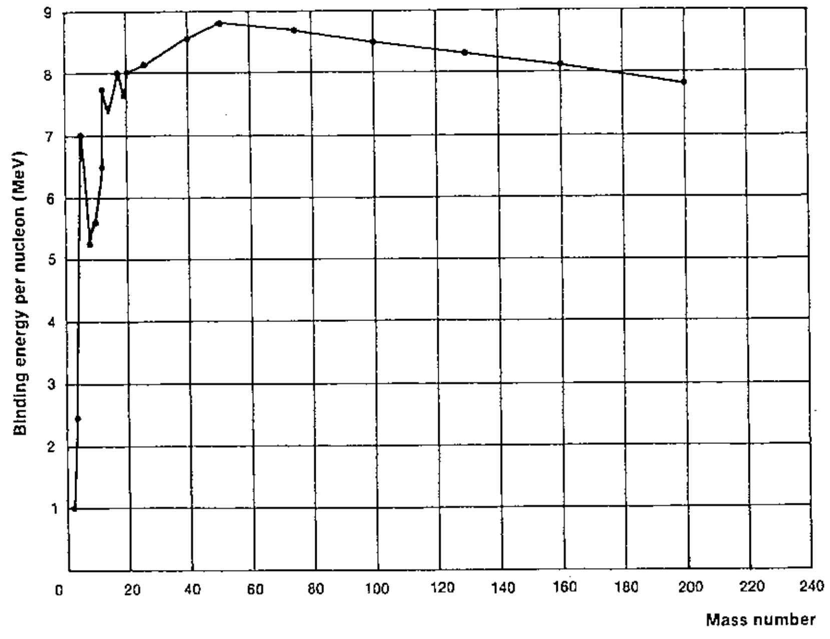


Fig. 65.1

17 Figure 65.1 shows how the binding energy per nucleon is related to the total number of nucleons in the nucleus. Consider the two nuclei ${}^{4}_{2}He$ and ${}^{57}_{26}Fe$.

(*a*) Which of the two nuclei would have the higher
 (*i*) binding energy per nucleon?
 (*ii*) total binding energy?

(*b*) Are the nucleons more strongly bound together in the ${}^{4}_{2}He$ or the ${}^{57}_{26}Fe$ nucleus?

(*c*) Both are stable nuclei. *Using only the evidence in Figure 65.1*, which do you think is the more stable? (There are other factors which influence nuclear stability.)

18 Consider the graph in Figure 65.1.

(*a*) If you drew a graph of *mass energy per nucleon* against *mass number*, would the graph have a minimum or maximum point around a mass number of about 50 to 60?

(*b*) Which of the following hypothetical nuclear transformations would generate energy?
 A Putting together two nuclei of mass number 50.
 B Breaking a nucleus of mass number 100 into two approximately equal parts.
 C Putting together two nuclei of mass number 10.
 D Breaking a nucleus of mass number 20 into two approximately equal parts.

19 If hydrogen gas is bombarded by neutrons, a neutron can be captured forming a deuteron ${}^{2}_{1}H$ with the emission of a gamma ray.

$$ {}^{1}_{0}n + {}^{1}_{1}H \rightarrow {}^{2}_{1}H + \gamma $$

The rest mass of a deuteron is 2.0136 u.

(*a*) What is the total rest mass of the neutron and proton in u?

(*b*) By how much does the total rest mass of the neutron and proton exceed that of the deuteron
 (*i*) in u;
 (*ii*) in MeV?

(*c*) If we regard the initial kinetic energy of the proton and the final kinetic energy of the deuteron as negligible, by how much does the energy of the gamma ray exceed the initial kinetic energy of the neutron?

(*d*) Actually the neutron requires very little energy to be captured by the proton. What would you expect the energy of the most common gamma ray to be in this neutron-capture experiment?

(*e*) A deuteron bombarded by gamma rays can be broken up into a proton and a neutron.

$$ \gamma + {}^{2}_{1}H \rightarrow {}^{1}_{1}H + {}^{1}_{0}n $$

What would be the minimum energy gamma ray required?

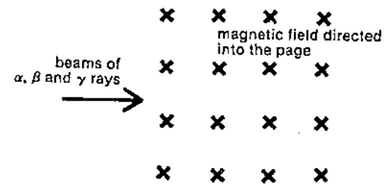


Fig. 65.2

20 Beams of alpha, beta and gamma rays, all of about the same energy, pass into a magnetic field as shown in Figure 65.2.

(*a*) Indicate whether each type will be deflected upwards, downwards or remain undeflected.

(*b*) Which type will undergo the largest deflection?

21 When passing through matter, alpha particles, beta particles and gamma rays of similar energy differ significantly in *ionising power* and *penetrating ability*. Ionising power is indicated by the number of ions produced per unit distance travelled by the beam, while penetrating power is the distance travelled by the beam before losing all its energy.

(*a*) Arrange the three types in decreasing order (i.e. starting with the highest) of
 (*i*) ionising power;
 (*ii*) penetrating ability.

(*b*) Why does the one with the highest ionising power have the lowest penetrating ability?

22 Using the Appendix and stating both atomic and mass numbers, write down the product nucleus when

(*a*) the polonium isotope ${}^{218}_{84}Po$ decays by alpha emission;

(*b*) the oxygen isotope ${}^{19}_{8}O$ decays by beta$^-$ emission;

(*c*) the bismuth isotope ${}^{210}_{83}Bi$ decays by gamma emission.

23 Rewrite and complete the following nuclear equations indicating radioactive decay. You may need to refer to the Appendix.

(*a*) ${}^{14}_{6}C \rightarrow {}^{14}_{7}N +$

(*b*) ${}^{217}_{85}At \rightarrow {}^{213}_{83}Bi +$

(*c*) ${}^{34}_{17}Cl \rightarrow \quad + \gamma$

(*d*) ${}^{228}_{90}Th \rightarrow {}^{4}_{2}He +$

(*e*) $\quad \rightarrow {}^{45}_{21}Sc + {}^{0}_{-1}e$

24 Using the Appendix, rewrite and complete the following radioactive decay equations, including all atomic and mass numbers:

(*a*) ${}^{241}Am \rightarrow {}^{4}He +$

(*b*) ${}^{90}Sr \rightarrow {}^{0}_{-1}e +$

(*c*) ${}^{137}Cs \rightarrow \gamma +$

(*d*) ${}^{198}Au \rightarrow {}^{198}Hg +$

25 If you plot all the stable isotopes on a graph of *neutron number N* against *proton* (or *atomic*) *number Z*, you will find that they fit roughly in the shaded region shown in Figure 65.3. Note the relation of this region to the line $N = Z$.

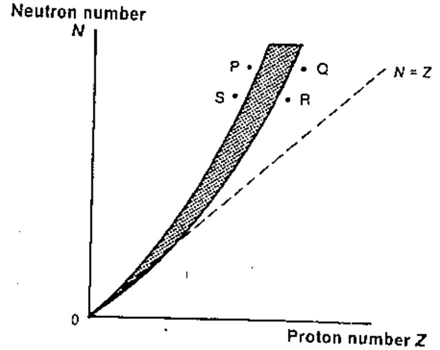


Fig. 65.3

(*a*) For small nuclei, an equal number of protons and neutrons gives rise to a stable nucleus, while for larger stable nuclei there is a *neutron excess*, $N - Z$. Can you suggest a reason why the additional neutrons make one of these larger nuclei more stable?

(*b*) Would an unstable isotope be more likely to become stable by emitting an alpha particle or a beta$^-$ particle, if it was positioned on the graph in Figure 65.3 at
 (*i*) point P?
 (*ii*) point Q?

26 Artificially produced radioactive nuclei sometimes decay by beta$^+$ emission, i.e. the emission of a positron or positive electron. Complete the following nuclear equations involving beta$^+$ decay, including all mass and atomic numbers. You may need to refer to the Appendix.

(*a*) ${}^{10}_{6}C \rightarrow \quad + {}^{0}_{1}e$

(*b*) ${}^{38}Ca \rightarrow \quad + {}^{0}_{1}e$

(*c*) $\quad \rightarrow {}^{101}Ag + {}^{0}_{1}e$

27 The shaded region in Figure 65.3 shows the approximate location of *stable* nuclei for a plot of neutron number against proton number. Would an unstable isotope be more likely to become stable by emitting a beta$^-$ or beta$^+$ particle, if it was positioned on the graph at

(*a*) point R?

(*b*) point S?

28 When a radioactive nucleus decays by beta$^-$ emission, the change is effectively

$$ {}^{1}_{0}n \rightarrow {}^{1}_{1}H + {}^{0}_{-1}e + \bar{\nu} $$

where $\bar{\nu}$ is an antineutrino.

When decay is by beta$^+$ emission the change is effectively

$$ {}^{1}_{1}H \rightarrow {}^{1}_{0}n + {}^{0}_{1}e + \nu $$

where ν is a neutrino.

(*a*) Using the neutron and proton masses at the beginning of this set and given that the rest masses of an electron and neutrino (or antineutrino) are 0.0005 u and zero respectively, decide whether beta$^-$ or beta$^+$ emission is likely to take place without an energy input.
(*b*) Should an energy term be added to the *right*-hand side of the *first* equation above and the *left*-hand side of the *second*, or vice versa?
(*c*) Why does beta$^+$ emission occur only with *artificially* produced rather than *naturally* occurring radioactive isotopes?

29 Source activity of radioisotopes used to be measured in *curie* (Ci), but is now measured in the SI unit of activity, the *becquerel* (Bq). 1 Ci (actually the activity of 1 g of radium-226) = 3.7×10^{10} disintegrations s^{-1}. 1 Bq = 1 disintegration s^{-1}. What is the activity in kBq of a school laboratory source marked "2.0 μCi"?

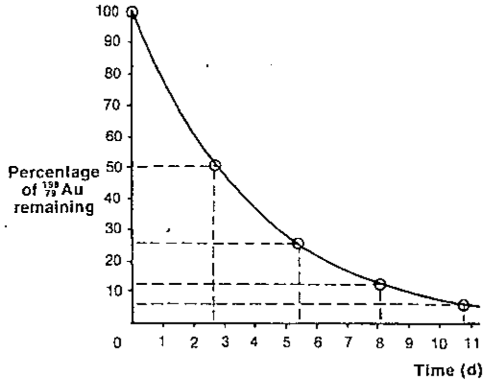


Fig. 65.4

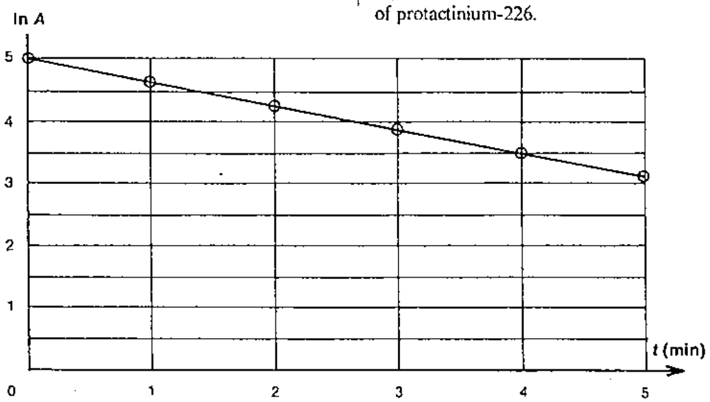


Fig. 65.5

30 Figure 65.4 shows a graph of the percentage remaining of a radioisotope of gold-198 used in the treatment of oral cancers as a function of time expressed in days. What is the half-life of gold-198?

31 Iodine-131, used in destroying malignant tumours of the thyroid, has a half-life of 8.07 d. If the initial activity of the radioisotope was 4.0 MBq, what would be its activity after
(*a*) 8.07 d?
(*b*) 16.14 d?
(*c*) 32.28 d?

32 A sample of technetium-99 is to be used in a diagnostic bone scan at 8.00 p.m. when its activity is 24 MBq. What was its activity when delivered to the diagnostic unit earlier in the day at 8.00 a.m.? ^{99}Tc has a half-life of 6.0 h.

33 Iridium-192 is used in the treatment of early cancer of the breast. It has a half-life of 74 d. If initially there is 3.6 mg of this isotope present, what time will elapse before it has been reduced to 0.90 mg?

34 Platinum-175 has a half-life of 2.1 s.
(*a*) What is its decay constant?
(*b*) If the activity of a platinum-175 source is 9.9×10^{15} Bq, how many platinum atoms are present?

35 If a treatment of cancer of the cervix employs a caesium-137 source of activity 4.0 MBq, what mass of caesium-137 is being used? Caesium-137 has a half-life of 30 y. The relative atomic mass of caesium is 133.

36 By means of a Geiger counter the activity *A* of a protactinium-226 source was measured at 1 min intervals for 5 min. A graph of ln *A* against time *t* was then drawn and is shown in Figure 65.5.
From the graph deduce values for
(*a*) the decay constant (in s^{-1}), and
(*b*) the half-life
of protactinium-226.

37 Carbon-14, with a half-life of 5730 y, is produced in the upper atmosphere by cosmic ray bombardment of carbon atoms in carbon dioxide. As a result, every living thing which is interacting with the carbon cycle in nature has within its carbon atoms a definite proportion of carbon-14, which produces an activity of 0.26 Bq per gram of *carbon in living materials*. When the organism dies, this interaction with the atmosphere ceases and the proportion of carbon-14 decreases. If 1.0 g of carbon from an ancient manuscript was found to have an activity of 0.19 Bq, approximately how old is the manuscript?

38 $^{238}_{92}$U is one of the radioactive nuclides that forms part of the "uranium decay series". When it decays, the nuclide it forms is also radioactive, and so is every nuclide until stable $^{206}_{82}$Pb is reached. For one atom of uranium-238 how many
(*a*) alpha decays, and
(*b*) beta$^-$ decays
take place in this chain of decay processes?

39 As one atom of $^{232}_{90}$Th decays and follows—over an extended period of time—the natural "thorium decay series", there are successive emissions of one alpha, two beta$^-$ and three alpha particles. Specify fully the nuclide that has been produced at this stage. You may need to refer to the Appendix.

40 Taking 200 MeV as the energy released during the fission of a single uranium nucleus, determine the approximate number of fission processes taking place per second in the UK Hartlepool Reactor operating at 666 MW.

41 A uranium-235 nucleus can fission in a number of ways. One case, in which lanthanum and bromine are produced, is represented as follows

$$^{235}_{92}\text{U} + ^{1}_{0}\text{n} \rightarrow ^{148}_{57}\text{La} + ^{85}_{35}\text{Br} + 3^{1}_{0}\text{n}$$

Calculate the energy (in joule) released in this fission process. Mass of uranium-235 = 235.124 u; mass of lanthanum-148 = 147.961 u; mass of bromine-85 = 84.938 u.

42 This question considers the major components of a fission reactor.
Answer key:
A It should be a good conductor.
B The nuclei in this material should be good absorbers of neutrons.
C The nuclei in this material should not be good absorbers of neutrons.
D It should be a fluid, i.e. a liquid or gas.
E It should be capable of undergoing a fission reaction when it absorbs a slow neutron.
F The atoms in this material should have small rather than large nuclei.
Which of the characteristics given in the answer key above are important for
(*a*) the fuel rods (two properties)?
(*b*) the control rods (one property)?
(*c*) the moderator (two properties)?
(*d*) the coolant (two properties)?

43 Plutonium-239, produced from uranium-238 in fast breeder reactors, is—like uranium-235—capable of undergoing a neutron-induced fission reaction, two possible simultaneous products being iodine-136 and niobium-100. Write a nuclear equation for this reaction. Use the Appendix.

44 Read or reread Problem 18 in this set. In (*b*) of that problem select from the nuclear transformations A to D the one which most closely approximates
(*a*) a fission process as used in nuclear reactors;
(*b*) a fusion process as occurs in the Sun.

45 A series of fusion reactions, known as the PP I reactions, which take place in the Sun, have the net effect of converting hydrogen into helium as follows:

$$4^{1}_{1}\text{H} \rightarrow ^{4}_{2}\text{He} + 2^{0}_{1}\text{e} + 2\nu + 2\gamma + 27 \text{ MeV}$$

This energy output seems small when compared with that of a typical fission reaction:

$$^{1}_{0}\text{n} + ^{235}_{92}\text{U} \rightarrow ^{92}_{36}\text{Kr} + ^{141}_{56}\text{Ba} + 3^{1}_{0}\text{n} + 200 \text{ MeV}$$

Determine the following ratios:

(*a*) $\dfrac{\text{energy emitted per PP I fusion reaction}}{\text{energy emitted per } ^{235}_{92}\text{U fission reaction}}$;

(*b*) $\dfrac{\text{energy emitted per kg of hydrogen in the fusion reaction}}{\text{energy emitted per kg of uranium in the fission reaction}}$.

46 When radiation is absorbed by matter, e.g. you or me, the *radiation absorbed dose* is measured in terms of the amount of energy absorbed per unit mass. The unit of radiation absorbed dose has for many years been the *rad*, which is the amount of radiation required for 10^{-5} J of energy to be absorbed per gram of tissue. The SI unit introduced in 1975 is the *gray* (Gy), which is the amount of radiation required for 1 J to be absorbed by 1 kg. If the rad is still in use, express 1 rad in terms of the gray.

47 The biological effect of radiation in animal or plant tissue is quantified as the *dose equivalent* and depends not only on the *radiation absorbed dose* but also on the *type* of radiation received. Each type of radiation has its own *quality factor* or *relative biological effectiveness*, e.g. the quality factor of gamma rays or x-rays is 1, while that of neutrons is 10.
Dose equivalent =
(radiation absorbed dose) $\times$ (quality factor)
Under the older system of units with radiation absorbed dose measured in rad, the dose equivalent was expressed in *rem*, whereas in the SI system, with the radiation absorbed dose in gray, the dose equivalent is given in *sievert* (Sv).
(*a*) 1 Gy = 1 J kg^{-1}. Write a similar statement for 1 Sv.

(b) If the rem is still in use, write an expression for 1 Sv in terms of rem.
(c) What would be the dose equivalent in Sv (or rem, if still in use) received by a plant in a laboratory, if the radiation absorbed dose was 0.12 Gy (or 12 rad) and the radiation used was
 (i) gamma rays?
 (ii) neutrons?

48 Since the dose of radiation received depends on the nature of the source, the type of radiation and its energy, one can determine for any gamma ray source a *specific gamma ray constant*, which is the radiation absorbed dose per unit time per unit distance from the source of unit activity. For example, under the older system of units, the specific gamma ray constant for cobalt-60 was 1.3 rad m^2 h^{-1} C_i^{-1}. (1 Ci = 1 curie)
(a) Write an expression for the specific gamma ray constant S, in terms of the radiation absorbed dose R in time t at a distance d from a source of activity A.
(b) Express the specific gamma ray constant for cobalt-60 in SI units, i.e. in Gy m^2 s^{-1} Bq^{-1}. (Even though this unit can be correctly written as Gy m^2, the more meaningful Gy m^2 s^{-1} Bq^{-1} tends to be used.) 1 Ci $= 3.7 \times 10^{10}$ Bq.
(c) What dose rate in μGy s^{-1} would be received by a medical package being sterilised by gamma radiation from a cobalt-60 source of activity 41 MBq situated 0.20 m from the package?

49 A physics teacher, known to his students as "Barney", spends about 8 h per weekday at an average distance of 5 m from three 74 kBq radium-226 sources used for radioactivity experiments. Radium-226 has a specific gamma ray constant of 6.2×10^{-17} Gy m^2 s^{-1} Bq^{-1}.
(a) For a 42-week school year, determine for Barney
 (i) the radiation absorbed dose in μGy y^{-1} (or mrad y^{-1}, if the rad is still in use);
 (ii) the dose equivalent in μSv y^{-1} (or mrem y^{-1}, if the rem is still in use).
(b) What percentage does this exposure represent of the 5 mSv y^{-1} (or 0.5 rem y^{-1}) recommended maximum exposure?

50 A Curietron machine used at the bed of a patient receiving radiotherapy for carcinoma of the cervix enables the 3.7 GBq (G = giga = 10^9) caesium-137 source to be withdrawn from the patient and retracted into a shielded section of the machine when medical personnel need to enter the patient's room. Determine whether it is necessary for a nurse to retract the source while speaking to the patient from the door of the patient's room—she is then 2.5 m from the source—if this occupies about 20 min during an eight-hour daily shift, which she works 5 days per week. The maximum occupational exposure limit for a nurse in a radiotherapy unit is 1 mSv $week^{-1}$ (or 0.1 rem $week^{-1}$). Specific gamma ray constant for caesium-137 is 2.4×10^{-17} Gy m^2 s^{-1} Bq^{-1}.

SET 66 ATOMIC ENERGY LEVELS, SPECTRA AND INTRODUCTORY WAVE MECHANICS

Topic	Problems
Frank-Hertz experiment, atomic energy levels, excitation and ionisation energies	1–4
Emission spectra and atomic energy levels	5–10
Absorption spectra and atomic energy levels	11–13
Bohr model of the hydrogen atom	14–19
Extension of the Bohr model to other atoms	20–21
de Broglie matter wavelength, wave properties of particles	22–27
Stable states of a particle in a "one-dimensional box"	28–30
Simple wave mechanical picture of a hydrogen atom	31–32

Where necessary in this set, use
speed of electromagnetic radiation in a vacuum or air,
$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$
Planck's constant, $h = 6.63 \times 10^{-34}$ J s
$$= 4.14 \times 10^{-15} \text{ eV s}$$
$$1 \text{ e} = 1.60 \times 10^{-19} \text{ C}$$
$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$$
Coulomb's law constant in a vacuum or air,
$$k = 1/4\pi\epsilon_0$$
$$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$
permittivity of a vacuum or air,
$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$
mass of an electron $= 9.11 \times 10^{-31}$ kg

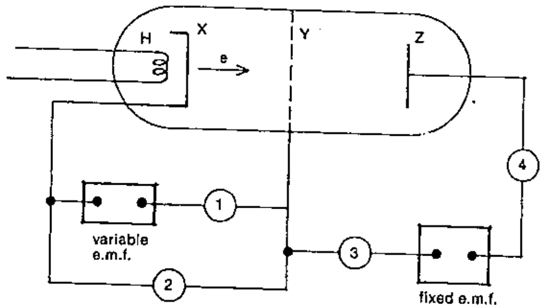


Fig. 66.1

1 The apparatus shown in Figure 66.1 comprises a tube containing a gas at low pressure. Electrode X, heated by heater H, emits electrons. A variable e.m.f. applied between X and Y, accelerates the electrons towards electrode Y which is made of open wire mesh which allows most of the electrons to pass through. A fixed e.m.f. provides a retarding potential difference between Y and Z. Measurements of electron current I flowing into Z are taken for different accelerating potential differences V. The results of these measurements are shown graphically in Figure 66.2. The experiment is somewhat similar to the experiment performed by Franck and Hertz.

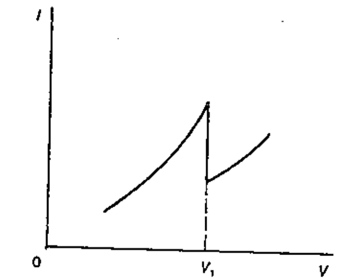


Fig. 66.2

(a) What meters of those labelled 1 to 4 in Figure 66.1 (which can be either voltmeters or milliammeters) are required in the circuit to obtain the desired measurements?
(b) What accounts for the drop in current when the e.m.f. across XY exceeds V_1?
(c) Sketch a graph of current against e.m.f. as the e.m.f. across XY is increased to a value slightly beyond $2V_1$.
(d) What is the approximate excitation energy of the gas in the tube?

2 Figure 66.3 shows a hypothetical energy level diagram for an atom of gas X. For simplicity assume there are no energy levels between levels 3 and 4. If a sample of gas X at low pressure was bombarded by a beam of electrons of known energy, what would be the possible energies of the *electrons* after passing through the gas, if the energy of the incident electrons was
(a) 6 eV?
(b) 7 eV?
(c) 9 eV?
(d) 11 eV?

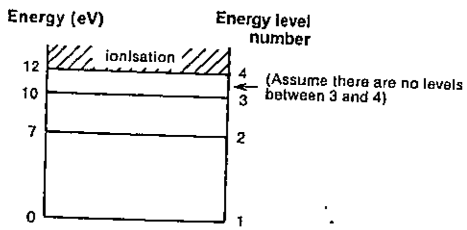


Fig. 66.3

3 For gas X whose hypothetical energy level diagram is shown in Figure 66.3, what is
(a) the smallest amount of energy an atom of gas X will accept from an electron passing through the gas?
(b) the minimum energy required to ionise an atom of gas X?
(c) the second smallest amount of energy an atom of gas X will absorb from an electron passing through the gas?
(d) the ionisation energy of the gas?
(e) the excitation energy of the gas?

4 If electrons of energy 11 eV are fired into a gas Y, electrons of energy 11, 5 and 2 eV are detected.
(a) What is the excitation energy of gas Y?

(*b*) What is the energy difference between the first and second excited states of the gas?

5 Refer again to gas X with the idealised energy level diagram shown in Figure 66.3. What *photon* energies would be observed coming from gas X if it was bombarded by electrons of energy
(*a*) 7 eV?
(*b*) 8 eV?
(*c*) 10 eV?
(*d*) 11 eV?
(*e*) 12 eV?
(*f*) 6 eV?

6 Some of the energy levels of a mercury atom are given in Figure 66.4.

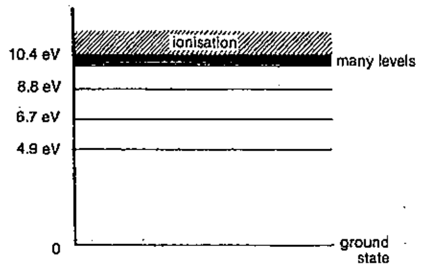


Fig. 66.4

(*a*) An electron with an energy of 5.8 eV, collides with a mercury atom. With what energy will it bounce off if the collision is
(*i*) elastic?
(*ii*) inelastic?
(*b*) A beam of electrons, each of which has an energy of 9.9 eV, passes through mercury gas.
(*i*) With what energies could electrons emerge?
(*ii*) Why is 0.1 eV a *possible* answer to (*b*) (*i*) above? (You may have omitted it.)
(*iii*) How many different photon energies could be observed coming from the gas?

7 Like many other questions in this set, this problem relates particle and wave properties of electromagnetic radiation.
(*a*) Write down an expression for photon energy E (a particle property) in terms of radiation frequency ν (a wave property).
(*b*) Deduce an expression for photon energy E (a particle property) in terms of radiation wavelength λ (a wave property).
(*c*) Your answer to (*b*) should be in the form $E = k/\lambda$, where k is a constant. If SI units are used, i.e. E is in joule and λ is in metre, what is the value of k?

(*d*) Often it is useful to be able to relate photon energy in electron volt to radiation wavelength in nanometre. Deduce the relationship for E (in eV) in terms of λ (in nm), including the numerical value of the constant. This relation is possibly worth memorising.

8 Consider again gas X with the energy level diagram shown in Figure 66.3.
(*a*) If it was excited by a beam of 8.00 eV electrons, what would be the
(*i*) wavelength (in nm), and
(*ii*) frequency
of the light emitted? Give your answers to three significant figures.
(*b*) Suppose now it was bombarded by electrons of energy 10.5 eV. Three spectral lines would be observed in the light emitted by the gas. The largest of the three wavelengths would correspond to a change in energy between two energy levels of element X. Which two levels?

9 Light of wavelength 500 nm is emitted by an atom as it falls from its third level, of energy 8.3×10^{-19} J, to its second energy level. What is the energy of the atom at this second level?

10 Refer to Figure 66.4, the energy level diagram for mercury. Determine the shortest wavelength (in nm) that will be emitted by mercury gas when it is excited.

11 Consider again gas X and its energy level diagram shown in Figure 66.3. If a sample of gas X at low pressure was bombarded by a beam of photons of known energy, what would be the possible energies of photons coming from the gas, if the incident photon energy was
(*a*) 6 eV?
(*b*) 7 eV?
(*c*) 9 eV?
(*d*) 10 eV?

12 If gas X—Figure 66.3 gives its energy level diagram—is bombarded by an ultraviolet laser of photon energy 13 eV,
(*a*) what energy electrons, and
(*b*) what energy photons
might be observed?

13 For this question refer to Figure 66.4, which shows the energy level diagram for mercury.
(*a*) How many spectral lines will be observed in the emission spectrum coming from mercury, if it is bombarded by 8.9 eV electrons?
(*b*) How many discrete dark lines will there be in the line absorption spectrum if light of all photon energies up to 8.9 eV is shone through mercury gas?
(*c*) Of the bright lines referred to in (*a*), what would be the highest frequency among them?

14 In the model of the hydrogen atom developed by Neils Bohr in 1912, an electron of mass m and carrying charge $-e$ is moving at a constant speed v in a stable circular orbit of radius r about a proton carrying charge $+e$, as indicated in Figure 66.5. Here the approximation will be made that the proton is so much more massive than the electron that it can be regarded as stationary.

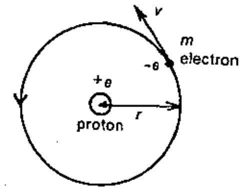


Fig. 66.5

The Coulomb law constant k, or ϵ_0 will appear with Planck's constant h in your answers to some of the questions that follow.
(*a*) Write *two* expressions for the magnitude of the electric force exerted on the electron by the proton.
(*b*) From your answer to (*a*) deduce an expression for the kinetic energy of the electron $\frac{1}{2}mv^2$ in terms of k (or ϵ_0), e and r.
(*c*) Express the electric potential energy U_e of the electron in terms of k (or ϵ_0), e and r.
(*d*) Deduce an expression for the total energy E of the electron.
(*e*) Bohr postulated that the angular momentum of the electron was quantised as whole number multiples of $h/2\pi$, where h is Planck's constant, i.e.

$$mvr = nh/2\pi,$$

where $n = 1, 2, 3, \ldots$

Use this expression and your answer to (*b*) above to deduce an expression for r in terms of n, h, k (or ϵ_0), m and e.
(*f*) Your answer to (*e*) suggests that there are only certain possible radii for the electron. Evaluate the smallest of these: it is known as the Bohr radius. Give your answer in μm.
(*g*) From your answers to (*d*) and (*e*) deduce an expression for the total energy E of the electron in terms of
(*i*) m, e, k (or ϵ_0), n and h;
(*ii*) n only, giving E in joule;
(*iii*) n only, but giving E in electron volt.

15 The expression deduced in Problem 14 gives the Bohr model predictions for the allowable energies of the hydrogen atom, i.e.

$$E = -13.6/n^2 \text{ eV},$$

where $n = 1, 2, 3, \ldots$
(*a*) Using this expression, determine the energy for the
(*i*) $n = 1$,
(*ii*) $n = 2$, and
(*iii*) $n = 3$
states for the hydrogen atom according to the Bohr model.
(*b*) The derivation assumed that the energy of the electron and proton was zero at infinite separation, i.e. when the atom is ionised. What value of n corresponds to ionisation?

16 Figure 66.6 shows the energy levels based on the Bohr model for the hydrogen atom.

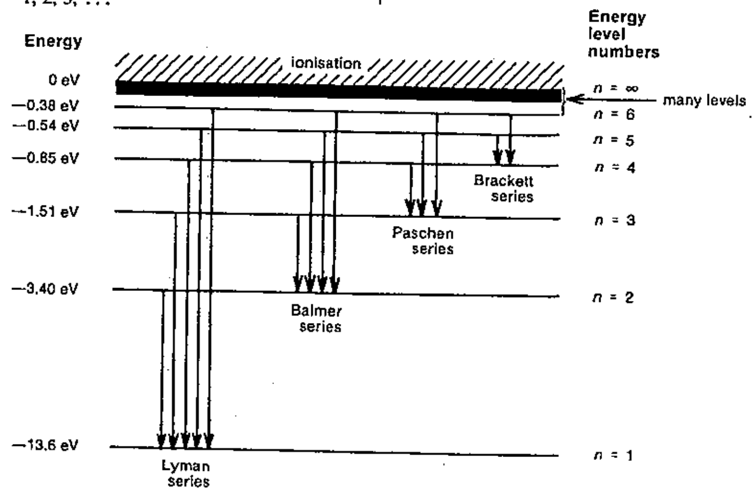


Fig. 66.6

(*a*) What is the excitation energy of hydrogen?
(*b*) What is the ionisation energy of hydrogen?
(*c*) What is the second least energetic photon that will excite a hydrogen atom?

17 The relation of the Lyman, Balmer, Paschen and Brackett spectral series to the allowable energy states for the hydrogen atom is shown in Figure 66.6.
(*a*) If the Balmer series occurs mainly in the visible spectrum, would the Lyman series be found in the infrared or ultraviolet region?
(*b*) The longest wavelength in the Paschen series would be produced by hydrogen atoms changing from the $n = a$ energy level to the $n = b$ level. What are the values of a and b?
(*c*) What is the energy of the photons giving rise to the second longest wavelength of the Balmer series?

18 A line in the spectrum of hydrogen has a wavelength of 1.88 μm. Is it a member of the Lyman, Balmer, Paschen or Brackett series?

19 Figure 66.7 shows the optical lines of atomic hydrogen which have frequencies greater than 4×10^{14} Hz. This range includes the whole of both the Balmer series and the Lyman series.
(*a*) Identify the upper and lower energy states involved in the transition which gives rise to the line marked B_3.
(*b*) The Balmer spectral lines approach a limit on the right-hand end of the series. What is the wavelength of this last line?
(*c*) Account for the fact that the frequency of the line marked B_1 is equal to the difference between the frequencies of the lines marked L_1 and L_2.

20 The expression giving the Bohr energy states for the hydrogen atom,

$$E = -13.6/n^2 \text{ eV } (n = 1, 2, 3, \ldots)$$

derived in Problem 14 needs only a minor modification to cover other simple cases.
(*a*) Determine the equivalent expression for the energy states of the singly charged helium ion He^+.
(*b*) What would be the excitation energy of this helium ion?

21 Calculate the ionisation energy of a doubly charged lithium ion Li^{2+}.

22 Imagine a dense flow of cars on a freeway passing at 72 km h^{-1} through solid lane separators (shown at PQ in Fig. 66.8), analogous to a parallel beam of monochromatic light passing through a diffraction grating. In the case of light, a diffraction pattern would be produced if the wavelength and spacing d of the slits are appropriately related. Take 1000 kg as the mass of a car.

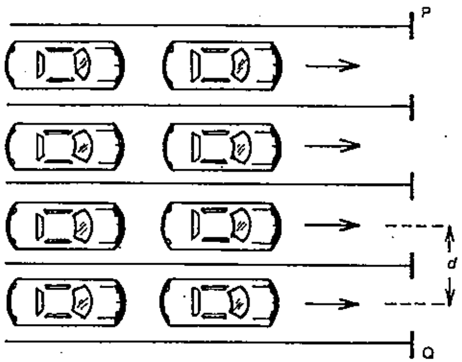


Fig. 66.8

(*a*) Determine the de Broglie wavelength of the cars.
(*b*) Which of the following would be an appropriate value for the lane width d for a diffraction pattern of cars to be observed on emerging from PQ?
A 4 m
B 1 m
C 10^{-15} m
D 10^{-30} m
E 10^{-38} m

23 (*a*) Estimate as an order of magnitude the de Broglie wavelength of
(*i*) a tennis ball that has just been served by an average player;
(*ii*) an average snail on the move;
(*iii*) an electron moving at 1 per cent of the speed of light.
(*b*) Of the tennis ball, the snail and the electron, which is most likely to exhibit *wave* properties?

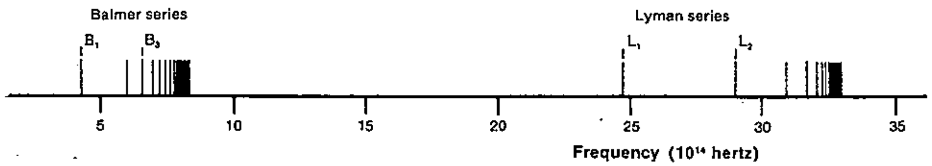


Fig. 66.7

24 Determine, in eV, the energy of a proton which would have a de Broglie wavelength of 50 pm. Mass of proton = 1.67×10^{-27} kg.

25 A neutron can enter an atomic nucleus only if its de Broglie wavelength is approximately equal to or less than the diameter of the nucleus. Assuming the diameter of an atomic nucleus to be 10^{-14} m, and taking the mass of a neutron to be 1.7×10^{-27} kg, find the least velocity for which a neutron is capable of entering the nucleus, giving your answer to one significant figure.

26 The following experiment illustrates the wave nature of the neutron. It is found that neutrons having a speed of 794 m s^{-1}, when directed at a particular angle to a certain set of crystal planes, will give rise to a strong "reflection", when the path difference for neutrons bouncing off successive layers is one wavelength. Neutron mass = 1.67×10^{-27} kg.
(*a*) X-rays are directed at the same angle to the same crystal planes. What must be their wavelength (in pm) if a strong reflection of exactly the same kind is to occur?
(*b*) Electrons are now directed at the same angle to the same crystal planes. What must be the speed of the electrons for a strong reflection of exactly the same kind to occur?

27 In a famous experiment in USA in 1927, Clinton Davisson and Lester Germer fired a beam of 75 eV electrons at a glancing angle of 18° to a crystal of nickel and thereby produced a strong reflection. No smaller angle produced a strong reflection.
(*a*) What is the de Broglie wavelength of the electrons in nm?
(*b*) Calculate the separation (in nm) of adjacent layers of atoms in the nickel.
(*c*) After 18° what was the next largest angle at which Davisson and Germer found a strong reflection of the electrons?

28 Consider the simple case of a particle of mass m free to move back and forth between the end walls of a "one-dimensional box" of length d as shown in Figure 66.9. Since the particle cannot escape from the box, wave mechanics suggest the most stable states would be the ones where the one-dimensional matter wave forms a standing wave within the box similar to those in a vibrating stretched string.

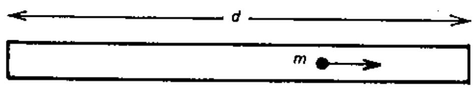


Fig. 66.9

(*a*) In its lowest energy state or ground state what is the particle's
(*i*) wavelength in terms of d?
(*ii*) momentum in terms of d and Planck's constant h?
(*iii*) energy in terms of d, h and m?
(*b*) For the particle in its *second* simplest stable state, express
(*i*) its wavelength in terms of d;
(*ii*) its energy in terms of d, h and m.
(*c*) Deduce expressions for the particle in its nth simplest stable state, giving its
(*i*) wavelength in terms of d, h and n;
(*ii*) energy in terms of d, h, m and n.

29 Consider further the particle in a "one-dimensional box" described in Problem 28 and Figure 66.9. For this question take the length d of the box as 12 mm.
(*a*) If the particle is in its ground or $n = 1$ state, how far from the left-hand end is the probability of finding it highest?
(*b*) If the particle is in its $n = 2$ energy state, is it more likely to be found
(*i*) at A, 3 mm from the left-hand end, or at B, 9 mm from the left-hand end?
(*ii*) at A or C, 6 mm from the left-hand end?
(*c*) In its $n = 3$ state, how far from the left-hand end would points be where the probability of finding the particle would be the same as that at D, 2 mm from the left-hand end?

30 This problem refers again to the particle in a "one-dimensional box" considered in Problem 28 and Figure 66.9. This time the particle is an electron.
(*a*) What is the difference between the electron energies in the first two levels, if the box is
(*i*) an evacuated glass tube with $d = 0.3$ m?
(*ii*) an approximation to an atom with $d = 0.3$ nm?
(*b*) In which case are we more likely to be able to detect the fact that the electron has discrete energy states?
(*c*) If the fall of the electron from its second to its lowest energy state, referred to in (*a*) (*ii*) above, gives rise to the emission of electromagnetic radiation, what would be its wavelength? Is this near or in the visible range?

31 A simple application of wave mechanics to the hydrogen atom might consider the stable states for the electron as those established when its matter wave formed a stationary wave around the electron's orbit. One example—not the simplest—is shown in Figure 66.10.

Fig. 66.10

(*a*) Write an expression for the circumference of the orbit in terms of λ the wavelength of the electron for
(*i*) the simplest or ground state;
(*ii*) the second simplest or first excited state;
(*iii*) the nth simplest state.

(*b*) Deduce an expression for the radius of the orbit r_n for the nth simplest state in terms of
   (*i*) λ and n;
   (*ii*) m the mass of the electron, v its speed, h Planck's constant, and n.

(*c*) Compare your relationship in (*b*) (*ii*) deduced from this simple wave mechanical model with Bohr's postulate that the angular momentum of the electron is quantised, i.e.

$$mvr = nh/2\pi$$

   (*i*) How do they compare?
   (*ii*) How then would the energy states for the hydrogen atom predicted by wave mechanics compare with those predicted by the Bohr model?

32 Consider a neutron of mass 2×10^{-27} kg bounding around inside a cobalt nucleus. Let us propose that a stable state for the nucleus is one where the de Broglie wavelength of the neutron forms a standing wave around the outer surface of the spherical nucleus, similar to that of an electron around its orbit in a hydrogen atom (Problem 31 and Fig. 66.10). Radius of the cobalt nucleus $= 4 \times 10^{-15}$ m.

(*a*) Deduce an expression for the wavelength associated with the nth energy state of the neutron in terms of n (and π).

(*b*) Determine an expression for the energy of the nth energy state of the neutron.

(*c*) Suppose the neutron falls from its first excited state to its ground state, giving out electromagnetic radiation. What is the photon energy in J?

(*d*) How does this compare with the energy of actual gamma ray photons from excited cobalt nuclei which tend to be about 1 or 2 MeV?

SET 67 REVISION PROBLEMS

Set	Topic	Problems
64	Structure of the atom	1-3
65	Structure of the nucleus and radioactivity	4-8
66	Atomic energy levels, spectra and introductory wave mechanics	9-13

Where necessary in this set, use
Coulomb's law constant in a vacuum or air,
$$k = 1/4\pi\epsilon_0 = 9.0 \times 10^9\ \text{N m}^2\ \text{C}^{-2}$$
permittivity of a vacuum or air,
$$\epsilon_0 = 8.85 \times 10^{-12}\ \text{C}^2\ \text{N}^{-1}\ \text{m}^{-2}$$
$$1\ \text{e} = 1.60 \times 10^{-19}\ \text{C}$$
$$1\ \text{eV} = 1.60 \times 10^{-19}\ \text{J}$$
mass of an alpha particle $= 6.65 \times 10^{-27}$ kg
speed of electromagnetic radiation in a vacuum or air,
$$c = 3.0 \times 10^8\ \text{m s}^{-1}$$
Planck's constant, $h = 6.63 \times 10^{-34}$ J s $= 4.14 \times 10^{-15}$ eV s

1 A beam of alpha particles is projected into a Wilson cloud chamber containing air and the tracks appear to be split into two branches when a collision occurs with a nucleus of the gas. Figure 67.1 illustrates one such event in which

$$\theta = 103.9°$$
$$\text{and } \phi = 134.8°$$

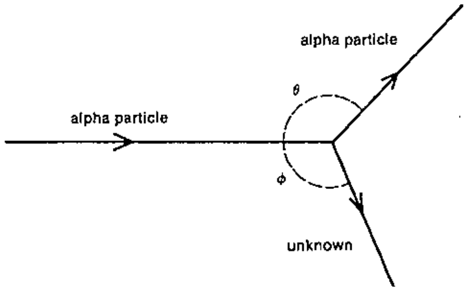


Fig. 67.1

(*a*) Determine the ratio of the speeds of the alpha particle before and after collision.

(*b*) Assuming that the masses of the alpha particle, the nitrogen atom, and the oxygen atom are in the ratio 4.00:14.0:16.0, determine whether the unknown atom is nitrogen or oxygen.

2 In an experiment similar to Rutherford's, energetic alpha particles were directed through a thin silver foil. Silver has an atomic number of 47. If an alpha particle approached to within 1.0×10^{-14} m of a silver nucleus, what then would have been the magnitude of
(*a*) the force of repulsion on, and
(*b*) the acceleration of
the alpha particle? Assume the more massive silver nucleus does not move.

3 If a 4.8 MeV alpha particle makes a head-on interaction with a platinum nucleus (atomic number 78), what is their minimum separation?

4 In terms of atoms composed of protons, neutrons and electrons, what structure is suggested for
(*a*) the isotopes of hydrogen (deuterium and tritium) whose respective masses are about twice and three times the mass of the simplest isotope of hydrogen?
(*b*) the two isotopes of lithium (atomic number 3) of masses respectively some 6 and 7 times the mass of the simplest hydrogen atom?

5 When a copper nucleus containing 36 neutrons is hit by a neutron, a new element is formed and a proton ejected. Write an equation for this nuclear reaction, showing all atomic and mass numbers. You will need to refer to the Appendix.

6 Cobalt-60 is a gamma ray source used for sterilising medical supplies. The source is produced by irradiating another substance with neutrons. What is this other nuclide?

7 A radioactive nuclide of radium $^{223}_{88}$Ra emits alpha particles and has a half-life of 11 d.
(*a*) If you started with 4.8×10^{21} atoms of radium-223 in an open test tube, after 44 d,
   (*i*) how many radium atoms, and
   (*ii*) how many atoms of any sort
   would you have in the bottom of the test tube?
(*b*) If, instead, the 4.8×10^{21} radium-223 atoms were placed in an otherwise evacuated and sealed glass phial, after 44 d how many atoms would be present in the phial?

8 Figure 67.2 shows how the count rate for a radioactive source varied with time after allowing for the background count. What is the half-life of the source?

9 The energy levels of atomic sodium are shown in Figure 67.3.

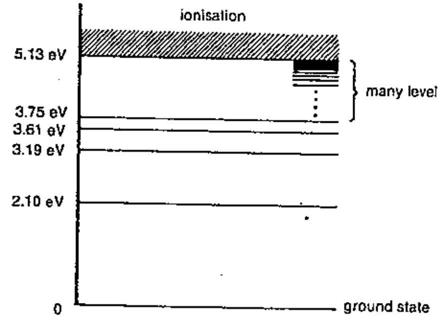


Fig. 67.3

(*a*) If sodium atoms in the ground state are bombarded with electrons, what is the minimum energy of a bombarding electron which will detach an electron from a sodium atom?
(*b*) Sodium atoms are bombarded with electrons of energy 3.30 eV. What will be the energy (or energies) of the photons emitted?
(*c*) The energy of the bombarding electrons is 3.50 eV. If a photon of energy 1.09 eV is produced in one particular collision with a sodium atom, what is the energy of the scattered electron?

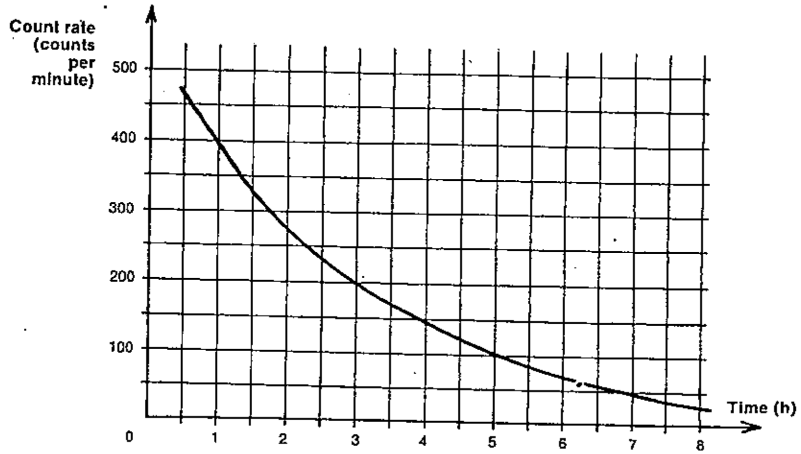


Fig. 67.2

10 Refer to Figure 67.3, which shows the energy levels for sodium.
(*a*) Determine the highest frequency radiation that sodium can emit.
(*b*) The characteristic yellow colour of a sodium street light is due to a strong 589 nm wavelength. This is caused by a change between two energy levels. Which levels?

11 10.0 eV electrons are fired through a gas and are found by experiment to retain 3.0 eV and 6.0 eV of energy.
(*a*) What is the excitation energy of the gas?
(*b*) What photon energies would be emitted by the gas?
(*c*) If white light was passed through the gas, what photon energies would be absorbed?

12 The wavelength of the red line in the visible (Balmer) series of the hydrogen spectrum is 656 nm. What is the difference in energy in eV between the two levels responsible for this line?

13 An electron of mass m is confined by electric forces between two rigid "walls" distance L apart.
(*a*) Write an expression for its de Broglie wavelength for its second excited state in terms of L.
(*b*) Deduce an expression for its energy in the second excited state in terms of m, L and Planck's constant h.

APPENDIX

APPENDIX ATOMIC NUMBERS AND CHEMICAL SYMBOLS OF THE ELEMENTS

Element	Chemical symbol	Atomic number
actinium	Ac	89
aluminium	Al	13
americium	Am	95
antimony	Sb	51
argon	Ar	18
arsenic	As	33
astatine	At	85
barium	Ba	56
berkelium	Bk	97
beryllium	Be	4
bismuth	Bi	83
boron	B	5
bromine	Br	35
cadmium	Cd	48
caesium	Cs	55
calcium	Ca	20
californium	Cf	98
carbon	C	6
cerium	Ce	58
chlorine	Cl	17
chromium	Cr	24
cobalt	Co	27
copper	Cu	29
curium	Cm	96
dysprosium	Dy	66
einsteinium	Es	99
erbium	Er	68
europium	Eu	63
fermium	Fm	100
fluorine	F	9
francium	Fr	87
gadolinium	Gd	64
gallium	Ga	31
germanium	Ge	32
gold	Au	79
hafnium	Hf	72
helium	He	2
holmium	Ho	67
hydrogen	H	1
indium	In	49
iodine	I	53
iridium	Ir	77
iron	Fe	26
krypton	Kr	36
lanthanum	La	57
lawrencium	Lr	103
lead	Pb	82
lithium	Li	3
lutetium	Lu	71
magnesium	Mg	12
manganese	Mn	25
mendelevium	Md	101
mercury	Hg	80
molybdenum	Mo	42
neodymium	Nd	60
neon	Ne	10
neptunium	Np	93
nickel	Ni	28
niobium	Nb	41
nitrogen	N	7
nobelium	No	102
osmium	Os	76
oxygen	O	8
palladium	Pd	46
phosphorus	P	15
platinum	Pt	78
plutonium	Pu	94
polonium	Po	84
potassium	K	19
praseodymium	Pr	59
promethium	Pm	61
protactinium	Pa	91
radium	Ra	88
radon	Rn	86
rhenium	Re	75
rhodium	Rh	45
rubidium	Rb	37
ruthenium	Ru	44
samarium	Sm	62
scandium	Sc	21
selenium	Se	34
silicon	Si	14
silver	Ag	47
sodium	Na	11
strontium	Sr	38
sulphur	S	16
tantalum	Ta	73
technetium	Tc	43
tellurium	Te	52
terbium	Tb	65
thallium	Tl	81
thorium	Th	90
thulium	Tm	69
tin	Sn	50
titanium	Ti	22
tungsten	W	74
unnilpentium	Unp	105
unnilquadium	Unq	104
uranium	U	92
vanadium	V	23
xenon	Xe	54
ytterbium	Yb	70
yttrium	Y	39
zinc	Zn	30
zirconium	Zr	40

ANSWERS

SET 1 MATHEMATICAL CONVENTIONS IN SCIENCE

1 (*a*) 2.23×10^{3} m (*b*) 1.2×10^{8} m (*c*) 8.4×10^{-4} m (*d*) 2.5×10^{-10} m
2 (*a*) 3 (*b*) 2 (*c*) 2 (*d*) 3 (*e*) Infinite
3 94.7 m
4 0.8 kilometre/second
5 1.06×10^{2} (or 106) cm^{2}
6 9.6×10^{2} volt
7 8.9×10^{-2} kg m^{-3}
8 6.28 cm^{3}
9 (*a*) 0.7912, 0.7923, 0.7934 (*b*) 0.792 (*c*) 0.7314, 0.7431, 0.7547 (*d*) 0.74 (*e*) (*i*) 3 significant figures (*ii*) 2 significant figures
10 (*a*) (*i*) 2 (*ii*) 3 (*iii*) 3 (*iv*) 2 (*b*) *A*: nearest degree; *B*: nearest tenth of a degree; *C*: nearest tenth of a degree
11 (*a*) 0.68 (*b*) 0.851 (*c*) 0.36 (*d*) 0.257 (*e*) 1.94
12 (*a*) 0.80 (*b*) 63° (*c*) 63.0° (*d*) 2.5 (*e*) 30.7°
13 (*a*) 10^{10} year (*b*) 10^{3} kg (*c*) 10^{-3} m (*d*) 10^{-18} coulomb (*e*) 10^{-31} m^{3}
14 (*a*) 3 (*b*) 10^{-3} m (*c*) 3.71×10^{-4} m (*d*) 7.44×10^{-4} m (*e*) 4×10^{2}
15 (*a*) 4×10^{3} N (*b*) 5×10^{-12} F (*c*) 2.2×10^{7} Ω (*d*) 1.2×10^{-2} s (*e*) 7×10^{-7} C (*f*) 3.65×10^{-7} m
16 (*a*) 5×10^{6} m^{2} (*b*) 2×10^{-6} m^{3} (*c*) 1.6×10^{-12} m^{2} (*d*) 2.5×10^{-8} m^{3}

SET 2 USING DIMENSIONAL ANALYSIS

1 (*a*) $[l]^{2}$ (*b*) $[l][t]^{-2}$ (*c*) $[m][l][t]^{-2}$ (*d*) $[m][l][t]^{-1}$ (*e*) $[m][l][t]^{-1}$ (*f*) $[m][l]^{2}$ (*g*) It is dimensionless (*h*) $[l][t]$ (*i*) $[m][l][l]^{-1}[t]^{-3}$
2 (*a*) 1 (*b*) −3 (*c*) −2 (*d*) 0 (*e*) −1 (*f*) −3
4 C
5 $[m][l]^{-1}[t]^{-2}$
6 Both (*b*) and (*d*) are *dimensionally* correct
7 (*a*) $[m]^{-1}[l]^{3}$ (*b*) kg^{-1} m^{3} s^{-2} (*c*) $[m][l]^{2}[t]^{-2}[n]^{-1}[T]^{-1}$ $[t]^{-2}$
8 (*a*) $a = -\frac{1}{2}$; $b = \frac{1}{2}$ (*b*) $v = k\sqrt{P/\rho}$
9 (*a*) $F = K\eta^{a}v^{2-a}r^{2-a}\rho^{1-a}$ (*b*) $F = K\eta rv$ and the force is independent of the density of the fluid
10 (*a*) The period *T* depends on both the length *l* and the gravitational field strength *g*, but is independent of the mass *m* (*b*) $T \propto \sqrt{l/g}$
11 (*a*) Yes (*b*) Yes (*c*) No

12 (*a*) Click zoom (or zoom click) (*b*) Acceleration (*c*) Lump click^{-3} zoom^{-3}
13 (*a*) Acceleration (*b*) Z H S^{-1} (zip hunk per shove)

SET 3 REDUCTIONS OF OBSERVATIONS

1 (*a*) The line of best fit is straight and intersects the origin of the graph (*b*) $F = 24l$
2 (*b*) *R* is not directly proportional to *θ*. Although the line of best fit is a straight line, it does not intersect the origin of the graph. (*c*) $R = m\theta + c$, where your value of *m* should be in the range 0.20 to 0.21, and *c* should be in the range 49 to 50.
3 (*a*) G (*b*) E (*c*) D (*d*) C (*e*) F (*f*) C (*g*) B (*h*) A
4 (*a*) E (*b*) B (*c*) A (*d*) D (*e*) A (*f*) C
5 $F = 144/r$
6 Coefficient of friction = 0.28
7 1.9×10^{11} newton metre^{-2} (or pascal)
8 (*a*) $f \propto 1/L$ (*b*) $f \propto \sqrt{T}$ (*c*) $f \propto 1/\sqrt{\mu}$
9 (*a*) Sets 1, 2, 3 and 4 (*b*) Direct proportionality, since while *D* increases, so does *P*. (*c*) P/D (*d*) $P \propto D$ (*e*) Sets 2, 5, 6 and 7 (*f*) $P \propto 1/x$
10 (*a*) *V* against $1/L$ (*b*) *V* against S^{2} (*c*) S^{2}/L
11 (*a*) *t* against $1/r^{2}$ (*b*) *t* against $1/(\rho_s - \rho_0)$ (*c*) $1/r^{2}(\rho_s - \rho_0)$
12 (*a*) $N_{10} \propto 1/M^{2}$ (*b*) $N \propto t^{2}$ (*c*) 80 rev (*d*) 5.0 rev (*e*) $N_{20} \propto 1/R^{2}$
13 (*a*) (*i*) D (*ii*) G (*iii*) F (*iv*) B (*v*) C (*b*) $\log d = n \log t + \log k$ (*c*) (*i*) *n* (*ii*) $\log k$
14 (*a*) (*i*) $\log_{10} P = -\log_{10} V + 2$ (*ii*) $P = 100/V$ (*b*) (*i*) $\log_{10} T = \frac{1}{2} \log m + 0.58$ (*ii*) $T = 3.8\sqrt{m}$
15 $R = 6 \times 10^{-8}\ T^{4}$
16 (*b*) $E = 0.0424\ t$, where *E* millivolt = e.m.f., and *t*°C = temperature difference (*c*) 65°C
17 (*a*) 209 Hz, 233 Hz, 256 Hz, 266 Hz, 276 Hz, 295 Hz (*b*) Frequency is directly proportional to the square root of the tensile force
18 (*a*) −3.5 cm (*c*) 48.5°C
19 (*b*) 5000 ohm

SET 4 CONCEPTS OF DENSITY AND PRESSURE

1 10^{3} kg m^{-3}
2 (*a*) 879 kg m^{-3} (*b*) $10^{3}\,x$ kg m^{-3} (*c*) $y/10^{3}$ g cm^{-3}
3 (*a*) 13.6 g cm^{-3} (*b*) 1.36×10^{4} kg m^{-3}

4 2 kg
5 (*a*) 10^{21} m^{3} (*b*) 10^{7} m
6 (*a*) 51 g (*b*) 63 g
7 14 cm^{3}
8 75 cm^{3}
9 8.0×10^{2} kg m^{-3}
10 3×10^{4} Pa
11 1.5×10^{4} N
12 0.13 m^{2}
13 6.4×10^{4} Pa
14 1.03×10^{4} N
15 1.0×10^{5} Pa
16 9.9 m
17 8.16 m
18 (*a*) 4.9×10^{5} Pa (*b*) 2.0×10^{10} N
19 9.0×10^{2} kg m^{-3}

SET 5 MEASUREMENT OF LENGTH, TIME AND FREQUENCY

1 (*a*) 5.794 mm (*b*) 4.737 mm
3 0.025 unit
4 (*a*) 50 (*b*) 37
5 (*a*) 9 mm (*b*) 0.9 mm (*c*) 0.1 mm (*d*) 0.1 mm (*e*) 9.8 mm
6 (*a*) 0.90 cm (*b*) 10 (*c*) 6th
7 (*a*) 19 mm (*b*) 0.95 mm (*c*) 0.05 mm (*d*) 4.35 mm
8 (*a*) 2.45 cm (*b*) 50
9 (*a*) 1/6 s (*b*) 8 glimpses (*c*) 1/48 s (*d*) 1/48 s (*e*) 48 Hz
10 (*a*) 5.0×10^{-2} (i.e. 1.20) s (*b*) 5.0×10^{-2} s (*c*) 20 Hz (*d*) 40 Hz, 60 Hz, 80 Hz
11 (*a*) 12 Hz (*b*) Disc is rotating twice between glimpses
12 (*a*) 1/160 s (*b*) 1/40 s (*c*) 40 Hz (*d*) 32 Hz
13 140 Hz
14 (*a*) 1/10 (*b*) 0.5 s
15 (*a*) $\frac{1}{4}$ or 0.25 s (*b*) 6 frames per second
16 54 pictures per second
17 30 min
18 2.0×10^{2} Hz

SET 6 MOTION IN A STRAIGHT LINE—Graphical treatment

1 (*a*) Zero (*b*) 40 m (*c*) 10 m
2 (*a*) 10 m s^{-1} (*b*) 2 m s^{-2} (*c*) 5 m s^{-1} (*d*) (*i*) 16 m (*ii*) 25 m (*e*) No
3 (*a*) 20 m (*b*) 16 m s^{-1} (*c*) 0, 6.7 and −15 m s^{-2} respectively
4 (*a*) 12 m (*b*) 6.0 m s^{-2} (*c*) −2.0 m s^{-2} (*d*) 6.0 m
5 (*a*) (*i*) B (*ii*) A (*iii*) B and D (*b*) 6.0 km (*c*) 42 km h^{-1} (*d*) -2.0×10^{2} km h^{-2}
6 (*a*) 1.2×10^{3}m (*b*) 200 km
7 (*a*) C (*b*) B (*c*) 20 m; 5 m min^{-1} (*d*) 10 m min^{-1}
8 (*a*) A to B and F to G (*b*) Between E and F (*c*) Zero (*d*) −0.86 m s^{-1}
9 (*a*) 3 m (*b*) P (*c*) Q (*d*) −1 m s^{-1} (*e*) 8 s
10 (*a*) 1.0×10^{-2} km s^{-1} (*b*) 2.0×10^{-2} km s^{-1} (*c*) 1.5×10^{-2} km s^{-1}; 1.6×10^{-2} km s^{-1} (*d*) 1.5×10^{-2} km s^{-1} (*e*) 2.5×10^{-4} km s^{-2}
11 (*a*) P, R, S, T (*b*) T (*c*) Q (*d*) 4 cm to the left (*e*) 16 cm (*f*) −30 cm s^{-1} (*g*) 27 cm s^{-1}
12 (*a*) 300 km h^{-2} (*b*) 12 km (*c*) 0.20 h
13 0.25 h
14 (*a*) (*i*) B (*ii*) A (*b*) 2.5 km (*c*) 7.6 km (*d*) 22 km h^{-1} (*e*) -1.2×10^{2} km h^{-2}
15 (*a*) 2.00 (*b*) 405 m (*c*) 135 m (*d*) (*i*) 0.900 m s^{-2} (*ii*) 1.58 m s^{-2}
16 (*a*) B (*b*) A and E (*c*) C and D (*d*) G (*e*) A and E (*f*) F
17 (*b*) 39 m
18 22 m s^{-1}; 0.73 m s^{-2}
19 $\frac{vn}{n+2}$; $\frac{v(n-1)}{n}$
20 5.0 m s^{-1}
21 (*a*) 29 mm $tick^{-1}$ (*b*) (*i*) 6.0 mm $tick^{-2}$ (*ii*) 2.4×10^{3} mm s^{-2} (*iii*) 2.4 m s^{-2}
22 (*a*) (*i*) C (*ii*) B (*iii*) A and E (*iv*) C (*v*) C (*b*) 140 mm s^{-1} (*c*) 800 mm s^{-2}
23 (*a*) 2.0 cm $pentatick^{-1}$ (*b*) (*i*) 2.0 cm $pentatick^{-2}$ (*ii*) 2.0 m s^{-2}. (*c*) 22 cm (*d*) The distance covered (or displacement) of the trolley from $t = 5.5$ to $t = 6.5$ pentatick; 5.5 cm
24 (*a*) 7 tick (*b*) 7.0×10^{-5} vel tick (*c*) 6×10^{-3} vel (*d*) 1.0×10^{-2} vel (*e*) 1.0 vel tick (*f*) 1.5×10^{-2} vel (*g*) 5.2 vel tick

SET 7 MOTION IN A STRAIGHT LINE—Algebraic treatment

1 30 m s^{-1}
2 1.0 m
3 4.0 m
4 5.0 m s^{-1}

5. (*a*) 1.5 m s^{-2} (*b*) 3.0 m s^{-2} (*c*) 4.0 m s^{-2} (*d*) 4.0 m s^{-2} (*e*) -5.5 m s^{-2}
6 (*a*) 32 m s^{-1} (*b*) 9.0 s (*c*) 10 s; 40 m s^{-1} (*d*) 10 m
7 8.0 cm s^{-2}; 2.0 $\times$ 10^{2} cm
8 0.17 m s^{-2}; 2.7 $\times$ 10^{3} m; 1.5 m s^{-2}; 3.0 $\times$ 10^{2} m
9 3 m s^{-2}
10 2.0 m s^{-2}; 4.0 m s^{-1}
11 28.0 m s^{-1}; 127 m
12 (*a*) 20 m s^{-1} (*b*) 1.5 m s^{-2}
13 0.80 m s^{-2}; 15 s
14 117 m
15 44 s
16 440 m
17 261 m; 38.0 s
18 $s = v_i t - \frac{1}{2}at^2$
19 Sports car: $s = 30t$; police van: $s = 15t + 0.6t^2$; 25 s
20 50.0 s; 1.75 $\times$ 10^{3} m
21 10^{3} m s^{-2}

SET 8 VECTORS AND SCALARS

1 (*a*) S (*b*) V (*c*) S (*d*) V (*e*) V (*f*) S (*g*) S (*h*) S (*i*) V (*j*) S (*k*) S (*l*) S (*m*) V
2 (*a*) S (*b*) S (*c*) V (*d*) S (*e*) S (*f*) V (*g*) S (*h*) S (*i*) V
3 A pull of 330 newton
4 300 newton in the direction of the pull
6 10 kN; 37° (to the nearest degree)
7 10 km, N 27° E (to the nearest degree)
9 55 m s^{-1} north
10 17 km h^{-1} south-west
11 (*a*) 12 m s^{-1} south (*b*) Zero (*c*) 8.5 m s^{-1} south-east (*d*) 4.1 m s^{-1}, E 20° S
12 69 (69.3) N; 40 N
13 (*a*) 0.22 km (*b*) 1.5 km h^{-1}
14 (*a*) 3.5 m s^{-1} (*b*) 13 s
15 (*a*) 60 N (*b*) 80 N
16 (*a*) 1.6 $\times$ 10^{2} N (*b*) 1.1 $\times$ 10^{3} N (*c*) 8.8 $\times$ 10^{2} N (*d*) Pulling
17 (*a*) 54 km h^{-1} north (*b*) 46 km h^{-1} north
18 250 km h^{-1}, N 16° E (to the nearest degree)
19 7.8 km h^{-1}, N 22° E
20 E 15.5° S; 374 km h^{-1}
21 63.2 km h^{-1}, N 18.4° E
22 464 km h^{-1}, S 62.7° W
23 24 km h^{-1} in direction E 30° N (i.e. from W 30° S)
24 (*a*) S 35.5° W (*b*) 608 km h^{-1}
25 Upstream at 60° to the bank; 8.7 min
26 (*a*) 10 km h^{-1}, upstream at an angle of 53° with the bank (*b*) 1.5 min
27 5000 m
28 (*a*) Row in a direction at right angles to the bank (*b*) In a direction which makes an angle of 73.7° with the bank

SET 9 INTRODUCTION TO NEWTON'S LAWS OF MOTION

1 1.8 m s^{-1}
2 (*a*) 4 m s^{-2} east (*b*) 2 m s^{-2} south
3 (*a*) 8 N upwards (*b*) 0.02 N west
4 0.14 kg
5 (*a*) P (*b*) R (*c*) 3 cm pentatick^{-2} (*d*) R
6 (*a*) 4.80 cm pentatick^{-1} (*b*) Half-way through the third tick (*c*) 11.3 cm pentatick^{-1} (*d*) 1.3 cm pentatick^{-2} (*e*) 20.8 cm s^{-2} (*f*) 62.4 cm s^{-2} (*g*) 0.936 N
7 (*a*) 10 cm s^{-1} (*b*) 5.0 m s^{-2} (*c*) 1.5 N east (*d*) Uniform retardation (acceleration of glider is 2.5 m s^{-2} west) (*e*) 29 cm s^{-1}
8 (*a*) Zero (*b*) 1/3 cm pentatick^{-2} (*c*) 2/3 cm pentatick^{-2} (*d*) 3 bricks
9 0.020 kg
10 9.1 $\times$ 10^{-27} N
11 0.28 kg
12 1.4 $\times$ 10^{2} N south
13 (*a*) 124 m s^{-1} east (*b*) 76 m s^{-1} east
14 (*a*) 6.3 $\times$ 10^{2} kg m s^{-1} north (*b*) 0.55 kg m s^{-1} downwards
15 (*a*) (*i*) 3.0 N s upwards (*ii*) 4.0 $\times$ 10^{2} N s south-west (*b*) (*i*) 3.0 kg m s^{-1} (or N s) upwards (*ii*) 4.0 $\times$ 10^{2} kg m s^{-1} (or N s) south-west
16 (*a*) 1.0 $\times$ 10^{2} N s east (*b*) 1.0 $\times$ 10^{2} kg m s^{-1} east (*c*) 80 kg m s^{-1} east (*d*) 10 m s^{-1} east
17 (*a*) 1.6 $\times$ 10^{2} N s (or kg m s^{-1}) west (*b*) 1.6 $\times$ 10^{2} kg m s^{-1} (or N s) west (*c*) 50 m s^{-1} west (*d*) 50 m s^{-1} west
18 2.25 s
19 7 N south
20 1.1 m s^{-2}
21 (*a*) 5 N (*b*) 1 N
22 (*a*) 17 (17.4) N along the bisector of the angle between the directions of the two forces (*b*) 5.8 m s^{-2} along the bisector of the angle between the directions of the two forces
23 14°
24 -1.73 m s^{-2}
25 (*a*) $200\sqrt{2}$ N to the right (*b*) 200 N (*c*) $400\sqrt{2}$ N to the right; acceleration to the right
26 (*a*) $A = 90.0°$, $B = 126.9°$, $C = 143.1°$ (*b*) As for (*a*), since the net force is still zero
27 (*a*) 8 N (*b*) 4 m s^{-2}
28 3.0 cm s^{-2}; 9.0 $\times$ 10^{-4} N
29 (*a*) 0.30 m s^{-2} (*b*) 3.2 $\times$ 10^{2} N
30 $(m_1 + m_2)f$ and $m_2 f$
31 5 m s^{-2}; 20 m s^{-1}; 40 m
32 8.5 $\times$ 10^{-12} N
33 40 m s^{-2}; 2.0 $\times$ 10^{3} N
34 0.49 N
35 (*a*) 8 N west (*b*) 8 N east
36 22.5 N

SET 10 NEWTON'S LAW AND GRAVITATIONAL EFFECTS NEAR THE EARTH'S SURFACE

1 9.818 N kg^{-1} downwards
2 39.2 N
3 -2.4 N
4 (*a*) 10^{4} N (*b*) 10^{-1} N (*c*) 10^{-3} N
5 (*a*) 1.30 $\times$ 10^{3} N (*b*) 217 N (*c*) 133 kg
6 (*a*) 4.00 N upwards (*b*) 200 N upwards
7 1.04 $\times$ 10^{3} (1036) N upwards
8 (*a*) 750 N (*b*) Zero (*c*) 750 N (*d*) 900 N (*e*) 600 N
9 (*a*) 5.88 $\times$ 10^{3} N (*b*) 5.61 $\times$ 10^{3} N (*c*) 6.15 $\times$ 10^{3} N (*d*) No. For example, in case (*b*) the lift could be speeding up while descending or slowing up while ascending
10 23.4 N
11 (*a*) He could be *accelerating* down the rope and there would then be a net *downward* force on him, in which case the upward force exerted on him by the rope (and therefore on the rope by him) would be *less* than the downward force on him, i.e. his weight (*b*) 2.25 m s^{-2} (*c*) 6.0 m s^{-1}
12 (*a*) 5.6 m s^{-1} upwards (*b*) 3.5 N upwards (*c*) 4.0 (3.99) N upwards
13 10 N
14 (*a*) 4.2 m s^{-2} (*b*) 2.8 N
15 (*a*) 1.4 m s^{-2} upwards (*b*) 2.5 s (*c*) 4.2 N upwards
16 49.0 N kg^{-1} downwards
17 $m_1 > km_2$
18 (*a*) 12 N (*b*) 6 m s^{-2} (*c*) 18 N downwards
19 5 s
20 1.47 N upwards
21 46.6 N; 110 N
22 0.196 N

SET 11 FLOTATION AND GRAVITATIONAL EFFECTS IN FLUIDS

1 (*a*) 0.53 N (*b*) 0.33 N (*c*) Zero (*d*) 0.20 N (*e*) 0.20 N (*f*) 20 g (*g*) 20 cm^{3} (*h*) 20 cm^{3} (*i*) 2.7 g cm^{-1}; 2.7 $\times$ 10^{3} kg m^{-3}
2 (*a*) 0.16 N (*b*) 16 g (*c*) 16 cm^{3} (*d*) 3.0 g cm^{-3}; 3.0 $\times$ 10^{3} kg m^{-3} (*e*) 20 g (*f*) 0.20 N (*g*) 0.27 N
3 (*a*) 0.24 N (*b*) 24 g (*c*) 30 cm^{3}
4 (*a*) 36 g (*b*) 0.35 N (*c*) 0.35 N (*d*) Zero (*e*) 0.35 N (*f*) 36 g (*g*) 0.60 g cm^{-3}
5 (*a*) 0.850 g cm^{-3} (*b*) 1.0 $\times$ 10^{3} cm^{3} (*c*) 8.5 $\times$ 10^{2} g (*d*) 8.3 N (*e*) 8.3 N (*f*) 8.3 N (*g*) 0.85 kg or 8.5 $\times$ 10^{2} g (*h*) 8.5 $\times$ 10^{2} cm^{3} (*i*) 8.5 $\times$ 10^{2} cm^{3} (*j*) 0.85 (*k*) 0.85 (*l*) 0.914
6 2.0 $\times$ 10^{2} cm^{3} or 2.0 $\times$ 10^{-4} m^{3}
7 2.7 $\times$ 10^{3} kg m^{-3}
8 24 g
9 (*a*) 0.40 cm^{3} (*b*) 4.0 cm (*c*) 1.2 $\times$ 10^{3} kg m^{-3} or 1.2 g cm^{-3}
10 6.0 $\times$ 10^{2} kg m^{-3} or 0.6 g cm^{-3}
11 Level will fall
12 2.0 kg
13 4.8 kg
14 1.67 $\times$ 10^{3} m^{3}
15 2.5 $\times$ 10^{2} kg m^{-3} or 0.25 g cm^{-3}
16 800 cm^{3}
17 0.074 g
18 -0.013 g
19 0.75; 1.5 N
20 792 kg m^{-3} or 0.792 g cm^{-3}
21 32:53 or 0.604

SET 12 VERTICAL MOTION NEAR THE EARTH'S SURFACE

1 (*a*) 39.2 m s^{-1} (*b*) 78.4 m (*c*) 24.5 m
2 (*a*) 3.2 s (*b*) 32 m s^{-1} (*c*) 27 m
3 9 s
4 (*a*) 30.6 m (*b*) 2.5 s (*c*) 24.5 m s^{-1} (*d*) 5.0 s (*e*) 9.80 m s^{-2} downwards
5 40.4 m
6 4.0 s
7 $\frac{1}{2}$
8 5.00 s; 34.3 m s^{-1}
9 12.1 m s^{-1}
10 Slightly more than 0.001 s

ANSWERS

12 (*a*) 1.00 $\times 10^{-2}$ N downwards (*b*) 2.45 $\times 10^{-2}$ N (*c*) 1.45 $\times 10^{-2}$ N upwards
13 (*a*) L (*b*) L (*c*) L (*e*) S
14 Shorter time

SET 13 PROJECTILE MOTION NEAR THE EARTH'S SURFACE

1 (*a*) Horizontal component (*b*) Vertical component (*c*) Remains the same
2 (*a*) Zero (*b*) 490 m downwards (*c*) 9.80 m s^{-2} downwards (*d*) 10.0 s (*e*) 160 m s^{-1} in the direction in which the plane was travelling (*f*) 160 m s^{-1} in the direction in which the plane was travelling (*g*) Zero (*h*) 1.60 km
3 (*a*) 1.14 s (*b*) 16.0 m (*c*) 17.9 m s^{-1} at an angle of 38.7° with the horizontal
4 (*a*) 5.00 s (*b*) 35.0 m (*c*) 49.5 m s^{-1} at an angle of 81.9° to the horizontal
5 60.0 m; 39.5 m s^{-1} at an angle of 82.7° to the horizontal
6 (*a*) $x = 160t$; $y = -4.90t^2$ (*b*) $y = -1.91 \times 10^{-4} x^2$ (*c*) A parabola
7 (*a*) 11 m s^{-1} (*b*) Zero (*c*) 9.8 m s^{-2} downwards (*d*) 1.1 s (*e*) 2.2 s (*f*) 6.5 m s^{-1} (*g*) 6.5 m s^{-1} (*h*) 14 or 15 m
8 (*a*) 2.6 s (*b*) 8.4 m (*c*) 40 m
9 (*a*) 0.95 m (*b*) 1.93 m
10 33.8 m
11 31 m s^{-1} at an angle of 45° to the horizontal
12 7.5 m
14 (Note that the stone is thrown at an angle of 45° to the horizontal to achieve the maximum horizontal range) Greatest vertical height is $\frac{1}{4}d$
15 (*a*) S (*b*) S (*c*) L (*d*) H (*e*) S (*f*) L (*g*) L (it will be negative)
16 (*a*) S (*b*) S (*c*) L (*d*) L (*f*) S (*g*) L (*h*) L
17 A longer horizontal distance during the rising part of the flight

SET 14 UPHILL AND DOWNHILL MOTION

1 (*a*) 245 N (*b*) 245 N (*c*) 424 N
2 1.13 N
3 1.1 $\times 10^3$ N
4 (*a*) Zero (*b*) 0.22 N
5 (*a*) 96 N downhill (*b*) 82 N (*c*) 18 N
6 1.17 $\times 10^3$ N
7 0.32° or 0°19′
8 1.4 m s^{-2}

SET 15 MOTION IN A CIRCLE

1 (*a*) 8.0 m s^{-2} (*b*) 8.0 $\times 10^3$ N (*c*) Towards the centre of the circular path
2 2.0 $\times 10^4$ N
3 0.034 m s^{-2}
4 (*a*) 40 Hz (*b*) 1.2 $\times 10^4$ m s^{-2}
5 4.5 Hz
6 1.26 $\times 10^3$ m
7 (*a*) 6.0 m s^{-1} along the tangent at A (*b*) 6.0 m s^{-1} along the tangent at B (*c*) 0.35 s (*d*) 6.0 m s^{-1} at an angle of 60° with the tangent at B (*e*) 17 m s^{-2} (*f*) 29 N
8 6.3 s
9 (*a*) 3.0 m s^{-2} north (*b*) 4.0 m s^{-2} west (*c*) 5.0 m s^{-2} west 37° north (*d*) 8.0 $\times 10^2$ N west 37° north
10 (*b*) 1.5 m (*c*) 0.92 N towards the centre of the circular path (*d*) 1.0 N (*e*) 1.8 s
11 (*a*) 1.5 (*b*) 1.5 (*c*) 6.0
12 41°
13 8°
14 (*a*) (*i*) 15.0 N downwards (*ii*) 15.0 N upwards (*b*) (*i*) 10.1 N downwards (*ii*) 19.9 N upwards
15 (*a*) 70.0 m s^{-1} (*b*) 2.0 $\times 10^3$ N upwards
16 (*a*) 329 N upwards (*b*) 1.20 $\times 10^4$ N upwards

SET 16 SIMPLE HARMONIC MOTION

1 (*a*) 25 N upwards (*b*) 9.0 m s^{-2} downwards
2 (*a*) 6.0 N m^{-1} (*b*) 1.8 N to the left (*c*) 0.40 m to the left (*d*) 1.0 $\times 10^2$ m s^{-2} to the right
3 0.8 s
4 (*a*) 5.0 N m^{-1} (*b*) 1.0 N to the right (*c*) Zero (*d*) 0.50 N to the left
5 (*a*) 114 N (*b*) 57.0 N
6 (*a*) C (*b*) A and E (*c*) 1.0 m s^{-1} (*d*) 0.87 m s^{-1}
7 1.61 m s^{-1}
8 0.693 m s^{-1}
9 0.359 s; 1.05 m s^{-1}
11 $2\pi\sqrt{(L-h)/g}$
12 0.90 s
13 $5h$ m; $4\pi\sqrt{h/g}$
14 1.27 s
15 1.678 $\times 10^{-2}$ N
16 44.937 s per day gain
17 (*a*) $v = \omega R \cos \omega t$ (*b*) $a = -\omega^2 R \sin \omega t$ (*c*) $F = -m\omega^2 R \sin \omega t$
18 (*a*) 0.10 m (*b*) 0.5 Hz (*c*) (*i*) $v = 0.10\pi \cos \pi t$ (*ii*) $a = -0.10\pi^2 \sin \pi t$ (*d*) 0.31 (or $\pi/10$) m s^{-1} (*e*) 0.58 m s^{-2}

SET 17 MOMENTUM AND ITS CONSERVATION

1 (*a*) 0.96 kg m s^{-1} east (*b*) 2.5 $\times 10^{-23}$ kg m s^{-1} upwards (*c*) 1.5 $\times 10^4$ kg m s^{-1} south
2 (*a*) 0.48 N s (or kg m s^{-1}) south (*b*) 0.48 kg m s^{-1} (or N s) south (*c*) 2.4 m s^{-1} south
3 (*a*) 10 kg m s^{-1} (*b*) 10^{-1} N s
4 (*a*) 450 N (*b*) 4.5 N s (or kg m s^{-1}) (*c*) 4.5 kg m s^{-1} (or N s) (*d*) 90 m s^{-1} (*e*) 450 kg m s^{-2} (or N) (The force equals the rate of change of momentum it produces)
5 (*a*) 39.2 N downwards (*b*) 39.2 kg m s^{-2} (or N) downwards
6 3 $\times 10^3$ kg
7 1 m s^{-1} horizontally
8 0.020 m s^{-1}
9 2.5 m s^{-1}
10 (*a*) 0.40 kg (*b*) 1.1 m (*c*) 5.0 $\times 10^{-2}$ m s^{-1}
11 (*a*) 12 cm s^{-1} (*b*) 18 cm s^{-1}
12 10 cm s^{-1} south
13 3.25 cm s^{-1}
14 (*a*) 6 m s^{-1} west (*b*) 4 m s^{-1} east (*c*) 9 m s^{-1} east
15 (*a*) Total momentum of the system before and after the collision = 0.40 kg m s^{-1} east (*b*) (*i*) 0.48 kg m s^{-1} west (*ii*) 0.48 kg m s^{-1} east (*c*) Change in momentum of A = −change in momentum of B (*d*) Impulse B exerts on A = −impulse A exerts on B (*e*) Force B exerts on A = −force A exerts on B, or, in other words, the force that B exerts on A is equal in magnitude, but opposite in direction, to the force that A exerts on B. (This is known as Newton's Third Law of Motion)
16 (*a*) 150 N north (*b*) No; to determine the net force *on the ball*, one must add vectorially the forces *acting on the ball* and not forces that the ball might exert on some other body
17 (*a*) 1.44 $\times 10^4$ kg m s^{-1} north (*b*) 240 kg m s^{-2} (or N) north (*c*) 240 N north (*d*) 240 N south (*e*) 1.2 $\times 10^5$ Pa (*f*) Higher
18 (*a*) $\frac{7}{4}NAmv^2$ (*b*) $\frac{7}{4}Nmv^2$
19 (*a*) 0.40 m from the 5.0 kg mass (*b*) 3.6 m from the 4.0 kg mass
20 3.0 cm from A on the perpendicular from A to BC
21 (*a*) 0.75 m (*b*) 7.5 m s^{-1} (*c*) 7.5 m s^{-1} (*d*) Zero
22 (*a*) 9.0 m s^{-1} east (*b*) 3.0 m s^{-1} east (*c*) 4.0 m s^{-1} west (*d*) Zero (*e*) 9.0 m s^{-1} east (*f*) 9.0 m s^{-1} east
23 (*a*) 3 m s^{-1} to the right (*b*) Zero (*c*) 3 m s^{-1} to the right
24 (*a*) 5.0 kg m s^{-1} (*b*) At an angle of 36.9° with the original direction of A's motion, on the side opposite to A (*c*) 1.0 m s^{-1} (*d*) 2.5 $\times 10^2$ N
25 (*a*) Zero (*b*) 1.3 $\times 10^{-22}$ kg m s^{-1} S 27° E (*c*) 3.9 $\times 10^2$ m s^{-1} S 27° E
26 (*a*) 0.500 kg m s^{-1} east (*b*) Zero (*c*) 0.500 m s^{-1} E 53.1° S

SET 18 WORK, KINETIC ENERGY AND POWER

1 (*a*) 72 J (*b*) 3.2 J
2 (*a*) 60 J (*b*) 2.94 $\times 10^6$ J (*c*) 4.2 $\times 10^2$ J
3 27 J
4 2.5 m s^{-2} south-west
5 54 J
6 (*a*) 24 J (*b*) 12 J (*c*) Zero
7 8.8 $\times 10^4$ J
8 (*a*) 4.0 $\times 10^2$ J (*b*) 23 N (*c*) 2.0 m s^{-2}
9 (*a*) 0.9 J (*b*) 1.6 $\times 10^2$ J
10 (*a*) 10^3 or 10^4 J (*b*) 10^{-9} or 10^{-8} J
11 (*a*) 10 J (*b*) 10 J (*c*) 18 J (*d*) 28 J (*e*) 3.7 m s^{-1}
12 1.25 $\times 10^6$ J
13 (*a*) 2.0 $\times 10^2$ J (*b*) 4.0 $\times 10^2$ J
14 (*a*) 8.0 J (*b*) 2.0 m s^{-1} (*c*) 32 J; 4.0 m s^{-1} (*d*) 12 J (*e*) 8.0 N s
15 (*a*) 10^4 J (*b*) 10^4 J (*c*) 2.0 $\times 10^3$ N
16 (*a*) 1.3 (4/3) m s^{-2} (*b*) 5.0 J (*c*) 2.0 m s^{-1}
17 (*a*) 9.0 $\times 10^2$ J (*b*) 1.5 $\times 10^2$ N towards the centre of the circle (*c*) Zero
18 (*a*) 0.40 kg m s^{-1} (*b*) $p = \sqrt{2mE_k}$ (*c*) $E_k = p^2/2m$
19 1.25 $\times 10^6$ J
20 (*a*) 0.45 (*b*) 1
21 (*a*) 6.0 $\times 10^5$ J, (*b*) 1.5 $\times 10^4$ J (*c*) 1.5 $\times 10^4$ W
22 7.0 $\times 10^3$ W
23 1.2 $\times 10^4$ N
24 37 W
25 1.76 $\times 10^5$ W

SET 19 GRAVITATIONAL POTENTIAL ENERGY IN A UNIFORM FIELD

1 (*a*) 11 J (*b*) 4.4 J
2 (*a*) 1.7 $\times 10^4$ J (*b*) 2.7 J (*c*) −4.1 $\times 10^3$ J
3 3.2 $\times 10^2$ J
4 111 kW
5 (*a*) 4.9 $\times 10^2$ J (*b*) 4.9 $\times 10^2$ J (*c*) 4.9 $\times 10^2$ J (*d*) 4.9 $\times 10^2$ J (*e*) 14 m s^{-1}
6 (*a*) $\frac{1}{2}mv_i^2$ (*b*) $\frac{1}{2}mv_i^2$ (*c*) $\frac{1}{2}mv_i^2$ (*d*) $v_i^2/2g$ or $0.051v_i^2$

7 $1.4\ m\ s^{-1}$
8 (a) 1.7×10^2 J (b) 46 J (c) 58 (or 57) J (d) $18\ m\ s^{-1}$
(e) 1.7×10^2 J
9 (a) 2.0×10^2 J (b) C (c) 78 J (d) 1.2×10^2 J
(e) $14\ m\ s^{-1}$
10 (a) 181 J (b) 81.0 J (c) $20\ m\ s^{-1}$ (d) 181 J (e) 24.6°
11 (a) $7.9\ m\ s^{-1}$ (b) 4.7×10^3 N (c) 1.6 m (d) M, E
(e) D
12 (a) 44 J (b) 3.7 N
13 (a) $J\ kg^{-1}$ (b) (i) 20.0 J (ii) 39.6 J (iii) 59.2 J
(c) (i) $10.0\ J\ kg^{-1}$ (ii) $19.8\ J\ kg^{-1}$ (iii) $29.6\ J\ kg^{-1}$
(d) $9.8\ J\ kg^{-1}\ m^{-1}$; they are equal, both being $9.8\ N\ kg^{-1}$ (or $J\ kg^{-1}\ m^{-1}$)
14 (a) (i) $29.4\ J\ kg^{-1}$ (ii) $49.0\ J\ kg^{-1}$ (b) $19.6\ J\ kg^{-1}$
(c) 196 J (d) Gravitational potential gradient or gravitational field strength; $9.8\ J\ kg^{-1}\ m^{-1}$ (or $N\ kg^{-1}$)
15 (a) $19.6\ J\ kg^{-1}$ (b) (i) Zero (ii) $9.8\ J\ kg^{-1}$
(iii) $19.6\ J\ kg^{-1}$ (iv) $19.6\ J\ kg^{-1}$ (c) 78.4 J
16 (a) 1.5×10^2 J (b) $25\ N\ kg^{-1}$ (or $J\ kg^{-1}\ m^{-1}$)

SET 20 ELASTIC POTENTIAL ENERGY

1 (a) 48 J (b) 48 J (c) Area between the plotted line and the compression axis
2 (a) 0.40 J (b) 0.40 J
3 40 J
4 (a) 0.10 J (b) 0.10 J (c) 0.10 J (d) 0.10 m
5 (a) 2.0 J (b) 5.6 J (c) 3.6 J (d) $6.0\ kg\ m\ s^{-1}$
6 (a) 0.48 J (b) 0.45 J (c) $6.1\ m\ s^{-1}$
7 (a) 1.23×10^{-2} J (b) (i) 7.84×10^{-3} J
(ii) 4.41×10^{-3} J
9 26 m
10 (a) Compressed air mortar (b) (i) 7.0×10^3 J
(ii) 5.6×10^3 J (c) (i) Zero (ii) 3.5×10^3 J (iii) 70 m

SET 21 ELASTIC AND INELASTIC COLLISIONS

1 (a) (i) A (ii) A (b) (i) C (ii) B (c) E
2 Z X Y
3 (a) $0.50\ m\ s^{-1}$ (b) $0.83\ m\ s^{-1}$ (c) K at $0.50\ m\ s^{-1}$, L at $1.0\ m\ s^{-1}$
4 (a) 9.2 (9.18) $\times 10^2$ J (b) $3.0\ m\ s^{-1}$ (c) 9.1 J
(d) 0.46 m (e) 99 per cent; the lost energy is converted into heat
5 (a) 15° (b) 0.44
6 White: 1.1 (1.08) $m\ s^{-1}$; black: 1.4 (1.38) $m\ s^{-1}$
7 (b) (i) Yes (ii) No
8 (a) $0.1\ m\ s^{-1}$ east (b) $0.6\ m\ s^{-1}$ east
9 (a) Momentum is conserved and the net force on the system is zero (b) Kinetic energy is conserved—the collision is elastic (d) The velocity of body 1 relative to body 2 before the collision equals the velocity of body 2 relative to body 1 after the collision. (The two bodies have exchanged relative velocities)
11 1.2 u
12 (a) $0.72\ m\ s^{-1}$ north (b) $0.02\ m\ s^{-1}$ north
13 (b) (i) 0.33 (ii) 1.0 (iii) 0.33 (c) 28 per cent, since the expression derived in (a) is symmetrical in m_1 and m_2
14 (a) 2 (b) 3 (c) $x = 0.51\ m\ s^{-1}$; $y = 0.65\ m\ s^{-1}$; $z = 0.34\ m\ s^{-1}$

SET 22 GRAVITATION IN A NON-UNIFORM FIELD AND SATELLITE MOTION

1 (a) 1 AU is the mean distance from the centre of the Earth to the centre of the Sun (b) (i) 3.16×10^7 s
(ii) 8.61×10^4 s (iii) 2.36×10^6 s
2 (a) (i) $R_p = a/2 - c/2$ or $R_p = \frac{1}{2}(a - c)$
(ii) $R_p = a/2 - ae/2$ or $R_p = \frac{1}{2}a(1 - e)$
(b) (i) $R_a = a/2 + c/2$ or $R_a = \frac{1}{2}(a + c)$
(ii) $R_a = a/2 + ae/2$ or $R_a = \frac{1}{2}a(1 + e)$
(c) (i) $R_{av} = \frac{1}{2}a$ (ii) $R_{av} = c/2e$ (d) (i) 4.43×10^{12} m
(ii) 7.37×10^{12} m (iii) 5.90×10^{12} m
3 (a) 5.07×10^9 m (b) 6 Sun radii
4 (a) (i) 3.63×10^8 m (ii) 4.05×10^8m (b) 4.2×10^7 m
5 $1.96 \times 10^{11}\ m^2\ s^{-1}$
6 (b) 1.66
7 (a) (i) $3.35 \times 10^{18}\ m^3\ s^{-2}$ (ii) $3.34 \times 10^{18}\ m^3\ s^{-2}$
(b) (i) $1.00\ AU^3\ y^{-2}$ (ii) $1.00\ AU^3\ y^{-2}$
8 (a) 1.46×10^8 s (b) 4.62 y
9 (a) 8.61×10^4 s (b) 4.22×10^7 m (c) 3.58×10^7 m
10 (a) 2.7×10^{12} N (b) With an aphelion distance of 5.3×10^{12} m, it moves further away from the Sun than all the planets except Pluto
11 (a) 7.8×10^2 N (b) 1.1×10^2 N
12 10^{-6} or 10^{-7} N
13 (a) 0.44 or 4/9 (b) Zero
14 3.48×10^8 m
15 (b) $9.80\ N\ kg^{-1}$ (c) $8.09\ m\ s^{-2}$
16 (a) 1.5 (b) 0.75
17 248 mm
18 (a) $GM/4\pi^2$ (b) $GM_S/4\pi^2$ (or $1.69 \times 10^{-12}\ M_S$)
19 $T \propto R$
20 1.89×10^{27} kg
21 (a) 2.66×10^7 m (b) 4.22×10^7 m
22 $M_C = 25$; $R_D = 16$; $F_B = 1/16$; $T_C = 125/64$
23 (a) (i) -6.59×10^{28} J (ii) -1.20×10^{31} J
(b) The Earth–Moon system.
24 (a) 7.00×10^3 kg (b) -4.12×10^{11} J
25 (a) GMm/R^2 (b) $-GMm/R$ (c) $GMm/2R$
(d) $-GMm/2R$ (e) $GMm/2R$
26 (a) -5.30×10^{33} J (b) 2.65×10^{33} J
(c) $2.98 \times 10^4\ m\ s^{-1}$ (d) -2.65×10^{33} J
(e) 2.65×10^{33} J
27 (a) 3.26×10^9 J (b) -6.52×10^9 J (c) 3.26×10^9 J
(d) 246 N
28 $\sqrt{GM/R}$
29 (a) $-E$ (b) $3R$ (c) E
30 (a) -3.11×10^{11} J (b) 2.79×10^{11}J
(c) -3.2×10^{10} J (d) 3.2×10^{10} J
31 (a) R (b) $4R$ (c) $\sqrt{4E/m}$ (d) Smaller
32 (a) -3.0×10^{11} J (b) 5.2×10^8 J (c) Gravitational potential energy (d) -3.0×10^{11} J (e) 3.0×10^{11} J
(f) $1.1 \times 10^4\ m\ s^{-1}$
33 (a) (i) GMm/R (ii) $\sqrt{2GM/R}$ (b) Escape speed is independent of the mass of the body
(c) $2.37 \times 10^3\ m\ s^{-1}$
34 (a) $g = GM/R^2$ (b) $U_g = -GMm/R$ (c) $V_g = GM/R^2$ (d) $dV_g/dR = GM/R^2$ (e) $g = dV_g/dR$, i.e. the magnitude of the gravitational field strength at a point equals the gravitational potential gradient at that point
35 (a) $J\ kg^{-1}$ (b) (i) $-6.25 \times 10^7\ J\ kg^{-1}$
(ii) $-5.44 \times 10^7\ J\ kg^{-1}$ (c) $8.1 \times 10^6\ J\ kg^{-1}$
(d) (i) 8.1×10^6 J (ii) 5.8×10^9 J (e) 1.95×10^{10} J of kinetic energy
36 (a) $-E/m$ (b) $-E/m$ (c) $120\ E/m$
(d) (i) $U_g = -4ER/r$ (ii) $V_g = -4ER/mr$
(iii) $dV_g/dr = 4ER/mr^2$ (e) (i) $F_g = 4ER/r^2$
(ii) $g = 4ER/mr^2$ (f) They are equal, as would be expected for the magnitude of the gravitational field strength and the gravitational potential gradient at a point
37 (a) (i) 1.1×10^{10} J (ii) 1.1×10^{10} J
(iii) $2.5 \times 10^5\ J\ kg^{-1}$ (b) $2.5 \times 10^5\ J\ kg^{-1}$
(c) $0.13\ J\ kg^{-1}\ m^{-1}$ (d) $0.129\ N\ kg^{-1}$ (e) At any point in a gravitational field the gravitational potential gradient and the gravitational field strength are equal in magnitude
(f) (i) $0.13\ m\ s^{-2}$ towards the centre of the Earth
(ii) 5.7×10^3 N towards the centre of the Earth

SET 23 ROTATIONAL KINEMATICS

1 (a) $\omega_f = \omega_i + \alpha t$ (b) $\theta = \omega_i t + \frac{1}{2}\alpha t^2$
(c) $\theta = \frac{1}{2}(\omega_i + \omega_f)t$ (d) $\omega_f^2 = \omega_i^2 + 2\alpha\theta$
2 (a) $66\ rad\ s^{-1}$ (b) 99 rad
3 (a) $199.0\ rad\ s^{-1}$ (b) 2.5 rotations s^{-1}
4 (a) $13\ rad\ s^{-2}$ (b) 1.5×10^4 rad
(c) 2.4 (or 2.3) $\times 10^3$ rotations
5 (a) $50.0\ rad\ s^{-1}$ (b) $100\ rad\ s^{-1}$ (c) $11.1\ rad\ s^{-2}$
6 $7.17 \times 10^{-2}\ rad\ s^{-2}$
7 (a) $0.1047\ rad\ s^{-1}$ (b) $1.454 \times 10^{-4}\ rad\ s^{-1}$
8 (a) Angular velocity (b) Angular acceleration
(c) Angular displacement (d) Change in angular velocity
9 (a) 3.0 s (b) $35\ rad\ s^{-1}$ (c) Slowing down (d) -35 rad
(e) 6.0 (5.97) rotations (f) (i) Zero (ii) $5.0\ rad\ s^{-2}$
10 (a) 22 (7π) $rad\ s^{-1}$ (b) $7.0\ m\ s^{-1}$
(c) 2.2×10^2 (70π) rad (d) 70 m
(e) -1.1 ($7\pi/20$) $rad\ s^{-2}$ (f) $-0.35\ m\ s^{-2}$
11 (a) $30\ rad\ s^{-1}$ (b) $3.0\ rad\ s^{-2}$
12 $10^2\ rad\ s^{-1}$

SET 24 ROTATIONAL STATICS AND DYNAMICS

1 4.9×10^2 N
2 40 cm from the end supporting the 180 g mass
3 (a) 375 N (b) 300 N
4 Cedric: 400 N; Edna: 300 N
5 750 N; 1.20 m from rope A
6 (a) $16\ kg\ m^2$ (b) $61\ kg\ m^2$
7 (a) $2.67\ kg\ m^2$ (b) $10.7\ kg\ m^2$ (c) $1.6 \times 10^{-3}\ kg\ m^2$
8 (a) (i) $\frac{1}{2}mr^2$ (ii) $\frac{2}{5}MR^2$ (b) (i) $r/\sqrt{2}$ (or $0.707r$)
(ii) $R\sqrt{2/5}$ (or $0.632R$)
9 $10^{38}\ kg\ m^2$
10 (a) $0.40\ kg\ m^2$ (b) 4.0 N m (or $kg\ m^2\ s^{-2}$)
(c) $10\ rad\ s^{-2}$
11 (a) 0.74 J s (or N m s or $kg\ m^2\ s^{-1}$) (b) $0.12\ kg\ m^2$
12 (a) 20 J s (or N m s or $kg\ m^2\ s^{-1}$) (b) 20 J s (or N m s or $kg\ m^2\ s^{-1}$) (c) $4.0\ kg\ m^2\ s^{-2}$ (or N m)
13 6.0π (or 19) $rad\ s^{-1}$
14 $1.9\ rad\ s^{-1}$
15 (a) 40 J (b) 40 J (c) 0.16 N m (d) 1.3 N
16 $1.5\ kg\ m^2$
17 10 or 1 J
18 2.0×10^3 W
19 (a) 150 J (b) $2.00\ m\ s^{-1}$ (c) $2.00\ m\ s^{-1}$ (d) 6.00 J
(e) 144 J

SET 25 SIMPLE MACHINES

1 8.00
2 (*a*) 48 (*b*) 26 N
3 70 per cent
4 12
5 (*a*) $p = 0.05$; $q = 5$ (*b*) 61.5 per cent
6 90 N
7 900 N
8 17.5 N
9 432 N
10 (*a*) 4.0 (*b*) 80 per cent (*c*) 8.0 N
11 (*a*) 3.0 (*b*) 83 per cent
12 (*a*) 0.43 kN (*b*) 88 per cent
13 2.5×10^2 N
14 (*a*) 28 (*b*) 1.8×10^2 (176) (*c*) 16 per cent
15 25 N
16 (*a*) 3.5×10^2 N (*b*) 1.4

SET 26 FLUID MECHANICS

1 (*a*) 5.0×10^5 Pa (*b*) 5.0×10^5 Pa (*c*) 2.5 N
2 (*a*) 2.8×10^5 Pa (*b*) 97 N (*c*) 0.13 cm
3 (*a*) 1.5×10^{-2} m^3 s^{-1} (*b*) 0.90 m^3 min^{-1}
4 1.1 m s^{-1}
5 0.60
6 0.52 m s^{-1}
7 (*a*) 1.6 m s^{-1} (*b*) 80 kPa
9 (*a*) 4 m s^{-1} (*b*) 0.6 m^3 min^{-1}
10 $N_R = 2V\rho/\pi r t\eta$
11 (*a*) Turbulent (N_R would probably be within the range 8000 to 40 000) (*b*) Laminar ($N_R \approx 14$)
12 1.1×10^{-2} N
13 (*a*) Laminar ($N_R = 2.0 \times 10^2$, which is less than 2000) (*b*) 1.2×10^{-5} m^3 s^{-1} (*c*) 48 Pa
14 (*a*) 1.1×10^{-12} N (*b*) 1.7×10^{-15} N (*c*) Zero (*d*) 1.1×10^{-12} N (*e*) 1.0 mm s^{-1}
15 8.3×10^{-2} Pa s
16 4×10^{-6} m
17 2.3×10^{-2} N
18 (*a*) 7.54×10^{-3} J (*b*) 1.20 Pa
19 1.3×10^3 Pa
20 0.85 mm
21 0.15 m
22 0.547 N m^{-1}
23 24 mm

SET 27 ELASTICITY

1 (*a*) 20 N m^{-1} (*b*) 0.10 m
2 (*a*) 1.6×10^8 Pa (*b*) 6.4×10^{-4} (*c*) 2.5×10^{11} Pa
3 (*a*) 2.0×10^8 Pa (*b*) 4.0 kg
4 Answers should be within these ranges: (*a*) (6.8 to 7.2) $\times 10^{10}$ Pa (*b*) (1.25 to 1.35) $\times 10^8$ Pa (*c*) (*i*) 25 to 27 N (*ii*) 4.5 to 4.7 mm
5 0.245 J
6 (*a*) 3.1×10^{-7} m (*b*) 7.7×10^{-6} J
7 (*a*) 1.4×10^4 N (*b*) 14 mm
8 (*a*) 2.8 km (*b*) At the top
9 (*a*) 5.6 mm (*b*) 7.0 mm
10 5 mm
11 (*a*) 0.20 rad (*b*) 14 mm
12 (*a*) No; the shearing stress is 1.10×10^8 Pa (*b*) 1.3×10^3 kg
13 10^{-1}
14 2.6×10^{10} Pa
15 1.4×10^{-9} m^3

SET 28 INTRODUCTORY RELATIVITY

1 (*a*) 3.0×10^8 m s^{-1} (*b*) 3.0×10^8 m s^{-1} (*c*) 3.0×10^8 m s^{-1}
2 (*a*) 9.1×10^{-31} kg (*b*) 9.2×10^{-31} kg (*c*) 9.1×10^{-30} kg
3 (*a*) 1.8×10^{-16} kg (*b*) 1.3×10^{-15} kg (*c*) 0.20 kg
4 1.1
6 (*a*) 1.0×10^{-14} J (*b*) 1.3×10^{-14} J
7 (*a*) 0.98 g (*b*) About 5500 (to the nearest 500)
8 8.2×10^{-14} J
9 (*a*) 0.025 402 u (*b*) 3.8×10^{-12} J

SET 29 REVISION PROBLEMS FOR PARTS 1 AND 2

1 (*a*) 1.3×10^{-5} m (*b*) 5.0×10^{-7} m
2 (*a*) luna tic (*b*) luna tic^{-1} (*c*) fran tic luna^{-1} (*d*) fran tic luna (*e*) fran luna
3 (*a*) 2.0×10^{-2} m^2 (*b*) 4.0×10^{-9} m^3 (*c*) 2.0×10^{-7} m (*d*) 5.0×10^2 (*e*) 10 m^2
4 (*a*) 12 Hz (*b*) 1/6 s or 0.17 s (*c*) A small black disc in each corner
5 (*a*) 1.0 cm pentatick^{-1} (*b*) (*i*) 1.0 cm pentatick^{-2} (*ii*) 1.0 m s^{-2} (*c*) Average velocity or average speed (*d*) 22 cm
6 (*a*) 2.64 km (*b*) 1.26 km (*c*) 840 m
7 (*a*) 0.2 cm tick^{-2} (*b*) 1.5 bricks (*c*) 3 bands
8 1.6 m s^{-2}
9 (*a*) 12 min (*b*) 2.36 p.m. (*c*) 283 km h^{-1} south-west (*d*) 12.7 km south
10 100 km h^{-1} S 36.9° W
11 E 33° S
12 (*a*) 2.50 m s^{-2} (*b*) 1.25 kN (*c*) 1.75 kN (*d*) Zero (*e*) 500 N
13 (*a*) 5.0×10^2 N (*b*) 5.4×10^2 N
14 49 m s^{-2}
15 (*a*) 78 m (*b*) 39 m s^{-1} (*c*) 20 m s^{-1} (*d*) 28 m s^{-1} (*e*) 2.0 s
16 (*a*) $x = 49t^2/16$; $y = 4.9t^2$ (*b*) $5y = 8x$, indicating the path is a straight line
17 (*a*) 1.5 N (*b*) 1.2 N (*c*) 22 N (*d*) 13 N
18 (*a*) 511 N (*b*) 9.87 m s^{-1}
19 (*a*) 0.540 s (*b*) 1.69 m s^{-2} (*c*) 115 N
20 (*a*) 0.710 s (*b*) 3.92×10^{-2} N (*c*) 12.5 cm
21 (*a*) 4.0 s (*b*) 2.0 kg (*c*) 1.5 N (*d*) 4.5 J (*e*) 9.0 m
22 (*a*) 5.0 m s^{-1} (*b*) E 37° S (*c*) 12 N s
23 (*a*) 14.5° (*b*) 29.4 J
24 (*a*) 0.490 N towards the support; 0.849 N along the tangent (*b*) 0.245 J; 0.221 kg m s^{-1} (*c*) None
25 (*b*) About 6.8 J (*c*) 4.0 J (*e*) About 6 J and 9.0 J
26 (*a*) 20.0 m s^{-1} (*b*) 120 m (*c*) 118 J
27 (*a*) Both balls have velocities of 10 cm s^{-1} in opposite directions (*b*) Larger ball at rest; smaller ball at 20 cm s^{-1}
28 2.83 m s^{-1} east
29 (*a*) $\frac{3}{4}$ (*b*) 9/16 (*c*) 1 (*d*) B (*e*) B (*f*) C (*g*) $GMN(q-p)/pq$
30 2.94×10^{-2} N m^{-1}
31 12.7 mm
32 5.5×10^{-2} s

SET 30 WAVES IN ONE DIMENSION

1 (*b*) Zero; 0.30 m s^{-1} downwards; 0.10 m s^{-1} upwards (*c*) An actual pulse could not have a vertical edge, and would have rounded edges
2 (*a*) A: up; B: down; C: up (*b*) B (*d*) 4.0 m s^{-1} downwards
3 (*a*) 250 Hz (*b*) 4.00×10^{-3} s
4 2.5 Hz
5 48 m s^{-1}
6 1.5 m
7 (*a*) 0.15 m (*b*) 2 Hz (*c*) 0.5 s (*d*) 3 m
8 (*a*) 2 mm (*b*) 0.25 m

9 (*a*) *x-z* plane (*b*) Negative *x*-direction (*c*) 0.4 m (*d*) 5 ms (*e*) 80 m s^{-1} (*f*) 3 mm (*g*) 2 mm
10 $y = 0.12 \sin(80\pi t - 10\pi z)$ or $y = 0.12 \sin(2.5 \times 10^2 t - 31z)$
11 (*a*) Same way up (*b*) Inverted (*c*) Reflected pulse
12 (*a*) C (*b*) Same way up (*c*) Inverted
13 (*a*) $x = V/2$ (*b*) $d = S/2$ (*c*) $x - d$ (*d*) $x + d$ (*e*) 0.67 or 2/3
15 (*a*) 10 cm (*b*) 60 cm (*c*) 25 Hz
16 3.5 Hz
17 (*a*) 10 m (*b*) 0.36 m
18 (*a*) M moves in simple harmonic motion (*b*) 3.0 cm (*c*) 1/30 s (or 0.033 s) (*d*) 1.1×10^5 cm s^{-2} (*e*) 5.7×10^2 cm s^{-1} (*f*) 80 cm

SET 31 WAVES IN TWO DIMENSIONS

1 (*a*) 0.10 s (*b*) 10 Hz (*c*) 40 mm (*d*) 0.40 m s^{-1}
2 (*a*) GO (*b*) (*i*) FO (*ii*) OD (*c*) (*i*) 20° (*ii*) 20°
3 (*a*) (*i*) 12 mm (*ii*) 12 mm (*iii*) 24 mm (*b*) 24 mm (*c*) (*i*) 12 mm (*ii*) 48 mm s^{-1}
4 (*a*) h_1/h_2 (*b*) 1 (*c*) h_1h_2
5 (*a*) 1.5 cm (*b*) 18 cm s^{-1} (*d*) 12 Hz (*e*) 12 cm s^{-1} (*f*) 1.5 (*g*) 2.0 Hz (*h*) A stationary pattern with crests half as far apart as before
6 (*a*) 1.3 (*b*) 1.3 (*c*) 9.0 Hz; 4.5 Hz (*d*) 36° (*e*) 48° (*f*) Yes
7 (*a*) Increased (*b*) Decreased
8 (*a*) 1.5 Hz (*b*) 0.60 m s^{-1} (*c*) Maximum disturbance (*d*) 19.1 m
9 (*a*) 0.20 m (*b*) Antinode (*c*) 0.10 m (*d*) 10 (*e*) 0.10 m
10 (*a*) 8.0 s (*b*) 2.4 m
11 (*a*) 0.091 (*b*) 6
12 0.18
13 (*a*) 4 (*b*) 25°
14 (*a*) 5.0 mm (*b*) 0.20 m
15 (*a*) The nodal lines (*b*) (*i*) Downwards (*ii*) Downwards (*iii*) Upwards

SET 32 REVISION PROBLEMS

1 (*a*) 20 mm (*b*) 0.2 s (*c*) 0.1 m
3 (*a*) 16 m (*b*) 48 m s^{-1} (*c*) 5.0 m (*d*) 5.0 Hz
4 (*a*) 5.0 Hz (*b*) 24 Hz (*c*) 0.75 cm (*d*) 18 cm s^{-1} (*e*) Same frequency as before, since the frequency of the water waves will be unchanged
5 (*a*) 1.2 (*b*) 1 (*c*) 1.2
6 (*a*) SQ (*b*) 1 (*c*) 0.80
7 (*a*) 20 m (*b*) N 6° E (*c*) 1.0×10^2 m

SET 33 TRANSMISSION OF SOUND

1 1.7 km
2 3.5×10^2 m s^{-1}
3 3.034×10^{-3} s
4 523.3 Hz
5 10^7 or 10^8 waves
6 0.840 m
7 (*a*) 575 Hz (*b*) 575 Hz (*c*) 2.59 m
8 4.0 kHz
9 344 m s^{-1}
10 58°C
11 (*a*) 1.49 km s^{-1} (*b*) 13°
12 (*a*) Greater than one (*b*) One (*c*) Greater than one
13 3.0×10^{-7} W m^{-2}
14 (*a*) 0.98 m (*b*) 3.5 mW
15 (*a*) 60 db (*b*) 3.2×10^{-3} W m^{-2}
16 (*a*) 83 db (*b*) 2.0×10^{-3} W m^{-2}
17 2.0×10^2
18 1.7×10^2 m
19 (*a*) 0.34 m (*b*) 0.04 m (*c*) 0.30 m (*d*) 1.1 kHz (*e*) 8.9×10^2 Hz
20 (*a*) 562 Hz (*b*) 470 Hz
21 (*a*) 1.0×10^3 waves (*b*) 1.0×10^2 waves (*c*) 1.1 kHz (*d*) 9.0×10^2 Hz
22 (*a*) 435 Hz (*b*) 365 Hz
23 28.3 m s^{-1}

SET 34 SOUND FROM VIBRATING STRINGS

1 1.2×10^2 m s^{-1}
2 0.10 g
3 82.5 Hz
4 123 Hz
5 0.75
6 0.45 mm
7 (*a*) 280 Hz (*b*) 560 Hz (*c*) 1.12 kHz (*d*) 1.40 kHz
8 $f = \frac{1}{2}n\sqrt{T/mL}$
9 256 Hz

SET 35 SOUND FROM VIBRATING AIR COLUMNS

1 (*a*) 4.80 m (*b*) (*i*) 70 Hz (*ii*) 210 Hz (*iii*) 350 Hz (*c*) (*i*) Third harmonic (*ii*) Fifth harmonic (*iii*) Thirteenth harmonic
2 (*a*) 1.04 m (*b*) 339 m s^{-1}
3 0.261 m
4 (*a*) 34.0 cm (*b*) 136 cm (*c*) 344 m s^{-1}
5 580 Hz
6 (*a*) 368 m s^{-1} (*b*) 435 Hz
7 (*a*) 0.80 m (*b*) (*i*) 430 Hz (*ii*) 860 Hz (*iii*) 1.29 kHz (*iv*) 2.58 kHz (*c*) (*i*) Second harmonic (*ii*) Third harmonic (*iii*) $(n + 1)$th harmonic
8 (*a*) 88.0 Hz (*b*) 1.88 m
9 353 m s^{-1}; 2.5 cm
10 2
11 1.6
12 Open pipe; 0.885 m

SET 36 INTERFERENCE EFFECTS WITH SOUND

1 (*a*) 1.60 m (*b*) 336 m s^{-1}
2 1.0 kHz
3 (*a*) 0.30 m (*b*) 11.7 m
4 (*a*) $\lambda/2$ (*b*) 1.28×10^3 m s^{-1}
5 (*a*) 3.1 m (*b*) 14° (*c*) 2.0 m
6 (*a*) 1.5 m (*b*) 6 (*c*) 10°
7 251.2 Hz; 248.8 Hz
8 303 Hz
9 501 Hz
10 5.2 Hz

SET 37 REVISION PROBLEMS

1 15 m
2 4.4 m
3 (*a*) 520 Hz (*b*) 480 Hz
4 0.41 s
5 (*a*) 480 Hz and 720 Hz (*b*) 9.00
6 15 N
7 0.27
8 339 m s^{-1}
9 (*a*) (*i*) 340 Hz (*ii*) 680 Hz (*b*) (*i*) 170 Hz (*ii*) 510 Hz
10 +12 Hz
11 1.7
12 (*a*) 0.60 m and 1.2 m (*b*) $0.24n - 0.12$
13 3.6 Hz
14 8.82×10^3 kg m^{-3}

SET 38 TRANSMISSION OF LIGHT

1 2.51 s
2 (*a*) 1.9×10^4 Hz (*b*) 2*N*, 3*N* and 7*N*
3 2.6 m
4 (*a*) B E (*b*) Both are $26\frac{1}{2}$° (*c*) 80 cm
5 0.80 m
6 B
7 8°
9 (*a*) 8.8×10^2 lm (*b*) 9.6 lx (*c*) 0.38 lm
10 (*a*) 1.5×10^3 lm (*b*) 5.4×10^2 lx
11 (*a*) 0.707 m (*b*) 42, 43 or 44 lx (depending on method) (*c*) Maximum
12 14 cm
13 10 per cent
14 27 s

SET 39 CURVED MIRRORS

1 (*a*) Same side (*b*) 30 cm (*c*) Real (*d*) Inverted (*e*) 0.50
2 4.0 cm high, real, inverted, 30 cm in front of the mirror
3 (*a*) Inverted (*b*) Real (*c*) 1.0 (*d*) 20 cm (*e*) About 40 cm
4 (*a*) Virtual, upright, enlarged, on the opposite side of the mirror to the whisker (*b*) 72 cm (*c*) 3.0
5 (*a*) C (*b*) B (*c*) D
6 YU
7 (*a*) Object: 24 cm; image: 12 cm (*b*) Object: 4.0 cm; image: 8.0 cm
8 Focal length: 1.5 cm
9 In the principal focal plane 15.0 m from the mirror; 0.139 m
10 (*a*) 15 cm (*b*) Opposite side (*c*) Virtual (*d*) Upright (*e*) 0.25
11 (*a*) 1.8 cm high, virtual, upright, 12 cm behind the mirror
12 (*a*) Virtual, upright (*b*) 50 cm (*c*) 10 cm behind the mirror
13 70 cm
14 Concave mirror; 26.7 cm
15 15 cm

SET 40 REFRACTION AT A PLANE INTERFACE

1 (*a*) 60° (*b*) 60° (*c*) 41° (*d*) 19°
2 (*a*) 1.60 (*b*) 1.60 (*c*) 0.625
3 40°
4 (*a*) 1.25 (*b*) 9.00 cm (*c*) 37.8° (*d*) 92.2°
5 1.49
6 19.5°; 10.5°
7 7.3°
8 8.6 mm
9 1.73
10 5.3 cm
11 1.6
12 (*a*) Violet light (*b*) 24.0° (*c*) 23.7° (*d*) 0.3°
13 (*a*) Blue (*b*) Greater (*c*) Blue (*d*) Orange (*e*) n_b, n_y, n_o
14 (*a*) Angle C (*b*) (*i*) 16° (*ii*) 29° (*iii*) 48° (*iv*) 28°
15 38°
16 37°
18 48.8°
19 62.5°
20 1.49
21 (*a*) OD, OB (*b*) OD (*c*) 1.52
22 (*a*) F C (*b*) C (*c*) J C (*d*) 71.2°
23 (*a*) (*i*) 42° (*ii*) 48° (*iii*) Only 90°, which is the maximum value for any angle of incidence (*b*) (*i*) Outside wall (*ii*) Thicker fibre (*iii*) Sharper curve
24 10 cm
25 Totally reflected

SET 41 LENSES

1 (*a*) Opposite side (*b*) (*i*) Inverted (*ii*) Diminished (*iii*) Real
2 (*a*) Inverted, enlarged, real (*b*) Inverted, diminished, real (*c*) Inverted, same size as the object, real
3 (*a*) Inverted, diminished, real (*b*) 15 cm (*c*) Opposite side (*d*) 0.50
4 (*a*) Real (*b*) Inverted (the colour slides are inserted in the projector upside down) (*c*) 6.50 m (*d*) 0.875 m
5 C
6 (*a*) 80 mm (*b*) 1.0
7 C E
8 (*a*) 16.5 cm (*b*) 1.65 cm
9 (*a*) 20 cm (*b*) 1.8 mm (*c*) 10^{-9}
10 (*a*) 2.0 cm (*b*) 2.2 cm
11 5.0 cm
12 (*a*) 10 cm (*b*) Slightly less than 10 cm
13 (*a*) 6.0 cm (*b*) Same side (*c*) Virtual (*d*) Upright (*e*) 0.40
14 (*a*) Virtual and upright (*b*) Same side (*c*) 12 cm (*d*) 8 cm
15 B C F H I K
16 (*a*) +4.0 m^{-1} (*b*) −5.0 m^{-1}
17 (*a*) The +20 m^{-1} lens (*b*) 40 cm

18 21 cm
19 74 cm
20 $R/(n-1)$
21 (*a*) Red light (*b*) 1.03
22 21.9 cm
23 Focal length is 40 cm; distance from A is 5 cm
24 10.3 cm
25 40 cm
26 15 cm

SET 42 SIMPLE OPTICAL INSTRUMENTS

1 (*a*) Between X and Y, 15 cm from X (Was your answer ambiguous?) (*b*) 0.50 (*c*) 40 cm from Y on the side opposite to X (*d*) 0.83 (*e*) Right way up
2 (*a*) 100 cm from P on the side opposite to the bulb (*b*) 23 cm from Q on the side opposite to the bulb (*c*) Real and inverted
3 8.8 cm from the concave lens on the side closer to the convex lens; 0.040
4 60 cm behind the lens
5 10 cm
6 (*a*) 0.24 cm (*b*) 0.48 cm
7 (*a*) 15 cm (convex) (*b*) 15 cm (concave) (*c*) A concave lens of focal length 30 cm
8 (*a*) 36 cm (*b*) 18 cm (*c*) Concave
9 (*a*) 3.0 cm (*b*) 2.7 cm
10 (*a*) Concave lenses (*b*) 50 cm
11 1.5 m
12 (*a*) Concave lens of focal length 1.8 m (*b*) Convex lens of focal length 50 cm
13 77 cm
14 21 cm
15 20 cm
16 10 cm from the mirror on the side opposite to the lens and 40 cm from the lens on the side opposite to the mirror
17 (*a*) 1.00 m (*b*) 10 cm (*c*) 1.10 m (*d*) 10
18 (*a*) 84 cm (*b*) 7 cm
19 (*a*) 7.1 cm (*b*) 1.00 m (*c*) 14
20 (*a*) 5.0 (*b*) 32 cm (*c*) 1.52 m (*d*) 5.0
21 (*a*) 20 (*b*) 2.50 m (*c*) The eyepieces are moved towards the objective
22 (*a*) 6.0 (*b*) 70 cm (*c*) 98 cm
23 (*a*) 2.4 cm (*b*) 3.6 cm (*c*) The opera glasses give an upright image whereas the simple telescope would give an inverted image
24 (*a*) 40 (*b*) 102.5 cm
25 (*a*) 9.2×10^{-3} rad; 0.53° (*b*) 0.15 m (*c*) 300 (*d*) 2.7 or 2.8 rad (dependent on method)
26 (*a*) 5.6 m (*b*) 6.8×10^{6} m
27 (*a*) 250 mm (*b*) 4.0 (*c*) 270 mm (*d*) 10 (*e*) 40
28 (*a*) 3.3 cm (*b*) 20
29 3.20 cm and 11.2 cm

SET 43 SIMPLE WAVE PROPERTIES OF LIGHT AND OTHER FORMS OF ELECTROMAGNETIC RADIATION

1 3.0×10^{8} m s^{-1}
2 (*a*) 4.69×10^{14} Hz (*b*) 2.13×10^{-15} s
3 Red, orange, yellow, green, blue, violet
4 Gamma rays, x-rays, ultraviolet light, visible light, infrared light, microwaves, radio waves
5 (*a*) Visible light (*b*) Microwaves (*c*) X-rays (*d*) Radio waves (*e*) All types of electromagnetic radiation (*f*) All types of electromagnetic radiation
6 (*a*) 5.20×10^{14} Hz (*b*) (*i*) 5.20×10^{14} Hz (*ii*) 356 nm (*iii*) 1.85×10^{8} m s^{-1}
7 (*a*) 7.41×10^{-7} s (*b*) 222 m (*c*) 2.0×10^{8} m s^{-1}
8 (*a*) (*i*) S (*ii*) S (*iii*) X (*iv*) X (*b*) 1.34
9 (*a*) Away from us (*b*) 1.40×10^{7} m s^{-1}
10 2.4 nm
11 Approximately 9 Hz
12 (*a*) Light waves between the threads of the handkerchief (*b*) Red light (*c*) Radio waves
13 (*a*) B (*b*) W (*c*) B (*d*) W (*e*) W
14 (*a*) 1.25×10^{8} m s^{-1} (*b*) 7.20×10^{8} m s^{-1}

SET 44 INTERFERENCE AND DIFFRACTION WITH ELECTROMAGNETIC RADIATION

1 1 km
2 (*a*) 3.0 cm (*b*) 10^{10} Hz
3 0.40 μm and 0.60 μm
4 (*a*) 38.9 cm (*b*) C D
5 (*a*) $7q$ (*b*) $7/9$
6 (*a*) A bright band centred on G (*b*) A dark line centred on K (*c*) 6.0×10^{-4} m
7 (*a*) 3.60 mm (*b*) 4.80 mm
8 2.21 mm
9 630 nm
10 420 nm
11 (*a*) $\frac{1}{2}$ (*b*) 19°
12 (*a*) 7 (*b*) 0.25
13 (*a*) 0.44 mm (*b*) $\frac{1}{2}$ (*c*) X is on the third bright band from C
14 (*a*) 4, including one at P (*b*) $5 \times 10^{3}\,\lambda$
15 (*a*) $D = A(n-1)$ (*b*) 0.12 mm
16 (*a*) Bright light (*b*) Bright light (*c*) Dark line (*d*) Bright light (*e*) Dark line
17 (*a*) (*i*) λ (*ii*) $5\lambda/2$ (*iii*) 3λ (*b*) 4.2 mm
18 (*a*) 1.40 μm (or 1400 nm) (*b*) 90 mm (*c*) 2.1×10^{-4} m
19 (*a*) 36 mm (*b*) $R/2$
21 36 μm
22 (*a*) 1.5×10^{10} m (*b*) 69 m
23 6 km
24 B C E
25 392 nm
26 Second order
27 600 lines mm^{-1}
28 4
29 393 nm
30 (*a*) Yes (*b*) (*i*) No (*ii*) No (*c*) (*i*) Blue (*ii*) Black (*d*) 90 nm (*e*) $3\lambda'/4$
31 (*a*) (*i*) Out of phase (*ii*) In phase (*b*) Reflected light (*c*) 590 nm (*d*) b and d
32 (*a*) 399 nm (*b*) 100 nm
33 (*a*) Dark band (*b*) 295 nm (*c*) 92 fringes (*d*) 27 μm
34 0.24 mrad, 0.014°
35 44 μrad
36 (*a*) (*i*) $\lambda/4$ (*ii*) $3\lambda/4$ (*iii*) $\lambda/2$ (*iv*) λ (*b*) $\lambda/2$ (*c*) 7λ (*d*) $d = m\lambda/2$ (*e*) $R = (r^2 + d^2)/2d$ or, better still, $R \approx r^2/2d$, since $r^2 \gg d^2$ (*f*) $R \approx r^2/m\lambda$
37 (*a*) 698 mm (*b*) 670 nm
38 391 mm
39 1.42
40 (*a*) $2d \sin\theta$ (*b*) $2d \sin\theta = m\lambda$, where $m = 0, 1, 2, \ldots$ (*c*) 22° and 48°
41 18°, 38°, 66°
42 0.314 nm
43 74.8 μm
44 176 μm

SET 45 WAVE-PARTICLE DUALITY OF ELECTROMAGNETIC RADIATION

1 (*a*) B C (*b*) D (*c*) D
2 (*a*) 4×10^{-15} eV s (*b*) 1.0 eV (*c*) (*i*) 2.0 eV (*ii*) 2.0 V
3 (*a*) B (*b*) C
4 (*a*) 0.12 eV (*b*) 4.78 eV
5 (*a*) 2.1 eV (*b*) 6.6×10^{5} m s^{-1}
6 (*a*) 3.3 eV (*b*) 1.2 eV (*c*) 6.8×10^{5} m s^{-1}
7 (*a*) 4.6×10^{-19} J (*b*) 2.0×10^{-19} J (*c*) 2.6×10^{-19} J (*d*) 7.7×10^{5} m s^{-1}
8 3.4×10^{-19} J
9 1.05 MHz
10 (*a*) 9.4×10^{9} Hz (*b*) 6.2×10^{-24} J
11 (*a*) 10^{-18} J (*b*) 10^{20} photons s^{-1}
12 (*a*) $E = h\nu$ (*b*) $E = hc/\lambda$ (*c*) $k = 1.99 \times 10^{-25}$ J m (*d*) $E = 1.24 \times 10^{3}/\lambda$
13 (*a*) Approximately 3.10 to 1.65 eV (*b*) 1.00 nm
14 1.05×10^{-27} kg m s^{-1} west
15 3.00×10^{-13} m
16 (*a*) $p = h\nu/c$ (*b*) 1.82×10^{-27} kg m s^{-1} (*c*) $p = E/c$
17 (*b*) 6.60×10^{-24} kg m s^{-1} east, 6.60×10^{-24} kg m s^{-1} north (*c*) Answer should be within the range (9.32 to 9.34) $\times 10^{-24}$ kg m s^{-1} south-east (*d*) Answer should be within the range (4.76 to 4.80) $\times 10^{-17}$ J (*e*) 1.98×10^{-15} J (*f*) 1.93×10^{-15} J (*g*) 2.5 per cent error (*h*) 3 pm
18 (*a*) 0.620 (or 0.622) pm (*b*) 1.07×10^{-21} kg m s^{-1} north (*c*) 5.35×10^{-22} kg m s^{-1} N 60° W (*d*) 1.24 pm (*e*) 0.62 pm
19 (*a*) (*i*) Einstein showed that for speeds approaching that of electromagnetic radiation the effective increase in mass with speed was given by this relation (*ii*) Conservation of energy (*iii*) Conservation of momentum in the x-direction (*iv*) Conservation of momentum in the y-direction
20 172 keV
21 (*a*) $h\nu/c$ east (*b*) $2h\nu/c$ west (*c*) $2NAh\nu/c$ west (*d*) $2NAh\nu/c$ west (*e*) $2NAh\nu/c$ east (*f*) $2Nh\nu/c$
22 (*a*) $E_T = Nh\nu$ (*b*) $2E_1/c$ This expression for the pressure would still hold if the average frequency of the polychromatic light equalled the frequency of the monochromatic light (*c*) 10^{-5} Pa (*d*) 10^{10} greater
23 (*a*) 1.21×10^{-27} kg m s^{-1} (*b*) 1.21×10^{-27} kg m s^{-1} (*c*) 3.62×10^{-19} J (*d*) 1.7×10^{20} photons s^{-1} (*e*) 1.7×10^{20} photons s^{-1} (*f*) 5.5×10^{-4} (*g*) 9.4×10^{16} photons s^{-1} (*h*) 1.1×10^{-10} kg m s^{-1} (*i*) 1.1×10^{-10} N (*j*) 1.1×10^{-10} N (*k*) 1.8×10^{-9} Pa
24 Approximately 10^{13} W
25 1.02 MeV
26 (*a*) 0.58 MeV (*b*) 1.60 MeV
27 B C D F
28 (*a*) F (*b*) D
29 (*a*) Lower frequency (*b*) Shorter wavelength
30 B C F

SET 46 REVISION PROBLEMS

1 268 Hz
2 *(a)* 26.7 W *(b)* 113 (112.5) cm *(c)* 1/4
3 *(a)* 16:1 *(b)* 256:1 *(c)* 2.8 m
4 *(a)* B C E H O *(b)* B C E G *(c)* A D F I K
5 *(a)* Inverted *(b)* 20 cm
6 *(a)* Larger *(b)* 1.6 cm
7 *(a)* Medium X *(b)* 0.58 *(c)* 1.6 *(d)* First interface
8 37 cm
9 1.096 cm
10 28.2°
11 *(a)* A D E I L *(b)* B C F G
12 Diverging; 7.5 cm
13 1.5
14 *(a)* 2.11 m *(b)* 0.031 *(c)* *(i)* In *(ii)* 0.20 cm
15 30 cm
16 *(a)* 1.34 m *(b)* 10.5 cm
17 17 cm
18 *(a)* *(i)* 30 cm and 1.0 cm *(ii)* 1.0 cm and 3.0 cm *(b)* *(i)* 30 *(ii)* 1.3×10^2
19 *(a)* Glass *(b)* S *(c)* Glass *(d)* S
20 A C D
21 *(a)* 0.25 m *(b)* If the point referred to is, say, P, then the path difference from the two sources = $PS_2 - PS_1$ = (5.0 − 3.0) m = 2.0 m, which is exactly two wavelengths. Thus there would be a maximum at P
22 *(a)* 0.10 *(b)* 9.0×10^{-7} m *(c)* 3.6×10^{-7} m *(d)* 0.063 (1/16)
23 0.19 mm
24 *(a)* 92 mm *(b)* 23 mm
25 Violet: 4.6°; red: 8.6°
26 *(a)* 3.0×10^{-7} m *(b)* 7.5×10^{-8} m
27 3.84 mm
28 1.50 eV
29 *(a)* 606 nm *(b)* 2.05 eV *(c)* 1.09×10^{-27} kg m s^{-1}

SET 47 THERMAL EXPANSION OF SOLIDS AND LIQUIDS

1 *(a)* *(i)* 273.15 K *(ii)* −182.96°C *(b)* $T = \theta + 273.15$
2 *(a)* 419.58°C *(b)* 373.15 K
3 264 K
4 20°C
5 1.9×10^{-5} K^{-1}
6 0.75 m
7 402.1 mm
8 13 mm
9 1999.5 m; a distance tolerance of 0.5 m in a 2000 m event would probably be acceptable
10 *(a)* *(i)* $T_2/T_1 = \sqrt{l_2/l_1}$ *(ii)* $T_2/T_1 = \sqrt{1 + \alpha\Delta\theta}$
11 *(a)* 8.2 s *(b)* Slow
12 *(a)* *(i)* $L(1 + \alpha\Delta\theta)$ *(ii)* $W(1 + \alpha\Delta\theta)$ *(iii)* $LW(1 + \alpha\Delta\theta)^2$
13 *(a)* *(i)* 0.19 mm *(ii)* 17 mm^2 *(b)* −0.67 per cent
15 *(a)* 1°C (or K) *(b)* 10^{-4} K^{-1} (you could possibly justify an answer of 10^{-3} K^{-1}) *(c)* 10^{-4} K^{-1} (or perhaps 10^{-5} K^{-1}) *(d)* 10^{-4} m (or 10^{-5} m) *(e)* Zero *(f)* 10^{-5} m^3 (or 10^{-4} m^3) *(g)* 10^{-1} kg m^{-3} (or 10^{-2} kg m^{-3})
16 *(a)* 0.017 per cent *(b)* 0.017 per cent *(c)* 0.017 per cent
17 1.354×10^4 kg m^{-3}
18 748.8 mm
19 *(a)* 6.0×10^{-7} m^3 (volume of mercury in the capillary tube at 20°C is only 1.6×10^{-9} m^3) *(b)* *(i)* 1.3×10^{-9} m^3 *(ii)* 1.3×10^{-9} m^3 *(iii)* 25 mm *(iv)* 5 mm *(c)* *(i)* 8.6×10^{-9} m^3 *(ii)* 165 mm *(iii)* 170 mm
20 0.45 litre
21 244 cm^3 (strictly 244.14 cm^3)

SET 48 THERMAL EXPANSION OF GASES AND THE IDEAL GAS LAWS

1 105 kPa
2 *(a)* 101 kPa *(b)* 0.133 kPa
3 *(a)* 810 mmHg *(b)* 840 mmHg *(c)* 270 mm
5 *(a)* 748 mmHg *(b)* 99.5 kPa
6 *(a)* 350 kPa *(b)* 0.098 m^3
7 15 cylinders
8 150 cm^3
9 *(a)* 273°C *(b)* −182°C
10 *(a)* Up *(b)* 33°C *(c)* 108 kPa
11 10 m^3
12 0.90 MPa
13 26°C
14 *(a)* 350 kPa *(b)* 376 kPa *(c)* 46°C
15 150 kPa
16 *(a)* 80 u *(b)* 2.4×10^{24} u *(c)* 4.0 g *(d)* 27 g
17 *(a)* *(i)* 20 u *(ii)* 20 u *(iii)* 80 u *(b)* *(i)* 20 g *(ii)* 90 g
18 *(a)* 32 *(b)* 96 u *(c)* *(i)* 16 g *(ii)* 32 g *(iii)* 8.0 g
19 10^{-21} mol of students, unless your school is much smaller or much larger than most
20 *(a)* 18 *(b)* *(i)* 5.00 mol *(ii)* 2.0 mol *(c)* 9.0×10^{24} atoms (or 15 mol of atoms)
21 *(a)* 29 (or 28.8) *(b)* 10 or 10^2 m^3 *(c)* 10 or 10^2 kg *(d)* 10^3 or 10^4 mol *(e)* 10^{27} or 10^{28} molecules
22 *(a)* 0.0620 mol *(b)* 20.0 *(c)* Neon
23 *(a)* $PV = mRT/M$ *(b)* $P = \rho RT/M$
24 3.9 kg
25 3.2×10^{-2} m^3
26 0.325 kg m^{-3}
27 0.13 g
28 72.9 kpa
29 *(a)* *(i)* 3.00 *(ii)* 3.00 *(b)* 1.38 *(c)* 6.00 *(d)* 0.778
30 117 kPa
31 *(a)* 105 kPa *(b)* 70 kPa
32 128 kPa and 64.0 kPa
33 20 kPa
34 0.305 g
35 *(a)* 76.6 per cent *(b)* 1.29×10^{-2} kg m^{-3}
36 74.5 per cent
37 -1.63×10^{-2} m^3
38 *(a)* 6.0 m s^{-1} *(b)* 8.0 m s^{-1} *(c)* 5.0 m s^{-1} *(d)* 6.8 m s^{-1}
39 6.21×10^{-21} J
40 *(a)* *(i)* 6.00×10^{-21} J *(ii)* 6.00×10^{-21} J *(iii)* 6.00×10^{-21} J *(iv)* 7.66×10^{-21} J *(b)* 5.32×10^{-26} kg *(c)* 475 m s^{-1} *(d)* Rotational kinetic energy, vibrational kinetic energy and vibrational potential energy
41 *(a)* 0.5 *(b)* 2 *(c)* 2 *(d)* 1 *(e)* 2
42 *(a)* 0.5 *(b)* 1 *(c)* 1.4 ($\sqrt{2}$)
43 *(a)* $\bar{v} = 0.921\sqrt{\overline{v^2}}$ or $\bar{v} = 2\sqrt{2\overline{v^2}/3\pi}$ *(b)* 6.62×10^{-26} kg *(c)* 1.81×10^5 m^2 s^{-2} *(d)* 426 m s^{-1} *(e)* 392 m s^{-1}
44 *(a)* 0.42 (5/12) *(b)* 0.60 (3/5) *(c)* 0.77 ($\sqrt{3/5}$) *(d)* 0.77 ($\sqrt{3/5}$) *(e)* 0.50
45 T
46 *(a)* A: 0.412 km s^{-1}; B: 1.37 km s^{-1} *(b)* Gas B, because at any time a higher proportion of its molecules will have speeds greater than the speed required to escape from the Earth's gravitational field. (Actually, gas A is carbon dioxide, while gas B is helium.)
47 *(a)* $3PV/2$ *(b)* 2.9×10^3 J
48 10^{-8} (or 10^{-9}) m
49 *(a)* 3.9×10^{19} m^3 *(b)* 1.8 mg
50 *(a)* 2.50×10^{25} molecules m^{-3} *(b)* 1×10^{-7} m *(c)* 5×10^2 m s^{-1} *(d)* 5×10^9 Hz
51 *(a)* $P = \frac{1}{3}\overline{v^2}/3$ *(b)* $\overline{v^2} \propto 1/T$, so T does not appear in the expression
52 462 m s^{-1}
54 4.0

SET 49 CHANGE OF TEMPERATURE AND CHANGE OF PHASE

1 5.0×10^2 J K^{-1}
2 30 kJ
3 370°C
4 21 kJ
5 4.1 kJ kg^{-1} K^{-1}
6 10^2 cups of tea
7 11 K or 11°C
8 40°C
9 6.0 MJ
10 1.2×10^3 m s^{-1}
11 *(a)* 0.632 K or °C *(b)* Lower, since on the way down some energy will be transferred to the air through which the water is falling, and some of the energy of the fallen water will be quickly shared with the large body of water at the bottom of the falls
12 *(a)* 21 J K^{-1} *(b)* 5.0 g
13 25°C
14 *(a)* 38.7 g *(b)* Greater
15 32°C
16 4.9×10^2 J kg^{-1} K^{-1}
17 1 or 10 K (or °C)
18 8.4×10^5 J kg^{-1}
19 *(a)* 2.26×10^6 J *(b)* 18 min
20 *(a)* 2.5×10^8 J *(b)* $900
21 0.60 kg
22 2.3×10^6 J kg^{-1}
23 *(a)* *(i)* 1.67×10^3 J min^{-1} *(ii)* 80.2 min *(b)* Higher
24 19 W
25 *(a)* 20°C *(b)* Specific latent heat of vaporisation *(c)* Specific heat capacity as a liquid *(d)* 20 kJ kg^{-1} *(e)* 1.2 kJ kg^{-1} K^{-1}
26 10°C
27 32 g
28 *(a)* 10°C *(b)* Higher
29 0°C
30 68°C
31 2
32 15.9°C
33 *(a)* 26.0°C *(b)* 36.7°C
34 *(b)* 30.0°C *(c)* 0.25°C min^{-1} *(d)* 0.13°C min^{-1} *(e)* +0.52°C *(f)* 2.4 kJ kg^{-1} K^{-1}
35 +0.094 (3/32) °C
36 *(a)* Amount of heat energy required to raise the temperature of one mole of a substance by one kelvin *(b)* $c_M = c_p M$ *(c)* J mol^{-1} K^{-1} *(d)* 24.1, 24.2, 25.0, 24.4, 25.1 *(e)* About 40 kJ
37 1.4×10^2 (138) J kg^{-1} K^{-1}

SET 50 THERMODYNAMICS

1 Temperature
2 Neither
3 55 J
4 (*a*) 600 K (*b*) The work done by the expanding gas (*c*) 10^2 J
5 10^5 or 10^6 J
6 (*a*) 2.0 kPa (*b*) 4.0 mm (*c*) The fact that the diameter of the piston was 24 mm
7 (*a*) (*i*) IV (*ii*) II (*iii*) III (*iv*) I (*b*) (*i*) 3.0×10^3 J (*ii*) Zero (*iii*) -2.9×10^3 J
8 340 mJ
9 (*a*) 835 J (*b*) 11.3 kJ (*c*) 10.4 kJ
10 (*a*) The joule
12 (*a*) 6.0 kJ (*b*) 6.0 kJ (*c*) -0.16 J (*d*) 6.0 kJ
13 Answer (*b*)
14 (*a*) $U_b = U_c$ (*b*) (*i*) $\Delta U = nc_V\Delta T$ (*ii*) $\Delta U = nc_P\Delta T - P\Delta V$ (*c*) $c_P = c_V + R$ (*d*) (*i*) $c_V = 3R/2$ (*ii*) $c_P = 5R/2$
15 (*a*) 20.8 J mol^{-1} K^{-1} (*b*) 12.5 J mol^{-1} (*c*) 1.66 (*d*) 1.67
16 (*a*) (*i*) $5R/2$ (*ii*) $5kT/2$ (*b*) (*i*) $7R/2$ (*ii*) $7kT/2$ (*c*) 3, translational kinetic energy associated with movement in three mutually perpendicular directions (*d*) 2, rotational kinetic energy associated with motion about the two mutually perpendicular directions which are perpendicular to the axis of the diatomic molecule (*e*) 2, vibrational kinetic and potential energies along the axis of the molecule
17 (*a*) 21 J mol^{-1} K^{-1} (*b*) 29 J mol^{-1} K^{-1} (*c*) 5 degrees of freedom
18 (*a*) 12.5 J mol^{-1} K^{-1} (*b*) 20.8 J mol^{-1} K^{-1} (*c*) 3 degrees of freedom (*d*) 20 J
19 (*a*) (*i*) $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$ or $\left(\frac{V_1}{V_2}\right)^{\gamma-1} = \frac{T_2}{T_1}$
(*ii*) $\frac{P_1^{\gamma-1}}{T_1^{\gamma}} = \frac{P_2^{\gamma-1}}{T_2^{\gamma}}$ or $\left(\frac{P_1}{P_2}\right)^{\gamma-1} = \left(\frac{T_1}{T_2}\right)^{\gamma}$
or $P_1^{(1-\gamma)/\gamma}T_1 = P_2^{(1-\gamma)/\gamma}T_2$ or $\left(\frac{P_1}{P_2}\right)^{(1-\gamma)/\gamma} = \frac{T_2}{T_1}$
(*b*) (*i*) Temperature increases (*ii*) Temperature decreases
20 2.4
21 (*a*) 4500 kPa (*b*) 900 K
22 (*a*) (*i*) $\Delta U = \Delta Q$ (*ii*) $\Delta U = \Delta Q - \Delta W$ (*iii*) $0 = \Delta Q - \Delta W$ or $\Delta Q = \Delta W$ (*iv*) $\Delta U = -\Delta W$ (*b*) (*i*) $\Delta U = \Delta Q - P\Delta V$ (*ii*) $\Delta U = -P\Delta V$
23 (*a*) CD (*b*) AB (*c*) DA (*d*) BC
24 (*a*) Isochoric change (*b*) 168°C (*c*) 92.6 J (*d*) Zero (*e*) +92.6 J
25 (*a*) Isobaric (*b*) 97°C (*c*) 2.4×10^{-3} mol (*d*) 5.1 J (*e*) 1.5 J (*f*) 6.6 J
26 (*a*) Zero (*b*) 14 J (*c*) 14 J
27 (*a*) -33°C (*b*) 29 J (*c*) 29 J (*d*) 80°C (*e*) 61°C
28 (*a*) (*i*) 300 J (*ii*) Zero (*iii*) -200 J (*iv*) Zero (*b*) 100 J
29 (*a*) 275 J (*b*) 175 J (*c*) +175 J (*d*) 700 J (*e*) 20 per cent
30 (*a*) 102°C or 375 K (*b*) 25 per cent
31 15 K
32 (*a*) The cup of coffee after stirring (*b*) Uncombed hair (*c*) Two glasses of water, both at 20°C (*d*) The diver entering the water (*e*) The Solar System in 1997 (*f*) The shuffled pack
33 (*a*) 12 J K^{-1} (*b*) The change in entropy should be greater for the boiling than for the melting process, since the change in the degree of disorder is much greater for the change from liquid to gas than it is for the change from solid to liquid (*c*) 61 J K^{-1}
34 2.1 mJ K^{-1}
35 (*a*) 0.97 kJ K^{-1} (*b*) 3.5 kJ K^{-1} (*c*) 0.30 kJ K^{-1}
36 (*a*) 3.3 kJ (*b*) +12 J K^{-1} (*c*) -11 J K^{-1} (*d*) +1 J K^{-1} (*e*) The total entropy of the system and its surroundings has increased
37 (*a*) (*i*) Q_{ab}/T_1 (*ii*) Zero (*iii*) $-Q_{cd}/T_2$ (*iv*) Zero (*v*) $Q_{ab}/T_1 - Q_{cd}/T_2$ (*b*) Zero
38 (*a*) 65.3 per cent (*b*) 38 per cent
39 (*a*) (*i*) ab (*ii*) da (*iii*) cd (*b*) $Q_{bc} - Q_{da}$
40 (*b*) 25 per cent (*c*) 8 kW
41 (*a*) (*i*) b (*ii*) d
42 (*a*) $W = Q_1 - Q_2$ (*b*) (*i*) $\eta = Q_2/(Q_1 - Q_2)$ or $\dfrac{1}{\dfrac{Q_1}{Q_2} - 1}$ (*ii*) $\eta = T_2/(T_1 - T_2)$ or $\dfrac{1}{\dfrac{T_1}{T_2} - 1}$
43 (*a*) 6.6 (*b*) 4.4×10^2 W
44 (*a*) 4.4 (*b*) 1.5 (*c*) (*i*) 2.9×10^5 J (*ii*) 1.9×10^5 J (*d*) 14 min
45 (*a*) 4.0 (*b*) 1.0 kW
46 (*a*) 16 (*b*) 180 W
47 (*a*) 0 K (*b*) 1 (or 100 per cent)

SET 51 TRANSFER OF HEAT

1 (*a*) 5.0×10^2 K m^{-1} (*b*) 1.1×10^3 W
2 (*a*) 2.4×10^2 W m^{-1} K^{-1} (*b*) 30 W (*c*) Greater than your answer to (*b*), since *some* heat would be lost through the insulation
3 (*a*) 2.3 kJ (*b*) 6.8 g
4 (*a*) 11 kW (*b*) 11 kW
5 (*a*) 1.8 kW (*b*) 39 min
6 10^7 J
7 (*a*) 20.0 W (*b*) 61.4°C (*c*) 61.3°C (*d*) In the copper
8 (*a*) 15 W (*b*) 0.038 W m^{-1} K^{-1}
9 37 kJ
10 56°C
11 (*a*) 2.0 kW (*b*) 0.12 kW, slightly more efficient!
12 (*a*) 11 kW (*b*) 1.5×10^2 W
13 The sheet of rubber
14 (*a*) R-value $= l/k$ (*b*) m^2 K W^{-1} (*c*) Brick: 0.092 m^2 K W^{-1}; pine weatherboard (the better insulator): 0.18 m^2 K W^{-1}
15 2.0 m^2 K W^{-1}
16 (*a*) Bottom (*b*) Top (*c*) Bottom (*d*) Top
17 (*a*) A D (*b*) The cooling rate will decrease. When the temperature of the water is above 4°C, heat will leave by both *conduction* and *convection*, whereas once the temperature has dropped below 4°C, heat will leave the water by *convection* only
18 (*a*) R (*b*) Anticlockwise (*c*) SQPR or possibly SQRP
19 (*a*) B and D open; A, C and E closed (*b*) A, C and E open; B and D closed
20 16
21 8.6×10^5 W m^{-2}
22 (*a*) 7.9×10^{-5} m^2 (*b*) 75 W
23 (*a*) 3.9×10^{26} W (*b*) 3.9×10^{26} W (*c*) 4.58×10^{-10} (*d*) 1.8×10^{17} W (*e*) 1.4×10^3 W m^{-2} (*f*) A significant proportion of the solar energy is reflected by the Earth's atmosphere and does not reach the surface of the Earth
24 1.5×10^{13} y (It will be around for a while yet!)
25 10^2 or 10^3 W
26 (*a*) 0.13 μm (*b*) Predominantly blue
27 3.7×10^3 K

SET 52 ENERGY TRANSFORMATIONS
(Mainly involving heat)

1 (*a*) 1.2 kJ (*b*) 1.2 kJ (*c*) Some of the work done would give rise to *kinetic energy* to *loose* pieces of gravel under the skidding tyre, rather than produce internal (heat) energy in that part of the road directly beneath the tyre
2 (*a*) 1.1×10^2 J (*b*) 1.9 K or °C
3 58 per cent
4 4.8×10^2 J
5 (*a*) 230 N (*b*) 38 kW
6 (*a*) $\pi d^2v/4$ (*b*) $\pi d^2v\rho/4$ (*c*) $\pi d^2v^3\rho/8$ (*d*) $\pi d^2v^3\rho/8$ (*e*) 2.2×10^2 W
7 500 W
8 (*a*) (*i*) 20 m s^{-1} (*ii*) 1.7×10^5 J (*b*) 1.7×10^5 J
9 (*a*) 1.2 kJ (*b*) 2.7×10^2 K or °C
10 10^{-1} K or °C
11 (*a*) 2.0×10^6 J (*b*) (*i*) 1.4×10^6 J (*ii*) 1.4×10^7 J (*c*) 78 s (*d*) 0.18 MW
12 (*a*) 6.86 kJ (*b*) 140 J (*c*) 6.72 kJ
13 10^{-2} K or °C
14 1.1×10^{14} J
15 (*a*) 1.7×10^5 kg s^{-1} (*b*) 87 per cent
16 2.8×10^8 W
17 (*a*) 1.5 m^2 (*b*) 1.5 kW (*c*) 5.8 per cent
18 (*a*) 49 MJ (*b*) 39 K or °C
19 (*a*) (*i*) 72 MJ (*ii*) 67 MJ (*b*) 19 K or °C
20 20 per cent
21 13 slices
22 (*a*) 6.0 MJ (*b*) 9 tins—the last few kilometres are not easy!
23 Your answer (depending on your mass) should be somewhere between 900 m and 1700 m. (Obviously high jumpers cannot organise their metabolism this easily!)
24 34 per cent
25 (*a*) (*i*) 1.21×10^{-3} m^3 (*ii*) 0.85 kg (*iii*) 4.0×10^7 J (*iv*) 9.6×10^6 J (*b*) 8.0×10^3 m (*c*) 1.2×10^3 N (*d*) 1.2×10^3 N
26 (*a*) 2046 (*b*) The world is not using it at a constant but at an increasing rate (*c*) 2020
27 (*a*) (*i*) 2.1×10^{10} J (*ii*) 2.9×10^{10} J (*iii*) 8.7×10^{10} J (*b*) It will have doubled between two and three times
28 (*a*) The total energy used between 1900 and 2000 (*b*) About 2144
29 (*a*) 10^{16} or 10^{17} J (*b*) 10^5 or 10^6 houses

SET 53 REVISION PROBLEMS

1 86°C
2 413.4 mm
3 0.14
4 0.30 m
5 113 kPa
6 (*a*) 318 kPa (*b*) 53.0 kPa
7 232 mmHg
8 (*a*) 1 (*b*) $7P/3$ (*c*) It will be less
9 (*a*) 2.5 (*b*) 2.5 (*c*) 1 (*d*) 2.5 (*e*) 2 (*f*) 1 (*g*) 1.3 (4/3)
10 (*a*) 1.0 (*b*) 2.0 (*c*) 8.0 (*d*) 4.0 (*e*) 4.0 (*f*) 0.71 $(1/\sqrt{2})$
11 (*a*) 1 (*b*) 3.0 (*c*) 1
12 (*a*) 42.8°C (*b*) Below your answer to (*a*)
13 18.5°C
14 -227°C or 46 K
15 (*a*) 16°C (*b*) 43°C
16 (*a*) C (*b*) B (*c*) A (*d*) D (*e*) C (*f*) C (*g*) A
17 10^2 K or °C
18 (*a*) 4.5 kJ (*b*) 18 per cent
19 A C D F

20 (*a*) 86°C (*b*) 4 h 30 min
21 135 K or °C
22 2.4×10^{10} J
23 29.2°C
24 15 K or °C
25 (*a*) 7.98×10^{25} W (*b*) 8.9×10^{8} kg s^{-1}
26 5.9×10^{3} K
27 194 m s^{-1}
28 8.4×10^{2} W
29 (*a*) 0.47 K or °C (*b*) 1.3 MW
30 0.10 K s^{-1} or °C s^{-1}
31 8.0×10^{2} W

SET 54 STATIONARY CHARGED PARTICLES IN ELECTRIC FIELDS

1 (*a*) Positive (*b*) Positive (*c*) Negative (*d*) Uncharged
2 (*a*) D (*b*) A (*c*) F (*d*) B
3 C
4 (*a*) D (*b*) A (*c*) A
5 3 N
6 4 km!
7 9.0 μC
8 (*a*) 1.6×10^{-4} N (*b*) 1.0×10^{-5} N (*c*) 4.0×10^{-5} N (*d*) 3.6×10^{-4} N
9 (*a*) Upward (*b*) 0.18 N (*c*) 0.30 N
10 (*a*) 4.0 μN (*b*) 5.0 μN (*c*) 53° (*d*) 45 μN
11 (*a*) 7.5 μN (*b*) Downwards (*c*) 7.5 μN
12 5.3×10^{-8} C
13 3 N C^{-1} west
14 (*a*) 6.0×10^{-16} N (*b*) West
15 $+3.2 \times 10^{-19}$ C
16 (*a*) 7.2×10^{-15} N west (*b*) 1.8×10^{3} N C^{-1} west (*c*) 1.8×10^{3} N C^{-1} west
17 4.8 N C^{-1} downwards
18 (*a*) 32 N C^{-1} towards the Earth's centre (*b*) 6.5×10^{6} m
19 (*a*) Downwards (*b*) (*i*) E (*ii*) E (*iii*) X (*iv*) X
20 (*a*) To the left (or in the direction from P to A) (*b*) To the right (or in the direction from P to B)
21 (*a*) 3.24×10^{3} N C^{-1} towards the −500 pC charge (*b*) 360 N C^{-1} towards the 400 pC charge
22 8 cm
23 (*a*) 51 N C^{-1} (*b*) North-east
24 2.25×10^{-8} J
25 15 V
26 6.5×10^{-9} J
27 9 V
28 (*a*) Uncharged side (*b*) 1.8×10^{-9} J
29 3.6×10^{2} V
30 U/x
31 (*a*) 9.6 V (*b*) −4.8 V (*c*) Zero
32 (*a*) A (*b*) C (*c*) B and D (*d*) −1
33 (*a*) F (*b*) Zero (*c*) (*i*) $-Q/8\pi\epsilon_0 r$ (or $-kQ/2r$) (*ii*) $Q/2\pi\epsilon_0 r^2$ (or $2kQ/r^2$) to the right (or towards $-Q$)
34 (*a*) C (*b*) E (*c*) E (*d*) C
35 (*a*) −2.7 V (*b*) 78 N C^{-1} in the direction E 40° S
36 (*a*) (*i*) E (*ii*) X (*iii*) D (*b*) (*i*) 3.0 V (*ii*) 6.0 V
37 (*a*) Spherical (*b*) Equal (*c*) Spacing between L_1 and L_2 would be less than that between L_2 and L_3 (*d*) At right angles

SET 55 MOVING CHARGED PARTICLES IN ELECTRIC FIELDS

1 (*a*) (*i*) Vq (*ii*) Vq (*iii*) Vq (*b*) Fd (*c*) V/d (*d*) Volt per metre, i.e. V m^{-1} (From this point on, electric fields will be expressed in the more commonly used V m^{-1} rather than N C^{-1}. However, it should be understood that the latter is quite acceptable)
2 (*a*) B D (*b*) Negative (For this reason many texts write $E = -\Delta V/\Delta d$)
3 (*a*) 4.0×10^{3} V m^{-1} (or N C^{-1}) (*b*) To the right (or towards the negatively charged plate)
4 (*a*) B (*b*) C
5 (*a*) 6.0×10^{3} V m^{-1} (or N C^{-1}) to the right (or towards Q) (*b*) -2.0×10^{-6} C (*c*) 9.0 mN (*d*) 96 V
6 (*a*) Positive (*b*) 5.0×10^{4} V m^{-1} (or N C^{-1}) (*c*) 4.0×10^{-14} N upwards (*d*) Zero (*e*) 4.0×10^{-14} N downwards (*f*) 4.1×10^{-15} kg
7 (*a*) 3.08×10^{-18} m^3 (*b*) 2.90×10^{-15} kg (*c*) 2.84×10^{-14} N (*d*) 2.84×10^{-14} N (*e*) 5.80×10^{4} V m^{-1} (or N C^{-1}) (*f*) 4.90×10^{-19} C (*g*) 1.63×10^{-19} C
8 (*a*) Positive (*b*) Upwards (*c*) Upwards (*d*) v (*e*) $3v$ downwards
9 (*a*) 4.5×10^{2} V (*b*) 2.7×10^{2} V (*c*) 2.4×10^{-4} m s^{-1} (*d*) (*i*) +4 (*ii*) −2
10 X: −1.0 pC, Y: +1.5 pC, Z: +1.5 pC
11 Either +1 e or +2 e, not +1.5 e (Remember Millikan showed charge is quantised in nature)
12 (*a*) 4×10^{-5} J (*b*) 2×10^{-5} N (*c*) 5 V m^{-1} (or N C^{-1}) (*d*) 5 V m^{-1} (or N C^{-1})
13 (*a*) 8 V (*b*) 5.0 kV
14 1.2 kJ
15 21 V
16 (*a*) 2.0×10^{-3} N (*b*) 2.0×10^{-4} J (*c*) 50 V
17 (*a*) 200 V (*b*) (*i*) 3.20×10^{-17} J (*ii*) 3.20×10^{-17} J (*c*) Doubling the separation has halved the electric field strength ($E = V/d$) and therefore the electric force. So the work done ($W = Fs$) has not changed, since while the displacement has been doubled the force has been halved
18 (*a*) 2.4×10^{-15} J (*b*) 7.3×10^{7} m s^{-1} (*c*) 0.55 s—it *is* travelling!
19 (*a*) 1.44×10^{-17} J (*b*) 1.32×10^{5} m s^{-1} (*c*) 1.65×10^{-27} kg
20 (*a*) (*i*) 2.4×10^{-18} J (*ii*) 15 eV (*b*) 1.6×10^{-19} J
21 40 V
22 2.4×10^{7} m s^{-1}
23 (*a*) (*i*) 56 V (*ii*) 54 V (*b*) 2 V (*c*) 2×10^{-12} m (*d*) 1×10^{12} V m^{-1} (*e*) 1.1×10^{12} V m^{-1}
24 (*a*) $E = Q/4\pi\epsilon_0 R^2$ (or kQ/R^2) (*b*) $U_e = Qq/4\pi\epsilon_0 R$ (or kQq/R) (*c*) $Q/4\pi\epsilon_0 R$ (or kQ/R) (*d*) $dV_e/dR = -Q/4\pi\epsilon_0 R^2$ (or $-kQ/R^2$) (*e*) $E = -dV_e/dR$
25 40 A
26 12 s
27 (*a*) 0.080 C (*b*) 5.0×10^{17} electrons
28 (*a*) 6.25×10^{15} electrons s^{-1} (*b*) 1.6×10^{6} electrons
29 (*b*) 1.0×10^{19} electrons s^{-1} (*c*) (*i*) 1.0×10^{-10} m^3 (*ii*) 1.0×10^{-4} m (*d*) 1.0×10^{-4} m s^{-1} (i.e. 0.1 mm s^{-1}, possibly lower than you expected)
30 (*a*) 1.5 V (*b*) 9.0 V
31 (*a*) 1.9×10^{-18} J (*b*) 12 eV
32 (*a*) 20 W (*b*) 6.0 kJ
33 0.5 mA
34 (*a*) Charge (*b*) 2.3×10^{5} C (*c*) 2.8 MJ
35 (*a*) 8×10^{20} (*b*) No
36 (*a*) 6.25×10^{18} (*b*) 2 (*c*) 1
37 (*a*) 2.00×10^{22} atoms (*b*) 4.00×10^{22} electrons (*c*) 2.00×10^{22} molecules (*d*) 66.4 mg
38 1.2 A
39 −0.3 A
40 (*a*) 2.4×10^{-15} N upwards (*b*) Zero (*c*) 2.6×10^{15} m s^{-2} upwards (*d*) 3.3×10^{-2} m upwards (*e*) 1.3×10^{7} m s^{-1} upwards (*f*) 1.4×10^{7} m s^{-1} (*g*) 65° above horizontal
41 (*a*) 5.1×10^{-17} J (*b*) 1.1×10^{7} m s^{-1} (*c*) 2.5×10^{3} V m^{-1} vertically downwards (*d*) 2.0×10^{-9} s (*e*) 8.8×10^{-4} m vertically upwards (*f*) 1.1×10^{7} m s^{-1}
42 9.3×10^{-31} kg
43 (*a*) $Qq/4\pi\epsilon_0 r^2$ (or kQq/r^2) (*b*) $-Qq/4\pi\epsilon_0 r$ (or $-kQq/r$) (*c*) $Qq/8\pi\epsilon_0 r$ (or $kQq/2r$) (*d*) $-Qq/8\pi\epsilon_0 r$ (or $-kQq/2r$) (*e*) $Qq/8\pi\epsilon_0 r$ (or $kQq/2r$)
44 (*a*) (*i*) -4.36×10^{-18} J (*ii*) 2.18×10^{-18} J (*iii*) -2.18×10^{-18} J (*iv*) 2.18×10^{-18} J (*b*) 10^{39}
45 2.7×10^{-11} m

SET 56 SIMPLE D.C. CIRCUITS

1 5.0 V m^{-1}
2 1.5 V
3 1.2 V
4 (*a*) (*i*) 6.0 V (*ii*) 4.0 V (*b*) 10 V (*d*) AB (*e*) (*i*) E (*ii*) I
5 (*a*) 4.5 V m^{-1} to the right (*b*) 12 V (*c*) Positive terminal
6 (*a*) E (*b*) C (*c*) B
7 9.6×10^{2} Ω
8 5.0 mA
9 2.4×10^{2} V
10 (*a*) Yes (*b*) No
11 1.0 kΩ
12 (*a*) 27 Ω, 10 per cent (*b*) 47 kΩ, 20 per cent (*c*) Orange, white, red, gold
13 (*a*) (*i*) 5.6 Ω, 5 per cent (*ii*) 150 Ω, 20 per cent (*iii*) 4.3 MΩ, 2 per cent (*b*) 6K8K
14 1.69×10^{-8} Ω m
15 0.60 m
16 10^{11} or 10^{12} Ω
17 (*a*) $S = \Omega^{-1}$ (*b*) The unit of conductance is the reciprocal of ohm or, if you like, "ohm upside down", so they simply wrote ohm back to front! (*c*) S m^{-1}
18 (*a*) 5 S (*b*) 27 Ω (*c*) 49 S
19 (*a*) 5.0 C s^{-1} (*b*) 240 J C^{-1} (*c*) 1.2×10^{3} J s^{-1} (or W) (*d*) 1.2×10^{3} W
20 (*a*) 7.1 A (*b*) 2.4×10^{5} J
21 (*a*) $P = I^2 R$ (*b*) $P = V^2/R$
22 6.8 A
23 (*a*) X: 5.8×10^{2} Ω, Y: 1.40 kΩ. The lower power bulb has the higher resistance. Did you guess correctly? (*b*) 30 kJ
24 (*a*) 2.00 kW (*b*) $44
25 (*a*) 8.4×10^{3} J (*b*) 20 W (*c*) 8.0 V
26 Energy
27 (*a*) 3.6 MJ (*b*) (*i*) 29 kW h (*ii*) 1.0×10^{2} MJ
28 (*a*) $I_T = I_A = I_B$ (*b*) $V_T = V_A + V_B$
29 (*a*) 3.2 mA (*b*) 1.6×10^{2} V (*c*) 3.0×10^{2} V
30 (*a*) (*i*) 5.0 mA (*ii*) 2.5 V (*iii*) 4.0 V (*iv*) 6.5 V (*b*) (*i*) 3.0 mA (*ii*) 3.0 mA (*iii*) 1.5 V (*iv*) 4.5 V
31 (*a*) (*i*) 0.30 A (*ii*) 3.0×10^{2} V (*iii*) 5.0×10^{2} V (*b*) (*i*) 0.30 A (*ii*) 3.0×10^{2} V

ANSWERS

32 (*a*) 22 Ω (*b*) 36 Ω (*c*) 360 Ω (*d*) 47 MΩ (to two significant figures)
33 (*a*) $I_T = I_A + I_B$ (*b*) $V_T = V_A = V_B$
34 (*a*) (*i*) 4.0 mA (*ii*) 2.0 V (*iii*) 1.5 mA (*iv*) 5.5 mA (*b*) 2.0 mA
35 (*a*) (*i*) 100 V (*ii*) 100 V (*iii*) 0.20 A (*iv*) 0.30 A (*b*) 400 V
37 (*a*) 16 Ω (*b*) 18 Ω (*c*) 12 Ω (*d*) 1.8 Ω (*e*) 2.2 Ω (to two significant figures)
38 15 Ω
39 (*a*) C (*b*) A
40 (*a*) 2 Ω (*b*) 1.6 Ω
41 (*a*) Connect four in series (*b*) Connect three in parallel (*c*) Connect a parallel group of four in series with one (*d*) Connect a parallel group of three in series with two others
42 (*a*) 20 V (*b*) 40 mA
43 (*a*) 2 A (*b*) 6×10^2 C (*c*) 18 V
44 52 Ω
45 (*a*) 9.0 V (*b*) 1.0 A (*c*) 1.5 A (*d*) 3.0 V (*e*) 12 V (*f*) 12 V (*g*) 3.0 A (*h*) 4.5 A
46 7.0 Ω
47 (*a*) 60 V (*b*) 50 V (*c*) Zero (*d*) 300 V
48 (*a*) 2.0 A (*b*) 16 V m^{-1} (*c*) 4.0 Ω (*d*) 4.0 A

SET 57 LESS SIMPLE D.C. CIRCUITS

1 (*a*) 0.50 A (*b*) 3.8 V
2 (*a*) (*i*) 10 A (*ii*) 6.0 A (*iii*) 1.0 A (*b*) (*i*) 2.0 V (*ii*) 6.0 V (*iii*) 11 V (*c*) The resistance of the external circuit must be much greater than the internal resistance of the source
3 (*a*) (*i*) 1.2×10^2 W (*ii*) 72 W (*iii*) 12 W (*b*) (*i*) 20 W (*ii*) 36 W (*iii*) 11 W (*c*) Developing heat inside the battery (heating its internal resistance) (*d*) Equal
4 (*a*) (*i*) 10 V (*ii*) 10 V (*b*) 9.6 V
5 (*a*) 1.54 V (*b*) 7.40 Ω
6 60 Ω
7 (*a*) (*i*) 4.32 V (*ii*) 0.80 Ω (*b*) (*i*) 1.08 V (*ii*) 0.050 Ω
8 0.020 Ω
9 (*a*) 8.0 V (*b*) 1.0 Ω
10 (*a*) (*i*) 2 (*ii*) 1 (*b*) (*i*) 3 (Did you spot the one around the outside?) (*ii*) 2
11 (*a*) 3.1 A (*b*) 2.9 A (*c*) 1.4 V
12 (*a*) 5.0 A (*b*) 4.8 V
13 (*a*) 4.7 V (*b*) 1.1 A
14 (*a*) 3.0 V (*b*) 1.2 V
15 (*a*) 16.0 V (*b*) (*i*) 1.00×10^{-2} A (*ii*) 1.44×10^3 Ω
16 8.0 V
17 1.53 V
18 2.13 V
19 0.50 A
20 (*a*) 2.0 V (*b*) 4.0 V
21 (*a*) Zero (*b*) (*i*) $12I_1 = 16I_2$ (*ii*) $21I_1 = RI_2$ (*c*) 28 Ω
22 (*a*) AB and BC: 0.4 A, AD and DC: $2/(4 + R)$ A (*b*) (*i*) 0.8 V (*ii*) $8/(4 + R)$ V (*c*) 6 Ω
23 (*a*) From Z to X (*b*) 37°C
24 8.45 mA
25 (*a*) (*i*) Parallel (*ii*) 0.505 Ω (*b*) (*i*) Series (*ii*) 150 Ω
26 A parallel resistor of 0.0200 Ω
27 A series resistor of 9.80 kΩ
28 (*a*) (*i*) 100 mV (*ii*) 100 mV (*b*) (*i*) 50.0 μA (*ii*) 2.00 kΩ
29 (*a*) 100 mV (*b*) (*i*) 900 mV (*ii*) 100 μA (*iii*) 9.00 kΩ (*c*) (*i*) 9.00 V (*ii*) 90.0 kΩ
30 (*a*) 50.0 μA (*b*) 950 μA (*c*) $R_W = 19R_X - 2000$ (*d*) (*i*) 1.80 kΩ (*ii*) 200 Ω
31 (*a*) 3.0 Ω (*b*) 2.0 A (*c*) 0.50 A (*d*) 12 V (*e*) 90 C (*f*) 56 W
32 750 Ω
33 5.0 km
34 (*a*) M_1: increase, M_2: decrease (*b*) M_1: increase, M_2: increase
35 2 Ω and 8 Ω
36 455 mV
37 (*a*) 1.5 V (*b*) 3.8 W (*c*) 3.3 A
38 210 Ω
39 B
40 11.6 V, 57.7 V

SET 58 MAGNETIC EFFECTS OF MOVING CHARGED PARTICLES

1 4.0×10^{-3} T to the right
2 (*a*) 4.0×10^{-3} N (*b*) East
3 (*a*) 3.0×10^{-3} N vertically upwards (*b*) 2.6×10^{-3} N vertically upwards
4 (*a*) Perpendicularly into the page (*b*) Perpendicularly into the page (*c*) No direction, as there is no magnetic force on YZ
5 (*a*) 2.51 m (*b*) 0.200 A
6 (*a*) Vertically downwards (*b*) 5.0×10^{-3} N
7 (*a*) 20 cm (*b*) Zero
8 (*a*) 0.12 N perpendicularly out of the page (*b*) Zero
9 (*a*) Clockwise (*b*) 90° (*c*) 0° or 180° (*d*) More spread out near the middle
10 (*a*) 1.8×10^{-2} N m (*b*) Zero
11 1.9×10^{-3} N m

12 (*a*) Vertically (*b*) Downwards
13 Towards the west
14 6.0×10^{-5} T
15 No, not unless your watch comes within 2.5×10^{-6} m of the overhead wire!
16 (*a*) Perpendicularly into the page (*b*) Roughly towards bottom right (*c*) Left-hand end
17 3.7×10^{-5} T
18 3×10^{-2} T vertically downward
19 4.8 A
20 (*a*) 10° west (*b*) 31 μT
21 (*a*) (*i*) −90° (*ii*) 0° (*b*) Hobart, Auckland, San Francisco, London
22 (*a*) (*i*) 2.0×10^{-5} T perpendicularly into the page (*ii*) 2.0×10^{-5} T perpendicularly into the page (*b*) 4.0×10^{-5} T perpendicularly into the page
23 (*a*) (*i*) Perpendicularly into the page (*ii*) Perpendicularly out of the page (*iii*) Perpendicularly out of the page (*b*) R
24 (*a*) (*i*) 2×10^{-5} T perpendicularly into the page (*ii*) Zero (*b*) To the right
25 (*a*) (*i*) From X to Y (*ii*) From A to B (*b*) 37°
26 (*a*) Anticlockwise (*b*) North-east (*c*) N 72° E
27 (*a*) 3.0×10^{-5} T (*b*) North (*c*) N 53° W
28 (*a*) 6.7×10^{-6} T (*b*) B_1 (*c*) Perpendicularly out of the page (*d*) (*i*) $20B_1$ (*ii*) 1.3×10^{-4} N (*e*) Down the page (*f*) 1.3×10^{-4} N up the page
29 (*a*) 2.0×10^{-2} N (*b*) Repulsion
30 1 A
31 3.5×10^{-5} m—a bit close!
32 (*a*) $F/2$ (*b*) Towards TU or to the right
33 (*a*) 4.8×10^{-11} N (*b*) 5.4×10^{18}
34 (*a*) Towards the observer (*b*) Upwards
35 (*a*) 0.10 m (*b*) 3.1×10^{-8} s
36 (*a*) 2.9×10^7 m s^{-1} (*b*) 4.5×10^{-14} N (*c*) 17 mm (*d*) 2.7×10^8 loops
37 8.3×10^{-16} J
38 282 mm
39 (*a*) Down (*b*) To the right
40 (*a*) Zero (*b*) North (*c*) $2.0 \times 10^4\, q$ N west (*d*) $4.0 \times 10^{-3}\, qv$ N east (*e*) 5.0×10^6 m s^{-1}
41 3.4×10^4 V m^{-1}
42 (*a*) $Vq = mv^2/2$ (*b*) $qvB = mv^2/R$ (*c*) $m = qB^2R^2/2V$ (*d*) $m \propto R^2$ (*e*) 6.6×10^{-27} kg
43 387 mm
44 (*a*) $v = E/B$ (*b*) $qvB = mv^2/R$ (*c*) $m = qB^2R/E$ (*d*) $m \propto R$
45 5.7 mm
46 (*a*) (*i*) R and T (*ii*) S (*b*) Q and S (*c*) 3.1×10^{-4} N A^{-1}
47 (*a*) 2.2×10^{-5} N A^{-1} (*b*) 5.5×10^{-5} T to the right (or in the direction from X to Y)
48 (*a*) 1.2×10^{-4} N A^{-1} (*b*) 5.5×10^{-4} T to the right

SET 59 ELECTROMAGNETIC INDUCTION AND ELECTROMAGNETIC WAVES

1 (*a*) (*i*) 0.45 Wb (weber) (*ii*) 0.44 Wb (*b*) −0.45 Wb
2 8.0×10^{-2} T
3 (*a*) 3.6×10^{-3} V (*b*) X
4 (*a*) 1.5 mV (*b*) West end
5 8.5×10^2 m s^{-1}.
6 (*a*) 4.0 V (*b*) From C to D (*c*) 1.0 A (*d*) 0.50 N to the right (*e*) Zero (*f*) 0.50 N to the left
7 (*a*) vLB/R (*b*) $v = mgR/L^2B^2$
8 1.3×10^{-4} V
9 2.2×10^{-3} T
10 (*a*) From S to R (or upwards) (*b*) Face R (*c*) (*i*) From S to R (or upwards) (*ii*) From R to S (or downwards) (*d*) $F_m = -F_H$ (*e*) $F_m = qvB$ (*f*) (*i*) $F_H = E_Hq$ (*ii*) $F_H = V_Hq/w$ (*g*) $V_H = vBw$ (*h*) $V_H = IB/ntq$
11 10^{-12} V
12 (*a*) 0.25 m s^{-1} (*b*) 2.9×10^2 millilitre min^{-1}
13 (*a*) (*i*) R (*ii*) S (*b*) 3.0×10^{20} conduction electrons m^{-3}
14 C
15 (*a*) 6.0×10^{-5} Wb (*b*) (*i*) 5.0×10^{-4} Wb s^{-1} (or V) (*ii*) 5.0×10^{-4} V (*iii*) 25 mV
16 0.04 V
17 0.4 V
18 6.28×10^{-2} C
19 (*a*) Anticlockwise (*b*) Clockwise
20 Clockwise
21 (*a*) Clockwise (*b*) Clockwise
22 Perpendicularly out of the page
23 (*a*) To the left (*b*) To the left
24 (*a*) To the right (*b*) 1.5×10^{-3} V
25 (*a*) A (*b*) D
26 (*a*) To the right (*b*) Opposite direction
27 (*a*) Q to P (*b*) Yes
28 The back e.m.fs would cancel
29 (*a*) Wb A^{-1} (*b*) V s A^{-1}
30 4.0 mH
31 9.5 mH
32 (*a*) 62 J (*b*) 5.6×10^2 V
33 (*a*) $Ew = Bwv$ (*b*) $B = E/v$ (*c*) $\dot{B}'w = \mu_0\epsilon_0 Evw$ (*d*) $B' = \mu_0\epsilon_0 Ev$ (*e*) $v = 3.00 \times 10^8$ m s^{-1}
34 (*a*) Downwards (*b*) No direction, as there will be a zero magnetic field at your position
35 (*a*) 211 m (*b*) Vertically downwards (*c*) North (*d*) North
36 (*a*) C (*b*) H (*c*) Q and S
37 v/c, where c = speed of electromagnetic radiation

SET 60 SIMPLE A.C. CIRCUITS

1 (a) 58 V (b) 116 V (c) Zero (d) 41 V (e) 0.2 s (f) 5 Hz
2 (a) 339 V (b) 20 ms
3 (a) 5.0 A (b) 7.1 A
4 (a) 9.9 V (b) 14 V
5 (a) 4.8 A (b) 3.4 A (c) 50 Hz
6 $V = 58 \sin 10\pi t$ or $58 \sin 31t$
7 (a) (i) Zero (ii) 2×10^{-4} Wb (b) From V to P (c) 8 mV
8 (a) C D (b) B
9 (a) 7.0 V (b) 14 V (c) 9.9 V
10 (a) 50 Hz (b) 0.10 T
11 (a) 50 turns (b) 0.20 A
12 (a) 1.2 A (b) Greater than your answer to (a)
13 (a) 25 turns (b) 10 A (c) 8.5 V
14 (a) 336 W (b) 7 A
15 (a) +20 μC (b) 5 μF (or 5×10^{-6} F)
16 1.9×10^{-4} m^2
17 (a) D (b) F (c) F (d) B (e) G
18 (a) 2.4×10^{-2} s (b) More rapidly
19 (a) Decreasing the resistance (b) 2.4×10^{-4} C (c) 20 μF
20 8.0 mA
21 1.8×10^{-6} J
22 (a) 6 μF (b) 24 pF (c) 36 pF (d) 120 pF (e) 500 μF
23 (a) 2.4 μF (b) 12 μF (c) 3.0 μF (d) 2 pF
24 $+C_2V/(C_1 + C_2)$, zero, zero, $-C_1V/(C_1 + C_2)$
25 2 series groups, each of 2 capacitors in parallel, or 2 parallel groups, each of 2 capacitors in series
26 2 series groups, each of 4 capacitors in parallel, or 4 parallel groups, each of 2 capacitors in series
27 0.67 μF
28 1.6×10^3 Ω
29 (a) 10 Ω (b) 5.0×10^{-4} (c) D
30 (a) 5×10^4 Ω (b) (i) 10^{-2} (ii) 10^2 (c) High pass filter
31 44 Ω
32 (a) 6.6 Ω (b) A
33 (a) (i) 8.8 Ω (ii) 88 kΩ (b) High pass filter
34 (a) A (b) E (c) (i) E (ii) B (d) B (e) D
35 Z
36 (a) C (b) (i) 50 Ω and 680 μF (ii) 300 Ω and 680 μF
37 (a) D or A (b) A (c) B
38 (a) 40 Ω (b) 50 Ω (c) 0.24 A
39 (a) D or A (b) A (c) C
40 (a) 1.2×10^2 Ω (b) 1.3×10^2 Ω (c) 20 mA (d) (i) 1.0 V (ii) 2.4 V (e) They do not add up to the total potential difference across them because they are not in phase with one another. They reach their peak values one-quarter of a period apart
41 (a) B (b) B or C
42 (a) 49 Ω (b) 69 Ω (c) 20 Ω (d) 25 Ω (e) 0.20 A (f) 4.0 V
43 (a) They are equal (b) 15 Ω (c) (i) $0.44\pi f$ (ii) $5.00 \times 10^3/\pi f$ (d) 34 Hz (e) Greater at f, since the current is a maximum when the impedance is a minimum
44 (a) 1.3 μF (b) 30 V

SET 61 HOUSEHOLD A.C. SUPPLY

1 26.3 per cent
2 (a) (i) 11.9 (ii) 0.022 (b) 2.6×10^3 A (c) Slightly lower (d) HR, because low power losses in the transmission wires are most important for the longest section, and this requires low current and therefore high voltage
3 (a) (i) 3 MW (ii) 3 per cent (iii) 97 MW (b) (i) Make the cables of a metal with lower resistivity, e.g. copper instead of aluminium. Disadvantage: copper is more expensive, and also denser, therefore requiring more supporting towers (ii) Increase the thickness of the lines. Disadvantages: lines would be more expensive; supporting core of galvanised steel wires would need to be strengthened and/or the number of supporting towers would need to be increased (iii) Use a higher transmitting voltage and therefore lower current. Disadvantage: more difficult and expensive to prevent leakage of charge at supporting towers
4 (a) 1.0 (b) 56 (c) 15
5 B C E
6 C
7 (a) Zero (b) 240 V (c) 339 V (d) 20 ms
8 A D F
9 (a) A neutral; B earth; C none of these leads (b) A neutral; B earth; C active
10 (a) Black (b) Green (c) Brown (d) Blue
11 (a) Between T_2 and Q (b) T_1 to P (c) (i) Active (ii) Active
12 (a) 2 (b) 5 (c) 4
13 (a) 13 A (b) 3.188×10^9 J (c) 77 per cent
14 (a) Clock; 2.9×10^4 Ω (b) 0.15 A (c) Yes, the deep fryer and the fan heater would draw a total current of 16.0 A from the mains
15 Yes, since the current through the heart could be 0.22 A
16 3.6 MJ
17 (a) 3.0 kW h (b) 33 cents

SET 62 INTRODUCTORY ELECTRONICS

1 (a) Forward biased (b) 20 mA (c) 15 V (d) 20 V
2 (a) Reverse bias (b) 15 V (c) 40 mA (d) (i) 5 V (ii) 25 mA (iii) 2×10^2 Ω (e) (i) 6 V (ii) 30 mA
3 3 Ω
4 6.17 a.m.
5 Between A and B
6 (a) (i) 6 V (ii) 0 V (b) This circuit gives a *high* output when its input is *not* high (b) (i) Lo (ii) 1
7 The output would be the same as the input
8 (a) 5.0 V (b) 3.1 V (c) (i) 5.0 (ii) 5.0 (iii) 25 Ω (iv) 125 Ω (d) (i) 0.60 V (ii) 6.0 V (iii) 5.4 V (iv) 9.0 (e) (i) 1.4 V (ii) 4.6 V (iii) 3.3
9 (a) On (or high) (b) Off (or low) (c) Off (or low) (d) On (or high)
10 *Either* interchange the LDR and variable resistor, *or* insert an *inverter* between the electronic switch and the light
11 (a) Yes (b) An upper limit of 0.60 V (c) Increase R
12 B C
13 (b) "On" for a longer period (c) Out of phase (d) They are equal
14 (b) "On" for a longer period (c) In phase (d) They are equal
15 (b) (i) 5 V (ii) 1 V (c) Out of phase
16 (a) 1.0 V (b) (i) 1.6 (or 1.7) V (ii) 4.6 V (iii) 0.40 V (iv) 3.0 (or 2.9) V (v) 7.5 (or 7.3) (c) (i) No (ii) Yes (iii) No (iv) Yes
17 (b) (i) Approximately +3.8 V (ii) 0.5 V (c) Your answer should be between +0.28 V and +0.38 V
18 (b) Arrangement II
19 (a) 15 V (b) 3 kΩ
20 (a) Negative (b) (i) Reduces the gain (ii) Reduces distortion
21 (a) Negative (b) Positive
22 21
23 (a) 2.6 V (b) 50 mV
24 5.3×10^2 kΩ
25 (a) (i) Low (ii) Low (iii) Low (iv) High (b) See Table 62.10 (c) This circuit gives a *high* output when neither one *nor* the other input is high (d) NOR gate

Table 62.10

Input 1	Input 2	Output
1	1	0
1	0	0
0	1	0
0	0	1

26 (a) See Table 62.11 (b) This gate gives a *high* output when one *or* both of the inputs is high (c) OR gate

Table 62.11

Input 1	Input 2	Output
1	1	1
1	0	1
0	1	1
0	0	0

27 (a) See Table 62.12 (b) Each of the two inputs is fed into a separate NOT gate, the outputs of which are connected to a NOR gate (c) This circuit gives a *high* output when both one *and* the other of two inputs are high

Table 62.12

Input 1	Input 2	Output
1	1	1
1	0	0
0	1	0
0	0	0

28 (a) See Table 62.13 (b) An AND gate followed by a NOT gate

Table 62.13

Input 1	Input 2	Output
1	1	0
1	0	1
0	1	1
0	0	1

29 (a) NOR gate (b) AND gate
30 (a) For inputs of 1-0 and 0-1 a NOR gate gives outputs of 0, while a NAND gate gives outputs of 1 (b) NOR gate
31 (a) NOT gate (b) AND gate (c) NOR gate (d) NAND gate (e) OR gate
32 (a) See Table 62.14 (b) See Table 62.15 (c) (i) E (ii) C

Table 62.14

		A + B in decimal notation	A + B in binary notation		
A	B	The answer	The answer	"Put down" digit	"Carry" digit
0	0	0	0	0	0
1	0	1	1	1	0
0	1	1	1	1	0
1	1	2	10	0	1

Table 62.15

Input A	Input B	Output C	Output D	Output E
0	0	0	1	0
1	0	0	0	1
0	1	0	0	1
1	1	1	0	0

33 (*a*) See Table 62.16 (*b*) A SAME gate, a LIKE gate, a PARITY gate or an EQUALITY gate. (It gives a high output if the two inputs are both high or both low)

Table 62.16

A	B	C	D	E	F
0	0	0	1	0	1
1	0	0	0	1	0
0	1	0	0	1	0
1	1	1	0	0	1

34 (*a*) See Table 62.17 (*b*) It has *two* equally *stable* states, Hi/Lo and Lo/Hi (*c*) Positive feedback (*d*) They are equally feedback links, since the output of each is fed to the input of the other

Table 62.17

Location of flying leads L_1	L_2	Output 1	Output 2
+6 V	0 V	Lo	Hi
loose	0 V	Lo	Hi
loose	loose	Lo	Hi
loose	+6 V	Hi	Lo
0 V	+6 V	Hi	Lo
0 V	loose	Hi	Lo
loose	loose	Hi	Lo
+6 V	loose	Lo	Hi

35 (*a*) See Table 62.18 (*b*) Dividing by 2

Table 62.18

Number of input pulses to trigger	State of output 1	State of output 2	Progressive number of pulses from output 1	Progressive number of pulses from output 2
0	0	1	0	0
1	1	0	1	0
2	0	1	1	1
3	1	0	2	1
4	0	1	2	2
12	0	1	6	6

36 (*a*) A: off, B: off (*b*) A: on, B: off (*c*) A: on, B: on
37 (*a*) The pulse would be longer (*b*) The pulse would be shorter (*c*) The pulse would be shorter (*d*) The pulse would be longer
38 (*a*) From high to low (*b*) C
39 (*a*) (*i*) Increase the time constant of the input circuit of pulse producer A (*ii*) Decrease the time constant of the input circuit of pulse producer B (*b*) A and D
40 (*a*) 30 pulses (*b*) 0.5 Hz (*c*) 25 nF
41 (*a*) 6.0×10^{-6} s (*b*) 1.00×10^{-2} s (*c*) D (*d*) Differentiating circuit
42 (*a*) The time constant (1 s) is much greater than the period of the square wave (2×10^{-2} s) (*b*) C (*c*) Integrator
43 It would be unsatisfactory, since the time constant of 2.5 μs is much *smaller* than the period of the square waves (2 ms), whereas it should be much *larger*. This could be rectified by raising the time constant

SET 63 REVISION PROBLEMS

1 (*a*) Equal (*b*) Z (*c*) Equal
2 (*a*) Zero (*b*) 1.0×10^{-9} N (*c*) 4.5×10^{-9} N
3 (*a*) 1.2×10^{-9} C (*b*) Positive
4 (*a*) Zero (*b*) Greater than (*c*) Horizontally to the right
5 (*a*) Negative (*b*) 1.6×10^{-6} J (*c*) 360 V (*d*) 2.5×10^{-4} m s^{-1} (*e*) 4.2×10^{-5} m s^{-1} downwards
6 (*a*) 182 eV (*b*) 8.0×10^{6} m s^{-1}
7 0.80 W
8 (*a*) $2F/5$ (*b*) q (*c*) -25 V (*d*) k
9 (*a*) $\sqrt{2eV_1/m}$ (*b*) $x\sqrt{m/2eV_1}$ (*c*) md/T^2
10 (*a*) A (*b*) C
11 0.118 m
12 20 min
13 12 Ω and 3.0 Ω
14 (*b*) 0.650 Ω (*c*) 1.46 V
15 (*a*) 1.51 V (*b*) 0.40 Ω
16 (*a*) 400 Ω in series (*b*) 0.100 Ω in parallel
17 (*a*) 127 Ω (*b*) 1.53 V (*c*) The potential differences would add up if the three components being compared were coil AB, coil BC, and coils AB and BC as a series group. (This would be the case if the voltmeter's resistance were significantly higher than that of the coils, but it is not.) Actually, the comparison concerns three *different circuits*: (*i*) a parallel group of 200 Ω (coil AB) and 500 Ω (the meter), in series with 127 Ω (coil BC), (*ii*) a parallel group of 127 Ω and 500 Ω, in series with 200 Ω, and (*iii*) a series group of 200 Ω and 127 Ω, in parallel with 500 Ω.
18 (*a*) 2.5×10^{-6} T (*b*) South
19 (*a*) 0.12 mT upwards (*b*) 0.13 mT
20 (*a*) $H \propto M^2$ (*b*) $F \propto M^2$ (*c*) $H \propto F$
21 (*a*) 0.71 (*b*) 1.4 (*c*) V
22 (*a*) B and H (*b*) A and C
23 (*a*) To the right (*b*) To the right
24 (*a*) 18 mV (*b*) 13 mV
25 (*a*) 0.1 mV (*b*) (*i*) West (*ii*) East (*c*) Zero
26 (*a*) Zero (*b*) Zero (*c*) 4×10^{-4} V (*d*) 4×10^{-6} N
27 (*a*) 4.7×10^{-2} V (*b*) Downwards
28 (*a*) 6 μC (*b*) 5 mA

29 (*a*) 1.4×10^{-9} C, 5.6×10^{-9} C (*b*) 2.8×10^{2} V (*c*) 2.0×10^{-8} J
30 (*a*) 5.0 W (*b*) 8.0 kW
31 219 V
32 A
33 B D
34 (*a*) NOT gate (*b*) NOT gate (*c*) OR gate (*d*) NOR gate
35 D E
36 C

SET 64 STRUCTURE OF THE ATOM

1 (*a*) 27 mm^3 (*b*) 3 mm (*c*) 1800 mm^2 (*d*) Your friend's statement suggests a drop of water with a linear dimension of 3 mm consists of one molecule, but droplets much smaller than this are easily observed. Secondly, it is quite easy to spread a teaspoonful of water over an area larger than 1800 mm^2
2 (*a*) 2×10^{-8} m (*b*) One linear dimension of a gold atom is equal to *or less than* 2×10^{-8} m
3 (*a*) 5.0×10^{-3} cm^3 (*b*) 1.0×10^{-4} cm^3 (*c*) 7.1×10^{2} cm^2 (*d*) 1.4×10^{-7} cm (*e*) Maximum dimension
4 (*a*) 2.7×10^{3} kg (*b*) 1.6×10^{30} u (*c*) 5.9×10^{28} atoms (*d*) 1.7×10^{-29} m^3 (*e*) 2.6×10^{-10} m
5 (*a*) Bale A (*b*) Bale A (*c*) Yes, the spheres and cubes are only *about* the same size and insufficient trials have been performed (*d*) 35 per cent
6 (*a*) 180° (*b*) 0° (*c*) θ decreases
7 (*a*) Greater than (*b*) Less than (*c*) Greater than
8 (*a*) (*i*) About 20 in 10^4 (*ii*) Yes (*b*) (*i*) Yes; about 30 in 10^4 (*ii*) No, since many alpha particles are likely to undergo more than one interaction
9 Approximately 9×10^{2} m—you will need a big hall!
10 (*a*) 10^{-6} (*b*) 10^{-13}
11 (*a*) 8.5×10^{-13} J (*b*) 8.5×10^{-13} J (*c*) Zero (*d*) 8.5×10^{-13} J (*e*) 4.3×10^{-14} m
12 8×10^{-14} m
13 (*a*) $nq^2/4\pi\epsilon_0 r$ (or knq^2/r) (*b*) $E_0 - nq^2/4\pi\epsilon_0 r$ (or $E_0 - knq^2/r$) (*c*) They are equal (*d*) The external force keeping K in position (*e*) $nq^2/4\pi\epsilon_0 E_0$ (or knq^2/E_0)
14 (*a*) 8.5×10^{-13} J (*b*) 7.2×10^{-13} J (*c*) 1.3×10^{-13} J (*d*) 6.3×10^{6} m s^{-1}
15 (*a*) 2.5×10^{35}—the electrical force is slightly larger than the gravitational force! (*b*) It is the same at *any* separation

SET 65 STRUCTURE OF THE NUCLEUS AND RADIOACTIVITY

1 (*a*) 7 (*b*) 15 (*c*) 8
2 (*a*) 14 (*b*) 6 (*c*) 8 (*d*) 6 (*e*) 6
3 $^{207}_{82}Pb$
4 (*a*) Atomic number and chemical symbol (*b*) ^{64}Ni
5 (*a*) $^{1}_{0}n$ (*b*) $^{1}_{1}H$ (*c*) $^{4}_{2}He$ (*d*) $^{0}_{-1}e$ (*e*) $^{0}_{1}e$
6 69.72
7 69.204 per cent
8 (*a*) 27 (*b*) 15
9 (*a*) $^{14}_{7}N + ^{1}_{0}n \rightarrow ^{14}_{6}C + ^{1}_{1}H$ (*b*) $^{10}_{5}B(n, \alpha)^{7}_{3}Li$
10 (*a*) $^{14}_{7}N + ^{4}_{2}He \rightarrow ^{17}_{8}O + ^{1}_{1}H$ (*b*) $^{7}_{3}Li + ^{1}_{1}H \rightarrow ^{4}_{2}He + ^{4}_{2}He$ (*c*) $^{9}_{4}Be + ^{4}_{2}He \rightarrow ^{12}_{6}C + ^{1}_{0}n$
11 2 neutrons
12 (*a*) 9×10^{13} J (*b*) 3×10^{4} y
13 (*a*) 1.66×10^{-24} g (*b*) 1.66×10^{-27} kg (*c*) 1.49×10^{-10} J (*d*) 9.31×10^{8} eV (*e*) 931 MeV
14 The mass of the nucleus is *less than* the sum of the masses of its constituent nucleons, since some of the mass energy of the separate nucleons is being used to bind them together
15 $B = \{Zm_p + (A - Z)m_n - M\}c^2$
16 (*a*) 0.0305 u (*b*) 28.5 MeV (*c*) 7.10 MeV nucleon^{-1}
17 (*a*) (*i*) $^{57}_{26}Fe$ (*ii*) $^{57}_{26}Fe$ (*b*) $^{57}_{26}Fe$ (*c*) $^{57}_{26}Fe$
18 (*a*) A minimum (*b*) B and C
19 (*a*) 2.0160 u (*b*) (*i*) 0.0024 u (*ii*) 2.2 MeV (*c*) 2.2 MeV (*d*) 2.2 MeV (*e*) 2.2 MeV
20 (*a*) α: upwards; β: downwards; γ: undeflected (*b*) β
21 (*a*) (*i*) α, β, γ (*ii*) γ, β, α (*b*) If the particles are producing the largest number of ions per unit distance travelled, they are consuming their energy at the fastest rate and therefore penetrate the least distance before losing all their energy
22 (*a*) $^{214}_{82}Pb$ (*b*) $^{19}_{9}F$ (*c*) $^{210}_{83}Bi$
23 (*a*) $^{14}_{6}C \rightarrow ^{0}_{-1}e + ^{14}_{7}N$ (*b*) $^{217}_{85}At \rightarrow ^{213}_{83}Bi + ^{4}_{2}He$ (*c*) $^{34}_{17}Cl \rightarrow ^{34}_{17}Cl + \gamma$ (*d*) $^{228}_{90}Th \rightarrow ^{4}_{2}He + ^{224}_{88}Ra$ (*e*) $^{45}_{20}Ca \rightarrow ^{0}_{-1}e + ^{45}_{21}Sc$
24 (*a*) $^{241}_{95}Am \rightarrow ^{4}_{2}He + ^{237}_{93}Np$ (*b*) $^{90}_{38}Sr \rightarrow ^{90}_{39}Y + ^{0}_{-1}e$ (*c*) $^{137}_{55}Cs \rightarrow \gamma + ^{137}_{55}Cs$ (*d*) $^{198}_{79}Au \rightarrow ^{198}_{80}Hg + ^{0}_{-1}e$
25 (*a*) The additional neutrons enable the protons in the nucleus to be spaced more widely apart, thereby reducing the electric repulsion between the protons. (They also provide additional short-range forces—between protons and neutrons and between neutrons and neutrons—which are attractive forces) (*b*) (*i*) β^- particle (*ii*) α particle
26 (*a*) $^{10}_{6}C \rightarrow ^{10}_{5}B + ^{0}_{1}e$ (*b*) $^{38}_{20}Ca \rightarrow ^{38}_{19}K + ^{0}_{1}e$ (*c*) $^{101}_{48}Cd \rightarrow ^{101}_{47}Ag + ^{0}_{1}e$
27 (*a*) β^+ particle (*b*) β^- particle